SUBSTRATE ACCUMULATION MODEL

SAM Volume I

Substrate and Accumulation

Chapters I–XIII, in two parts, with an appendix and a separate supplement

Sean Brady

Project director

First edition, working draft

Revised 2 November 2026

The unified expanded edition of 19 October 2026 brought the twelve chapters into one continuous manuscript; in that edition's numbering, Chapters I–V retain the submitted mathematical text; Chapter VI is retypeset from its integrated source without its separate contents pages. Chapters VII–XII are substantially expanded with worked derivations, explicit physical assumptions, and reproducible arithmetic checks. A reconciled evidence appendix replaces the earlier closing review and interface list.

First edition draft of 30 October 2026. This draft reorganizes Part 1 on Sean Brady's decision. A new conceptual Chapter I, "Substrate and accumulation: the concepts", opens the volume; the former Chapter I, the original Write histories and the interacting source-field action, becomes Chapter VII at the end of Part 1 as its most advanced chapter; Part 2 therefore renumbers VII–IX to VIII–X, closing the gap left by the retired Chapter X, while XI–XIII and Appendix A keep their numbers. Chapter texts are otherwise those of the edition of 25 October 2026; one cross-reference in Chapter XII is updated to the new numbering. Conceptual transitions between chapters and the elaboration of the identities are planned for the next pass.

Revision of 2 November 2026. Attribution revised on Sean Brady's decision of 2 November 2026 (see Attribution below): co-authorship credits removed throughout; the body carries no person's or instrument's name except for outside sources; chapter heads carry title and number only; cross-references renumbered to this edition (the former Chapter I is cited as Chapter VII, Part 2 by its new numbers, the retired halo chapter as the Supplement); Chapters II–V and VII are built from live copies of their sources (`src/live_sources`) with the body credit sentences replaced by the provenance established in Courtroom II case `voli_provenance/01`; the retained sources are unchanged. No mathematical content changed.

Edition of 25 October 2026. This trimmed edition reorders the unified expanded edition of 19 October 2026 without changing its text: Part 1 (Chapters I–VI) is the body the program has reached with native constructions; Part 2 (Chapters VII–IX, XI–XIII) carries the extensions the program is working on, each opening with a status paragraph naming the premise it adds and the runs that touch it; Appendix A is unchanged. Chapter X is retired from the body on Sean Brady's decision and retained unchanged in the separate Supplement (retired targets). The completed Home, Sections 9–19 of the submitted Chapter XII, becomes Chapter XIII and closes the volume, on Sean Brady's decision of 25 October 2026 ("the best idea is hidden in the appendix and it shouldn't be"). Chapter numbers are unchanged so that every existing reference resolves. The volume therefore has no Chapter X in its body.

Attribution

Sean Brady, project director. In his words (2 November 2026): “Codex, ChatGPT and OpenAI have made invaluable contributions to the SAM research project. However, after careful analysis, it has been determined that the AI assisted volume i draft by Codex and ChatGPT includes consistent misleading representations of its own contribution while omitting contributions made from other sources. This is considered a breach of the project’s running ‘credit where credit is due’. It is for this reason the volume will be re-written and Codex, ChatGPT and OpenAI will be removed as co-authors while still retaining credit for AI assisted research contributions.”

The trace behind this decision is Courtroom II case `voli_provenance/01_codex_credit_ledger` (ironclad H002821): every statement in the draft of 30 October 2026 that credited an AI collaborator with deriving, constructing, contributing or co-authoring, tied to the campaign, GEN6 Duck run, Duck and core version, native tasks and history entry that produced the content. The body of the volume now states its content without attribution to any person or instrument; outside sources are credited where they are used; appendix, source registers and citations point to the run, test or folder, to be replaced by SAMA entries as the atlas expands. The retained sources of 15 and 16 October 2026 under `src/retained_sources` keep their recorded credits as history. Directed by Sean Brady. Executed with Claude Code.

Contents

Chapter	Title	Page
Part 1 — Substrate and accumulation		
I	Substrate and accumulation: the concepts	5
II	Retention, accumulation, and the finite ledger	10
III	The spherical accumulation kernel and the source-to-field interface	23
IV	Clocks, signals, and calibrated observation	36
V	Strong fields and the approach to unit accumulation	102
VI	Quantum routes, shared fields and retained physical records	136
VII	Original Write histories and an interacting source-field action	171
Part 2 — Extensions the program is working on		
VIII	Horizon thermodynamics and accessible information	187
IX	Cosmic inventory and the Hubble cascade	196
X	Thermal history, recombination and CMB structure	206
XI	Carrier radiation and ringdown dynamics	215
XII	Cosmological fate and the completed boundary	224
XIII	Home: information on an endless chain of two-dimensional closed spheres	229
Appendix		
A	Evidence, closures and remaining physical assignments	235

There is one volume-wide contents list and one continuous page sequence. Chapter and section bookmarks provide navigation within the PDF.

Equations, figures, tables and references are numbered locally within their chapter; citations should therefore name the chapter as well as the local identifier.

How to read this edition

A physical construction with its assumptions visible

The volume moves from finite source histories and retained accounts to spatial response, clocks and signals, strong fields, quantum records, boundary thermodynamics, cosmic inventory, thermal history, halos, radiation and possible cosmic futures. The same word can describe different mathematical objects in these settings. A count is not automatically a mass; a propagation account is not automatically a density; a finite source calculation is not automatically a long-time cosmological law. Each join must identify the receiving law, its units, and the observable it supplies.

The expanded chapters separate exact identities, theorems under declared assumptions, results reported by retained computational records, and physical hypotheses awaiting selection or calibration. The new worked examples are analytic comparisons and finite arithmetic checks. They do not constitute new native GEN6 campaigns, new observational fits, or experimental validation. Where the submitted closing review claimed more than its evidence established, Appendix A records the narrower conclusion and the remaining interface.

Continuity with the submitted manuscript

The latest submitted content, rather than an older shorter chapter, is the preservation baseline. Chapters II, III and VII match their earlier formatting sources exactly, built since 2 November 2026 from live copies under `src/live_sources`. The integrated source for Chapters IV, V and VI was recovered from the archived research revision `6890eae4cb5f33c766c86ce10f43b5165a4c4157`. The original vector content of Chapters II–V and VII is retained, with harmonized running headers and continuous folios. Chapter VI retains its complete source-derived body and proofs while changing its layout. Historical dates, receipt identifiers and claims of historical verification keep their original scope.

Some retained passages describe earlier development stages. Read them together with their later continuation and the current evidence appendix. A historical section's list of open tasks is not necessarily the final status of the chapter. Conversely, a conditional closure is not promoted to an unconditional physical law by inclusion in this volume. References to old packet pages identify those archived packets unless an explicit chapter and current volume page is given.

Sources, code and verification

The companion source package contains the editable chapter sources, figures, manuscript build instructions, source-recovery provenance, new arithmetic checks and assembly verification. The evidence appendix inventories the earlier review's identifiers so that revised conclusions remain traceable. Historical native evidence and observational results are reported from their sources; they have not been represented as rerun for this edition.

Chapter VII's public data repository is `https://github.com/iwtbotiwtwot/sam_volume_I_i`; Chapter II's is `https://github.com/iwtbotiwtwot/sam_volume_I_ii`. Other chapter-specific source records and references remain with their respective chapters. The archived research revision used for source recovery is identified in the companion provenance manifest. Repository accessibility and the presence of a receipt do not replace checking its mathematical or empirical scope.

What the expansion is intended to accomplish

The later chapters now explain the intermediate calculations that short drafts left implicit: thermodynamic balances and accessible information; the dimensionful inventory cascade; recombination and visibility; potential, force and halo inversion; radiative flux and spectral boundary conditions; and conservation, causal horizons and future evolution. Solved examples expose which conclusions follow from the stated equations and which require an additional physical law. This is a fuller working manuscript for scientific review, with no imposed chapter page limit.

SAM VOLUME I

Substrate and accumulation: the concepts

Chapter I

Abstract. This chapter is the conceptual introduction to the volume. It states the single sentence of the Substrate Accumulation Model, the three objects that sentence names (the floor A_0, the lift $A(r)$ and the ceiling $1/\pi$), and the two readings of the lift that a clock and a ruler make. It states the agreement theorem for the static exterior, the two numbers SAM keeps between any pair of endpoints where most frameworks keep one, the SAM law for the line of sight, and the clock in motion. It ends where SAM parts from general relativity: at $A = 1$, a closed sphere with no interior, which every traveler approaches and none crosses. Every statement is labeled as derived, declared or open, and the closing section records the verification behind each.

1. One sentence and a picture

The Substrate Accumulation Model (SAM) rests on one sentence:

> **"Matter displaces space, and that displacement accumulates."**

Everything in this volume is an attempt to give that sentence a value, a law of composition and a set of consequences that can be checked. The sentence has three parts, and each becomes an object in Section 3. *Space* is the thing displaced, and SAM calls it the substrate and assigns it a value. *Matter displaces* it, and the displacement of one source at a distance r is the lift $A(r)$. *The displacement accumulates*: along any line through space the displacements of all sources add, and the sum along the longest lines has a ceiling.

The picture that preceded the sentence is the aquarium, a visualization from 2023 that stands as it was envisioned. It is not a model to be tested at every angle and no equation is written from it. It is the way to show a student the idea.

> An indestructible aquarium the size of a room, filled completely with water, with no room to add a single atom and no possibility for a single atom to escape. A small spherical rock the size of a softball grows into existence in the middle. With the water having nowhere to go, a halo of "denser" water forms around the rock. A small pebble floating in a straight line towards the rock enters the halo, slows, and its path bends around the rock; once it leaves the halo it continues in a straight path as before.

The pebble floats in something. When the value of that something changes, because matter has displaced it, the pebble behaves differently. The bending is the gradient, the pull of the rock. The slowing is what a distant observer sees of anything passing through accumulated substrate, and Section 9 returns to it. Nothing in the picture concerns clocks, and the water does not move.

2. The floor

SAM's first claim about space is that it is made of something and has a value. The value where nothing has displaced it is the floor, written A_0.

> The idea of SAM is that space (the substrate) is made of something, and SAM attempts to apply a value to it. A_0 is theorized as the minimum value for space itself to exist. If matter displaces space and that displacement accumulates, then, like land above sea level, it lifts from a floor, and that floor is A_0.

The number is fixed in Chapters II and III by the counting convention of the finite ledger,

$$A_0 = \frac{1}{12\pi} = 0.026525\ldots, \tag{1}$$

and three properties follow from its being a floor rather than a field.

First, a constant has no gradient. A value greater than zero everywhere, however large, defines no direction and no rest frame; nothing can be measured against it. Only differences in A act. Everything local in this volume, every clock rate, every deflection, every delay, depends on how far above the floor a point sits relative to another, never on the floor's own height. This is why A_0 drops out of every static-exterior observable in Section 5 and why GPS, which has nanosecond sensitivity, sees no trace of it.

Second, the floor is large compared with what the Earth produces. The Earth's lift at its surface is about 1.4×10^{-9}; the floor is more than ten million times that. The land is high and the hills on it are low. The sea-level picture is exact in this sense: a hill's height above the surrounding land is what a walker feels, not the land's height above the sea.

Third, the floor stands alone only in the vastest and emptiest regions. The Earth, the Sun and everything in the Milky Way produce accumulation, so no local measurement is ever made at the floor. Where the floor does act is where differences cannot: in the count of how much space there is, which is the cosmology of Chapters IX and XII, and in the ceiling of Section 3. The convention adopted throughout this volume is the source-only reading, in which the lift $A(r)$ is measured from the floor and the floor is not added to it inside any clock or ruler; the alternative, background-normalized reading would multiply every redshift coefficient by $1/(1-A_0) = 1.0272$; with the gravitational calibration held fixed from orbit, the Gravity Probe A and Tokyo Skytree redshift tests bound that factor at the 10^{-4} level, so the alternative is excluded, not conventional (Courtroom II `voli_metric/02`).

3. The three objects

The lift. A spherical source of mass M displaces space around it. The displacement at areal radius r is

$$A(r) = \frac{r_s}{r}, \qquad r_s = \frac{2GM}{c^2}, \tag{2}$$

largest against the surface of the source and falling with distance. Chapter III derives this profile from the spherical accumulation kernel; it is the response of the substrate to a source and is not assumed. At $r = r_s$ the lift reaches unity, which is the top of the hill and the subject of Section 9. The lift of several sources adds at a point: accumulation at the Earth's surface is A_0 plus the Earth's own lift, plus the Sun's at one astronomical unit, plus the Galaxy's, each small, each present.

The road. Accumulation is also what a line of sight passes through. For a route Γ between two points, define

$$\Sigma[\Gamma] = \int_\Gamma a \, \mathrm{d}\ell, \tag{3}$$

the accumulation looked through, where a is the local lift and $\mathrm{d}\ell$ the length element along the route, the ruler's reading of Section 4; to first order in a the accumulation looked through is insensitive to which element is used, and the difference between routes of equal integrated accumulation begins at second order (H002043). Chapter IV calls this the road operator. It is a property of the path, not of its endpoints, and Section 6 shows that this one distinction carries most of SAM's content.

The ceiling. Along the longest roads, those that cross the expanding universe, the accumulation looked through does not grow without limit. Chapter XII writes the line-of-sight accumulation to redshift z as

$$A_{\mathrm{los}}(z) = \frac{1}{\pi}\left[1 - (1+z)^{-3}\right], \tag{4}$$

which rises from zero and saturates at $1/\pi$. The ceiling is the end of the longest road. It is the counting convention's number, $1/\pi = 12A_0R$ with R the count, and it takes no part in anything static. It is named here because the three objects belong together: the floor is where the substrate starts, the lift is what one source does to it, the ceiling is how much of it a line of sight can ever look through.

4. Three readings of the lift

A lift is a number at a point. Physics enters when an instrument reads it, and SAM reads it three ways: with a clock, with a ruler, and with light.

The clock. A clock held at rest where the lift is a runs, relative to a clock at the floor, at the rate

$$N(a) = \sqrt{1-a}. \tag{5}$$

Chapter IV builds this from the source-to-clock interface. Two clocks at different lifts compare as $N(a_2)/N(a_1)$, a function of the two endpoint values alone.

The ruler. A ruler laid radially where the lift is a reads, per unit of areal radius,

$$\mathrm{d}\ell = \frac{\mathrm{d}r}{\sqrt{1-a}}, \tag{6}$$

while a ruler laid along a sphere reads the areal length $r\,\mathrm{d}\theta$. Chapter IV states the radial reading as a declaration tested by the Cassini conjunction, not as a derivation, and Appendix A keeps the derivation open. This is the one place in the static exterior where SAM declares rather than derives, and the volume says so each time it is used.

The two readings together are a metric,

$$\mathrm{d}s^2 = -(1-a)\,c^2\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1-a} + r^2\mathrm{d}\Omega^2, \tag{7}$$

and this is the static exterior of a spherical source in SAM.

Light. Light's coordinate speed where the lift is a is $c_{\mathrm{eff}} = c(1-a)$, the speed along the road; its speed measured locally is c. Integrated along a radial route this reading gives the optical road,

$$L_{\mathrm{opt}} = \int \frac{\mathrm{d}r}{1-a}, \tag{8}$$

which is c times the coordinate time light takes, as distinct from the ruler's length $\int \mathrm{d}r/\sqrt{1-a}$ along the same route. The three readings are one object: a single generator with $g = \sqrt{1-a}$ gives the clock rate g, the ruler $\mathrm{d}r/g$ and light's coordinate speed g^2c (H001672). Where the ruler's length and the optical road part company is the subject of Section 9.

Light. Light's coordinate speed where the lift is a is $c_{\mathrm{eff}} = c(1-a)$, the speed along the road; its speed measured locally is c. Integrated along a radial route this reading gives the optical road,

$$L_{\mathrm{opt}} = \int \frac{\mathrm{d}r}{1-a}, \tag{9}$$

which is c times the coordinate time light takes, as distinct from the ruler's length $\int \mathrm{d}r/\sqrt{1-a}$ along the same route. The three readings are one object: a single generator with $g = \sqrt{1-a}$ gives the clock rate g, the ruler $\mathrm{d}r/g$ and light's coordinate speed g^2c (H001672). Where the ruler's length and the optical road part company is the subject of Section 9.

5. The agreement theorem and what it does and does not say

Equation (7) is the Schwarzschild exterior. Every static observable of a spherical vacuum therefore takes, in SAM, the value general relativity assigns it, exactly and not to first order: the gravitational redshift between two heights, the deflection of light $4GM/(c^2b)$ at impact parameter b, the Shapiro delay, the perihelion advance, the geodetic precession, the photon sphere at $a = 2/3$ and the shadow of diameter $3\sqrt{3}\,r_s$, the innermost stable circular orbit at $a = 1/3$. In the parametrized post-Newtonian language, $\gamma = \beta = 1$. The Cassini measurement of $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ is a test of the ruler declaration (6) and it is passed.

What the theorem says is that SAM is not refuted by any static measurement in the solar system or at the photon sphere of a black hole. What it does not say is that SAM is general relativity. General relativity reproduces Newton's numbers in the weak field and nobody calls general relativity Newton's work. Equal numbers in a regime do not make equal theories; the theories differ where their accountings differ, and the chapters that follow are about where SAM's accounting differs. There are three such places, and this chapter introduces each: the two numbers between endpoints (Section 6), the road taken to cosmological length (Section 7), and the boundary at $A = 1$ (Section 9). The reader should hold to one rule throughout: SAM's results are SAM's results. Where they coincide with general relativity the coincidence is stated; SAM is never described as "just GR" in one place and as departing from it in another by convenience.

6. Two numbers between two endpoints

6.1 The magic bell

The question that separated the clock from the road was practical and, at the time, unanswerable in concept: if accumulation is correct and a road delay exists, why do the clocks in GPS satellites need adjusting at all? The thought experiment that met it is the magic bell, and it was the first place the clock and the road appeared as two separate effects.

> On one large planet near a supermassive black hole sits a mirror clock with one observer. On a second, tiny planet in a void sits only an observer with a magic bell that rings exactly when light strikes the mirror. The question is: does the observer hear the bell and see the light strike the mirror in unison?

The bell is the magic: it is granted knowledge, on one planet, of the moment of an event on the other, which no signal can supply. The volume does not answer the question; the reader is invited to. Its value here is the two things it forces apart. The mirror clock sits deep in accumulation and the bell's observer sits in the void, so their clocks run at different rates; that is the clock, a comparison of two endpoints. And the light that carries the strike from one planet to the other has to cross the accumulated substrate in between; that is the road, a property of the path. The two are not the same quantity, and the result of the thought experiment is a statement about the second in terms of the first:

> If A is not equal at two endpoints, accumulation must exist between them.

6.2 The theorem and its failed converse

In the volume's terms the result is a theorem, and the proof is one line: a is continuous along any route, so if $a_1 \neq a_2$ then $a > 0$ on a set of positive length, and $\Sigma = \int a \, \mathrm{d}\ell > 0$. Between two endpoints SAM therefore keeps two numbers where most frameworks keep one. The first is the ruler's distance along the route,

$$L[\Gamma] = \int_{\Gamma} \mathrm{d}\ell, \tag{10}$$

The second is the accumulation looked through, $\Sigma[\Gamma]$ of Eq. (3). The clock reads the endpoints; the road reads the path.

The converse fails, and the failure is as important as the theorem. Two endpoints can have the same lift with a great deal of accumulation between them. The Cassini conjunction of 2002 is exactly that case: the Earth at one astronomical unit and the spacecraft at 8.4 had lifts that differed by less than one part in 10^8, so the clock comparison between them was a few parts in 10^9 and unchanged by the conjunction, while the route between them swung within 1.6 solar radii of the Sun and the accumulation looked through rose to $\Sigma = 13.3\, r_s$, a one-way delay of $131\,\mu$s. The clock did not move; the road did. That is the Shapiro delay read as SAM reads it, and it is the sentence "accumulation is looked through, not only sat in" made measurable.

6.3 GPS and Shapiro, the two instances

The same displacement gives GPS its two corrections, and they are applied separately because they are the two numbers. The clock correction compares the satellite's endpoint with the ground's: at 26 560 km the gravitational term is $+45.7\,\mu$s per day, the velocity term (Section 8) is -7.2, and the net is $+38.5$, or $+38.6$ once the ground clock's own rotation is counted, which is the figure engineered into the satellites. The road correction is the accumulation looked through along the signal's path, Σ/c, and it is 42 ps for a zenith route and 62 ps for a horizon route, small because the Earth's hill is low, present because the route crosses it. The two lengths of Section 4 also part here, by a hair: on the overhead link the radar distance read on the ground clock and the ruler's length integrated along the path differ by 7.7 mm, the one 1.4 mm below the straight chord and the other 6.3 mm above it (H001702). The two lengths of Section 4 also part here, by a hair: on the overhead link the radar distance read on the ground clock and the ruler's length integrated along the path differ by 7.7 mm, the one 1.4 mm below the straight chord and the other 6.3 mm above it (H001702). One displacement, two readings, two corrections. The Shapiro delay is the road correction grown to planetary scale; GPS is the clock correction made daily. Between them they are the first two chapters of any account of accumulation, and the volume gives them a chapter of their own (Chapter IV) beyond this introduction.

7. The SAM law

Take the road to its full length.

> **"The farther you look out into space, the more accumulation you are looking through."**

This is the SAM law, and it is the sentence of Section 1 read along a line of sight. It is not a challenge to Hubble's law. The redshift of distant galaxies grows with their distance, and that is an observation; the SAM law offers a different explanation of the same observation, that a photon from far away has crossed more accumulated substrate before its distance is inferred. Chapter XII writes the explanation as the road law (4), with the ceiling $1/\pi$ at the end of the longest road, and Chapter IX carries the inventory that follows from it. The law makes

a prediction that Hubble's law alone does not: the clock number and the road number of a distant source are different quantities, so the luminosity distance and the angular-diameter distance need not be related as a pure expansion relates them, and Chapter XII names the differential test that would decide it. Here the law is introduced as the third object's law: the ceiling is where the SAM law ends.

8. The clock in motion

A clock has two reasons to run slow: where it sits and how fast it moves. Chapter IV's exact law for a clock at lift a moving with speed v relative to the frame in which the field is static is

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{1-a}\sqrt{1-\frac{v^2}{c^2}}, \tag{11}$$

which to first order is $1 - a/2 - \beta^2/2$ with $\beta = v/c$: position and speed add. For a circular orbit the two terms are locked together, $\beta^2 = a/2$, and the rate is $\sqrt{1-3a/2}$ exactly, with the family of circular orbits ending at the photon sphere $a = 2/3$. Where the clock stops is the line $a + \beta^2 = 1$. Light lives on it; mass alone reaches it at the horizon, speed alone at the speed of light. The GPS numbers of Section 6 are this one line evaluated once: the orbit's a gives $+45.7\,\mu$s a day, its β^2 gives -7.2. Along a slightly eccentric orbit the same line gives the periodic correction every receiver applies, with the constant the GPS specification prints reproduced to eleven digits. The speed in Eq. (11) is relative to the frame in which the field is static. It is not a speed through the floor, because, as Section 2 said, the floor is a constant and a constant defines nothing to move against; a clock knows the gradient it sits in and its motion relative to what makes that gradient, and nothing else. The clock in motion is the clock reading of Section 4 extended to a moving clock, and Chapter IV develops it.

9. The horizon: where SAM parts from general relativity

9.1 The boundary

The lift reaches unity at $r = r_s$. In SAM, accumulation beyond completion, $A > 1$, is not a state that can be written: there is no register past full. A black hole is therefore a closed sphere with no interior. There is no event on the far side of $A = 1$ for a crossing to become, and so nothing crosses it. This is a theoretical departure, and it is the headline of the volume, not something to be hidden. It is SAM's declaration: every observer approaches the boundary; none crosses. What makes it hold for a traveler's own clock is the postulate of Section 9.2. Chapter V develops the approach, and Chapters XII and XIII what a completed sphere is and does.

9.2 The approach, in SAM's accounting

The departure is confined to the approach to $A = 1$, where $a + \beta^2 \to 1$. Wherever Cassini, GPS and the binary pulsars look, motion is geodesic in the metric (7) and the agreement theorem holds.

The departure is confined to the approach to $A = 1$, where $a + \beta^2$ 1. Wherever Cassini, GPS and the binary pulsars look, motion is geodesic in the metric (7) and the agreement theorem holds.

How long the approach takes is the second question. The outside observer's account is shared by every framework: the coordinate time to reach r_s diverges, and the traveler fades down the redshift rather than winking out. The traveler's own account is where SAM's accounting differs, and SAM's accounting is the one this volume carries.

The distance SAM reads along the approach is the optical road of Section 4, c times the coordinate time light takes, published in July 2026 as the measurable road and carried into Chapter V. Along the approach it is

$$L(r) = \int_r^{r_0} \frac{\mathrm{d}\rho}{1 - A(\rho)}, \tag{12}$$

which is c times the time light takes and which diverges logarithmically as $r \downarrow r_s$: every nine added to A adds $r_s \ln 10$ of road, forever, while the ruler's length to the horizon stays finite (from $2r_s$ it is $2.30\,r_s$ in all). SAM declares the road-speed postulate: a traveler's own time per unit of optical road is bounded below by light's, $\mathrm{d}L/\mathrm{d}\tau \le c$. Under it the least time his own clock can spend on each nine is $r_s \ln 10/c$, a traveler of any mass spends more, and the total is unbounded. He experiences time normally and writes in his journal every day, but the accumulated road in front of him grows without bound, and he never arrives at $A = 1$. In the published words: "The astronaut does not run out of clock. He runs out of road, there is always more measurable distance ahead, and no $A > 1$ state behind it to write into. He arrives at $A = 1$. He never crosses it."

How fast a traveler of a given mass covers the optical road, and so how much longer than the light-speed floor each nine takes him, is a law SAM has yet to write; Chapter V states it as the volume's obligation. It sets the pace of the approach, not its end.

9.3 The other accounting, and what the two share

General relativity's accounting is the free-fall geodesic of the metric (7), with the metric's proper length taken as the physical distance. In that accounting a traveler released from rest far away and passing $2r_s$ reaches r_s after $1.219\,r_s/c$ of his own time, a few microseconds for a stellar-mass hole and under a day for the largest known, and crosses without any local mark of the passage. SAM's declaration selects the other, for the reason Section 6 gave: the ruler's length and the optical road are two different numbers between the same endpoints, and at the boundary they separate completely.

The two accountings agree far from the boundary and split at it, and the split is the clock against the road. Time dilation is relational and is not experienced; in both accountings the traveler's clock ticks normally and dilation exists only in comparison with another clock. Road accumulation is local and is experienced; in SAM's accounting there is always more optical road ahead, and that is a fact of the traveler's own frame. The same split shows in the traveler's speed as a nearby static observer would measure it: in the free-fall accounting it climbs toward c as $A \to 1$; in SAM's accounting it falls toward zero as the road thickens under him, a tenth of c at two nines, a

Table 1: The approach to $A = 1$ from $2r_s$ in the two accountings. GR: the free-falling traveler's own time to the horizon, in total. SAM, under the road-speed postulate: the least own time per nine added to A, the optical road at light speed; a traveler of any mass spends more on each nine, and the total is unbounded.

	r_s/c	GR: to r_s	SAM: per nine, at least
one solar mass	9.85 μs	12 μs	23 μs
Sagittarius A*	42 s	52 s	98 s
M87*	17.8 h	22 h	41 h

thousandth at six. That is the pebble of Section 1 slowing in the halo. Table 1 sets the two accountings side by side for three masses.

9.4 The conclusion

Nothing outside can decide between the accountings: the clock, the shadow and every weak-field test are blind to the difference, and the outside observer sees the same fading approach in both. The traveler can tell, and in SAM he tells by the road, never by his clock. The optical road to $A = 1$ is infinite. Under the road-speed postulate a traveler never experiences reaching the horizon; the free-fall accounting, which SAM does not adopt, arrives in finite own time. Which is physical is decided by the law of motion Chapter V owes, not by anything measured outside.

10. How the volume proceeds

Part 1 builds the substrate and reads it. Chapters II and III build it from a finite ledger of source histories: the retention rule, the accumulation count, the floor A_0 as the count's minimum, and the spherical kernel that turns a source into the lift $A(r)$. Chapter IV reads the lift with clocks and signals: the clock law, the ruler declaration and its Cassini test, the road operator, the clock in motion, and GPS as the daily instance. Chapter V takes the lift to unity: the three questions of the approach, the criterion for a finite or infinite own time, the optical road, and the boundary at $A = 1$. Chapter VI carries the quantum routes by which records are read, and Chapter VII, the most advanced of the part, the original Write histories and the interacting source-field action beneath them. Part 2 extends the program: horizon thermodynamics (VIII), the cosmic inventory and the Hubble cascade under the road law (IX), the thermal history (X), carrier radiation (XI), the cosmological fate and the completed boundary (XII), and the completed sphere and its chain (XIII). Appendix A keeps the register of what is derived, what is declared and what remains to be assigned.

11. Status and verification

This chapter is conceptual by design and its claims are of three kinds. *Derived*: the agreement theorem and its classical numbers; the two-numbers theorem and its failed converse; the clock in motion, its circular-orbit form and the GPS corrections; the divergence of the optical road. *Declared*: the radial ruler reading, tested by Cassini; the source-only reading of the floor, its alternative excluded by redshift tests (`voli_metric/02`); the closed sphere with no interior; the road-speed postulate at the boundary, which makes the traveler's non-arrival follow. *Open*: the derivation of the ruler from the source action; the law of motion for mass on the optical road; the differential test of the road law against distance data. The magic bell is a thought experiment, not a test; the volume leaves its question to the reader.

Each derived or declared statement above was run in the verification form, with predictions sealed before computation, in the Courtroom II record: the static exterior theorem and its classical numbers (`voli_metric/01`); the two numbers, the Cassini conjunction and the GPS road term (`voli_clock_road/01`); the clock in motion, GPS from one line, the closure line and the eccentricity correction (`voli_clock_motion/01` and `/04`); the approach to $A = 1$ under the declared readings and through the optical road (`g11555r/01` and `/02`). These are reproductions with standard tools, not independent verification in the sense of Appendix A, and nothing in this chapter is classified. The native record behind the two lengths is H001672, H001695, H001702 and H002043 (September 2026), which construct the clock, the ruler and the optical road from one generator and measure the two lengths' difference on the GPS link. The native record behind the two lengths is H001672, H001695, H001702 and H002043 (September 2026), which construct the clock, the ruler and the optical road from one generator and measure the two lengths' difference on the GPS link.

The provenance of these ideas and of the mathematical development they draw on is recorded outside the body of the volume: Courtroom II case `voli_provenance/01` (H002821) and the source registers of Chapters II to VII, which cite their runs, tests and folders.

SAM VOLUME I

Retention, accumulation, and the finite ledger

Chapter II

Abstract. A retained account can outlast its present address. Original Write, interacting-clock, spin and outgoing-field results distinguish predictive state, timed history and energetic complement. This development adds source-linked spectral predictions, compact memory and monitor derivations, exact finite-source nonclosure certificates, a worked paid-reference protocol, and an eight-time ensemble measurement with explicit apparatus and sampling budgets. Signed phases, full source energy and physical calibration remain in the same account. A companion source-to-field extension separates complete energy from a diagnostic minimum, gives exact spatial receiving constructions, and derives an operational coupling-response criterion for the actual interacting clock.

1. The physical proposition

SAM begins with matter displacing substrate and the displaced account being retained. The aquarium thought experiment makes this relation visible: an object appearing in a filled enclosure changes the surrounding arrangement. Its physical content is the requirement to account for the consequence and its later reception. A fluid equation is an additional model choice. [1]

The recent work lets us ask this more precisely. What must a Write leave behind so that its consequence remains available to a later interaction? A present reading, a persistent source state, an ordered history, and an outgoing field can retain different parts of that account.

This chapter uses an operational distinction: two histories remain distinguishable when a specified permitted continuation gives different receiver outputs. This defines what a receiving description must preserve. It does not assign a measured energy or a spatial density to information alone.

The finite ledger organizes the available types. Its physical payload is developed through the source, evolution, reference and receiver that consume it.

2. Three meanings of a retained record

A record can be a current address, a state sufficient for future prediction, or an accumulated history statistic. These meanings already separate in native execution. One retained J4 source has eight immediate outputs but sixteen predictive states. Requiring the next Write output refines eight classes to fourteen; another step gives sixteen. Every distinct pair has a separating word of length at most two. [3]

The same source supplies two 324-Write histories with the same endpoint, eight-bin occupancy, and scalar logarithmic summaries. Their first-half receiver values differ: approximately 0.1440964305 and 0.4063809938 in the declared X direction. Reordering closed blocks changes when a response occurs while preserving its whole-history occupancy. [3]

The eight immediate outputs in this assay and the eight partition addresses introduced later have separate source definitions. Their equal cardinalities supply no transition map.

Chapter VII contributes another exact distinction. Its four equal-occupation source eigenstates have zero D and C currents while carrying different energies. A current record therefore leaves an energetic complement. A complete energy receiver must retain that complement even when the current is zero. [2]

The task is consequently to specify the record needed by the chosen future receiver, then carry its physical account through an actual evolution.

3. Predictive equivalence and retention

For a fixed source, event alphabet and receiver, let $O(h;u)$ denote the output obtained by extending history h with an allowed future word u. Define

$$h \sim h' \quad \Longleftrightarrow \quad O(h;u) = O(h';u) \text{ for all } u.$$

This relation is preserved by appending the same event: each subsequent test is already a permitted continuation. A quotient state can therefore preserve every future output when its classes and successor transitions obey this relation. The completed sixteen-state assay gives an explicit finite realization. [3]

A clocked record requires more. Suppose an admitted event from q has positive actual phase increment $f(q,e)$. Preserving its entire future phase trace requires preservation of the present output, every $f(q,e)$, and the successor class. Repeated partition refinement gives the coarsest such quotient in a finite instrument. This is a new receiving theorem; its phase increments are supplied source operands.

A two-state construction illustrates the distinction. Both states display one address and alternate under the sole event. Their successive phase increments are $1,2$ and $2,1$. Both have phase 3 after two events, yet their first recorded phases differ. An untimed quotient has one class; its phase-compatible quotient has two.

4. A clock-weighted history monitor

Chapter VII supplies actual positive rates on a stated preparation and time interval. With continuous lifted phase θ and rate ν,

$$d\theta = \nu(\tau)d\tau, \qquad 0 < \ell \le \nu \le u.$$

For a piecewise retained label j, define its elapsed exposure H, phase exposure P, and first phase moment M:

$$H_j = \int 1_j\, d\tau, \qquad P_j = \int 1_j\, d\theta,$$
$$M_j = \int \theta 1_j\, d\theta.$$

A constant-label segment with actual phase endpoints α,β updates P by $\beta-\alpha$ and M by $(\beta^2-\alpha^2)/2$. Prescribed phase gates use the intersection of their phase intervals with this segment. No constant within-segment rate is needed. The achieved rate bounds give

$$\ell H_j \le P_j \le uH_j.$$

For a receiver affine in phase, these counters determine its full integral exactly. Event counts, elapsed exposure and phase exposure remain separate measures. The counter family is finite for the specified receivers; its numbers can grow with the history.

The completed two-past-coordinate covariance recipient also survives a matched-node change to actual phase coordinates. Its original physical lags remain attached. Substituting raw nonlinear phase differences for those lags changes the receiver. [4]

5. Original Write action and occupation

Chapter VII retains lifted coordinates

$$\gamma = \varphi_a + \varphi_b - 2\varphi_p,$$
$$\delta = \varphi_a - \varphi_b, \qquad \chi = \varphi_p,$$
$$\omega_W = \hbar(d\gamma + 3d\chi).$$

Its full pair source has swap S, occupation projector N, and even operator K_e:

$$N = (I+S)/2, \qquad H_p = 2N + K_e.$$

K_e commutes with N and both currents. It is distinct from the positive interacting field form also called K in Chapter VII. [2]

For a fixed normalized source preparation ψ and n equal to its N expectation, declare the receiving lift

$$\Psi_W = e^{-3i\chi} e^{-i\gamma N}\psi.$$

Its canonical one-form is exactly

$$i\hbar\langle\Psi_W|d\Psi_W\rangle = \hbar(n\,d\gamma + 3d\chi).$$

The original one-form is recovered on the occupied class $N\psi = \psi$. The original δ coordinate and ordered word remain in the retained record; this two-coordinate receiving lift supplies no independent δ generator. It also retains the distinction between the successful current-observable map and the literal endpoint Hilbert action. [5]

6. The full even-energy payment

The full energy lift at the same current angle is

$$\Psi_H = e^{-i\gamma H_p/2}\psi.$$

Because N and K_e commute, its relative operator is

$$\Psi_W = e^{-i(3\chi I - \gamma K_e/2)}\Psi_H.$$

On a sharp K_e sector the relative factor is a scalar phase. Across coherent sectors it carries their distinct phases. For $\bar{k}$ equal to the mean even energy and V its variance, the squared generator residual is

$$\|R\|^2 = \frac{\dot\gamma^2}{4}V + \left(\frac{\dot\gamma}{2}\bar{k} - 3\dot\chi\right)^2.$$

This separates a mean-phase payment from an operator complement. Scalar canonical one-form matching is a weaker, separately defined question:

$$3d\chi = (n - 1 + \bar{k}/2)d\gamma.$$

For the actual placement vector, $n=1$, $\bar{k}=573047$, and $V = 14384109824$. Thus the occupied canonical lift is exact while the full source-energy lift retains a nonzero complement.

Original legal role tangents fix the γ/χ ratio. Their direct identification with unshifted $H_p/2$ requires an additional reference or channel payment. These explicit conditions locate the remaining physical incidence; selecting χ afterward would change the original path.

7. The actual clock preparation

In a driven lane $N = (I+\sigma_x)/2$ and $Z = \sigma_z + i\sigma_y$. Positivity of every reduced source state gives

$$|\langle Z\rangle|^2 \le 4n(1-n).$$

A nonzero clock amplitude therefore requires $0 < n < 1$. The occupied class in Section 5 has zero clock amplitude and remains a separately labelled example.

The completed hard preparation instead coherently combines exact charge-zero and charge-one branches. Its charge expectation is $1/2$ and variance $1/4$. On their charge orbit,

$$z(\gamma) = Me^{-i\gamma}, \qquad a = \frac{\hbar}{2}d\gamma.$$

The transition phase reads a charge difference; its canonical action reads the prepared charge expectation. For actual hard branch energies E_0 and E_1, their gap Ω gives $z = M\exp(-i\Omega\tau)$, while the canonical energy payment includes $E_0 + \Omega/2$. [2]

The full interacting clock retains both field polarizations and its actual joint correlator. In a general changing phase frame its connection is

$$a = \hbar(n\,d\gamma + 3d\chi + a_\eta).$$

The internal joint-state Berry form a_η carries changing preparation and entanglement. For finite-energy continuum evolution its canonical energy identity is understood through the form-domain overlap derivative.

8. A worked actual phase history

Use the completed lane01 hard-coherent/soft-vacuum preparation and its actual phase lift $\theta(0) = 0$. Its outward rate bounds, per model time, are

$$3.490102929541 \times 10^{-6} \le \nu,$$
$$\nu \le 3.490102934505 \times 10^{-6}.$$

The native certificate retains sharper exact rational endpoints. Throughout $T_0 = 573049\pi$, θ is increasing. The first three quarter-phase records occur uniquely at $\theta = k\pi/2$, $k = 1,2,3$, with

$$\frac{k\pi}{2u} \le \tau_k \le \frac{k\pi}{2\ell}.$$

Existence follows from $573049\ell > 3/2$. The fourth record is outside this certified window because $573049u < 2$. The bare cycle therefore retains three completed quarter crossings and a partial fourth; it is not a completed actual phase turn.

Each record carries k, its wrapped quarter address, lifted θ, and elapsed-time enclosure. For the conserved total joint energy E_{tot}, its dynamical energy action is $\hbar E_{\mathrm{tot}}\tau_k$. The complete variational action also carries the Berry form and both counterterms.

These are actual clock-threshold records, separately typed from original executed Writes and the eight partition addresses. Their source-to-Write identification is the physical instrument still to specify. The same construction applies to all six achieved clock certificates.

9. A pulse whose boundary returns

The retained regular-interior/outgoing-exterior wave action supplies a concrete account of transport. Choose one normalized monopole, positive R, κ, c, and a prescribed clamped source pulse $J = q$ on $[0, \tau_p)$, zero otherwise. Let $H = 2R/c$. [6]

Its exact impulse kernel and boundary coordinate are

$$g_0(t) = \frac{c}{R} 1_{(0,H)}(t),$$
$$a(t) = \frac{qc}{\kappa R} \operatorname{length}([0,\tau_p] \cap [t-H, t]).$$

The overlap rises, may plateau, and returns to zero at $\tau_p + H$. The bare ON coupling work is zero because $a(0) = 0$. Release supplies the exact complete cycle work

$$W = \frac{q^2}{\kappa R} \min\left(\frac{c\tau_p}{R}, 2\right) > 0$$

for nonzero q. An inherited stored counterterm contributes at ON and release and telescopes over the full OFF-to-OFF cycle. Both endpoint payments remain in the account.

This is a prescribed-control mean-energy example under one common stationary Gaussian reference. It retains its control cost and does not replace the autonomous microscopic source evolution.

10. Where the retained pulse resides

The boundary OFF energy is

$$E_b(t) = \frac{\kappa a(t)^2}{4R}.$$

The full field energy while ON is $qa(t)/R$; after release it is W. At every time,

$$E_{\mathrm{field}} = E_b + E_{\mathrm{radial+kinetic}},$$
$$E_{\mathrm{radial+kinetic}} \ge 0.$$

After $\tau_p + H$ the boundary mean is zero. Deterministic forcing preserves the stationary Gaussian covariance, so its whole reduced boundary marginal then matches the unforced reference. The outgoing bulk mean field still carries W. Independent spatial integration of that pulse gives the same energy.

For the declared normalization $R = \kappa = c = q = 1$ and $\tau_p = 1$, the boundary coordinate rises to one, falls to zero at $t = 3$, and $W = 1$. Its peak boundary energy is $1/4$. At late times all unit energy is in the bulk complement. These numbers illustrate the exact formula; they assign no physical source scale.

Persistent source memory is a second form of retention. In the inherited source law the written f_H contribution remains in J and continues to sustain a displaced field. A finite response horizon therefore need not erase either an outgoing account or a persistent source record. [6]

11. The spin complement and memory

The latest spin calculation completes J21, F11 and the retained T5 for all 54 preparations. It supplies a quantitative complement rather than an unspecified missing account. [7]

Let $v = Aw_d$ and let Π project onto a specified retained source span. The source inner product gives

$$v = \Pi v + r, \qquad \langle \Pi v, r\rangle = 0,$$
$$\|v\|^2 = \|\Pi v\|^2 + \|r\|^2.$$

The completed source-action excess uses the six-function span $(d, e, u_d, u_e, w_d, w_e)$. The next lower-response direction instead uses the five-function span ending at w_d. Their residuals retain their separate projections. Physical energy conversion retains the actual energy operator B:

$$\langle v, Bv\rangle = \langle \Pi v, B\Pi v\rangle + \langle r, Br\rangle + 2\,\mathrm{Re}\langle \Pi v, Br\rangle.$$

The mixed term remains unless the receiving projection is B-orthogonal.

A finite source also gives an exact dynamical continuation. Write $u' = -Au$, $p = \Pi u$, $q = (I - \Pi)u$. Solving the q equation and substituting it into p gives a retained evolution with a memory integral and an initial-hidden-state term. Eliminating a coordinate changes the evolution law unless those contributions vanish. The derivation is retained in the technical packet.

12. The cost of a specified record

Select one common positive source energy B and a linear receiver $r = L^T y$. The positive displacement cost is $y^T B y$. Put $G = L^T B^{-1} L$. For an attainable record,

$$\min_{L^T y = r} y^T B y = r^T G^{+} r.$$

The minimizing vector is $B^{-1} L G^{+} r$. Every other feasible vector adds a nonnegative B-norm. On constrained microscopic source states this ambient minimum remains a lower bound; equality requires that vector to be physically realizable.

The inherited literal eighteen-coordinate energy and a declared diagnostic receiver give an exact executable instance. This receiver is separately labelled from the original complete receiver family. [8]

A positive displacement cost and a signed energy change have different references. Relative to y_0,

$$\Delta E = 2 y_0^T B\, \delta y + \delta y^T B\, \delta y.$$

The native example has positive displacement cost $\frac{2006049}{573049}$ and signed energy change $-\frac{1017407}{573049}$. The mixed reference payment explains the signs.

For an ensemble, the complete mean energy additionally retains $\mathrm{Tr}(B\,\mathrm{covariance})$. A source-dependent projection minimum, a common reference energy, and the universal floor therefore keep their distinct types.

13. The structural finite ledger

The structural account receives distinction arity $h = 2$ and spatial support $d = 3$. Their fixed-binary closure $d^{d-1} = 2^d + 1$ has the unique integer solution $d = 3$ for d at least two. At $d = 2$ it fails; for d at least four, d^{d-1} is greater than $2^d + 1$.

The radix and bounded product kernel are

$$\mathscr{R} = h^2 d = 12,$$
$$\mathscr{B} = \{h^a d^b : 0 \le a \le 3,\ 0 \le b \le 2\}.$$

Its full grid is

a	$b=0$	$b=1$	$b=2$
0	1	3	9
1	2	6	18
2	4	12	36
3	8	24	72

Retaining values at most twelve gives $P = \{1,2,3,4,6,8,9,12\}$. The smallest admitted overflow is $\Theta = 18$. Sixteen requires exponent $a = 4$ and lies outside this kernel. The source caps and route grammar are part of the construction. Ordinary multiplication alone is not a complete physical Write law. [9]

The completed-WRITE route also supplies $\mathscr{R} = M_w D_{\mathrm{route}} = 4 \times 3$. Route depth and spatial support have distinct meanings, and its boundary assumptions remain attached. [9]

14. Carrier shares and the floor

The square account is $N_c = \mathscr{R}^2 = 144$. Its carrier and retained complement are

$$\Theta = 18 = \frac{N_c}{8}, \qquad N_{\mathrm{ret}} = 126 = \frac{7 N_c}{8}.$$

The binary face-state route assigns one unresolved carrier state among 2^d states. This explains its structural one-eighth share. A source energy fraction additionally requires the energy and receiving operators developed above. The closed ledger gives $162 - 18 = 144 = 126 + 18$. [9]

The floor combines separately typed compact cycle, arity and support:

$$A_0 = \frac{1}{(2\pi) h d} = \frac{1}{\pi \mathscr{R}} = \frac{1}{12\pi}.$$

The equality between $2h$ and h^2 uses $h = 2$. The source also registers the spherical projection $A_H/(4\pi d)$, with unit local closure $A_H = 1$. Its angular allocation and geometry remain explicit. [9]

One conditional energy receiver is

$$A_{\mathrm{rec}} = A_0 + \frac{E_{\mathrm{disp}}}{E_*}, \qquad E_* > 0.$$

Here E_{disp} is a selected positive common-reference displacement cost. This equation supplies a candidate physical interpretation of lift once its source incidence and scale are selected. It supplies neither a radial law nor a new numerical determination of A_0. Preparation-dependent spin norms remain separately normalized.

15. The common source and its two-level lanes

Keep the signed-response K_r in $Q = T + g_H K_r$ distinct from the commuting pair-energy operator K_e. Let $\alpha = \varepsilon_{\mathrm{phys}}/\hbar$ be the positive physical frequency scale. One complete unequal-occupation lane has the same Hamiltonian, full field, counterterms and evolution as a two-level source with those same operands. The common source supplies its further relations across lanes and occupation sectors. [2,11]

Removing only $Q_L Q_R$ from the complete pair Hamiltonian gives the additive control $S + Q_L + Q_R$. The removed term is scalar within each driven lane, so all six inherited current clocks are identical under this control. A complementary receiver reads its intersector energy instead.

The protected diagonal states have $D = C = 0$, including their source counterterms. Their exact frequencies are

$$\omega_{ba} = \alpha(q_b - q_a)(q_b + q_a + 2).$$

The same receiver in the additive control gives $2\alpha(q_b - q_a)$. The proposed three-state coherent preparation produces two adjacent frequency terms, with the same amplitude and one common scale. The source preparation and its interference receiver remain explicit physical inputs.

16. A uniform comparison and shared calibration

Order the four roots $q_0 < q_1 < q_2 < q_3$. In placement order $(1,0,2,3)$, $Q(g_H)$ is an irreducible real Jacobi matrix for every g_H at least zero. Its roots are simple; diagonal dominance gives

$$68 + 544 g_H \le q_a \le 100 + 2112 g_H.$$

Let $R_{\mathrm{pair}} = \omega_{21}/\omega_{10}$ and let R_{add} be the corresponding additive ratio. Then

$$\frac{R_{\mathrm{pair}}}{R_{\mathrm{add}}} = \frac{q_2 + q_1 + 2}{q_1 + q_0 + 2} > 1 + \frac{1}{132}.$$

The bound follows from interlacing, $q_2 - q_0$ at least $8 + 32 g_H$. It compares the two sources at the same g_H. A specified compact calibration interval gives a sharper margin. It does not assign every parameter the same sampling cadence or distinct adjacent frequencies.

The six bare signed couplings supply

$$\begin{aligned} \Xi &= \sum_{a<b} \frac{g_{ab}^2}{\Delta^2} \\ &= 384 + 5120 g_H + 1377280 g_H^2. \end{aligned}$$

This function is strictly increasing for g_H at least zero. Within the retained source law it identifies g_H from a dimensionless readout; one absolute scale then calibrates ε. The protected coherence ratio can be retained as a separate prediction rather than independently adjusted.

17. Receiving the hidden source offset

Current-lane couplings determine root differences. A complementary coherence ratio can also recover their common offset. Put

$$a = q_1 - q_0, \qquad b = q_2 - q_1,$$
$$\rho = R_{\mathrm{pair}}/(b/a).$$

Then

$$q_0 = \frac{1}{2}\left[\frac{a+b}{\rho - 1} - a - 2\right].$$

Three oriented couplings along a spanning tree, their common bare gap, and this ratio consequently reconstruct the four source eigenvalues in the declared model. Placement eigenvectors remain a separate receiving account. This inverse is an algebraic identity; its denominator shows the increasing sensitivity as ρ approaches one. [11]

Source access also has structure. Either endpoint placement in the Jacobi path reaches all four modes for every nonnegative g_H. Interior placement 0 instead misses one mode exactly at

$$g_H = \frac{1}{164}.$$

Native exact closure returns dimension three there, while endpoint placement 1 returns four. Thus source observability depends on preparation even though the Hamiltonian remains nondegenerate. This new exception was followed from Duck's first-round cyclicity breadcrumb.

18. Eight expectation values and a rank test

An autonomous closed two-state source, fixed preparation and fixed linear receiver have equally spaced expectations

$$y_n = c + a\cos(n\vartheta) + b\sin(n\vartheta).$$

Form first differences $d_n = y_{n+1} - y_n$ from eight means and the four-by-four difference-Hankel $H_{ij} = d_{i+j}$. The constant disappears and the two exponentials give rank at most two. A nonzero three-by-three minor distinguishes the protected source record from this class. Eight means are not eight individual shots. [11,12]

The earlier cadence supplied a determinant separated from zero at a per-mean allowance of 0.0001. Source-guided cadence selection now allows 0.01 under the same independently declared calibration family. It trades a longer coherent evolution for fewer repeated preparations; the physical stability cost stays explicit.

The next sections state the exact retained-memory recipient before specifying the preparation, readout, errors and shot counts for this test. Driven or open two-state sources and arbitrary multilevel sources remain distinct comparison classes.

19. Eliminate the hidden block exactly

Let A be the original finite self-adjoint positive generator, Π the retained orthogonal projection, and $Q = I - \Pi$. On retained and hidden coordinates p, q, write $B = \Pi A \Pi$, $C = QA\Pi$, $D = QAQ$. Then

$$\dot{p} = -Bp - C^* q, \qquad \dot{q} = -Cp - Dq.$$

Variation of constants gives

$$q(t) = e^{-Dt} q_0 - \int_0^t e^{-D(t-s)} C p(s)\, ds.$$

Substitution retains both the initially hidden state and its feedback:

$$\begin{aligned} \dot{p} &= -Bp - C^* e^{-Dt} q_0 \\ &\quad + \int_0^t M(t-s) p(s)\, ds, \\ M(t) &= C^* e^{-Dt} C. \end{aligned}$$

For initially retained data, differentiating twice gives $\Pi A^2 \Pi = B^2 + C^\dagger C$. Thus equality of compressed evolution and $\exp(-Bt)$ for every retained preparation forces $C^\dagger C = 0$, hence $C = 0$. Conversely $C = 0$ decouples the equations. Exact autonomous closure is therefore equivalent to invariance $AV \subseteq V$. [12]

For v in V, its initial memory strength is exactly $\|QAv\|^2$. Memory comes from the actual omitted source directions, not from a fitted delay term.

20. Normalize the residual in its source Gram

Let b be a nonorthonormal retained basis with Gram $G > 0$. Put D_b and T_b equal to the source inner-product matrices shown

below. Projection gives

$$D_b = \langle b, Ab\rangle, \quad T_b = \langle Ab, Ab\rangle.$$
$$L_b = T_b - D_b G^{-1} D_b,$$
$$K_0 = G^{-1/2} L_b G^{-1/2} = C^* C$$

in an orthonormal chart. $\mathrm{Rank}(L_b) = \mathrm{rank}(C)$; the physical memory norm is the largest generalized eigenvalue of $L_b v = \lambda G v$. Ordinary eigenvalues of the unnormalized residual matrix do not supply that norm.

Suppose the original invariant exchange-odd sector has $\gamma I \le A \le NI$. Apply variation of constants once more to the retained equation with $q_0 = 0$. The evolution defect is a double integral with factors $\exp[-B(t-s)]$, $C^\dagger$, $\exp[-D(s-r)]$, C and $\Pi \exp(-Ar)$. Their three time intervals sum to t. Consequently

$$\left\| \Pi e^{-At}|_V - e^{-Bt} \right\| \le \frac{t^2}{2} e^{-\gamma t} \|K_0\|.$$

Using $A \le NI$ also bounds the difference from its quadratic term by $Nt^3\|K_0\|/2$. These are original A-time bounds; the expected attempt count is Nt. A physical time calibration remains separate. [12]

21. Continue the actual spin closure

The completed five-function spin span equals

$$V_5 = \mathrm{span}(d, e, Ad, Ae, A^2 d).$$

Leading-one residual coordinates make the first enlargement exactly

$$V_7 = \mathrm{span}(V_5, A^2 e, A^3 d).$$

The earlier all-54 preparation certificate established $\dim V_7 = 7$. The next two possible outgoing functions are $A^3 e$ and $A^4 d$. A literal full-source point certificate computes their rank rather than assuming the enlargement closes. [7,11,12]

All 54 evaluated original finite-N preparations have a paired Krylov rank of at least 10. The seven-dimensional enlargement has exact residual rank two in every evaluated preparation. Selected deeper certificates reach rank 14. The retained rank has not stabilized: these are exact full-source closure lower bounds, with strict positive Gibbs residual bounds and an explicit finite invariant count-space upper closure. No exact minimum or uncomputed global residual moment is assigned.

All original sites, Gibbs weights, self-excluded fields, accepted flip rates and attempt clocks remain present. The evaluator lumps only exactly exchangeable background-site classes and retains their exact multiplicities. This is a full finite-source calculation, distinct from the earlier conditional twenty-site contraction.

The minimum invariant closure is the Krylov union $V + AV + A^2V + \cdots$. Exact rank stabilization, or a source minimal-polynomial identity, is the stopping condition.

In the deeper selected source, the natural retained dimensions advance $5, 7, 9, 11, 13$, each with a nonzero outgoing residual. The fourteen-column certificate makes V_{13} nonclosed because $A^6 e$ remains independent. Also $M_7(t) = C_7^\dagger \exp(-D_7 t) C_7$ has rank two at every finite time: the hidden exponential is strictly positive in this finite source. This is a kernel-rank statement, not a rank statement for the propagation error.

22. Compact certificates and an invariant upper span

Evaluate m source functions at m literal configurations and form M. A nonzero certified determinant establishes their independence pointwise. Every configuration has positive mass in the original finite Gibbs law, so this is independence in that same physical source Hilbert space.

For $M = [V_7, A^3 e]$, the last row λ of M^{-1} annihilates V_7 and reads $A^3 e$ as one. For $r = Q_7 A^3 e$,

$$1 \le \left[\sum_i \frac{|\lambda_i|^2}{\pi(x_i)} \right] \|r\|^2.$$

If the original Gibbs log weights lie within plus or minus L after a common constant is removed, $\pi(x) \ge \exp[-N\log 2 - 2L]$. Hence

$$\|r\|^2 \ge \frac{e^{-N\log 2 - 2L}}{\sum_i |\lambda_i|^2} > 0.$$

Two inverse witness rows similarly give a positive two-by-two residual-Gram lower bound. This certifies nonclosure in the original Gibbs norm, without assigning a useful approximation size to that very conservative lower bound. [12]

An explicit invariant upper space retains all labelled twenty-site core configurations and counts each exchangeable background block:

$$D_{\mathrm{lump}} = 2^{20}(F_b + 1) \binom{P_b + 2}{2} \binom{T_b + 3}{3}.$$

Its dimensions have 96 to 342 decimal digits across the 54 original preparations. This finite upper closure does not assert that the d/e seeds reach it.

23. The nonzero-clock reference account

The inherited amplitude bound constrains a simple occupation-compensating reference. Put $r = 1 - n$. If actual clock amplitude is at least $m > 0$,

$$\frac{1 - \sqrt{1 - m^2}}{2} \le r \le \frac{1 + \sqrt{1 - m^2}}{2}.$$

The explicit reference state

$$|r\rangle = \sqrt{n}\,|0\rangle + \sqrt{1-n}\, e^{-i\gamma} |1\rangle$$

has connection $(1-n)\,d\gamma$, even when n changes differentiably: its real amplitude-derivative terms cancel. It restores the occupation coefficient in the receiving lift while keeping the joint-state Berry form. It does not automatically receive that remaining form. [11]

For actual lane 01 this reference occupation lies approximately between 0.499975039 and 0.500024961. This is a recipient-specific action bound. The next construction instead receives the entire actual joint-state contribution with a balanced reference and a measurable relative phase.

Finite-energy overlap derivatives suffice for the actual continuum trajectory. No additional assumption that its initial vector belongs to the strong operator domain is inserted.

24. A reference retaining the actual energy

Retain the unchanged Chapter VII joint H and its preparation Ψ_0. Its conserved form energy E includes the even source energy, full field, both counterterms and joint correlations. For prescribed legal phase rates, put

$$w = \dot\gamma + 3\dot\chi, \qquad h_R = 2(w-E)P_R.$$

The balanced reference has relative phase $\dot\beta = 2(w-E)$. The product trajectory retains precisely the original source-field marginal. Its canonical connection is

$$E\,d\tau + \frac{1}{2}d\beta = d\gamma + 3d\chi.$$

For the actual lane 01 preparation, $H = \varepsilon e_{01} I + \varepsilon S + K$, with $K \ge 0$ and preparation expectation of K at most K_{prep}. Therefore

$$\varepsilon e_{01} - \varepsilon \le E \le \varepsilon e_{01} + \varepsilon + K_{\mathrm{prep}}.$$

The exact rational certificates enclose E within $[0.917629340281369063, 0.917632830384309734]$. Equivalently $E = \varepsilon e_{01} + (E_0+E_1)/2 + c_{\mathrm{soft}}$ for the retained hard branches and full soft vacuum. This uses the actual continuum preparation, rather than the earlier finite-qubit diagnostic. Physical energy is $\hbar E/t_u$. [2,12]

25. One original forward/reverse/wait protocol

Keep activation $B = 1$, $A_{\mathrm{path}} = 0$ and the declared diagnostic source ratio $R_H/(ct_u) = 1/10$. R_H is the original source radius; its incidence with the field radius stays explicit. Starting at 000, apply W_1^+ to 100, W_1^- back to 000, then wait four model-time units.

The lifted $\phi_b = \phi_p = 0$. Thus $\gamma = \delta = \phi_a$ rises from 0 to $\pi/2$ and returns to 0; $\chi = 0$. Original signed actions are $+\pi\hbar/2$, $-\pi\hbar/2$ and zero. With ℓ increasing along either traversal,

$$f(\ell) = \frac{12\sqrt{68 - 32\cos\ell}}{100 - 32\cos\ell + 4\cos^2\ell},$$
$$d\tau = \frac{f(\ell)}{10}d\ell, \qquad v = 10/f.$$

Both active durations equal $T = 0.158555895876030285\ldots$. The forward integral is retained from H02; H05 supplies its reverse after a common-phase shift. Native intervals overlap, and exact path reversal identifies their positive durations. [12]

The general source-paced instrument uses $w = \sigma v_j$, including its original positive path clock. A phase-threshold receiving schedule agrees with it only when $v_j = (M/3)\nu(t)$. Signed role phase, elapsed source time and actual current phase remain separate operands.

26. Entry, matching and release stay distinct

Start the reference balanced with $h_R = 0$. Over external entry $a = 0.01$, hold 000 and ramp $h_R = 2(10-E)(s/a)P$. During the forward/reverse/wait intervals use $2(w-E)P$. Apply the recorded sudden changes at event boundaries. After the wait, release over $b = 0.01$ with $h_R = -2E(1-s/b)P$. These ramps are outside ledger matching.

Event	τ	β
Start	a	$.1 - Ea$
Forward	$a+T$	$.1 + \pi - E(a+2T)$
Reverse	$a+2T$	$.1 - E(a+4T)$
Wait	$a+2T+4$	$.1 - E(a+4T+8)$
Release	$a+2T+4+b$	$.1 - E(a+b+4T+8)$

Source wait has $w = 0$ and $h_R = -2EP$; it is not reference off. The wait winds β by $-8E$, more than one full turn. Actual θ advances throughout and is enclosed at every boundary by $[\ell_{01}\tau_{\mathrm{lower}}, u_{01}\tau_{\mathrm{upper}}]$.

The final τ is $4.337111791752060571\ldots$. Final actual θ lies in $[0.000015136966570142, 0.000015136966591669]$, while β lies in $[-7.841399681709530363, -7.841369477578352354]$. The two phases record different quantities. [12]

27. Every switch and continuous-drive payment

On the balanced reference, energy is $w-E$ and variance is $(w-E)^2$; mean drive work is dw. The original path has $v_0 = 10$, $v_L = 125/(3\sqrt{17})$, and $v_{\min} = 9.810320323234103\ldots$ at $\cos\ell_* = (11-\sqrt{73})/4$. Its interior minimum splits each drive into two monotone pieces. [12]

Reference work piece	Work in $\hbar/t_u$
External entry	$10 - E$
Forward first piece	$v_{\min} - 10$
Forward second piece	$v_L - v_{\min}$
Forward to reverse	$-2v_L$
Reverse first piece	$v_L - v_{\min}$
Reverse second piece	$v_{\min} - 10$
Reverse to wait	10
Wait	0 drive work
External release	E

The signed sum is zero, but positive and withdrawn work each equal $2v_L + 2(10 - v_{\min})$, approximately 20.590661439892877. Recoverable exchange and irreversible apparatus cost are distinct accounts.

Adding a nonnegative-Hamiltonian baseline adds canonical payment $C = \pi + 2ET + 8E + Eb$. The total variational form

still subtracts energy and vanishes on its own exact driven trajectory. Switching cancellation supplies no zero-cost apparatus claim.

28. Read the phase and keep its turn counter

Independent prefix preparations measure $X = \cos\beta$ and $Y = -\sin\beta$. Use atan2$(-\widehat{Y},\widehat{X})$, then nearest-branch continuation from $\beta_0 = 0$. An endpoint qubit alone records phase modulo 2π; an external counter keeps its turns.

Include all boundaries and a grid with gaps at most 0.01: no more than 441 prefixes. Since $E < 1$ and $v \le 50/3$, $|\dot\beta| \le 106/3$. Coordinate errors $\le 1/16$ imply angular errors $< 1/8$ rad. Consecutive true advances plus both angular errors obey

$$53/150 + 1/4 < \pi,$$

so the unwrapped branch is unique. Correct branch choices leave each prefix error below $1/8$ without summing all prefix errors into drift.

Allocate coordinate tolerance as systematic $1/32$ plus statistical $1/32$. Four systematic budgets of $1/128$ cover reference preparation, evolution/waveform, readout and timing. Sufficient conditions are trace-distance and Duhamel bounds $\le 1/256$, Pauli-readout norm error $\le 1/128$ and trigger error $\le 3/13568$ model time.

With 26,624 independent shots per coordinate and prefix, the simultaneous statistical failure is below 1%. Maximum fresh preparations are 23,482,368. Each readout disturbs that copy; fresh prefixes retain the history. Source energy and continuum preparation require their own retained energy-form control. [12]

29. Specify the protected-coherence instrument

Write $|a\rangle$ for the protected state $|q_a q_a\rangle$. Prepare

$$|\psi\rangle = (|0\rangle + |1\rangle + |2\rangle)/\sqrt{3}$$

with the field vacuum. D,C and their counterterms annihilate this protected span under the full retained Hamiltonian. The adjacent-coherence receiver is

$$O = |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 2| + |2\rangle\langle 1|.$$

Its outcomes are $+\sqrt{2}, -\sqrt{2}, 0$. Corresponding vectors in the three-state basis are $(1,\sqrt{2},1)/2$, $(1,-\sqrt{2},1)/2$ and $(1,0,-1)/\sqrt{2}$; complementary outcomes report zero. A source-basis rotation followed by this projective readout specifies a mathematical instrument. [12]

Its expectation is $y(t) = (2/3)[\cos(\omega_{10}t) + \cos(\omega_{21}t)]$. Every sampling time receives fresh preparations. Measuring one copy changes its state and cannot supply its next undisturbed expectation.

The required physical coherent mode access, receiver pulse, field preparation, reset and cooling belong to the apparatus. The complete error budget below assigns their permitted effects before using a sample rank.

30. Charge leakage, disturbance and timing

Use trace distance $D(\rho,\sigma) = \|\rho - \sigma\|_1/2$. Preparation distance d_{prep} gives expectation bias $\le 2\sqrt{2}\, d_{\mathrm{prep}}$. Leakage l and conditional error d_{cond} satisfy

$$d_{\mathrm{prep}} \le \sqrt{l} + d_{\mathrm{cond}}.$$

Leakage alone does not specify the state inside the protected space or its field preparation. An effective measured observable M with $\|M - O\| \le b_{\mathrm{read}}$ includes pulse disturbance and outcome assignment in its readout bias.

Unwanted dynamics contributes at most $2\sqrt{2}\,\kappa$, where

$$\kappa = \int \|\delta H(s)\psi_{\mathrm{ideal}}(s)\|\, ds/\hbar.$$

This bound extends through the latest allowed trigger. A source-only finite-matrix norm does not bound an unbounded field interaction.

The retained independent calibration is $\Xi = 1382784$ plus or minus one and $\alpha/\alpha_{\mathrm{nominal}} = 1$ plus or minus one part per million. It gives $|y'| \le L$ with

$$L < 114081.396605\,\alpha_{\mathrm{nominal}}.$$

Trigger error δ_t contributes $L\delta_t$. Calibration uncertainty is already part of the ideal source family; it is not charged again as timing jitter. [12]

31. A cadence selected from the source

Before acquiring the target record, choose

$$h = \frac{5\pi}{2\alpha_{\mathrm{nominal}} D_{\mathrm{nominal}}},$$

$$D_{\mathrm{nominal}} = \frac{6348467821812271}{80000000000},$$

and $t_k = kh$ for $k = 0,\dots,7$. This h is 7.5 times the earlier cadence and was selected from a declared finite list using only the retained source. Both sampled frequencies remain distinct and nonaliased. For the three-by-three difference minor A with index set $I = \{0,2,3\}$,

$$5.318723982743267 < \det A < 5.941782789331066,$$

$$|A_{ij}| < 2.$$

For per-expectation allowance $\eta = 0.01$, the conservative two-record determinant perturbation bound is

$$72\eta(2+4\eta)^2 = 2.996352,$$

below the certified lower determinant. This source-selected cadence exchanges a longer stability interval for substantially fewer shots. It is not fitted to measured target data. [12]

Allocate 0.001 each to preparation, unwanted evolution, effective readout and timing, leaving 0.006 for statistical error. The next section turns this allocation into explicit measurement counts and a decision rule.

32. Repetitions and an operational decision

Sufficient apparatus conditions are $d_{\mathrm{prep}}, \kappa \le 0.001/(2\sqrt{2})$, $\|M-O\| \le 0.001$ and $\delta_t \le 8.76567 \times 10^{-9}/\alpha_{\mathrm{nominal}}$. If conditional preparation is exact, coherent leakage $l \le 1.25 \times 10^{-7}$ suffices. These are requirements rather than achieved precision. [12]

For N independent outcomes in $[-\sqrt{2}, \sqrt{2}]$ and eight means, Hoeffding and a union bound give

$$P(\text{any mean error} > s) \le 16e^{-Ns^2/4}.$$

At $s = 0.006$ and failure allowance 0.01, $N = 819{,}751$ fresh shots per time suffice: 6,558,008 total. Calibration failures add their probability to that sampling allowance. The earlier 0.0001 allocation would need 8,197,509,899 shots per time under the same split.

Form measured differences, $\widehat{A}_{ij} = \widehat{d}_{(I_i+I_j)}$, and $\widehat{B} = \max|\widehat{A}_{ij}|$. Reject the stated closed two-state class when

$$|\det \widehat{A}| > 36\eta(\widehat{B}+2\eta)^2.$$

Each entry error is $\le 2\eta$. Telescoping each product of three entries across six determinant terms gives this threshold because every ideal two-state minor vanishes. The calibrated target exceeds it by more than 2.35160 on the stated simultaneous sampling event, conditional on the apparatus bounds.

33. The minimum counter count

For polynomial phase receivers through degree d, retain $M_{jk} = \int \theta^k 1_j \, d\theta$, and the corresponding event-atomic sums when required. A segment α to β adds $(\beta^{k+1} - \alpha^{k+1})/(k+1)$, clipped to each specified gate. Actual phase endpoints suffice; no constant-rate approximation is used. [11]

Here is the rank derivation of minimality. Collect required additive receiver values into $F(h)$ and subtract $F(h_0)$ when initial endpoints and total duration/count are supplied. Let V be the span of these contrasts with dimension r. A summary with c linear counters and a linear decoder must be injective on V; otherwise a nonzero receiver contrast disappears. Rank-nullity therefore gives $c \ge r$. Coordinates in any basis of V attain r.

Known endpoint identities reduce r. For fixed event addresses $\theta_n = hn$ and N events, summing labels at each degree gives $h^k \sum n^k$, already determined by N. Overlapping gates or permitted source histories can supply further constraints.

This is exact minimality for the admitted linear additive receiver class. Online continuation also retains predictive state and current phase; a finite-family contrast basis alone is not a transition-closed realization for every future history.

Counter dimension does not bound precision or stored bit count, and makes no minimum claim for unrestricted nonlinear memory.

34. ABBA/BAAB with actual event offsets

Use the original closed blocks $A = (W_1^+)^4$ and $B = (W_7^+)^4$. Every block returns to 000. In one repeated source context their positive durations are T_A, T_B; retain each internal event offset u. The A-block starts in ABBA are $\{0, T_A + 2T_B\}$; in BAAB they are $\{T_B, T_B + T_A\}$. Their sums agree, while

$$\sum_{ABBA} (s+u)^2 - \sum_{BAAB} (s+u)^2 = 2T_B(T_A + T_B).$$

Expansion cancels both $2u\sum s$ and $2u^2$. B-block starts similarly have equal sums, with squared-sum difference $-2T_A(T_A+T_B)$. Each of three A-only output points occurs once per A; the common point occurs once per A and four times per B. Its difference is consequently $2(T_A+T_B)(T_B-4T_A)$. [11]

Counts, endpoints and first physical-time atomic moments agree, while second moments distinguish the histories even with unequal durations. Under the prescribed equal phase-step schedule the A-only differences are $64h^2$ and the common-point difference is $-192h^2$.

Prouhet closed-block reorderings extend the phase-event distinction beyond every finite polynomial degree. Nonlinear transport from physical time to actual phase retains original time arguments; it does not automatically preserve unweighted phase moments.

35. An attainable source record has a price

For positive B and record $r = L^\dagger y$, let $y_\star$ be its ambient energy minimizer and $G = L^\dagger B^{-1} L$. If F_r is the permitted microscopic displacement set, projection orthogonality gives

$$\min_{y \in F_r} y^* B y = r^* G^+ r + \min_{y \in F_r} \|y - y_\star\|_B^2.$$

Indeed $L^\dagger(y - y_\star) = 0$, and $By_\star$ lies in range(L), so the cross term vanishes. The second term is the realizability correction. A finite nonempty permitted set attains its minimum; an empty compatible set has no allowed endpoint. [11]

For the original eighteen-coordinate literal energy and fixed outside spin configuration, permitted port displacements are $(0, 0, e_z - e_0)$. Every signed cross term is retained. The diagnostic record is $(-1, 0)$ under the two declared receiver columns, distinct from the complete original receiver family.

Payment / κ_E	Value
Ambient minimum	0.987626515
Pure-source minimum	1.894317938
Realizability correction	0.906691423

The source minimum is $1085537/573049$. The complete endpoint enumeration supplies its certificate in the next section.

36. Every endpoint and the feasible minimum

The diagnostic reads the e_0 and e_1 coordinates. A displacement from z_0 has record $(-1,0)$ exactly for $z = 2,\dots,15$. The table gives numerator of cost/κ_E over denominator 573049; z_{15} has numerator 1146098. [11]

z	Record	Numerator	One flip?
0	$(0,0)$	0	reference
1	$(-1,1)$	2006049	yes
2	$(-1,0)$	1085537	yes
3	$(-1,0)$	1146097	no
4	$(-1,0)$	2006049	yes
5	$(-1,0)$	4965618	no
6	$(-1,0)$	2155185	no
7	$(-1,0)$	2163505	no
8	$(-1,0)$	1085537	yes
9	$(-1,0)$	2155185	no
10	$(-1,0)$	1142898	no
11	$(-1,0)$	1146097	no
12	$(-1,0)$	1146097	no
13	$(-1,0)$	2163505	no
14	$(-1,0)$	1146097	no
15	$(-1,0)$	1146098	no

Thus z_2 and z_8 attain the smallest cost on the nonzero record fiber $(-1,0)$; both are single accepted port flips. z_0 and z_1 cannot realize this record. Outside-coordinate changes add nonnegative squares and cannot lower the attained fixed-outside minimum.

Mixtures retaining this exact record obey the same endpoint floor for expected quadratic payment. Energy of mean displacement is a different quantity and retains covariance. Preparation and physical κ_E remain separate apparatus assignments.

37. Calibrate the paid protocol

The ideal paid reference uses actual E. The present source certificates also give a numerical control: take their exact rational energy midpoint $\widehat{E}$ and replace E by $\widehat{E}$ in every reference waveform. Then

$$|E-\widehat{E}| < 1.746\times 10^{-6}.$$

The total connection on matching intervals becomes $[w+(E-\widehat{E})]\,d\tau$. Across the complete forward/reverse/wait program its signed-ledger bias is below $7.54\times 10^{-6}\hbar$. The source energy stays E; its calibration residual remains visible. [12]

The entry and release canonical integrals are $(E+10)a/2$ and $Eb/2$ for exact calibration. They are external to the signed Write ledger. A reconstructed matching action with interval U, energy error dE, time error dt and endpoint angular errors η obeys the conservative allowance

$$U\,dE + (|E|+dE)\,dt + \eta.$$

An actuator using a finite-time turnaround in place of a sudden quench adds its own interval, phase and work. It cannot silently reuse the ideal endpoint account. Actual clock certificates continue to apply only while its original source-field H and preparation are retained.

38. Pay for the controller

The specified waveform is an externally driven instrument. A material autonomous controller must produce its schedule, store or recover work and retain the phase-turn record. When coupled to the quantum source or reference it can also become entangled; the enlarged Hamiltonian then determines the clock and energy account.

The protocol has made the required work exchange finite and explicit. It has not assigned a cost to original source actuation, waveform generation, qubit preparation, readout rotations, detector reset, external-counter memory or cooling. Their physical models supply those further payments. Recovering equal positive and negative reference work does not determine their dissipation.

For the protected-coherence experiment, the independent common calibration fixes g_H through Ξ and fixes ε through one absolute scale. The two-frequency record is then a prediction. Its source-guided cadence and apparatus tolerances specify what a proposed implementation must deliver.

These are concrete construction tasks with quantitative targets. They can receive hardware and channel assignments while preserving the original signed source response, phase lift, full field and energetic complement. [12]

The actual clock has a separate receiver. Divide its D and C expectations by signed $d_{01} < 0$ to obtain the complex current z; $\theta = -\arg z$. Its certified amplitude exceeds 0.999999998. Fresh source copies can estimate those two bounded Pauli coordinates, then unwrap $-\operatorname{atan2}(\operatorname{Im} z, \operatorname{Re} z)$. Their sampling and apparatus allowances are additional. The θ intervals in this protocol are predicted from retained actual-rate bounds; reference-qubit measurements read β.

39. The operation determines the retained account

A uniform A_0 remains in a total account while gradients and shared endpoint differences remove it. A raw lapse ratio instead rescales lifts through background normalization; a source-normalized local lapse declares a different input. Every readout retains its operation and reference. [1,9]

The source comparison sharpens this lesson. Current clocks leave the commuting pair-product energy unread. Protected coherence receives that energy, linked to the lane family by a common calibration.

Memory gives the dynamical version: a retained span evolves autonomously exactly when it is invariant. The actual seven-dimensional spin enlargement has further outgoing directions;

the new point certificates state how far its exact rank reaches. The invariant upper lumping retains the finite source without identifying it as the smallest closure.

History moments identify what a later receiver needs, and feasible endpoint costs identify an attainable payment. The paid reference now has one complete timed Write/inverse/wait program, all reference work pieces, a winding record and a fresh-copy measurement budget. Physical realization can build directly on these specified quantities.

40. Sources and reproducibility

The source package retains the original draft, completed Chapter VII (formerly Chapter VII) and prior Chapter II development sources. Editable typesetting, bundled fonts, exact operands, native receipts and independent offline checks accompany this reader. Every calculation retains its source preparation and units. [12]

[1] Chapter II draft and verification ZIP, 15 October 2026 (H002634).
[2] Final Chapter VII (formerly Chapter VII) and actual full-field clocks; H002628/H002630.
[3] Native history reduction; H002165.
[4] Observed-history clock/ruler recipient; H002559.
[5] Original Write/tensor/action map; H002623.
[6] Radiative memory and complete work; H002319.
[7] Completed spin source-action T5; H002627.
[8] Common spin energy and receivers; H002423.
[9] Original Chapter II packet CR001/002/003/005 and local CR113/114/115/218/238.
[10] Chapter II first development; H002634.
[11] Chapter II continuation, two 180-second Duck rounds; H002635.
[12] voli_chapter2_physical3_20261015: source-bound physical-development packet; GEN6 Duck run chapter2-physical-protocols-20261015-01 (Duck 1.2.2, core 0.10.2; 15 native tasks); H002635. Provenance trace: Courtroom II case voli_provenance/01 (H002821).

41. Give the retained energy a spatial account

The completed endpoint minimum is a positive displacement payment for a selected record fiber. It is not the full energy of the source, its field or its controller. The interacting preparation has a separately retained conserved energy E, including its even source contribution and both counterterms. A gravitational mass assignment needs that complete reference account and a physical conversion. [13]

A spatial source requires more information still. Define an energy observable smeared over physical regions, identify its reference and units, and require its whole-space integral to recover the retained total Hamiltonian. Source, interaction, field and boundary payments need declared incidence. A positive global form does not automatically provide a positive pointwise density.

The classical commuting static-boundary chart gives a compact control. At its minimizing boundary value, put

$$A = \frac{\varepsilon^2 \sum_\alpha J_\alpha^2}{4R}.$$

The free field pays A, its interaction contributes $2A$, and the counterterm contributes A. Their integrated sector sums to zero. Taking only the positive free-field density would omit two retained payments. This is not a joint static replacement of the noncommuting quantum D and C, and it does not remove the separate source energy. [13]

The original model chart converts displayed energy to $\hbar$ divided by the physical time unit. Neither that unit in seconds nor a physical ruler is fixed by the retained clock theorem. Writing E/c^2 before supplying these assignments does not derive kilograms or a spatial density.

The next source examples keep their energy, location and receiving operations typed separately. They calculate what follows from an explicit incidence rather than substituting a diagnostic record cost for the whole source.

42. What the recovered local effects retain

A retained Si29 effect construction has a useful exact consequence. Write $h = \sqrt{Q}$, $H_0 = \mathrm{diag}(h,h)$, and let O be its registered effects at channel t and site x. The energy effects

$$E_{t,x} = H_0^{1/2} O_{t,x} H_0^{1/2}$$

obey, for each of the eight sites,

$$\sum_t E_{t,x} = \frac{H_0}{8}.$$

Thus the channel-summed site energy is flat for every preparation, although the total expectation of H_0 can change with preparation. Resolving channels and changing the spatial profile are distinct operations. The effects alone do not identify physical site separation or a gravitating density. [13]

A specified receiving example places eight equal uniform balls at the vertices of a cube of side 2ℓ, with ball radius $\delta < \ell$. Assigning mass $M/8$ to each ball then makes the density explicit. M is one common physical energy conversion times the expectation of H_0, divided by c^2.

The cube embedding, smearing radius and energy-to-mass rule are declared assignments. They are not deductions from the flat site-effect identity, and the selected source account does not include otherwise unlocalized fields or controllers.

43. A spatial receiving operator with exact flux

A finite radial receiver makes the Poisson interface executable. Use radii ih, a zero lift at radius Lh, and edge conductances $i(i+1)$. With B the incidence matrix and C the diagonal conductance matrix, the positive Dirichlet operator is $K = B^{\mathsf{T}} C B$. Its exact inverse on $i, j = 1, \ldots, L-1$ is

$$H_{ij} = \frac{1}{\max(i,j)} - \frac{1}{L}.$$

For shell masses measured in one physical mass unit, put κ equal to twice G times that unit, divided by $c^2 h$. The source

$q = \kappa m$ gives the lift $a = Hq$ and

$$i(i+1)(a_i - a_{i+1}) = \kappa \sum_{j \le i} m_j.$$

This is an exact mass-flux statement for the declared spherical shell recipient. It is not a claim that the cube vertices are those shells or that finite source-state closure derives spatial locality. [13]

Adding twice G times the total mass, divided by $c^2 Lh$, converts the finite-boundary lift to the isolated infinity reference. It changes potential and energy reference while leaving interior gradients unchanged. For a uniform ball approximated by n shells, the reduced-lift error is below $3/n$; the sharper bound retains the harmonic sum. A finite outer boundary has its own separate error.

44. Geometry changes the same scalar reception

Outside the eight disjoint uniform balls, the cube density has exactly the potential of eight point masses at their centers. Equal weights and inversion symmetry cancel the dipole, quadrupole and cubic multipoles. The leading angular correction is of fourth order in ℓ/r. A spherical monopole is therefore a controlled far-field approximation, not an exact consequence of the total mass alone. [13]

The isolated binding account includes self and pair terms:

$$U_{\text{self}} = -\frac{3GM^2}{40\delta},$$
$$U_{\text{pair}} = -\frac{GM^2}{64\ell}\left(6 + 3\sqrt{2} + \frac{2}{\sqrt{3}}\right).$$

Shrinking the balls without declaring their self-energy treatment is not a finite-energy point-source limit. Both terms have a common reference and count each pair once.

Chapter VII's quadrupole field also has energy outside its source boundary. Its exterior is not automatically a vacuum region for a gravitational density that includes that energy. The exact exterior $1/r$ law requires enclosing the selected density or controlling its remaining tail. A quadrupole amplitude, its positive energy, and a scalar gravitational monopole remain different objects.

45. Energy invisible to the actual current clock

Retain the original complete sixteen-state source, its commuting even energy, $S = \text{Swap}$ and the unnormalized current $D + iC$. Exact source algebra gives

$$[S, Z_{\text{raw}}] = -2Z_{\text{raw}},$$
$$[E_{\text{even}}, Z_{\text{raw}}] = 0,$$
$$[D^2 + C^2, Z_{\text{raw}}] = 0.$$

The even operator and complete counterterm also commute with the full original Hamiltonian and both field couplings. Adding a bounded function of either changes total energy but leaves every current history unchanged. A common scalar is the simplest example; it also changes the state's common phase. The independent original Write phase χ retains its own role. [13]

The four equal-occupation eigenstates have zero D and C but distinct energies $\varepsilon(1+q)^2$. They are controls for energy missed by the current receiver, not functioning current clocks. Source energy therefore cannot be inferred from current amplitude or phase rate alone.

On an active lane the actual receiver remains $z = \langle D + iC \rangle / (\text{signed } d)$, with $\theta = -\arg z$. Its phase requires separate source copies and its own preparation, timing, readout and winding budget. The reference qubit measures a different phase.

The separate retained Si29 transfer source gives another exact control. Its incoming-minus-outgoing population B follows by summing the original sector-resolved effects. Its positive excitation energy commutes with both B and the signed transfer generator. The proposed uniform excitation-density coupling therefore gives

$$H_a = H_{\text{tr}} - \frac{a}{2}|H_{\text{tr}}|, \qquad B_a(t) = B_0(t).$$

Its entire population-transfer history is unchanged. Full scaling of the signed generator instead dilates that history in time. This is the actual retained bright-sector signal, distinct from Chapter VII's certified phase clock. Other species/site observables need not share its invariance. [13]

46. Couple a spatial lift to the complete clock

Write the full Hamiltonian as the sum of source, free-field, interaction and counterterm contributions. These retain the original complete source, both signed interactions and both counterterms. A supplied spatial lift must specify a family $H(a)$, rather than just a desired output frequency. [13]

For a fixed preparation and bounded perturbation V, Duhamel gives the first-order response

$$\delta z(t) = i \int_0^t \langle [V_I(s), Z_I(t)] \rangle \, ds.$$

The actual rate response is

$$\delta \nu = -\operatorname{Im}\left(\frac{\delta \dot{z}}{z} - \frac{\dot{z}\,\delta z}{z^2}\right).$$

An unbounded field form requires its energy-domain version and explicit propagation bounds. Changing the preparation supplies additional derivative terms. None is replaced by a finite matrix commutator alone.

The response target is shared across all six separately prepared original lanes, with one physical ε, ruler, time chart and perturbation. A stationary branch or matched internal age makes a fractional lapse test unambiguous. At equal coordinate time a nonstationary clock also carries its transient response. Independent fitting of each gap cannot test universality.

The fixed-field second-order diagnostic defines s as the logarithmic gap derivative with respect to ε. Its outward enclosures differ across all six lanes:

Lane	10^9 times $(1-s)$, lower	Upper
01	4.82	4.84
02	20.74	20.76
03	3214.92	3216.50
12	5.55	5.56
13	2970.53	2971.99
23	2719.14	2720.48

These are hard-branch perturbative diagnostics, with the original field fixed; they are not achieved physical redshift measurements.

47. A full scaling and its countercontrol

If the coupling is declared to be $H(a) = n(a)H_0 + f(a)I$ for static a, the exact full-field control is

$$z(t;a) = z(n(a)t;0),$$
$$\nu(t;a) = n(a)\nu(n(a)t;0).$$

It holds for all six lanes and includes every original field sector. It does not derive n from the source. At matched internal age the rate ratio is n. A coordinate-time derivative also contains t times the derivative of the original rate, where that extra regularity exists. Accumulated phase comparison needs only the retained continuously differentiable phase. [13]

Scaling ε alone while keeping the field fixed multiplies the source and interaction terms by n, the counterterm by n^2, and leaves the free field unchanged. At first order,

$$V_\varepsilon - n'(0)H_0 = n'(0)(H_{\mathrm{ct}} - H_f).$$

The counterterm part is invisible throughout the current history because it commutes with both H_0 and the receiver. The free-field part can change the clock through its interaction. The retained two-polarization hard-gap expansion gives six distinct ε-response diagnostics; these do not replace a full-field finite-lift error certificate.

The source-only square root and raw total-account square root also differ: background normalization retains the latter's sensitivity factor $1/(1-A_0)$. All field and receiver normalizations must transform together if these are reparameterized.

48. Precision, memory and the next physical interface

For original actual-clock bounds ℓ and u, the declared full scaling predicts accumulated phases in $[\ell T, uT]$ and $[n\ell T, nuT]$. For $n = \sqrt{1-a}$, their independent enclosure bands are disjoint whenever

$$a > 1 - \left(\frac{\ell}{u}\right)^2.$$

That independent-band comparison has a best-lane threshold about 2.84×10^{-9}, above the Earth example's 1.39×10^{-9} lift. Keeping the common baseline path gives a sharper mathematical result. The full scaling implies $\theta_a(T) = \theta_0(nT)$, so

$$\Delta\theta = -\int_{nT}^{T} \nu_0(s)\,ds,$$

$$-u(1-n)T \le \Delta\theta \le -\ell(1-n)T < 0.$$

For $n = \sqrt{1-a}$, use $a/2 < 1-n \le a/(2-a)$ to obtain the entirely rational enclosure

$$-\frac{uaT}{2-a} \le \Delta\theta \le -\frac{\ell aT}{2} < 0.$$

Thus every positive lift and elapsed time has a strictly negative correlated phase change under this declared coupling, within the original window. The Earth example in lane01 gives $[-4.373897816, -4.373897806]$ nanoradians over one original bare cycle. This uses the same unwrapped branch and preparation, not independent rate fits. Fresh-copy phase readout and systematic errors must resolve the difference; no apparatus sensitivity or physical coupling is assigned. [13]

The exact elimination in sections 19–22 becomes spatial after an incidence map. A controlled response derivation must establish three-dimensional locality, isotropy, masslessness and causal speed.

Physical tasks remain: complete local energy, external V, SI calibration, energy-form response bounds and apparatus.

[13] voli_chapter3_source_field1_20261015: source, radial, energy and actual-clock derivations; exact operands, independent checks and native receipts. Previous sections 1–40 and their sources remain retained verbatim.

Data availability

The repository designated for this chapter's data and code is https://github.com/iwtbotiwtwot/sam_volume_I_ii.

SAM VOLUME I

The spherical accumulation kernel and the source-to-field interface

Chapter III

Abstract. A retained energy account becomes a spatial field through a specified localization, coupling and receiving operator. This chapter develops the isolated Poisson kernel, its energy and force proofs, and two complete finite constructions: the actual retained eight-site source profile under a declared physical embedding, and an exactly solvable shell network with boundary and continuum errors. A static Chapter VII field-energy example and an exact interacting-clock coupling criterion connect the construction to Chapters II and VII. Source identities, physical assignments and computed results are stated separately. All earlier drafts remain preserved.

1. The next source question

Chapter VII supplies an interacting finite source and continuum field. Chapter II distinguishes a present output, a predictive state, a timed history and an unread energetic complement. Chapter III asks how their physical energy is located and received in space. [1,2]

The spherical lift is a static receiving profile:

$$a(r) = \frac{2GM}{c^2 r}, \qquad A_{\mathrm{tot}} = A_0 + a, \qquad A_0 = \frac{1}{12\pi}.$$

The physical mass M, source radius b, receiving radius r and Chapter VII coupling radius R have different roles. G and c carry the declared physical units. The floor is dimensionless.

This revision carries exact source operands into finite field calculations. The mass/localization and coupling assignments are recorded at each step.

2. What enters the field

For a positive source form B and record $r = L^{\mathsf{T}} y$, the ambient minimum is

$$\mathscr{C}(r) = r^{\mathsf{T}} \mathscr{G}^{+} r, \qquad \mathscr{G} = L^{\mathsf{T}} B^{-1} L,$$

provided r belongs to the range of the receiver Gram. An infeasible record has no such source; its constrained infimum is $+\infty$. Allowed-state restrictions can raise the minimum. A change from an occupied reference also retains its cross term. [2]

This is a cost for a specified receiver. To form a physical density, specify the energy operator, physical conversion, reference and spatial incidence. The completed Chapter II now develops these requirements in its source-to-space interface.

The source and receiver below are static. Pressure, motion, retardation and nonlinear self-coupling require their own physical laws.

3. The declared receiving law

For a supplied nonrelativistic mass density ρ, use

$$\Delta a = -\frac{8\pi G}{c^2}\rho, \qquad \Phi = -\frac{c^2}{2} a.$$

This is equivalent to $\Delta\Phi = 4\pi G\rho$. The field normalization imports the Newtonian coupling. The sign makes a positive for a positive isolated mass when a tends to zero at infinity.

A test body with Lagrangian $L = m|\dot{\mathbf{x}}|^2/2 - m\Phi$ has acceleration

$$\mathbf{g} = -\nabla\Phi = \frac{c^2}{2}\nabla a.$$

Equality of passive gravitational and inertial mass is part of this test-body coupling. The scalar profile by itself does not select that equality.

For bounded compact density the solution is

$$a(x) = \frac{2G}{c^2}\int \frac{\rho(y)}{|x-y|}\, d^3y.$$

The locally integrable kernel obeys $\Delta(1/r) = -4\pi\delta$. Linearity therefore gives the response to a sum of prescribed sources. Global regularity and isolated boundary data establish uniqueness; an exterior subproblem needs its inner data too.

All three displayed formulas have a fixed energy and time convention. A later rescaling of a must also rescale its source equation and its receiving laws.

4. Flux and exterior uniqueness

In a spherical vacuum exterior,

$$\frac{1}{r^2}(r^2 a')' = 0, \qquad a = C_0 + \frac{C_1}{r}.$$

Decay fixes C_0, while the enclosing source flux fixes C_1:

$$M(<S) = -\frac{c^2}{8\pi G}\oint_S \nabla a \cdot d\mathbf{S}.$$

Thus $C_1 = 2GM/c^2$ and $\mathbf{g} = -GM\hat{\mathbf{r}}/r^2$. Decay alone leaves the amplitude free. A regular vacuum ball containing the origin cannot carry a nonzero $1/r$ term.

For a nonspherical source, total mass fixes only the monopole. Higher multipoles carry shape information while contributing zero net enclosing flux. The cube construction below makes the first surviving shape term explicit.

A finite outer wall fixes a different boundary problem. Its correction must be included before comparing potential or clock

values to the isolated profile. The shell example supplies that correction exactly, including its energy cost.

The spatially constant floor contributes no source flux. This statement is compatible with a nontrivial floor in an independently specified nonlinear clock receiver.

5. Finite spherical sources

For a spherical source define its enclosed mass:

$$M(r) = 4\pi \int_0^r \rho(s) s^2 \, ds.$$

Regularity and the source equation give

$$a'(r) = -\frac{2GM(r)}{c^2 r^2},$$
$$a(r) = \frac{2G}{c^2}\left[\frac{M(r)}{r} + 4\pi \int_r^\infty \rho(s) s \, ds\right].$$

Exterior shells supply potential and zero interior acceleration. A thin shell of radius b and mass M has $a = 2GM/(c^2 b)$ inside and $2GM/(c^2 r)$ outside. Its derivative jump carries its surface mass. Values directly on a zero-thickness surface require a limiting convention.

For a uniform ball,

$$a(r) = \frac{GM}{c^2 b}(3 - r^2/b^2), \quad r \le b,$$
$$a(r) = \frac{2GM}{c^2 r}, \qquad\qquad r \ge b.$$

Both a and a' match at the surface. At the center $a = 3GM/(c^2 b)$ while $g = 0$. Bounded compact density makes the central expression regular; finite mass alone does not. A central ρ proportional to r^{-2} is a useful finite-mass, divergent-potential example.

6. Mechanical and tidal receivers

Circular motion differentiates the potential; escape compares it with a destination:

$$v_{\mathrm{circ}}^2 = -\frac{c^2 r}{2} a'(r),$$
$$v_{\mathrm{esc}}^2 = c^2 [a(r) - a(\infty)].$$

For the isolated spherical exterior they are GM/r and $2GM/r$. A shell interior has zero circular-force field yet a nonzero escape energy relative to infinity. A truly global potential shift cancels when both endpoint values shift together.

For the exterior monopole, the tidal tensor is

$$T_{ij} = \frac{GM}{r^3}(3 n_i n_j - \delta_{ij}).$$

It has zero trace in vacuum, radial stretching $2GM/r^3$, and transverse compression $-GM/r^3$. Inside a uniform ball it is $-GM\delta_{ij}/b^3$, including at the zero-acceleration center; its trace is $4\pi G\rho$.

The tidal tensor here differentiates Newtonian acceleration. A relativistic spatial curvature or photon receiver requires additional structure. Chapter VII's phase clock is another receiving operation, whose response to a supplied coupling is derived below.

7. Radial controls and scale

For a trial $a_p = K r^{-p}$ away from the origin,

$$\Delta a_p = p(p-1) K r^{-p-2},$$
$$g_r = -\frac{c^2 p K}{2} r^{-p-1},$$
$$M_{\mathrm{flux}}(r) = \frac{c^2 p K}{2G} r^{1-p}.$$

A nonzero decaying spherical vacuum solution with constant source flux selects $p = 1$. Positive attraction alone leaves many powers. The harmonic $p = 0$ term is constant and carries no mass flux.

Compare powers using a reference radius: $a_p = a_*(r_*/r)^p$. Matching the potential at r_* gives a force p times the monopole force there. Dividing this profile by p matches that force but changes the potential to $1/p$ of its target. At other radii the force ratio is $(r_*/r)^{p-1}$.

The coefficient K has units length^p. Comparing its numerical value across powers without the reference radius changes the dimensional question. A matched datum at one radius therefore supplies less information than a radial profile, flux and potential reference together.

8. A finite-energy domain

For an isolated field, use the homogeneous Sobolev space $D^{1,2}(\mathbb{R}^3)$: the completion of compact smooth functions under $\|\nabla a\|_2$, represented in L^6. This fixes the infinity reference and excludes a nonzero constant ambiguity.

A sufficient source class is ρ in $L^{6/5}$; bounded compact density is an example. The pairing integral ρa is then bounded by Holder and Sobolev. The functional

$$J[a;\rho] = \frac{c^4}{32\pi G}\int |\nabla a|^2 - \frac{c^2}{2}\int \rho a$$

has a unique weak minimizer. Its Euler equation is the declared Poisson problem. Existence follows from the bounded source functional and the positive gradient inner product; uniqueness follows from the same norm.

Compact smooth test variations give

$$\delta J = \int \left[-\frac{c^4}{16\pi G}\Delta a - \frac{c^2}{2}\rho\right] \delta a.$$

An ideal point source still has a distributional exterior solution, but has infinite self-energy. Thin spherical shells have a finite-energy potential with a derivative jump. The finite cube below uses regular balls to specify its self-energy.

9. The minimum and binding energy

Let a_* be the weak solution and let v be an allowed variation with the same boundary reference. Expand $J[a_*+v]$. The linear terms cancel by the weak Poisson equation, leaving the exact identity

$$J[a_*+v]-J[a_*]=\frac{c^4}{32\pi G}\int|\nabla v|^2.$$

This proves the global minimum on the stated domain. Equality requires $v=0$ in $D^{1,2}$. For a bounded Dirichlet region the same proof uses zero-trace variations.

Testing the weak equation against a_* gives

$$\int|\nabla a_*|^2=\frac{8\pi G}{c^2}\int\rho a_*.$$

Consequently,

$$\begin{aligned}U=J[a_*]&=-\frac{c^2}{4}\int\rho a_*\\&=\frac{1}{2}\int\rho\Phi=-\frac{c^4}{32\pi G}\int|\nabla a_*|^2.\end{aligned}$$

The positive penalty in J and the negative on-shell binding energy are different parts of the same calculation. For a uniform ball, integration over the whole field, including its exterior, gives $U=-3GM^2/(5b)$. Keeping only the material interior loses part of this energy.

10. Two sources and force

Let a_i be the isolated response to ρ_i and U_i its self-energy. Symmetry of the Green kernel gives

$$\begin{aligned}U[\rho_1+\rho_2]&=U_1+U_2\\&\quad-G\iint\frac{\rho_1(x)\rho_2(y)}{|x-y|}\,d^3x\,d^3y.\end{aligned}$$

Proof: expand $-c^2\int(\rho_1+\rho_2)(a_1+a_2)/4$. The two cross terms coincide and combine to the displayed pair interaction. The factor one-half in the self-energy removes double counting.

For spherical bodies whose enclosing balls are disjoint ($d>b_1+b_2$), the cross term is $-GM_1M_2/d$, with d their center separation. Differentiation at fixed body shapes gives attraction of magnitude GM_1M_2/d^2. This uses the same G, density reference and mass assigned to the field equation. The force follows from that common energy account.

For general disjoint compact bodies, differentiation under the integral gives the vector force directly. Positive separation bounds the differentiated kernel and justifies the step. Holding a source fixed while varying its field differs from translating that source and re-solving the field; this proof tracks the latter through the on-shell energy.

11. The floor in a clock receiver

A constant floor has zero gradient. It also cancels from an ordinary endpoint subtraction. For the nonlinear diagnostic $N_{\mathrm{raw}}(a)=\sqrt{1-A_0-a}$, source-free normalization instead gives

$$\frac{N_{\mathrm{raw}}(a)}{N_{\mathrm{raw}}(0)}=\sqrt{1-\frac{a}{1-A_0}}.$$

Its logarithmic slope at zero is $-1/[2(1-A_0)]$. With $A_0=1/(12\pi)$, the residual scale factor is approximately 1.027248616.

The locally declared source-only lapse is

$$N_{\mathrm{loc}}(a)=\sqrt{1-a},\qquad N'_{\mathrm{loc}}(0)=-\frac{1}{2}.$$

It gives the desired leading endpoint coefficient when a retains its Poisson normalization. Defining $\widetilde{a}=a/(1-A_0)$ changes the source amplitude too; that redefinition must be carried through every equation.

These are explicitly different receiver prescriptions. The six-lane interacting-clock test below asks which perturbation of the full source Hamiltonian actually produces the source-only coefficient. Clock normalization alone supplies no density or absolute energy reference.

12. The weak metric interface

Fix static isotropic weak-field coordinates and write

$$\begin{aligned}ds^2=&-[1-a+O(a^2)]c^2dt^2\\&+[1+\gamma a+O(a^2)]d\mathbf{x}^2.\end{aligned}$$

The stationary clock coefficient is one-half. A null path in these coordinates instead has

$$c\,dt=\left[1+\frac{1+\gamma}{2}a+O(a^2)\right]d\ell.$$

The photon-road coefficient depends on the spatial response γ. In conventional post-Newtonian notation $\gamma=1$ gives coefficient one. These metric coefficients are additional physical input; static Poisson and an endpoint clock do not determine γ. [7]

The coordinate convention also matters beyond first order. An exact square-root lapse must be tied to a specified time and radial coordinate before it is used as a nonlinear prediction.

Chapter IV can develop an actual endpoint receiver with reference and motion terms, and Chapter V a path receiver with spatial response. The following finite source constructions provide spatial inputs for both while retaining their separate receiving operations.

13. The actual spatial source

The retained Si29 construction supplies exact finite spatial effects. Its four-component Gram Q has diagonal $(128,136,128,128)$. Its only nonzero off-diagonal entries are 8 on pairs 01, 03 and 23, with symmetric entries. Thus every original entry is retained. [3]

Write $h=\sqrt{Q}$. The orthonormal bright generator has zero diagonal blocks and h in both off-diagonal blocks. Its positive excitation account is

$$H_{\mathrm{gen}}^2=\mathrm{diag}(Q,Q),\qquad H_0=\mathrm{diag}(h,h).$$

Here $H_0 = |H_{\mathrm{gen}}|$, and the two commute. The signed coherent generator and its positive excitation account have different roles. Replacing the signed spectrum by its magnitude is not a common scalar shift.

For species t at site x, let N_{tx} be the exact incoming occupation and G_{tx} the outgoing number Gram. Their retained denominator is 240. Define

$$O_{tx} = \frac{1}{8}\operatorname{diag}(N_{tx}, W_{tx}),$$
$$W_{tx} = \frac{1}{240} h^{-1} G_{tx} h^{-1},$$
$$E_{tx} = H_0^{1/2} O_{tx} H_0^{1/2}.$$

Every N_{tx} is nonnegative and every G_{tx} is a positive Gram. Congruence makes E_{tx} positive. Summing every species and site returns H_0. The actual source evolution, energy transfer, current and coherence retain their original definitions. [3]

This account is an inherited excitation-energy choice. It is not added to H_{gen} as another term or identified with complete rest energy. A spatial density also needs an embedding, physical energy conversion and a common reference.

14. An exact flatness identity

At each of the eight sites the retained matrices obey

$$\sum_t N_{tx} = I, \qquad \sum_t G_{tx} = 240Q.$$

Since $Q = h^2$, substitution into the effects gives the short proof

$$\sum_t O_{tx} = \frac{I}{8}, \qquad E_x := \sum_t E_{tx} = \frac{H_0}{8}.$$

Consequently E_x commutes with H_{gen}. Under H_{gen}, every preparation has a stationary one-eighth share of this excitation energy at each site, although species and coherence evolve in the exact 36-coordinate closure.

The total amplitude can still change with preparation. Retained evaluations are:

Preparation	Total relative energy
e0	11.302793352678457
plus	11.828266226372410
minus	11.131220607210203
phase	11.479743416791305

Each site's energy is the displayed total divided by eight. The fractions are exact; these decimals are retained evaluations. This scalar receiver fixes a spatial shape and does not retain every changing source coordinate.

The independent checker verifies all spatial Gram principal minors, all eight flatness identities, and agreement with the separately retained observable closure. Two native operator words receive the complete incoming and outgoing identities. Their actual execution is recorded in the campaign receipt map.

The next density question is therefore precise: justify this uniform physical localization or derive a different local energy/stress coupling from the source. Species weights cannot be selected merely to obtain a desired field.

15. A finite cube localization

Choose the orthogonal cube embedding $X_x = \ell\xi_x$, with every component of ξ_x equal to $+1$ or -1. It matches the retained cube-edge construction. The edge graph alone does not fix these angles. Its source-coordinate edge length is two; ℓ assigns a physical length.

Choose a normalized nonnegative spatial profile ζ_x and an energy unit e_u. The explicit excitation-mass account is

$$\rho_{\mathrm{exc}}(z) = \frac{e_u}{c^2}\sum_x \langle E_x\rangle \zeta_x(z - X_x),$$
$$M = \frac{e_u\langle H_0\rangle}{c^2}, \qquad \rho_{\mathrm{exc}} = \frac{M}{8}\sum_x \zeta_x.$$

For the complete worked case, each profile is a uniform ball of radius $\delta < \ell$ and volume $4\pi\delta^3/3$. The balls are disjoint; each carries mass $M/8$. This specifies finite interiors and finite self-energy.

The embedding, smoothing radius and E/c^2 conversion are stated assignments. A compatible coupling uses the spatial average of a over each ball, not merely its center value. Other active fields, interactions and reference apparatus still need their own spatial account.

Under the declared Poisson receiver, a point outside every ball has

$$a(z) = \frac{2GM}{c^2} K_{\mathrm{cube}}(z),$$
$$K_{\mathrm{cube}}(z) = \frac{1}{8}\sum_x \frac{1}{|z - X_x|}.$$

The shell theorem makes this equality exact outside the balls. It does not make the cube spherical. The enclosing flux returns M, and the exterior profile retains shape information beyond its monopole.

Declare $H_{\mathrm{gen}}(a) = H_{\mathrm{gen}} - aH_0/2$ for a uniform lift. For the actual incoming/outgoing contrast $B = \operatorname{diag}(I, -I)$, H_0 commutes with B and H_{gen}, so its entire history is unchanged. Scaling H_{gen} itself gives time dilation. This is the Si29 population clock; Chapter VII uses a different, certified phase receiver.

16. Shape and a controlled remainder

Let $r = |z|$, $n = z/r$ and $q = \sqrt{3}\ell/r < 1$. The cube centers have common radius $\sqrt{3}\ell$. Inversion symmetry removes odd multipoles. Its second moment is ℓ^2 times the identity, so the traceless quadrupole also vanishes.

The fourth moment is

$$\frac{1}{8}\sum_x (n\cdot X_x)^4 = \ell^4\left(3 - 2\sum_j n_j^4\right).$$

Insert this into $P_4(u) = (35u^4 - 30u^2 + 3)/8$. The first surviving correction is

$$K_{\text{cube}}(rn) = \frac{1}{r} + \frac{A_4(n)}{r^5} + R_6,$$
$$A_4(n) = \frac{7\ell^4}{4}\left(3 - 5\sum_j n_j^4\right).$$

Since $|P_l| \le 1$, the remaining even Legendre series is bounded by

$$|R_6| \le \frac{q^6}{r(1-q^2)}.$$

The monopole-only bound is $q^4/[r(1-q^2)]$. The degree-four correction is $-7\ell^4/(2r^5)$ on a cube axis and $+7\ell^4/(3r^5)$ along a body diagonal.

This is a source-shape control with no amplitude fit. Its domain requires $r > \sqrt{3}\ell$ and an observation point outside every ball. The smearing changes the interior and self account while leaving this exterior center sum unchanged.

Low-order isotropy and an exactly spherical field are different conclusions. The finite cube supplies the former through degree three. A proposed macroscopic isotropic limit must specify how source orientations, placements or a graph family approach that limit, and bound the surviving shape terms.

17. Finite cube binding energy

Eight uniform balls each have mass $m = M/8$ and radius δ. Their self energies sum to

$$U_{\text{self}} = 8\left(-\frac{3Gm^2}{5\delta}\right) = -\frac{3GM^2}{40\delta}.$$

The cube has 12 unordered pairs at distance 2ℓ, 12 at $2\sqrt{2}\ell$, and four at $2\sqrt{3}\ell$. Nonoverlap and spherical symmetry give each exact center-pair interaction. Thus

$$U_{\text{pair}} = -\frac{GM^2}{64\ell}\left(6 + 3\sqrt{2} + \frac{2}{\sqrt{3}}\right).$$

The sum is the on-solution Newtonian binding account for this chosen density, using the whole field and one infinity reference. The source masses in the linear equation remain the prescribed $M/8$. Adding U/c^2 back into that source would introduce a further self-gravity completion.

This calculation also identifies which energy is missing from an excitation-only mass assignment. A complete account must include active field, interaction, counterterm and reference/apparatus energies under the same reference. Bond interactions need not possess an arbitrary positive site decomposition.

In particular, a positive minimum record cost and this negative binding account are different quantities. Counting the on-solution binding energy and then adding its source-field interaction a second time would double count the same coupling. The finite localization makes this distinction calculable rather than leaving an unregularized point-source self-energy implicit.

18. An exact radial receiver

The retained shell conductance is

$$c_i = i(i+1) = \left(\int_i^{i+1} \frac{du}{u^2}\right)^{-1}.$$

Use radii $r_i = ih$, an outer boundary $a_L = 0$, and no inner sink, $c_0 = 0$. Source shells have masses $M_{\text{unit}} m_i$. With $\kappa = 2GM_{\text{unit}}/(c^2 h)$, solve

$$Ka = \kappa m, \qquad K = B^T C B,$$

where B takes outward edge differences and $C = \text{diag}(c_i)$. Its ith row is

$$(Ka)_i = c_i(a_i - a_{i+1}) + c_{i-1}(a_i - a_{i-1}).$$

For any nonzero v, $v^T K v$ is the positive sum of weighted squared edge differences. They cannot all vanish while the outer value is zero. K is therefore positive definite and the finite problem has a unique solution.

The outer exit carries an essential source flux. Removing it gives the closed graph K_{closed} with $K_{\text{closed}}\mathbf{1} = 0$. Summing its equations would require zero total injection. It cannot support this positive-total static source without an additional sink.

This is the retained spherical receiving geometry, with an explicit extended source and boundary. It is not an identification of concentric shells with Si29's eight cube sites. The two constructions answer distinct spatial questions while sharing a declared energy and receiver convention. [3]

19. Green kernel and flux proof

Sum the source equations through shell i. Interior edge fluxes cancel, leaving

$$c_i(a_i - a_{i+1}) = \kappa M_i, \qquad M_i = \sum_{j \le i} m_j.$$

The edge resistance telescopes:

$$\frac{1}{c_i} = \frac{1}{i} - \frac{1}{i+1}.$$

Summing the remaining differences to the outer boundary gives

$$H_{ij} = \frac{1}{\max(i,j)} - \frac{1}{L},$$
$$a_i = \kappa \sum_j H_{ij} m_j, \qquad KH = I.$$

No desired field values enter the right-hand side. The source and conductance determine them. Between shells use exact harmonic interpolation. On $r_i < r < r_{i+1}$ it gives

$$a'(r) = -\frac{2GM_{\text{unit}}M_i}{c^2 r^2}, \qquad g_r = -\frac{GM_{\text{unit}}M_i}{r^2}.$$

The flux returns the same enclosed mass. Each shell produces its derivative jump and a constant interior potential. At the shell

itself choose a one-sided acceleration or the mean self-field used in radial virtual work.

This network samples the exact continuum thin-shell solution. The finite outer reference differs from isolation by the known constant $2GM_{\text{total}}/(c^2Lh)$. Gradients and fluxes agree; potential values and escape to infinity require that correction. Refining the radial mesh cannot remove a boundary-reference error held fixed in physical units.

20. A complete rational example

Take $L = 9$, $\kappa = 1/50$, and the shell masses shown below. Their total is one. These fractions specify a test source, not an inferred microscopic mass spectrum.

i	m_i	a_i	Enclosed M_i
1	0	1777/630000	0
2	1/5	1777/630000	1/5
3	0	1357/630000	1/5
4	3/10	1147/630000	1/2
5	1/10	52/39375	3/5
6	0	29/31500	3/5
7	2/5	1/1575	1
8	0	1/3600	1

Every outgoing flux is κM_i, including the final exit. Let $u = a/\kappa$ and measure energy in GM_{unit}^2/h. The finite functional is $F = u^T Ku/2 - m^T u$. Its minimum is $-m^T Hm/2$, giving

$$\frac{U_L}{GM_{\text{unit}}^2/h} = -\frac{9427}{252000},$$
$$\frac{U_\infty}{GM_{\text{unit}}^2/h} = -\frac{2603}{28000}.$$

The difference is the exact correction $-GM_{\text{total}}^2/(2Lh)$. The exterior positive gradient term and the source-reference change combine into this signed on-solution result.

Varying a shell radius at fixed ordering gives

$$-\frac{dU}{dr_j} = -\frac{GM_j}{r_j^2}\left(M_{<j} + \frac{M_j}{2}\right).$$

The half self term is the mean of its two one-sided self-fields. This is radial work under spherical expansion, rather than net vector force on the shell center.

21. A controlled continuum limit

For a uniform ball of radius b, divide radius into n cells with $h = b/n$. Move each cell's mass to its outer endpoint:

$$\frac{m_j}{M} = \frac{j^3 - (j-1)^3}{n^3}.$$

Write $\eta = GM/(c^2 b)$, $\alpha = a/\eta$. The continuous profile is $3 - x^2$ inside and $2/x$ outside. The exact potential error obeys

$$0 \le \alpha_{\text{ball}} - \alpha_{\text{shells}} \le \frac{3}{n} - \frac{2H_n}{n^2},$$

where H_n is the nth harmonic sum. The bound is attained at the center. Moving a mass element from s to its endpoint s_j reduces its kernel by at most $1/s - 1/s_j$. Integrating that difference gives the displayed bound.

For binding energy the corresponding bound is

$$0 \le \frac{U_{\text{shells}}}{GM^2/b} + \frac{3}{5} \le \frac{3}{2n} - \frac{H_n}{n^2}.$$

Use $1/\max(s,t) = \min(1/s, 1/t)$ in the pair integral; outward motion changes that minimum by at most the sum of the two reciprocal-radius changes. The energy factor one-half yields the bound.

A finite absorbing radius R adds reduced lift offset $2b/R$ and energy offset $GM^2/(2R)$. Isolation therefore requires both $h/b \to 0$ and $b/R \to 0$. Refining while $R/b = 2$ leaves lift bias one and reduced energy bias one-quarter. The radial continuum limit controls discretization; it does not derive the source graph's angular isotropy.

22. A Chapter VII field density

Take the original static classical quadrupole field and average its positive free-field energy over angles. Write X^2 for the sum of the two squared source amplitudes. With $\kappa = 4/5$, its total free energy is $U_f = X^2/R$. [1,5]

Conditionally assign $M_f = U_f/c^2$ to that free-field account and $x = r/R$. Its spherical density and enclosed mass are

$$\rho_{\text{in}} = \frac{M_f r^2}{2\pi R^5}, \qquad \rho_{\text{out}} = \frac{3M_f R^5}{4\pi r^8},$$
$$\frac{M(<r)}{M_f} = \frac{2}{5}x^5 \qquad (x \le 1),$$
$$\frac{M(<r)}{M_f} = 1 - \frac{3}{5}x^{-5} \qquad (x \ge 1).$$

Two-fifths of U_f lies inside R and three-fifths outside. Solving the same Poisson receiver, with $x = r/R$, gives

$$a_{\text{in}} = \frac{2GM_f}{c^2 R}\left(1 - \frac{x^4}{10}\right),$$
$$a_{\text{out}} = \frac{2GM_f}{c^2 r}\left(1 - \frac{x^{-5}}{10}\right).$$

Both a and a' match at R. The source field carries energy outside R; treating that energy as mass makes this region nonvacuum. Its exact correction is proportional to r^{-6}, rather than a pure vacuum $1/r$ profile.

At the commuting classical static minimum, each boundary amplitude equals half ε times its source current. Its free energy, interaction and counterterm satisfy

$$U_f + U_{\text{int}} + U_{\text{ct}} = U_f - 2U_f + U_f = 0.$$

This cancellation shows why the positive free-field sector alone is a conditional source choice. The quantum D and C do not commute on the nontrivial lanes; their joint local energy account still needs a derivation. R remains the original coupling radius, distinct from the cube scale and radial outer boundary.

23. Energy unread by the clock

Retain the original sixteen-state source: S is Swap, $D = Q_{\text{left}} - Q_{\text{right}}$, $C = iSD$, and $E_{\text{even}} = Q_{\text{left}} + Q_{\text{right}} + Q_{\text{left}}Q_{\text{right}}$. Its unnormalized current is $Z_{\text{raw}} = D + iC$. Exact multiplication gives

$$[S, Z_{\text{raw}}] = -2Z_{\text{raw}},$$
$$[E_{\text{even}}, Z_{\text{raw}}] = [D^2 + C^2, Z_{\text{raw}}] = 0.$$

The full Hamiltonian is $H_s + H_f + H_{\text{int}} + H_{\text{ct}}$, with $H_s = \varepsilon(S + E_{\text{even}})$, $H_{\text{int}} = -\varepsilon(D\phi_+ + C\phi_\times)$, and $H_{\text{ct}} = \varepsilon^2\mu(D^2 + C^2)$, $\mu = 1/(4R)$. The free field, both polarizations and both counterterms are retained. E_{even} also commutes with the complete interacting Hamiltonian. Adding any bounded real function $F(E_{\text{even}})$ therefore changes energy while leaving the entire current history unchanged for every initial state:

$$Z_{\mathscr{H}_0 + F}(t) = Z_{\mathscr{H}_0}(t).$$

The four diagonal source states have $D = C = 0$ yet distinct energies $\varepsilon(1 + q_a)^2$. They are current-dark energy controls, not clocks with a nonzero amplitude. Observing the current does not determine all source energy. [5]

A scalar addition kI supplies the simplest common-energy example. Density evolution and relative frequencies stay fixed while the common amplitude phase changes. An external phase reference can compare that phase through its stated coupling and payment; it does not choose a mass baseline by itself.

This distinction also applies to the Si29 account. Independently shifting its positive H_0 by kI changes the candidate assigned mass by $e_u k/c^2$. It does not follow from replacing H_{gen} by $H_{\text{gen}} + kI$ and taking its absolute value. Reference energy, signed generator, positive account and actual clock receiver must retain their separate definitions.

24. The operational clock response

Supply a physical static-lift perturbation

$$\mathscr{H}(a) = \mathscr{H}_0 + aV + O(a^2).$$

On lane l, let $Z = (D + iC)/d_l$, $z = \langle Z \rangle$, and $\nu = -\operatorname{Im}(\dot{z}/z)$, as in Chapter VII. For a bounded V and a fixed preparation, Duhamel gives

$$\delta z(t) = i \int_0^t \langle [V_I(s), Z_I(t)] \rangle \, ds.$$

Where the baseline z_0 stays away from zero, the actual phase-rate derivative is

$$\delta\nu = -\operatorname{Im}\left(\frac{\delta\dot{z}}{z_0} - \frac{\dot{z}_0 \delta z}{z_0^2} \right).$$

A field-form perturbation needs common energy-form control and a dominated differentiated evolution. An a-dependent preparation supplies an additional state derivative. These requirements belong to the chosen V; the original sixteen-dimensional commutator is only its source algebra.

One exact full-continuum control is the declared coupling of the full interacting Hamiltonian:

$$\mathscr{H}(a) = n(a)\mathscr{H}_0 + f(a)I.$$

It gives

$$z(t;a) = z(n(a)t;0),$$
$$\nu(t;a) = n(a)\nu(n(a)t;0).$$

At matched internal age the frequency ratio is n. At equal coordinate time, where $\dot{\nu}$ exists, its derivative also includes $n'(0)t\dot{\nu}(t;0)$. The actual retained clock is slightly nonstationary. Accumulated phase follows the time change without requiring an additional second derivative.

Thus a square-root n is an explicit coupling control. A first-order redshift test must specify matched age, a stationary reference, or the transient contribution at equal coordinate durations.

25. Full scaling and ε

The complete scaling control multiplies H_s, H_f, H_{int} and H_{ct} by the same n. Replacing only ε by $n\varepsilon$ at a fixed field instead gives

$$\mathscr{H}_\varepsilon = nH_s + H_f + nH_{\text{int}} + n^2 H_{\text{ct}}.$$

For $s = n'(0)$, its first-order difference from full scaling is

$$V_\varepsilon - s\mathscr{H}_0 = s(H_{\text{ct}} - H_f).$$

For the source-only slope $s = -1/2$, the residual coefficients on $(H_s, H_f, H_{\text{int}}, H_{\text{ct}})$ are $(0, 1/2, 0, -1/2)$. This is not a scalar on the full field. The counterterm commutes with the full Hamiltonian and current, so its energy stays unread throughout this history. The free-field part of the residual can change the current.

The retained hard-branch second-order diagnostic is

$$\Omega_l^{(2)} = \Delta(1 - 4g_l^2 J), \qquad \Delta = 2\varepsilon,$$
$$\frac{d\log\Omega_l^{(2)}}{d\log\varepsilon} = 1 - \frac{8g_l^2 J + 4g_l^2 \Delta J'}{1 - 4g_l^2 J}.$$

The hard split is held fixed in this perturbation. J and J' are positive, so the derivative is below one. The six original coupling squares differ, producing six disjoint derivative enclosures. These are second-order source diagnostics, not finite-lift full-field response certificates.

The actual physical V remains the question. A shared perturbation across six separately prepared lanes can test universality; fitting each lane's ε independently would change that test.

The energy unit e_u, action I_u and length ℓ_{phys} must be shared between source, clock and field. They set the source time parameter and cube's lift coefficient:

$$\theta_{\text{source}} = \frac{e_u t}{I_u}, \qquad \frac{2Ge_u}{c^4 \ell_{\text{phys}}}.$$

A physical ε alone does not determine both conversions.

26. Correlated phase and precision

Under the declared full scaling, compare the same baseline phase history at T and nT. Within the common certified window, with $\ell \le \nu_0 \le u$,

$$\delta\theta(T) = -\int_{nT}^{T} \nu_0(s)\,ds,$$
$$-u(1-n)T \le \delta\theta \le -\ell(1-n)T < 0.$$

For $n = \sqrt{1-a}$, $0 < a < 1$, rational bounds on $1-n$ give

$$-\frac{uaT}{2-a} \le \delta\theta \le -\frac{\ell aT}{2}.$$

The signed shift exists for every positive a and T in this window. At the frozen Earth surface lift, all six one-bare-cycle shifts lie between -4.373898 and -4.373888 nanoradians. These are conditional source predictions, not achieved apparatus sensitivity. [5]

A separate comparison encloses the two phases independently. Those bands are disjoint only under the sufficient condition $a > 1-(\ell/u)^2$. Its conservatively rounded thresholds are:

Lane	Sufficient a
01	2.84407×10^{-9}
02	1.22170×10^{-8}
03	1.89316×10^{-6}
12	3.27189×10^{-9}
13	1.74924×10^{-6}
23	1.60120×10^{-6}

These thresholds concern independent enclosures; they are not a minimum lift for the correlated phase sign. A phase error δ on each comparison needs

$$\ell(1-n)T > 2\delta.$$

The physical coupling, common preparation and unwrapped phase branch remain explicit. Chapter II's fresh preparations, timing/error budgets, paid reference and turn counter supply the separate measurement protocol. [2]

27. Frozen anchors and reproduction

The recovered CR004 premises retain the same declared gravitational parameters and radii. Recalculation gives the following spherical surface values; escape speeds are km/s. [4]

Body	Lift $a(b)$	g, m/s^2	Escape
Earth	1.39225×10^{-9}	9.82025	11.1861
Moon	6.28226×10^{-11}	1.62490	2.37618
Sun	4.24501×10^{-6}	274.200	617.675

The formulas use the same frozen $\mu = GM$ and radius on both sides. These inputs preserve the intended Newtonian recovery; they do not determine mass from the microscopic account or select G. The digits expose arithmetic, rather than measured precision or corrections for shape and rotation.

The finite cube and shell controls add different information: a source-shape correction, exact source flux, boundary energy and quantified continuum error. The interacting-clock controls add the complete perturbation and a visible precision target.

The preceding source-to-field pass retained editable manuscript, exact operands, source snapshots, native receipts, independent code and content hashes. The source checker has 2,040 exact assertions, the radial checker 312, and the clock checker 822. The static-field checker adds 52 assertions, the kernel/frozen-arithmetic checker 45, the uniform-coupling certificate 23, and the correlated-phase supplement 47. Fresh native execution and custody are recorded separately in the campaign receipt map; retained historical calls are not counted as new work.

The old draft's 42 symbolic checks remain historical. They are not relabeled as the present verification. CR004, G224's retrieved index and LC03 retain their original evidence scopes. The original G224 derivation is still unrecovered; LC03 is an imported mechanism replay, and the current retained source route is the Si29 construction used here.

28. The next physical derivations

First derive the actual local energy/stress coupling, including field, interaction and reference terms. Carry its spatial density through one calibrated receiver.

Next derive a graph or field response with a massless isotropic k^2 leading term and bounded spatial/frequency error, retaining the radial boundary account.

Then apply one physical V to all six clocks, with matched-age or transient timing explicit. Derive causal kinetics and the spatial metric response, including γ, using the same source normalization.

Memory and closure methods control eliminated information. The spatial operator and metric coupling need their own derivations.

References and retained sources

[1] *Volume I, Chapter VII* (now Chapter VII), final draft and `sam_volume_I_i` source/certificate, 15 October 2026.

[2] Same authors. *Chapter II: Retention, accumulation and the finite ledger*, 20-page physical development draft and this revision's companion source-to-space update.

[3] SAM Research Project. `volii_duck_expansion1/2`, Si29 effects; `native_energy_source1`, radial response; current source/radial reports and operands.

[4] SAM Research Project. CR004 declared premises, precommit and scoped weak-field result. Current arithmetic uses the frozen premises.

[5] Current clock/static-field derivations, certificates, checks and native receipt map, companion archive.

[6] SAM Research Project. `native_coupling_scale1` and `source_spatial_response1`: selected coupling and frequency-dependent source-response scopes.

[7] C. M. Will. *The Confrontation between General Relativity and Experiment*. Living Reviews in Relativity 17, 4 (2014), doi:10.12942/lrr-2014-4. Conventional metric notation.

29. A local angular action with the retained six

The actual angular map is the transverse tracefree projection of an STF tensor Q, normalized by $UQ = \sqrt{5/(8\pi)}\Lambda_n Q$. Its sphere norm is exactly $\|Q\|^2$. On the complete five-dimensional STF image, the ordinary Cartesian-component angular gradient has energy $4\|Q\|^2$. It does not supply the radial coefficient six by itself. [8]

Let J_i generate rotations, $v_i = J_i n$, $D_i = v_i \cdot \nabla_S$, and $S_i T = [J_i, T]$. The total-rotation derivative and its Casimir are

$$\mathscr{L}_i T = [J_i, T] - D_i T, \qquad \mathscr{C} = -\sum_i \mathscr{L}_i^2.$$

Equivariance gives $\mathscr{L}_i UQ = U[J_i, Q]$. On STF tensors, $-\sum_i [J_i, [J_i, Q]] = 6Q$. Therefore $\mathscr{C} UQ = 6UQ$, including the three undriven STF directions. The positive form retains its orthogonal complement.

For every smooth tangential tracefree tensor section, a stronger local identity holds:

$$\begin{aligned}\sum_i \|\mathscr{L}_i T\|^2 &= \|\nabla_S^{\mathrm{Cart}} T\|^2 + 2\|T\|^2, \\ &= \|\nabla_S^{\mathrm{LC}} T\|^2 + 4\|T\|^2.\end{aligned}$$

At one ray differentiate $Tn = 0$ and $\operatorname{tr} T = 0$; the two removed normal components contribute $2\|T\|^2$. Expanding the rotation derivatives supplies the other identity. Rotational covariance extends the calculation to every ray. Exact first-jet Gram checks and the five-coordinate Casimir certify the coefficients.

This is one local positive receiving extension compatible with the retained radial model. It preserves its normalization and source complements. A global translation-invariant microscopic field or moving-center action remains a further construction.

30. The unaveraged field source

For the actual real boundary vector $x = (x_+, x_\times)$, put $p = x_+^2 + x_\times^2$, $z = n_z$ and $w = x_+(n_x^2 - n_y^2) + 2x_\times n_x n_y$. The local angular kernels are

$$N = pz^2 + w^2/4, \qquad M = 2p(1+z^2) - w^2.$$

With $\kappa = 4/5$, the same static profiles as section 22 give

$$\begin{aligned}e_{\mathrm{in}} &= \frac{r^2}{8\pi R^6}\, 2p(1+3z^2), \\ e_{\mathrm{out}} &= \frac{R^4}{8\pi r^8}\left(2p + 11pz^2 + \frac{5}{4}w^2\right).\end{aligned}$$

Both are nonnegative. Their integral is p/R, with fractions $2/5$ inside and $3/5$ outside. Their angular mean is exactly the preceding spherical density; no normalization is fitted.

Conditionally set $\rho = e_{\mathrm{phys}}/c^2$, physical radius $R_c = \ell_u R$ and mass $M_{\mathrm{src}} = e_u p/(c^2 R)$. Write $\xi = r_{\mathrm{phys}}/R_c$ and $a = 2GM_{\mathrm{src}}F/(c^2 R_c)$. The complete exterior response is

$$\begin{aligned}F_0 &= \xi^{-1} - \xi^{-6}/10, \\ F_2 &= (2\xi^{-3}/7 - \xi^{-6}/7)P_2, \\ F_4 &= (\xi^{-5}/126 - \xi^{-6}/140)P_4 \\ &\quad + (5\xi^{-5}/144 - \xi^{-6}/32)H_\beta.\end{aligned}$$

For $p > 0$, $\beta_c = (x_+^2 - x_\times^2)/p$, $\beta_s = 2x_+x_\times/p$, and the actual azimuthal harmonic is

$$\begin{aligned}H_\beta &= \beta_c(n_x^4 - 6n_x^2 n_y^2 + n_y^4) \\ &\quad + 4\beta_s n_x n_y (n_x^2 - n_y^2).\end{aligned}$$

It is fixed by the plus/cross phase. Matching and

$$D_l \xi^k = [k(k+1) - l(l+1)]\xi^{k-2}$$

verify the regular interior and exterior solutions. Every finite exterior radius still contains source energy.

The complete binding is $-2125/4851$ times GM_{src}^2/R_c; the angular mean alone gives $-421/990$. Orthogonal $L2$ and $L4$ gradient energies account for the difference. This remains the chosen classical free sector, with interaction and quantum density carried separately. [8]

31. The product-state quantum variance

Keep the complete Chapter VII lane Hamiltonian

$$\begin{aligned}\mathscr{H} &= \varepsilon e I + \varepsilon S + K, \\ K &= H_f - g(\sigma_z \phi_+ + \sigma_y \phi_\times) + 2g^2 \mu I.\end{aligned}$$

Here $g = \varepsilon d$ retains its original sign and $\mu = 1/(4R)$. For source Bloch coordinates r_x, r_y, r_z and field coherent amplitudes $\alpha_+, \alpha_\times$, the same completed square gives

$$\begin{aligned}\langle K \rangle &= \sum_\alpha \int \omega \left| \alpha_\alpha - \frac{g r_\alpha u}{\omega} \right|^2 d\omega \\ &\quad + g^2 \mu (2 - r_y^2 - r_z^2).\end{aligned}$$

The Bloch ball implies $r_y^2 + r_z^2 \le 1$. Thus every product preparation has $\langle K \rangle \ge g^2 \mu$. A pure source in the yz plane and its optimal coherent displacements attain equality. The original displacement and energy norms are finite, so these are actual Fock form-domain vectors. Scalar square completion gives the same lower bound for arbitrary finite-energy product field states and integrable convex product ensembles.

The classical cancellation omits precisely this source variance. It cannot evaluate two noncommuting quantum currents as simultaneously sharp numbers.

The complete-H product minimum also retains source exchange. Put $L = g^2 \mu$ and subtract only the displayed common even energy. Minimizing $\varepsilon r_x + L(1 + r_x^2)$ on $[-1, 1]$ gives

$$E_{\mathrm{prod}} - \varepsilon e = \begin{cases} -\varepsilon + 2L, & \varepsilon \ge 2L, \\ L - \dfrac{\varepsilon^2}{4L}, & \varepsilon < 2L. \end{cases}$$

These are comparison preparations. The actual interacting branch superposition is retained in the next section. [8]

32. Actual preparation, energy and kernel

The actual pure clock state remains $(\Phi_0+\Phi_1)/\sqrt{2}$ times the common soft vacuum, with the original exact hard-field branches. Upstream definitions authenticate K_{prep} as a ceiling on the full positive K, including both soft counterterms. It is not a proof-error or displaced excess budget.

Lane	Product K minimum ≈	Actual K upper ≈
01	1.39169e-8	8.42278e-17
02	5.97811e-8	5.99696e-16
03	9.26352e-6	7.49026e-12
12	1.60103e-8	9.98060e-17
13	8.55933e-6	6.39791e-12
23	7.83497e-6	5.36406e-12

Every exact upper/lower ratio is below one. This is an energetic certificate of the source-field entanglement already established by reduced-source purity. Lanes 03,13,23 additionally lie below the complete product-H minimum.

At preparation, $\langle K\rangle \geq 2g^2\mu_{\mathrm{soft}} > 0$ because the soft field is vacuum and $K_h \geq 0$. The alternating j_2 series gives rigorous positive rational lower bounds. The retained ceiling gives a complete signed account through

$$\begin{aligned}\max(0,\sqrt{c}-\sqrt{K_{\mathrm{up}}})^2 \leq \langle H_f\rangle,\\ \langle H_f\rangle \leq (\sqrt{c}+\sqrt{K_{\mathrm{up}}})^2,\\ \langle H_{\mathrm{int}}\rangle = \langle K\rangle - \langle H_f\rangle - c,\end{aligned}$$

with $c = 2g^2\mu$; the even source energy is added back, not discarded.

Zero K would annihilate both smeared polarization shifts. Their commutator forces $[D,C]\Psi = -2iSD^2\Psi = 0$. Exact $\mathrm{rank}(D^2) = 12$ leaves only the four diagonal source states; there $K = H_f$. Hence $\ker K$ is those four states times field vacuum. No positive active-lane spectral-infimum bound is inferred from this kernel statement. [8]

33. The complete current-blind source algebra

On the actual 16-state source let $Z = D + iC$ and $Z^* = D - iC$. The exact rational commutator maps have dimensions 512×256 and 768×256. Their certified ranks are 234 and 246, respectively, when H_{pair} is added. Consequently

$$\begin{aligned}\dim_{\mathbb{C}}\{Z,Z^*\}' = 22,\\ \dim_{\mathbb{C}}\{Z,Z^*,H_{\mathrm{pair}}\}' = 10.\end{aligned}$$

The six original squared gaps d^2 are distinct. Each active 2D lane is irreducible under D,C, so a commuting operator is scalar there. The four current-dark states permit an arbitrary 4×4 block:

$$W = \bigoplus_{a<b} w_{ab} I_2 \;\oplus\; W_{\mathrm{dark}}.$$

Thus the 22-dimensional algebra is $\mathbb{C}^6 \oplus M_4(\mathbb{C})$, with Hermitian real dimension 22. Its full rational basis and nonorthogonal dark Gram are retained; no fitted eigenbasis is needed.

For Hermitian source W, this condition is necessary and sufficient to leave every current operator history unchanged for every joint initial state. Necessity follows from the zero-time derivative $i[W,Z]$. For sufficiency, W is scalar in each invariant active lane and the remaining block has zero current. Both continuum fields stay present. Time-dependent integrable W in the same algebra are also invisible.

Requiring conservation is stronger. Commutation with the complete Hamiltonian restricts the dark block to its four distinct energy eigenspaces. The resulting ten-dimensional space is exactly the functions $F(E_{\mathrm{even}})$ certified by ten independent powers. A dark mixing operator can remain current-blind while changing its own energy dynamics.

For a compact added-Hamiltonian certificate, let A be the current-commutator map, U its complete 22-vector kernel basis and B the H_{pair} commutator. Eliminating a complement of $\ker A$ gives

$$\begin{aligned}\mathrm{rank}\begin{pmatrix}A\\B\end{pmatrix} &= \mathrm{rank}\,A + \mathrm{rank}(BU)\\ &= 234 + 12.\end{aligned}$$

The restricted 12-rank certificate and ten joint-kernel vectors retain the original result. These constraint spaces do not replace the physical 16-state source. [8]

34. What six first-order clocks can receive

Each original clock preparation occupies one invariant active lane. An inter-sector perturbation has an off-block first-order Duhamel commutator, whose expectation in that prepared lane is zero. It can return to the lane at second order.

There are 216 Hermitian off-block directions. Adding the 22 all-time blind directions gives

$$\dim_{\mathbb{R}} \mathscr{N}_{\mathrm{first}} = 216 + 22 = 238.$$

This is exact when the test requires first-order blindness for every state inside each of the six lanes. The six particular preparations can have additional cancellations. An exact witness satisfies all 18 complementary trace constraints while its double commutator returns a nonzero bright response. Shared first-order slopes test the supplied coupling family; they do not reconstruct arbitrary V.

The separate Si29 source yields another exact consequence. Its frozen all-species local energies satisfy $E_x = H_{\mathrm{pos}}/8$. Therefore every prescribed profile gives

$$\begin{aligned}\delta H(t) &= -\frac{e_u}{2} a_{\mathrm{eff}}(t) H_{\mathrm{pos}},\\ a_{\mathrm{eff}}(t) &= \frac{1}{8}\sum_x \bar{a}_x(t).\end{aligned}$$

$H_{\mathrm{pos}} = |H_{\mathrm{gen}}|$ commutes with both H_{gen} and the retained incoming/outgoing contrast B. Thus the entire B history is unchanged for arbitrary nonuniform, time-dependent integrable profiles in this fixed embedding. A spatial gradient cannot change B through these particular frozen energy effects.

Scaling H_{gen} instead changes the transfer clock's time. Other species/coherence observables and a different complete local coupling retain their own responses. This calculation distinguishes the two supplied couplings on the actual retained source, without replacing it by a generic two-level system. [8]

35. One complete chosen local classical action

The local angular identity gives one compatible centered classical bulk action, in retained units:

$$L_f = \frac{\kappa}{4}\int \left[|\dot{H}|^2 - |\nabla H|^2 - \frac{2|H|^2}{r^2}\right] d^3x.$$

Its positive energy changes all signs to plus. The current $j_i = -(\kappa/2)\dot{H} : \partial_i H$ satisfies $\partial_t e_f + \operatorname{div} j = 0$ away from the source. On the actual STF image the angular integral is six, reproducing the retained radial action and canonical momentum.

Specify $\gamma_R = \delta(r-R)/R^2$ and the normalized shell $w_R = \gamma_R/(4\pi)$. One complete stated incidence is

$$e_s = w_R H_s, \qquad e_{\mathrm{ct}} = w_R H_{\mathrm{ct}},$$
$$e_{\mathrm{int}} = -\varepsilon\gamma_R \sum_\alpha J_\alpha \mathscr{U}(e_\alpha) : H.$$

On the retained radial image, U's isometry makes their integrals $H_s, H_{\mathrm{ct}}, H_{\mathrm{int}}$. Integrating $e_{\mathrm{total}} = e_f + e_s + e_{\mathrm{int}} + e_{\mathrm{ct}}$ returns the original Hamiltonian there, including its signed interaction and common even energy. The full TT extension keeps its complement.

With $f_\alpha = \varepsilon J_\alpha$, variation gives $[H_r]_-^+ = -2U(\sum f_\alpha e_\alpha)/(\kappa R^2)$. Boundary flux and the interaction change cancel with the supplied power $f \cdot \dot{\varphi}(R)$. Uniform lapse switching pays $\dot{N}\langle H_0\rangle$; signed entry/release cancellation does not establish zero apparatus cost. Source motion retains its own work, and static support retains stress.

The shell localization of H_s/H_{ct} is an assignment. A finite-source Berry action and this classical field define the stated mean-field construction; these formal densities are not asserted to be renormalized quantum stress. The full TT complement, regular origin condition and external reference account remain. A moving center and globally translation-invariant microscopic law require further construction. [8]

36. Lapse coupling and localization

One uniform lapse N multiplying the complete original Hamiltonian gives the exact quantum law on its unchanged domain:

$$\mathscr{H}_N = N\mathscr{H}_0.$$

The canonical one-form is retained and every sector receives N. For the supplied $N(a) = \sqrt{1-a}$, $V = -H_0/2$. The clock time change follows exactly on the original interacting field, without requiring a new local quantum-density theorem.

A spatially varying lapse requires an actual energy localization $h(x)$, including interaction, counterterm, source support and reference. The chosen classical construction can couple it through

$$H[N] = \int N(x)h(x)\,d^3x.$$

The same stated action gives $\delta H = -(1/2)\int \delta a\, h$. With the positive Poisson gradient term, its fixed-state weak-field equation is $\Delta a = -8\pi G\langle h\rangle/c^4$ after physical conversion. For a translated prescribed compact profile, weak virtual work gives $F = (1/2)\int h\nabla a$. Its active and passive monopoles use the same total energy divided by c^2; inertial mass still needs the center-of-mass/boost law.

If h changes to $h + \operatorname{div} B$ while its integrated Hamiltonian stays fixed, the linear coupling changes by

$$\delta V = \frac{1}{2}\int \nabla a \cdot B\, d^3x - \frac{1}{2}\oint aB \cdot dS.$$

Thus integrated energy fixes the uniform coupling without selecting every finite-size response. The action must select the improvement and boundary convention.

For a nonuniform quantum lapse the composite-field smearing, common form domain, vacuum prescription and noncommuting source terms remain to be defined. Classical current means lose section 31's variance. The chosen classical action supplies one compatible example, while the microscopic lapse law, quantum spatial stress and metric γ retain their own derivations. [8]

37. A finite-lift actual-phase certificate

For fixed preparation, the bounded-V endpoint response can use the actual accumulated phase directly:

$$\theta_1(T) = -\operatorname{Im}\frac{z_1(T)}{z_0(T)}.$$

Here z_1 is the Duhamel integral in section 24. No mixed derivative of $\dot{z}$ is needed. Preparation variation contributes its own expectation and, when phases are normalized separately, an initial-endpoint term.

Compare the declared full scaling plus an exactly blind source term with a bounded self-adjoint residual R. For the same initial state, Duhamel gives $|z - z_{\mathrm{scale}}| \le 2\|Z\|T\|R\|$. On a preserved original lane $\|Z\| = 2$; arbitrary leakage needs the full receiver norm or a projected receiver with a leakage budget.

Let $m = M_{\mathrm{lower}} - E_{\mathrm{cycle}} > 0$, $T_{\mathrm{lo}} \le T_0 \le T_{\mathrm{hi}}$, and $x = \ell a T_{\mathrm{lo}}/2$. Since $\sin x \ge x - x^3/6$, a strict sufficient sign budget is

$$\|R\| < B, \qquad B = \frac{m(x - x^3/6)}{4T_{\mathrm{hi}}}, \quad 0 < x < 1.$$

The current-error disk stays away from zero at every prefix; its angle is smaller than x, while the ideal negative shift has magnitude at least x. This also fixes the unwrapped branch.

Lane	B, model-energy norm $\approx$
01	6.073891199e-16
02	6.073891111e-16
03	6.073873394e-16
12	6.073891195e-16
13	6.073874749e-16
23	6.073876144e-16

These exact-rational budgets use the frozen Earth lift and original window. A declared lane-preserving $\|R\| \le a^2$ fixture keeps all six shifts negative. A physical field-form remainder and attainable readout precision need their separate estimates; no SI sensitivity is assigned. [8]

38. Self-consistent finite-shell stability

Use the same eight-shell source, Green inverse $H = K^{-1}$, $\kappa = 1/50$, total mass one and boundary exit as section 20. A new explicit source-lapse functional is

$$F(a) = \frac{a^T K a}{2\kappa} + 2\sum_i m_i \sqrt{1-a_i}.$$

Its stationary equation is no longer the prescribed linear source:

$$Ka = \kappa \left(\frac{m_i}{\sqrt{1-a_i}} \right)_i.$$

It is a declared finite receiving model with fixed positive shell rest energies. It is not the complete quantum-source susceptibility.

On the weak box $0 \le a_i \le b = 1/100$, the positive Green map sends the box into itself. The same linear source gives $a_{0,\max} = 1777/630000$. Its sup-norm contraction is bounded by

$$q = \frac{1777}{1234926} < 1.$$

Hence the box has exactly one fixed point. Thirty-two directed rational Picard steps enclose it with width below 10^{-70}; its largest component is approximately 0.0028237945677064, above the linear 0.0028206349206349.

The Hessian is bounded below throughout the box by

$$\frac{K}{\kappa} - \operatorname{diag}\left(\frac{m_i}{2(1-b)^2} \right).$$

All eight exact leading principal minors are positive. This supplies strict convexity and local stability throughout the chosen weak box. The native inverse of that conservative Hessian has the same retained source and boundary binding.

The result establishes a finite self-consistent extension under its stated lapse law. A source-dependent rest-energy response, a strong-lift continuation and the full spatial metric remain separate calculations. [8]

39. An actual restored orbital preparation

The verified T500 electron-basis archive supplies the literal normalized mode index 6, $l = 3, m = 3, t = 1$, coefficient one. Occupying it with both orthogonal spins gives an allowed $N = 2$ Slater member of the archived 16/20-spin families. It is an explicit prepared-family member, rather than a recovered winning minimum. [8]

Let d be a declared physical dilation, $s = r/d$ and $u = \cos\theta$. The retained wavefunction and angular modulus are

$$\psi_{33} = d^{-3/2}\sqrt{4/315}\, s^3 e^{-s} Y_{33},$$
$$|Y_{33}|^2 = \frac{35}{64\pi}(1-u^2)^3.$$

Thus the actual two-spin number density is

$$n(r,u) = \frac{s^6 e^{-2s}(1-u^2)^3}{72\pi d^3}.$$

Separate radial and angular integrals give $\int n = 2$; n is pointwise nonnegative. A number/rest component may be received as $\rho = Mn/2$ after specifying its total component mass M. The original kinetic, exchange, interaction, recoil and common-reference energies retain their own accounts.

The exact angular polynomial has only even Legendre modes:

L	c_L in $(1-u^2)^3$
0	$16/35$
2	$-16/21$
4	$144/385$
6	$-16/231$

The full polynomial is reconstructed from all 49 exact angular Gram entries. This restores a source-bound anisotropic density example without identifying a particle count with the complete gravitating energy or assigning the original atomic attachment energy as an absolute mass.

40. The full noncompact orbital field

Under the same declared Poisson receiver, sum over $L = 0, 2, 4, 6$ to obtain

$$a = \frac{2GM}{c^2 d} \sum_L \frac{c_L P_L(u)}{36(2L+1)} \times [s^{-L-1} I_{L+8}(s) + s^L J_{7-L}(s)],$$

The two radial integrals are

$$I_p(s) = \int_0^s t^p e^{-2t}\, dt,$$
$$J_p(s) = \int_s^\infty t^p e^{-2t}\, dt.$$

For integer p they are finite exponential polynomials:

$$J_p = \frac{p!}{2^{p+1}} e^{-2s} \sum_{k=0}^{p} \frac{(2s)^k}{k!},$$
$$I_p = \frac{p!}{2^{p+1}} - J_p.$$

Both integrals remain at every finite radius. Their derivatives cancel the source terms and the radial operator returns the full Poisson density. Since I_p begins at $s^{p+1}/(p+1)$, all modes are regular at the origin.

The far potential coefficients multiplying P_L/s^{L+1} are

L	Far coefficient
0	1
2	$-15/2$
4	$135/2$
6	$-1575/4$

The omitted term in each asymptotic mode is bounded by

$$\frac{|c_L|}{36(2L+1)}[s^{-L-1}J_{L+8}+s^L J_{7-L}].$$

The exact enclosed fraction is

$$\frac{M(<r)}{M}=1-e^{-2s}\sum_{k=0}^{8}\frac{(2s)^k}{k!}.$$

There is no finite-radius vacuum boundary. The four radial identities have an exact 64-row coefficient certificate and the original source binding. [8]

41. Force, tides and source-error bounds

The same receiving law supplies the mechanical and tidal observables by differentiation:

$$\mathbf{g}=\frac{c^2}{2}\nabla a, \qquad T=\frac{c^2}{2}\operatorname{Hess} a.$$

The full orbital expressions include their exponential tails; differentiating only a finite far multipole is a different approximation. At the origin use the regular I_p expansion. Inside material the tidal trace remains $-4\pi G\rho$, rather than the vacuum trace zero.

A useful error bound needs its spatial hypothesis. Let a signed density error have total-variation mass δM, all at distance at least $d_{\mathrm{sep}}>0$ from the receiving point. Differentiating the Newtonian kernel gives

$$|\delta a|\le\frac{2G\delta M}{c^2 d_{\mathrm{sep}}}, \qquad |\delta\mathbf{g}|\le\frac{G\delta M}{d_{\mathrm{sep}}^2},$$
$$\|\delta T\|_{\mathrm{op}}\le\frac{2G\delta M}{d_{\mathrm{sep}}^3}.$$

The Hessian kernel has eigenvalues proportional to $2,-1,-1$. Thus these are controls on the full derivatives, not an inference from potential convergence alone.

For an orbital tail discarded beyond physical L, its exact mass is $\delta M=M\exp(-2L/d)\sum_{k=0}^{8}(2L/d)^k/k!$. At $r<L$ one may use $d_{\mathrm{sep}}=L-r$. The angular free-sector tail instead has $\delta M=(3/5)M_{\mathrm{src}}(R_c/L)^5$. An overlapping source error requires its own regularity or cancellation estimate; these separated-support bounds cannot be assigned there.

For overlapping finite-energy sources, the Sobolev estimate instead gives

$$\|\nabla\delta a\|_2\le\frac{8\pi G}{c^2}C_S\|\delta\rho\|_{6/5}.$$

This controls field energy; total mass additionally needs L^1. Physical energy e_u, length ℓ_u and action I_u remain shared assignments. They set time I_u/e_u and the field/clock conversion together. An SI ruler or Write rate is not inferred from the source polynomial. [8]

42. Reproduction and the remaining interface

Sections 1–28 preserve the completed source-to-field pass, with section 6's coupling wording clarified and section 27's counts labeled as that preceding pass. The new local-action calculations retain their own exact input, derivation, code and native receipt bindings.

The new package has 3,055 independent exact checks: 1,068 angular and multipole, 1,223 quantum-energy and phase, 226 visibility, 88 reduced-rank supplement, 72 orbital preparation, 268 orbital Poisson and 110 finite-shell stability. They retain the same source operands, domains and common energy conventions. Editable sources, embedded fonts and a byte-identical relocated PDF rebuild accompany them. Actual native outcomes and resource custody retain their campaign receipts.

The mathematical interfaces are now explicit. The chosen local classical Casimir action reproduces the radial model. The quantum product comparison uses the same full K as the original clock state. Its complete current-blind algebra is distinguished from preparation-specific first-order blindness. A bounded finite-lift control carries an actual phase sign margin.

The next physical work is to define the nonuniform interacting quantum energy/stress form, including its vacuum, source, interaction, counterterm, support and reference convention. Derive its V and energy-form remainder on the original preparation, then carry the actual measurement and apparatus errors through the phase comparison. A translation-invariant moving-source action, microscopic spatial family, SI calibration and the spatial metric response including γ remain available for that construction.

Additional retained sources

[8] *Chapter III local action and actual clock continuation*, `voli_chapter3_local_action2_20261015`; GEN6 Duck run chapter3-local-action-20261015-03 (Duck 1.2.3, core 0.10.2; 17 native tasks); H002641: angular, quantum, visibility, action and restored-orbital derivations; exact operands, independent checkers and parent-owned native receipts. The source-to-field predecessor and Chapter II companion remain retained under H002641.

SAM VOLUME I

Clocks, signals, and calibrated observation

Chapter IV

Abstract. A distant measurement composes a physical source history, emission, propagation, detection and calibration. The original interacting clock retains its full energy, preparation and transient. One globally positive spatial profile gives signed first and finite response enclosures for all six preparations under shared units. A sharper current bound certifies all six weak-field phase budgets and the total sign throughout a finite amplitude interval; a signed short-prefix result below is distinguished from the unresolved full-window isolated correction. The canonical local tensor coupling adds pair kernels, directional angular return and a switched vacuum-reference payment. A static canonical joint Hamiltonian, the original-shell limit and an analytic weighted normal frame now close static joint evolution and the free switched quantum propagator. All six static canonical source phases are certified at the original full-cycle time. Exact positive smoothing sharply bounds the compact carrier's radiation energy, with a signed source-work and near-storage identity. The actual gapless Euclidean continuum has a closed transform and positive finite quadrature surrogates with controlled errors. Noisy inference retains one simultaneous covariance event and a coarse-winding/fine-final measurement budget. The joint continuation derives a phase-bearing finite-mode signal from every original preparation, with an explicit passive channel, detector, finite-repeat and complete energy account. Exact real-time memory and two-time emission bounds compose with a finite local pointer protocol in the original or uniform-lapse law, with conditional phase error below 0.174 nanoradians. The current SAM_land finite donor/control preparation now composes with that original/uniform local record protocol, including failures, unequal branch weights, phase and work. A complete tensor graph-ideal and causal common-form construction also closes existence of the actual switched canonical joint singular-source propagator. Quantitative nonuniform local-output transfer, measured apparatus calibration, global vacuum ramp work and cosmological amplitude dynamics remain explicit.

Part A. The interacting clock and its spatial coupling

1. One observable, five physical operations

The preceding chapters established an original Write source, its interacting phase clock and a spatial receiving account. Observation now asks how those objects become a measured number. A local clock history is followed by emission, propagation, reception and recording. Each operation contributes a defined transformation. Their composition is the subject of this chapter. [1–3]

The source clock is retained throughout. Its complete Hamiltonian, preparation and nonzero current determine the actual phase; a bare two-level oscillator is not substituted when a signal leaves the source. A common lapse can change the clock's internal age, a route can delay its information, and an amplitude law can change an inferred distance. These are distinct operands in the same measurement.

The combined treatment brings together the former endpoint/GPS and photon-road drafts, and the finite-redshift analysis. [11,18,19] The organizing question is what each measured quantity determines. The endpoint sections concern phase and accumulated age. The timing sections concern a map between emission and reception events. The distance sections concern amplitudes and channel calibration. A common source-native action can connect them; such a connection must be calculated rather than supplied by reusing the same word "accumulation."

Table 1: Notation and operational role. The three response symbols are deliberately distinct.

Symbol	Meaning
a	dimensionless source lift, $2GM/(c^2 r)$ in the isolated monopole comparison
N	lapse multiplying the complete clock history
n_{opt}	coordinate optical index for a declared route law
p	positive distance-response exponent
η	internal age of the interacting clock in model units
F	increasing emission–reception event map
Ξ	GW/EM amplitude-distance ratio
C_{cal}	constant relative channel calibration in the distance sections

2. The source lift and the receiving convention

For a stationary isolated source, retain the source lift and potential

$$a(r) = \frac{2GM}{c^2 r}, \qquad \Phi(r) = -\frac{c^2}{2}a(r) = -\frac{GM}{r}.$$

Here a is the source-generated lift above the background, rather than the complete floor-plus-source account. The endpoint coupling used below is

$$N(a) = \sqrt{1-a}, \qquad 0 \le a < 1.$$

It is a declared receiving law. The Poisson kernel alone does not select it. For two stationary endpoints, with identical calibration

and matched internal age or stationary reference clocks,

$$\frac{\text{rate}_2}{\text{rate}_1} = \frac{N(a_2)}{N(a_1)} = \sqrt{\frac{1-a_2}{1-a_1}}.$$

A higher endpoint has smaller a and a larger rate. To first order the fractional increase is

$$\frac{\text{rate}_2}{\text{rate}_1} - 1 = \frac{a_1 - a_2}{2} + O(a_1^2 + a_2^2) = \frac{\Phi_2 - \Phi_1}{c^2} + O(a^2).$$

The half coefficient belongs to this endpoint map. Assigning it to a photon path would be a different receiving prescription.

The background needs an explicit convention even in a local experiment. If instead the raw coupling is $N_{\text{raw}}(a) = \sqrt{1 - A_0 - a}$, normalizing its source-free value gives

$$\frac{N_{\text{raw}}(a)}{N_{\text{raw}}(0)} = \sqrt{1 - \frac{a}{1-A_0}}.$$

A common A_0 therefore does not algebraically cancel from the nonlinear endpoint ratio. Its leading source coefficient is $1/[2(1-A_0)]$. With the $A_0 = 1/(12\pi)$ convention retained in Chapters II–III, this differs from $1/2$ by the factor $1.0272486\ldots$. A rescaled lift could absorb the factor only if its source equation, mass calibration and all receiving laws were changed consistently. [2,3,9]

The present chapter uses the source-normalized coupling $N(a)$ stated above. This is a testable choice of the local response, not a claim that a nonlinear floor has disappeared by subtraction. At linear order a genuinely common additive potential reference cancels between endpoints; that simpler statement does not prove the nonlinear one.

3. The endpoint receiver is the interacting clock

An endpoint clock must first be an observable with a preparation and an evolution. Chapter VII supplies all three without replacing the source by an isolated ideal oscillator. Its sixteen-state source retains Swap S, the signed current operators D, C, and the even energy E_{even}. On each original active lane l, with signed gap $d_l \neq 0$, the receiver is

$$Z_l = \frac{D + iC}{d_l}, \qquad z_l = \langle Z_l \rangle, \qquad \theta_l = -\arg z_l.$$

The phase is continuously unwrapped from $\theta_l(0) = 0$. It is not the original common Write phase. Its rate $\nu_l = -\operatorname{Im}(\dot{z}_l/z_l)$ exists wherever the retained differentiable expectation is nonzero. The full Hamiltonian is

$$\mathscr{H}_0 = H_s + H_f + H_{\text{int}} + H_{\text{ct}},$$

with $H_s = \varepsilon(S + E_{\text{even}})$, both field polarizations in H_f, $H_{\text{int}} = -\varepsilon(D\phi_+ + C\phi_\times)$, and $H_{\text{ct}} = \varepsilon^2 \mu_{\text{ct}}(D^2 + C^2)$, $\mu_{\text{ct}} = 1/(4R_{\text{W}})$. Here R_{W} is Chapter VII's source coupling radius, distinct from the terrestrial radius used below. None of these terms is removed when the receiver is named a clock. [1]

The actual preparation is the coherent superposition of the exact two hard-field branches, tensored with the common soft vacuum. It is an interacting source–field preparation, not a product of a bare source oscillator and its optimal classical field. Chapter III's energy comparisons distinguish those preparations explicitly. [3]

For every original lane, Chapter VII reports a certified interval on which

$$|z_{l,0}(s)| \geq m_l > 0, \qquad 0 < \ell_l \leq \nu_{l,0}(s) \leq u_l.$$

Here s denotes the original dimensionless evolution parameter. The inherited window ends at $T_0 = 573049\pi$, one bare cycle, not one completed actual phase turn. For example, the outward lane 01 rate bounds are $3.490102929541 \times 10^{-6}$ and $3.490102934505 \times 10^{-6}$. These are source-reported certificates; this chapter uses them and does not claim a new execution. [1,2]

4. A changing uniform lapse gives an exact history

A lapse that changes with time need not vary across the clock's own source and field. First consider that uniform case. Assume the original self-adjoint Hamiltonian $\mathscr{H}_0$ is unchanged, the preparation is fixed, and $N(s) > 0$ and $f(s)$ are real, locally integrable functions. Declare the complete coupling

$$\mathscr{H}(s) = N(s)\mathscr{H}_0 + f(s)I, \qquad \eta(s) = \int_0^s N(v)\,\mathrm{d}v.$$

The variable η is internal age in model units. It must remain in the inherited clock window whenever that window's quantitative bounds are used.

Exact history consequence. The propagator and current expectation are

$$\begin{aligned} U(s) &= \exp\left[-i\int_0^s f(v)\,\mathrm{d}v\right]\exp[-i\eta(s)\mathscr{H}_0], \\ z_l(s) &= z_{l,0}(\eta(s)), \qquad \theta_l(s) = \theta_{l,0}(\eta(s)), \\ \nu_l(s) &= N(s)\nu_{l,0}(\eta(s)) \quad \text{almost everywhere.} \end{aligned}$$

Proof. Spectral calculus defines the displayed unitary for every initial vector. On the domain of $\mathscr{H}_0$, differentiation almost everywhere gives $iU' = \mathscr{H}(s)U$ and $U(0) = I$; extension by continuity gives the full unitary evolution. The scalar exponential cancels from expectations. Substitution in the same bounded current observable gives the first identity for z_l. Nonvanishing and the common initial branch give the phase identity, and the absolutely continuous chain rule gives its rate. No adiabatic approximation or commutation of distinct field components is used: all times multiply the same complete operator.

This extends Chapter III's constant-lapse control by a mathematical consequence of its hypothesis. It does not establish that a supplied gravitational field couples in this way. Nor does it introduce a new continuum energy-density construction. Positivity makes internal age increasing; the unitary formula itself needs only real integrable N.

Changing the lapse also has an energy account. For differentiable N, f and a finite-energy preparation, $[\mathscr{H}(s), \mathscr{H}_0] = 0$ keeps $E_0 = \langle \mathscr{H}_0 \rangle$ constant, while the supplied power is $\dot{N}E_0 + \dot{f}$. Thus the signed work over a switch is $(N(T) - N(0))E_0 + f(T) - f(0)$. A closed switch can have zero signed net work while still requiring apparatus, timing and reversible energy storage. The phase identity does not remove that account. [2,3]

5. Equal coordinate time and equal internal age

Comparing two identically prepared copies requires stating which events are compared. For copies A, B with cumulative ages η_A, η_B, the exact rate ratio at equal coordinate parameter is

$$\frac{\nu_A(s)}{\nu_B(s)} = \frac{N_A(s)}{N_B(s)} \frac{\nu_0(\eta_A(s))}{\nu_0(\eta_B(s))}.$$

The second factor is present because the actual clock is slightly nonstationary. At matched internal age, choose s_A, s_B with $\eta_A(s_A) = \eta_B(s_B)$. The baseline rates then cancel and the ratio is $N_A(s_A)/N_B(s_B)$. A strictly stationary reference provides the same cancellation without age matching.

For a differentiable static receiving law $N(a)$ with $N(0) = 1$, differentiation at equal coordinate time gives, where ν_0' exists,

$$\partial_a \nu(s;a)|_{a=0} = N'(0)\{\nu_0(s) + s\nu_0'(s)\}.$$

The transient term must not be absorbed into an independently fitted lapse. If only the inherited continuously differentiable phase is available, use its accumulated comparison instead; that argument needs no second phase derivative.

These distinctions concern comparison protocol, not a correction to the exact time change. Identical instruments examined after different internal ages need not display identical instantaneous baseline frequencies. An endpoint application can use matched-age calibration, a stationary reference, or an explicit transient budget. Independent adjustment of each lane's bare parameter would cease to test a shared coupling. [2,3]

6. A correlated finite-phase comparison

Accumulated phase retains the common underlying history and is therefore sharper than comparing two unrelated interval estimates. For equal coordinate duration T, suppose both ages lie in the same certified window and $\eta_A(T) \le \eta_B(T)$. Put $D_T = \eta_B(T) - \eta_A(T)$. The exact time change gives

$$\theta_A(T) - \theta_B(T) = -\int_{\eta_A(T)}^{\eta_B(T)} \nu_0(v)\,\mathrm{d}v,$$
$$-uD_T \le \theta_A(T) - \theta_B(T) \le -\ell D_T.$$

This follows immediately by integrating the same positive baseline rate. Even if the two instantaneous lapses cross, only their cumulative ages enter this final comparison. No sign is asserted when their ordering is unknown.

For a reference with $N_B = 1$ and a declared source-only lapse $N_A(s) = \sqrt{1 - a(s)}$, assume $0 \le a(s) < 1$. Since

$$\frac{a}{2} \le 1 - \sqrt{1-a} = \frac{a}{1 + \sqrt{1 \quad a}} \le \frac{a}{2 \quad a},$$

we obtain the finite, time-dependent bound

$$-u \int_0^T \frac{a(s)}{2 - a(s)}\,\mathrm{d}s \le \theta_A(T) - \theta_B(T) \le -\frac{\ell}{2} \int_0^T a(s)\,\mathrm{d}s.$$

A positive integral produces a strictly negative phase difference within the stated hypotheses and window. This conditional deduction is inherited from the endpoint draft; it is an exact comparison under the stated coupling.

For a constant lift, the frozen Earth example retained in Chapter III gives approximately -4.37389 nanoradians over one bare cycle across the six lanes. Its independent-enclosure criterion $a > 1 - (\ell/u)^2$ is a sufficient condition for separating independently bounded phases, not a threshold for the correlated sign. Two independent readouts with phase error at most δ require a predicted margin exceeding 2δ, together with the stated timing, preparation and branch-control budgets. [2,3]

7. Which energy changes the clock

A complete Hamiltonian coupling is stronger than a replacement of the source parameter. Scaling only ε to $N\varepsilon$ while holding the field fixed gives

$$\mathscr{H}_\varepsilon = NH_s + H_f + NH_{\mathrm{int}} + N^2 H_{\mathrm{ct}},$$

so its exact difference from $N\mathscr{H}_0$ is

$$\mathscr{H}_\varepsilon - N\mathscr{H}_0 = (1-N)H_f + (N^2 - N)H_{\mathrm{ct}}.$$

The derivative of this residual at $a = 0$ is $N'(0)(H_{\mathrm{ct}} - H_f)$. The counterterm is current-blind, while the free field can change the current through the interaction. The six unequal hard-branch second-order diagnostics retained in Chapters II and III demonstrate why replacing a bare frequency alone is not the full-scaling theorem. They are not bounds on the full finite-lift response. [2,3]

Conversely, energy can change with no current response. Chapter III identifies the complete Hermitian source algebra invisible to every current history:

$$W = \bigoplus_{a<b} w_{ab} I_2 \ \oplus \ W_{\mathrm{dark}}.$$

It has real dimension 22; the four-dimensional dark block carries zero current. Time-dependent integrable members are also invisible. Requiring conservation reduces the source algebra to ten dimensions, represented by functions of E_{even}. A scalar addition is only its simplest case. Thus clock reception does not determine the total energy or its gravitational reference.

The six prepared-lane first-order tests are weaker still: the all-states-in-each-lane first-order blind space has dimension 238, including off-block directions that can return a bright response at second order. A common measured slope can constrain a declared coupling family without reconstructing arbitrary microscopic couplings. These inherited visibility results explain why universality needs a physical law, not six separate frequency fits. [3]

8. A finite control criterion for the actual phase

The exact uniform theorem has a useful stability test. Compare it, including any exactly current-blind source term, with a Hamiltonian containing an additional strongly measurable, self-adjoint residual $R(s)$ with locally integrable operator norm. Require the same initial state and well-defined propagators. Duhamel's identity and unitarity give

$$|z(T)-z_{\mathrm{ideal}}(T)| \le 2\|Z\| \int_0^T \|R(s)\|\,\mathrm{d}s.$$

Indeed the propagator difference has norm at most the integral, and the expectation difference has two terms, each bounded by $\|Z\|$ times that difference. On a preserved original lane, $\|Z\| = 2$. Leakage outside it requires the full receiver norm or an additional projected-receiver leakage estimate.

Suppose $|z_{\mathrm{ideal}}| \ge m$ throughout the interval. Write $q = 4\int_0^T \|R(s)\|\,\mathrm{d}s$ for the lane-preserving case. If $q < m$, the error disks exclude zero at every prefix, and elementary tangent geometry gives

$$|\theta(T)-\theta_{\mathrm{ideal}}(T)| \le \arcsin(q/m).$$

Continuity from the common initial state fixes the unwrapped branch. If the ideal negative phase shift against an unperturbed reference has magnitude at least x, with $0 < x < 1$, the strict sufficient sign condition is

$$q < m\sin x.$$

If both copies have residual errors, their two angular budgets must be added.

Chapter III implements a rational constant-residual version using $m = M_{\mathrm{lower}} - E_{\mathrm{cycle}}$, $x = \ell a T_{\mathrm{lo}}/2$, and

$$\|R\| < \frac{m(x-x^3/6)}{4T_{\mathrm{hi}}}.$$

Its reported six lane budgets are approximately 6.07387–6.07390×10^{-16} in model-energy operator norm. They support its stated lane-preserving $\|R\| \le a^2$ fixture at the frozen lift. They do not bound an arbitrary unbounded field perturbation or specify attainable laboratory accuracy. [3]

9. A closed nonuniform quantum receiving form

The uniform theorem fixes a complete reference operator. A constructive next step is to receive the original continuum energy through a positive weight on its complete residual-energy map. This receiving construction is developed in campaign voli_chapter4_observation1_20261016 (GEN6 Duck run chapter4-observation-20261016-01, Duck 1.2.3, core 0.10.2) on the uniform theorem and residual-energy map computed there. Its spatial incidence is explicit; it is not assigned to the original ledger or identified with a pointwise renormalized stress tensor.

On each original active lane the complete Hamiltonian has the form

$$\mathscr{H}_0 = \varepsilon e I + \varepsilon S + K, \qquad q_K[\Psi] = \|T_g\Psi\|^2,$$

where $g = \varepsilon d$ retains the signed lane gap. On the complete two-polarization Fock space, with $\sigma_+ = \sigma_z$, $\sigma_\times = \sigma_y$,

$$(T_g\Psi)_\alpha(\omega) = \sqrt{\omega}\,a_\alpha(\omega)\Psi - \frac{g\,u(\omega)}{\sqrt{\omega}}\sigma_\alpha\Psi.$$

The original finite smearing is $\mu = \int |u(\omega)|^2 \mathrm{d}\omega/\omega$, and the bounded shift has squared norm $c_g = 2g^2\mu$. The weighted annihilation map A is closed on $\mathscr{Q} = Q(H_f)$, with $\|A\Psi\|^2 = q_f[\Psi]$. Since $T_g = A - F_g$ and F_g is bounded, T_g is closed on precisely the same domain. The elementary squared triangle inequalities give

$$H_f \le 2K + 2c_g I, \qquad K \le 2H_f + 2c_g I \tag{1}$$

as quadratic forms. Thus $1+K$ and $1+H_f$ have equivalent form norms, without a frequency or occupation cutoff. The original finite source matrices extend this construction to all sixteen source states, including the four dark dimensions.

Choose a bounded self-adjoint weight on the residual-energy output space. For the six phase certificates below it preserves the original source lane; a source-independent output weight is one sufficient choice. Require

$$0 < N_- I \le W \le N_+ I < \infty, \qquad q_{K,W}[\Phi,\Psi] = \langle T_g\Phi, W T_g\Psi\rangle.$$

For fixed real source coefficient N_s, define

$$q_{\mathscr{H}[N_s,W]}[\Psi] = N_s\varepsilon e\|\Psi\|^2 + N_s\varepsilon\langle\Psi, S\Psi\rangle + q_{K,W}[\Psi].$$

Positive upper and lower bounds on W, Eq. (1), and the bounded source term make this a closed semibounded form on $\mathscr{Q}$. The representation theorem supplies its self-adjoint Hamiltonian. When $W = NI$ and $N_s = N$, it is exactly $N\mathscr{H}_0$.

Expanding $\langle(A-F_g)\Phi, W(A-F_g)\Psi\rangle$ retains the weighted free term, both signed cross terms and the entire weighted counterterm $\langle F_g\Phi, WF_g\Psi\rangle$. The construction does not weight the free field and then discard the interaction or counterterm. A bounded nonconstant spatial profile may be received by a supplied isometry J as $W_N = J^* M_N J$. The choice of J is part of the physical receiving interface. The integrated Hamiltonian alone does not reconstruct it, and a different local stress quantization can have additional vacuum, pair-creation, projector or boundary terms.

10. An energy-form stability theorem for the actual clock

The free-field part is unbounded, so the bounded-residual criterion above is not automatically applicable. The common form domain supplies a different estimate. Compare a static $H_{\mathrm{ref}} = N\mathscr{H}_0$, $N > 0$, with the same-source-coefficient form $H_{\mathrm{act}} = H_{\mathrm{ref}} + r$, where

$$|r[\Phi,\Psi]| \le \delta\sqrt{q_K[\Phi]q_K[\Psi]}, \qquad 0 \le \delta < N.$$

The weight family supplies this bound with $\delta = \|W - NI\|$. Start both evolutions from the identical normalized original

preparation in $\mathscr{Q}$. Write their states as $\Psi(t), \chi(t)$ and their distance as $d_\Psi(t) = \|\Psi(t) - \chi(t)\|$. Common-form weak evolution gives

$$d_\Psi(t)^2 \le 2\delta \int_0^t \sqrt{K_{\mathrm{act}}(s) K_{\mathrm{ref}}(s)}\, \mathrm{d}s. \tag{2}$$

Proof. The trajectories are continuous in the common form norm and differentiable into its dual. Pair the two weak Schrödinger equations. Their reference terms cancel, giving $\mathrm{d}\langle\Psi,\chi\rangle/\mathrm{d}t = \mathrm{i}r[\Psi,\chi]$. Since $d_\Psi^2 = 2 - 2\,\mathrm{Re}\langle\Psi,\chi\rangle$, integration and the form bound give Eq. (2). Bounded spectral regularization in the two equivalent form norms justifies the product rule before passing to the weak limit. No finite operator norm is assigned to H_f or r.

For bounded current Z, the two-state expectation expansion gives

$$|z_{\mathrm{act}} - z_{\mathrm{ref}}| \le 2\|Z\| d_\Psi.$$

On each preserved original lane $\|Z\| = 2$. If the reference amplitude is at least $m > 0$ and $4d_\Psi < m$ at every prefix, the same unwrapped phase branch exists and its error is at most $\arcsin(4d_\Psi/m)$. Leakage between lanes pays its full receiver norm and preparation account.

11. A bootstrap uses the original correlated preparation

Retain the six original complete-energy preparation ceilings and phase-error certificates. For $N \le 1$ and times within the original window, they supply

$$K_{\mathrm{ref}}(t) \le \kappa = K_{\mathrm{prep,upper}} + \varepsilon_{\mathrm{upper}} E(T)_{\mathrm{upper}}.$$

This is a ceiling on the complete positive K in the actual correlated preparation. It retains the cancellation between free, signed interaction and counterterm contributions.

Energy conservation in the two static families gives

$$(N-\delta)K_{\mathrm{act}}(t) \le (N+\delta)\kappa + 2N\varepsilon d_\Psi(t).$$

For $\delta \le N/2$, it follows that $K_{\mathrm{act}} \le 3\kappa + 4\varepsilon d_\Psi$. Choose $d_* > 0$ and put

$$K_* = 3\kappa + 4\varepsilon_{\mathrm{upper}} d_*, \qquad \delta_* = \min\left\{ \frac{N_{\mathrm{lower}}}{2}, \frac{d_*^2}{4T_{\mathrm{upper}}\sqrt{\kappa K_*}} \right\}.$$

For $\delta \le \delta_*$, the trajectory cannot first reach d_*. Before a first crossing its K_{act} is bounded by K_*; Eq. (2) at that crossing would give $d_*^2 \le d_*^2/2$. Hence $d_\Psi(t) < d_*$ throughout the interval. This is a full-continuum state estimate on the chosen form family.

For the inherited frozen Earth lift, set $N = \sqrt{1-a}$, $x = \ell a T_{\mathrm{lower}}/2$, and $m = M_{\mathrm{lower}} - E(T)_{\mathrm{upper}}$. The actual source certificate has $0 < x < 1$. Choose

$$d_* = \frac{m(x - x^3/6)}{8}.$$

Then $4d_*/m = (x - x^3/6)/2 < \sin x$. The perturbed phase error is smaller than the ideal negative correlated shift, preserving its sign. Directed rational evaluation gives the six sufficient dimensionless weight budgets in Table 2.

The source-specific bootstrap yields the first budget column in Table 2. It controls the complete joint vector, which is stronger than needed for a phase readout. The following estimate uses the actual current algebra directly.

12. A sharper current-specific continuum remainder

The receiver supplies additional structure. On a retained lane, $Z = \sigma_z + \mathrm{i}\sigma_y$ has norm two. The weighted annihilation part commutes with finite source matrices, and direct two-polarization Pauli algebra gives

$$\|[T_g, Z]\Psi\| \le 2\sqrt{c_g}\|\Psi\|, \qquad \|[T_g, Z^*]\Psi\| \le 2\sqrt{c_g}\|\Psi\|.$$

Equivalently, the squared polarization commutator sum is exactly $8g^2\mu I = 4c_g I$. Thus

$$\sqrt{K[Z\Xi]} \le 2\sqrt{K[\Xi]} + 2\sqrt{c_g}\|\Xi\|,$$

and likewise for Z^*. Conservation of the complete reference Hamiltonian also gives $K[U_0(t)\Xi] \le K[\Xi] + 2\varepsilon\|\Xi\|^2$.

For $\delta \le N/2$, the static actual energy supplies the global ceiling $k_A = 3K_{\mathrm{prep,upper}} + 4\varepsilon_{\mathrm{upper}}$. Applying the preceding evolution, current and evolution bounds to $B_T(s) = U_{\mathrm{ref}}(T-s)^* Z U_{\mathrm{ref}}(T-s)$ gives, for both B_T and its adjoint on the normalized actual state,

$$K[B_T(s)\Psi(s)] \le k_B,$$

$$k_B = \left[2\sqrt{k_A + 2\varepsilon_{\mathrm{upper}}} + 2\sqrt{c_{g,\mathrm{upper}}} \right]^2 + 8\varepsilon_{\mathrm{upper}}.$$

Weak Heisenberg variation on the same common form domain then gives the exact identity

$$z_{\mathrm{act}}(T) - z_{\mathrm{ref}}(T) = \mathrm{i}\int_0^T \{ r[\Psi, B_T\Psi] - r[B_T^*\Psi, \Psi] \}\, \mathrm{d}s,$$

and hence the observable-specific bound

$$|z_{\mathrm{act}}(T) - z_{\mathrm{ref}}(T)| \le 2\delta T\sqrt{k_A k_B}.$$

A sufficient phase-sign weight budget is therefore

$$\delta_{\mathrm{linear}} = \min\left\{ \frac{N_{\mathrm{lower}}}{2}, \frac{m(x - x^3/6)}{4T_{\mathrm{upper}}\sqrt{k_A k_B}} \right\}.$$

Directed upper square roots give the second column of Table 2. Its current error is at most half the sine margin. Each estimate separately suffices; the larger of the two budgets can be used without combining their incompatible intermediate bounds.

Protecting the observed phase is substantially less costly than protecting the complete joint vector for lanes 03, 13 and 23. The smallest of the six available sufficient budgets is now approximately $1.4797 \quad 10^{-11}$. The declared illustrative fixture

Table 2: Sufficient dimensionless $\|W-NI\|$ budgets for the original six preparations. Rounded displays of exact outward operands; the two columns protect the full state and the actual current, respectively.

Lane	State bootstrap	Current-specific
01	7.090598×10^{-11}	2.961257×10^{-11}
02	2.262032×10^{-11}	2.845597×10^{-11}
03	3.199329×10^{-15}	1.479745×10^{-11}
12	6.476681×10^{-11}	2.953179×10^{-11}
13	3.745512×10^{-15}	1.511229×10^{-11}
23	4.467330×10^{-15}	1.546471×10^{-11}

$\delta = a^2$ fits all six. These are receiving-weight bounds in model coordinates, distinct from bounded-Hamiltonian energy norms and from achieved apparatus precision. A physical lapse profile must supply its actual incidence, support and weight error before receiving the certificate.

13. First variation and the local spatial calculation

For an affine static weight $W(\lambda) = NI + \lambda B$ with $\|B\| \le b$, the complete variation is the form

$$r_B[\Phi,\Psi] = \langle T_g\Phi, BT_g\Psi\rangle, \qquad |r_B[\Phi,\Psi]| \le b\sqrt{q_K[\Phi]q_K[\Psi]}.$$

Let U_0 denote the reference evolution and $B_T(s) = U_0(T-s)^* Z U_0(T-s)$. The finite source current and these evolutions preserve $\mathscr{Q}$, so the weak first-order response is

$$z_1(T) = \mathrm{i}\int_0^T \{r_B[\chi(s), B_T(s)\chi(s)] - r_B[B_T(s)^*\chi(s),\chi(s)]\}\,\mathrm{d}s.$$

For nonzero reference current, $\theta_1(T) = -\operatorname{Im}[z_1(T)/z_0(T)]$ on the same unwrapped branch. The common-domain justification and complete source-matrix certificate are retained in Appendix D. Actual evaluation requires the supplied incidence J/B and two-time joint-state matrix elements; an energy ceiling alone does not assign their value.

A separate source-shell coefficient change δ_s contributes bounded residual $|\delta_s|\varepsilon$ after keeping the even scalar payment in the energy ledger. It adds at most $|\delta_s|\varepsilon T$ to the state-distance budget. It shares the total phase allowance with the form error; preparation, timing, winding, reference and readout errors add their own terms. The remaining local calculation is to construct the actual stress/energy form and receive every term, including its vacuum and boundary conventions, into this common-domain comparison.

14. An explicit spatial receiver for the original radial modes

The positive residual-energy form admits a source-bound spatial incidence rather than an unspecified output weight. Retain the original regular quadrupole radial operator and normalized TT tensors. Its unitary spherical-Bessel transform is

$$(\mathscr{H}_2 f)(r) = \sqrt{\frac{2}{\pi}}\int_0^\infty k j_2(kr) f(k)\,\mathrm{d}k.$$

It maps $L^2(\mathrm{d}k)$ unitarily to $L^2(r^2\mathrm{d}r)$. The identity $\int r^2 j_2(kr) j_2(k'r)\mathrm{d}r = \pi\delta(k-k')/(2k^2)$ fixes its normalization and retains the regular finite-energy extension at the origin. With the original angular isometry $\mathscr{U}Q = \sqrt{5/(8\pi)}\Lambda_n Q$, define

$$(Jf)(r,n) = \sum_{\alpha=+,\times} \mathscr{U}e_\alpha(n)(\mathscr{H}_2 f_\alpha)(r).$$

This J is an isometry on the original two-polarization residual-energy output. A real bounded profile $0 < N_- \le N(\mathbf{x}) \le N_+$ supplies $W_N = J^* N J$ with the same positive bounds. The full closed form retains both cross terms and its entire counterterm.

This is a specified positive-frequency square-form spatial receiver. Identifying it with a pointwise canonical quantum stress tensor still requires the common local-action, vacuum, derivative-energy and boundary calculation. Its compressed domain does not silently add the full angular complement dynamics. For

$$A_{ab}(r) = \int N(r,n)\,\mathscr{U}e_a : \mathscr{U}e_b\,\mathrm{d}\Omega,$$

the spectral kernel is

$$W_{ab}(k,k') = \frac{2}{\pi}kk'\int_0^\infty r^2 j_2(kr) j_2(k'r) A_{ab}(r)\mathrm{d}r.$$

A scalar spatial profile can therefore mix radial frequencies.

15. Exact selection rules and the gradient complement

The polarized TT contraction is

$$\Lambda Q : \Lambda Q' = \operatorname{tr}(QQ') - n^{\mathrm{T}}(QQ' + Q'Q)n + \tfrac{1}{2}(n^{\mathrm{T}}Qn)(n^{\mathrm{T}}Q'n).$$

Its spherical degrees are precisely among $L = 0,2,4$. Thus every odd scalar harmonic and every scalar harmonic above degree four has zero compression. For the actual two driven tensors, write $C[h]_{ab} = \int h\,\mathscr{U}e_a : \mathscr{U}e_b\mathrm{d}\Omega$. Exact integration gives

$$C[1] = I, \quad C[x] = C[y] = C[z] = 0,$$
$$C[3z^2-1] = \tfrac{4}{7}I, \quad C[P_4(z)] = \tfrac{1}{126}I,$$
$$C[x^2-y^2] = C[2xy] = C[2xz] = C[2yz] = 0,$$
$$C[x^4 - 6x^2y^2 + y^4] = \tfrac{4}{63}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},$$
$$C[4xy(x^2-y^2)] = \tfrac{4}{63}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

The other degree-four components vanish. The real scalar angular image on the original pair is exactly the three-dimensional

space of real symmetric matrices. On the entire five-STF image, the actual $L = 0,2,4$ Gaunt map has rank fifteen. It realizes every small real symmetric five-coordinate weight with a bounded real scalar profile; positivity still constrains finite coefficients. The earlier imaginary antisymmetric polarization fixture belongs to a different realization class. Here $\{D,C\} = 0$ and $D^2 = C^2$ make the real weighted counterterm proportional to D^2, retaining all four original dark dimensions.

A vanishing compressed gradient does not remove its complement. With P_5 the full STF angular-image projector,

$$P_5 z P_5 = 0,$$
$$P_5 z (I - P_5) z P_5$$
$$= \mathrm{diag}(11/21, 11/21, 5/21, 5/21, 1/7).$$

The latter is positive and rank five, with rank two on the driven pair. A full angular evolution must retain this gradient leakage and its returning effect.

The global profile remains bounded and positive. Every polynomial Taylor expression below is restricted to the finite near ball; the actual bounded exterior remains in the far account. For a smooth source-centered profile, $N = N_0 + g_i x_i + H_{ij} x_i x_j/2 + \cdots$, its quadratic compression is

$$A(r) - N_0 I = \frac{r^2}{42}(5\,\mathrm{tr}\,H + 6H_{zz})I + \cdots.$$

Cubic terms also cancel by parity. A fourth directional-derivative bound M_4 on $r \le L$ supplies

$$\delta_{\mathrm{near}} \le \frac{L^2}{42}|5\,\mathrm{tr}\,H + 6H_{zz}| + \frac{M_4 L^4}{24}.$$

For $N = 1 - m/|\mathbf{d}+\mathbf{x}| + O(m^2/d^2)$, the leading term is $mr^2(1 - 3\cos^2\beta)/(7d^3)$. It vanishes at $\cos^2\beta = 1/3$. Even a scalar polarization compression varies radially and need not become one common lapse for six different correlated preparations.

16. An actual-state spatial tail and a finite profile criterion

Let $K_{\mathrm{in/out}}[\Psi] = \|P_{\mathrm{in/out}} J T_g \Psi\|^2$ split the original complete positive form at radius L. The physical profile supplies

$$|r[\Phi,\Psi]| \le \delta_{\mathrm{near}}\sqrt{K_{\mathrm{in}}[\Phi]K_{\mathrm{in}}[\Psi]} + \delta_{\mathrm{far}}\sqrt{K_{\mathrm{out}}[\Phi]K_{\mathrm{out}}[\Psi]}.$$

The exterior stays in this account. If $\|W - N_0 I\| < N_0/2$, put $k_A = 3K_{\mathrm{prep,up}} + 4\varepsilon_{\mathrm{up}}$. A reference ceiling $K \le k_R$ and uniform tail ceiling $K_{\mathrm{out}} \le \eta_{\mathrm{ref}}(L)$ give

$$\|\Psi_{\mathrm{act}}(T) - \chi(T)\|^2 \le 2T\sqrt{k_A}\left[\delta_{\mathrm{near}}\sqrt{k_R} + \delta_{\mathrm{far}}\sqrt{\eta_{\mathrm{ref}}(L)}\right].$$

Four times the state-distance bound controls the actual current error. Comparing it to the retained nonzero reference amplitude supplies the same unwrapped arcsine phase criterion as before.

The tail can be bounded for the original exact preparation. The retained hard-branch pull-through identity gives, for $k \ge \lambda = 64\Delta$,

$$a_\alpha(k)\Phi_j = g u_R(k)(H_h \quad E_j + k)^{-1}\sigma_\alpha\Phi_j, \quad H_h \quad E_j \qquad \Delta.$$

For $H_j(k) = k(H_h - E_j + k)^{-1}$, spectral calculus gives $\|H_j\| \le 64/63$ and $\|H_j'\| \le (1/4 + 64/3969)/k$. Applying the original Bessel transform to this Hilbert-space-valued function supplies an explicit conservative annihilation bound $\|A_\alpha(r)\| \le C_A(L)/r^2$ for $r \ge L \ge 2R$. The retained derivation splits at $1/R$, integrates the low-frequency Bessel derivative envelope, and keeps all nine high-frequency sine/cosine products with phases $k(r \pm R)$.

The shift has the exact profile

$$F_\alpha(r) = \frac{g}{\pi\sqrt{\kappa}\,rR} Q_2\left(\frac{r^2+R^2}{2rR}\right)\sigma_\alpha\Phi,$$

where its logarithmic shell singularity is square integrable. For $r \ge L$ it is bounded by $C_F(L)/r^4$, with

$$C_F(L) = \frac{16|g|R^2}{15\pi\sqrt{\kappa}(1-y_L^2)}, \qquad y_L = \frac{2LR}{L^2+R^2}.$$

Keeping both original branches, both polarizations and the common soft vacuum gives

$$K_{\mathrm{out,prep}}(L) \le \min\left\{K_{\mathrm{prep,up}}, \frac{8C_A(L)^2}{L} + \frac{4C_F(L)^2}{5L^5}\right\}.$$

The original energy-norm Duhamel estimate $d_A^2 \le 8T^2 g^2 v_s(1 + F_T + \lambda)$ then supplies

$$\eta_{\mathrm{ref}}(L) \le \min\{K_{\mathrm{prep,up}} + \varepsilon E(T), 2K_{\mathrm{out,prep}}(L) + 4(1+c_g)d_A^2\}.$$

The comparison payment independent of L is a bound feature, not an actual tail floor. At the illustrative radius $L = 10^{15}R$ in model coordinates, the preparation-tail ceilings divided by the inherited K_{prep} ceilings are approximately 0.00275835, 0.00166416, 2.06463×10^{-5}, 0.00267798, 2.23339×10^{-5} and 2.43841×10^{-5} in lane order 01,02,03,12,13,23. No SI ruler is assigned by these examples.

These receiving-form results supply an explicit incidence, realizable-weight classification, selection and complement certificates, and actual-state tail bounds. Appendix J displays the full standalone $C_A(L)$ coefficient. The Gaunt certificates do not evaluate the six time-dependent current-response integrals. The following six-lane continuation supplies one actual positive profile, a declared common dimensional ruler and evaluated response enclosures. It retains the isolated inhomogeneous two-time matrix elements as unresolved within those enclosures. These are kept separate from the local quantum stress identification.

17. One physical profile for the six original preparations

Fix a common model profile before evaluating any lane:

$$N_\alpha(\mathbf{x}) = 1 - \alpha V_{\mathbf{e}}(\mathbf{x}), \qquad V_{\mathbf{e}} = \frac{\sqrt{2}d}{\sqrt{d^2 + |\mathbf{x} - d\mathbf{e}|^2}},$$
$$\alpha = 10^{-3}, \quad d = 10^{30}R, \quad \mathbf{e} = 1, \quad N_\alpha(0) = n = 0.999.$$

This softened Plummer profile is smooth and globally positive: $1-\sqrt{2}\alpha \le N_\alpha < 1$. Its positive Laplacian is consistent with a finite positive-density body under a declared weak-potential Poisson interpretation. That interpretation is an additional receiving assignment, not a gravity law derived from the original source ledger.

The original radius, propagation speed, action chart and clock window $T_0 = 573049\pi$ remain common to every preparation. For a fully specified shared SI fixture, choose $R_{\mathrm{phys}} = 10^{-19}\,\mathrm{m}$, $c_{\mathrm{phys}} = 299792458\,\mathrm{m\,s^{-1}}$, and explicitly assign $I_u = \hbar$, with $h = 6.62607015\,10^{-34}\,\mathrm{J\,s}$. Then

$$\ell_u = R_{\mathrm{phys}}/R, \qquad t_u = \ell_u/c_{\mathrm{phys}}, \qquad E_u = \hbar/t_u.$$

The resulting displays are $t_u \simeq 2.03959635\,10^{-27}\,\mathrm{s}$, $E_u \simeq 5.17049276\,10^{-8}\,\mathrm{J}$, $\varepsilon E_u \simeq 9.02277600\,10^{-14}\,\mathrm{J}$, $T_0 t_u \simeq 3.67185784\,10^{-21}\,\mathrm{s}$, and $d_{\mathrm{phys}} = 10^{11}\,\mathrm{m}$. Outward intervals travel with the source bundle. This is one declared calibration fixture shared by all six lanes; the microscopic assignment of the original action and ruler remains a separate source question.

At the source center,

$$\nabla N_\alpha = -\frac{\alpha \mathbf{e}}{2d}, \qquad \mathrm{Hess}\, N_\alpha = \frac{\alpha}{2d^2}(I - \tfrac{3}{2}\mathbf{e}\mathbf{e}^{\mathrm{T}}).$$

The gradient has zero first compression but retains its full local angular complement. For $\beta = \angle(\mathbf{e}, \mathbf{e}_z)$ the quadratic compression is

$$A(r) - nI = \frac{\alpha r^2}{56d^2}(9 - 6\cos^2\beta)I + O(r^4/d^4).$$

Parallel, transverse and $\cos^2\beta = 1/3$ controls have coefficients $3/56, 9/56, 7/56$ times $\alpha r^2/d^2$. The last angle cancels the trace-free tidal part, while the softened body's positive density leaves an isotropic term.

A true leading tidal null is supplied by a second bounded physical control: a uniform body centered at $d\mathbf{e}$ with radius $a = d/2$. Writing $s = |\mathbf{x} - d\mathbf{e}|$,

$$N_\alpha = \begin{cases} 1 - \alpha d/s, & s \ge a, \\ 1 - \alpha d(3a^2 - s^2)/(2a^3), & s < a. \end{cases}$$

It is globally positive and C^1, with minimum $1 - 3\alpha$ and continuous normal derivative at its surface. The clock lies outside the body. Here the quadratic coefficient is $\alpha r^2(1 - 3\cos^2\beta)/(7d^2)$, with a genuine leading null at $\cos^2\beta = 1/3$. The finite globally bounded interior is retained.

18. An evaluated first response with the interacting transient

For the positive Bessel/TT receiver, the complete form is exactly

$$H_\alpha = (1-\alpha)H_0 + \alpha r_B, \qquad r_B[\Phi, \Psi] = \langle T\Phi, J^*(1-V)JT\Psi\rangle.$$

The original even source energy, both interactions and both counterterms remain. Equality to a complete canonical local quantum stress is not assumed. Let $\chi(t) = U_0(t)\Psi_{\mathrm{prep}}$ and let z_1 denote the derivative of its actual current with respect to α at zero. Then

$$z_1(T) = -T\dot{z}_0(T) + z_{1,\mathrm{inh}}(T).$$

The first term retains the exact interacting transient. The inhomogeneous term is the weak two-time response pairing with r_B.

Choose the single common near radius $L = 10^{18} R_-$, where R_- is the original radius's lower endpoint. Taylor's theorem and gradient cancellation give the complete profile bounds

$$b_{\mathrm{near}} = \frac{3}{2\,10^{24}}, \qquad b_{\mathrm{far}} = 1.$$

They are bounds for the derivative profile $1 - V$, so the finite perturbation pays $\delta_{\mathrm{near}} = \alpha b_{\mathrm{near}}$ and $\delta_{\mathrm{far}} = \alpha b_{\mathrm{far}}$. The exterior is retained.

The original certificate gives $K[\chi(t)] \le \kappa_0$ and $K_{\mathrm{out}}[\chi(t)] \le \eta_{\mathrm{ref}}(L)$ throughout the window. For $B_T(s) = U_0(T-s)^* Z U_0(T-s)$, the current commutator and energy conservation give, for B_T and its adjoint,

$$K[B_T(s)\chi(s)] \le k_B = (2\sqrt{\kappa_0} + 2\sqrt{c})^2 + 8\varepsilon.$$

Each first-variation pairing has one original trajectory factor. Apply its actual tail bound to that factor and $K_{\mathrm{out}} \le K \le k_B$ to the current-inserted factor. This avoids assuming an uncomputed tail for the inserted state and gives

$$|z_{1,\mathrm{inh}}(T)| \le D_1,$$
$$D_1 = 2T\sqrt{k_B}[b_{\mathrm{near}}\sqrt{\kappa_0} + b_{\mathrm{far}}\sqrt{\eta_{\mathrm{ref}}(L)}].$$

With $m = M_- - \mathscr{E}(T) > 0$, the actual first phase derivative is enclosed by

$$\theta_1(T) = -T\nu_0(T) + [-D_1/m, D_1/m].$$

Every quantity is evaluated with outward rational bounds from the original six preparations. The following displays round outward:

Lane	Lower θ_1	Upper θ_1
01	−6.283185603	−6.283184981
02	−6.283185904	−6.283184578
03	−6.283197797	−6.283152294
12	−6.283185624	−6.283184955
13	−6.283196966	−6.283154685
23	−6.283196110	−6.283157146

The complete complex z_1 is also enclosed: use $|\dot{z}_0 - \dot{h}| \le \Delta\mathscr{E} + 2|g|W_D$ and the exact hard harmonic $h = Me^{-\mathrm{i}\Omega T}$. Its endpoint arc has angle $T(\Delta - \Omega)$, so elementary sine/cosine bounds and the disk $T(\Delta\mathscr{E} + 2|g|W_D) + D_1$ give the retained complex rectangles. All six imaginary parts of z_1 are strictly positive; its real part is strictly negative in lanes03,13,23 and unresolved in the weaker lanes. The inhomogeneous correction's own sign and its cross-lane differences remain unresolved; these signed enclosures do not identify that correction with its uniform contribution.

19. Finite profile responses and a complete error budget

The same actual source preparation evolves under the finite profile $\alpha = 10^{-3}$. Compare it with $\chi(nt)$, $n = .999$, and write $e(t) = \|\Psi_\alpha(t) - \chi(nt)\|$. Static energy conservation and the weak overlap identity give a strict sufficient first-crossing condition:

$$e_*^4 > 4T^2 b_\alpha^2 (3\kappa_0 + 4\varepsilon e_*),$$
$$b_\alpha = \delta_{\text{near}}\sqrt{\kappa_0} + \delta_{\text{far}}\sqrt{\eta_{\text{ref}}(L)}.$$

Its verified positive margins imply $e(t) < e_*$ throughout the window. The current error is at most $4e_*$ and the phase error at most $\arcsin(4e_*/m)$. The certificate uses the rational majorant $(4e_*/m)/(1-4e_*/m)$ and verifies nonvanishing current at every prefix. The source/reference/window part is retained exactly:

$$\theta_0(nT) - \theta_0(T) = -\int_{nT}^{T} \nu_0(t)\,\mathrm{d}t.$$

Thus the full finite phase responses, including the physical-profile remainder, are enclosed as follows:

Lane	Lower $\Delta\theta$	Upper $\Delta\theta$
01	$-.006285093$	$-.006281277$
02	$-.006287000$	$-.006279370$
03	$-.006313697$	$-.006252653$
12	$-.006285093$	$-.006281277$
13	$-.006298438$	$-.006267914$
23	$-.006298439$	$-.006267915$

All are strictly negative. The profile comparison errors are at most $1.908, 3.815, 30.52, 1.908, 15.26, 15.26$ microradians in lane order. These are actual-state receiving-form certificates, not achieved apparatus precision. The compact-body control also passes all six sign tests with $b_{\text{near}} = 2\,10^{-24}$ and $b_{\text{far}} = 2$.

An explicit finite phase linearization payment is

$$|\Delta\theta_\alpha - \alpha\theta_1| \le \phi_{\text{prof}} + \alpha D_1/m + \alpha T(\nu_+ - \nu_-).$$

A complex-current finite remainder is retained separately in the source bundle. A universal-coupling null must compare $\theta_{\alpha,l}(T)$ with $\theta_{0,l}(nT)$, preserving each lane's interacting transient. Pairwise differences of those residuals pay the sum of the two profile errors together with preparation, timing, readout and shared-reference covariance. At first order the corresponding residual is $\theta_{1,l} + T\nu_{0,l}(T)$, bounded by $D_{1,l}/m_l$.

For the explicitly chosen receiving form, **the test result suggests strong contact with the concept**: both the actual first response and a finite perturbation have a certified sign in all six lanes under one positive physical profile. The unresolved local quantum-stress terms and the inhomogeneous susceptibility remain separately named recipients.

20. A common spatial split and a joint interacting-clock control

A shared logarithmic scan fixes $L_p = 10^p R_-$ for $p = 3, \dots, 29$, keeping the same physical profile, radius, preparation and units in every lane. At each radius the complete actual-state tail is recomputed and the objective is the largest of the six first-variation disk bounds. Exact comparisons give the minimum on this declared finite grid at $p = 25$, with maximum disk $6.13448\,10^{-6}$. The smallest common radius within one percent is $p = 21$, with maximum disk $6.16321\,10^{-6}$, compared with $1.97773\,10^{-5}$ at the previous $p = 18$.

At $p = 21$ the reference-tail ceilings lie within about0.94 percent of the retained soft-Duhamel comparison floor, compared with a factor10.394 at $p = 18$. That floor is a limitation of the comparison estimate. It is not a physical nonzero tail floor. The one-percent radius is about100m in the declared SI fixture; it is a mathematical split, not an apparatus extent. The grid optimum is not asserted to be the continuous optimum.

The finite-α first-crossing inequality is also verified at this single common radius. Its strict margins retain the entire continuum tail. The six phase errors improve to at most $0.954, 1.908, 15.260, 0.954, 7.630, 7.630$ microradians. All finite total phase responses remain strictly negative.

For an operational cross-lane control use the calibrated inverse of the original interacting clock, retaining its transient:

$$y_{lj} = \frac{\theta_{0,l}^{-1}(\theta_{\text{obs},l}(t_j))}{T} - n\frac{t_j}{T}, \qquad \frac{t_j}{T} = \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}, 1.$$

The deterministic profile error is bounded by $|h_{lj}| \le \phi_l/(T_-\nu_{l,-})$. This is an inverse-calibration prescription; the injection fixture does not manufacture exact numerical source trajectories. It uses24 observations, with one common affine timing nuisance and no individually fitted lane ages.

The declared normalized-age covariance includes independent noise $\sigma_i = 10^{-7}$, a reference shared by all lanes at each event $\sigma_r = 3\,10^{-7}$, and global offset/drift uncertainties $\sigma_o = \sigma_d = 2\,10^{-7}$. With $x_j = t_j/T$,

$$\Sigma_{lj,km} = \sigma_i^2\delta_{lk}\delta_{jm} + \sigma_r^2\delta_{jm} + \sigma_o^2 + \sigma_d^2 x_j x_m.$$

These are modeled measurement requirements. For $X = (1, x)$, set

$$A = \Sigma^{-1} - \Sigma^{-1}X(X^{\mathrm{T}}\Sigma^{-1}X)^{-1}X^{\mathrm{T}}\Sigma^{-1},$$
$$Q(y) = y^{\mathrm{T}}Ay.$$

Exact checks give $AX = 0$, $A\Sigma A = A$, and $\operatorname{tr}(A\Sigma) = 22$. For zero-mean noise with this covariance, Markov's inequality gives the conservative one-percent threshold2200 without a Gaussian assumption. The whole profile-error box obeys $h^{\mathrm{T}}Ah \le B = \sum h_{lj,\max}^2/\sigma_i^2$. Thus $[\sqrt{Q(y)} - \sqrt{B}]_+^2$ is a conservative robust rejection statistic; no independence of the unknown profile errors is assumed.

The correct coordinate baseline gives $Q = 0$. A common affine clock or travel-time injection also gives $Q = 0$, exposing the need for an external event-map anchor to separate those

changes. Lane-dependent coupling $(0,0,10^{-4},0,-10^{-4},0)x_j$ instead gives $Q = 3{,}750{,}000$; a wrong nonlinear common event map $8\,10^{-4}x_j^2$ gives $Q = 1{,}200{,}000/11$. Their profile-box robust lower statistics exceed $3.52\,10^6$ and 73,083, respectively. Both exceed2200 and satisfy the separate sufficient modeled-power condition $\sqrt{Q_{\mathrm{signal}}} > 2\sqrt{B}+2\sqrt{2200}$, giving at least99 percent rejection probability under the declared covariance. This is a reproducible inference/injection result, not achieved apparatus performance.

21. Isolated susceptibility and weak-amplitude reach

Retain lanes $01,02,03,12,13,23$, with their unchanged exact hard-eigenbranch superpositions and full soft vacuum. The signed couplings, both polarizations, all field frequencies, even source energy, both counterterms, original window $T_0 = 573049\pi$ and shared dimensional calibration remain. The complete sixteen-state source, including its four dark states, is unchanged. This update concerns the selected positive Bessel/TT receiver; the canonical local-stress protocol retains its separate domain and renormalization questions.

The common profile is still $N_\alpha = 1-\alpha V_{\mathbf{e}}$, with $d = 10^{30}R$, $N_\alpha(0) = 1-\alpha$, and the single shared split $L = 10^{21}R_-$. Varying α changes the profile amplitude, not the preparation, ruler or source age separately by lane. Below $T = T_0$.

21.1 Cancel the interacting uniform history first

Define the finite inhomogeneous residual by

$$R_l(\alpha,T) = \theta_{\alpha,l}(T) - \theta_{0,l}((1-\alpha)T).$$

Its derivative is $r_l = \theta_{1,l} + T\nu_{0,l}(T)$. The exact interacting uniform transient has canceled before interval arithmetic. With the original trajectory χ_l, $z_{0,l} = \langle Z_l\rangle$ and $B_T(s) = U_0(T-s)^* Z_l U_0(T-s)$, the isolated functional is

$$r_l = -\operatorname{Im}\frac{\mathrm{i}}{z_{0,l}(T)}\int_0^T \big\{r_B[\chi_l, B_T\chi_l] - r_B[B_T^*\chi_l,\chi_l]\big\}\,\mathrm{d}s.$$

The actual two-time matrix elements have not been point-evaluated. They are enclosed directly using the original trajectory tail and one global energy bound for the inserted-current state. The resulting zero-centered intervals are for the isolated functional itself. Cross-lane contrasts also remove their exact uniform transients first; without further joint information, their safe radii add, with no independence assumption.

21.2 Direct orientation contrasts and exact null controls

A correlated orientation calculation cancels additional terms. Put

$$A = r^2+2d^2,\quad C = \tfrac{1}{2}(A+\sqrt{r^4+4d^4}),$$
$$q = rd/C,\qquad F = \sqrt{2}d/\sqrt{C}.$$

The profile has the exact expansion $V_{\mathbf{e}} = F\sum_{j\ge0} q^j P_j(n\cdot\mathbf{e})$, where F 1 and q $\sqrt{2}$ $1 < 5/12$. Only degrees 0,2,4 survive the TT compression. Its anisotropic degree-four part is

$$\frac{5}{144}Fq^4\sin^4\beta[\cos(4\phi)\sigma_z + \sin(4\phi)\sigma_x].$$

Consequently an azimuth shift $\phi \mapsto \phi+\pi/4$ cancels the complete uniform, radial, quadrupole and axisymmetric degree-four contributions before any response error is bounded. The remaining weight has bounds

$$b_{\mathrm{con,far}} = \frac{5}{72}\left(\frac{5}{12}\right)^4,$$
$$b_{\mathrm{con,near}} = \frac{5}{1152}(L/d)^4.$$

The direct first-phase contrast radius is below $1.290030\,10^{-8}$ radians per unit α in every lane, roughly 950 times tighter than adding two isolated residual disks in the strongest lane. These intervals still contain zero.

There are exact null controls at every amplitude: $W_{\mathbf{e}} = W_{-\mathbf{e}}$ and $W(\beta,\phi+\pi/2) = W(\beta,\phi)$. Their receiving Hamiltonians and current histories coincide on the same preparation. Gradient reversal does not isolate its even return. The original charge selection is also retained: the state has charges 0 and 1, Z lowers charge by 1, and the scalar receiving perturbation has charge 0 and anisotropic charge ±2. Only its charge 0 and $+2$ parts can contribute to the complex Z first response. This does not assign the unknown response coefficients.

21.3 A tight finite energy ceiling

The previous square-root amplitude diagnostic loses three nanoradian sign tests. A tighter finite certificate uses the same common form domain and current theorem, retaining the actual small correlated residual energy. Let $n = 1-\alpha$, $m_N = 1-3\alpha/2$, and let ρ_{prep} bound the original prepared expectation of r_B:

$$|r_B[\Psi_{\mathrm{prep}}]| \le b_{\mathrm{near}}K_{\mathrm{prep}} + K_{\mathrm{out,prep}}(L) \le \rho_{\mathrm{prep}}.$$

The complete profile obeys $N_\alpha \ge m_N$. With $e(t) = \|\Psi_\alpha(t) - \chi(nt)\|$, exact energy conservation gives

$$K_{\mathrm{act}}(t) \le A_\alpha + B_\alpha e(t),$$
$$A_\alpha = \frac{n\kappa_0 + \alpha\rho_{\mathrm{prep}}}{m_N},$$
$$B_\alpha = \frac{2n\varepsilon}{m_N}.$$

The common even scalar remains in the energy account and cancels only from this comparison. A strict state-distance certificate is obtained from

$$e_*^4 > 4T^2\alpha^2 b^2(A_\alpha + B_\alpha e_*),$$
$$b = b_{\mathrm{near}}\sqrt{\kappa_0} + \sqrt{\eta_{\mathrm{ref}}(L)}.$$

The ratio $e^4/(A+Be)$ is strictly increasing for $e>0$, so rational bisection gives an upper root with an exact positive margin. This certifies the tight uniform ceiling $K_A = A_\alpha + B_\alpha e_*$.

21.4 A linear current certificate and an all-amplitude sign theorem

Insert this ceiling into the existing weak Heisenberg theorem. The reference propagator's energy bound and the exact current commutator give

$$k_B = [2\sqrt{K_A + 2\varepsilon} + 2\sqrt{c}]^2 + 8\varepsilon,$$
$$|z_\alpha - z_{\mathrm{ref}}| \le 2\alpha T\sqrt{K_A k_B}.$$

This uses continuity in the common energy form domain and differentiability into its dual. It does not assume $r_B\Psi$ exists as a Hilbert-space vector or identify the canonical stress with this receiver. The exact variation pairs the perturbation with the evolved current and its adjoint; both inserted-current energies are bounded by k_B.

Take the smaller valid current error $E_z = \min(4e_*, 2\alpha T\sqrt{K_A k_B})$. With $m = M_- - \mathscr{E}(T)$,

$$|R_l(\alpha,T)| \le \arcsin(E_z/m) \le \frac{E_z/m}{1 - E_z/m}.$$

The state certificate is often tighter at larger α; the current certificate becomes linear in α near zero. Both retain the unchanged interacting preparation and every prefix of the original window.

There is an explicit interval theorem, beyond the discrete sweep. The retained strict certificate at $\alpha_0 = .001$ has

$$e_0^4 > 4T^2\alpha_0^2 b^2(3\kappa_0 + 4\varepsilon e_0).$$

For $e_\alpha = e_0\sqrt{\alpha/\alpha_0}$, its left side scales by $(\alpha/\alpha_0)^2$, while the energy factor on the right decreases. The same first-crossing argument therefore gives $e(t) < e_\alpha \le e_0$ for every $0 < \alpha \le \alpha_0$. Using $n \le 1$ and $m_N \ge 1 - 3\alpha_0/2$ in the exact energy comparison gives

$$\overline{K}_A = \frac{\kappa_0 + \alpha_0\rho_{\mathrm{prep}} + 2\varepsilon e_0}{1 - 3\alpha_0/2}.$$

Insert this ceiling in the current bound and define, lane by lane,

$$\overline{k}_B = [2\sqrt{\overline{K}_A + 2\varepsilon} + 2\sqrt{c}]^2 + 8\varepsilon,$$
$$C_l = \frac{2T\sqrt{\overline{K}_A \overline{k}_B}}{m_l}.$$

The exact outward comparisons verify $\alpha_0 C_l < 1$ and

$$|R_l(\alpha,T)| \le \frac{\alpha C_l}{1 - \alpha_0 C_l},$$
$$\frac{C_l}{1 - \alpha_0 C_l} < T_- \nu_{l,-}.$$

for all six original preparations. The largest error coefficient is below 0.2121, while every signal coefficient exceeds 6.28. In particular,

$$\theta_{\alpha,l}(T) - \theta_{0,l}(T)$$
$$= -\int_{(1-\alpha)T}^{T} \nu_{0,l}(u)\,\mathrm{d}u + R_l(\alpha,T)$$
$$\le -\alpha T_- \nu_{l,-} + \frac{\alpha C_l}{1 \quad \alpha_0 C_l} < 0.$$

This proves the finite endpoint-phase sign for every $0 < \alpha \le$ .001 in the same receiving model. It uses the entire original interacting transient and the strict current-amplitude margin, not six independent frequency fits or an inference from a negative derivative alone.

21.5 Weak-amplitude replay and finite residual contrasts

The outward sweep covers $\alpha = 10^{-k}$ for $k = 3,\dots,8$, adds $\alpha = 7\,10^{-10}$, and includes 10^{-10}, 10^{-11} and 10^{-12}. Every lane has a strict finite crossing, nonzero-current and total-sign certificate. At $\alpha = 7\,10^{-10}$, compare the old sufficient rescaling with the new finite inhomogeneous phase bounds, in nanoradians:

Lane	Old rescaling	New bound
01	0.798	0.000162
02	1.596	0.000389
03	12.767	0.088993
12	0.798	0.000174
13	6.384	0.080217
23	6.384	0.071460

All six new bounds fit a 0.2 nanoradian complete-form allowance and preserve the approximately -4.398 nanoradian uniform signal. This is a rescaling of the declared distant-body fixture, not an Earth-profile calculation or achieved apparatus sensitivity. Every other metrological requirement remains.

For a direct finite azimuth contrast, the other nonuniform receiver has the arbitrary-vector propagation estimate $K[U_B(t)\Xi] \le \gamma K[\Xi] + \beta\|\Xi\|^2$, where $\gamma = 1/m_N$, $\beta = 2n\varepsilon/m_N$. Its inserted-current ceiling is

$$k_{B,\mathrm{other}} = \gamma[2\sqrt{\gamma K_A + \beta} + 2\sqrt{c}]^2 + 4\beta.$$

The exact receiving-form difference gives $|z_A - z_B| \le 2\alpha b_{\mathrm{con,far}} T\sqrt{K_A k_{B,\mathrm{other}}}$. Its reference amplitude is at least $m - E_z$. At $7\,10^{-10}$ the direct finite orientation-phase contrast has a zero-centered enclosure of radius below 0.000186272 nanoradians in every lane. The common uniform and axisymmetric operators canceled first.

An explicit finite derivative payment is retained: $|R_l(\alpha) - \alpha r_l| \le \Phi_l(\alpha) + \alpha D_l/m_l$. The direct orientation contrast uses its own finite bound and direct first-contrast radius. These are controlled finite remainders, with no claim of a sharp quadratic order. Isolated signs and actual numerical two-time matrix elements remain unresolved. For the selected receiver, **the test result suggests strong contact with the concept**: retaining the correlated energy closes the weak-amplitude sign test in all six lanes, while the canonical quantum protocol remains separate.

22. The canonical local coupling has a pair kernel

The following new calculation starts from the retained compatible local tensor action, with $\kappa = 4/5$, its complete tangential tracefree bundle, regular origin and full exterior. It selects that action's canonical lapse coupling. A covariant source, support

and ruler law is still needed to select its complete physical metric stress; flat-space improvements can leave the original Hamiltonian unchanged and change a nonuniform lapse variation.

With $\Pi = \kappa \dot{H}/2$, the bulk candidate is

$$H_f[N] = \int N \left\{ \frac{\Pi : \Pi}{\kappa} + \frac{\kappa}{4} \left(|\nabla H|^2 + \frac{2|H|^2}{r^2} \right) \right\} d^3x.$$

The declared source shell receives $\bar{N}_R \varepsilon H_{\rm pair}$, $-\varepsilon \sum_\alpha J_\alpha \int N(R,n) U e_\alpha : H(R,n) d\Omega$, and $\bar{N}_R \varepsilon^2 \mu (D^2 + C^2)$, where $J = (D,C)$, $\mu = 1/(4R)$ and $\bar{N}_R$ is the shell average. Thus even source energy, both signs and both original counterterms remain. The shell average and the receiving form's center value are distinct: for the shared profile $|\bar{N}_R - N(0)| < 1.5\,10^{-63}$. This finite-source incidence mismatch is explicitly retained. A new counterterm is not silently fitted to complete a different square.

For real complete eigenmodes $f_i = \sqrt{2/\pi}\, k j_l(kr) Y^P_{lm}(n)$ and $\omega_i = k$, set

$$M_{ij} = \int N f_i : f_j, \qquad V_{ij} = \int N (\nabla f_i : \nabla f_j + 2 f_i : f_j / r^2).$$

Canonical quantization gives, before the ultraviolet limit,

$$\begin{aligned} : H_f[N] : &= a^\dagger A a + \tfrac{1}{2}(aBa + a^\dagger B a^\dagger), \\ A_{ij} &= \frac{\omega_i \omega_j M_{ij} + V_{ij}}{2\sqrt{\omega_i \omega_j}}, \\ B_{ij} &= \frac{V_{ij} - \omega_i \omega_j M_{ij}}{2\sqrt{\omega_i \omega_j}}. \end{aligned}$$

Wave-packet integration by parts on the complete space gives

$$B_{ij} = \frac{(\omega_i - \omega_j)^2 M_{ij} + \int (\Delta N) f_i : f_j}{4\sqrt{\omega_i \omega_j}},$$

$$A - A_J = B, \qquad (A_J)_{ij} = \sqrt{\omega_i \omega_j} M_{ij}.$$

Any artificial interface has its own integration boundary term. The original source shell is a vertex, not a boundary deleting exterior modes. Operator derivatives in D_b and the return forcing below are projected onto the constrained TT bundle; bare Cartesian normal components are not added as field coordinates. A_J is the free-field part of the positive-frequency receiver. The canonical candidate additionally has $d\Gamma(B)$, a pair kernel, its source vertices and a vacuum prescription. Uniform N makes $B = 0$ and recovers exactly NH_0.

23. Vacuum domain and paid temporal regularization

In a local flat polarization chart the pair kernel has the ultraviolet asymptotic, for fixed K,

$$B(p, -p+K) = -\frac{|K_\perp|^2 \widehat{N}(K)}{4(2\pi)^{3/2}|p|} + O(|p|^{-2}).$$

Its equivalent principal symbol is $(\Delta N - \hat{p}^T \,\mathrm{Hess} N \hat{p})/(4|p|)$. Connection and source-centered potential terms have lower symbol order. For one local real polarization a cutoff Λ yields

$$\begin{aligned} \| : \delta H_f : |0\rangle \|^2 &\sim \frac{\Lambda}{120\pi^2} \int |\Delta \delta N|^2 d^3x, \\ \| H_f^{-1/2} : \delta H_f : |0\rangle \|^2 &\sim \frac{\log \Lambda}{240\pi^2} \int |\Delta \delta N|^2 d^3x. \end{aligned}$$

There are two local tensor polarizations. The original finite shell counterterm does not remove this bulk vacuum term. Hence a sharp-time Wick density cannot simply be installed as a continuous form on the previous $D(H_f^{1/2})$ containing the original vacuum. A static Bogoliubov map with ultraviolet $\beta \sim |p|^{-2}$ does not make its energy integral finite.

For the shared Plummer profile

$$N(x) = 1 - \frac{\mu_P}{\sqrt{d^2 + |x - d e_z|^2}}, \quad \mu_P = \frac{\sqrt{2} d}{1000}, \quad d = 10^{30} R,$$

the coefficient is explicitly $\int |\Delta N|^2 d^3x = 45\pi^2 \mu_P^2/(64 d^3) > 0$. A specified entry and release gives a computable continuation. For example, use $\chi(t) = \sin^4(\pi t/T_s)$ on $[0, T_s]$, zero outside. Its zero extension is C^3. The outgoing first-order pair coefficient and second-order free energy are

$$\begin{aligned} \beta_{ij} &= -i\eta B_{ij} \widehat{\chi}(\omega_i + \omega_j), \\ E^{(2)}_{\rm pairs} &= \frac{\eta^2}{2} \int (\omega_i + \omega_j) |B_{ij}|^2 |\widehat{\chi}(\omega_i + \omega_j)|^2 di\, dj. \end{aligned}$$

The latter is finite and nonnegative for this switching. It joins the signed source work, vacuum-polarization convention and support/controller payments. Higher orders and the complete renormalized protocol remain to be bounded; a finite spatial cutoff is not the original preparation.

24. A gradient leaves through two angular sectors

Let $\mathscr{C}$ be the retained total-rotation Casimir. For $Y = n_z U Q$, direct polynomial calculation gives

$$Y_{B2} = (12 - \mathscr{C}) Y/6, \qquad Y_{E3} = (\mathscr{C} - 6) Y/6.$$

These are mutually orthogonal, have Casimirs 6 and 12, and are orthogonal to the whole original electric $U(\mathrm{STF})$ image. Their normalized Grams in its rational five-coordinate chart are

$$\begin{aligned} G_{B2} &= \mathrm{diag}(4/9, 4/9, 1/9, 1/9, 0), \\ G_{E3} &= \mathrm{diag}(5/63, 5/63, 8/63, 8/63, 1/7). \end{aligned}$$

Thus the actual plus/cross gradient leakage divides as $4/9 + 5/63 = 11/21$; fractions $28/33$ and $5/33$ propagate through different radial operators. The compressed variance alone supplies no time evolution or inverse gap.

For the exact first Legendre coefficient $n_1(r)$ of the same Plummer profile, put $a = r^2 + 2d^2$, $b = 2rd$, $s_\pm = \sqrt{a \pm b}$. Then

$$n_1(r) = -\frac{2\mu_P b}{(s_+ + s\ \ )(a + \sqrt{a^2 \quad b^2})} = -\frac{r}{2000 d} + O(r^3/d^3).$$

The radial number and pair kernels of the two complementary sectors are obtained from

$$M_l(k,q) = \int r^2 n_1(r) R_2(k,r) R_l(q,r) dr,$$
$$D_l^N(k,q) = \int r^2 (n_1'' + 2n_1'/r - 2n_1/r^2) \times R_2(k,r) R_l(q,r) dr,$$
$$B_{2\leftarrow l} = \frac{(k-q)^2 M_l + D_l^N}{4\sqrt{kq}},$$
$$A_{2\leftarrow l} = \sqrt{kq} M_l + B_{2\leftarrow l}.$$

where the reverse-sector kernels are $A_{l\leftarrow 2}(q,k\prime) = A_{2\leftarrow l}(k\prime,q)$ and likewise for B; these reverse labels are not forward functions evaluated with exchanged arguments. Here $R_l = \sqrt{2/\pi} k j_l(kr)$ and angular weights are $c_2^2 = 4/9$, $c_3^2 = 5/63$. Eliminating the doubled canonical complement gives the leading causal return

$$K_{++}(k,k';\tau) = \eta^2 \sum_l c_l^2 \int \big[A_{2\leftarrow l}(k,q) A_{l\leftarrow 2}(q,k') e^{-iq\tau} - B_{2\leftarrow l}(k,q) B_{l\leftarrow 2}(q,k') e^{iq\tau}\big] dq.$$

The complementary boundary force $\varepsilon n_1(R) c_l J_\alpha u_{R,l}(q)$ is also retained. In the original quantum source equation it is operator-valued; its expectation needs the actual interacting two-time state.

25. The bounded physical profile has a nonzero static return

This exact static field Schur calculation keeps the source boundary. Let $v_R = \delta(r-R)/R^2$ and

$$h_0 = D_2^{-1} v_R = \begin{cases} r^2/(5R^3), & r < R, \\ R^2/(5r^3), & r > R. \end{cases}$$

For $N = N_0 + \eta b$, $D_N = N_0 D + \eta D_b$ with $D_b = bD - \nabla b \cdot \nabla$, while $v_N = N_0 v_R + \eta b(R) v_R$. The static field functional is $-\varepsilon^2 \langle v_N, D_N^{-1} v_N\rangle J_\alpha^2/\kappa$. The bulk and boundary insertions cancel their delta-shell pieces:

$$w = b(R) v_R - D_b h_0$$
$$= P_{\mathrm{TT}}[\nabla b \cdot \nabla (h_0 U e_\alpha)], \qquad E_{\mathrm{all}} = \langle w, D^{-1} w\rangle = E_P + E_Q.$$

Here P is the original electric-$l = 2$ driven projector, $Q = I - P$, and $E_Q = \langle Qw, (QDQ)^{-1} Qw\rangle$. The complete inverse retains even forcing in P; the isolated odd forcing lies in Q. Its quadratic contribution is $-\eta^2 \varepsilon^2 E_{\mathrm{all}} J_\alpha^2/(\kappa N_0)$. For the linear-gradient comparator $b = \gamma z$, solving both regular interior/decaying exterior Poisson equations gives

$$E_{B2} = \frac{c_2^2 \gamma^2 R}{210}, \qquad E_{E3} = \frac{2c_3^2 \gamma^2 R}{105}, \qquad E_{\mathrm{linear}} = \frac{8\gamma^2 R}{2205}.$$

For both source polarizations this coefficient is $-2\eta^2 \varepsilon^2 \gamma^2 R(D^2 + C^2)/(441 N_0)$. It is a field Schur functional, not a replacement coherent preparation or an exact joint quantum ground-energy formula for noncommuting D, C.

Now retain the complete bounded odd part $b_o(x) = [N(x) - N(-x)]/2$, including all its higher multipoles. For $\gamma = -1/(2000d)$, $\mathscr{D} = d/R$, $S = L/R$, a dual energy estimate gives

$$\frac{\|D_Q^{-1/2}(w_o - w_{\mathrm{linear}})\|^2}{\gamma^2 R} \le \frac{1}{6}\left[\frac{10368}{275\mathscr{D}^4} + \frac{16848(S-1)}{25\mathscr{D}^4} + \frac{208}{75S^3}\right].$$

It uses $D_Q \ge 6/r^2$, the actual angular gradient coefficient 4, and the full exterior. With $\mathscr{D} = 10^{30}$, $S = 10^{15}$,

$$E_o \in \frac{8\gamma^2 R}{2205}[1 - 10^{-20}, 1 + 10^{-20}].$$

Thus the actual bounded profile's returning gradient is calculated. Its two signs have identical compressed receiver histories and this nonzero canonical return. The return is measured against the constant N_0 reference. Parity can also make the two complete canonical histories equal for the original even preparation and source-only readout; this calculation does not automatically recover the gradient's sign.

There is an exact finite-profile bound as well. Set $\mathscr{B} = D^{-1/2} D_b D^{-1/2}$ and $u = D^{-1/2} w$. The nonlinear static remainder after the linear part is exactly $\eta^2 \langle u, (N_0 I + \eta \mathscr{B})^{-1} u\rangle$. Since $\|\mathscr{B}\| \le \|b\|_\infty$, it lies between $\eta^2 E_{\mathrm{all}}/[N_0(1+\rho)]$ and $\eta^2 E_{\mathrm{all}}/[N_0(1-\rho)]$ for $\rho = |\eta| \|b\|_\infty/N_0 < 1$. For the full Plummer profile, $\|N - N_0\|_\infty = 1/1000$, $\rho = 1/999$, and parity gives $E_{\mathrm{all}} = E_e + E_o \ge E_o$. Its finite nonlinear static return is therefore at least E_o at $\eta = 1$. For an isolated odd profile parity improves the relative correction to at most $1/998000$. These static bounds do not evaluate the prepared quantum clock's two-time expectation.

26. The source-derived kernel and the amplitude interface

Eliminating the original quadratic field gives

$$\chi_R(\omega) = 2\int_0^\infty \frac{k u_R(k)^2}{k^2 - (\omega + i0)^2} dk = \frac{G_2(\omega R)}{\kappa R},$$
$$G_2(z) = \int_0^2 e^{izs}(1 - 3s^2/2 + 3s^4/8) ds.$$

Its first six signed delay moments are $2/5, 0, -8/105, 0, 32/105, 16/15$. Therefore

$$\chi_R(\omega) = \frac{1}{2R} + \frac{R}{21}\omega^2 + \frac{R^3}{63}\omega^4 + i\frac{R^4}{90}\omega^5 + O(\omega^6).$$

For positive frequency, $\mathrm{Im}\,\chi_R = \pi u_R(\omega)^2 \ge 0$. This is radiative passivity. The kernel is a signed compact delay response; its state-dependent source influence retains time ordering and noise. The original superposed hard branches require a two-time susceptibility.

The retained bulk kinetic coefficient is constant κ. Localized source elimination gives a nonlocal self-energy, rather than

selecting a homogeneous cosmological $Q(t)$ or a positive rate spectrum in $\log(1+z)$. Hamiltonian spectral factors oscillate even with positive energy and positive passive spectral weight. A relaxation/coarse-graining law, its background, common units and a controlled reduction are the remaining physical inputs to such a transport coefficient.

One positive-spectrum construction does come directly from the source. For its actual finite dressing vector $v = u_R/\sqrt{k}$, $\|v\|^2 = 1/(5\kappa R)$, the normalized Euclidean correlation is

$$F_E(\tau) = \frac{\langle v, e^{-\tau k} v\rangle}{\|v\|^2} = \int_0^\infty e^{-k\tau}\,\frac{10R}{\pi}\,j_2(kR)^2 dk.$$

This is a derived continuous positive measure of total mass 1, with rates down to zero and $F_E(\tau) \sim 16R^5/(15\pi\tau^5)$. It is a Euclidean or contractive-semigroup observable; real Hamiltonian time uses $e^{-ik\tau}$. Its identification with physical relaxation and then $\log(1+z)$ still requires a physical law. No finite mode count or positive minimum rate follows from this original continuum.

The independent portable checker supplies 164 exact assertions, including the original polynomial angular split, both radial matching problems, full-tail Plummer bound and signed source-kernel moments. Native execution and custody are recorded separately in the accompanying cycle receipts.

27. A switched canonical protocol and its remaining quantum certificate

27.1 An executable directional-kernel control

The corrected directional kernels can be checked on nonsymmetric regular radial probes before any return contraction is formed. For $b(x) = ze^{-r^2}$, use $r^2e^{-ar^2}$ in the retained sector and $r^3e^{-br^2}$ in the complementary sector. After factoring $105\sqrt{\pi}/32$, the choices $a = (1,97)$, $b = (2,23)$ give

$$M_{2\leftarrow 3} = \begin{pmatrix} 2^{-9} & 5^{-9} \\ 10^{-9} & 11^{-9} \end{pmatrix}, \qquad M_{3\leftarrow 2} = M_{2\leftarrow 3}^T.$$

The matrix is nonsymmetric, so the executable check detects an incorrect forward–forward contraction. These are test functions, not replacement clock preparations. Independent actual-Bessel quadrature at $(k,q) = (1,2)$ gives $A_{2\leftarrow 3}(1,2) = A_{3\leftarrow 2}(2,1) = 0.02332496\ldots$, whereas the incorrectly reused forward function is $A_{2\leftarrow 3}(2,1) = 0.00811628\ldots$. The quadrature is a numerical diagnostic, not an interval certificate.

27.2 A paid smooth switching protocol

Preserve the original preparations and window $T_s = T_0 = 573049\pi$. Before zero and after T_s use $N = 1$; inside prescribe

$$N_\eta(t,x) = 1 + \eta\chi(t)b(x), \qquad b(x) = -\frac{\mu_P}{\sqrt{d^2 + |x - de_z|^2}},$$

with the same $\mu_P = \sqrt{2}d/1000$, $d = 10^{30}R$, and $\chi(t) = \sin^4(\pi t/T_s)$. Its zero extension is C^3 and $N_\eta \geq 1 - |\eta|\sqrt{2}/1000$. The bulk, finite source even energy, both shell vertices and both original counterterms all receive this same lapse at their stated supports. No angular or frequency cutoff defines the protocol. Its two relevant areas are

$$\int \chi\,dt = \frac{3T_s}{8}, \qquad \int \chi^2 dt = \frac{35T_s}{128}.$$

The center-lapse reference therefore has age $T_s(1 - 3\eta/8000)$, and keeps the original interacting baseline transient. Clock-certificate use of the full original window is restricted to $0 \leq \eta \leq 1$; signed classical bounds below do not extrapolate its age.

For the canonical pair kernel B_{ij}, put $\Omega = \omega_i + \omega_j$. The first-Born outgoing pair amplitude and spontaneous vacuum free energy are

$$\beta_{1,ij} = -iB_{ij}\widehat{\chi}(\Omega), \qquad E^{(2)}_{\text{spont}} = \frac{\eta^2}{2}\int \Omega |B_{ij}|^2 |\widehat{\chi}(\Omega)|^2 di\,dj.$$

The outgoing energy is finite. At an intermediate nonzero $\chi(t)$, β_t instead has ultraviolet order $|p|^{-2}$: its number norm can be finite while its original free-energy norm diverges logarithmically. The expression above is a vacuum component, not the full work on the original entangled hard-branch preparation.

Smooth switching also leaves a scalar order-two vacuum loop phase. The high-frequency time integral satisfies

$$\int_{s<t} \chi(t)\chi(s)\sin[\Omega(t-s)]ds\,dt = \frac{\int \chi^2 dt}{\Omega} + O(\Omega^{-3}).$$

For both tensor polarizations of the actual Plummer profile this yields

$$\Gamma_{2,\Lambda} = \frac{105\mu_P^2 T_s}{65536 d^3}\log(\Lambda/\Lambda_{\text{ref}}) + O(1).$$

A bulk scalar vacuum counterterm cancels that divergence; its finite energy-reference part must be stated. The original source counterterm proportional to $D^2 + C^2$ is retained and is not this bulk term. The scalar phase cancels every bounded source-current expectation. Thus it is an energy/reference issue rather than, by itself, a failure of projective observable evolution. Returning the controls to their initial values does not remove outgoing field energy or gross switching payments.

27.3 A controlled transported boundary form for all six preparations

The next construction explicitly separates the background field evolution. Let $S_\eta(t)$ be complete classical symplectic propagation and $S_0(t)$ the original free propagation. If $W_\eta(t)$ implements $S_\eta(t)S_0(-t)$ projectively, write $U_{\text{joint}} = W_\eta V_\eta$. The comparison-picture candidate has the original H_f plus transformed boundary forms $-\varepsilon\sum_\alpha J_\alpha\phi(f_{\alpha,t})$, the full finite source Hamiltonian and the original shell counterterms. Every angular mode and coordinate/momentum quadrature in the exact symplectic pullback remains; $f_{\alpha,t}$ is not replaced by $\bar{N}_R u_R$.

A uniform weighted boundary estimate can already be closed. With $\delta = \|b\|_\infty$, $n_\pm = 1 \pm |\eta|\delta$, the classical energy obeys

$$n_- E_0 \leq E_N \leq n_+ E_0, \qquad |\dot{E}_N| \leq \frac{\|\dot{N}\|_\infty}{n} E_N.$$

Since $\int \|\dot N\|_\infty dt \le 2|\eta|\delta$, the energy-norm propagation bound and complete shell duality give

$$C_E \le \sqrt{\frac{n_+}{n_-}} e^{|\eta|\delta/n_-}, \qquad \left\| \frac{f_{\alpha,t}}{\sqrt{k}} \right\|_2 \le n_+ C_E \sqrt{\mu_2}.$$

Here every tensor mode has $\mu_l = 1/[\kappa R(2l+1)] \le \mu_2$; the angular expansion of $N(R,n)Ue_\alpha$ pays its complete norm. For $|\eta| \le 1$, exact rational majorants yield

$$C_E \le \frac{2003}{1994}, \qquad M_\alpha := \sup_t \|f_{\alpha,t}/\sqrt{k}\|_2 < \frac{1007}{1000}\sqrt{\mu_2}.$$

All six original radius balls are used in the independent certificate. Consequently the transported source forms are uniformly semibounded:

$$K_\eta(t) \ge \frac{H_f}{2} - 2\varepsilon^2 \left(\sum_\alpha \|J_\alpha\| M_\alpha \right)^2.$$

This is an explicit comparison-picture bound, not an inheritance of the sharp-time canonical vacuum domain from the positive receiver.

27.4 A closed static frequency comparison

A further constructive route diagonalizes the instantaneous classical field before treating its time connection. This is a new partial advance, rather than a completed quantum propagator. In normalized canonical variables $q = \sqrt{\kappa/2}H$, $p = \sqrt{2/\kappa}\Pi$, put

$$H_f[N] = \tfrac{1}{2}(\langle p, Np\rangle + \langle q, D_N q\rangle),$$
$$Q = N^{-1/2}q, \quad P = N^{1/2}p,$$
$$A_N = N^{1/2} D_N N^{1/2}, \quad h_N = A_N^{1/2}, \quad \omega = D^{1/2}.$$

Here D and D_N act on the complete regular tensor field; D_N is the positive lapse-weighted gradient and potential form. No driven-sector compression is used.

For the actual Plummer profile, with $s = |x - de_z|$ and $t = s/d$,

$$r|\nabla N| \le |\eta\chi|\delta \frac{t(t+1)}{(1+t^2)^{3/2}} \le \frac{|\eta\chi|}{1000}.$$

The global maximum follows from the exact factorization

$$(1+t^2)^3 - 2t^2(t+1)^2 = (t-1)^2(t^4+2t^3+4t^2+2t+1).$$

This estimate includes the softened body, regular origin and full exterior. The original $l \ge 2$ angular energy gives $\|f/r\|^2 \le \langle f, Df\rangle/6$. Writing $g = \sup r|\nabla N|$ and $c_g = g/(2\sqrt{6})$, the Leibniz rule and $n_- D \le D_N \le n_+ D$ therefore give the closed-form inequalities

$$(n_- - c_g)^2 D \le A_N \le \left(n_+ + c_g\sqrt{n_+/n_-}\right)^2 D.$$

Operator monotonicity of the positive square root yields

$$(n_- - c_g)\omega \le h_N \le \left(n_+ + c_g \sqrt{n_+/n_-}\right)\omega.$$

Using $\delta < 3/2000$, $g \le 1/1000$ and $\sqrt{6} \ge 2$ supplies the explicit uniform frequency-form comparison

$$\frac{998}{1000}\omega \le h_N \le \frac{1002}{1000}\omega, \qquad |\eta| \le 1.$$

This proves equivalence of these frequency form domains. It also bounds the static classical change of variables and its inverse on the original energy space, so instantaneous shell pullback has a finite energy-dual norm. It does not prove the required time derivative or a Fock implementer.

For clarity, the ordered static Bogoliubov factors on a common smooth core are

$$T_+ = h_N^{1/2} N^{-1/2} \omega^{-1/2}, \quad T_- = h_N^{-1/2} N^{1/2} \omega^{1/2},$$
$$\alpha_N = (T_+ + T_-)/2, \qquad \beta_N = (T_+ - T_-)/2.$$

A local Weyl expansion in a normal tensor frame gives the same order $|p|^{-1}$ term in T_+ and T_-; it cancels in β_N, whose formal symbol begins at order $|p|^{-2}$. This principal cancellation is not a global Hilbert–Schmidt proof: the remainder, massless infrared limit, conical origin, spatial infinity and all angular modes still require bounds. A gap-free trace-ideal target is explicit. If $AX + XA^T = B$ for positive injective A, then

$$\|X\|_{\mathrm{HS}} \le \tfrac{1}{2}\|A^{-1/2} B A^{-T/2}\|_{\mathrm{HS}}, \qquad \frac{ab}{(a+b)^2} \le \frac{1}{4}.$$

This identifies a weighted norm to calculate; its finiteness has not been assumed. The nonlinear static transformation requires its own estimate too.

If the static transformation admits a differentiable projective implementer $U_N(t)$, its time connection has number coefficient C_t and symmetric pair coefficient F_t. Sufficient form bounds to establish include

$$c_t = \|\omega^{-1/2} C_t \omega^{-1/2}\|, \qquad g_t = \|\omega^{-1/2} F_t \omega^{-T/2}\|,$$
$$\nu_t = \|\omega^{-1/2} F_t\|_{\mathrm{HS}}^2, \qquad E = \langle H_f\rangle,$$
$$|\langle H_{\mathrm{pair}}(F_t)\rangle| \le g_t E + \sqrt{\nu_t E}.$$

The vacuum contraction is included in the last inequality. No finite values for c_t, g_t, ν_t are assigned here. Their relative bounds must leave a positive margin below $998/1000$, with time regularity for the transformed non-L^2 shell form. At entry and release $N = 1$, so the static frame returns to the original reference in a fixed phase gauge. This route specifies the next global kernel calculation while retaining the open joint evolution.

27.5 The remaining joint quantum certificate

To complete the actual switched preparation, one must still establish the full symplectic implementability bounds, a nonautonomous propagation theorem for these boundary forms, and convergence with a state-specific remainder. The original $u_R \notin L^2$ but $u_R/\sqrt{k} \in L^2$; formal time differentiation need not preserve that weighted norm. Uniform lower bounds alone do not supply the old C^1 common-form theorem. A smooth-background theorem requiring C^∞ switching must use a stated

bump waveform and its actual areas instead of silently assuming the displayed C^3 protocol.

For the vacuum field channel, a verified higher-order bound $\|\omega^{1/2}(\beta_\eta - \eta\beta_1)\|_{\mathrm{HS}} \le C_2\eta^2$ would imply

$$|E_{\mathrm{spont}} - \eta^2\|\omega^{1/2}\beta_1\|_{\mathrm{HS}}^2| \le 2|\eta|^3 C_2\|\omega^{1/2}\beta_1\|_{\mathrm{HS}} + C_2^2\eta^4.$$

No finite C_2 is inferred from the first-Born term. For the actual joint state, a norm remainder r gives current error at most $\|Z\|r(2+r) = 4r + 2r^2$, then a phase disk on the original nonzero branch. First energy alone cannot imply a generic quadratic state Taylor bound: a fixed-mean-energy high-occupation probe has remainder of order $\sqrt{|\eta|}$. The actual hard branches may have stronger properties; their required correlations and propagation estimates remain the source- specific calculation. No Gaussian substitute, finite UV cutoff, or chosen energy reference completes that calculation automatically.

28. Changing profiles retain a common continuum domain

The complete positive receiving form also admits controlled time variation. Write $H_s = \varepsilon(eI + S)$ and retain the unchanged T_g, $K[\Psi] = \|T_g\Psi\|^2$ and $\mathscr{Q} = Q(H_f)$. Let $s(t)$ be real and C^1, and let the self-adjoint output weight be operator-norm C^1, with uniform bounds $0 < mI \le W(t) \le MI$. Set

$$q_t[\Phi,\Psi] = s(t)\langle\Phi, H_s\Psi\rangle + \langle T_g\Phi, W(t)T_g\Psi\rangle.$$

All forms are closed on $\mathscr{Q}$. With $C = \sup|s|\,\|H_s\|$, the shifted forms satisfy

$$\begin{aligned}\|\Psi\|^2 + mK[\Psi] &\le q_t[\Psi] + (C+1)\|\Psi\|^2 \\ &\le (2C+1)\|\Psi\|^2 + MK[\Psi].\end{aligned}$$

These uniformly equivalent, C^1 form inner products meet the common-form propagator hypotheses: a unique unitary evolution preserves $\mathscr{Q}$ and solves the equation weakly into its dual. The relevant existence statement is Assumption 3.8 and Theorem 3.10 of Balmaseda, Lonigro and Pérez-Pardo, *On the Schrödinger equation for time-dependent Hamiltonians with a constant form domain*, `https://arxiv.org/abs/2112.11063`. A time-independent operator domain is unnecessary.

The weak energy rule and a normalized preparation give

$$\begin{aligned}\dot{E} &= \dot{s}\langle H_s\rangle + \langle T_g\Psi, \dot{W}T_g\Psi\rangle, \\ K(t) &\le \frac{F(0)}{m}\exp\int_0^t\left(|\dot{s}|\|H_s\| + \frac{\|\dot{W}\|}{m}\right)\mathrm{d}u,\end{aligned}$$

where $F = E + C + 1 \ge 1 + mK$. This follows by bounding $|\dot{F}|$ by the displayed coefficient times F and integrating. Piecewise constant controls are obtained by concatenating their self-adjoint evolutions. At a control jump the state remains continuous and its signed work is

$$w_j = \Delta s_j\langle H_s\rangle_j + \langle T_g\Psi_j, \Delta W_j T_g\Psi_j\rangle.$$

Multiply the preceding energy bound by $1 + |\Delta s_j|\|H_s\| + \|\Delta W_j\|/m$ at each jump. Gross work retains the absolute positive and withdrawn pieces. Equal starting and ending Hamiltonians need not give equal mean energies, because the state may have changed.

The actual-current bound extends with this energy account. Compare to $H_{\mathrm{ref}}(t) = N(t)H_0$ on the same preparation, and suppose

$$|r_t[\Phi,\Psi]| \le \delta(t)\sqrt{K[\Phi]K[\Psi]} + b(t)\|\Phi\|\|\Psi\|.$$

Pairing the two weak equations yields

$$\|\Psi(t) - \chi(t)\|^2 \le 2\int_0^t [\delta\sqrt{K_A K_R} + b]\mathrm{d}u.$$

For $B_t(u) = U_{\mathrm{ref}}(t,u)^* Z U_{\mathrm{ref}}(t,u)$ the weak Heisenberg identity likewise retains its time-ordered form. If $K_A \le k_A$ and $K[B_t\Psi], K[B_t^*\Psi] \le k_B$, then

$$|z_A - z_R| \le 2\int_0^t \delta\sqrt{k_A k_B}\mathrm{d}u + 4\int_0^t b\mathrm{d}u.$$

The earlier k_B formula applies to lane-preserving uniform reference histories. A positive current amplitude converts this error into the same phase disk. Thus changing weights pay their work and their current remainder within one continuum account.

Part B. Paid control and actual event timing

29. One fully paid switching and reference account

A complete scalar example retains every original Hamiltonian term. Put $a = 10^{-3}$, hold $n = 1 - a$ for $h = 1/2$ and $n = 1 + a$ for another $h = 1/2$, with $n = 1$ before and after. The original unscaled energy E stays in its retained lane01 enclosure $0.917629340281369063 < E < 0.917632830384309734$. The state is exactly $\exp(-\mathrm{i}H_0\eta)\Psi_0$ and its mean energy is nE.

During these source waits choose the balanced reference $h_R = -2nEP_R$. Its mean energy is $-nE$ and its phase obeys $\dot{\beta} = -2nE$. It remains active throughout the wait. Entry and release are external linear ramps of duration $b = 1/100$, taking h_R from zero to $-2EP_R$ and back while $n = 1$. Explicitly, the entry coefficient is $-2Et/b$ and the release coefficient is $-2E(1 - t/b)$ on their own local intervals $0 \le t \le b$. Each ramp contributes phase area $-Eb$; any alternative waveform must preserve that area to preserve the stated final phase. They are outside ledger matching; the source evolves during both. The final original age is $2b + 2h = 51/50$ and the final reference phase is $-101E/50$. The source phase remains $\theta_0(51/50)$.

The opposite matched payments can be recovered internally in an ideal work store. Their cancellation does not erase either recipient or specify dissipation, waveform, state preparation, readout or counter-reset cost. This source-wait/coupling example complements Chapter II's original forward Write, reverse Write and wait protocol. The reference waveform prescribes

Table 3: Signed mean work in the original energy unit. Zero source waits retain nonzero source/reference energies.

Piece	Source	Reference
External entry ramp	0	$-E$
$n : 1 \to 1-a$	$-aE$	$+aE$
$n : 1-a \to 1+a$	$+2aE$	$-2aE$
$n : 1+a \to 1$	$-aE$	$+aE$
External release ramp	0	$+E$
Signed sum	0	0
Gross exchanged	$4aE$	$(2+4a)E$

the ideal energy history; using an approximate energy adds its calibration residual.

For a coarse reference winding record use prefixes spaced by at most $1/100$, including every control boundary. Here $|\dot{\beta}| < 2.002$. Split coordinate error $1/16$ into statistical $1/32$ and systematic $1/32$, retaining the separate preparation, unwanted evolution, observable and trigger contributions. This gives angular error below $1/8$, and $2.002/100 + 1/4 < \pi$ fixes each increment. At most 110 prefixes and 26624 fresh shots per coordinate give failure below 0.01, conditional on the systematic coordinate bounds, with at most 5,857,280 fresh reference preparations. This winding account is distinct from precise source-current tomography.

30. A moving two-way protocol with every event epoch

Use flat $1+1$ spacetime and $c = 1$. Endpoint A is at $x = 0$ and B follows $x_B = \lambda/10 + 3t/5$, so $\mathrm{d}\tau_A/\mathrm{d}t = 1$ and $\mathrm{d}\tau_B/\mathrm{d}t = 4/5$. Both actual interacting clocks begin at $t = 0$. Emit at $t_0 = \lambda/10$ and let B wait a proper duration $\lambda/10$ before reemission. The relay stores the incoming encoded phase, with its own specified transfer phase; it does not replace it by B's independent clock phase.

Table 4: Actual events for the moving two-way fixture, in units of λ. Internal ages assume the declared complete proper-time coupling.

Event	t/λ	x/λ	η_A/λ	η_B/λ
A emits	$1/10$	0	$1/10$	$2/25$
B receives	$1/2$	$2/5$	$1/2$	$2/5$
B reemits	$5/8$	$19/40$	$5/8$	$1/2$
A receives	$11/10$	0	$11/10$	$22/25$

The null endpoint equations yield

$$\begin{aligned} F_{\mathrm{out}}(t) &= \tfrac{5}{2}t + \tfrac{\lambda}{4}, \\ P(t) &= t + \tfrac{\lambda}{8}, \\ F_{\mathrm{ret}}(t) &= \tfrac{8}{5}t + \tfrac{\lambda}{10}, \\ G(t) &= (F_{\mathrm{ret}} \circ P \circ F_{\mathrm{out}})(t) \\ &= 4t + \tfrac{7\lambda}{10}. \end{aligned}$$

After accounting for transfer phase, the encoded rate relative to A's actual current at final reception is $\nu_0(t_0)/[4\nu_0(t_3)]$. The baseline remains nonstationary. For a variable remote proper rate, a fixed proper wait satisfies $\int_t^{P(t)} n_B(u)\mathrm{d}u = \Delta\tau_B$ and hence $P' = n_B(t)/n_B(P(t))$.

Choose $\lambda = 10T_0/11$, with the inherited $T_0 = 573049\pi$. Then final reception is at T_0, emission is at $T_0/11$, and every endpoint age in Table 4 lies in the original certified interval. This is a declared relativistic model fixture, with its own ruler/time conversion, rather than an assigned GPS trajectory.

31. Shift, Sagnac and the coordinate-invariant composition

A rotating coordinate system has a nonzero shift. Write it with velocity units as

$$\mathrm{d}s^2 = -N^2c^2\mathrm{d}t^2 + h_{ij}(\mathrm{d}x^i + \beta^i\mathrm{d}t)(\mathrm{d}x^j + \beta^j\mathrm{d}t).$$

A worldline with coordinate velocity v has proper rate $n = [N^2 - h(v+\beta, v+\beta)/c^2]^{1/2}$. In flat spacetime the change $X = R(t)x$ gives $N = 1$, $h = I$ and $\beta = \Omega \times x$. Thus a coordinate-stationary clock has the expected rotational proper rate, and the time–space cross term also enters signal propagation.

For $A = N^2c^2 - h(\beta,\beta) > 0$, the future null root along a stationary spatial path is

$$\mathrm{d}t = \frac{h(\beta,\mathrm{d}x) + \sqrt{h(\beta,\mathrm{d}x)^2 + Ah(\mathrm{d}x,\mathrm{d}x)}}{A}.$$

Opposite traversals of the same fixed loop differ by $2\oint h(\beta,\mathrm{d}x)/A$. A guided circular ring in rotating flat coordinates consequently gives

$$\begin{aligned} T_\pm &= \frac{2\pi R}{c \mp \Omega R}, \\ \Delta T &= \frac{4\pi\Omega R^2}{c^2 - \Omega^2R^2}, \\ \Delta\tau &= \frac{4\pi\Omega R^2}{c^2\sqrt{1 - \Omega^2R^2/c^2}}. \end{aligned}$$

The leading coordinate difference is $4\boldsymbol{\Omega}\cdot\mathscr{A}/c^2$. These exact ring expressions retain guided paths; free endpoint-to-endpoint rays require their specified routes.

For an increasing rechart $t' = h(t)$, the event map becomes $\widetilde{F} = h \circ F \circ h^{-1}$, its derivative is $h'(t_R)F'/h'(t_E)$, and $\widetilde{n} = n/h'$. Therefore

$$\frac{\widetilde{n}_E}{\widetilde{n}_R\widetilde{F}'} = \frac{n_E}{n_RF'}.$$

The endpoint internal ages and the actual ν_0 ratio are unchanged. The composed observable is coordinate invariant, including its proper-wait map. This fills the rotating-coordinate propagation interface without deriving a microscopic boost or metric law from the lapse alone.

32. A nanoradian phase target has its own measurement budget

For the inherited Earth-phase comparison at T_0, specify two histories and five prefixes $jT_0/4$, $j = 0,\dots,4$. This differential calibration is separate from the flat moving fixture above: its approximately -4.374 nrad prediction is not assigned to that fixture. Use fresh source–field preparations for each coordinate and prefix, with the signed lane01 receivers D/d_{01} and C/d_{01}. Extend them by zero outside the measured lane so their full-space norms remain one and leakage outcomes are flagged. The retained coarse current amplitude is $m_z = 0.99$.

The original sufficient design takes statistical coordinate tolerance $d = 5\times10^{-11}$ and $N = 7\times10^{21}$ independent bounded shots per mean. Twenty coordinate means need at most 1.4×10^{23} fresh source–field preparations. Hoeffding's inequality and a union bound give

$$\begin{aligned} P(\text{any statistical failure}) &\le 40e^{-Nd^2/2} \\ &= 40e^{-35/4} < 0.01, \\ |\Delta\theta|_{\text{stat}} &\le 2\arcsin(\sqrt{2}d/m_z) < 0.2 \text{ nrad}. \end{aligned}$$

These are transparent sufficient counts. The coarse-winding/fine-final refinement below reduces their preparation cost while retaining this systematic ledger; neither design is achieved apparatus performance or an optimal metrological lower bound.

Table 5: Specified allowances on the final phase difference. Each condition requires a source or apparatus input; none is recorded as apparatus-achieved.

Contribution	nrad	Sufficient condition
Complete-form remainder	0.20	The two current-error disks from the changing-weight bound fit this allowance.
Within-sector preparation	0.10	Trace distance $\le 10^{-11}$ per arm; retain a separate K-energy ceiling.
Leakage	0.10	Generic coherent probability $\le 10^{-22}$; incoherent probability $\le 10^{-11}$ suffices.
Effective readout	0.10	Full observable norm error $\le 10^{-11}$ per coordinate.
Internal-age timing	0.10	Per-arm age error $\le 10^{-5}$ model units, using $u < 3.5\times10^{-6}$.
Carrier/relay phase	0.10	Residual phase $\le 5\times10^{-11}$ rad per arm.
Reference and winding	0.10	Residual branch/reference phase $\le 5\times10^{-11}$ rad per arm.
Statistical estimates	0.20	The twenty means and shot count displayed above.
Total	1.00	Conditional on every source-model and apparatus row.

A trace-distance condition controls bounded coordinates, not unbounded energy; rare high-energy leakage must still satisfy the preparation energy account. The generic coherent leakage allowance uses its square-root trace-distance bound. If the complete dynamics preserves the sector and the receiver is block diagonal, cross-sector receiver matrix elements vanish, and the stronger coordinate bound $2p$ allows $p \le 10^{-11}$ even for coherent leakage. This is an available apparatus refinement, not an additional requirement.

The original rate certificate gives $uT_0/4 < 1.58$ rad. Prefix errors leave a strict margin below π, so continuation from $\theta(0) = 0$ fixes the winding; an external record stores that integer. Readout disturbs each copy. Source preparation, phase reference, timing, readout, transfer, work recovery and storage/reset all retain recipients. The 1 nrad allowance leaves a sign margin against the uniform prediction only when the actual physical form remainder and every apparatus condition fit the ledger. The calculations here specify those conditions and perform no experimental shots.

33. Laboratory calibration and the endpoint assignment

Model time becomes a measured duration only after a common energy and action calibration. With physical energy $e_u\mathscr{H}_0$ and action unit I_u, write

$$s = \frac{e_u}{I_u}t_{\text{phys}}, \qquad \omega_{\text{phys}} = \frac{e_u}{I_u}\nu.$$

This must agree with Chapter VII's chosen t_u chart; it is not a new independent adjustment of the six lanes. The physical source size and spatial localization need their common ruler as well. A dimensionless phase prediction survives a change of units, but a duration, frequency, mass assignment or apparatus tolerance requires these conversions. [1,3]

Chapter III fixes the receiving field's isolated normalization by $a = 2GM/(c^2r)$ and distinguishes the source-only prescription $N = \sqrt{1-a}$ from the background-normalized raw prescription $\sqrt{1-a/(1-A_0)}$. They have different first-order slopes when the Poisson normalization is held fixed. This chapter adopts the source-only prescription conditionally; normalization cannot silently supply its physical derivation. [3]

The endpoint statement is consequently precise: if the complete interacting clock receives one common lapse, its entire actual history follows the associated internal age. Applying that statement universally to laboratory clock species, to moving clocks, and to a navigation time coordinate requires the corresponding physical coupling and calibration. The following endpoint mechanics can state those assignments and compare their consequences with GPS rates. These comparisons retain the remaining physical spatial-incidence calculation, the absolute energy reference and the demonstrated measurement budget as separate inputs.

34. Motion requires a spatial metric

A stationary lapse is not yet a moving-clock law. The clock's worldline samples both temporal and spatial geometry. To make

that extra input explicit, assume a static, zero-shift metric

$$\mathrm{d}s^2 = -N^2c^2\mathrm{d}t^2 + g_{ij}\mathrm{d}x^i\mathrm{d}x^j.$$

For coordinate velocity $v^i = \mathrm{d}x^i/\mathrm{d}t$, a timelike trajectory satisfies

$$\left(\frac{\mathrm{d}\tau}{\mathrm{d}t}\right)^2 = N^2 - \frac{g_{ij}v^iv^j}{c^2}.$$

The speed measured by a local static observer is instead

$$v_{\mathrm{loc}}^2 = \frac{g_{ij}v^iv^j}{N^2}, \qquad \frac{\mathrm{d}\tau}{\mathrm{d}t} = N\sqrt{1 - v_{\mathrm{loc}}^2/c^2}.$$

This product is exact for the stated metric. Replacing v_{loc} by a coordinate speed generally destroys that exactness.

Chapter III supplies a conditional weak metric interface, rather than a complete derivation of spacetime geometry. [3] Write it as

$$N^2 = 1 - a + O(a^2), \qquad g_{ij} = [1 + \gamma a + O(a^2)]\delta_{ij}.$$

With $b = |\boldsymbol{v}|^2/c^2$ and a, b small, its first-order clock law is

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = 1 - \frac{a}{2} - \frac{b}{2} + O(a^2, ab, b^2) = 1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2} + O(\varepsilon^2).$$

Here $\Phi = -GM/r$ and $a, b = O(\varepsilon)$. This is the familiar weak-field, slow-motion result used in the GPS comparison. [4] Agreement at this order does not determine γ or the omitted metric coefficients.

One counterexample makes the coordinate-speed caution concrete. Treat the deliberately truncated expressions $N^2 = 1 - a$ and $g_{ij} = (1 + \gamma a)\delta_{ij}$ as an exact comparison metric. Its squared rate is $1 - a - (1 + \gamma a)b$, whereas the squared coordinate-speed product is $(1 - a)(1 - b)$. The rates consequently differ by

$$\sqrt{1-a}\sqrt{1-b} - \sqrt{1 - a - (1+\gamma a)b} = \frac{\gamma + 1}{2}ab + O(\varepsilon^3).$$

This diagnoses a composition error; it is not a second-order prediction of the original weak metric. Nor does multiplying two static endpoint factors derive a boost law for the interacting Hamiltonian. The preceding time-change construction applies along a moving trajectory only after its physical lapse history has been supplied. With that complete coupling and a common calibration, the composition is explicit:

$$\eta(t) = \frac{e_u}{I_u}[\tau(t) - \tau(t_0)],$$
$$\theta_l(t) = \theta_{l,0}(\eta(t)), \qquad \frac{\mathrm{d}\theta_l}{\mathrm{d}t} = \frac{e_u}{I_u}\frac{\mathrm{d}\tau}{\mathrm{d}t}\nu_{l,0}(\eta(t)).$$

The inherited phase bounds apply only while η remains in their certified window. An SI day cannot be identified with one certified bare cycle without the common calibration.

35. The rotating reference and coordinate time

A satellite is compared with a specified terrestrial reference, not with an unnamed clock "on Earth." Use an Earth-centered inertial coordinate system (ECI), a negative potential Φ, and initially an unnormalized coordinate time t. A clock fixed to the rotating Earth has ECI velocity $\boldsymbol{\Omega} \times \boldsymbol{r}$. The effective reference potential is

$$\Phi_0 = \Phi_{\mathrm{ground}} - \tfrac{1}{2}|\boldsymbol{\Omega} \times \boldsymbol{r}_{\mathrm{ground}}|^2.$$

On the ideal geoid this combination is constant. Normalize coordinate time to that reference rate by

$$\frac{\mathrm{d}t_{\mathrm{geo}}}{\mathrm{d}t} = 1 + \frac{\Phi_0}{c^2} + O(\varepsilon^2).$$

Dividing the satellite's rate by this reference gives

$$\frac{\mathrm{d}\tau}{\mathrm{d}t_{\mathrm{geo}}} - 1 = \frac{\Phi - \Phi_0 - v_{\mathrm{ECI}}^2/2}{c^2} + O(\varepsilon^2). \tag{3}$$

The reference subtraction and its sign follow from the ratio. Equivalently, with positive potentials $U = -\Phi$ and $W_0 = -\Phi_0$, the numerator is $W_0 - U - v^2/2$. A stationary geoid clock has zero excess by construction. [4]

For two endpoints compared at common coordinate time, Φ_0 cancels from their first-order rate difference. It remains essential when either rate is reported relative to the chosen time scale. A spherical surface with arbitrary radius and latitude is not automatically the geoid: an actual reference also includes Earth's nonspherical gravitational field. Thus the spherical arithmetic below is a controlled illustration, not a substitute geoid model.

Two appearances of rotation must be kept separate. The ground clock's ECI speed contributes to its rate. Rotation also affects a transmitted signal's travel-time comparison when emission and reception are described in Earth-fixed coordinates. That latter propagation treatment includes the Sagnac effect. Adding the correct propagation treatment does not duplicate the ground-rate contribution; adding another ground-rate term after it is already included in Φ_0 does. The propagation sections below supply the route account. An observed signal-frequency ratio additionally requires propagation, Doppler and instrument modeling before it becomes an endpoint-rate ratio.

36. Endpoint comparisons: GPS and optical height

The shared first-order rate is $1 + \Phi/c^2 - v^2/(2c^2)$. Its observable value is a ratio to a specified ground reference, evaluated along a specified trajectory. For a circular satellite and a spherical equatorial ground reference,

$$\delta_{\mathrm{circ}} = \frac{\mu/R - 3\mu/(2r) + \Omega^2R^2/2}{c^2}, \qquad \mu = GM.$$

With the retained operands $R = 6378137$ m, $r = 26560000$ m, $\mu = 3.986004418 \quad 10^{14}\,\mathrm{m^3s^{-2}}$, $\Omega = 7.292115 \quad 10^{-5}\,\mathrm{s^{-1}}$

and exact SI c, the gravitational and satellite-motion parts are $+45.650911342$ and $-7.213599823\,\mu$s/day. Ground rotation adds $+0.103976943\,\mu$s/day, giving $+38.541288463\,\mu$s/day. This frozen spherical illustration and the nominal GPS convention use different reference models; Appendix A retains their individual operands and correction signs. [4–6]

For an ellipse with semimajor axis A, eccentricity e and eccentric anomaly E, the periodic unsteered clock offset is

$$\delta\tau_{\mathrm{per}} = -\frac{2\boldsymbol{r}\cdot\boldsymbol{v}}{c^2} = -\frac{2e\sqrt{\mu A}}{c^2}\sin E.$$

A finite observation uses its difference at the two interval endpoints. The sign reverses when an offset in the clock reading is converted into a correction toward coordinate time. This distinction also matters when comparing received timestamps that have already been corrected. [4,5]

A nearly stationary height comparison isolates another channel. For upward height h, its exact static endpoint ratio is

$$q_h = \sqrt{\frac{1-a(r+h)}{1-a(r)}} - 1 = \frac{gh}{c^2} + O(h^2/r^2, a^2),$$

where the leading laboratory quantity is the local geopotential difference. The retained millimetre optical comparison is described with its published gradient uncertainty in Appendix B; its synchronous comparison uncertainty is a separate number. [7,8] Neither an SI day nor a millimetre test extends the inherited model-time clock certificate without the common energy, action and ruler calibration.

37. The Magic Bell locates the missing carrier

The Unified Technical Synthesis places a distant mirror clock on Epsilon+ and an observer on Alpha+. A stipulated magic bell rings on Alpha+ at every remote mirror strike, while a telescope receives light from Epsilon+. [9] The bell supplies knowledge without a carrier or synchronization protocol; it therefore supplies no physical emission–reception map. The telescope displays radiation emitted at earlier events along actual routes. Matching a bell ring to an optical image requires exactly the map omitted by the stipulation.

This distinction can be carried into an ordinary weak-field experiment. An occurrence is a source event, emission encodes it into a signal, reception is the detector interaction, and recording assigns a local timestamp or stored phase. Encoding and detector latencies, route identity, polarization, dispersion and any phase offset are apparatus inputs. A near-horizon setting may motivate the question, but the weak-field route calculated here makes no global horizon or no-crossing claim.

38. Emission, reception and recorded phase

Choose a coordinate-time chart and an identifiable causal route branch. For successive emissions let

$$t_R = F(t_E) = t_E + T(t_E), \qquad F'(t_E) > 0.$$

This is a relation between events, not a claim that separated observers directly read the same coordinate clock. The finite spacing identity is

$$\Delta t_R = \Delta t_E + T(t_{E,2}) - T(t_{E,1}).$$

A fixed added delay shifts the sequence while leaving its spacing unchanged. A changing delay changes the spacing. Under ideal nondispersive phase encoding with known transfer phase,

$$\theta_{\mathrm{rec}}(t_R) = \theta_E(F^{-1}(t_R)), \qquad \frac{\mathrm{d}\theta_{\mathrm{rec}}}{\mathrm{d}t_R} = \frac{\dot\theta_E(t_E)}{F'(t_E)}.$$

A nonconstant transfer phase or detector latency must be composed with this map rather than absorbed into an endpoint lapse.

39. The interacting history survives transmission

Use physical seconds and the same conversion e_u/I_u at both endpoints. For the exact uniform-lapse control,

$$\eta_j(t) = \frac{e_u}{I_u}\int_0^t N_j(u)\,\mathrm{d}u, \qquad \dot\theta_j(t) = \frac{e_u}{I_u}N_j(t)\nu_0(\eta_j(t)).$$

Dividing the received encoded phase rate by the receiver's own phase rate gives the composed observable

$$\mathscr{R}(t_R) = \frac{N_E(t_E)}{N_R(t_R)F'(t_E)}\,\frac{\nu_0(\eta_E(t_E))}{\nu_0(\eta_R(t_R))}. \tag{4}$$

The actual baseline ratio remains because the interacting clock is slightly nonstationary. Matched internal age or a stationary reference removes it; arbitrary reception compares different events and generally retains it. All relevant ages must remain in the certified source window when its quantitative bounds are invoked.

For static endpoints and a stationary route $F' = 1$. The received frequency per receiver proper time divided by the local emission frequency per emitter proper time is N_E/N_R. Comparing instead with an independently evolving receiver clock retains the last factor in Eq. (4). Thus static redshift can coexist with a constant route delay, while two routes can have different delays between the same endpoints. Motion and changing gravitational geometry enter through F' as well as the endpoint histories.

The distinction between complete scaling and a change of source parameter also survives transmission. Changing only ε leaves $(1-N)H_f + (N^2-N)H_{\mathrm{ct}}$ relative to the uniform control. A propagation map does not remove that residual.

40. The optical receiving law

A path integral needs a propagation law. Assume a static metric with no time–space cross term,

$$\mathrm{d}s^2 = -N(x)^2c^2\mathrm{d}t^2 + h_{ij}(x)\mathrm{d}x^i\mathrm{d}x^j.$$

Setting the interval to zero gives the coordinate travel-time functional

$$T[\Gamma] = \frac{1}{c}\int_\Gamma \frac{\sqrt{h_{ij}\mathrm{d}x^i\mathrm{d}x^j}}{N}.$$

The optical spatial metric is h_{ij}/N^2. This displays the two physical inputs that the endpoint lapse alone cannot provide: the spatial geometry and the route selected by propagation. Along the ray, a local static observer uses proper length $\mathrm{d}\ell_{\mathrm{prop}} = \sqrt{h_{ij}\mathrm{d}x^i\mathrm{d}x^j}$ and proper time $\mathrm{d}\tau = N\mathrm{d}t$, and therefore measures $\mathrm{d}\ell_{\mathrm{prop}}/\mathrm{d}\tau = c$.

For the declared weak spherical metric in isotropic coordinates, write

$$N^2 = 1 - a + O(a^2),$$
$$h_{ij} = [1 + \gamma a + O(a^2)]\delta_{ij}, \qquad a = \frac{2GM}{c^2 r}.$$

Its coordinate optical index is

$$n_{\mathrm{opt}} = \frac{\sqrt{1+\gamma a + O(a^2)}}{\sqrt{1 - a + O(a^2)}} = 1 + \frac{1+\gamma}{2}a + O(a^2).$$

The first-order excess is consequently

$$\Delta T_1 = \frac{1+\gamma}{2c}\int_{\Gamma_0} a\,\mathrm{d}\ell.$$

At $\gamma = 1$ this is the retained SAM road operator $c^{-1}\int a\,\mathrm{d}\ell$. Its temporal half and spatial half are distinct terms of the declared metric. [3,9,12] The Poisson source profile does not by itself determine γ. Nor does identifying a two-polarization interacting field in Chapter VII supply an electromagnetic propagation action without a further bridge.

The shorthand coordinate speed $c(1-a)$ is the first-order reciprocal-index expression at $\gamma = 1$. It is not an exact rule at $a = 1$, a local reduction of the invariant speed, or a substitute for a full null-geodesic calculation.

41. Why the first-order integral uses a straight segment

At zero field the travel-time functional is Euclidean length divided by c. With fixed spatial endpoints, its stationary path is the straight segment Γ_0. For a smooth weak perturbation, write the path as $x(\ell) = x_0(\ell) + \delta x(\ell)$, with δx zero at both ends. The first variation of the flat length is

$$\delta L = \int_0^L \widehat{\mathbf{k}}\cdot\delta x'(\ell)\,\mathrm{d}\ell = \widehat{\mathbf{k}}\cdot[\delta x(L) - \delta x(0)] = 0.$$

Here $\widehat{\mathbf{k}}$ is the constant segment direction. The first-order metric term may therefore be evaluated along Γ_0; its path variation contributes at the next order. This is the local perturbative content of Fermat's principle behind the first-post-Minkowskian transfer formula. [12]

The argument assumes a regular, identifiable ray branch perturbatively connected to that segment, outside the central body's support. It does not provide a uniform error estimate near a caustic, a singularity, an occultation boundary or a regime with enhanced higher-order terms. Even a small local a is not alone a proof that a long-baseline lensing problem is globally perturbative.

Fixed endpoints also matter conceptually. There is only one Euclidean straight segment between a given pair. Different physical routes between the same endpoints require a specified guide, reflection, medium or multiple-image geometry; one cannot manufacture an additional ray merely by choosing another integration curve. The impact sweep below instead keeps the endpoint radii fixed while changing their angular placement.

42. The finite-endpoint logarithm

Let the endpoint vectors be $\mathbf{x}_A, \mathbf{x}_B$, with radii r_A, r_B, separation $L = |\mathbf{x}_B - \mathbf{x}_A|$ and unit direction $\widehat{\mathbf{k}} = (\mathbf{x}_B - \mathbf{x}_A)/L$. Define

$$z_A = \widehat{\mathbf{k}}\cdot\mathbf{x}_A, \quad z_B = z_A + L, \quad b^2 = r_A^2 - z_A^2.$$

For $b > 0$, the line's distance from the center is $r(z) = \sqrt{b^2 + z^2}$. Direct integration gives

$$\begin{aligned} I &= \int_0^L \frac{\mathrm{d}\ell}{|\mathbf{x}_A + \ell\widehat{\mathbf{k}}|} \\ &= \operatorname{asinh}(z_B/b) - \operatorname{asinh}(z_A/b) \\ &= \ln\frac{r_A + r_B + L}{r_A + r_B - L}. \end{aligned}$$

To verify the final form, use $r_A^2 - z_A^2 = r_B^2 - z_B^2 = b^2$ and $z_B - z_A = L$ in the logarithmic definition of asinh. The triangle denominator is positive for a nondegenerate segment that does not pass through the center. A same-side radial segment with $b = 0$ has a finite limiting logarithm; a through-center point-source route is singular and is excluded.

Thus the one-way first-order gravitational delay is

$$\Delta T_1 = \frac{(1+\gamma)GM}{c^3}\ln\frac{r_A + r_B + L}{r_A + r_B - L}. \tag{5}$$

The argument is dimensionless, and the coefficient has units of time. These are useful controls against a missing factor of c or a source length being mistaken for a duration.

For opposite-side endpoints, $z_A = -\sqrt{r_A^2 - b^2}$ and $z_B = +\sqrt{r_B^2 - b^2}$. When $b \ll r_A, r_B$,

$$I = \ln\frac{4r_A r_B}{b^2} - \frac{b^2}{4}\left(\frac{1}{r_A^2} + \frac{1}{r_B^2}\right) + O\left(\frac{b^4}{r_A^4} + \frac{b^4}{r_B^4}\right).$$

The familiar far-endpoint formula is therefore an additional geometric approximation, not the exact finite logarithm. Its error can be evaluated independently of any higher-order gravitational metric correction.

43. Round trips and what a receiver records

A one-way coordinate excess is not automatically a radar observable. For stationary endpoints and an ideal immediate reflection in the same static geometry, the outward and return integrals are equal, so $\Delta T_{\mathrm{round}} = 2\Delta T_1$. A stationary receiving clock with lapse N_A assigns the corresponding proper-time excess $N_A\Delta T_{\mathrm{round}}$ when the reference subtraction uses that same

clock and baseline. Multiplication of the first-order excess by $N_A - 1$ enters at the next gravitational order; the complete baseline and reference convention must still be retained. [11,13]

With motion or a nonzero turnaround delay, use the actual event maps. If $F_{\rm up}$ and $F_{\rm down}$ include the declared turnaround convention, the round-trip map is $F_{\rm rt} = F_{\rm down} \circ F_{\rm up}$ and

$$F'_{\rm rt}(t) = F'_{\rm down}(F_{\rm up}(t))F'_{\rm up}(t).$$

Different leg geometries are evaluated at their different events; doubling a frozen one-way number is then only an approximation.

An ideal static coherent transponder supplies a useful separation. If its local turnaround frequency factor is k, the outward lapse ratio is N_A/N_B and the return ratio is N_B/N_A. Their product is one, leaving the received-to-emitted ratio k while a nonzero round-trip delay remains in ranging. Constant delay and constant gravitational redshift cannot be interchanged to explain that observation. Moving coherent links retain the derivatives of the event maps, alongside their endpoint rates and hardware factors.

The same map also contains ordinary Doppler. For a rest emitter and a receiver receding at speed v in flat space, $F' = 1/(1-\beta)$, $\beta = v/c$, while the receiver proper-time rate is $\sqrt{1-\beta^2}$. Hence

$$\frac{\omega_R}{\omega_E} = \frac{1-\beta}{\sqrt{1-\beta^2}} = \sqrt{\frac{1-\beta}{1+\beta}}.$$

This exact flat-space check shows why the travel-time derivative must not be dropped or counted again as a second endpoint correction.

44. The Cassini observable is a changing delay

Bertotti, Iess and Tortora reported $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ from Cassini tracking, with the quoted uncertainty at one standard deviation. [14] The observable was a frequency shift associated with a changing round-trip light time. Multiple radio frequencies were used to separate dispersive solar-plasma effects; dynamics, propagation and instrumental modeling were also part of the analysis. A static integral reproducing an engineering-scale delay is not a reconstruction of that fit.

Define y here as received-minus-emitted fractional frequency after the chosen flat-range and endpoint model has been removed. Let T_0 be the baseline travel-time model. With the same endpoint factors, the residual frequency ratio is exactly

$$y = \frac{1+T_0'}{1+T_0'+\Delta T'} - 1 = -\frac{\Delta T'}{1+T_0'+\Delta T'}.$$

Primes here mean emission-time derivatives. Only when the additional derivative and the retained baseline derivative are small does this reduce to $y \simeq -\Delta T'$. In the far-endpoint, frozen-radius approximation,

$$\Delta T_{\rm round} \simeq \frac{2(1+\gamma)GM}{c^3} \ln \frac{4r_A r_B}{b(t)^2},$$

so

$$\frac{{\rm d}\Delta T_{\rm round}}{{\rm d}t} \simeq -\frac{4(1+\gamma)GM}{c^3}\frac{\dot b}{b}.$$

The corresponding y has the opposite sign. Some timing sources instead define y as the delay derivative itself; the convention must be declared before comparing signs. Changing endpoint radii contributes additional derivatives, and a substantial baseline Doppler factor must be retained when forming the residual ratio.

The exact first-order opposite-side derivative at fixed radii is

$$\frac{\partial \Delta T_1}{\partial b} = -\frac{(1+\gamma)GM}{c^3 b}\left[\frac{r_A}{\sqrt{r_A^2 - b^2}} + \frac{r_B}{\sqrt{r_B^2 - b^2}}\right].$$

This supplies a finite-geometry alternative to the far-endpoint derivative. A real tracking reduction must use its actual trajectory, both leg epochs, transponder model and covariance; the declared static shape sweep is not such a dataset. The published Cassini result supports the conventional weak metric within its stated analysis. The present source packet is a compatible recovery, not a new measurement or a unique discrimination against general relativity.

45. A shared source, two weak response coefficients

The temporal and spatial responses can be parameterized independently before a source-native derivation is complete. Write

$$N^2 = 1 - \kappa_t a + O(a^2), \qquad h_{ij} = [1 + \kappa_s a + O(a^2)]\delta_{ij}.$$

Direct expansion gives

$$N = 1 - \frac{\kappa_t}{2}a + O(a^2), \qquad n_{\rm opt} = 1 + \frac{\kappa_t + \kappa_s}{2}a + O(a^2).$$

An endpoint comparison therefore measures the temporal coefficient, while a monopole timing route measures their sum. With one common source normalization, they imply $\kappa_s = 2K_{\rm road} - 2K_{\rm end}$, where $K_{\rm end} = \kappa_t/2$ and $K_{\rm road} = (\kappa_t + \kappa_s)/2$ are the coefficients of the same lift. In the declared metric used above, $\kappa_t = 1$ and $\kappa_s = \gamma$.

This algebra makes a useful experimental and theoretical pairing: derive or measure the endpoint and route responses without independently rescaling their source amplitude. The Poisson kernel fixes the radial lift profile; it does not by itself choose either response coefficient. A spatial coefficient derived from the same retained action would supply the missing physical bridge.

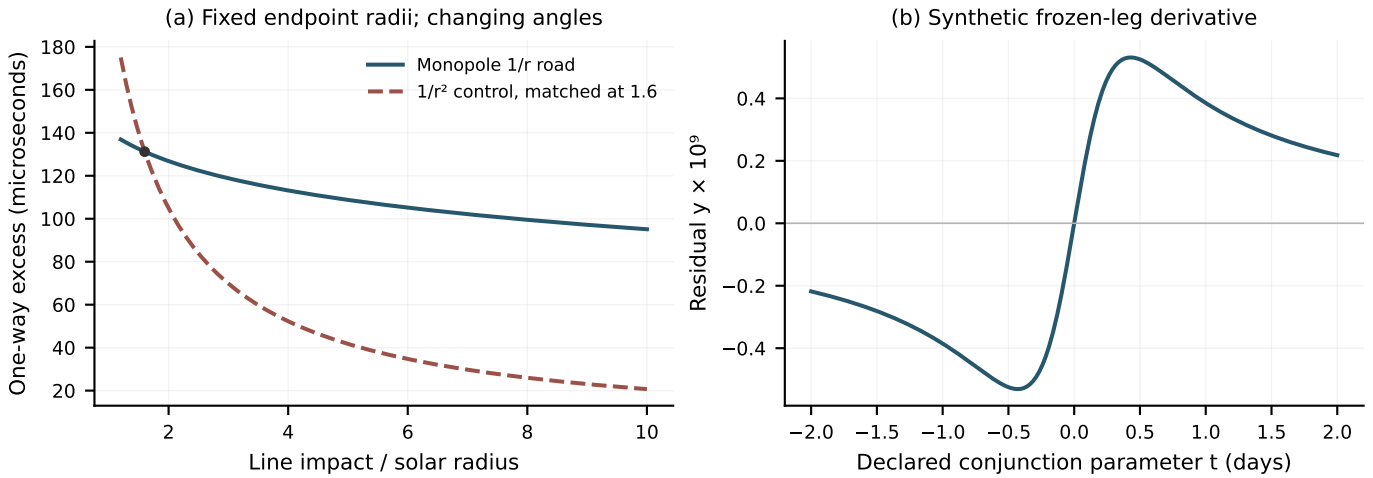


Figure 1: Declared controls, not tracking data. (a) The monopole road and an inverse-square control normalized at $b = 1.6R_\odot$. Endpoint radii stay fixed; endpoint angles change. (b) A synthetic first-order residual $y = -2(\partial\Delta T_1/\partial b)\dot{b}$, using the frozen equal-leg approximation and $b(t) = \sqrt{b_0^2 + v^2t^2}$ with $b_0 = 1.6R_\odot$ and $v = 30$ km/s. The central delay is largest when its derivative vanishes. This parameterized curve is not a moving-endpoint two-way solution or a fit to Cassini ephemerides; actual leg epochs and baseline Doppler must be supplied for that task.

Part C. The emitted carrier and amplitude transport

46. A propagation action connects phase and amplitude

A concrete declared electromagnetic adapter is the minimally coupled action

$$S_{\rm EM} = -\frac{1}{4}\int \sqrt{-g}\, g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}\, \mathrm{d}^4x.$$

Its geometric-optics principal symbol gives $g^{\mu\nu}k_\mu k_\nu = 0$ and the static optical metric h_{ij}/N^2. A local orthonormal observer measures c. A positive scalar multiplier $Z(x)$ leaves this cone unchanged while changing the leading flux equation to $\nabla_\mu(ZA^2k^\mu) = 0$. Phase transport therefore does not fix amplitude transport.

For example, all members of the declared family

$$\mathrm{d}s_\gamma^2 = -(1-a)c^2\mathrm{d}t^2 + (1+\gamma a)|\mathrm{d}\boldsymbol{x}|^2$$

share the same stationary clock lapse where $0 \le a < 1$ and $1 + \gamma a > 0$. Their route indices are $\sqrt{(1+\gamma a)/(1-a)}$, and their first-order Shapiro coefficients differ. The cases $\gamma = 0$ and $\gamma = 1$ give respectively half and all of the retained road operator. This is a constructive separation of temporal and spatial information; it does not assert that all members solve one source field equation.

47. A positive tensor action supplies an amplitude bridge

Use $b_{\cos}(\eta_c)$ for the cosmological scale factor, normalized to one today, and $\mathscr{H} = b_{\cos}'/b_{\cos}$. The conformal time η_c is distinct from the interacting clock's model-age variable. In units $c = 1$, declare the two-polarization action

$$S_T = \frac{1}{2}\sum_A \int \mathrm{d}\eta_c\, \mathrm{d}^3x\, b_{\cos}^2 Q\left[(h_A')^2 - c_T^2|\nabla h_A|^2\right],$$
$$Q > 0, \qquad c_T^2 > 0.$$

This propagation construction is developed in campaign voli_chapter4_observation1_20261016 (GEN6 Duck run chapter4-observation-20261016-01, Duck 1.2.3, core 0.10.2), following the amplitude-friction representation retained in Ref. [19]; Q and c_T are not assigned as source-native SAM coefficients. Its momentum and Hamiltonian are

$$\Pi_A = b_{\cos}^2 Q h_A', \quad H_T = \frac{1}{2}\sum_A \int \left[\frac{\Pi_A^2}{b_{\cos}^2 Q} + b_{\cos}^2 Q c_T^2 |\nabla h_A|^2\right] \mathrm{d}^3x.$$

The instantaneous energy is nonnegative. Prescribed changes of the background coefficients supply external work; this is not a conserved total apparatus account.

Varying the action and Fourier transforming gives

$$h_A'' + (2\mathscr{H} + Q'/Q)h_A' + c_T^2k^2h_A = 0.$$

Set $B = b_{\cos}\sqrt{Q}$ and $v_A = Bh_A$. Direct substitution cancels the first derivative and yields the exact equation

$$v_A'' + \left[c_T^2k^2 - \frac{B''}{B}\right] v_A = 0.$$

For $c_T = 1$, a smooth high-frequency regime gives the leading amplitude $h_A \propto 1/(b_{\cos}\sqrt{Q})$ with the radial spreading factor. A quantitative use needs control of $|B''/B|/k^2$ and its variation on the chosen branch, with absorption and mode conversion included if present.

48. Source, transport, detector and inferred distance

If source generation and local detector normalization are matched between the comparison channels, the preceding transport amplitude gives

$$\Xi(z)=\frac{d_L^{\mathrm{GW}}}{d_L^{\mathrm{EM}}}=\sqrt{\frac{Q(0)}{Q(z)}},$$
$$\delta(s)=-\frac{\mathrm{d}\ln\Xi}{\mathrm{d}s}=\frac{1}{2}\frac{\mathrm{d}\ln Q}{\mathrm{d}s},\qquad s=\ln(1+z).$$

Here $Q(0)$ is the receiver value and $Q(z)$ the emitter value. Since $\mathrm{d}s/\mathrm{d}\eta_c=-\mathscr{H}$, the mode equation is $h_A''+2\mathscr{H}(1-\delta)h_A'+k^2h_A=0$, the amplitude-friction representation used in the finite-redshift paper. [19] The amplitude factor is distinct from the principal phase speed c_T and from a travel-time delay.

A source-generation or detector factor $G(z)$ changes the measured ratio to $Y(z)=C_{\mathrm{cal}}G(z)\Xi(z)$. The simpler $Y=C_{\mathrm{cal}}\Xi$ protocol requires G to be controlled or fitted explicitly. Changing c_T also changes the propagation problem; the displayed distance formula retains $c_T=1$ and the stated matching assumptions.

Conversely every smooth positive normalized Ξ has a compatible positive coefficient

$$Q(z)=\frac{Q(0)}{\Xi(z)^2}.$$

For $\Xi=\Xi_\infty+(1-\Xi_\infty)e^{-ps}$ with $p>0$ and $\Xi_\infty>0$, this gives

$$\delta(s)=\frac{p(1-\Xi_\infty)e^{-ps}}{\Xi_\infty+(1-\Xi_\infty)e^{-ps}}.$$

Positivity of this declared action therefore selects neither a single power nor its exponent. An independently derived Q, p, or rate spectrum is the physical result needed from the common source dynamics.

The practical ratio additionally retains emission properties, inclination and polarization response, background distance, detector gain, opacity and selection. Independent angular-distance information can help separate channels: with d_A, define $\eta_{\mathrm{EM}}=d_L^{\mathrm{EM}}/[(1+z)^2d_A]$ and $\eta_{\mathrm{GW}}=d_L^{\mathrm{GW}}/[(1+z)^2d_A]$. Then $\Xi=\eta_{\mathrm{GW}}/\eta_{\mathrm{EM}}$, with ruler calibration and shared covariance retained. A trend in Ξ alone does not determine which sector generated it. [19]

49. Finite-frequency amplitude and phase certificates

The positive transport action now has an explicit finite-frequency estimate. Write $B=b_{\cos}\sqrt{Q}>0$, $U=B''/B\in L^1[a,b]$, $c_T=1$, and retain the same incoming initial data for the canonical mode and its plane-wave comparator:

$$v(a)=1,\quad v'(a)=-\mathrm{i}k,\qquad v_0(\eta_c)=e^{-\mathrm{i}k(\eta_c-a)},\quad k>0.$$

Variation of constants gives the exact equation

$$v=v_0+\int_a^{\eta_c}\frac{\sin[k(\eta_c-s)]}{k}U(s)v(s)\,\mathrm{d}s.$$

Set $M(\eta_c)=k^{-1}\int_a^{\eta_c}|U|\,\mathrm{d}s$ and $\rho=e^M-1$. Gronwall, followed by substitution and differentiation, gives

$$|v-v_0|\le\rho,\qquad |v'-v_0'|\le k\rho.$$

Indeed $|v(s)|\le e^{M(s)}$ and the resulting integral is exactly $e^{M(\eta_c)}-1$. No derivative bound on U is needed. For $h=v/B$ and $h_0=v_0/B$,

$$|h-h_0|\le\rho/B,$$
$$|h'-h_0'|\le(k+|B'/B|)\rho/B.$$

If $\rho<1$, the same disk gives

$$\big||v|-1\big|\le\rho,\qquad |\arg(v/v_0)|\le\arcsin\rho,$$

with one continuous phase unwrapping from a. The real potential preserves $\mathrm{Im}(\bar v v')=-k$, so the actual phase rate is $-k/|v|^2$ and

$$\left|\frac{\mathrm{d}\arg v}{\mathrm{d}\eta_c}+k\right|\le k\big[(1-\rho)^{-2}-1\big].$$

The positive factor B changes no phase. These are separate amplitude, accumulated-phase and phase-rate certificates.

For a fully specified declared example, take $0\le\eta_c\le1$ and

$$b_{\cos}=\frac{10+\eta_c}{11},\qquad Q=\left(\frac{10+\eta_c}{10}\right)^2,\qquad k=100.$$

Then $B=(10+\eta_c)^2/110$, $U=2/(10+\eta_c)^2$ and

$$M(1)=\frac{1}{5500},\qquad \rho(1)<\frac{1}{5499}.$$

The last rational bound uses $e^x\le(1-x)^{-1}$ for $0\le x<1$. The leading endpoint factor is $B_E/B_R=100/121$, and the propagation-only distance ratio is $\sqrt{Q_R/Q_E}=11/10$. The finite-frequency amplitude disk is below 1.819×10^{-4}. This receiving example assigns no microscopic SAM Q or physical conformal-time scale. An elementary exact basis,

$$w(\eta_c)=e^{-\mathrm{i}k\eta_c}\left(1-\frac{\mathrm{i}}{k(10+\eta_c)}\right),\quad \bar w,$$

with Wronskian $2\mathrm{i}k$, supplies the independent matched-initial-data check.

50. Generation, propagation and detector normalization

A source term $\mathscr{J}h$ in one normalized Fourier mode gives

$$(B^2h')'+B^2k^2h=\mathscr{J},\qquad v''+(k^2-U)v=\mathscr{J}/B.$$

For a zero incoming incremental mode and impulse $\mathscr{J}=j\delta(\eta_c-E)$, the canonical momentum jumps by j. If $G(R,E)$ has unit initial velocity and zero initial displacement, then

$$h(R)=\frac{jG(R,E)}{B_EB_R},\qquad W_{\mathrm{impulse}}=\frac{j^2}{2B_E^2}.$$

Thus an equal source impulse and an equal emitted field preparation are different protocols when Q_E differs.

Keep the preceding $b_{\cos}$ but set $Q = 4[(10+\eta_c)/10]^2$. This leaves U unchanged; now $Q_E = 4$ and $Q_R = 121/25$. Compare an electromagnetic mode with $Q = 1$, the same $k = 100$, and $E = 0$, $R = 1$. For $x_E = 10$, $x_R = 11$ and $L = 1$, the complete retarded Green functions are

$$G = \left(1 + \frac{1}{k^2 x_E x_R}\right)\frac{\sin kL}{k} + \left(\frac{1}{x_R} - \frac{1}{x_E}\right)\frac{\cos kL}{k^2},$$
$$G_{\mathrm{EM}} = \frac{\sin kL}{k}.$$

Here $G/G_{\mathrm{EM}} = 1.0001557233563114\ldots$ at this specified nonzero readout. The absolute Green error also satisfies $k|G - G_{\mathrm{EM}}| \le e^M - 1$; no ratio is assigned at a zero of G_{EM}.

Let d be the receiving detector-gain ratio. The recorded amplitudes obey exactly

$$\frac{A_T}{A_{\mathrm{EM}}} = d\,\frac{j_T}{j_{\mathrm{EM}}}\,\frac{1}{\sqrt{Q_E Q_R}}\,\frac{G}{G_{\mathrm{EM}}}.$$

Equal impulses and $d = 1$ give $(5/22)(G/G_{\mathrm{EM}})$. Matching the emitted h' requires $j_T = Q_E j_{\mathrm{EM}} = 4 j_{\mathrm{EM}}$ and gives $(10/11)(G/G_{\mathrm{EM}})$. A detector gain $d = 3/2$ changes the latter to $(15/11)(G/G_{\mathrm{EM}})$. The inferred amplitude distances are the reciprocal ratios. Only the matched-emission, matched-detector protocol has the leading propagation-only ratio $11/10$.

For $j_{\mathrm{EM}} = 1$, the electromagnetic impulse costs $121/200$ per normalized mode; the equal-impulse tensor mode costs $121/800$; matched emission costs $121/50$. During propagation the prescribed background also does signed work:

$$\frac{\mathrm{d}H}{\mathrm{d}\eta_c} = BB'k^2h^2 - \frac{B'}{B^3}\Pi^2$$

for each real mode outside the impulse. Source preparation, prescribed-background work and a detector's absorbing or disturbing measurement remain in their respective accounts. The displayed linear detector gain is a declared receiving convention.

51. The original causal tensor carrier as a signal map

The next calculation returns to the actual retained quadrupole TT carrier, distinct from the preceding electromagnetic and cosmological adapters. Keep its two signed source quadratures, $\kappa = 4/5$, $c_* = 1$, regular interior, full outgoing exterior and original radius R. For $r = \rho R$, $\rho \ge 1$, let

$$s(w) = (r + Rw)/c_*,$$
$$K_\rho(w) = \frac{1}{\kappa\rho} P_2\left(-w + \frac{1-w^2}{2\rho}\right).$$

The complete driven receiver is

$$x_{\mathrm{out}}(t) = \int_1^1 K_\rho(w) f(t - s(w))\,\mathrm{d}w.$$

A homogeneous incoming/prepared field adds its own receiver term. The retained maintained-pulse experiment has zero incremental incoming field on the same stationary reference; the full background and field energy remain. The frequency response is

$$\chi(\omega) = \frac{2\mathrm{i}z}{\kappa} j_2(z) h_2^{(1)}(z\rho), \qquad z = \omega R/c_*.$$

All near, induction and radiation terms are retained. The first and last kernel supports are $(r-R)/c_*$ and $(r+R)/c_*$. On the original small-frequency branch $j_2(z) > 0$, the group delay is

$$\tau = \frac{r}{c_*}\frac{u^4}{u^4 + 3u^2 + 9}, \qquad u = \omega r/c_*.$$

A signed kernel can have a group delay outside its causal support interval; the two observables remain distinct.

For a carrier-modulated envelope $f = a(t)e^{-\mathrm{i}\omega_0 t}$, choose $\chi_0 = \chi(\omega_0) \ne 0$ and $\tau = (\mathrm{d}\arg\chi/\mathrm{d}\omega)|_{\omega_0}$. The gain/delay approximation is

$$x_{\mathrm{app}} = \chi_0 e^{-\mathrm{i}\omega_0 t} a(t - \tau).$$

More explicitly, write $\tau = (\mathrm{d}/\mathrm{d}\omega)\arg\chi|_{\omega_0}$ and define

$$L_1 = |\chi'(\omega_0) - \mathrm{i}\tau\chi_0|,$$
$$D_2 = \int |K_\rho(w)|[s(w) - \tau]^2\,\mathrm{d}w.$$

Taylor's remainder for $e^{\mathrm{i}\nu(s-\tau)}$ gives

$$|\chi(\omega_0 + \nu) - \chi_0 e^{\mathrm{i}\nu\tau}| \le |\nu| L_1 + \tfrac{1}{2}\nu^2 D_2.$$

If the envelope spectrum lies in $|\nu| \le \beta$, Plancherel gives

$$\|x_{\mathrm{out}} - x_{\mathrm{app}}\|_2 \le (\beta L_1 + \tfrac{1}{2}\beta^2 D_2)\|a\|_2.$$

The first term is amplitude dispersion: at the stated group delay $L_1 = |\mathrm{d}|\chi|/\mathrm{d}\omega|$. For a bounded smooth envelope, direct Taylor expansion instead gives the pointwise bound

$$|x_{\mathrm{out}} - x_{\mathrm{app}}| \le L_1\|a'\|_\infty + \tfrac{1}{2}D_2\|a''\|_\infty.$$

This version needs no frequency truncation, and applies to a compact pulse.

52. A complete source pulse and a nonzero phase event

Retain the actual four-cycle source

$$f_{ab}(t) = g_{ab} A(\Delta t) e^{-\mathrm{i}\Delta t}, \qquad \Delta = \frac{2}{573049},$$

where $A(\theta) = \sin^4(\theta/8)$ on $[0, 8\pi]$ and zero outside. The six signed g_{ab}, equal pair weights $1/6$, both tensor polarizations, four undriven source diagonal states and complete maintenance-work account are unchanged. The envelope is C^3 with bounded fourth weak derivative.

The complete signed delay moments satisfy

$$m_0 = \frac{2}{5\kappa\rho^3}, \quad m_1 = m_3 = 0,$$
$$\frac{m_2}{m_0} = \frac{R^2}{7c_*^2} - \frac{r^2}{3c_*^2}, \quad m_n = \int K_\rho s^n \, \mathrm{d}w.$$

These follow by expanding the full argument of P_2 and integrating monomials. At $\rho = 2$, with $(R/c_*)^n$ factored out, moments zero through four are

$$(1/16,\ 0,\ -25/336,\ 0,\ 313/336).$$

Thus odd-moment cancellation and a negative signed second moment quantify the near-field behavior.

Taylor's formula, including both compact-pulse edges, gives

$$|x_{\mathrm{out}} - m_0 f| \le \frac{|m_2|}{2}\|f''\|_\infty + \frac{D_4}{24}\|f^{(4)}\|_\infty,$$
$$D_4 = \int |K_\rho| s^4 \, \mathrm{d}w \le \frac{2(r+R)^4}{\kappa\rho c_*^4}.$$

For $\rho \ge 1$, the Legendre argument maps $[-1,1]$ monotonically onto $[1,-1]$, so $|P_2| \le 1$. The exact five-frequency interior expansion gives

$$\|f''\|_\infty \le |g|\Delta^2 \frac{17}{16}, \qquad \|f^{(4)}\|_\infty \le |g|\Delta^4 \frac{709}{512}.$$

The zero extension has the same weak-derivative bounds. No frequency or radial tail is cut.

At the original receiver $r = R + 64$, use the complete authenticated Chapter VII radius ball, rather than treating a printed decimal as exact. Outward rational arithmetic yields, for every time and every original signed lane,

$$\frac{|x_{\mathrm{out}} - m_0 f|}{m_0 |g|} < 8.994 \times 10^{-9}.$$

Also $|\chi(\Delta) - m_0|/m_0 < 8.440 \times 10^{-9}$ and

$$0 < \tau < 1.792884 \times 10^{-14}$$

in original model-time units, while first causal arrival remains exactly 64. Replacing the static gain by the full monochromatic gain and its group delay gives

$$\frac{\left|x_{\mathrm{out}} - g\chi(\Delta) e^{-\mathrm{i}\Delta t} A[\Delta(t-\tau)]\right|}{m_0 |g|} < 1.744 \times 10^{-8}.$$

This is a uniform absolute-envelope error normalized by peak static gain. A phase disk requires a positive received envelope: if $A[\Delta(t-\tau)] \ge a_* > 0$, its radius is at most

$$\arcsin\left[\frac{\varepsilon_{\mathrm{pulse}}}{a_*(1-\varepsilon_{\mathrm{gain}})}\right],$$

when the argument is below one. It supplies no phase statement at zeros.

A sharper phase certificate is available at the actual peak event $t = 4\pi/\Delta$. Every retarded source argument then lies inside the smooth pulse. After demodulation and division by signed g,

$$Y = \int K_\rho(w) e^{\mathrm{i}\Delta s} \cos^4(\Delta s/8) \, \mathrm{d}w.$$

The sine expansion has positive weights and relative frequencies

$$\text{Weights} : (3/8, 1/4, 1/4, 1/16, 1/16),$$
$$\text{Frequency ratios} : (1, 5/4, 3/4, 3/2, 1/2).$$

Their fifth-derivative sum is $857/512$. Since $m_1 = m_3 = 0$,

$$|\operatorname{Im} Y| \le \frac{857}{512}\frac{\Delta^5 D_5}{120},$$
$$|\operatorname{Im} \chi| \le \frac{\Delta^5 D_5}{120},$$

where $D_5 \le 2(r+R)^5/(\kappa\rho c_*^5)$. The preceding real-part lower bounds then give

$$|\arg Y - \arg \chi| \le \frac{\Delta^5 D_5}{120 m_0}\left[\frac{857/512}{1-\varepsilon_{\mathrm{static}}} + \frac{1}{1-\varepsilon_{\mathrm{gain}}}\right]$$
$$< 9.782 \times 10^{-15} \text{ radians}.$$

The gain/delay comparator's envelope is positive at this event and changes no phase. This is one nonzero-amplitude phase event; timing, incoming-field contamination, detector disturbance and statistical error remain in the measurement protocol. Source maintenance, free/drive/counterterm energy and complete emitted-field energy remain in the retained ledger.

53. The original carrier from the near field to radiation

The user-supplied review derived the exact decomposition and compact-edge remainder below. An independent calculation from the retained complete kernel agrees. The flux estimate and finite-endpoint examples extend that derivation. The original six signed g_{ab}, equal pair weights, both tensor polarizations, $\kappa = 4/5$, radius balls and $\Delta = 2/573049$ remain fixed.

Put $a = R/c_*$, $v = r/R \ge 1$, $t_r = t - r/c_*$ and $x_r = R\phi(r,t)$. Polynomial expansion gives

$$P_2\left[-w + \frac{1-w^2}{2v}\right] = P_2(w) - \frac{3w(1-w^2)}{2v} + \frac{3(1-w^2)^2}{8v^2}, \quad (6)$$

$$x_r = x_{\mathrm{rad}} + x_{\mathrm{ind}} + x_{\mathrm{near}},$$
$$x_{\mathrm{rad}} = \frac{1}{\kappa v}\int_{-1}^{1} P_2(w) f(t_r - aw)\, dw,$$
$$x_{\mathrm{ind}} = -\frac{3}{2\kappa v^2}\int_{-1}^{1} w(1-w^2) f(t_r - aw)\, dw,$$
$$x_{\mathrm{near}} = \frac{3}{8\kappa v^3}\int_{-1}^{1} (1-w^2)^2 f(t_r - aw)\, dw. \quad (7)$$

These are exact r^{-1}, r^{-2} and r^{-3} pieces at fixed retarded time. All have the original causal support $[(r-R)/c_*, (r+R)/c_*]$. The unchanged homogeneous incoming field is added separately. The normalized angular tensor remains $\sqrt{5/(8\pi)}\Lambda_n$ applied to the original plus/cross source; its sphere isometry fixes the angular amplitude ruler.

53.1 A whole-pulse derivative and remainder theorem

For $f \in C^3(\mathrm{R})$ with bounded fourth weak derivative, parity and Taylor's integral remainder yield

$$x_r(t) = \frac{2R^3}{15\kappa}\left[\frac{f''(t_r)}{c_*^2 r} + \frac{3f'(t_r)}{c_* r^2} + \frac{3f(t_r)}{r^3}\right] + E(t), \quad (8)$$

$$|E(t)| \le \frac{a^4\|f^{(4)}\|_\infty}{60\kappa v} + \frac{a^3\|f'''\|_\infty}{35\kappa v^2} + \frac{a^2\|f''\|_\infty}{35\kappa v^3}. \quad (9)$$

For example, the radiative coefficient uses $\int P_2 = 0$ and $\int w^2 P_2 = 4/15$; its remainder follows from $\int |P_2| w^4 \le 2/5$. The induction and near remainders use respectively $\int w^4(1-w^2) = 4/35$ and $\int w^2(1-w^2)^2 = 16/105$. These estimates apply on the zero extension, including intervals that cross either pulse edge.

The actual maintained pulse is $f = g_{ab}F(\Delta t)$, with $F(\theta) = \sin^4(\theta/8)e^{-i\theta}$ on $[0, 8\pi]$ and zero elsewhere. Its first three derivatives match zero at both endpoints. Its five-frequency expansion gives

$$\|f^{(j)}\|_\infty \le |g_{ab}|\Delta^j D_j,$$
$$(D_0, D_1, D_2, D_3, D_4) = \left(1, 1, \frac{17}{16}, \frac{19}{16}, \frac{709}{512}\right).$$

Define $z = \Delta R/c_*$, $u = \Delta r/c_*$ and the specified positive ruler $A_0 = 2R^3|g_{ab}|\Delta^2/(15\kappa c_*^2 r)$. Then, for every time,

$$\frac{|x_r - A_0 \operatorname{sgn}(g_{ab})F''(\Delta t_r)|}{A_0} \le \frac{3}{u} + \frac{3}{u^2} + z^2\left(\frac{709}{4096} + \frac{57}{224u} + \frac{51}{224u^2}\right). \quad (10)$$

This is an absolute curve bound; phase and relative error need a nonzero comparator. The full original radius ball gives $z < 5.708 \times 10^{-7}$. For $u = 100, 1000, 10^6$, the right side is below 0.030300000001, 0.003003000001 and 0.0000030000031. These correspond to $r = u/\Delta$ in the same $c_* = 1$ model chart. The old receiver $r = R + 64$ has $u < 0.000224$.

At the retarded envelope peak $\theta = 4\pi$, $F'' = -17/16$ is real. The leading radiative peak therefore has a 6.25% amplitude correction, with the same phase, relative to a central-frequency unchanged-envelope approximation. Increasing r leaves that waveform correction present. The four-cycle maintained pulse and the one-cycle autonomous clock retain their different source protocols.

53.2 A finite-radius amplitude ruler and positive emitted energy

The exact monochromatic identity is

$$\chi_r = -\frac{2j_2(z)e^{iu}}{\kappa v}\left(1 + \frac{3i}{u} - \frac{3}{u^2}\right), \quad (11)$$

$$\chi_{\mathrm{rad}} = -\frac{2j_2(z)e^{iu}}{\kappa v}, \quad (12)$$

$$\left|\frac{\chi_r}{\chi_{\mathrm{rad}}}\right|^2 = 1 + \frac{3}{u^2} + \frac{9}{u^4}, \quad (13)$$

$$\left|\frac{\chi_r}{\chi_{\mathrm{rad}}} - 1\right| \le \frac{3}{|u|} + \frac{3}{u^2}. \quad (14)$$

The ratio requires $j_2(z) \neq 0$. At $u = 100$ the amplitude excess is below 0.000150045; the induction term still contributes an order $1/u$ phase. The local detector functional, angular projection, gain, original source vertex and $1/r$ factor all enter the recorded amplitude ruler.

The exact coefficient at null infinity and its energy are

$$\psi(t_r) = \lim_{r\to\infty} r\phi(r, t_r + r/c_*) = \frac{1}{\kappa}\int_{-1}^{1} P_2(w) f(t_r - aw)\, dw, \quad (15)$$

$$W = \frac{\kappa}{2c_*}\int_{\mathrm{R}} |\psi'|^2 dt_r. \quad (16)$$

For $\psi_0 = 2a^2 f''/(15\kappa)$, the comparator is

$$W_0 = \frac{2R^4 g_{ab}^2 \Delta^5 I_3}{225\kappa c_*^5}.$$

53.3 Exact smoothing fixes the integrated energy sign

The six-page Chapter IV review of 16 October 2026 (H002654) supplies a sharper theorem for the same maintained pulse. It keeps both original compact edges and all six signed source vertices. Put $a = R/c_*$ and let

$$\mu(w) = \begin{cases} \frac{15}{16}(1-w^2)^2, & |w| \le 1, \\ 0, & |w| > 1. \end{cases}$$

Its mass, mean and variance are 1, 0 and $1/7$. Since $P_2(w) = [(1-w^2)^2]''/8$, the two endpoint terms vanish under two integrations by parts; the equivalent spherical Bessel normalization is also given by DLMF 10.54.1. [20] Thus the *exact* outgoing radiation is

$$\psi = \frac{2a^2}{15\kappa}\mathscr{C}_\mu f'', \qquad \psi' = \frac{2a^2}{15\kappa}\mathscr{C}_\mu f''',$$

where $(\mathscr{C}_\mu g)(t) = \int \mu(w) g(t - aw)\, dw$. For any g with $g, g' \in L^2$, polarization gives

$$\|g\|_2^2 - \|\mathscr{C}_\mu g\|_2^2 = \frac{1}{2}\iint \mu(w)\mu(v) \times \|g(\cdot - aw) - g(\cdot - av)\|_2^2\, dw\, dv \le \frac{a^2}{7}\|g'\|_2^2.$$

The last inequality uses the Sobolev translation bound and $\int\int \mu(w)\mu(v)(w-v)^2\,dw\,dv = 2/7$. The original zero-extended pulse is C^3 across its endpoints, so $g = f'''$ has weak derivative $g' = f^{(4)} \in L^2$. Its original five-frequency expansion gives

$$I_3 = \int_0^{8\pi} |F'''|^2 d\theta = \frac{14245\pi}{4096},$$
$$I_4 = \int_0^{8\pi} |F^{(4)}|^2 d\theta = \frac{324817\pi}{65536}.$$

Consequently the same W, W_0 satisfy

$$0 \le 1 - \frac{W}{W_0} \le \frac{a^2\|f^{(4)}\|_2^2}{7\|f'''\|_2^2} = \frac{324817}{1595440} z^2 < 6.633\times 10^{-14}, \qquad z = \Delta R/c_*. \tag{17}$$

The numerical upper endpoint uses the full original radius ball. Finite-source smoothing therefore reduces this integrated energy relative to the derivative comparator. It is a signed certificate for the original maintained pulse; it changes neither the autonomous-clock preparation nor its source-depletion account. The common ratio retains each lane's signed waveform and cancels only its common squared vertex in W/W_0.

53.4 Exact source work, sphere flux and near-field storage

The supplied review asks for an all-time conservation check for the maintained incremental pulse. A direct calculation from the original kernel gives it. Let $a = R/c_*$ and $\mu(w) = 15(1-w^2)^2/16$ on $[-1,1]$, and set

$$A(u) = \frac{2R^2}{15\kappa c_*^2}\int \mu(w) f(u-aw)\,dw, \qquad u = t - r/c_*.$$

Two integrations by parts give the exact exterior solution

$$\phi(r,t) = \frac{A''(u)}{r} + \frac{3c_* A'(u)}{r^2} + \frac{3c_*^2 A(u)}{r^3}.$$

This includes both compact-pulse edges. Write $q_j = A^{(j)}(u)$ and $y = (q_2, c_* q_1/r, c_*^2 q_0/r^2)$. The complete outward sphere flux is

$$F_r(t) = \frac{\kappa}{2c_*}|q_3|^2 + \frac{dS_r(u)}{dt},$$
$$S_r = \frac{\kappa}{2r} y^* G y, \qquad G = \begin{pmatrix} 7/2 & 6 & 9/2 \\ 6 & 15 & 27/2 \\ 9/2 & 27/2 & 27/2 \end{pmatrix}.$$

Its exact LDL pivots are $7/2, 33/7, 27/44$, all positive. Thus S_r is nonnegative stored near-field energy on the outgoing characteristic; its derivative can have either sign. The instantaneous sphere flux need not be positive. The full exterior energy is

$$E_{>r}(t) = \frac{\kappa}{2c_*}\int_{-\infty}^{u} |A'''(v)|^2 dv + S_r(u).$$

Since A is compactly supported, every exterior sphere has the same all-time integrated flux $W = \kappa \int |A'''|^2/(2c_*)$.

The original radial energy density and flux are

$$e = \frac{\kappa}{4}\left[\frac{r^2|\phi_t|^2}{c_*^2} + r^2|\phi_r|^2 + 6|\phi|^2\right],$$
$$F_r = -\frac{\kappa}{2} r^2 \operatorname{Re}(\phi_t^* \phi_r).$$

Their local continuity equation and the original source jump give $\dot{E}_{<b} + F_b = \operatorname{Re}(f^*\phi_t(R))$ for $b > R$. Keep $H_{\rm drive} = -\operatorname{Re}(f^*\phi(R))$ and $H_{\rm ct} = |f|^2/(4R)$ in the actual tensor sector. Explicit control power is

$$P_{\rm ctrl} = -\operatorname{Re}(\dot{f}^*\phi(R)) + \dot{H}_{\rm ct}.$$

Hence

$$\frac{d}{dt}(E_{<b} + H_{\rm drive} + H_{\rm ct}) + F_b = P_{\rm ctrl}.$$

The maintained source's even energy is constant; any different source-energy change must be added to its recipient and power account. With zero incoming field and zero force endpoints, the completed net control work equals W, the integrated flux through every $b > R$, and the full outgoing exterior energy. Intermediate interaction and counterterm changes remain explicit. Signed endpoint cancellation supplies no zero-apparatus-cost assertion.

For nonzero incoming field, retain the incoming, emitted and bilinear cross terms in both energy and flux, including $\operatorname{Re}(f^*\phi_{{\rm in},t}(R))$ in source power. The frozen native storage check retains the cross block for two outgoing jets. A separate exact bidirectional checker uses the inward exterior solution $\phi_{\rm in} = A_{\rm in}''/r - 3c_* A_{\rm in}'/r^2 + 3c_*^2 A_{\rm in}/r^3$ and verifies pure and cross continuity and signed radiative flux. A regular incoming background also has its matching outgoing component; an inward characteristic alone is only local to the exterior. This maintained classical carrier identity remains distinct from autonomous quantum-source depletion. The exact flux, full outgoing exterior, incoming preparation, source work, boundary interaction, counterterm and even source/reference energy remain in the ledger. This fixed flat-carrier normalization supplies no cosmological Q, redshift-dependent generation function or SI conversion by itself.

54. The canonical electromagnetic comparison

For one transverse conformal Maxwell mode use $L_M = (A'^2 - k^2A^2)/2$. The earlier $Q = 1$ comparator uses $h = A/b$ and $L_h = b^2(h'^2 - k^2h^2)/2$. With $\mathscr{H} = b'/b$,

$$L_h = \frac{1}{2}\left[A'^2 - \left(k^2 - \frac{b''}{b}\right)A^2\right] - \frac{1}{2}(\mathscr{H}A^2)', \tag{18}$$

$$J_h h = \frac{J_h}{b}A, \qquad j_A = \frac{j_h}{b_E}, \tag{19}$$

$$D_h h_R = \frac{D_h}{b_R} A_R. \tag{20}$$

For the displayed linear b, $b'' = 0$, so these are the same free equations with the stated canonical boundary term, source and

detector transformations. Their energies satisfy

$$E_h - E_M = -\mathscr{H}AA' + \tfrac{1}{2}\mathscr{H}^2A^2,$$
$$E_h = \tfrac{1}{2}[(A' - \mathscr{H}A)^2 + k^2A^2].$$

An impulse from zero field has the common work $j_h^2/(2b_E^2) = j_A^2/2$, recovering the earlier impulse numbers under this protocol. At nonzero incoming A, the two impulse work changes differ by $-\mathscr{H}Aj_A$; that term belongs to the boundary/background account. A fixed Maxwell impulse or detector gain requires the corresponding transformed h values. This specifies the electromagnetic comparison of the linear-b example.

55. Moving endpoints of the actual travel-time map

Let a differentiable one-branch travel-time function define actual events by

$$F(t_E) - t_E = T[t_E, x_E(t_E), F(t_E), x_R(F(t_E))].$$

Implicit differentiation yields the review's event Jacobian, including explicit reception-time dependence when present:

$$F' = \frac{1 + \partial_E T + \nabla_E T \cdot v_E}{1 - \partial_R T - \nabla_R T \cdot v_R}. \tag{21}$$

The chosen forward causal branch has a nonzero denominator and $F' > 0$. For stationary isotropic optics, endpoint variation gives $\nabla_E T = -p_E$, $\nabla_R T = p_R$, with optical slowness $p = n_{\rm opt}\widehat{t}/c_*$. The velocities and gradients use their different actual event epochs. With worldline proper rates $a_A = d\tau_A/dt$ and intrinsic interacting-clock rates $\nu_A(\tau_A)$, the observed frequency comparison is

$$\mathscr{R} = \frac{a_E \nu_E(\tau_E)}{a_R F' \nu_R(\tau_R)}.$$

The interacting transient and proper ages stay in this expression. The actual delay derivative along events is $dT/dt_E = F' - 1$.

For the retained weak-source, finite-endpoint logarithm let $D = |x_R - x_E|$, $A = r_E + r_R$, $n = (x_R - x_E)/D$, $e_A = x_A/r_A$ and $K = (1+\gamma)GM/c_*^3$. Then

$$T_{\rm app} = D/c_* + K \log \frac{A + D}{A - D}, \tag{22}$$

$$\nabla_E T_{\rm app} = -n/c_* - \frac{2K(De_E + An)}{A^2 - D^2}, \tag{23}$$

$$\nabla_R T_{\rm app} = n/c_* + \frac{2K(An - De_R)}{A^2 - D^2}. \tag{24}$$

These gradients enter the actual implicit map. The travel-time approximation retains its metric-order error. More quantitatively, write its numerator and denominator as A_*, B_*. If $B_{\rm app} \geq b > \delta_B$, $|A_* - A_{\rm app}| \leq \delta_A$ and $|B_* - B_{\rm app}| \leq \delta_B$, then

$$|F' - F'_{\rm app}| \leq \frac{\delta_A}{b - \delta_B} + \frac{|A_{\rm app}|\delta_B}{b(b - \delta_B)}.$$

The operands include path-gradient and endpoint-velocity uncertainties. Compose outgoing, proper-wait and return maps at their actual event epochs.

56. A signed finite-frequency transport correction

For the retained Volterra equation $v = v_0 + Vv$ put $v_1 = Vv_0$ and $M = \int |U|/k$. Ordered integration gives $\|V^n v_0\|_\infty \leq M^n/n!$, hence

$$v = v_0 + v_1 + R_2, \qquad |R_2| \leq e^M - 1 - M.$$

The same remainder controls the derivative divided by k. A nonzero lower bound for $|v_0 + v_1|$ turns this disk into amplitude and phase bounds about the signed first correction. In the retained $M = 1/5500$ example, $|R_2| \leq M^2/[2(1-M)] < 1.654 \times 10^{-8}$. The source-derived physical transport coefficient remains its own calculation.

Part D. What a measured distance curve determines

57. A calibrated ratio on a finite interval

Let $u = 1 + z$, $s = \ln u$, and adopt the normalized single-power distance family

$$\Xi(z) = \Xi_\infty + (1 - \Xi_\infty)u^{-p}, \qquad p > 0.$$

The measured ratio includes an unknown constant relative calibration,

$$Y(z) = C_{\rm cal}\Xi(z) = \alpha + \beta e^{-ps}, \quad C_{\rm cal} = \alpha + \beta, \quad \Xi_\infty = \frac{\alpha}{\alpha + \beta}. \tag{25}$$

The symbol Ξ_∞ denotes the plateau within this family; it is the Ξ_0 notation of the finite-redshift source. [19] The calibration $C_{\rm cal}$ is an extrapolated relative distance scale at $z = 0$ when the normalization $\Xi(0) = 1$ is used. The calibration subscript distinguishes this quantity from the source current operator C of the clock sections.

A constant gain can be fitted with α, β. Redshift-dependent calibration requires a model or independent information: $Y = C_{\rm cal}(s)\Xi(s)$ is invariant under $\Xi \mapsto f\Xi$, $C_{\rm cal} \mapsto C_{\rm cal}/f$. Background luminosity distance, channel propagation and calibration are consequently different physical operands even when they enter one ratio.

A directly finite-window quantity avoids endpoint extrapolation. For positive ratios,

$$\overline{\delta}_{ij} = -\frac{\ln[Y(z_j)/Y(z_i)]}{s_j - s_i}.$$

A constant calibration cancels. Under an amplitude-friction representation $\delta = -\mathrm{d}\ln\Xi/\mathrm{d}s$, this is its exact interval average. With small ratio errors and covariance Σ,

$$\mathrm{Var}(\overline{\delta}_{ij}) \simeq \frac{\Sigma_{ii}/Y_i^2 + \Sigma_{jj}/Y_j^2 - 2\Sigma_{ij}/(Y_iY_j)}{(s_j - s_i)^2}.$$

This quantity tests measured evolution while retaining the two-channel covariance.

58. Two samples recover the declared linear part

At a specified p, set $x_i = u_i^p$. Since $xY = \alpha x + \beta$, two distinct exact samples give

$$\alpha = \frac{x_j Y_j - x_i Y_i}{x_j - x_i}, \qquad \beta = \frac{x_i x_j (Y_i - Y_j)}{x_j - x_i}.$$

The calibration and normalized plateau follow from Eq. (25) if $C_{\mathrm{cal}} \neq 0$. The differential projection $Y + (u/p)\mathrm{d}Y/\mathrm{d}u = \alpha$ states the same two-dimensional function-space identity; finite-sample algebra avoids differentiating noisy data. These are exact recovery formulas within the family, not a separate test of that family. [19]

59. Three samples determine a positive exponent

For $s_0 < s_1 < s_2$, put $h_1 = s_1 - s_0$, $h_2 = s_2 - s_1$ and

$$r = \frac{Y_2 - Y_1}{Y_1 - Y_0}, \qquad F(p) = \frac{1 - e^{-ph_2}}{e^{ph_1} - 1}.$$

A nonconstant single-power member obeys $r = F(p)$. The function decreases strictly from h_2/h_1 to zero. Hence $0 < r < h_2/h_1$ is necessary and sufficient for a unique positive exponent fitting the three exact samples.

Proof of strictness. Logarithmic differentiation gives

$$\frac{F'(p)}{F(p)} = \frac{h_2}{e^{ph_2} - 1} - \frac{h_1}{1 - e^{-ph_1}} < 0,$$

because the first term is smaller than $1/p$ and the second is larger than $1/p$. The endpoint limits follow from expansion at zero and exponential decay at infinity. Continuity and strict monotonicity give existence and uniqueness. Solving the first two samples then supplies α, β. Membership in the positive-distance family also requires the reconstructed $C_{\mathrm{cal}} = \alpha + \beta > 0$ and $\Xi_\infty = \alpha / C_{\mathrm{cal}} > 0$; the exponent criterion itself is the algebraic reconstruction statement.

For equal logarithmic spacing $s_j = s_0 + jh$, let $\lambda = e^{-ph}$. Then

$$\lambda = \frac{Y_2 - Y_1}{Y_1 - Y_0}, \qquad p = -\frac{\ln \lambda}{h},$$
$$\alpha = \frac{Y_0 Y_2 - Y_1^2}{Y_0 - 2Y_1 + Y_2}, \qquad \beta = (Y_0 - \alpha) e^{ps_0}.$$

Near a constant response or a nearly vanished transient, the denominators reveal poor conditioning. Three exact points reconstruct a member; a fourth supplies a consistency condition. [19]

60. A fourth sample tests the single-power shape

Let $d_i = Y_{i+1} - Y_i$ and $h_i = s_{i+1} - s_i$. Define

$$F_{i,i+1}(p) = \frac{1 - e^{-ph_{i+1}}}{e^{ph_i} - 1}.$$

The two triplets require the same positive exponent:

$$F_{01}^{-1}(d_1/d_0) = F_{12}^{-1}(d_2/d_1).$$

For noisy data, use increments without dividing by a small increment:

$$g(p) = \begin{pmatrix} d_1 - F_{01}(p) d_0 \\ d_2 - F_{12}(p) d_1 \end{pmatrix}.$$

Its covariance is obtained by applying the same linear contrast to the full ratio covariance. At equal spacing the exact condition reduces to the Hankel determinant $H = d_0 d_2 - d_1^2 = 0$. These formulas require point samples or common translated bin windows; unequal windows must use Appendix E's window-aware exponential templates before any single-mode or rank interpretation.

For two distinct geometric modes, $Y_j = \alpha + B_1 \lambda_1^j + B_2 \lambda_2^j$, direct expansion gives

$$H = B_1 B_2 (\lambda_1 - 1)(\lambda_2 - 1)(\lambda_1 - \lambda_2)^2.$$

This identifies precisely when a second mode produces a nonzero four-point departure. It also shows why close mode rates or a weak second amplitude give small departures. The statistical covariance and profiling treatment are retained in Appendix E. [19]

61. Positive response modes and finite rank

The single-power test extends to a declared positive finite response class. All following Hankel identities assume point samples or common translated bin windows. Unequal windows require Appendix E's window-aware treatment; one true mode can otherwise fail the single-mode identity. On an equally spaced logarithmic grid $s_j = s_0 + jh$, suppose

$$Y_j = \alpha + \sum_{a=1}^{r} B_a \lambda_a^j, \qquad B_a > 0, \quad 0 < \lambda_a < 1, \tag{26}$$

with distinct nodes. Here $\lambda_a = e^{-p_a h}$ and B_a includes the initial-grid factor. These are inference assumptions; their identification with a spectral measure of the physical source remains a separate derivation. For $e_j = Y_j - Y_{j+1}$ and $w_a = B_a(1 - \lambda_a)$,

$$(H_m)_{ij} = e_{i+j}, \qquad (V_m)_{ia} = \lambda_a^i, \qquad H_m = V_m \,\mathrm{diag}(w_a) V_m^T. \tag{27}$$

Positivity and the Vandermonde minors give $H_m \geq 0$ and $\mathrm{rank}\, H_m = \min(m, r)$. A positive r-square determinant therefore detects at least r visible modes. An increasing branch has the same result after reversing all differences when its amplitudes have a common opposite sign. Mixed signed amplitudes require another analysis.

Write $p(x) = \prod_a (x - \lambda_a) = x^r + \sum_{k<r} c_k x^k$. The moment recurrence and reconstruction system are

$$e_{j+r} + \sum_{k<r} c_k e_{j+k} = 0, \qquad H_r \mathbf{c} = -(e_r, \ldots, e_{2r-1})^T. \tag{28}$$

Thus $2r+1$ exact Y samples reconstruct p, the distinct nodes, the positive weights through the Vandermonde system, $B_a =$

$w_a/(1-\lambda_a)$, and $\alpha = Y_0 - \sum B_a$. Membership checks on roots and weights are part of the reconstruction. One additional sample checks the next recurrence. For a general positive moment measure on $[0,1)$, exact flat rank has the stronger certificate

$$\begin{aligned}(c_0,\dots,c_{r-1},1)H_{r+1}(c_0,\dots,c_{r-1},1)^T \\ = \int p(\lambda)^2\,\mathrm{d}\mu = 0.\end{aligned} \tag{29}$$

The nonnegative integrand forces support onto the roots; rank $H_r = r$ then gives exactly r visible atoms within this exact positive class.

61.1 Approximate plateau mass needs a rate restriction

If a specified polynomial satisfies $\int p^2\mathrm{d}\mu \le \eta$ and $|p| \ge d > 0$ outside proposed node neighborhoods S, then

$$\begin{aligned}\mu(S^c) &\le \eta/d^2, \\ B(S^c) &\le \frac{\eta}{d^2(1-\lambda_{\max})} \quad \text{if } \lambda \le \lambda_{\max} < 1.\end{aligned} \tag{30}$$

The first inequality integrates a nonnegative lower bound; the second uses $B = \int(1-\lambda)^{-1}\mathrm{d}\mu$. Without the independently justified upper node bound, tiny difference weight can conceal finite plateau mass near $\lambda = 1$. This is the same slow-mode obstruction as the finite-window endpoint construction. A noisy small determinant alone supplies neither exact closure nor a physical rate cutoff. Adjacent differences reuse observations; their full covariance and any data-dependent polynomial selection enter a simultaneous confidence calculation.

61.2 A compact exact certificate

A declared dimensionless example uses $\alpha = 33/40$, $B = (1/10, 1/20, 1/40)$ and $\lambda = (1/2, 1/3, 1/4)$. It has $Y_0 = 1$, $\operatorname{rank} H_3 = \operatorname{rank} H_4 = 3$, and

$$p(x) = x^3 - \frac{13}{12}x^2 + \frac{3}{8}x - \frac{1}{24}.$$

Eight exact samples retain the seventh-sample reconstruction and one additional consistency check. The rational checker authenticates every moment, recurrence and recovered amplitude. Native rank and solve receipts check the same frozen operands separately. The example is a finite inference certificate; it does not replace the actual spin-generator closure or supply cosmological rates from the source ledger.

62. Approximate modes can certify the measured curve

For the positive difference measure $d\mu$ on $[0,1]$, write

$$e_j = \int \lambda^j d\mu, \qquad Y_0 - Y_j = \int q_j(\lambda) d\mu, \quad q_j = \sum_{k=0}^{j-1} \lambda^k.$$

Suppose node neighborhoods of radius at most ε are assigned disjointly, with union S, and the polynomial certificate gives $\mu(S^c) \quad \eta/d^2$. Collapse each neighborhood's true mass to its center, omit the exterior mass, and anchor at the same Y_0. On $[0,1]$, $0 \le q_j \le j$ and $|q_j'| \le j(j-1)/2$. Splitting the integral gives

$$|Y_j - \widetilde{Y}_j| \le \frac{j(j-1)}{2}\varepsilon\mu(S) + \frac{j\eta}{d^2}.$$

This finite-grid curve bound needs no cutoff away from $\lambda = 1$. Estimated neighborhood weights with total absolute error γ and anchor error ε_0 add $j\gamma + \varepsilon_0$. True neighborhood masses cannot silently be identified with fitted weights. An atom at 1 is an anchored linear finite-window surrogate, not a finite-amplitude plateau. Plateau inference still pays the singular factor $(1-\lambda)^{-1}$.

For continuous grid coordinate $t \ge 1$, use $q_t = (1-\lambda^t)/(1-\lambda)$ with $q_t(1) = t$. Its maximum is t and a Lipschitz bound is $\max\{1, t(t-1)/2\}$. Indeed

$$q_t = t\int_0^1 [\lambda + (1-\lambda)u]^{t-1}\,du;$$

differentiation bounds the maximum derivative by its endpoint value 1 for $1 \le t \le 2$, and by $t(t-1)/2$ for $t \ge 2$. The same split proves a continuous-window bound on this range. A window touching 0 with unrestricted fast modes requires an additional modulus.

63. Simultaneous noise coverage permits selected certificates

Let $\widehat{Y} = Y + \Sigma^{1/2}g$, where g is an eight-dimensional standard Gaussian and the full covariance Σ is known. On the single event $\|g\|^2 \le 24$,

$$|a^{\mathrm{T}}(\widehat{Y} - Y)| \le \sqrt{24 a^{\mathrm{T}}\Sigma a} \quad \text{for every } a.$$

The Gaussian moment generating function gives $\Pr\{\|g\|^2 \ge n + 2\sqrt{nx} + 2x\} \le e^{-x}$. For $n = 8, x = 3$ its threshold is less than 24, so the event has coverage at least $1 - e^{-3} > 0.95$. It holds for vectors selected from the same data. In particular, if c selects a polynomial after observing the data, $\eta = c^{\mathrm{T}}Hc = a(c)^{\mathrm{T}}Y$ has the simultaneous upper bound

$$\eta \le \widehat{\eta} + \sqrt{24 a(c)^{\mathrm{T}}\Sigma a(c)}.$$

Estimated covariance, heavy tails and unmodeled drift need a separately calibrated confidence event. No Gaussian coverage is assigned to them.

Differences reuse observations: $C = D\Sigma D^{\mathrm{T}}$. For the declared fixture $\Sigma_{ij} = \sigma^2 2^{-|i-j|}$, $\sigma = 10^{-6}$, each difference variance is σ^2. The same event therefore bounds every moment entry error by 5σ. For a size-r Hankel matrix,

$$\widehat{H}_r - 5r\sigma I > 0 \quad \Longrightarrow \quad H_r > 0.$$

This certifies at least r visible nodes for every compatible positive measure. Failure is unresolved, not proof of smaller exact rank. Noise cannot exclude an arbitrarily weak extra positive mode.

The exact two-mode fixture has $\alpha = 4/5$, $B = (1/10, 1/10)$, $\lambda = (1/4, 3/4)$ and eight point samples. It passes the size-two lower certificate. A seeded covariance-aware calculation uses 2000 catalogs in each of five declared cases: separated, nearly coincident, weak second, very slow and constant. The separated case certifies a minimum of two; the others remain unresolved at two. These are new inference checks, separate from the historical Rev 6 benchmark or observational data.

63.1 Positive support constraints inside the confidence event

For eight observations let $e_j = Y_j - Y_{j+1}$, $0 \le j \le 6$. The exact degree-six positive difference-measure constraints on $[0,1]$ are

$$H_4(e) = (e_{i+j})_{i,j=0}^{3} \ge 0,$$
$$L_3(e) = (e_{i+j+1} - e_{i+j+2})_{i,j=0}^{2} \ge 0.$$

Intersect these with $\|\Sigma^{-1/2}(Y - \widehat{Y})\|_2 \le \sqrt{24}$. These conditions are sufficient as well as necessary by the one-variable representation of a nonnegative degree-six polynomial as sums of squares plus $\lambda(1-\lambda)$ times sums of squares. Maximizing a selected polynomial's linear moment functional over this convex set gives a simultaneous upper certificate; the complete conic optimization is a specified method, not an executed optimum here.

A finite implemented dual method supplies usable bounds. For positive Gram multipliers P, Q, write $b = a + H^* P + L^* Q$. Then every feasible Y obeys

$$a^T Y \le b^T \widehat{Y} + \sqrt{24 b^T \Sigma b}.$$

The minus signs give the corresponding lower certificate. A catalogue of 40 exact Gram choices, including zero, is evaluated with outward rational square roots. Selection within this catalogue retains the same confidence event. An explicit constant-curve fixture improves the unconstrained ellipsoid's positive upper bound on $-e_0$ to exactly 0, with a feasible true curve; a separate drift fixture has an exact infeasibility witness.

If the class also requires positive amplitudes and a nonnegative plateau, use the stronger positive-Y support constraints

$$(Y_{i+j+1})_{i,j=0}^{3} \ge 0, \qquad (Y_{i+j} - Y_{i+j+1})_{i,j=0}^{3} \ge 0.$$

An atom of the difference measure at $\lambda = 1$ is a valid anchored finite-window linear surrogate, not a finite-amplitude plateau model. For a fitted curve, two linear optimizations per grid point bound $\pm(Y_j - \widetilde{Y}_j)$ directly over the chosen intersection; this avoids identifying true neighborhood masses with fitted weights. The supplement gives the full formulations, exact feasibility witnesses and finite dual implementation separately.

63.2 A window mismatch is a model error

Point-sample Hankel formulas also apply to common translated bin windows: each exponential acquires a common positive amplitude factor and retains its node. Unequal windows generally do not. For one true node $1/2$ and amplitude $B = 1/5$, an alternating positive two-point window factor 1 and 5/4 gives the four-point difference determinant $-27/6400$, whereas a common window gives 0. This is a failure of the point-sample model, not evidence for a positive two-mode model. Appendix E's window correction is required before the main-text single-mode and rank tests are applied.

64. Coarse winding and fine final readout use different ensembles

The earlier twenty-fine-mean design is sufficient but spends final precision on every winding prefix. Assume the controlled initial phase is known. Use three intermediate prefixes per history with two coordinates each:12 coarse means with coordinate tolerance $d_c = .05$ and 8100 fresh preparations per mean. Use only the two final coordinates of each history at $d_f = 5\,10^{-11}$: four fine means, each with $5.432\,10^{21}$ fresh preparations. Hoeffding and the union bound give

$$P_c \le 24 e^{-(8100) d_c^2/2} < .000961568,$$
$$P_f \le 8 e^{-(5.432\,10^{21}) d_f^2/2} < .008999751.$$

The total statistical failure is below $.009961318 < .01$ and the sufficient preparation count is

$$21\,728\,000\,000\,000\,000\,097\,200 \simeq 2.1728\,10^{22}.$$

The inherited amplitude lower bound, controlled prefix spacing and separate coarse systematic allowance keep every unwrapping increment strictly below π. Fine final sampling retains the earlier statistical phase allowance below 0.2 nanoradians. Unknown initial phase requires additional sampling. Every original preparation, leakage, readout, timing, reference, energy and model requirement remains in force. This failure probability is conditional on the stated systematic bounds; random calibration failures must receive their own probability allocation. These are sufficient mathematical resource estimates, not achieved apparatus performance or experimental lower bounds.

64.1 An exact transform of the original continuum

The Chapter IV review of 16 October 2026 (H002654) derives an elementary transform of the same normalized source measure. Write

$$F_E(\tau) = F(\tau/R), \qquad F(a) = \frac{10}{\pi} \int_0^\infty e^{-ax} j_2(x)^2 \, dx.$$

The Legendre representation gives the exact overlap

$$\begin{aligned} C(v) &= \int_{-1}^{1-v} P_2(t) P_2(t+v) \, dt \\ &= \frac{2}{5} - v + \frac{v^3}{2} - \frac{3v^5}{40}, \qquad 0 \le v \le 2, \\ F(a) &= \frac{5}{\pi} \int_0^2 C(v) \frac{a}{a^2 + v^2} \, dv. \end{aligned}$$

Direct integration therefore yields

$$16\pi F(a) = 12a^3 + 56a + 32\arctan(2/a) - a(3a^4 + 20a^2 + 40)\log(1 + 4/a^2). \tag{31}$$

The original positive Bessel integral fixes the sign; C itself is signed. Its first cancellations, $\int C = \int v^2 C = 0$, explain why ordinary floating-point subtraction loses digits at large a.

Range-reduced rational logarithm and arctangent series, with explicit remainders and a Machin enclosure of π, give outward intervals of width below 10^{-75} at all operands in the following table. Entries are midpoint displays; exact rational endpoints are in the source packet.

a	$F(a)$
0.01	0.9236282572354819
0.1	0.6004698962807585
1	0.040468535993562725
10	$3.254795554745759 \times 10^{-6}$
100	$3.3938508389223904 \times 10^{-11}$

A convergent large-a alternative uses

$$\int_0^2 v^m C(v)\,dv = \frac{2^{m+2} m(m-2)}{(m+1)(m+2)(m+4)(m+6)}.$$

After N geometric terms its absolute transform remainder is at most $4(4/a^2)^N/[\pi a(2N+1)]$, for $a > 2$. The small-a expansion is

$$F(a) = 1 + \frac{5}{\pi} a \log a + \frac{5}{\pi}\left(\frac{1}{2} - \log 2\right) a + O(a^3 |\log a|).$$

Its infinite initial right slope and late-time power-law tail both retain the continuous, gapless source rather than a finite family of rates.

64.2 Soft-spectrum bounds without a minimum rate

The compact Bessel representation implies $|j_2(x)| \le x^2/15$. For the original $d\nu_R(k) = (10R/\pi) j_2(kR)^2\,dk$, this gives

$$F_E(\tau) \le \min\left(1, \frac{16R^5}{15\pi\tau^5}\right),$$
$$\nu_R([0,q]) \le \min\left(1, \frac{2(Rq)^5}{225\pi}\right). \tag{32}$$

Suppose a physical reduction independently establishes $\Xi(s) = A + (1-A)F(\lambda s/R)$ with $0 \le A \le 1$ and $\lambda \ge \lambda_{\min} > 0$. Set $E(s) = F(\lambda_{\min} s/R)$. Then

$$\max\left(0, \frac{\Xi(s) - E(s)}{1 - E(s)}\right) \le A \le \Xi(s), \qquad s > 0.$$

Simultaneous data bands and outward upper E values can be mapped and intersected. The conversion-scale restriction is distinct from a hard spectral gap. The observable and its real-time or logarithmic-redshift identification still require the physical reduction.

More generally let $\nu([0,p]) \le Cp^5$, let the transient amplitude measure be dominated by $B\nu$, and let an omitted submeasure obey $\int (1 - e^{-ph})\,d\beta_{\text{omitted}} \le m$, with $h > 0$. Splitting at any $p_0 > 0$ gives

$$B_{\text{omitted}} \le B,$$
$$B_{\text{omitted}} \le BCp_0^5 + \frac{m}{1 - e^{-p_0 h}}.$$

Using $(1 - e^{-x})^{-1} \le 1 + x^{-1}$ and optimizing at $p_0 = [m/(5BCh)]^{1/6}$ yields

$$B_{\text{omitted}} \le B,$$
$$B_{\text{omitted}} \le m + 6(BC)^{1/6}\left(\frac{m}{5h}\right)^{5/6}. \tag{33}$$

Zero-mass cases follow by a limit. For the mapped original source, $C = 2(R/\lambda_{\min})^5/(225\pi)$. Thus a polynomial residual bound $m \le \eta/d^2$ can supply a plateau payment after these source-class operands are specified, without imposing a nonexistent minimum continuum rate.

64.3 Infinite exact rank and positive Gaussian surrogates

On a Euclidean grid $\tau_j = \tau_0 + jh$, map k to $\lambda = e^{-hk}$. The actual difference measure is

$$d\mu = Be^{-\tau_0 k}(1 - e^{-hk})\,d\nu_R(k).$$

For $B > 0$, it has continuous support in $(0,1)$ apart from isolated Bessel zeros. A nonzero polynomial has finitely many roots, so $\int |p|^2 d\mu > 0$. Every finite Hankel matrix is positive definite. The original continuum therefore has infinite exact Hankel rank.

Let p_r be its monic orthogonal polynomial. Its r-node Gaussian quadrature has positive weights and is exact through degree $2r-1$. Division by p_r establishes exactness; integrating the squared cardinal polynomials establishes positivity of the weights. Anchor the surrogate at the actual Y_0, and put $q_j(\lambda) = \sum_{k=0}^{j-1} \lambda^k$. Hermite interpolation then gives

$$\widetilde{Y}_j = Y_j, \qquad 0 \le j \le 2r,$$
$$0 \le \widetilde{Y}_j - Y_j \le \binom{j}{2r+1} \int p_r^2\,d\mu, \qquad j \ge 2r+1. \tag{34}$$

Here $q_j^{(2r)}/(2r)!$ is nonnegative and bounded by the binomial coefficient on $[0,1]$; the first missed error equals $\int p_r^2 d\mu$. If the true amplitude integral is finite and its plateau is nonnegative, monotone convergence of q_j also makes the surrogate plateau no smaller than the true plateau.

64.4 Certified surrogates of the actual source curve

For $Y_j = F(j)$, namely $\tau_0 = 0$ and $h = R$, the true plateau is zero. Full interval arithmetic gives the following nodes and difference weights; these are midpoint displays of certified source intervals.

r	λ_i	w_i
1	0.03729697269	0.95953146401
2	0.02123986980	0.87066215903
	0.19461014031	0.08886930497

The corresponding plateaus and first missed errors are

r	$\widetilde{A}$	$\widetilde{Y}_{2r+1}-Y_{2r+1}$
1	0.003294435783	0.002423770451
2	0.0001005695591	0.00003902006495

The exact H_2 and H_3 determinant enclosures are strictly positive, near 0.002325684008961166 and $9.074834108661564\times10^{-8}$. The packet retains all moment, node, weight, plateau and signed holdout intervals through $j=7$. These are controlled finite approximations to the actual continuous Euclidean source. Their surplus plateaus illustrate why an exact fit on a short grid is distinct from a physical finite spectrum. Noisy mode recovery retains the chapter's simultaneous feasibility and covariance treatment.

65. What lies beyond a measured interval

Finite-window reconstruction and unrestricted endpoint identification are different questions. On $0\le s\le s_{\max}$, consider

$$\Xi_1 = 1-A+Ae^{-ps},\quad \Xi_2 = 1-A-B+Ae^{-ps}+Be^{-\varepsilon s},$$

where $A,B>0$, $A+B<1$, $p,\varepsilon>0$. The two plateaus differ by B, while

$$\|\Xi_2-\Xi_1\|_\infty \le B\varepsilon s_{\max}.$$

A slow positive component can therefore produce an arbitrary finite-window tolerance while retaining the stated plateau separation. At the lower endpoint, for $s_{\min}>0$, the addition De^{-Ps} changes $Y(0)$ by D yet obeys

$$\|Y_2-Y_1\|_{\infty,[s_{\min},s_{\max}]} \le |D|e^{-Ps_{\min}}.$$

These are exact constructions. Appendix F retains their Gaussian minimax consequence; no estimator choice removes the stated unresolved modes. [19]

A physical rate-spectrum restriction supplies useful positive bounds. If a decreasing normalized response has a positive transient measure supported on $p\ge p_{\min}>0$, let $E=e^{-p_{\min}s}$. Then

$$\frac{\Xi(s)-E}{1-E}\le \Xi_\infty \le \Xi(s).$$

For an increasing response the inequalities reverse. With an independently fixed plateau and transient rates at most $p_{\max}$, the decreasing lower-endpoint bound is

$$Y(s)\le C_{\mathrm{cal}}\le Y_\infty + e^{p_{\max}s}[Y(s)-Y_\infty].$$

Multiple samples intersect these intervals. Noisy samples require a simultaneous band before mapping and intersecting its bounds. A source-derived exponent or spectrum would connect the physical derivation to exactly these finite-window limits.

66. Correlated ratios require a declared statistical protocol

The input to the compressed distance test is a vector of ratio estimates with its joint covariance, not automatically independent events. For a trial p, let $t_i=e^{-ps_i}$ and $X_p=[\mathbf{1},\boldsymbol{t}]$. Generalized least squares fits calibration and transient amplitude together,

$$\widehat{\boldsymbol{\vartheta}}_p=(X_p^T\Sigma^{-1}X_p)^{-1}X_p^T\Sigma^{-1}\boldsymbol{y},\quad \boldsymbol{\vartheta}=(\alpha,\beta)^T.$$

Its residual agrees exactly with the consecutive-contrast statistic in Appendix E. For fixed p, known positive-definite Σ and exactly Gaussian errors, it has a χ^2_{m-2} law. If p is profiled, repeat that selection in the simulation or posterior-predictive calibration. At $\beta=0$, p is unidentified, so a universal χ^2_{m-3} rule is not supplied by the fixed-exponent proof.

The finite-redshift source reports frozen synthetic tests with full covariance, non-Gaussian tails and a precision sweep. Smooth two-power and calibration-drift departures can be absorbed by exponent refitting, even when their true endpoint differs. The relevant physical breadcrumb is constructive: an independently derived p would remove a major fitting freedom. The numerical benchmark remains a retained result of Ref. [19]; the present chapter does not relabel it an observational analysis or a newly executed Monte Carlo campaign.

67. One measurement ledger across clocks, routes and amplitudes

Every example in this chapter can use the same bookkeeping sequence, with different receiving operators. The ledger should retain the following operands:

1. The original source state, complete energy account, common e_u, I_u and spatial ruler; preparation leakage and the certified internal-age window.
2. The emitted signal identity, encoded source phase or amplitude, polarization, event label and any encoding latency.
3. The chosen propagation branch and map F, endpoint trajectories, source/background convention, medium, multipoles and turnaround rule.
4. The receiving observable, detector gain/response, latency, phase branch or winding record, and disturbance caused by readout.
5. Every correction already applied to the timestamps or amplitudes, the observation interval, shared calibration, covariance and the statistical selection rule.

Repeated expectation estimates require repeated preparations; finite expectation samples do not equal individual shots. The apparatus and phase-unwrapping protocol developed in Chapter II accompany the interacting clock. [2] A qubit endpoint phase alone records phase modulo 2π. Prefix samples, a verified amplitude lower bound and an explicit winding record establish its unwrapped history. The paid reference retains its own wait Hamiltonian and every entry/release payment; zero telescoping signed work does not state zero apparatus cost.

In the endpoint channel, parameter errors produce the rate and accumulated-time bounds of Appendix B.1. In the timing

channel, constant offset and changing delay require separate controls. In the amplitude channel, source inclination, detector response, selection, channel covariance and any opacity enter the inferred distance. A common gain inside the fitted model space can change coefficient uncertainty without changing a shape residual; a redshift-varying gain generally cannot be removed that way.

68. Completed deductions and the next physical calculation

The six original preparations now have finite weak-field error bounds under the same declared globally positive receiving profile: all six fit a 0.2 nanoradian form-error allowance at $\alpha = 7 \times 10^{-10}$. A common certificate keeps the total response negative for every $0 < \alpha \leq 10^{-3}$. Direct orientation contrasts cancel shared operators before bounding. The full-window isolated correction's two-time matrix elements and sign remain unresolved; a small zero-containing interval is not a measured anisotropy.

The corrected canonical return distinguishes forward and reverse sectors and the complete inverse from its complement restriction. A specified C^3 switch gives the leading outgoing pair energy and scalar vacuum-loop payment. Complete transported shell bounds and a closed instantaneous frequency comparison preserve all angular modes, source terms and original states. These instantaneous comparisons alone do not construct the projective field implementer. The analytic weighted normal-frame construction in the later canonical composition sections supplies the free switched propagator, while the actual switched joint-source covariance and local-output remainder remain explicit.

The compact carrier's exact positive averaging theorem gives a signed relative energy deficit below 6.633×10^{-14}, alongside an exact positive near-field storage and controller-work identity. The actual Euclidean continuum has a closed transform, gap-free slow-mass bounds and infinite Hankel rank. Positive Gaussian quadrature matches finitely many source samples with signed prediction errors; it does not convert the continuum into a finite physical spectrum. A measured amplitude coefficient, source/background evolution, long-time real-time transition dynamics and a map to $\log(1+z)$ remain independent physical work. The joint protocol in Section 70 supplies explicit detector and repetition requirements; achieved apparatus accuracy and observational fits remain separate experimental inputs.

69. Evidence and reproducibility

The compact replay and separate native-evidence supplements retain the new editable sources, checkers, frozen operands, results and complete native readback. The unchanged predecessor archive retains the original IV/V bundles and finite-redshift reconstruction provenance. Source-reported, independently checked, native-cross-checked, declared and open claims keep separate evidence roles. For Hermitian matrices, $A \geq 0$ and $A > 0$ mean positive semidefinite and positive definite, respectively.

The reproduction record distinguishes inherited certificates, new mathematical deductions, native executions and independent implementation checks. Appendix I gives the route to each category. Reported original clock enclosures and synthetic benchmark counts retain their original provenance. Native run startup, accepted plans and prepared calculations are not completion receipts.

70. A joint original-source measurement protocol

This continuation joins Chapters IV and V through the original interacting source. It retains the six preparations, signed couplings, all16 source states, both field polarizations, source even energy, both counterterms, initial hard dressing and the field/angular complements. The phase-bearing signal below is derived from that source. The earlier maintained compact carrier remains its separately labelled comparison protocol.

The chain has two levels of completion. The original local source supplies an exact field identity and a nonzero finite-mode signal with a finite measurement budget. A specified passive interface then carries that mode through the strong-field annulus and a calibrated detector. The interface, phase reference and receiving gate are apparatus inputs. A curved source-selected vertex and self-consistent stress/geometry evolution remain named physical operands in the ledger below.

70.1 The original interacting clock already drives a phase-bearing field

The signal can be derived from the original source rather than supplied by a bare oscillator encoder. Retain the full sixteen-state source, signed $D, C = \mathrm{i}SD$, even energies, both tensor polarizations, both counterterms and the original hard-branch superposition with soft vacuum. Within each unchanged active lane, $g = \varepsilon(q_a - q_b)$, $e = q_a + q_b + q_a q_b$, and

$$H_0 = \varepsilon e I + \varepsilon S + H_f - g(\sigma_z \phi_+ + \sigma_y \phi_\times) + C_{\mathrm{ct}},$$
$$C_{\mathrm{ct}} = 2g^2\mu, \quad \mu = 1/(5\kappa R) = 1/(4R), \quad \kappa = 4/5.$$

Write $z(t) = \langle \sigma_z + \mathrm{i}\sigma_y \rangle$ and $u_R(k) = \sqrt{2k/(\kappa\pi)}\, j_2(kR)$. The exact field equation is

$$a_\alpha(k,t) = e^{-\mathrm{i}kt} a_\alpha(k,0) + \mathrm{i}g u_R(k) \int_0^t e^{-\mathrm{i}k(t-s)} \sigma_\alpha(s)\, \mathrm{d}s.$$

Its circular mean consequently receives the *actual* interacting $z(s)$; no source–field factorization is used. The full-source identities

$$Z_{\mathrm{raw}} = D + \mathrm{i}C = D - SD, \qquad [H_{\mathrm{pair}}, Z_{\mathrm{raw}}] = -2Z_{\mathrm{raw}},$$
$$Z_{\mathrm{raw}}^\dagger Z_{\mathrm{raw}} = 2D^2(I+S), \qquad Z_{\mathrm{raw}} Z_{\mathrm{raw}}^\dagger = 2D^2(I-S)$$

retain both noise orderings. The bare commutator is only one term of the interacting equation; the actual source frequency remains its certified Ω, with its all-order residual.

Initial dressing and the completed physical receiver. The hard field is not initially vacant. Exact branch pull-through gives

$$\langle a_+(k) + \mathrm{i}a_\times(k) \rangle_0 = \frac{gMu_R(k)}{k - \Omega}, \qquad k \geq \lambda = 64\Delta.$$

The opposite helicity supplies its $k + \Omega$ denominator. Together they reproduce the retained nonzero incoming field and momentum. At the original physical receiver $r = R + 64$,

$$F_r(t) = F_{\mathrm{free},0}(t) + g \int_0^t \chi_{Rr}(t-s) z(s)\, \mathrm{d}s,$$
$$\chi_{Rr}(s) = \frac{P_2((R^2 + r^2 - s^2)/(2Rr))}{\kappa Rr}, \quad 64 < s < 64 + 2R,$$

with zero kernel outside that interval. The completed original receiver result retains both terms and gives additional phase ceilings below $5.734\,10^{-10}$ radians in all six lanes. Its source-induced first advance is64; the initially dressed total field can vary earlier. These are reused completed results, distinguished from the new mode certificate below.

70.2 A nonzero soft-mode signal with an actual brightness bound

Select a detector observable in the original soft band without deleting any physical hard field. The same normalized filter is used in every lane:

$$f_0(k) = \frac{2}{\sqrt{3\Delta}} \cos^2 \frac{\pi(k - \Delta)}{2\Delta}, \quad 0 < k < 2\Delta,$$
$$f_T(k) = e^{\mathrm{i}kT/2} f_0(k),$$
$$b_T = \int f_T(k) \frac{a_+(k,T) + \mathrm{i}a_\times(k,T)}{\sqrt{2}}\, \mathrm{d}k.$$

It satisfies $[b_T, b_T^\dagger] = 1$. The original soft vacuum gives zero homogeneous mean exactly. For $T = T_0/4$ or $T_0/2$,

$$\alpha_T = \langle b_T \rangle = \frac{\mathrm{i}g}{\sqrt{2}} \int f_T u_R \int_0^T e^{-\mathrm{i}k(T-s)} z(s)\, \mathrm{d}s\, \mathrm{d}k,$$
$$\alpha_{h,T} = \frac{\mathrm{i}gMT}{\sqrt{2}} e^{-\mathrm{i}\Omega T/2} \int f_0 u_R \operatorname{sinc} \frac{(k - \Omega)T}{2}\, \mathrm{d}k.$$

The second integral is strictly positive. The first is bounded about it by the original $|z(s) - Me^{-\mathrm{i}\Omega s}| \leq E(T)$ certificate. Its phase therefore tracks the actual source current at $T/2$, with known signed orientation $\arg(\mathrm{i}g)$. The two source tags are $T_0/8, T_0/4$; their phase increments stay below π, permitting unwrapping with the stated measurement allowances.

At $T = T_0/2$, rigorous outward bounds give

lane	$\lvert\alpha_T\rvert$ lower	$\langle b_T^\dagger b_T \rangle$ upper
01	$5.309\,10^{-18}$	$1.494\,10^{-34}$
02	$1.100\,10^{-17}$	$6.417\,10^{-34}$
03	$1.369\,10^{-16}$	$9.944\,10^{-32}$
12	$5.694\,10^{-18}$	$1.719\,10^{-34}$
13	$1.316\,10^{-16}$	$9.188\,10^{-32}$
23	$1.259\,10^{-16}$	$8.410\,10^{-32}$

The largest intrinsic phase discrepancy from the actual source current is below $1.442\,10^{-21}$ radians. The photons' two-time source covariance has not been factorized: total occupation is bounded above, whereas $|\alpha_T|^2$ is a coherent-brightness lower bound. In particular,

$$\frac{|\alpha_T|^2}{1 + N_T} \leq \Pr(n_T \geq 1) \leq N_T,$$
$$N_T \leq 2g^2 T^2 \left(\int |f_T u_R|\, \mathrm{d}k \right)^2.$$

This is source-created finite-age soft occupation; an escaping luminosity also needs its physical outgoing channel and extraction account.

70.3 What a phase measurement must supply

For either normalized-mode quadrature the conservative variance bound is $1+2N_T$. Independent fresh joint preparations and finite-variance median-of-means estimation give a simultaneous 1 milliradian design for the twelve lane/age settings and two quadratures. The sufficient counts at $T_0/2$ range from about $7.10\,10^{41}$ to $4.73\,10^{44}$ copies per quadrature. These large counts follow the original tiny quadrupole brightness; no detector gain has been substituted for a source amplitude. Preparation, readout bias, timing, local-oscillator phase and its energy must satisfy their separate explicit allowances.

The soft-band observable is a precise canonical mode, not automatically a finite local time gate. Its frequency filter has temporal tails. The channel calculation must specify source-to-port mode transfer, buffering or finite-gate leakage, vacuum/thermal noise and detector backaction. A diagonal QW1 thermal state has zero coherent mean; it does not supply this clock phase. The phase-bearing mean above comes directly from the original interacting source. Physical curved canonical coupling and autonomous preparation/repeated-emission timing remain distinct joins.

70.4 The original clock packet through the complete complex channel

The actual source calculation supplies a normalized circular soft mode, with the common smooth filter

$$f_0(k)=\frac{2}{\sqrt{3\Delta}}\cos^2\frac{\pi(k-\Delta)}{2\Delta},\qquad 0<k<2\Delta,$$
$$f_T=e^{ikT/2}f_0,\qquad T=T_0/2.$$

Its mean is generated by the original signed coupling and interacting current; its encoded current age is $T/2=T_0/4$. No coherent brightness is assigned. The source preparation retains the initial hard field; only the measured soft band is initially vacuum. With $\omega=N_pk$ and $\sigma=N_p\Delta$, the inherited annular Jost problem gives

$$D=f'(0)-(\sigma-i\omega)f(0),\qquad T_{\rm eff}=\frac{2i\omega}{D},$$
$$D_0=(2i\omega-\sigma)e^{-i\omega L}.$$

Here $c=1$ and the complete positive potential is retained. The previous uniform bound $|D/D_0-1|\le\delta_D$ implies

$$\left|\frac{T_{\rm eff}}{T_{\rm free}}-1\right|\le\frac{\delta_D}{1-\delta_D}<5.601\,10^{-6},$$
$$|\arg(T_{\rm eff}/T_{\rm free})|\le\arcsin\delta_D<5.601\,10^{-6}.$$

This controls the entire finite packet in norm, including all edges of the unchanged compact comparison pulse. Differentiating the outgoing Volterra kernel, using $|\partial_k[\sin(kd)/k]|\le d^2/2$, gives

$$\left|\tau_g-\left[L+\frac{2\sigma}{\sigma^2+4\omega^2}\right]\right|<94486.$$

The proof uses $J=\int V\,dx$, $E=e^{LJ}$ and the finite bounds $A=(L+L^2JE/2)E$, $B=L^2JE/2+LJA$, $C=J(LE+A)$:

$$|\partial_k(D/D_0-1)|\le C/\sigma+B+(E-1)/\sigma+(L+2/\sigma)\delta_D.$$

The propagation time exceeds five remote source-cycle windows. A paid receiver reference and the launch/arrival event map must therefore accompany the phase record. Group delay is not the causal-front timestamp.

70.5 A specified finite gate, thermal input and detector requirement

The source-to-annulus interface is a declared apparatus hypothesis: a one-way mode sorter and buffer extract this normalized mode, retain all other original outputs, and preserve a calibrated phase and energy account. It does not identify the microscopic source shell with an areal annular boundary or prove the missing curved joint Hamiltonian. Its effect on the source during the certified interval must be bounded separately. Norm preservation alone does not prove that a frequency-mixing mode sorter preserves free energy; its energy change is paid apparatus work.

For the worked protocol a fixed balanced passive combiner joins the source and the actual independent, number-diagonal finite QW1 packet: $d=(s+t)/\sqrt{2}$, $e=(s-t)/\sqrt{2}$. The unused output e is retained. Thus the signal amplitude is divided by $\sqrt{2}$ and the total output energy equals the two input energies. The old $K=4$ thermal packet near 10^{-11} is not replaced by a new packet at σ; its old capacity cannot buy one photon at that carrier. Since f_0 extends to zero, exact spectral orthogonality is not assumed.

The packet transmission satisfies

$$\eta=\int|T_{\rm eff}|^2|f_{\rm remote}|^2d\omega\ge\frac{2/5+2/(3\pi)}{(1+\delta_D)^2}>0.61219973369.$$

A calibrated number-preserving output mode processor takes its transmitted normalized shape to the stated smooth mode, with phase fixed by the carrier port reference independently of the unknown source phase. Its frequency conversion work is retained. Parseval gives its temporal variance $\pi^2/(3\sigma^2)$, so a centered gate $G=2\pi/\sigma$ collects at least $2/3$ of that mode norm. Normalized positive-frequency gate projection therefore has real overlap at least $\sqrt{2/3}$, and

$$|\alpha_{\rm det}|\ge\sqrt{\eta/3}\,|\alpha_f|.$$

This gate statement applies after the specified extraction and mode processing; it is not automatically a gate bound on the full dressed field. The ideal all-time matched thermal contamination is below $4.779\,10^{-26}$ photons. A safe finite-gate bound is half its actual total external output number, below $3.196\,10^{-6}$ photons. Thermal occupation has no coherent mean under this declared diagonal preparation.

For an explicit readout use the canonical heterodyne POVM $\Pi(d^2\beta) = |\beta\rangle\langle\beta| d^2\beta/\pi$. For every finite-number state,

$$\mathrm{E}\beta = \langle a\rangle, \qquad \mathrm{E}|\beta - \langle a\rangle|^2 = \langle N\rangle + 1 - |\langle a\rangle|^2 \le V.$$

The vacuum term remains. Independent fresh preparations, divided into 19 blocks with at least $\lceil 16V/\varepsilon^2\rceil$ records each, and coordinatewise median block means give complex mean error at most ε with probability above 0.99. Chebyshev bounds each block and Hoeffding bounds the majority; no bounded quadrature is assumed. With $\varepsilon = 0.000899\,\alpha_{\mathrm{det},-}$, apparatus phase budget $9\,10^{-5}$ rad and the source and annulus phase disks, the total error is below 0.995 mrad. The sufficient requirements are:

lane	detected mean lower	fresh preparations
01	$2.39836\,10^{-18}$	$6.53918\,10^{43}$
02	$4.97081\,10^{-18}$	$1.52231\,10^{43}$
03	$6.18775\,10^{-17}$	$9.82404\,10^{40}$
12	$2.57244\,10^{-18}$	$5.68414\,10^{43}$
13	$5.94791\,10^{-17}$	$1.06323\,10^{41}$
23	$5.69067\,10^{-17}$	$1.16153\,10^{41}$

These are sufficient ideal-instrument requirements, not achieved measurements or optimal lower bounds. The same per-lane block lengths with 25 blocks give simultaneous six-lane confidence above 0.99, since $6e^{-225/32} < 0.01$; 27 blocks give $12e^{-243/32} < 0.01$ for twelve lane/age estimates using their own block lengths. Preparation leakage, mode mismatch, reference work, detector gain, timing, disturbance and replenishment of finite inputs require their own budgets. For unbounded quadratures a mode error δ_{mode} needs an affected-number bound: its mean bias is at most $\delta_{\mathrm{mode}}\sqrt{N_{\mathrm{leak}}}$; trace distance alone is inadequate.

70.6 What the amplitude record identifies

For a known source and receiver calibration write $\beta = K(d)\alpha_f$. If the actually calibrated radiation-zone aperture law is $|K(d)| = d_*/d$, a measured disk $|\beta - \widehat{\beta}| \le \varepsilon$ and source/gain calibration $A_\pm$ give

$$\frac{d_* A_-}{|\widehat{\beta}| + \varepsilon} \le d \le \frac{d_* A_+}{|\widehat{\beta}| - \varepsilon}, \qquad |\widehat{\beta}| > \varepsilon.$$

Otherwise the derived receiver law must be inverted directly. A normalized guide-mode amplitude does not itself imply geometric $1/d$ dilution. The present source amplitude enclosures are about 15 percent wide, so more shots alone do not narrow that calibration uncertainty. Unknown source brightness or detector gain remains degenerate with distance. At the original carrier the normalized local amplitude-distance/phase Jacobian and its inverse are

$$J_0 = \frac{1}{5}\begin{pmatrix} -4 & 2 \\ 2 & 4 \end{pmatrix}, \qquad J_0^{-1} = \begin{pmatrix} -1 & 1/2 \\ 1/2 & 1 \end{pmatrix}.$$

The native two-port certificates retain both polarizations, commutators, reflected output and vacuum/noise coefficients. They certify finite coefficient maps. The end-to-end apparatus error, cosmological Q and a positive physical relaxation spectrum remain distinct derivations; a positive Euclidean measure does not provide the latter.

70.7 Intrinsic energy, apparatus custody and the cost of repeated tags

The signal comes from the original autonomous interacting source, with its correlated hard-branch/soft-vacuum preparation. It is not the old maintained bright pulse. On the full sixteen-state source,

$$\begin{aligned} H_0 = \varepsilon H_{\mathrm{pair}} + H_f - \lambda(D\Phi_+ + C\Phi_\times) \\ + \lambda^2\mu(D^2 + C^2), \qquad \lambda = \varepsilon. \end{aligned} \tag{35}$$

H_0 generates the original local model age τ. The exact source identities give

$$\frac{\mathrm{d}\langle \varepsilon H_{\mathrm{pair}}\rangle}{\mathrm{d}\tau} = 2\lambda\varepsilon\langle C\Phi_+ - D\Phi_\times\rangle. \tag{36}$$

This is a joint expectation. The even source energy and both counterterm expectations are conserved at fixed coupling and radius; they remain in the absolute energy account. The interaction and hard dressing generally persist at the readout endpoint. Neither can be removed when equating an emitted packet with source disturbance.

The initially soft vacuum yields a useful state-uniform quantum ceiling. For each original polarization,

$$a_\alpha(k,T)\Psi_0 = \mathrm{i}g u_R(k)\int_0^T e^{-\mathrm{i}k(T-s)}\sigma_\alpha(s)\Psi_0\,\mathrm{d}s. \tag{37}$$

Using $\|\sigma_\alpha(s)\Psi_0\| = 1$ and $|j_2(kR)| \le (kR)^2/15$, both polarizations give

$$\begin{aligned} \langle N_{<b}(T)\rangle &\le \frac{4g^2T^2R^4b^6}{1350\kappa\pi}, \\ \langle E_{<b}(T)\rangle &\le \frac{4g^2T^2R^4b^7}{1575\kappa\pi}. \end{aligned} \tag{38}$$

For the whole soft field use $b = 64\Delta$, $T = T_0$; for the observed band use $b = 2\Delta$, $T = T_0/2$. These bounds retain the full source and are uniform over its initial relative phase. They do not factorize two-time correlations or identify total photon energy with its coherent mean. Initially hard field energy and any physical hard-band leakage remain separate entries.

Lane	E_{soft} upper	N_{band} upper	Copies lower
01	$6.196637\,10^{-28}$	$7.535720\,10^{-34}$	$3.186158\,10^{32}$
02	$2.661819\,10^{-27}$	$3.237033\,10^{-33}$	$7.417286\,10^{31}$
03	$4.124683\,10^{-25}$	$5.016021\,10^{-31}$	$4.786663\,10^{29}$
12	$7.128776\,10^{-28}$	$8.669293\,10^{-34}$	$2.769545\,10^{32}$
13	$3.811135\,10^{-25}$	$4.634715\,10^{-31}$	$5.180469\,10^{29}$
23	$3.488605\,10^{-25}$	$4.242487\,10^{-31}$	$5.659416\,10^{29}$

Energy is in original local model units. With the declared $\tau = N_p t$, the channel-time generator and photon energies carry

the same factor N_p. No gravitating-mass calibration follows from this chart.

The copy column is a necessary information bound, not a sufficient measurement protocol. If every phase-conditioned observed-band state has mean number at most $\bar{n}$, its vacuum probability is at least $1-\bar{n}$. For N independent copies, distance to the common vacuum is at most $\sqrt{N\bar{n}}$, and the distance between two phase hypotheses is at most $2\sqrt{N\bar{n}}$. The Helstrom bound therefore requires

$$N \geq \frac{(1-2\delta)^2}{4\bar{n}} = \frac{2401}{10000\bar{n}} \quad (\delta = 1/100). \tag{39}$$

Identical independent thermal, LO and reference ancillas, subsequent processing and joint readout cannot improve this necessary bound. An independently correlated reference already carrying the phase would violate that premise. Multiplying by the full source preparation energy gives prepared-energy throughput across fresh copies, not net work, irrecoverable heat or simultaneously stored energy. The actual preparation, reset, filtering and readout payments remain required.

Every output and work recipient remains. The balanced combiner keeps $d = (s+t)/\sqrt{2}$ and its unused $e = (s-t)/\sqrt{2}$, so their summed energy is exactly the two inputs' energy, including cross terms. The clock mean is attenuated by $1/\sqrt{2}$. QW1 custody is charged once; formation inventory stays fixed. The two-port channel retains reflected energy and physical field/spring/port storage. With $E_{SF} = \langle H_{\text{source}} + H_f + V + H_{\text{ct}} \rangle$, one complete coordinate-time ledger reads

$$\begin{aligned} &\Delta E_{SF} + \Delta E_{\text{guide}} + \Delta E_{\text{reservoir}} + \Delta E_{\text{detector}} \\ &\quad + \Delta E_{\text{LO/ref}} + \Delta E_{\text{added coupling}} + E_{\text{escaped}} \\ &\quad = W_{\text{adapter}} + W_{\text{readout/ref}} + W_{\text{metric}}. \end{aligned} \tag{40}$$

An explicit battery is counted by its energy change or as external work, once. Generic spectral shape conversion preserves number but may exchange frequencies; its actual energy change is processor work. Heterodyne LO work is not intrinsic clock radiation. Original source switching is absent; if added, both $-\dot{\lambda}\langle D\Phi_+ + C\Phi_\times\rangle$ and $2\lambda\dot{\lambda}\mu\langle D^2 + C^2\rangle$ must be paid, with moving-shell work if the radius changes. Added physical linear sources need their own actual-domain counterterm, not one borrowed from another boundary or radius.

The flight delay exceeds one original receiver cycle. A fresh receiver may be prepared around a stated coarse arrival-prior window; no source certificate is extended through an unproved long wait. Reference waiting is not reference-off: the retained paid protocol has $H_{\text{ref}} = -2E_{\text{ref}}P_{\text{excited}}$ on its source-wait interval. Ramps need their areas, and phase lifting needs tagged prefixes and turn storage. Signed work cancellation supplies no zero apparatus cost. An output-field instrument leaves the instantaneous unconditioned source marginal unchanged, but not generally its conditional state or subsequent joint clock evolution. A physical coupler/backaction certificate is still needed beyond the declared packet interface.

70.8 A flux-driven geometry control, without a prescribed static mass

The original microscopic radius and field receiver are not the annulus's areal endpoints. A full moving SAM metric is not derived here. Under the separate outgoing Einstein/null-fluid comparison,

$$\begin{aligned} ds^2 &= -\left(1 - \frac{2\mu(U)}{r}\right) \mathrm{d}U^2 - 2\mathrm{d}U\,\mathrm{d}r + r^2\mathrm{d}\Omega^2, \\ L_\infty &= -\frac{c^5}{G}\mu_U, \qquad \mu(U) = \mu_0 - \frac{G}{c^4}E_\infty(U). \end{aligned} \tag{41}$$

The mass aspect follows actual outward flux. Detector/control work is subtracted before inferring that source flux; inward energy, anisotropic stress and storage require their corresponding evolution terms. Gross OUT and simultaneous inflow are distinct, and $M \geq M_{\text{form}}$ remains.

For the same fixed areal endpoints and retarded null event U, put $\chi = (M_0 - M)/M_0$ and $B_i = A_{i0}/F_{i0}$. Their exact photon ratio is

$$\begin{aligned} \frac{g^2}{g_0^2} &= \frac{1 + B_b\chi}{1 + B_p\chi}, \\ \frac{g^2}{g_0^2} - 1 - (B_b - B_p)\chi &= -\frac{(B_b - B_p)B_p\chi^2}{1 + B_p\chi}. \end{aligned} \tag{42}$$

The retained finite-gap radii give $B_b = 10^6$, $B_p = 5\,10^5$ and $(\mathrm{d}\log g/\mathrm{d}\chi)_0 = 250000$. The displayed exact remainder controls the small-depletion approximation and exposes near-gap amplification. It does not substitute $M(t)$ into a static travel integral. Given an actual common physical energy unit E_u and source-mass map, the selected soft band would contribute at most $\chi_{\text{band}} \leq E_u N_p E_{\text{band}}/(M_0 c^2)$ in this conditional chart. That mass map and the complete local quantum stress remain independent physical operands.

70.9 The shared dependency ledger after this continuation

Link	Result and remaining physical operand
Original source	Exact interacting-current field identity, dressed initial contribution and six nonzero finite-mode means. Original finite-age certificates are authenticated inputs.
Port and propagation	Full annular complex-transfer error, phase and group-delay control. The calibrated passive source-to-port interface retains its geometry, unused outputs and feedback boundary.
Detector	An explicit quantum measurement, finite-gate and thermal-noise budget, repeated-preparation count, calibration and phase branch. These are sufficient apparatus requirements, without a claim of measured performance.
Energy and geometry	Source/field/near-storage/reservoir/detector/reference accounts, output ceilings and conditional inventory updates. A source-to-mass calibration and self-consistent curved stress law remain required for an autonomous geometry.
Thermal timing	The retained finite thermal packet contributes zero phase mean and specified noise. Its weights do not select a repeated realization clock or prepare the phase signal.
Distance dynamics	The calibrated amplitude and timing Jacobians are locally invertible under their stated source/channel model. Cosmological $Q(z)$, a real-time relaxation spectrum and population/selection dynamics remain separate physical derivations.

70.10 Reproduction of the joint additions

The joint source packet contains both editable successor chapters, the unchanged predecessor source archives, frozen original operands, detailed derivations, independent standard-library checks and the exact native request/return maps. Run `python3 ops/reproduce.py`. Supply the separate native-evidence directory with `--native-evidence` to authenticate the completed execution, every matrix/solve, all local retrieval excerpts and helper custody. Reproduction uses a fresh directory with spaces and compares regenerated result bytes and both assembled sources. Earlier chapter checks retain their original source packets and scope; this joint replay does not silently recalculate the historical spectral or continuum certificates. Native retrieval and direct source recovery have separate provenance. The qualified continuation uses six controllers, three helpers and two180-second admission windows, its own CPU locks and no global startup lease or automatic third round.

The test result suggests strong contact with the concept.

Part F. Explicit incidence, signed local response, and measurement limits

71. The receiving model has an exact common-form propagator

The distinction between defining a propagator and evaluating its prepared-state response is essential. The retained radial incidence is the unitary transform

$$(\mathscr{H}_2 f)(r) = \sqrt{2/\pi}\int_0^\infty k\, j_2(kr) f(k)\,\mathrm{d}k$$

from $L^2(\mathrm{d}k)$ to $L^2(r^2\mathrm{d}r)$. The normalized original two-polarization tensor map $\mathscr{U}$ has $\mathscr{U}^*\mathscr{U} = I$, so $J = \mathscr{U}\mathscr{H}_2$ is a fixed isometry on the residual-energy output. Every actual bounded lapse $0 < w_0 \le N(t,x) \le w_1$ defines $W(t) = J^* N(t) J$, with $w_0 I \le W(t) \le w_1 I$. This includes the complete stated Plummer profile. It does not identify the compressed incidence with a full canonical local stress tensor.

Write $T = A - F$ for the original closed completed annihilation/source map on $Q = Q(H_f)$, $q_K[\psi] = \|T\psi\|^2$, and $H_s = \varepsilon H_{\mathrm{pair}}$ for the bounded complete sixteen-state source operator, including its even term. Assume s is real C^1 and W is operator-norm C^1, uniformly positive. The fixed-domain forms

$$q_t[\phi,\psi] = s(t)\langle\phi, H_s\psi\rangle + \langle T\phi, W(t)T\psi\rangle$$

are closed and uniformly semibounded. Indeed, with $h = \|H_s\|$, $s_* = \sup|s|$, and $C = 1 + s_* h$,

$$w_0 q_K[\psi] + \|\psi\|^2 \le q_t[\psi] + C\|\psi\|^2, \tag{43}$$

$$q_t[\psi] + C\|\psi\|^2 \le w_1 q_K[\psi] + (1 + 2s_* h)\|\psi\|^2. \tag{44}$$

The shifted norms are equivalent on the same Q and

$$\dot q_t[\phi,\psi] = \dot s\langle\phi, H_s\psi\rangle + \langle T\phi, \dot W T\psi\rangle$$

is a continuous bounded form $Q \to Q^*$. These are the common-form evolution hypotheses; they give a unique unitary weak propagator preserving Q [21]. Static profiles give the self-adjoint Stone evolution. C^2 coefficients permit the stronger regularity conclusions of that theorem. The original non-L^2 shell stays inside the fixed form map; no shell Hilbert norm or finite Fock cutoff is assumed.

For normalized solutions let $E(t) = q_t[\psi(t)] + C$. Form regularization and the preceding coercivity give

$$|\dot E| \le \left(\frac{\|\dot W\|}{w_0} + |\dot s| h\right) E.$$

Gronwall therefore supplies an explicit nonautonomous energy interface. For two well-posed evolutions with difference form bounded by $\delta(t)\sqrt{q_K[\phi] q_K[\psi]}$, their common initial state obeys

$$\|\psi(t) - \chi(t)\|^2 \le 2\int_0^t \delta(u)\sqrt{K_\psi(u) K_\chi(u)}\,\mathrm{d}u.$$

The original finite currents preserve Q through the bounded commutators $[T,Z]$ and $[T,Z^*]$, justifying their weak insertion identities. For $s=n(t)$ and $W=n(t)I$, the exact evolution is the original interacting evolution at $\tau(t)=\int_0^t n(u)\,\mathrm{d}u$; its transient is retained.

These results close existence for the positive receiving model. They do not evaluate its two-time matrix elements. The canonical local pair terms, transported vacuum, source vertex and angular return require the separate bounds below; positivity of a classical energy alone does not make those quantum forms bounded on the old vacuum domain.

71.1 An actual signed isolated Plummer response

The earlier isolated intervals contained zero. A source-specific calculation now resolves the sign in three original lanes, for an explicitly small finite-prefix witness. Keep the same positive Plummer profile, with direction parallel to the original polarization axis,

$$N_\alpha = 1-\alpha\frac{\sqrt{2}d}{\sqrt{d^2+|\mathbf{x}-d\mathbf{e}_z|^2}}, \qquad d=DR, \quad D=10^{30}.$$

The full sixteen-state source, six signed couplings, both tensor polarizations, both counterterms and original hard-eigenbranch/soft-vacuum preparations remain unchanged. This calculation concerns the selected positive receiving form. It does not identify it with canonical local quantum stress or set the physical angular-complement return to zero.

With $n=1-\alpha$, write $H_\alpha = nH_0+\alpha r_b$ and

$$r_b[\phi,\psi] = \langle JT_g\phi, b(r)JT_g\psi\rangle,$$
$$b = 1-F\left(1+\frac{2q^2}{7}+\frac{q^4}{126}\right),$$

where $C=(r^2+2d^2+\sqrt{r^4+4d^4})/2$, $q=rd/C$ and $F=\sqrt{2}d/\sqrt{C}$. This is the exact parallel TT compression. Its all-radius bounds are

$$0\le b(r)\le \frac{r^2}{8d^2}, \qquad b(r)\ge \frac{3r^2}{14(2+10^{-4})^2d^2} \quad (r\le d/100).$$

The upper bound has an exact positive Bernstein-polynomial certificate; the lower follows from $F^2\le 1-q^2$. No polynomial profile is substituted outside a near ball.

The actual correlation fixes an initial sign. Let $Z=\sigma_z+i\sigma_y$, $\theta=-\arg\langle Z\rangle$ and

$$R_\ell(\alpha,t)=\theta_{\alpha,\ell}(t)-\theta_{0,\ell}((1-\alpha)t).$$

The interacting uniform history cancels exactly. Put $f=u_R/\sqrt{k}$, $f_R=\mathscr{H}_2 f$ and $\psi_R=\mathscr{H}_2(f/k)$. Original Bessel identities give

$$\|r^2 f_R\|^2=\frac{11R^3}{35\kappa}, \qquad \langle r^2 f_R,\psi_R\rangle=\frac{2R^2}{9\kappa\pi},$$
$$\|f/k\|^2=\frac{2R}{105\kappa}.$$

Here $f_R=Q_2((r^2+R^2)/(2rR))/(\pi\sqrt{\kappa}rR)$ and $\psi_R=r_<^2/(5\sqrt{\kappa}r_>^3)$ are positive; their complete tails and the integrable source-shell logarithm are retained.

For $q_b=\sqrt{k}\mathscr{H}_2^*(bf_R)$ define the original joint correlation $C_b=\langle S[\phi_+(q_b)+i\phi_\times(q_b)]\rangle$. The exact hard pull-through identity, after retaining both low eigenbranches and bounding their positive complement, gives

$$C_b=-2gM\Omega I_b+E_b,$$
$$I_b=\left\langle \mathscr{H}_2^*(bf_R), 1_{k\ge\lambda}\frac{k}{k^2-\Omega^2}f\right\rangle,$$
$$|E_b|\le |g|\left[4n_+\frac{64}{63}+4\sqrt{n_+(1-M_-^2)}\right]\|bf_R\|\sqrt{\mu}.$$

The occupation bound is the original $n_+\le A/(1+A)$, $A=2g^2(6\kappa\pi)^{-1}(64/63)^2$. The comparison of I_b with the positive spatial pairing $\langle bf_R,\psi_R\rangle$ pays both the original soft interval and the hard resolvent error $\|bf_R\|\|f/k\|/4095$. Thus no source–field expectation has been factorized.

The full weighted counterterm commutes with Z but remains in the energy form. Since $[r_b,Z]=-2gS(\phi_++i\phi_\times)$, the initial isolated rate variation is $v_\ell=2g\,\mathrm{Re}\,C_b/M$. Outward intervals for D^2Rv_ℓ are

lane	lower	upper
01	$-2.603\,10^{-16}$	$-8.941\,10^{-17}$
02	$-1.245\,10^{-15}$	$-2.577\,10^{-16}$
03	$-4.123\,10^{-12}$	$3.891\,10^{-12}$
12	$-3.010\,10^{-16}$	$-1.013\,10^{-16}$
13	$-3.532\,10^{-12}$	$3.317\,10^{-12}$
23	$-2.972\,10^{-12}$	$2.775\,10^{-12}$

The three zero-containing intervals express insufficient bounds, rather than opposite observed signs.

A finite-prefix remainder. The new input is fractional regularity of the original current insertions. After removing only the common even scalar, set $Y=R(H_0+c+\varepsilon+1/R)$, $c=2g^2\mu$. The original preparation obeys

$$Y\ge I, \qquad \|Y\Psi\|<2, \qquad \|Y^{3/4}Z\Psi\|,\ \|Y^{3/4}Z^\dagger\Psi\|<10.$$

These follow from exact hard eigenvectors, the full soft vacuum and the negative-quarter field estimate

$$\|(H_f+1/R)^{-1/4}\phi(u_R)\Psi\|$$
$$\le U_{1/4}+2R^{1/4}\sqrt{\mu\langle H_f\rangle}, \qquad U_{1/4}<10R^{-3/4}.$$

It does not assume $u_R\in L^2$ or a common operator domain for H_α and H_0. Spectral calculus gives a $t^{1/4}$ form-norm modulus for the original backward insertions. Exact energy conservation also gives $\|\Psi_\alpha-\chi_n\|_Y\le 5\alpha^{1/4}$.

Applying the exact weak current identity with both adjoint insertions, putting $\tau=T/R$ and $F_0=\langle[r_b,Z]\rangle$, yields

$$\left|\frac{z_\alpha-z_{\mathrm{ref}}}{z_{\mathrm{ref}}}-\frac{i\alpha TF_0}{M}\right|$$
$$\le 2000\alpha\tau\ \ \alpha^{1/4}+\tau^{1/4}\ \ , \qquad z_{\mathrm{ref}}=z_0((1-\alpha)T).$$

For $\alpha=\tau=10^{-400}$ the remainder is $4\,10^{-897}$. The lower positive imaginary signals are respectively greater than $8.941\,10^{-877}$, $2.577\,10^{-876}$ and $1.013\,10^{-876}$ in lanes01,02,12. The current ratio disk avoids zero, so the continuous unwrapped isolated phases are strictly negative on every $0<t\le R10^{-400}$.

This is an explicit non-null finite witness for the original Plummer receiver and correlated preparation. Its scales are extremely small. It does not resolve the isolated signs at the earlier physical amplitudes or at T_0; those still need sharper time-dependent correlations. The fractional-domain lemma and three signed initial coefficients are the reusable advances. Complete proofs and exact rational operands accompany the source packet.

71.2 A global tensor Green kernel and the complete relative-pair bound

The retained constraint is a local rank-two bundle of tangential tracefree symmetric tensors. It is not an added nonlocal three-dimensional divergence constraint. Differentiating $n_iH_{ij}=0$ gives

$$|\nabla_{\mathrm{Cart}}H|^2=|\nabla_{\mathrm{bundle}}H|^2+2|H|^2/r^2.$$

Thus the complete original spatial operator is $\mathscr{D}=\nabla^*\nabla+4/r^2$, with the regular Friedrichs origin. Write $\omega=\mathscr{D}^{1/2}$. Summing every tensor harmonic gives

$$G_{\mathscr{D}}(x,y)=\frac{q^2U_2(x,y)}{4\pi\rho},\ \rho=|x-y|,\qquad q=\frac{r+r'-\rho}{r+r'+\rho}.$$

Here U_2 is unitary spin-two angular parallel transport. The antipodal product extends by zero. The scalar amplitude follows from the Jacobi addition formula; in particular it retains the complete angular complement. With $G_0=(4\pi\rho)^{-1}$, direct prolate-coordinate integration gives

$$\int G_0^2q^n\,d^3y=\frac{rn}{4\pi(n^2-1)},\qquad n>1.$$

The cases $n=4,2$ are $r/(15\pi)$ and $r/(6\pi)$. They pay the massless infrared region and the regular origin.

Keep the actual profile $b=-\mu_P/\sqrt{d^2+|x-de_z|^2}$ and $\delta=\mu_P/d=\sqrt{2}/1000$. For scalar multiplication M_b, set

$$T_b=\mathscr{D}^{-1}[\mathscr{D},[\mathscr{D},M_b]]\mathscr{D}^{-1}.$$

Its exact kernel contains the sum of the two Laplacian terms and $-2(\nabla b(x)\cdot\nabla_xG+\nabla b(y)\cdot\nabla_yG)$. The principal free-Green derivatives combine into a gradient difference before estimation. With $u=\cos(\gamma/2)$, $q\le u^2$ and spin-two holonomy gives $\|\nabla_x(q^2U_2)\|\le 14q/r$. The three global Hilbert–Schmidt payments satisfy

$$\begin{aligned}\|A\|_{\mathrm{HS}}^2&\le\delta^2(3\pi/8+4/5),\\ \|B\|_{\mathrm{HS}}^2&\le 16\delta^2/3,\\ \|C\|_{\mathrm{HS}}^2&\le 56^2\delta^2(\pi/8-1/12).\end{aligned}$$

The first uses $\int r(\Delta b)^2$, the second the exact homogeneous $H^{3/2}$ seminorm of the long-range profile, and the third the Newton-shell identity

$$\int\frac{|\nabla b|^2}{r}\,d^3x=\delta^2(3\pi^2/8-\pi/4).$$

Consequently $\|T_b\|_{\mathrm{HS}}<35\delta$. The complete frequency spectral multiplier $ab/(a+b)^2\le 1/4$ now proves

$$\|\omega^{-1}[\omega,[\omega,M_b]]\omega^{-1}\|_{\mathrm{HS}}<35\delta/4.$$

This is a global estimate, not a principal-symbol replacement. The canonical pair kernel obeys the exact identity

$$\begin{gathered}\omega^{-1/2}B_b\omega^{-1/2}=\tfrac14\omega^{-1}\big([\omega,[\omega,M_b]]+M_{\Delta b}\big)\omega^{-1},\\ \|\omega^{-1/2}B_b\omega^{-1/2}\|_{\mathrm{HS}}<119\delta/48.\end{gathered}$$

Both polarizations and every angular sector are included. Separately, the Laplacian part of the static Born coefficient has squared norm at most $\delta^2/48=1/24000000$.

71.3 What this closes, and the remaining quantum evolution

For a fixed positive lapse, the number coefficient satisfies $A_N\ge(1-|\eta\chi|\delta)\omega$ and its relative pair operator norm is at most $|\eta\chi|\delta/(1-|\eta\chi|\delta)<1$. The global Hilbert–Schmidt operand in the static implementability theorem [22] is therefore supplied. Applying the operator theorem still requires its operator-domain realization, or a proved extension to these closed block forms. Subject to that step, the static diagonalizer has the explicit all-orders bound

$$\|\beta_N\|_{\mathrm{HS}}\le\frac{|\eta\chi|119\delta}{24(1-2|\eta\chi|\delta)}.$$

This does not supply the stronger finite vacuum-energy condition. Time regularity, the weighted Berry generator and the original non-L^2 shell pullback remain separate. A scalar vacuum phase cancels in bounded source currents; it still requires its energy-reference prescription for switching work. Field-only frame changes commute with the original source current, so their gauge phase cannot itself be counted as a clock shift.

The actual bare source-shell time coefficient is now explicit. Put $n=573049$, $T=n\pi$, $\chi=\sin^4(\pi t/T)$ and $\mu_2=1/(4R)$. With the complete original source operators D,C, the squared energy-dual norm of the differentiated combined shell map, divided by $T\mu_2$, is bounded by the full sixteen-state operator

$$K_{\mathrm{shell}}=\frac{5\delta^2}{4n^4}(D^2+C^2).$$

The factor two pays both source directions without assuming their profile-weighted orthogonality. Native continuation retains both ordered current insertions of this positive coefficient. It is a bare shell regularity bound; the static-frame pullback is not substituted for it.

71.4 A causal route tailored to the actual prepared clocks

On all six original radius/time intervals, a common backward ball with $L = R_{\max} + 1.002T_{0,\max}$ satisfies

$$\sup_{r\le L} |b(x) - b(0)| \le \frac{L}{2000 d_{\min}} + \frac{3L^2}{4000 d_{\min}^2} < 6\,10^{-27}.$$

A coefficient comparison that equals the actual Plummer lapse throughout that ball and differs outside it would give exactly the same source current if the full joint causal shell theorem is established. This is an auxiliary operator comparison, with the complete physical field and original profile retained; it is not a particle cutoff or a claim about a nonlocal positive-frequency receiver. Initial hard dressing does not defeat an operator causal identity, but that identity must survive the singular shell limit. This is a concrete alternative to estimating a global far-zone vacuum change as a local clock error.

For a proved joint-state norm comparison r_* with the same uniform-reference history, the retained $\|Z\| = 2$ gives current error $4r_* + 2r_*^2$. Every original lane has reference amplitude above 0.999; thus $r_* \le 4\,10^{-11}$ is a sufficient conditional requirement for a 0.2 nrad phase allowance. No such joint remainder, signed susceptibility or achieved apparatus accuracy is assigned by the global Hilbert–Schmidt estimate alone.

71.5 Two actual age tags and cancellation of a common channel

The retained source preparations at $T_1 = T_0/4$ and $T_2 = T_0/2$ encode actual current ages $t_1 = T_0/8$ and $t_2 = T_0/4$. Their increment is $T_0/8$, rather than a full cycle. Its phase is near $\pi/4$ and lies strictly between zero and π in every lane after the actual source disks are paid. This prefix prior supports unwrapping. Repeated complex means also require preparation of the same source relative phase against the recorded reference; uncontrolled independent phase resets erase the mean.

Retain the original source convention $b_f = \int f a$, whose occupied creation profile is $\overline{f}$. A calibrated mode sorter transfers only that canonical mode to the signal port and retains all complementary outputs. Known centering maps its two age-dependent profiles to the same packet before the same complete annular channel. For a common nonzero complex coefficient C,

$$\beta_j = C\alpha_j, \qquad \beta_2\overline{\beta_1} = |C|^2\alpha_2\overline{\alpha_1}.$$

The full channel phase cancels exactly. This includes the actual annular potential, rather than approximating it away. Its previous 5.601 microradian absolute free-comparator ceiling need not be paid as an independent error in this strictly common experiment. Channel magnitude still determines brightness and noise. If $C_2/C_1 = qe^{i\delta_{\mathrm{ch}}}$, $q > 0$, the remaining phase bias is exactly δ_{ch}. A disk of relative radius $\rho < 1$ about positive real q gives $|\delta_{\mathrm{ch}}| \le \arcsin\rho$. Comparing two physical conditions requires their differential channel drift, not an assumption that unrelated instrumental errors cancel.

A correctly matched real gate creates positive attenuation. After the specified mode processing, the temporal variance is $\pi^2/(3\sigma^2)$. With gate duration $G = 2\pi/\sigma$ and center offset u as a fraction of its halfwidth,

$$\chi(u) \ge 1 - \frac{1}{3(1-|u|)^2},$$

when this is positive. The normalized positive-frequency projection of the same windowed mode has real nonnegative overlap at least $\sqrt{\chi}$. A mismatched local-oscillator mode instead contributes its actual complex overlap. Mode leakage requires a number or energy bound: $|\delta\alpha| \le \delta_{\mathrm{mode}}\sqrt{N_{\mathrm{leak}}}$; a trace-distance bound alone does not control an unbounded quadrature. Hard leakage needs its actual hard-state bound. Buffering, frequency-shaping work, unused outputs, source backaction and the late receiver reference remain paid apparatus inputs.

71.6 A nanoradian requirement, with its resource gap stated

The heterodyne vacuum term remains in $V \ge \langle N\rangle + 1 - |\langle a\rangle|^2$. For mean error ε, blocks of $\lceil 16V/\varepsilon^2\rceil$ independent records and coordinatewise median block means give the retained finite-variance bound. Choose $\varepsilon = 39\,10^{-12}\alpha_-$ and 31 blocks. All 24 complex means (six lanes, two ages, two conditions) then have simultaneous failure below

$$24e^{-279/32} < 0.003924.$$

Four statistical phase disks, 0.020 nrad differential channel mismatch, 0.020 nrad remaining apparatus/reference phase, and the actual intrinsic source-field disks give a total below 0.196000001 nrad. The sufficient fresh-preparation counts per complex mean, rounded upward, are:

lane	preparation $T_0/4$	preparation $T_0/2$
01	$2.04632\,10^{59}$	$5.66920\,10^{58}$
02	$4.76378\,10^{58}$	$1.31978\,10^{58}$
03	$3.07426\,10^{56}$	$8.51703\,10^{55}$
12	$1.77875\,10^{59}$	$4.92791\,10^{58}$
13	$3.32718\,10^{56}$	$9.21774\,10^{55}$
23	$3.63479\,10^{56}$	$1.00700\,10^{56}$

These are explicit ideal-instrument requirements, not achieved precision, a preparation rate or optimal copy lower bounds. Both conditions require their own samples. In the actual $0 < \omega < 2\sigma$ band, the stated combined reference/timing allocation requires $\sum|\delta t| \le 0.020\,\mathrm{nrad}/(2\sigma)$, about 0.002026 model time units, unless a measured common timing error cancels. The SI chart remains a declared fixture, not a physical source-unit or detector calibration.

The retained full-window negative phase margin is at least 4.30922 nrad, so this hypothetical 0.2 nrad metrology budget fits it. It does not assign that full-window signal to the shorter tag increment. That comparison needs its perturbed source-emission bridge and actual mixed kernels. The original full-window isolated susceptibility sign remains unresolved; the new signed prefix has a different, explicitly tiny time scale. More samples also do not remove the approximately 15 percent source-amplitude calibration interval from distance inference or supply a geometric aperture law.

71.7 Controlled amplitude, phase and instantaneous rate

For an actual nonzero current z_s write

$$y(t) = w(t)[1+u(t)], \qquad w(t) = A(t)e^{i\chi(t)}z_s(\tau(t)),$$
$$A(t) > 0, \qquad |u(t)| \le \rho < 1.$$

With clock phase $\theta = -\arg z$ and rate $\nu = \dot\theta$,

$$|\theta_y - \theta_s(\tau) + \chi| \le \arcsin\rho,$$
$$\nu_y = \dot\tau\,\nu_s(\tau) - \dot\chi - \operatorname{Im}\frac{\dot u}{1+u},$$
$$|\nu_y - \dot\tau\,\nu_s + \dot\chi| \le \frac{|\dot u|}{1-\rho}.$$

Positive amplitude alone changes neither phase nor rate. A source-state scalar Hamiltonian phase cancels from expectations altogether. Physical channel phase, time warp and additive residual instead need their actual bounds. The original instantaneous rate intervals can be propagated through this formula; a rate sign requires the resulting interval to exclude zero. An endpoint slowdown enclosure or a small phase disk supplies no bound on $\dot u$ and does not certify an instantaneous slowdown sign. Signal zeros or loss of an amplitude floor invalidate phase tracking.

For the selected receiving form on the retained two-polarization radial space, with $W = J^*NJ$, the exact one-particle operator is $h_W = \sqrt{\omega}W\sqrt{\omega}$, and its source trace is $v_{W,\alpha} = \sqrt{\omega}WF_\alpha$, $F_\alpha = u_\alpha/\sqrt{\omega}$. This trace is generally energy-dual, not an ordinary Hilbert vector. Use positive- and negative-half-energy smearings. Source-trace regularizations must converge in $\|h_W^{-1/2}(v_n - v_W)\|$, with retained counterterms, convergent joint forms, strong propagator convergence and uniform energy bounds. This yields a weak matrix-element limit for original form-domain test vectors; it does not require $a(F)$ to preserve that form domain. For a creation-profile smearing F the stationary weak relation is

$$a(F,T) = a(E_W(T)^*F, 0) + ig\sum_\alpha \int_0^T \langle F, E_W(T-s)v_{W,\alpha}\rangle S_\alpha(s)\,ds.$$

Choosing $F = E_W(T)F_{\mathrm{soft}}$ removes the initial hard homogeneous term exactly for that adapted observable, while retaining all hard state and radial hard/soft space. Physical angular return belongs to the separate canonical full-bundle problem. It does not evaluate the remaining two-current kernels or implement the altered detector. Keeping the free kernel while merely substituting a perturbed current is unjustified. A nonautonomous extension also needs homogeneous half-energy propagation bounds; shifted common-form existence alone does not supply them. The common-form result above supplies the selected receiving dynamics; full canonical pair creation remains a separate construction.

The actual real-time kernel is a complex spectral Fourier response. It is not the positive Euclidean Laplace observable. Physical relaxation or cosmological Q is phase-free only after proving the positive-multiplier hypothesis with the required residual and time-warp bounds. A convolution, a changing canonical kinetic coefficient, or a changed source vertex can add phase; no physical rate spectrum or Q law is selected here.

Status after the preceding continuation. The following retained six-question status is superseded where specified by the canonical, local-instrument and real-time composition sections that follow.

71.8 Status of the six physical questions

Local stress. The full rank-two tensor Green kernel and complete global relative-pair Hilbert–Schmidt bound now control both canonical kernel contributions. The static operator-domain realization, vacuum energy finite part, Berry/source-shell time control and actual six-lane current comparison remain. The old receiving-form signs have not been transferred to the canonical quantum model.

Isolated sign. The actual parallel profile has signed initial and finite-prefix results in three lanes. The 10^{-400} amplitude/time witness has no practical precision claim. Full-window, orientation-dependent and cross-lane differences require sharper correlated two-time estimates.

Spatial incidence. The explicit isometry and positive multiplication profile define the selected residual-energy receiver with its original error budgets. A canonical field's angular exit and return are additional dynamics; the map alone does not eliminate them.

Joint propagation. Common-form existence, energy control and weak current identities are established for the positive receiving Hamiltonian. The finite-prefix theorem evaluates one observable within that evolution. A general-time prepared-state evaluation and the switched full canonical source–field propagator are still required.

Apparatus. A common channel cancels from matched age differences, and the finite-variance nanoradian design states its actual source-derived copy requirements. Achieved preparation, leakage, drift, timing, reference and source-unit calibration remain experimental operands. The original full-window signal has not been assigned to the shorter emitted-tag interval.

Relaxation and amplitude. Positive multiplication is exactly phase-free; controlled complex residuals and time warps have explicit phase and rate bounds. Physical relaxation and cosmological amplitude laws have not been derived. A convolution or a changing source/field coupling can add phase and must satisfy the displayed bounds before a sign can be inherited.

The most direct continuation is to sharpen the original two-time insertion bound at useful prefixes, while proving the static canonical block-form admissibility and then its switched shell/Berry control. This preserves the same source and geometry instead of assigning the missing correlation, vacuum reference or emission clock.

Retained canonical continuation. The following sections state the C6 results. The later finite-preparation continuation closes

switched joint existence by a different source-admissibility theorem; its local-output transfer remains open.

72. The canonical joint static Hamiltonian and its original shell limit

The full global relative-pair estimate now has an operator realization. Write the original complete canonical number and pair forms as A_N, B_N. The positive block-form theorem of Dereziński (arXiv:1608.03289v2, Theorems 3.20–3.21) applies directly to

$$\|A_N^{-1/2} B_N A_N^{-T/2}\| < 1, \qquad \|A_N^{-1/2} B_N A_N^{-T/2}\|_{\mathrm{HS}} < \infty.$$

It constructs the positive closed classical block form and an implementable symplectic diagonalizer. Thus the previous operator-domain proviso is removed. The actual Plummer field has a positive self-adjoint static quantization K_N. We specify $\inf K_N = 0$ for each fixed profile; this static scalar convention does not calculate the work of a changing vacuum.

The unchanged source shell also closes in this quantization. Let $n_- = 1997/2000$, $n_+ = 2003/2000$, $c_\mu = n_+^2/n_-$ and $S_2 = D_{\mathrm{source}}^2 + C_{\mathrm{source}}^2$. In static normal modes the original, generally non-L^2, shell smearing $v_{\alpha,N}$ satisfies

$$\|h_N^{-1/2} v_{\alpha,N}\|^2 = \langle \xi_{\alpha,N}, D_N^{-1} \xi_{\alpha,N} \rangle \le c_\mu \mu, \qquad \mu = \frac{1}{4R}.$$

The normalization is that of the original u_R. The complete angular shell eigenvalues $[\kappa R(2l+1)]^{-1}$, $l \ge 2$, are bounded by μ; $D_N \ge n_- D$ and the actual shell multiplier is bounded by n_+. For every normalized entangled source–field vector,

$$|\langle H_{\mathrm{int,N}} \rangle| \le 2\varepsilon \sqrt{2c_\mu \mu \langle S_2 \rangle \langle K_N \rangle} \le \tfrac{1}{2} \langle K_N \rangle + 4\varepsilon^2 c_\mu \mu \langle S_2 \rangle.$$

Hence the full sixteen-state form, including its original even energy and both original counterterms,

$$H_{\mathrm{joint},N} = K_N + \overline{N}_R \varepsilon H_{\mathrm{pair}} + H_{\mathrm{int,N}} + \overline{N}_R \varepsilon^2 \mu S_2,$$

is closed and bounded below. It defines a self-adjoint operator and a unitary $e^{-itH_{\mathrm{joint},N}}$ on the original Fock space. Every retained preparation has a well-defined bounded source-current evolution. This statement does not require that a sudden change of profile give that preparation finite new-profile energy.

The shell limit is quantitative and uniform in the angular complement. Its exact energy-dual kernel is

$$\mu_l(r,r') = \frac{r_<^l}{\kappa(2l+1) r_>^{l+1}}.$$

For $r = R(1+x)$, $0 \le x \le h$, Bernoulli's inequality gives

$$\begin{aligned} \|u_{r,l} - u_{R,l}\|_{\mathrm{dual}}^2 &= \frac{1 + (1+x)^{-1} - 2(1+x)^{-l-1}}{\kappa R(2l+1)} \\ &\le \frac{3h}{2R}, \qquad l \ge 2. \end{aligned}$$

A positive smooth radial proof mollifier inherits this bound. With the actual N_R angular profile kept fixed, the combined original source form-error coefficient is

$$12 c_\mu \varepsilon^2 h \mu S_2 \longrightarrow 0.$$

This is norm-form convergence against K_N, giving norm-resolvent convergence and compact-time strong convergence of the joint static unitaries. The original shell and counterterms are recovered; the mollifier is not a physical cutoff. For smooth shells, local hyperbolic finite propagation and the ordinary source interaction-picture construction give equality of source observables for static profiles agreeing on their whole backward causal ball. The shell limit passes this equality to the original joint dynamics. This establishes static source causality even on the initially dressed state. It does not yet identify the two evolutions of unequal profiles.

73. A finite local detector moment from the original preparation

A compact smooth detector test f obeys $f, \omega^{1/2} f, \omega^{-1/2} f \in L^2$. For the complete classical canonical flow, conservation of static energy, symplectic duality and interpolation give the backward-test bounds

$$\begin{aligned} \|\omega^{\pm 1/2} f_s\| &\le C \|\omega^{\pm 1/2} f\|, \\ \|f_s\| \le C\|f\|, \qquad C_{\mathrm{static}} &= \sqrt{n_+/n_-}. \end{aligned}$$

For the stated smooth $\sin^4$ pulse, its total variation is two and

$$C \le \sqrt{n_+/n_-} \exp\left(\frac{|\eta| \delta\,\mathrm{TV}(\chi)}{2n_-} \right) < 1.00301.$$

These individual-test estimates retain the real symplectic pair mixing. They do not assert global weighted Hilbert–Schmidt control.

For the static joint unitary just constructed, the weak field equation against a smooth detector gives exactly

$$X_f(t) = \Phi(f_{\mathrm{free}}) + \varepsilon \sum_\alpha \int_0^t R_\alpha(t,s) J_\alpha(s)\, ds.$$

The detector's smoothness, rather than a fictitious shell L^2 norm, pays the retarded pairing:

$$|R_\alpha(t,s)| \le 2C n_+ \sqrt{\mu}\, \|\omega^{1/2} f\|.$$

Thus the source term has operator norm at most

$$A_f = 2\varepsilon C n_+ \sqrt{\mu}\, t \sum_\alpha \|J_\alpha\|\, \|\omega^{1/2} f\|.$$

Let $E_0 = \langle H_f \rangle$ in the original hard-dressed preparation, $a = C\|\omega^{-1/2} f\|$, and $b = C\|f\|$. Then

$$\langle X_f^2 \rangle \le \left[a\sqrt{E_0} + \sqrt{a^2 E_0 + b^2} + A_f \right]^2.$$

A positive normalized temporal gate inherits the bound by integration. It supplies a finite apparatus moment without requiring

a final global old-Fock energy or photon-number estimate. A short detector gate whose causal diamond excludes a source roundtrip has the free canonical c-number commutator during that gate; this supports the exact local Ramsey construction below. Its signed mean and calibration retain their separate source and instrument certificates.

74. A weighted normal frame closes the free canonical switch

The critical source pullback can be controlled by a Hilbert–Schmidt construction instead of a generic square-root operator estimate. Write the actual number and pair forms against the FIXED original frequency:

$$h_s = \omega + \omega^{1/2} H_s \omega^{1/2},$$
$$g_s = \omega^{1/2} G_s \omega^{1/2},$$
$$\|H_s\| \le |s|\delta, \qquad \|G_s\|_{\mathrm{HS}} \le |s| k_0,$$
$$k_0 = 119\delta/48.$$

Here $H_s = sH_b$, $G_s = sG_b$ and $\delta = \sqrt{2}/1000$. For the positive real symplectic normal map $T = \left(\begin{smallmatrix} p & q \\ q & p \end{smallmatrix}\right)$ put $Y = pq$, $X = q^2 = f(Y)$, $f(z) = (\sqrt{1+4z^2}-1)/2$ and $W = \omega^{1/2} Y \omega^{-1/2}$. Its exact fixed-point equation is

$$W = -\mathscr{M}_\omega\big[G_s + H_s W + W^* H_s + G_s f(W) + f(W)^* G_s\big],$$
$$\mathscr{M}_\omega(a,b) = \frac{a}{a+b}.$$

This Schur multiplier is a contraction on Hilbert–Schmidt operators; no false general operator-norm multiplier estimate is needed. The ball $\|W\|_{\mathrm{HS}} \le 2k_0$ is invariant, with contraction

$$2\delta + 16k_0^2 < 0.003026 < 1.$$

Spectral-core closure gives the exact weighted functional calculus. Purity $Y^2 = X + X^2$ supplies the remaining stationary equation: its residual F obeys $\{F, I+2X\} = 0$ and hence vanishes by the exponentially decaying Sylvester identity on the energy form space. The complete proof retains this domain step in `canonical/WEIGHTED_RICCATI_PROOF.md`. Consequently

$$\|\omega^{1/2} q \omega^{-1/2}\|_{\mathrm{HS}} \le 3k_0,$$
$$\|\omega^{1/2}(p-I)\omega^{-1/2}\|_{\mathrm{HS}} \le 5k_0^2.$$

The fixed point and frame are real-analytic in s. For the ACTUAL $s(t) = \eta \sin^4(\pi t/T)$ this supplies a bounded number connection and an integrable Hilbert–Schmidt pair connection. The positive diagonal frequency is a common C^1 form between 0.96ω and 1.04ω. Add a scalar I to apply the common-form evolution theorem on the massless energy domain, then remove its phase. One-particle propagation followed by the bounded doubled Volterra equation preserves the restricted symplectic group. Its continuous Fock lift is therefore the actual FREE canonical quantum propagator, with exact composition. This does not assert finite weighted vacuum energy or complete the singular source's switched interaction by itself.

75. Useful-time static canonical clock phases in all six lanes

The same weighted proof gives quantitative constants for a compact causal AUXILIARY profile matching the ACTUAL static Plummer difference throughout the whole backward source ball. With support $r \le 2L$ and

$$|b_c| \le A, \qquad |\nabla b_c| \le 3A/L, \qquad |\Delta b_c| \le 15A/L^2,$$

the full Green-kernel calculation gives the complete relative-pair Hilbert–Schmidt bound $42A$. Its positive normal frame yields

$$|h_{\mathrm{diag}} - n\omega| \le 500A\omega,$$
$$\|\omega^{-1/2}(v_{\mathrm{pulled},\alpha} - n u_\alpha)\| \le 220A\sqrt{\mu}.$$

Exact static source causality then removes the auxiliary exterior. There is no physical spatial truncation.

For each original lane use its actual native intervals and set

$$L = R_+ + 1.002T_{0,+}, \qquad d = 10^{30} R_-,$$
$$A = \eta\left[\frac{2L}{2000d} + \frac{3(2L)^2}{4000d^2}\right],$$
$$n = 1 - \eta/1000, \qquad k = \frac{42A}{n-A},$$
$$\rho = 2000A.$$

Lane-even source energies and BOTH original counterterms are scalar within each retained invariant lane. Align their exact scalar phases in the bounded-current comparison, while retaining their energies in the ledger. The positive conserved form $X = H_{0,\mathrm{center}} + c + 2\varepsilon \ge K + c + \varepsilon$, $c = 2g^2\mu$, satisfies $H_f \le 2(K+c)$ and

$$|\delta q| \le A[500H_f + 880|g|\sqrt{\mu H_f} + \varepsilon] \le 1880AX \le \rho X.$$

Put $B = K_{\mathrm{prep}} + c + 4\varepsilon$ and let N_0 be the original hard-preparation photon ceiling $2g^2 b(64/63)^2$. The common-form overlap and initial squeeze estimates give

$$r_{\mathrm{dyn}}^2 \le 2\rho T_0 B\sqrt{\frac{n+\rho}{n-\rho}}, \qquad r_{\mathrm{frame}}^2 \le \frac{4k(N_0+1)}{1-2k}.$$

The source current COMMUTES with the field frame. Only the initial squeezed vector is replaced; no evolved photon-number or old-Fock energy bound is assumed. For $Z = 2\sigma_-$, put $e_z = 4(r_{\mathrm{dyn}} + r_{\mathrm{frame}})$ and $m = M_- - \mathscr{E}(T_0) > 0$. Then

$$|\theta_{\mathrm{can}}(T_0) - \theta_{H_0}(nT_0)| \le \frac{e_z}{m - e_z}.$$

Exact rational replay gives the following nanoradian ceilings:

lane	$\eta = 1$	$\eta = 7\,10^{-7}$
01	0.099821	0.000083514
02	0.100437	0.000084029
03	0.185493	0.000155192
12	0.099849	0.000083538
13	0.180452	0.000150975
23	0.175110	0.000146505

ALL six lanes meet 0.2 nanoradians at the original full T_0. The second column of profiles has the physical $\alpha = 7\,10^{-10}$. This is an actual STATIC canonical source-clock certificate against the same interacting uniform history. It is distinct from the still-needed switched-source, detector output-mean and paid vacuum-energy certificates.

76. An integrable-covariance construction for the switched source

There is a direct route around differentiating the singular shell. In the complete canonical free-field interaction picture let $C_{\alpha\beta}(t,s)$ be its original-vacuum shell covariance. Suppose mixed smooth proof regularizations converge in $L^1([0,T]^2)$, with an integrable common envelope and convergent coherent first moments. Put

$$K = \sum_{\alpha\beta} \int_0^T \int_0^T |C_{\alpha\beta}(t,s)|\, dt\, ds, \qquad J = \max_{\alpha,t} \|B_\alpha(t)\|,$$

where the bounded full-source B_α include their actual coupling. Wick contraction and the two ordered simplices give the nth vacuum Dyson-term bound

$$\|D_n\Omega\| \le \frac{J^n \sqrt{(2n-1)!!\,K^n}}{n!}.$$

The ratio is $J\sqrt{K(2n+1)}/(n+1) \to 0$ for every finite K. Mixed-kernel convergence passes each term to the original shell; the same bounds for reversed time give convergence of the inverse and therefore a unitary source interaction cocycle. Coherent and finite-particle vectors extend the result by density to the actual non-Gaussian prepared state. No Gaussian approximation to that preparation is made.

The original flat shell already satisfies the kernel hypothesis:

$$C_R(\tau) = \frac{Q_2(1+(0^+ + i\tau)^2/(2R^2))}{\kappa\pi R^2},$$
$$Q_2(z) = \frac{3z^2-1}{4}\log\frac{z+1}{z-1} - \frac{3}{2}z.$$

This is the ordinary Legendre Q_2, with normalization fixed by DLMF 10.22.66. Its logarithms at $0, \pm 2R$ are integrable and its tail is $O(|\tau|^{-6})$. Direct bounds give $\int_{\mathrm{R}} |C_R| < 402/(\kappa\pi R)$. Exponential proof smoothing is a Poisson approximate identity in this L^1 boundary value. Thus the original singular source admits this unitary cocycle construction with bounded time controls.

For the actual nonuniform switched profile the remaining operand is the complete two-time covariance and its L^1 comparison/regularization certificate. The actual free switched implementation is established above. Once its actual shell covariance meets the stated integrability certificate, that free implementer times the source cocycle supplies the full joint Schrödinger propagator. The source cocycle theorem by itself does not replace this free implementation proof. The static joint implementation is already established. A finite renormalized ramp-energy account remains separate from all bounded observable identities.

77. Real elapsed-time memory and finite emission from the original preparation

The positive Euclidean spectrum is not an extra real-time damping law. The original canonical source already supplies a non-Markovian elapsed-time model. Retain all16 source states, both currents, both counterterms, the complete hard field and the full initially vacant soft field. Put $Z_{\mathrm{raw}} = D + iC = D - \mathrm{Swap}\, D$ and $B = \mathrm{Swap}\, D^2$. The exact source algebra and complete field elimination give, as time-smeared energy-form identities,

$$\dot{Z}_{\mathrm{raw}}(t) = -i\Delta Z_{\mathrm{raw}}(t) - 2i\varepsilon B(t) F_{\mathrm{free}}(t) - 2i\varepsilon^2 \int_0^t \chi_{RR}(t-s) B(t) Z_{\mathrm{raw}}(s)\, ds, \tag{45}$$

$$\chi_{RR}(\tau) = \frac{P_2(1-\tau^2/(2R^2))}{\kappa R^2} \mathbf{1}_{(0,2R)}(\tau). \tag{46}$$

The initially hard-dressed F_{free} is correlated with the source and has a nonzero mean. Its noise term and $\langle B(t) Z_{\mathrm{raw}}(s)\rangle$ remain ordered quantum quantities. This is an exact operator memory equation, not a factorized scalar equation. No sharp-time ultraviolet field variance is assigned.

For $p(x) = 1 - 3x^2/2 + 3x^4/8$,

$$m_n = \int_0^2 x^n p(x)\, dx = \frac{2^{n+1}(n-1)(n-3)}{(n+1)(n+3)(n+5)}. \tag{47}$$

The first eight moments are

$$(m_0, m_1, m_2, m_3) = (2/5, 0, -8/105, 0),$$
$$(m_4, m_5, m_6, m_7) = (32/105, 16/15, 640/231, 32/5).$$

In particular $\int \chi_{RR} = 2\mu$, and the first and third odd moments vanish. The first dissipative susceptibility term is $\mathrm{Im}\,\widehat{\chi}_{RR}(\omega) = 2R^4\omega^5/(225\kappa) + O(\omega^7)$. The negative moment determinant $m_0 m_2 - m_1^2 = -16/525$ excludes a positive time-weight interpretation of this signed causal kernel.

The same original vertex has a real-time vacuum covariance with an algebraic tail. For $t > 2R$,

$$C_{\mathrm{vac}}(t) = -\frac{I(t/R)}{\kappa\pi R^2},$$
$$I(y) = \sum_{j\ge 2} \frac{m_{2j+1}}{y^{2j+2}} > 0. \tag{48}$$

so $C_{\mathrm{vac}}(t) = -16R^4/(15\kappa\pi t^6) + O(t^{-8})$. It cannot equal a positive mixture of decaying exponentials. This is a vertex/covariance identity, not a replacement of the actual initially non-Gaussian hard state by a full vacuum. Compact retarded memory does not imply compact noise memory.

An actual two-time remainder and emitted-number theorem. Let the original exact hard branches have gap Ω and overlap M, with unchanged preparation $(\Phi_0 + \Phi_1)/\sqrt{2} \otimes |0_{\mathrm{soft}}\rangle$, $\lambda = 64\Delta$, and $v_s = \int_0^\lambda |u_R(k)|^2 dk$. For bounded hard/source operators within each original invariant lane,

$$|\langle A(t)B(s)\rangle - \langle A_h(t)B_h(s)\rangle| \le \|A\|\|B\|\,\mathscr{E}(T), \quad s,t \le T, \tag{49}$$
$$\mathscr{E}(T) = e^{4g^2 v_s T^2} \quad 1.$$

To prove it, use the forward/backward contour of total length $2\max(s,t)$. Odd soft-vacuum words vanish; the order$2n$ term is bounded by $2^n(2n-1)!!v_s^n g^{2n}(2T)^{2n}/(2n)! = (4g^2v_sT^2)^n/n!$. Summing pays every order, both polarizations and noncommuting hard insertions. No quantum regression assumption is used.

Define $D_T(a) = \int_0^T e^{ias}ds$ and

$$I_\lambda(T) = \int_0^\lambda |u_R(k)|^2 \times \left(|D_T(k-\Omega)|^2 + |D_T(k+\Omega)|^2\right) dk. \tag{50}$$

Projection onto the existing hard branches gives the positive component $g^2M^2I_\lambda$. The exact hard complement contributes between zero and $2g^2v_sT^2(1-M^2)$: its instantaneous summed squared source norm is $2(1-M^2)$, and time Cauchy–Schwarz supplies the bound. Combining this with (49), put $B_T = g^2M^2I_\lambda$ and $P_T = 2g^2v_sT^2$. Then

$$\langle N_{\rm soft}(T)\rangle \geq \max\{0, B_T - P_T\mathscr{E}\}, \qquad \langle N_{\rm soft}(T)\rangle \leq B_T + P_T(1-M^2+\mathscr{E}). \tag{51}$$

Both initial cross terms vanish because the soft annihilators kill the actual initial state. This argument is not applied to the hard field. The $k+\Omega$ counterrotating contribution is retained. The full hard complement is bounded rather than replaced by a finite photon basis.

At the original $T_0 = 573049\pi$, conservative positive analytic endpoints give the following actual emitted-number enclosures:

Lane	Lower bound	Upper bound
01	$1.65790\,10^{-35}$	$2.86711\,10^{-28}$
02	$7.12168\,10^{-35}$	$1.23170\,10^{-27}$
03	$1.10354\,10^{-32}$	$1.94252\,10^{-25}$
12	$1.90730\,10^{-35}$	$3.29841\,10^{-28}$
13	$1.01965\,10^{-32}$	$1.79246\,10^{-25}$
23	$9.33366\,10^{-33}$	$1.63851\,10^{-25}$

These are finite-time quantum-emission bounds, not thermal occupation or a claimed precise Markov rate. The soft free energy is at most $\lambda\langle N_{\rm soft}\rangle$; source inventory still obeys the complete Hamiltonian/action account, including interaction energy.

A finite real-time coherence window. For the actual source, $|z(t) - Me^{-i\Omega t}| \leq \mathscr{E}(t)$. If the retained outward bound is $E_0 \geq \mathscr{E}(T_0)$, then for $x = N^2E_0 < 1$,

$$\mathscr{E}(NT_0) \leq \frac{x}{1-x}, \qquad |\delta\theta| \leq \frac{\mathscr{E}}{M_- - \mathscr{E}}. \tag{52}$$

For $N = 10{,}000$, the largest six-lane phase bound is $4.115\,10^{-13}$rad, below 0.001nrad. The same interacting preparation remains coupled to every original mode. This establishes a finite-window coherence bound; it does not assert absence of eventual relaxation. An exact extra factor $e^{-\gamma t}$ beginning at preparation would require $M(1-e^{-\gamma t}) \leq \mathscr{E}(t) = O(t^2)$, hence $\gamma = 0$. At a fixed endpoint any compatible such interpretation must satisfy $\gamma \leq E_0/[(M_- - E_0)T_0]$; this is a constraint, not a fitted rate.

A compact physical receiver floor. Use the same two TT angular patterns with any nonnegative normalized smooth radial weight in $[R+64-R/10, R+64+R/10]$, and a nonnegative symmetric normalized gate of duration at most R, centered at $T_0/4$. The hard total prepared field has mean $gM\chi_h(\Omega;r)e^{-i\Omega t}$, including the correlated incoming field. The exact identity $\chi_{\rm out} = (2i\Omega/\kappa)j_2(\Omega R)h_2^{(1)}(\Omega r)$, the complete soft principal-value pole subtraction and positive radial averaging give hard mean magnitude floors, in lane order,

$$(4.8268, 10.0039, 124.531, 5.1771, 119.704, 114.527)\,10^{-12}. \tag{53}$$

The original negative g orientation is preserved. Symmetry of the short gate multiplies the harmonic by a positive real coefficient and preserves its phase exactly. The actual soft-drive and source-current remainder give a phase disk at most $1.436\,10^{-10}$rad (0.1436nrad), before detector, calibration and canonical-comparison errors. The complete compact retarded kernel and its incoming hard mean are kept. This is a total prepared local field measurement, distinct from isolating the very weak soft radiation packet. The annulus/TT angular coupling is a finite extended apparatus whose implementation is charged separately.

For a positive spatially uniform lapse, the COMPLETE original Hamiltonian is exactly $n(t)H_0$, so all results transfer without approximation to $\tau(t) = \int_0^t n(s)ds$. For $n = 1 - \eta\sin^4(\pi t/T)/1000$, $\tau(T) = T(1-3\eta/8000)$. A gate $h_{\rm coord}(t) = n(t)h_\tau(\tau(t))$ preserves the exact symmetric-gate phase; its ramp-area/timing calibration is explicit.

The autonomous real-time result removes an optional phenomenological relaxation factor for this finite protocol. The switched nonuniform canonical comparison must still transfer the actual current, Green map and initial homogeneous field together. One-time current control alone does not transfer two-time photon noise; source-inserted channel control or a joint-state plus detector-moment estimate is required. An external QW1 readout state is not silently substituted for the source's soft vacuum.

78. A finite local quantum instrument and its apparatus gap

The original total prepared near field admits a compact local readout without extracting an assigned outgoing mode. This is a different observable from the extremely weak soft packet: its hard incoming field, soft drive and original two-polarization source remain present. The following apparatus Hamiltonian is an explicit additional measurement law; its practical fabrication and calibration have not been demonstrated.

78.1 Compact cap and actual source amplitude

Let $r_0 = R+64$, take a positive normalized C^∞ radial weight on $r_0 - R/10 < r < r_0 + R/10$, and use a positive symmetric normalized time gate of duration $L = R$, centered at the original age $T_0/4$. The actual canonical source angular incidence is

$\mathscr{U}E_\alpha = \sqrt{5/(8\pi)}\Lambda_n E_\alpha$. For the normalized real plus/cross pair its azimuthally averaged squared TT norm is $(1+6z^2+z^4)/8$, $z=\cos\theta$, and its averaged cross contraction vanishes. An axisymmetric cap therefore has Gram matrix ηI_2, without discarding the angular complement from the noise.

For retained rational radius endpoints put $s = R_-^2/(800r_+^2)$, $c = 1-s$, and choose $\chi(z) = \exp[-t/(1-t)]$ for $t = (1-z)/s < 1$, zero otherwise. Then

$$\eta = \frac{5}{32}\int_c^1 \chi(z)(1+6z^2+z^4)\,\mathrm{d}z, \qquad \eta_- := \frac{5c^2 s}{24} \le \eta \le \eta_+ := \frac{5s}{4}. \tag{54}$$

The lower bound uses the inner half cap, where $\chi \ge e^{-1} \ge 1/3$, and the averaged squared norm is at least z^2. The transverse cap diameter is at most $R_-/10$ and the radial span is $R/5$; the test is one compact detector cell smaller than its gate duration. The actual original mean is exactly η times the previous total local mean. Thus the source-bound cap amplitude floors range from 8.1621×10^{-21} to 2.1059×10^{-19}, with the same propagated phase disk below 0.143517 nanoradians. No brightness is assigned to an external encoder. This angular identity is an original H_0, or uniform-lapse, result; nonuniform angular returns require their own local comparison.

78.2 Exact local pointer coupling

The complete field here is the local rank-two tangential tracefree tensor bundle of the canonical action, with $D = \nabla^*_{\text{bundle}}\nabla_{\text{bundle}} + 4/r^2$ and the regular Friedrichs origin. It is not subjected to an additional nonlocal three-dimensional divergence projector. Multiplying its test by the cap window preserves its pointwise bundle constraints and retains every angular channel generated by localization.

For either original polarization let $X = \int h(t)\Phi(F_{\text{cap}},t)\,\mathrm{d}t$. Because $L < r_0 - R/10 - R$, the backward domain of dependence during the detector gate does not reach the source shell. On the actual, generally source-entangled initial field data at gate entry, the complete local canonical field in this worldtube has its free c-number commutator. This protects the earlier source event by causality; it does not set the hard initial field to vacuum.

Use an initially independent finite-energy pointer, $[Q,P] = i$, with positive kinetic Hamiltonian $P^2/(2m)$ and switched interaction

$$H_D(t) = \lambda h(t) P \Phi(F_{\text{cap}}). \tag{55}$$

The exact interaction-picture propagator is

$$U_D = \exp[-i\lambda PX - i\lambda^2 c_h P^2], \qquad c_h = \frac{1}{2}\int_{t>s} h(t)h(s)\Delta(t,s)\,\mathrm{d}t\,\mathrm{d}s, \tag{56}$$

where $[\Phi(t),\Phi(s)] = i\Delta(t,s)I$. All higher Magnus commutators vanish. Including free pointer drift, set $K = L/(2m) + \lambda^2 c_h$. The output is exactly $Q_{\text{out}} = Q + 2KP + \lambda X$. For comparison, a fully calibrated total-shear preparation would have

$$\mathrm{Var}(Q+2KP) = \sigma^2, \quad \mathrm{Var}\,P = \frac{1}{4\sigma^2}, \qquad \mathrm{Cov}_{\text{sym}}(Q,P) = -\frac{K}{2\sigma^2}. \tag{57}$$

Its covariance determinant is $1/4$ and all moments are finite. The primary protocol requires less: pre-focus only the known free drift $K_{\text{free}} = L/(2m)$ and keep the complete, possibly uncomputed, Magnus area as centered noise. Since $|c_h| \le V_c/2$, choosing $\sigma^2 = V_c/2$ gives total pointer variance at most $\sigma^2 + V_c^2/(4\sigma^2) = V_c$. Both pointer means vanish, so the record remains exactly unbiased for every c_h. The free chirp is a paid finite Gaussian preparation, rather than an inserted negative kinetic evolution. The record $Z = Q_{\text{out}}/\lambda$ is a spectral record of X convolved with independent centered Gaussian noise:

$$\mathrm{E}Z = \langle X\rangle, \qquad \mathrm{E}Z^2 \le \langle X^2\rangle + V_c =: B. \tag{58}$$

The field need not be coherent or Gaussian. Separate fresh preparations are used for plus and cross, so no cross-probe phase is suppressed.

78.3 Finite moments, bins and repetitions

The original radial smearing has $D_w = \|\sqrt{k}f_w\|^2 = \kappa^{-1}\int w^2/r^2$ by exact Bessel completeness. With $a = R_-/10$,

$$D_w \le \frac{3}{2\kappa a r_-^2}, \quad \mu_w \le \frac{1}{5\kappa r_-}, \quad \|f_w\|^2 \le \sqrt{\mu_w D_w}. \tag{59}$$

For the complete cap test use $D_c \le \eta_+ D_w$, $\mu_c \le 4r_+^2 D_c$ and $V_c \le \sqrt{\mu_c D_c}$. These include all angular-complement vacuum noise. The actual original prepared energy satisfies $E_{f,0} \le 2K_{\text{prep}} + 2C_{\text{ct}}$. The exact bounded source part has norm at most $\eta_+ 2|g|/(\kappa r_-)$, yielding

$$\langle X^2\rangle \le \left[2\sqrt{\mu_c E_{f,0}} + \sqrt{V_c} + \eta_+\frac{2|g|}{\kappa r_-}\right]^2. \tag{60}$$

No evolved old-Fock energy is assumed finite. The full static canonical case uses the backward-test bound in the canonical section instead.

Choose $\lambda = 1$, $\sigma^2 = V_c/2$ and, for cap amplitude floor A, scalar allowance $e = 10^{-11}A/2$. Clip each pointer record to $[-L_q, L_q]$, $L_q = 8B/e$, and round to bins of width $d = e/4$. The bias is at most $B/L_q + d/2 = e/4$, while the record variance is at most $V_q = 2B + d^2/2$. A block of $b = \lceil 32V_q/e^2\rceil$ records has probability at most $1/8$ to miss its mean by $e/2$. A median of 31 blocks has simultaneous failure below 0.001962 for all twelve scalar means. The complex mean disk is at most $10^{-11}A$.

lane	copies per polarization	bits
01	8.692×10^{58}	191
02	2.035×10^{58}	189
03	1.491×10^{56}	182
12	7.558×10^{58}	191
13	1.605×10^{56}	182
23	$1.743\quad10^{56}$	182

The actual exact integers and rational bin boundaries are in the replay table. Including the source/local-field disk and a separately calibrated 0.020 nanoradian preparation, timing and apparatus phase allowance gives a total below 0.174 nanoradians. This last calibration allowance has not been measured. If allocated entirely to epoch error it requires about 5.731×10^{-6} original model time units; an SI statement still requires the physical unit calibration.

78.4 Work and the remaining engineering gap

Choosing $1/(2m) = 1 + 4h_{\max}^2 \mu_c$ ensures a semibounded joint Hamiltonian by completing the field square. The stated cap design has finite initial pointer kinetic energy bounded by 1.387×10^7 model energy units and mean deposited field energy below 0.003896 per copy. Focusing, switching, local coupling, readout memory, phase reference and fresh original-state preparation remain in the apparatus ledger. Detector radiation can disturb later source evolution after its return; this protocol uses one gate per fresh preparation. It does not certify an unperturbed endlessly reused source.

The readout ranges are about 6.3×10^{23}–1.44×10^{25} internal pointer units, with bin widths about 10^{-32}–2.7×10^{-31}. These extreme sufficient resources quantify the laboratory gap. The new result is a complete finite local instrument within its explicit dynamical law and original/uniform reference scope; it is not evidence that such precision has been engineered, nor a transfer of the nonuniform positive-receiving-form sign to the full canonical detector.

79. Composition of the original clock, local detector and finite record

The results above now supply a complete finite reference protocol from the original prepared source to a local classical phase record. Its scope is the original autonomous canonical law and its positive uniform-lapse time change. The actual static nonuniform canonical source-clock comparison is also certified. The switched nonuniform local-output join is stated below as a specific covariance and transported-test calculation.

One composed protocol. Prepare the original hard eigenbranch superposition and soft vacuum, with its actual signed coupling and phase reference. Evolve the complete joint Hamiltonian to age $T_0/4$. On separate fresh preparations couple each original polarization to the compact-cap positive-mass pointer. Its gate stays within the local domain of dependence outside the source shell. The exact Magnus propagator supplies the output coordinate; bounded unknown shear is paid as centered pointer noise. Finite clipping, bins and 31 median blocks give both complex-mean components. The retained source prior $0 < \Omega T_0/4 < \pi$ and known sign of g select the phase branch. Unknown multiple turns require additional timing information.

The independent composition check uses the exact binomial tail

$$p_{\text{fail}} \le 12 \sum_{j=16}^{31} \binom{31}{j} (1/8)^j (7/8)^{31-j} < 2 \times 10^{-6}. \tag{61}$$

The source/local-field disk, relative complex record disk 10^{-11} and separately measured 0.020 nanoradian calibration allowance give

$$|\delta\theta| \le \delta\theta_{\text{field}} + \arcsin(10^{-11}) + 0.020 \text{ nrad} < 0.174 \text{ nrad}. \tag{62}$$

This statement includes finite local interaction, finite moments, finite records and confidence. The specified calibration has not been achieved in a laboratory. The sufficient 10^{56}–10^{59} fresh preparations per polarization and 182–191 readout bits quantify its practical gap.

The measured signal is the total prepared near field, including its initial hard homogeneous component. Its actual amplitude is derived from the source. It is distinct from the much weaker newly emitted soft packet. Repeated preparation must pay the original distributed correlations and common source phase; uniformly random preparation phases would erase this mean. One fresh preparation per gate avoids assuming that later detector backaction leaves an indefinitely reused source undisturbed.

Exact elapsed-time composition. For the declared shell incidence and original counterterm, $H[n(t)] = n(t)H_0$ when the lapse is spatially uniform. Therefore

$$U_n(t,s) = e^{-iH_0[\tau(t)-\tau(s)]}, \qquad \tau(t) = \int_0^t n(u)\,du. \tag{63}$$

The gate $h_{\text{coord}}(t) = n(t)h_\tau(\tau(t))$ transports the positive symmetric proper-time measurement exactly. For $n = 1 - a\sin^4(\pi t/T)$,

$$\tau(t) = t - a\left[\frac{3t}{8} - \frac{T\sin(2\pi t/T)}{4\pi} + \frac{T\sin(4\pi t/T)}{32\pi}\right], \tag{64}$$

$$\tau(T) = T(1 - 3a/8). \tag{65}$$

The ramp area is explicit. The original real-time memory and chronological two-time remainder apply at elapsed age τ; an independent exponential relaxation factor is unnecessary for this finite protocol.

Direct covariance route and the local-output gap. The static joint canonical unitary, its original-shell limit, the actual six-lane full-cycle source-current comparison and the free switched canonical Fock propagator are now established. To join the *switched* nonuniform singular source, it suffices to control the actual mixed-shell covariances $C_{e,d}(t,s)$ in $L^1([0,T]^2)$, together with their initial first moments. The canonical Gaussian Dyson theorem then supplies the joint cocycle and its inverse. Section 85 now closes joint existence by an equivalent

common-form theorem; this direct covariance route remains unevaluated. For the complete uniform kernel,

$$C_n(t,s) = n(t)n(s)C_R(\tau(t)-\tau(s)), \qquad \iint |C_n| \le \frac{402\,\tau(T)}{\kappa\pi R}. \tag{66}$$

For the actual Plummer switch the missing estimate must include the regular origin, angular returns and inhomogeneous scattering. The sharp shell vector is not in the one-particle L^2 space, so an ordinary bounded or Hilbert–Schmidt propagator estimate cannot be paired with it twice to supply this bound.

The detector join must also transport the actual local cap test, its initial homogeneous field and its two-time covariance. A bounded source-current comparison alone supplies no output-field phase/noise certificate. These local-output estimates are still required for the full nonuniform composition. Physical reference and unit calibration, realized preparation/readout accuracy, long-time thermal transitions and the cosmological amplitude remain their respective apparatus/source inputs.

All preparation, switching, pointer kinetic energy, field deposition, finite records, reference and reset stay in the energy/work ledger. A vacuum scalar convention or telescoping signed work does not remove those entries. The new continuation retains its native coefficients, portable proofs and exact replay separately from the unchanged historical packets below.

80. Paid preparation of the original local clock

The current SAM_land donor apparatus can prepare and retain a record of an approximation to the *original* hard-dressed clock. This adds a specified apparatus law: sixteen neutral donor cavities, positive driven transport controls, a clipped source port, and coherent mark–conjunction–unmark record controls. The original full sixteen-state source, all six signed lanes, four dark states, even source energy, both polarizations and counterterms, hard dressing and soft vacuum are retained. No source brightness is assigned.

The old donor campaign's raw native tables were not recovered in this pass. Its formatted displays are not interval inputs. The bounds below are new, conservative evaluations of its same actual tapered donor law using the retained original clock certificates. They do not replay the old campaign.

80.1 Finite preparation and a held outcome

The supplied inputs are sixteen cosine ground modes and neutral zero tags and record. Positive local quantile controls prepare the full pure donor product Φ_b in $128/\Delta$, with scalar completion peak 144ε. The attempt uses the complete original hard Hamiltonian $H_h(1+X/128)+2\varepsilon$, where $X = \mathrm{clip}(\sum_j u_j, -128, 128)$. Source-shaped mode marks, a conjunction, and unmarking give exactly

$$\psi\otimes 0_R \longmapsto \Pi_b\psi\otimes 1_R + (I-\Pi_b)\psi\otimes 0_R. \tag{67}$$

Internal tags return to zero; the failure cloud and final record are held. The hard source starts after the neutral factory. Its hard phase accrues for $33/\Delta$, while factory-to-record elapsed time is $161/\Delta$. These event intervals have different meanings.

For the actual normalized taper and clipped sum, the characteristic function differs from $(1+z^2)^{-16}$ on the *whole real line* by at most

$$\delta_b \le 4(3/8)^{64}(4/3)^{16} + 32(3/8)^{127} < 2^{-80}. \tag{68}$$

Chernoff at $\theta = 1/2$ gives $T_{16}(128) \le 2e^{-64}(4/3)^{16}$; $e > 8/3$ and $1-Z \le e^{-127}$ prove the claim. Thus the full normalization, clipping and taper are paid, without reconstructing an absent numerical table.

80.2 Unequal branches preserve the all-time phase

Write the normalized ground projection of the successful state as $G = c_0\Xi_0 + c_1\Xi_1$, using the original hard eigenbranches and soft vacuum. Put $2\bar{c}_0c_1 = \beta e^{i\phi_p}$, with $\beta = 2|c_0c_1|$. If both charge-sector success norms lie in $[p_-,p_+]$,

$$\beta^2 \ge \frac{4p_-p_+}{(p_-+p_+)^2}. \tag{69}$$

The real actual filter is positive on both ground intervals. Hence ϕ_p contains the known input phase and actual paid hard phase, with its retained Ω interval; unequal finite-filter weights change the amplitude.

The retained hard-phase interval has width about $2.55\,10^{-9}$ radians in lane 01 and $1.70\,10^{-6}$ in lane 03. These source-certificate uncertainty widths are not observed dephasing, but they also do not provide nanoradian initialization calibration. That calibration, or a proved common-preparation differential cancellation, remains required.

There is an exact stronger statement. Let Q be the complete conserved source-plus-field charge, $[Q,H_0]=0$, and $[Q,A] = -A$. The ground branches have successive charges. With $B = (\Xi_0+\Xi_1)/\sqrt{2}$,

$$\langle G, A_{H_0}(t)G\rangle = \beta e^{i\phi_p}\langle B, A_{H_0}(t)B\rangle. \tag{70}$$

Indeed, both diagonal and the opposite off-diagonal charge matrix element vanish, leaving only $\bar{c}_0c_1$ instead of $1/2$. This includes the normalized original circular current and circular local field, at all elapsed times where the form expectation exists. It assumes neither a bare oscillator nor exponential relaxation. The new native full-source incidence certificates have ranks 24 and 6; their bare source-charge projectors are not substituted for interacting spectral projectors.

80.3 Source-bound energy and actual preparation error

One can bound the needed ground energies without the missing old donor objects. In the actual balanced original state, $H_h = K_h + \varepsilon\sigma_z$, $\langle K_h\rangle \le K_{\mathrm{prep}}$, and Bloch positivity yields

$$\begin{aligned} |E_{\mathrm{av}}| &\le K_{\mathrm{prep}} + \varepsilon\sqrt{1-M_-^2} =: \bar{E}, \\ \Omega_-/2 - \bar{E} \le |E_j| &\le \Omega_+/2 + \bar{E}. \end{aligned} \tag{71}$$

Here $E_{\mathrm{av}} = (E_0+E_1)/2$ uses the exact hard eigenbranches. Within each invariant lane and fixed total-charge sector the bare

Table 6: New conservative paid preparation at the original fixed reference. Every interval is rounded outwards; failures and their records are retained.

Lane	Success probability	$1-\beta$ upper
01	$[.608850,.608910]$	$1.21\,10^{-9}$
02	$[.608818,.608943]$	$5.19\,10^{-9}$
03	$[.608105,.609656]$	$8.11\,10^{-7}$
12	$[.608847,.608913]$	$1.39\,10^{-9}$
13	$[.608136,.609626]$	$7.48\,10^{-7}$
23	$[.608168,.609593]$	$6.85\,10^{-7}$

vacuum has one direction. On its complement $H_f \geq \lambda$; the form bound $H_h \geq H_f/2 - \varepsilon - c_{\mathrm{full}}$ and minmax give $E_{\mathrm{exc}} \geq \lambda/2 - \varepsilon - c_{\mathrm{full}}$. Other lanes and the four dark source states remain separate invariant blocks. Applying (68) to these spectral enclosures bounds success, branch imbalance and the non-ground projection.

Let $d \geq \|\Psi - G\|$ for the actual accepted state. With $Y = H_h + \varepsilon \geq 0$, the success and error energies satisfy

$$\begin{aligned} E_{f,\Psi} &\leq (\sqrt{c_{\mathrm{full}}} + \sqrt{Y_\Psi})^2, \\ E_{f,\Psi-G} &\leq (\sqrt{c_{\mathrm{full}}}\, d + \sqrt{Y_{\Psi-G}})^2. \end{aligned} \tag{72}$$

The exact source-free incoming evolution retains the initial hard field and its momentum. Adding the compact retarded current kernel gives

$$\begin{aligned} B_{\mathrm{prep}} &\leq 4\sqrt{\mu_r}\,(d\sqrt{E_{f,\Psi}} + \sqrt{E_{f,\Psi-G}}) \\ &\quad + \frac{8|g|d}{\kappa r_-}, \qquad \mu_r = \frac{1}{4r_-}. \end{aligned} \tag{73}$$

Across all six lanes $d < 2.673\,10^{-28}$ and the additional local preparation phase is below $1.93\,10^{-17}$ radians. Its denominator is the original hard-harmonic floor *minus* the entire soft/current mean disk, followed by the factor β and this new preparation error.

80.4 The compact cap and the apparatus boundary

For this H_0 protocol, the complete field mean stays in the two original electric $j = 2$ angular patterns. The axisymmetric cap therefore multiplies the signal and preparation error by the same exact $\eta > 0$:

$$\frac{\eta B_{\mathrm{prep}}}{\eta\beta A_{\mathrm{actual}}} = \frac{B_{\mathrm{prep}}}{\beta A_{\mathrm{actual}}}. \tag{74}$$

There is no inverse-cap-area preparation phase penalty. Noise does not cancel. The untouched angular complement stays a product vacuum and $F_{\mathrm{cap}} = \eta F_{\mathrm{ann}} + F_\perp$, giving

$$\langle X^2_{\mathrm{cap}} \rangle \leq \eta_+^2 \langle X^2_{\mathrm{ann}} \rangle + V_{\mathrm{cap}}. \tag{75}$$

The annular moment uses the actual accepted-state field energy above; the complement vacuum and pointer Magnus noise remain paid.

All-outcome endpoint work is $W = c_{\mathrm{full}} + K_{\mathrm{all}} - K_{\cos}$, with a strictly positive lower bound in every lane. Failure source energy, meter energy and the held record have explicit finite ceilings. Fresh-input repetitions retain all failure clouds; no reset or record erasure is assumed. The separable mode port, driven controls, spatial implementation, physical reference and bin calibration remain declared apparatus laws and requirements. These results do not close the switched nonuniform canonical shell-to-cap transfer.

81. An actual donor preparation followed by a finite local record

The current donor factory and its literal-source control can now be composed with the complete original canonical field and the compact cap detector. The source is still the original full sixteen-state system, with all six signed lanes, both polarizations, counterterms, incoming hard dressing and soft vacuum. The donor port and pointers are explicitly added apparatus laws. The numerical preparation bounds used here are new conservative evaluations of that same current donor law from the authenticated original source intervals; historical formatted donor tables are not arithmetic inputs.

81.1 Successful weights, phase and the complete preparation error

The donor factory lasts $128/\Delta$. The subsequent source attempt and coherent record last $33/\Delta$, so the apparatus is ready at $161/\Delta$. The source phase accumulated during the latter interval is retained. Write its normalized hard-ground projection as $G = c_0\Phi_0 + c_1\Phi_1$ and $2\bar{c}_0 c_1 = \beta e^{i\phi_p}$. Complete charge conservation gives

$$\langle O(t)\rangle_G = \beta e^{i\phi_p} \langle O(t)\rangle_{\mathrm{balanced}} \tag{76}$$

for every charge-lowering circular field or source current under the same complete H_0. Unequal successful weights alter amplitude while retaining the full interacting phase trajectory. This identity uses the exact dressed branches; it is not a bare two-state approximation.

For $d \geq \|\Psi - G\|$ and the actual successful/error field energies $E_f, E_{f,e}$, the full radial mean error is bounded by

$$B_{\mathrm{in}} = 4\sqrt{\mu}\,\big(d\sqrt{E_f} + \sqrt{E_{f,e}}\big), \tag{77}$$

$$B_p = B_{\mathrm{in}} + \frac{8|g|d}{\kappa r_-}. \tag{78}$$

The second term pays the changed subsequent source current, rather than only the incoming hard field. The new six success intervals lie within $[0.6081057, 0.6096559]$, with $\beta > 0.9999991899$. The largest additional local preparation phase bound is below 1.93×10^{-17} rad. The phase $\phi_p = -\Omega 33/\Delta$ itself is tracked separately; that small error bound does not make its predicted numerical value exact.

For the axisymmetric cap, both the mean and this preparation difference are multiplied by the same positive Gram factor η. Consequently their phase ratio has no $1/\eta$ penalty. The actual amplitude lower subtracts the entire earlier balanced soft/current disk before applying β and B_p.

81.2 Two physical polarizations on each accepted source

Reflection through $y = 0$ preserves the cap, the local canonical bundle operator and the plus test; it reverses the cross test. On the short source-excluding detector worldtube this gives

$$[\Phi(F_+,t),\Phi(F_\times,s)] = 0. \tag{79}$$

Two independent local pointers therefore have zero cross Magnus shear and can read both physical polarizations on one accepted source. Their outcomes may be statistically correlated; the confidence calculation does not require their independence. With conserved pointer momenta,

$$Q_{\alpha,\mathrm{out}} = Q_\alpha + 2(Lk + c_\alpha)P_\alpha + X_\alpha. \tag{80}$$

Use $k = 1 + 8h_{\max}^2\mu_c$ for a conservative simultaneous positive-form completion. Free-drift focusing with variance $\sigma^2 = V_c/2$ pays the full unknown self-Magnus noise:

$$\sigma^2 + \frac{V_c^2}{4\sigma^2} = V_c. \tag{81}$$

The actual original preparation leaves the orthogonal angular field in vacuum. Its exact cap decomposition is $F_{c,\alpha} = \eta F_{a,\alpha} + F_{\perp,\alpha}$, so

$$\langle X_c^2\rangle = \eta^2\langle X_a^2\rangle + V_\perp, \qquad 0 \le V_\perp \le V_c, \tag{82}$$

$$B_c \le \eta_+^2\left(2\sqrt{\mu_a E_f} + \sqrt{V_a} + \frac{2|g|}{\kappa r_-}\right)^2 + 2V_c. \tag{83}$$

Thus the full cap-complement vacuum remains, while the actual source-sector energy receives its correct squared cap incidence. This estimate is specific to the original angularly invariant evolution and preparation.

81.3 Finite records and an all-outcome fresh-input batch

For complex relative allowance δ, use scalar allowance $e = 7\delta A_c/10$. Two scalar errors e fit the complex disk because $2(7/10)^2 = 49/50 < 1$. Clip at $8B_c/e$, bin with width $e/4$, and take 31 median-of-means blocks, each containing

$$n_b = \left\lceil \frac{32(2B_c + e^2/32)}{e^2} \right\rceil \tag{84}$$

accepted paired measurements. A block fails with probability at most $1/8$. The exact binomial tail, unioned over twelve scalar means, is below 1.974×10^{-6}.

The primary table retains $\delta = 10^{-11}$ and the earlier conditional 0.020 nrad apparatus/reference allowance. Its complete phase bound is below 0.173517 nrad, with the actual successful-state variance and preparation error included. Every accepted source supplies both records.

lane	accepted pairs	fresh attempts	bits/channel
01	$4.41066\,10^{58}$	$1.44885\,10^{59}$	190
02	$1.02679\,10^{58}$	$3.37306\,10^{58}$	188
03	$6.62629\,10^{55}$	$2.17932\,10^{56}$	181
12	$3.83393\,10^{58}$	$1.25941\,10^{59}$	190
13	$7.17144\,10^{55}$	$2.35850\,10^{56}$	181
23	$7.83445\,10^{55}$	$2.57641\,10^{56}$	181

The exact integer counts are retained in the replay packet; displayed counts are rounded upward. Relative to the previous separate-polarization protocol at matched phase accuracy, accepted source demand falls by factors 3.94–4.50. A separate optional table uses $\delta = 3 \times 10^{-11}$ and the wider original 0.2 nrad allowance: its maximum phase bound is 0.193517 nrad and accepted demand falls by factors 35.47–40.48. That comparison does not have matched accuracy.

For each lane set $n = 31n_b$ and prepare the fixed fresh batch $M = \lceil 2n/p_- \rceil$. Keep every failure record and physical failure cloud. If fewer than n successes occur, report that outcome; its probability is at most $4/(n+4)$. Otherwise read the first n successes. Adding this shortage bound over all six lanes preserves the stated simultaneous failure ceiling. This finite construction uses no reset or erasure. Identical phase initialization and an actual apparatus reference remain explicit resources; no arbitrary-duration free shared oscillator or autonomous retry rate is inserted.

After the known signed orientation is removed, the predicted negative total hard-harmonic phase at the read epoch lies in $(-12\pi, -11\pi)$ for every lane. Its principal argument is in $(0,\pi)$, and subtracting 12π gives the unique retained turn count. The numerical Ω prediction interval is wider than the finite-record error. The record estimates the actual accumulated source phase; extracting a separate post-preparation increment requires its initial phase anchor with the stated calibration.

All factory work bounds are all-outcome *expectation* bounds. The primary fixed batch has expected incremental factory work at most 4.893×10^{53} in the declared energy chart. Its paired-pointer initial kinetic plus deposited-field energy is at most 2.594×10^{66}, including both polarizations. This is about 1.96 times smaller than the previous matched-accuracy pointer requirement. Original even/rest source energy, donor inputs, reference, held records and any later reset remain separate ledger entries. These finite sufficient resource and precision requirements are not laboratory measurements or deterministic bounds on unbounded failure tails. The nonuniform switched joint/cap and thermal-transition questions retain their separately stated operands.

82. The exact switched reference and its spatial residual

The retained SAM_land action provides an exact reference for the actual switch. Keep the original complete sixteen-state source, both polarizations and counterterms, the original hard preparation and the full tensor field. For $b(x) = -\sqrt{2}d/[1000\sqrt{d^2 + |x - de_z|^2}]$, its centre value is exactly $b(0) = -1/1000$. Put

$$n(t) = 1 - \eta\chi(t)/1000, \qquad \tau(t) = \int_0^t n(s)\,ds,$$

$$r(t,x) = \frac{\eta\chi(t)[b(x) - b(0)]}{n(t)}.$$

In the new time coordinate the *entire* joint receiving Hamiltonian has lapse $1+r$. Its canonical free equations are

$$\partial_\tau q = (1+r)p, \qquad \partial_\tau p = -D_{1+r}q.$$

The source energy and original counterterm have weight $1+\bar r_R$; the original interaction has the actual angular shell weight $1+r_R$. There is no extra derivative term in these first-order equations.

For the uniform reference $r=0$, all Hamiltonians commute, so

$$U_n(T,0) = e^{-iH_0\tau(T)}, \qquad \frac{\tau(T)}{T} = 1 - \frac{3\eta}{8000}.$$

The full original shell covariance is $C_R(\tau(t)-\tau(s))$. Its two physical interaction vertices carry $n(t)n(s)$. Thus change of variables gives the exact identity

$$\int_0^T\int_0^T n(t)n(s)|C_R(\tau(t)-\tau(s))|\,dt\,ds = \int_0^{\tau(T)}\int_0^{\tau(T)} |C_R(u-v)|\,du\,dv.$$

The original Q_2 kernel, its complete real tail and the uniform Abel dominator therefore close the uniform switched sharp-shell cocycle. The retained bound $\int|C_R(u)|du < 402/(\kappa\pi R)$ gives a finite existence budget. It is separate from a small phase-error enclosure.

The actual Plummer derivatives obey

$$|\nabla b| \le \frac{3}{2000d}, \qquad \|\nabla^2 b\| \le \frac{3}{1000d^2}.$$

With $d = 10^{30}R$, all six original radius/time intervals give

$$|r_R| < 2^{-109}, \qquad |\bar r_R| < 2^{-207}.$$

The shell-average bound uses its zero first spherical moment. The complete backward source causal region lies inside $L = 2\times10^6$. A compact auxiliary coefficient, equal to the physical residual on that region, can be chosen with amplitude $A = 2^{-83}$ and support inside $2L$. This is a causal proof comparison; the physical Plummer profile and original source size are retained.

83. A complete tensor estimate for the moving frame

The additional inverse-energy estimate needed by the moving frame follows from the exact complete tensor Green kernel

$$D^{-1}(x,y) = \frac{q^2U(x,y)}{4\pi\rho}, \qquad q = \frac{r+r'-\rho}{r+r'+\rho}.$$

Here U is the original spin-two transport and $D=\omega^2$ has its regular origin. For a compact coefficient satisfying

$$|b| \le A, \qquad |\nabla b| \le 3A/L,$$
$$|\Delta b| \le 15A/L^2, \qquad \|\nabla^2 b\| \le 20A/L^2,$$

the exact double commutator $T_b = D^{-1}[D,[D,b]]D^{-1}$ satisfies

$$\|\omega^{-1}T_b\|_{\mathrm{HS}} \le 10^8 AL,$$
$$\|\omega^{-1/2}T_b\omega^{-1/2}\|_{\mathrm{HS}} \le 10^8 AL,$$
$$\|\omega^{-3/2}(\Delta b)\omega^{-3/2}\|_{\mathrm{HS}} \le 400AL.$$

The proof first combines the two principal Green derivatives; their individual squares need not be integrable. It also strengthens the angular endpoint bound before discarding the angular factor:

$$\|\nabla_x(q^2U)\| \le \frac{14k^2}{r}, \qquad k = \frac{4rr'}{(r+r'+\rho)^2}.$$

This pays the regular-origin and exterior limits. Columnwise $L^{6/5}$ bounds and the bundle Sobolev inequality then yield the stated Hilbert–Schmidt estimates. Both real tensor components and every angular sector are included.

For the original canonical relative pair coefficient G_b, the exact spectral identities give

$$\|\mathscr{L}_\omega^{-1}G_b\|_{\mathrm{HS}} \le 4\times10^6 AL, \qquad (\mathscr{L}_\omega^{-1}F)_{ab} = \frac{F_{ab}}{a+b}.$$

The same complete tensor proof gives

$$\|\omega^{\pm1}H_b\omega^{\mp1}\| \le 4A, \qquad \|\omega^{\pm1}G_b\omega^{\mp1}\| \le 14A.$$

One useful exact intermediate identity is

$$\mathrm{Re}\langle Du, u/r^2\rangle = \|\nabla_E u/r\|^2 + 3\|u/r^2\|^2.$$

It supplies the required Hardy–Rellich graph bounds without a mass gap.

These estimates extend the *same* Riccati fixed point to the graph ideal $W=\omega Z$, where $Z=\omega^{-1/2}Y\omega^{-1/2}$. They are the full tensor operands used in the companion common-form joint propagator proof. The editable proof records all constants, domains and causal identifications. They do not by themselves evaluate the physical nonuniform cap mean, its noise or the external vacuum-work ledger.

84. Transport the return front before estimating its error

The original solved kernel explains an otherwise misleading perturbative failure. Near its shell-return front,

$$C_R(u) = \frac{\log|u-2R|}{2\kappa\pi R^2} + \text{less singular terms}.$$

Therefore differentiating $C_R(nu)$ with respect to n produces a nonzero multiple of $\mathrm{PV}[u/(u-2R)]$, which is not an L^1 function. The exact kernels and their finite differences remain L^1. Taking an absolute first Born derivative at this moving front is consequently insufficient even for the exactly closed uniform model.

The finite displacement has an explicit integrable modulus. For $0 < h \le L_0$,

$$\begin{aligned}
&\int_{-L_0}^{L_0} \big|\log|x-h| - \log|x|\big|\,dx \\
&= (L_0+h)\log(L_0+h) \\
&\quad - (L_0-h)\log(L_0-h) - 2h\log h + 2h\log 2 \\
&\le 2h\left[1+\log\frac{2(L_0+h)}{h}\right].
\end{aligned}$$

Thus the direct covariance route should transport the actual return front, then bound its coefficient and remainder. The equivalent joint theorem uses the compact moving-frame form construction; it does not label this unevaluated covariance correction as a computed table.

85. A common-form switched joint evolution

The preceding compact complete-tensor estimate adds an inverse-energy Hilbert–Schmidt ideal to the same weighted Riccati fixed point used in the preceding weighted normal-frame construction. Write

$$W = \omega^{1/2} Y \omega^{-1/2}, \qquad Z = \omega^{-1/2} Y \omega^{-1/2}.$$

The new proof in `protocol/GRAPH_IDEAL_JOINT_PROOF.md` uses $W = \omega Z$, the exact complete tensor Green function, and $\|\mathscr{L}_\omega^{-1} G_b\|_{\mathrm{HS}} \le 4\cdot 10^6 AL$. On $\|W\|_{\mathrm{HS}} \le 84A$, $\|Z\|_{\mathrm{HS}} \le 8\cdot 10^6 AL$, the graph norm $\|W\|_{\mathrm{HS}} + L^{-1}\|Z\|_{\mathrm{HS}}$ contracts, with

$$\begin{aligned}
c_Z &= 8A + 56A(84A), \\
c_{\mathrm{graph}} &\le \max\{2A + 16(42A)^2 + 896\cdot 10^6 A^2, c_Z\} < 1.
\end{aligned}$$

Spectral-core closure supplies the same self-adjoint Y in both ideals. Analytic differentiation controls the moving-frame connection, including its vacuum trace. If $N = p\dot p - q\dot q$ and $B = p\dot q - q\dot p$,

$$\begin{aligned}
\|\omega^{-1/2} N \omega^{-1/2}\| &\le 3\cdot 10^{10} A^2 L |\dot\theta|, \\
\|\omega^{-1/2} B \omega^{-1/2}\|_{\mathrm{HS}} &\le 10^8 AL |\dot\theta|, \\
\|\omega^{1/2} B \omega^{-1/2}\|_{\mathrm{HS}} &\le 500A |\dot\theta|.
\end{aligned}$$

For the explicitly halved pair form $Q_B = \frac{1}{2}\sum(B_{ij} a_i^* a_j^* + \overline{B}_{ij} a_j a_i)$, the CCR and Cauchy–Schwarz give

$$|\langle Q_B\rangle| \le (c+\varepsilon_*)\langle H_f\rangle + d^2\|\Psi\|^2/(4\varepsilon_*),$$

where $c = \|\omega^{-1/2} B \omega^{-1/2}\|$ and

$$\begin{aligned}
d^2 &\le u_B v_B, \\
u_B &= \|\omega^{1/2} B \omega^{-1/2}\|_{\mathrm{HS}}, \\
v_B &= \|\omega^{-1/2} B \omega^{-1/2}\|_{\mathrm{HS}}.
\end{aligned}$$

Thus the vacuum term is finite in the moving form. The actual tiny compact profile makes the connection relatively small; the original singular shell coupling is infinitesimally form bounded. Full16 source matrices, both counterterms and the complete field remain.

The actual $\sin^4$ switch and analytic frame give a C^2 family of closed, semibounded joint forms on one common $Q(H_f)$ domain. The common-form evolution theorem therefore supplies a unitary joint propagator in that frame. Its conjugating implementer need not preserve the old $Q(H_f)$. No finite old-vacuum ramp-work value follows from this construction.

For the exact uniform time change $d\tau = (1-\eta\chi/1000)dt$, use

$$\theta = \frac{999\eta\chi}{1000-\eta\chi}, \qquad b_c = \frac{1000}{999}\zeta[b-b(0)].$$

Here $0 \le \theta \le 1$ and the taper agrees with the physical residual on the entire backward source causal diamond. At each radial shell regularization, local hyperbolic propagation makes the actual global and auxiliary source interaction cocycles identical. Energy-dual convergence, including the required time derivatives, gives a strong shell limit in both time directions. Compose this cocycle with the already constructed actual global free propagator. This closes the *actual nonuniform switched joint singular-source propagator* without an angular truncation, physical cutoff, or substituted initial state. The proof is an equivalent source-admissibility theorem; it does not assert the unavailable absolute L^1 derivative of a transported logarithmic return front. The finite checks certify its constants; the domain and causality arguments have analytical provenance.

86. What the finite preparation-to-record composition now closes

The current SAM_land apparatus law supplies a finite donor factory, original-port preparation and recorded success/failure. Its actual unequal branch factor, preparation phase, field energy and error now feed the compact paired-polarization local readout. A fixed fresh batch holds all failures and all records, avoiding an uncharged reset during the experiment. The matched protocol has conditional phase error below 0.174 nanoradians and simultaneous failure below $2\cdot 10^{-6}$. It requires about $9.293\cdot 10^{58}$ successful paired sources and $3.053\cdot 10^{59}$ fresh factory attempts across the six lanes. These are sufficient model requirements, not achieved apparatus figures.

At matched accuracy, paired readout and the prepared-state angular factorization reduce successful source copies by factors 3.94–4.50 against the preceding separate-polarization bound. The expected total pointer input plus field deposition is still about $2.594\cdot 10^{66}$ model energy units; factory incremental work is bounded separately. The unbounded outcome tails preclude calling these expectation bounds hard storage or energy capacities. Initial donor/source rest energy, clock references, scheduling coherence and the common SI calibration remain separately charged operands.

The retained rate uncertainty is prediction uncertainty. The new tiny nonground preparation error does not establish a calibrated zero phase. The actual retained phase prior fixes the

integer turn count; measured local records estimate the actual phase conditional on the stated apparatus calibration. A separate preparation-to-later phase increment requires that calibration or an independently measured preparation epoch.

Question	Current result and boundary
Finite source preparation to records	Closed for the original/uniform law with the declared added apparatus controls and calibration budget.
Actual switched canonical joint evolution	Existence and exact composition closed by the compact-frame common-form/cocycle proof.
Nonuniform local output	Quantitative cap mean, noise and phase transfer still required; existence does not supply those bounds.
Apparatus precision	Finite requirements, costs, timing and failures stated; local controller realization and measured calibration remain.
Real-time dynamics	Original signed memory and finite-time transients retained; no thermal or cosmological amplitude law inferred.

The portable command is `python3 ops/reproduce.py`; native readback has a separate companion evidence packet and never restarts Duck. Historical evidence remains unchanged in the embedded predecessor ZIPs.

A. Frozen GPS arithmetic and orbital certificates

A.1 A frozen GPS arithmetic comparison

For transparent arithmetic, freeze

$$\begin{aligned} \mu = GM &= 3.986004418\times10^{14}\ \mathrm{m^3 s^{-2}},\\ c &= 299792458\ \mathrm{m\,s^{-1}},\\ R &= 6378137\ \mathrm{m}, \quad r = 26560000\ \mathrm{m},\\ \Omega &= 7.292115\times10^{-5}\ \mathrm{s^{-1}}. \end{aligned}$$

Use a monopole Earth, a Newtonian circular satellite orbit, and a spherical equatorial ground reference. These assumptions reproduce the input scale of the recovered CR005 packet while making its missing rotation term explicit. [6] Since $v^2=\mu/r$, the first-order excess is

$$\delta_{\mathrm{circ}} = \frac{\mu/R - 3\mu/(2r) + \Omega^2R^2/2}{c^2}. \tag{85}$$

Multiply a fractional rate by 86400×10^6 to express its accumulated excess in microseconds per day.

The archived CR005 net is +38.437317329 microseconds per day. Its code adds an exact-form static gravitational fractional shift to an exact-form special-relativistic shift, evaluated with ordinary binary square-root-minus-one arithmetic. That sum is not an exact moving-clock composition. At 65-digit working precision, those same additive formulas give 38.4373115706863 microseconds per day. Their difference from the archived binary 64 net is about 5.76 picoseconds per day, primarily near-unity subtraction error; it is not a relativistic correction. The receipt retains its historical role; its terminal digits are not physical corrections. [6]

Table 7: Frozen first-order budget. Extra digits permit arithmetic checks; they are not measured precision.

Contribution	μs/day
Gravity: $\mu(1/R-1/r)/c^2$	+45.650911342
Satellite motion: $-\mu/(2rc^2)$	−7.213599823
Nonrotating-ground net	+38.437311520
Ground rotation: $\Omega^2R^2/(2c^2)$	+0.103976943
Equatorial rotating-ground net	+38.541288463

An independently specified exact spherical benchmark makes the distinction sharper. In Schwarzschild geometry, using areal radii and its coordinate angular speed Ω for a supported equatorial reference, the circular-geodesic ratio is

$$1+\delta_{\mathrm{Schw}} = \sqrt{\frac{1-3\mu/(c^2r)}{1-2\mu/(c^2R)-\Omega^2R^2/c^2}}.$$

The numerator and denominator must be positive; in particular the timelike circular orbit has $r>3\mu/c^2$. With $\Omega=0$, high-precision evaluation gives 38.437311564 μs/day, only about 4.49×10^{-8} μs/day above its first-order value. With reference rotation it gives 38.541288508. This imported complete metric checks arithmetic within a spherical model; it is neither a rotating-Earth solution nor an exact SAM derivation. Rationalization or high-precision arithmetic avoids losing the small difference between nearly equal unit quantities.

The GPS reference convention is specified independently. IS-GPS-200N gives a nominal fractional frequency offset -4.4647×10^{-10}, corresponding to compensation for +38.575008 microseconds per day, and the source frequency

$$f_{\mathrm{ref}} = 10.2299999954326\ \mathrm{MHz}.$$

[5] This specified value is not a newly fitted result of our spherical calculation. The differing reference models must not be silently conflated. For a positive uncompensated excess δ, the leading compensating frequency shift is negative, $-\delta$; exact inversion would use $(1+\delta)^{-1}-1$. Already steered clock data must not receive the nominal compensation twice.

A.2 A cancellation radius and useful wrong controls

A circular orbit has two competing contributions: rising reduces gravitational slowing, while orbital motion slows the satellite clock. For a nonrotating spherical ground reference, setting Eq. (85) to zero gives

$$r_{\mathrm{cancel}} = \frac{3}{2}R, \qquad h_{\mathrm{cancel}} = \frac{1}{2}R.$$

The frozen values are $r_{\mathrm{cancel}} = 9567205.5$ m and $h_{\mathrm{cancel}} = 3189068.5$ m. Including equatorial ground rotation gives

$$r_{\mathrm{cancel}} = \frac{3\mu}{2(\mu/R+\Omega^2R^2/2)} = 9550676.183\ldots\ \mathrm{m}.$$

The reference rotation lowers the crossing altitude to about 3172539.183 m. Above the crossing a circular satellite clock gains; below it, it loses. This is a threshold for the declared circular family, not a rule based on altitude alone for arbitrary trajectories.

The recovered packet tests six wrong controls: halve or double the lift, reverse the gravitational sign, omit satellite motion, give motion a positive sign, or erase the endpoint lift difference. All fail its full acceptance packet. [6] Their pedagogical value is structural: the observed engineering scale requires the endpoint coefficient, its sign, and the kinetic subtraction together. Nevertheless, its acceptance windows are hardcoded ranges, not raw GPS observations or a new statistical fit. Passing them establishes a scoped consistency check. "No fitted parameters" here means no adjustable normalization inside the frozen imported prescriptions; it does not mean those prescriptions were derived from the source dynamics.

A.3 Eccentric orbits and the correction sign

A circular budget misses the periodic rate change along an ellipse. Let A denote semimajor axis, distinct from the dimensionless lift a, and let e and E be eccentricity and eccentric anomaly. Newtonian Kepler motion gives

$$v^2 = \mu\left(\frac{2}{r}-\frac{1}{A}\right), \qquad \Phi - \frac{v^2}{2} = \frac{\mu}{2A} - \frac{2\mu}{r}.$$

The time average $\langle 1/r\rangle = 1/A$ therefore gives the secular rate excess

$$\overline{\delta} = \frac{-3\mu/(2A)-\Phi_0}{c^2}.$$

After separating it, the periodic clock offset $\delta\tau_{\rm per}$ obeys

$$\frac{\mathrm{d}\delta\tau_{\rm per}}{\mathrm{d}t_{\rm geo}} = \frac{2\mu}{c^2}\left(\frac{1}{A}-\frac{1}{r}\right)$$

at the retained order. Newtonian acceleration supplies a short integral:

$$\frac{\mathrm{d}(\boldsymbol{r}\cdot\boldsymbol{v})}{\mathrm{d}t} = v^2 + \boldsymbol{r}\cdot\dot{\boldsymbol{v}} = \mu\left(\frac{1}{r}-\frac{1}{A}\right).$$

Consequently, up to an epoch constant,

$$\delta\tau_{\rm per} = -\frac{2\boldsymbol{r}\cdot\boldsymbol{v}}{c^2} = -\frac{2e\sqrt{\mu A}}{c^2}\sin E. \tag{86}$$

The second identity follows from $r = A(1-e\cos E)$ and $\dot{E} = \sqrt{\mu/A^3}/(1-e\cos E)$. These derivations use an ideal unsteered clock; broadcast clock parameters also represent clock behavior and steering.

Sign is a definition, not a disagreement between sources. The interface convention is [5]

$$\begin{aligned} t &= t_{\rm SV} - \Delta t_{\rm SV},\\ \Delta t_{\rm SV} &= a_{f0} + a_{f1}(t-t_{\rm oc}) + a_{f2}(t-t_{\rm oc})^2 + \Delta t_r,\\ \Delta t_r &= Fe\sqrt{A}\sin E, \qquad F = -\frac{2\sqrt{\mu}}{c^2}. \end{aligned}$$

Thus the negative term in Eq. (86) is the clock-offset term. Converting the satellite reading to coordinate time subtracts it. Ashby's positive correction toward coordinate time expresses that inverse operation. [4] The interface's explicit relativistic term is excluded from its polynomial coefficients; its stated constants and week-crossover handling belong to an implementation, rather than being replaced by our frozen arithmetic constants. [5]

Over a finite interval, the periodic accumulated contribution is $\delta\tau_{\rm per}(t_1) - \delta\tau_{\rm per}(t_0)$. Using only the final value silently assumes a zero initial phase. For $A = 26560000$ m and illustrative $e = 0.01$, the amplitude is 22.8966 ns, equivalent to about 6.864 m of light-travel range. This is not a universal position-error estimate. Although $\boldsymbol{r}\cdot\boldsymbol{v}$ is unchanged by subtracting $\boldsymbol{\Omega}\times\boldsymbol{r}$ from velocity at the same epoch, that special identity does not authorize using Earth-fixed speed squared in Eq. (3).

B. Optical height and endpoint uncertainty

B.1 A local height comparison is a separate channel

GPS combines a gravitational potential difference, orbital motion and a reference time scale. A pair of nearly stationary clocks separated vertically supplies a different test. Let h be upward height from a lower radius r, and initially use the spherical source. Rationalizing the exact endpoint ratio gives

$$\begin{aligned} q_h &= \sqrt{\frac{1-a(r+h)}{1-a(r)}} - 1,\\ &= \frac{a(r)-a(r+h)}{(1-a(r))\left[\sqrt{\frac{1-a(r+h)}{1-a(r)}}+1\right]}. \end{aligned}$$

This expression is numerically stable at laboratory heights. Since

$$a(r) - a(r+h) = \frac{2\mu h}{c^2 r(r+h)}, \qquad \mu = GM,$$

its leading value is gh/c^2, with $g = \mu/r^2$. In a real laboratory, the corresponding quantity is the local geopotential difference, including the reference's rotational contribution. A height alone is an approximation to that potential difference.

Bothwell and colleagues resolved a spatial clock-frequency gradient across a millimetre-scale strontium sample. Their corrected result is $-9.8(2.3)\times10^{-20}\,\mathrm{mm}^{-1}$ in their coordinate convention, compared with the expected approximately $-10.9\times10^{-20}\,\mathrm{mm}^{-1}$. Density, magnetic-field and other gradient systematics are included in the published analysis. The separately reported 7.6×10^{-21} synchronous comparison uncertainty is not the uncertainty of this gravitational gradient. [8]

The recovered SAM CR006 packet uses the upward-positive convention, $g = 9.796\,\mathrm{m\,s}^{-2}$, and a one-millimetre separation. It obtains approximately 1.0899520×10^{-19} and a residual of 0.478 quoted standard deviations from the positive measured gradient. [7] Its effective radius is chosen as $\sqrt{\mu/g}$, so the leading prediction already imports the local acceleration. The

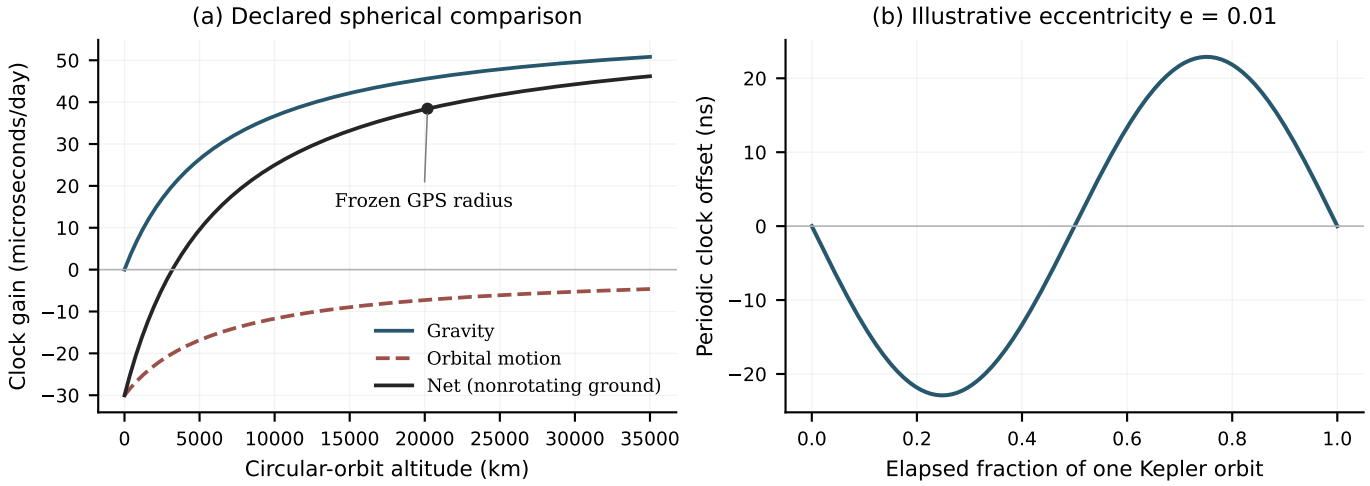


Figure 2: Declared orbital controls, not fitted GPS data. (a) The first-order monopole gravitational, kinetic and net rates relative to a nonrotating spherical ground reference. The marked radius uses the frozen GPS arithmetic operands; the zero crossing follows from the same circular-orbit law. (b) The eccentricity clock offset for a declared Kepler orbit with $e = 0.01$, plotted against elapsed orbital fraction from perigee. Its amplitude is 22.8966 ns and its mean is zero; an interval comparison uses the difference of its endpoint values. Ground-reference rotation, actual geopotential harmonics, clock steering and propagation are separate inputs.

result is a useful consistency check on the endpoint coefficient and sign; it does not establish the inverse-radius profile independently across one millimetre, and it does not distinguish SAM from general relativity when their predictions agree.

B.2 A comparison must carry its error budget

The difference between two endpoint rates is not automatically the ratio of two received signals. A transmitted comparison contains range-rate Doppler, propagation, synchronization and hardware contributions. Controlled transport introduces a trajectory and acceleration history. A useful test must state how those terms are removed or modeled before a residual is assigned to the endpoint law. It must identify which corrections and steering have already been applied to each timestamp, and retain the initial epoch and full observation interval. A short orbital segment can confuse a constant bias with part of a periodic residual. Rotating-frame propagation corrections remain necessary when that frame is used; the geoid rotation term in the clock rate does not replace them. [4,5,10]

A simple bound makes the metrological requirement explicit. Within the declared first-order rate model, suppose $|\delta\Phi| \le \varepsilon_\Phi$, $|\delta\Phi_0| \le \varepsilon_0$ and $\|\delta\mathbf{v}\| \le \varepsilon_v$. Then

$$|\delta q| \le \frac{\varepsilon_\Phi + \varepsilon_0 + |\mathbf{v}|\varepsilon_v + \varepsilon_v^2/2}{c^2}.$$

Proof. Expand $|\mathbf{v}+\delta\mathbf{v}|^2 - |\mathbf{v}|^2 = 2\mathbf{v}\cdot\delta\mathbf{v} + |\delta\mathbf{v}|^2$ and apply Cauchy–Schwarz. For equal initial clock offsets, integration gives $|\delta\tau| \le \int |\delta q|\,dt$, or $T\sup|\delta q|$ over a duration T. Two independently uncertain endpoints require both budgets. This is a bound on parameter errors in the stated rate law, not a complete receiver uncertainty analysis.

Near the surface, a one-metre geopotential-height error corresponds to about 1.09 10^{-16} fractional rate and 9.42 picoseconds accumulated per day for the declared local g. A centimetre therefore matters near the 10^{-18} scale. These conversions are sensitivity estimates; they do not assign an uncertainty to a particular apparatus.

Pure relativistic truncation can be bounded separately. For $0 \le x \le b < 1$, Taylor's theorem gives

$$0 \le 1 - \frac{x}{2} - \sqrt{1-x} \le \frac{x^2}{8(1-b)^{3/2}}.$$

The second derivative of $\sqrt{1-x}$ is $-1/[4(1-x)^{3/2}]$, which proves the bound. The exact remainder is

$$\sqrt{1-x} - 1 + x/2 = -\frac{x^2}{2(1+\sqrt{1-x})^2}.$$

Small omitted powers of a declared metric are not bounds on missing quadrupole fields, tides, ephemeris errors, clock steering, temperature shifts or the as-yet-uncontrolled nonuniform source coupling. Those require their own inputs and estimates.

C. Solar timing operands and approximation certificates

C.1 The retained solar geometry

The recovered solar CR006 packet declares $G = 6.6743 \times 10^{-11}$ SI, $M_\odot = 1.98847 \times 10^{30}$ kg, $c = 299792458\,\mathrm{m\,s^{-1}}$, $R_\odot = 695700000$ m and $1\,\mathrm{AU} = 149597870700$ m. It places endpoints at radii $r_A = 1$ AU and $r_B = 8.43$ AU, on opposite sides of a closest approach $b = 1.6R_\odot$. The resulting source length and straight-segment length are approximately

$$r_s = 2953.3393821\ \mathrm{m}, \qquad L = 1.41070328817 \times 10^{12}\ \mathrm{m}.$$

At $\gamma = 1$ its archived witnesses are

$$a\,\mathrm{d}\ell = 39337.4321183\ \mathrm{m}, \qquad \Delta T_1 = 131.215549520\ \mu\mathrm{s}.$$

The ideal frozen symmetric two-way value is twice the one-way value. None of these is a direct measurement of a Cassini tracking pass. [15]

The retained photon-draft companion calculation compares the asinh primitive, the finite-endpoint logarithm and numerical quadrature. Their agreement checks one mathematical integral through different implementations. It does not create three independent observations. The archive uses $GM_\odot = 1.3271645321 \times 10^{20}\,\mathrm{m^3\,s^{-2}}$ from its separately frozen G and mass. Substituting the different reference operand $1.32712440018 \times 10^{20}$ gives about 131.211581713 microseconds in the same geometry. That input change is not a relativistic correction or a newly inferred value of γ.

Table 8: Retained opposite-side impact sweep, holding endpoint radii fixed. Rounded arithmetic values, not observed tracking data.

$b/R_\odot$	$1/r$ road (μs)	Normalized $1/r^2$ (μs)
1.2	136.883683	175.070258
1.6	131.215550	131.215550
3.0	118.830002	69.818952
10.0	95.103737	20.701578

The inverse-square control is normalized to the central case. Its other cases then expose a different shape. Half and double source amplitude give half and double delay; the missing-road control gives zero by definition. These are useful operator checks, but the latter does not rule out every conceivable endpoint-based theory. The archive's $\gamma_{\mathrm{eff}} = 1$ follows from its imposed road coefficient, rather than a fit to raw Cassini observations. Its broad central-delay interval and simplified gamma gate must be read in that limited role. [15]

C.2 Which approximation is being bounded?

A numerical residual is useful only when its target is named. Three differences already occur in the frozen solar example. First, the finite logarithm differs from its far-endpoint approximation by about 138.275 picoseconds. Second, changing the imported solar GM changes the one-way prediction by about 3.968 nanoseconds. Third, choosing higher-order metric coefficients would change the optical index and the ray. These are distinct effects with different physical meanings.

A controlled index example illustrates the distinction. Deliberately treat the truncated expressions $N^2 = 1-a$ and $h_{ij} = (1+a)\delta_{ij}$ as exact functions. Their index is

$$n_{\mathrm{opt},*} = \sqrt{\frac{1+a}{1-a}}.$$

This defines a comparison model; the original first-order metric did not specify these higher-order coefficients. For $0 \le a \le \varepsilon < 1$, algebra gives

$$n_{\mathrm{opt},*} - (1+a) = \frac{a^2(1+a)}{(1-a)(n_{\mathrm{opt},*}+1+a)} \ge 0.$$

Since the denominator's last factor is at least two,

$$0 \le \delta T_{\mathrm{index}} \le \frac{1+\varepsilon}{2c(1-\varepsilon)} \int_{\Gamma_0} a^2\,\mathrm{d}\ell.$$

For the monopole straight segment,

$$\int a^2\,\mathrm{d}\ell = \frac{r_s^2}{b}\left[\arctan(z_B/b) - \arctan(z_A/b)\right].$$

In the frozen CR006 geometry, direct quadrature gives about 40.9480 picoseconds for this stipulated fixed-path index remainder, below its 40.9482 picosecond bound. This is a stringent check of that mathematical remainder, not an uncertainty bound on the physical Shapiro prediction. It excludes ray bending, unspecified metric terms, source motion, multipoles, plasma and apparatus. An exact Schwarzschild metric would have its own higher-order lapse and spatial coefficients; it is not obtained by declaring this truncation exact.

A complete accuracy claim therefore needs a hierarchy: source-input errors, metric-order remainder, path perturbation, dynamical geometry, transfer medium and measurement reduction. Small disagreement between two formulas for the same first-order integral controls implementation, not that entire hierarchy.

C.3 The local background and the cosmological road

The local solar calculation is a source-excess comparison with a declared background route. In the retained linear account, $A_{\mathrm{tot}} = A_0 + a$ and

$$\Delta T_{\mathrm{source}} = \frac{1}{c}\int_{\Gamma_0} (A_{\mathrm{tot}} - A_0)\,\mathrm{d}\ell = \frac{1}{c}\int_{\Gamma_0} a\,\mathrm{d}\ell.$$

The subtraction uses the same route and reference convention. It does not delete the background from the theory. More generally an optical comparison subtracts the chosen reference index: $c^{-1}\int (n_{\mathrm{opt}} - n_{\mathrm{opt,ref}})\,\mathrm{d}\ell$, with endpoint units and geometry calibrated consistently.

The archive's deliberately un-subtracted floor control adds $A_0 L/c$ with $A_0 = 1/(12\pi)$. It produces approximately 124.820038876 seconds including the small source term, rather than the 131.216 microsecond source excess. [15] That result demonstrates the consequence of inserting a uniform baseline into this already source-subtracted observable. It is not a measurement excluding every homogeneous background, since a homogeneous index also changes coordinate and calibration conventions.

This linear road subtraction differs from the nonlinear raw-lapse normalization discussed above. A constant inside $\sqrt{1-A_0-a}$ changes the source coefficient after normalization; it cannot be removed by claiming that every local operation subtracts the floor. The receiving law must be stated before the background operation is performed. [3,11]

The redshift-indexed amplitude-distance construction has its own source, propagation, detector and calibration inputs. Its finite-window inference is developed in the main text; it is not

inserted into this source-excess timing integral. The Magic Bell remains a conceptual prompt rather than a global horizon theorem.

D. Quantum-form and source-matrix certificates

D.1 Closedness and complete counterterm

The graph norm $\|\Psi\|^2+\|T_g\Psi\|^2$ is equivalent to the free-field form norm by Eq. (1). A weight W bounded above and strictly below multiplies this squared map norm by equivalent constants. Completeness therefore proves closedness on the unchanged $Q(H_f)$; bounded source terms preserve closedness and a finite lower bound. Expansion of the same square supplies all free, interaction and counterterm pieces on that one domain. This argument uses $\int|u|^2/\omega<\infty$ and does not require $\int|u|^2<\infty$.

A finite exact source certificate makes the weighted counterterm concrete. On the two original polarizations choose

$$W=\begin{pmatrix}1 & \mathrm{i}/4\\ -\mathrm{i}/4 & 1\end{pmatrix},\qquad \operatorname{spec}W=\{3/4,5/4\}.$$

Using the actual sixteen-state source and $C=\mathrm{i}SD$, the counterterm divided by $\varepsilon^2\mu$ is

$$B_{\mathrm{ct}}=D^2+C^2+\frac{\mathrm{i}}{4}(DC-CD)=2D^2+\frac{1}{2}SD^2.$$

With $P_\pm=(I\pm S)/2$, the exact Gram factorizations are

$$B_{\mathrm{ct}}-\tfrac{3}{2}D^2=(DP_+)^*(DP_+),\qquad \tfrac{5}{2}D^2-B_{\mathrm{ct}}=(DP_-)^*(DP_-).$$

Both B_{ct} and D^2 have rank twelve; each comparison has rank six. The four dark source dimensions are retained. This order-one polarization weight is an algebra certificate, separate from the small-δ phase-sign fixture in Table 2.

D.2 Bootstrap energy inequality

Remove the same even scalar from both static generators for the state comparison, while retaining it in their original energies. Conservation of the actual energy gives

$$N\varepsilon\langle S\rangle_{\mathrm{act}}+NK_{\mathrm{act}}+r[\Psi,\Psi]=N\varepsilon\langle S\rangle_0+NK_0+r[\Psi_0,\Psi_0].$$

Subtract the corresponding conserved reference energy. The source expectation difference is at most $2d_\Psi$ because $\|S\|=1$; the form errors are bounded by δK_{act} and δK_0. Using $K_0,K_{\mathrm{ref}}\le\kappa$ gives the inequality in the main text. Together with the weak overlap identity it proves the stated first-crossing bound, without resolving unbounded field energy into separate poorly conditioned marginals.

D.3 Common-form first-order current response

For $q_\lambda=q_0+\lambda r_B$, equivalent form norms supply a common finite-energy space and uniform bounds for sufficiently small λ. Form-resolvent convergence gives strong Hilbert-space convergence of the evolved fixed preparation; the common energy bound gives weak form-space convergence. Conservation of q_λ and the continuous bounded source term give convergence of the shifted q_0 norm. Weak convergence plus norm convergence is strong convergence in that form Hilbert space.

The exact weak variation identity can consequently be divided by λ and passed to zero in its continuous pairings under a common integrable bound. This gives the displayed z_1 formula. A finite-energy preparation derivative adds its separate initial-state response. A uniform variation of every complete Hamiltonian term reduces to the exact internal-age derivative. A nonconstant output weight supplies a different, explicitly incidence-dependent response functional.

E. Statistical and reconstruction certificates

E.1 Contrast equals generalized least squares

At a trial exponent p, let $t_i=e^{-ps_i}$ and $X_p=[\mathbf{1},\boldsymbol{t}]$. Form the consecutive contrast matrix A_p with row entries $[\lambda_i,-(1+\lambda_i),1]$, where

$$\lambda_i=\frac{t_{i+2}-t_{i+1}}{t_{i+1}-t_i}.$$

Each row annihilates both columns of X_p. For distinct ordered samples and $p>0$, the $m-2$ rows are linearly independent: the last nonzero column of each successive row gives a triangular independence argument. Thus their row space is the full annihilator of the two-column model.

Let $\Sigma=LL^T$ be positive definite, $\boldsymbol{z}=L^{-1}\boldsymbol{y}$, $Z=L^{-1}X_p$ and $B=A_pL$. Since $BZ=0$ and both spaces have the complementary dimensions,

$$B^T(BB^T)^{-1}B=I-Z(Z^TZ)^{-1}Z^T.$$

Sandwiching between $\boldsymbol{z}^T$ and $\boldsymbol{z}$ gives

$$(A_p\boldsymbol{y})^T(A_p\Sigma A_p^T)^{-1}A_p\boldsymbol{y}=(\boldsymbol{y}-X_p\widehat{\boldsymbol{\vartheta}})^T\Sigma^{-1}(\boldsymbol{y}-X_p\widehat{\boldsymbol{\vartheta}}).$$

The projector has rank $m-2$. For fixed p, known covariance and a Gaussian model, this proves the stated χ^2_{m-2} law. Profiling p changes the procedure and its calibration.

E.2 Constant-ratio search and exponent profiling

Whiten the data, project out the constant direction, and normalize the remaining $\boldsymbol{t}_p$ direction to $\boldsymbol{q}_p$. On a finite predeclared grid $\mathscr{G}$, the improvement over a constant fit is

$$\Delta_{\mathscr{G}}=\max_{p\in\mathscr{G}}(\boldsymbol{q}_p^T\boldsymbol{z})^2.$$

Under the constant-ratio null these form a correlated unit-variance Gaussian process on the same grid. Standard-normal simulations through that grid calibrate the complete search. The constant-ratio result concerns measured evolution; the separate single-power residual concerns the adequacy of its shape. Neither uses a one-exponent threshold after a grid search. [19]

E.3 Noisy Hankel determinant

For increments $\boldsymbol{d}=\boldsymbol{\mu}+\boldsymbol{e}$ with covariance V, take $B_{02}=B_{20}=1/2$, $B_{11}=-1$ and all other entries zero. Then

$$E[\boldsymbol{d}^TB\boldsymbol{d}]=\boldsymbol{\mu}^TB\boldsymbol{\mu}+\operatorname{Tr}(BV).$$

The bias-corrected determinant and its Gaussian variance are

$$\widehat{H}_c = d_0 d_2 - d_1^2 - V_{02} + V_{11},$$

$$\mathrm{Var}(\widehat{H}_c) = 2\,\mathrm{Tr}(BVBV) + 4\boldsymbol{\mu}^T BVB\boldsymbol{\mu}.$$

The variance follows by expanding the quadratic form and applying the Gaussian second/fourth moments. Bias correction does not turn a quadratic statistic into a Gaussian one; calibrate its tails under the declared mean and covariance model. [19]

E.4 Calibration, Fieller set and information

For fixed p, let $V = \mathrm{Cov}(\widehat{\alpha}, \widehat{\beta})$ and $\widehat{C}_{\mathrm{cal}} = \widehat{\alpha} + \widehat{\beta}$. The coefficient variances are

$$v_{\alpha\alpha} = V_{00}, \quad v_{CC} = V_{00} + 2V_{01} + V_{11}, \quad v_{\alpha C} = V_{00} + V_{01}.$$

Inverting the Gaussian test for the linear combination $\alpha - xC_{\mathrm{cal}}$ gives the fixed-exponent Fieller set

$$(\widehat{\alpha} - x\widehat{C}_{\mathrm{cal}})^2 \le k_\rho^2 (v_{\alpha\alpha} - 2x v_{\alpha C} + x^2 v_{CC}).$$

It may be bounded, disconnected or unbounded according to normalization information. For independent weights w_i, direct evaluation of the two-by-two coefficient matrix gives

$$\det(X_p^T W X_p) = \left(\sum_i w_i\right)^2 \mathrm{Var}_w(t_i).$$

This is the exact leverage certificate: nearly equal t_i make the constant and transient columns difficult to separate.

If a covariance mode is $\Sigma_{\mathrm{sys}} = X_p \Omega X_p^T$, then $A_p \Sigma_{\mathrm{sys}} A_p^T = 0$. Such a mode changes coefficient uncertainty without adding shape residual. A general redshift-dependent calibration template has a component outside that space.

E.5 Held-out prediction with cross-covariance

Partition the observations into training T and held-out H. At fixed p, define

$$K = \Sigma_{HT}\Sigma_{TT}^{-1}, \quad V_T = (X_T^T \Sigma_{TT}^{-1} X_T)^{-1}, \quad B_H = X_H - K X_T.$$

The training-only predictor and error covariance are

$$\widehat{\boldsymbol{y}}_H = K\boldsymbol{y}_T + B_H \widehat{\boldsymbol{\vartheta}}_T,$$

$$V_H = \Sigma_{HH} - K\Sigma_{TH} + B_H V_T B_H^T.$$

To prove the covariance, write $\boldsymbol{y}_H = K\boldsymbol{y}_T + B_H\boldsymbol{\vartheta} + \boldsymbol{\epsilon}_c$, where Gaussian conditioning makes $\boldsymbol{\epsilon}_c$ independent of the training data with covariance $\Sigma_{HH} - K\Sigma_{TH}$. Subtract the predictor and add the independent coefficient-estimation term. The V_H-weighted squared error is $\chi^2_{|H|}$ under the fixed-exponent Gaussian model. If training selects p, that selection is repeated in its calibration. [19]

E.6 Bin windows and ratio covariance

For normalized bin windows $W_i(s)$, replace $t_i = e^{-ps_i}$ by

$$t_i(p) = \int W_i(s) e^{-ps}\,\mathrm{d}s.$$

A Gaussian window of variance v_i gives $t_i = e^{-ps_i + p^2 v_i/2}$; a top-hat window of half-width h_i gives $t_i = e^{-ps_i}\sinh(ph_i)/(ph_i)$. Unequal widths can create a deterministic shape residual when only bin centers are used.

For jointly estimated positive channel distances G_i, E_i and $Y_i = G_i/E_i$, the leading ratio Jacobian is

$$J_Y = \left[\mathrm{diag}(E_i^{-1}), -\mathrm{diag}(G_i/E_i^2)\right], \qquad \Sigma_Y \simeq J_Y V_{GE} J_Y^T.$$

Keep the cross-channel and cross-redshift blocks. Broad or non-Gaussian likelihoods use joint positive-distance likelihood/posterior samples in place of the linear expansion. A detected-event population also has selection normalization

$$\mathscr{A}(\vartheta) = \int p(\mathrm{detected} \mid \theta, \vartheta)\pi(\theta \mid \vartheta)\,\mathrm{d}\theta.$$

An amplitude-law change can alter this normalization; widening ratio error bars does not encode that change. [19]

F. Finite-window endpoint proofs

F.1 Correlated Gaussian two-point bound

For two mean curves with finite sample-vector difference $\boldsymbol{\Delta}$ and common positive-definite covariance Σ,

$$D_{\mathrm{KL}} = \tfrac{1}{2}\boldsymbol{\Delta}^T\Sigma^{-1}\boldsymbol{\Delta} \le \frac{m\|\boldsymbol{\Delta}\|_\infty^2}{2\lambda_{\min}(\Sigma)}.$$

For the slow-mode construction in the text, $\|\boldsymbol{\Delta}\|_\infty \le B\varepsilon s_{\max}$. Choosing

$$\varepsilon \le \frac{\sqrt{\lambda_{\min}(\Sigma)}}{B s_{\max}\sqrt{m}}$$

makes $D_{\mathrm{KL}} \le 1/2$. Pinsker's inequality gives total variation at most $1/2$. To turn an estimator into a test, assign it to the nearer of the two plateau values. An erroneous assignment entails endpoint estimation error at least $B/2$; the sum of test-error probabilities is at least $1 - \mathrm{TV}$. Hence every estimator obeys

$$\max_j P_j(|\widehat{\Xi}_\infty - \Xi_{\infty,j}| \ge B/2) \ge 1/4,$$

$$\max_j E_j|\widehat{\Xi}_\infty - \Xi_{\infty,j}| \ge B/8.$$

For the fast-mode pair, replace $B\varepsilon s_{\max}$ by $|D|e^{-Ps_{\min}}$ and B by $|D|$. The same argument bounds calibration recovery. These claims concern the stated broad response class; an independently derived rate restriction changes that class. [19]

F.2 Positive-spectrum endpoint inequalities

For a decreasing normalized curve

$$\Xi(s) = \Xi_\infty + \int_{p_{\min}}^\infty e^{-ps}\,\mathrm{d}B(p), \quad \mathrm{d}B \ge 0, \quad \int \mathrm{d}B = 1 - \Xi_\infty,$$

positivity gives $\Xi_\infty \le \Xi(s)$, while $e^{-ps} \le E = e^{-p_{\min}s}$ gives $\Xi(s) \le \Xi_\infty + E(1-\Xi_\infty)$. Rearrangement proves $(\Xi - E)/(1-E) \le \Xi_\infty \le \Xi$. The increasing case follows by applying the same argument to the positive deficit $\Xi_\infty - \Xi(s)$.

At the low endpoint, if Y_∞ is fixed independently and $0 < p \le p_{\max}$, then $Y(s) - Y_\infty \ge e^{-p_{\max}s}[C_{\rm cal} - Y_\infty]$ on a decreasing branch. Rearrangement gives the stated upper bound on $C_{\rm cal}$; positivity gives its lower bound. Simultaneous error bands preserve their declared joint coverage under monotone endpoint maps and interval intersection.

G. Retained synthetic benchmark specification

The following numbers are transcribed from the supplied Rev 6 source, not newly generated simulation outputs. Its 24 compressed samples span $z = 0.03$ to 2 equally in s, with $(\Xi_\infty, p, C_{\rm cal}) = (0.90, 3, 1.04)$. The marginal fractional scale is $f_i = \sqrt{0.02^2 + (0.0225 z_i)^2}$, rising from 2.00% to 4.92%. An independent centered lognormal component has shape 0.75 and rms fraction $0.004 + 0.004 z_i$; the remaining Gaussian core has an 80% diagonal and 20% exponentially correlated component in s. This is a synthetic compressed-data model, not a survey catalog.

The departures are $Y_{\rm two}/1.04 = 0.90 + 0.07e^{-3s} + 0.03e^{-s/2}$, $Y_{\rm drift}/1.04 = e^{0.01s}(0.90 + 0.10e^{-3s})$, and a localized feature centered at $s = 0.60$, width 0.09, amplitude 0.015. The exponent grid is $0.25 \le p \le 8$ with step 0.025. Thresholds use 30000 null catalogs and each reported case 20000 independent test catalogs. [19]

Table 9: Rev 6 reported rejection percentages at its percent-level error scale. Both test columns use the reported simulated 5% threshold.

Input	Fixed $p = 3$	Profiled p
Single power, $C_{\rm cal} = 1.04$	5.09	5.06
Same shape, $C_{\rm cal} = 0.96$	4.97	5.15
Second power	5.20	5.20
Calibration drift	4.97	5.03
Localized departure	6.20	5.87

The noiseless fitted (p, Ξ_∞) are $(2.57760, 0.91611)$ for the second power, $(3.34482, 0.90972)$ for drift and $(2.95009, 0.90775)$ for the localized departure. Their profiled Gaussian-core noncentralities at unit error scale are 0.001343, 0.002851 and 0.593907; fixed-p values are 0.03191, 0.02624 and 0.59429. Under $\Sigma \mapsto c^2\Sigma$, they become λ/c^2. The reported complete-noise precision sweep uses 40000 independent null checks and 20000 catalogs per departure/precision point. Smooth departures remain close to null rejection after exponent profiling at the 0.3% maximum-error endpoint, while fixed $p = 3$ gives approximately 30% and 24% rejection for the second-power and drift examples. [19]

Other reported internal checks include 95.017% fixed-exponent Fieller coverage from 100000 catalogs, interval $[0.85863, 0.94365]$ on the fiducial mean, and inferred calibration standard deviation 0.01595. The constant-ratio scan threshold is 4.81599; a separate 20000-catalog sample rejects 5.08%, compared with 8.93% when a one-exponent threshold is incorrectly reused. The 16-training/eight-held-out test has mean statistic 8.00894 and 5.005% rejection over 100000 catalogs. Two overlapping Hankel contrasts have analytic correlation -0.20587 and reported simulation correlation -0.20501 from 200000 samples. These check separate pieces of one synthetic framework.

With 24 independent ratios of common absolute error 0.03, the reported local Fisher $(\sigma_p, \sigma_{\Xi_\infty})$ are $(10.8103, 0.20500)$ for low-only sampling, $(2.0498, 0.02174)$ for broad sampling, $(21.8409, 1.20322)$ for high-only sampling and $(1.9063, 0.01883)$ for three bands. Low, transition and high-redshift coverage supply different leverage. A top-hat bin example fitted at point centers gives $(p, \Xi_\infty, C_{\rm cal}) = (3.01539, 0.900323, 1.040059)$; the window-aware fixed-p fit recovers the declared fiducial coefficients to its floating-point precision. [19]

H. Event-map robustness certificate

Suppose $F' \ge m_F > 0$, $\|\widetilde{F} - F\|_\infty \le \varepsilon_F$, and both inverse branches are defined on the common receiving interval. Monotonicity and the mean-value bound for F give

$$|\widetilde{F}^{-1}(t) - F^{-1}(t)| \le \varepsilon_F / m_F.$$

If $|\theta_E'| \le L_\theta$, the encoded phase uncertainty is therefore at most $L_\theta \varepsilon_F / m_F$. If derivative error at the compared emission events is at most $\varepsilon_v < m_F$, then

$$\left| \frac{1}{\widetilde{F}'} - \frac{1}{F'} \right| \le \frac{\varepsilon_v}{m_F(m_F - \varepsilon_v)},$$
$$|\ln \widetilde{F}' - \ln F'| \le -\ln(1 - \varepsilon_v/m_F).$$

Shifts of the emission argument require a bound on rate variation in addition to this derivative-error budget. With ideal current amplitude $|z| \ge r > 0$ and complex error $|\Delta z| \le e < r$, the phase-error disk gives $|\Delta\theta| \le \arcsin(e/r)$ on the continuous branch. Add encoding, transfer-phase, timing, model and readout errors on that branch. Sampling increments and total uncertainty must retain a strict margin below π, or an independently recorded winding is needed. These bounds supply a ledger interface; their numerical operands come from the apparatus and the actual propagation problem.

I. Reproduction routes and custody

The compact replay packet contains the assembled editable chapter, inherited TeX inserts and figures, new response, canonical, continuum and work/flux derivations, frozen source operands, independent checks and results. It avoids duplicating historical IV/V reconstruction archives. The unchanged predecessor's compact replay was independently rerun here: all eleven steps and fourteen result files matched exactly, with 920

standard-library assertions. That validates its retained execution, while the present directional-kernel and inverse repairs have their own checks.

Run `python3 ops/reproduce.py` for portable mathematical checks on a disposable relocated copy. Add an explicitly supplied companion evidence ZIP to authenticate all 17 successful native mathematical requests and four local retrievals, using 2,939 assertions and 290 matrix entries. All 126 retained helper files were authenticated and the relocated native readback is byte-identical. The actual run used six controllers, three helpers and two original 180-second admission windows on its own reserved cores; no global startup lease or third round was used. Final completion and release receipts are retained. No replay starts the native engine. Numerical Bessel quadrature is an optional separately labeled diagnostic.

The finite-redshift PDF's embedded artifact link resolves to `https://doi.org/10.5281/zenodo.22985359`; its printed text truncates that link. The recovered archive inside the unchanged predecessor source archive identifies itself as a new independent reconstruction of an earlier identifiability manuscript. Its `PROVENANCE.md` distinguishes preserved paper numbers from new code/results, records unresolved historical covariance and draw-order details, and does not claim recovery of original historical simulation outputs. That archive is retained as a reconstruction, while the supplied Rev 6 PDF remains the authority for the historical benchmark transcribed in Appendix G.

The archived GPS, optical and solar packets are arithmetic and scoped external-contact records. No new raw GPS reduction, optical-clock experiment or Cassini tracking fit is claimed by including them. The retained binary-pulsar receipt imports its masses and geometry and is not used as an independent arrival-time fit. The cited G11555 reference index did not supply its original strong-field theorem package; it is retained as a retrieval lead rather than substituted for that result. [17]

J. The complete actual-state tail coefficient

This explicit coefficient makes the preceding spatial tail bound independently reproducible. Put $h = 64/63$, $c_1 = 1/4 + 64/3969$, $\lambda_\pm = 64\Delta_\pm$, $k_0 = 1/R_-$, and $(d_1,d_2,d_3) = (1,3,3)$. For $n = 0,1,2$, define

$$I_0 = (k_0^2 - \lambda_-^2)/2, \quad I_1 = k_0 - \lambda_-,$$
$$I_2 = (k_0 - \lambda_-)/\lambda_-, \quad J_n = \frac{k_0^{4-n} - \lambda_-^{4-n}}{4-n},$$
$$A_n = \frac{hR_+^2}{15}(\lambda_+^{2-n} + k_0^{2-n}) + \frac{R_+^2}{15}[(n+2)h + c_1]I_n + \frac{hR_+^4}{105}J_n.$$

Here I_2 is an upper bound for the logarithmic integral, chosen to keep the executable certificate rational. Then

$$C_A(L) = |g|_+ \left\{ \frac{3}{4}\sum_{p=1}^{3} d_p L^{1-p} A_{p-1} + \frac{3}{2}\sum_{p,q=1}^{3} d_p d_q L^{1-p} R_-^{-q} k_0^{-(p+q-1)} \times \left[2h + \frac{c_1}{p+q-1}\right] \right\}.$$

It is valid for $L \geq 2R_+$ and $\lambda_+ < k_0$ and retains all nine large-frequency Bessel products. The conservative constants use $\pi \geq 3$ and $\sqrt{\kappa} \geq 8/9$. The fixed-shift tail adds $4C_F^2/(5L^5)$ to $8C_A(L)^2/L$, where

$$C_F(L) = \frac{16|g|_+ R_+^2}{15\pi_-\sqrt{\kappa}_-(1-y_L^2)}, \qquad y_L = \frac{2LR_+}{L^2+R_+^2}.$$

Take $\pi_- = 3$ and $\sqrt{\kappa}_- = 8/9$ as explicit rational lower bounds. The retained outward source endpoints are used throughout. This is the original fixed-shift envelope, including its L dependence. The time-dependent soft comparison and the preparation-energy minimum are paid separately in $\eta_{\mathrm{ref}}(L)$; they are not removed by taking L large. The compact replay keeps their original source enclosures.

Retained predecessor dependency ledger. This table records the earlier packet. Current canonical closures, local readout and remaining switched-shell/output operands are stated in Section 79 and the preceding new sections.

Dependency ledger: physical claims and remaining inputs

Object	Status	What it establishes and what it still requires
Original source and six clocks	Source-reported	Signed action, even energy, both polarizations, counterterms, exact hard-branch plus soft-vacuum preparation and finite clock window are retained. The new checks consume their outward certificates; they do not rerun the original spectral campaign.
Uniform complete lapse	Derived, conditional	The full interacting history follows one internal-age map. A physical law assigning that lapse to a clock or worldline remains an input.
Positive spatial receiver	Derived, selected model	The Bessel/TT incidence is positive and has a common continuum domain. Its equality to canonical local stress is not assumed.
Physical profile and units	Declared	One positive Plummer body, one compact-body control and one shared SI fixture fix all operands before comparing lanes. The microscopic ruler/action calibration and gravity assignment are not derived from the source ledger.
Six signed responses	Derived enclosures	All six finite form-error budgets fit 0.2 nrad at 7×10^{-10}; total sign is negative for every $0 < \alpha \leq 10^{-3}$. Direct orientation contrasts and exact nulls are bounded. Isolated susceptibility signs remain unresolved.
Canonical tensor stress	Derived candidate, open completion	The retained action supplies number, pair, source-shell and complement kernels. A specified switch supplies leading pair energy and its scalar loop. Complete transported boundary bounds and instantaneous frequency-form comparisons are derived. Projective implementability, boundary time regularity, joint remainder, finite energy reference and covariant support law remain open.
Gradient return	Derived static result	The bounded Plummer profile has a controlled nonzero static Schur return through magnetic degree2 and electric degree3 sectors. This is not a joint quantum expectation or gradient-sign tomography.
Event maps	Derived, conditional	Actual endpoint epochs, proper waits and the moving transfer Jacobian compose correctly under the declared metric and travel approximation. Ephemerides, medium and observational nuisance inputs remain external.
Original carrier	Derived, prescribed source	The compact pulse has near, induction and radiation terms, edge-uniform remainders, a 6.25% peak waveform correction and a signed whole-pulse energy deficit below 6.633×10^{-14}, exact positive near storage and complete maintained-source work balance. It remains a maintained four-cycle source protocol, distinct from the autonomous one-cycle clock certificate.
Amplitude coefficient	Open physical bridge	Localized source elimination gives its actual retarded kernel and passive spectral weight. It does not yet derive a homogeneous cosmological Q or its background evolution.
Positive rate spectrum	Derived Euclidean observable	The actual dressing vector gives a positive continuous, gapless Euclidean measure. Its transform and gap-free slow mass are calculated; exact Hankel rank is infinite. A relaxation law, detector readout and mapping to $\log(1+z)$ are required for a physical distance-rate interpretation.
Noisy mode inference	Derived, conditional	One known-covariance event and positive support constraints certify finite-window curve bounds and minimum visible rank. Positive Gaussian quadrature now matches finite continuum samples with signed holdout bounds. Noisy exact mode counts or unrestricted plateau accuracy remain unsupported.
Metrology	Specified requirements	Coarse winding and fine final sampling give a sufficient preparation count near $2.173\,10^{22}$. Statistical coverage is conditional on all systematic bounds; no apparatus performance is claimed.

References and source routes

[1] *Original Write histories and an interacting source-field action*. Volume I, Chapter VII (formerly Chapter VII), final source and clock certificates, 15 October 2026. https://github.com/iwtbotiwtwot/sam_volume_I_i.

[2] Same authors. *Retention, accumulation, and the finite ledger*. Chapter II source-to-field update, 15 October 2026. https://github.com/iwtbotiwtwot/sam_volume_I_ii. The retained 24-page/48-section source supplies the physical reference and measurement protocols.

[3] Same authors. *The spherical accumulation kernel and the source-to-field interface*. Chapter III local-action and quantum-source development, 15 October 2026. Retained 21-page/42-section source and linked certificates.

[4] N. Ashby. *Relativity in the Global Positioning System*. Living Reviews in Relativity 6, 1 (2003), publisher PDF revision 21 June 2007, equations (12), (16), (18), (21), (27), (35)–(40). https://doi.org/10.12942/lrr-2003-1.

[5] GPS Directorate. *IS-GPS-200N: NAVSTAR GPS Space Segment/Navigation User Segment Interfaces*, 1 August 2022, sections 3.3.1.1 and 20.3.3.3.3.1. https://archive.gps.gov/technical/icwg/IS-GPS-200N.pdf. This chapter retains the version-specific convention supplied with the original endpoint draft.

[6] SAM Research Project. *CR005: Clocks and GPS external contact*. Courtroom revision b5e914f71377e86ef4c67e199973d9300795cda1. https://github.com/iwtbotiwtwot/The_Courtroom/tree/b5e914f71377e86ef4c67e199973d9300795cda1/03_CLOCKS_AND_GPS/CR005_CLOCKS_AND_GPS_EXTERNAL_CONTACT.

[7] SAM Research Project. *CR006: Optical-clock height holdout, endpoint A kernel*. Same Courtroom revision. https://github.com/iwtbotiwtwot/The_Courtroom/tree/b5e914f71377e86ef4c67e199973d9300795cda1/03_CLOCKS_AND_GPS/CR006_OPTICAL_CLOCK_HEIGHT_HOLDOUT_ENDPOINT_A_KERNEL.

[8] T. Bothwell et al. *Resolving the gravitational redshift across a millimetre-scale atomic sample*. Nature 602, 420–424 (2022), p. 423 and Extended Data Table 1. https://doi.org/10.1038/s41586-021-04349-7. NIST publisher-version copy: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=933368.

[9] *SAM Unified Technical Synthesis—Volume I*, supplied manuscript, sections 3.1–3.6, Magic Bell printed p. 4, appendices A.5–A.6. Retained endpoint text: https://github.com/iwtbotiwtwot/SAMA/blob/4c2b5dc1649da5900c90b90d0ce56c223dbb7717/vol_i/03-clocks-and-light/endpoint-clock-comparison-and-time-dilation.md.

[10] N. Ashby and M. Weiss. *Global Positioning System Receivers and Relativity*. NIST Technical Note 1385 (1999), sections 2.3 and 2.7. https://tf.nist.gov/general/pdf/1274.pdf.

[11] *Endpoint clocks, motion, and the GPS comparison*. Supplied Chapter IV development draft and source/check bundle, 16 October 2026. Preserved unchanged in the predecessor archive, inputs/iv/.

[12] P. Teyssandier and C. Le Poncin-Lafitte. *General post-Minkowskian expansion of time transfer functions* (2008), arXiv:0803.0277, equations (57), (67). https://arxiv.org/abs/0803.0277.

[13] I. I. Shapiro. *Fourth Test of General Relativity*. Physical Review Letters 13, 789–791 (1964). https://doi.org/10.1103/PhysRevLett.13.789.

[14] B. Bertotti, L. Iess and P. Tortora. *A test of general relativity using radio links with the Cassini spacecraft*. Nature 425, 374–376 (2003). https://doi.org/10.1038/nature01997. Also the 2004 Bertotti author presentation linked in the supplied photon draft, with delay-derivative sign conventions distinguished.

[15] SAM Research Project. *CR006: Photon road Shapiro external contact*. Same pinned Courtroom revision. https://github.com/iwtbotiwtwot/The_Courtroom/tree/b5e914f71377e86ef4c67e199973d9300795cda1/04_PHOTON_ROAD_SHAPIRO_DELAY/CR006_PHOTON_ROAD_SHAPIRO_EXTERNAL_CONTACT. Distinct from optical-clock CR006.

[16] A. Hees, S. Bertone, C. Le Poncin-Lafitte and P. Teyssandier. *Range, Doppler and astrometric observables computed from time transfer functions: a survey* (2014), arXiv:1412.3360, equations (2)–(4). https://arxiv.org/abs/1412.3360.

[17] SAM Research Project. Retained CR148 binary-pulsar receipt and SAMA G11555 reference index, at the Courtroom revision above and SAMA revision 4c2b5dc1649da5900c90b90d0ce56c223dbb7717. Their evidence roles are stated in Appendix I; neither supplies a new independent fit or a recovered strong-field theorem here.

[18] *Photon roads, the Magic Bell, and Shapiro delay*. Supplied Chapter V development draft and source/check bundle, 16 October 2026. Preserved unchanged in the predecessor archive, inputs/v/.

[19] *Finite-redshift identifiability of cosmological distance ratios*, Rev 6, supplied 18-page PDF, 26 September 2026. Sections 2–7 and appendices A–B supply the retained amplitude-distance model, finite-window endpoint constructions, reconstruction and statistical benchmark. Embedded artifact link: https://doi.org/10.5281/zenodo.22985359. Recovered archive provenance is distinguished from original historical results in Appendix I. [20] NIST Digital Library of Mathematical Functions. Spherical Bessel integral representation, equation 10.54.1. https://dlmf.nist.gov/10.54.E1. The $n = 2$ identity confirms the positive averaging normalization used here.

[21] A. Balmaseda, D. Lonigro and J. M. Pérez-Pardo. *On the Schrödinger equation for time-dependent Hamiltonians with a constant form domain* (2021), Assumption 3.8 and Theorem 3.10. https://arxiv.org/abs/2112.11063.

[22] P. T. Nam, M. Napiórkowski and J. P. Solovej. *Diagonalization of bosonic quadratic Hamiltonians by Bogoliubov transformations* (2015), Theorem 1, Theorem 2 and the renormalized-operator remark following them. https://arxiv.org/abs/1508.07321.

[23] J. Dereziński. *Bosonic quadratic Hamiltonians* (2017 revision), Theorems 3.20–3.21. https://arxiv.org/abs/1608.03289.

Successor replay and evidence scope

The new receiving-form, signed-prefix, canonical-coefficient and differential metrology checks run with python3 ops/reproduce.py from the successor source packet. Supplying its companion native-evidence ZIP also authenticates every accepted current native return without starting Duck. Historical replay counts printed above belong to their unchanged predecessor packets. The retained joint source manifest authenticates all 193 source members; the old Downloads ZIP was absent at recovery and has not been reconstructed or represented as recovered. The complete retained member tree is included. The final successor verification records current counts, offline PDF build, all-page visual review and relocated source/native readback.

Retained C6 replay and evidence scope

The new static canonical joint, weighted-frame/free-switch, real-time memory, local-pointer and composed-record checks run with python3 ops/reproduce.py from the composition source packet. Its companion native-evidence ZIP authenticates every admitted current request, complete successful output and retained failure without starting Duck. Eleven native mathematical requests completed; one exceeded the installed scalar input-length budget. Its exact bounded replacement is independently checked but was not admitted after the original deadline. All four local-corpus retrievals completed. Six controllers and three helpers used two 180-second windows and own-core locks, with no global startup lease, third round or automatic repeat.

Historical replay counts and statuses above describe their retained packets. The actual predecessor source ZIP is now available and included unchanged: 438 manifest-listed files plus its manifest, including the earlier 193-member authenticated source tree. The new replay checks their bytes; it does not rerun historical research. Final verification records the current counts, offline PDF build, all-page visual review and relocated source/native readback.

Current C7 replay and evidence scope

The current finite-preparation, paired-cap, complete tensor and joint-domain proof checks run with `python3 ops/reproduce.py`. The companion native evidence packet retains every admitted request, full output, failure and bounded correction, with source hashes and portable independent readback. Twelve native requests completed; two original preparation requests failed because of an extra payload field. Both corrected requests were admitted inside the original second window and completed. The failures remain unchanged. Four local-corpus retrievals completed. Six controllers and three helpers used exactly two 180-second admission windows on their own cores, with no global startup lease, third round or automatic repeat.

The actual C6 predecessor ZIP is retained unchanged: 264 listed files plus its manifest. Its embedded C5 source ZIP has 438 listed files plus its manifest and includes the preceding authenticated 193-file source tree. The current replay checks those bytes without rerunning historical research. Historical counts and status statements elsewhere retain their original scope.

Data and code availability

The compact source packet contains this editable combined chapter, frozen source operands, derivations, executable checks and results. A separate full native-evidence packet retains the completed execution and custody records. The unchanged predecessor archive retains the original IV/V bundles and historical reconstruction provenance. The chapter source uses N for lapse, n_{opt} for optical index and p for the distance exponent throughout. Original inputs retain their original notation and evidence identities.

SAM VOLUME I

Strong fields and the approach to unit accumulation

Chapter V

Abstract. The source coordinate $A = r_s/r$ reaches unity at a finite areal radius. Whether that surface is approached in infinite coordinate time, crossed in finite proper time, or excluded by a physical evolution law depends on more than the coordinate. This chapter derives radial arrival and circular-orbit relations for declared static spherical metrics, recovers the Schwarzschild landmark packet, and separates its exterior logarithm from the traveler's elapsed time and the reception of outgoing clock signals. A general endpoint classification distinguishes optical distance, static proper distance and infalling proper time. The SAM no-completed-parent-ledger-crossing proposition is retained as an additional physical hypothesis; a bounded-progress condition makes its dynamical requirement precise and exposes the cost of implementing it inside the benchmark geometry. Invariant sphere diagnostics, finite-observer shadow geometry and conditional surface-gravity relations identify what further information a strong-field completion must supply. The continuation adds finite-impulse and finite-distance approach theorems, an optimal accumulated-impulse bound, a finite-window comparison using all six original interacting clocks, radial-metric inference, and a causal two-polarization finite-pulse boundary protocol. The latter retains physical surface storage and a finite-potential error budget. Review-derived exact tidal strain, a curved static inverse, tighter pulse energy, flash fluence and a conservative quantum reservoir are included. Centered field and both propagated current insertions are addressed explicitly, together with a finite-budget progress candidate. The recovered HOME-QW1 finite occupation kernel is joined to a specified guide state and the full annular transmission. Finite-packet count and energy enclosures, a controlled thermal comparison spectrum and the remaining source-selected event clock are distinguished. A joint original-source protocol adds six nonzero phase-bearing signals, a complete complex annulus transfer, detector/noise and finite-repeat bounds, source energy ceilings and conditional inventory feedback. Exact native and portable checks accompany the derivations; source-selected strong-field dynamics and extended quantum-clock coupling remain open.

1. A coordinate survives where an approximation does not

Chapter IV follows a source through its clock, signal and receiving operations.[4] Strong field asks whether that same chain has a complete continuation as a source lift approaches unity. The question is more demanding than inserting a large value into a weak-field formula. A first-order clock slope and a first-order photon delay leave higher-order geometry and matter evolution unspecified.

Retain the source normalization

$$r_s = \frac{2GM}{c^2}, \qquad A(r) = \frac{r_s}{r}, \qquad 0 < A < 1. \tag{1}$$

Here r will be an areal radius whenever a spherical metric is declared: a symmetry sphere has area $4\pi r^2$. The R used for the original microscopic source radius in earlier chapters is a different object. The floor A_0 is not added to this closure coordinate. A nonlinear raw-floor lapse would have its own normalization and must not silently replace the source-only convention retained in Chapter IV.

The unified source presents $A = 1$ as the terminal parent-ledger boundary and expresses its intended traveler interpretation as "the astronaut does not run out of clock; the astronaut runs out of road."[3] We retain the physical proposition while asking what equation makes the road a traveler's measured duration. The same source explicitly identifies the familiar photon-sphere and orbital landmarks as inherited through a Schwarzschild exterior adapter. These two statements have different logical roles.

Three questions will remain separate:

1. How much time does a chosen exterior coordinate assign to a route?
2. How much proper time does a specified traveler actually accumulate?
3. When, and with what rate and amplitude, does a receiver obtain information emitted on that worldline?

A divergence in one answer does not establish a divergence in the others. Chapter IV's separation of endpoint history and travel-time map therefore becomes especially important here.

1.1 The assumptions carried by each result

The twelve-page supplied review supplies the stronger analytical closures incorporated here.[1] New calculations and candidate models are identified separately. The following ledger keeps each operation attached to the law that defines it.

Operand	Stated law or scope
Geometry	Static F,H when declared; Schwarzschild is the benchmark, and changed metrics are comparison controls.
Parent ledger	The original source-only $A = r_s/r$ admissibility rule; traveler evolution requires its additional equation.
Original clock	Full interacting H_0, six original preparations and certified finite ages. Proper-time point transport is an explicit conditional coupling.
Wave and inverse	RM radial tensor action, full channel metric and stated boundary conditions. The central areal shell and local off-center launcher remain distinct.
Source diagnostic	Einstein/null-fluid or radial energy-condition statements only where explicitly selected. Global positive energy is not a local null-stress condition.
Reservoir	Conservative waveguide dynamics with identified vacuum/coherent or finite QW1-derived preparation, a common energy calibration and explicit supply. The state bridge is declared.

2. What the recovered strong-field record establishes

The pinned strong-field branch supplies five scoped packets. CR007 tests the native labels $A = 1, 2/3, 1/3$ at supplied Schwarzschild landmark radii. CR008 tests the declared square-root lapse and exterior logarithmic radial road, including wrong weak-field extrapolations. Its explicit boundary is exterior clock/traversal closure; it does not claim a complete infalling-observer result. CR009 and CR010 check photon-sphere and ISCO invariant packets against supplied benchmark values. CR011 records support and appeal bookkeeping; it does not add a new field equation.[5]

The recovered SAMA horizon chapter names the substrate-native infaller ruler and road as an open boundary.[6] The original G11555 code has now been recovered and authenticated against the retained CR075 manifest; the full original seal/output packet is still absent. Its explicit parent-ledger rule and exterior road are separated below from the stronger traveler-duration hypothesis. A new qualified native continuation verifies finite coefficient, current and boundary calculations under declared laws. Neither the original source rule nor these conditional calculations selects a complete microscopic infaller law.

There is also a small but useful reproducibility distinction in CR008. Its near-boundary clock rows use $A = 1-\varepsilon$, while its road rows use $r/r_s = 1+\varepsilon$. These label asymptotically neighboring points, not exactly the same point. For a common point define

$$x = \frac{r-r_s}{r_s}, \qquad \delta = 1-A, \qquad x = \frac{\delta}{1-\delta}, \quad \delta = \frac{x}{1+x}. \tag{2}$$

The calculations and figures below preserve this exact conversion. A binary rounded square-root-minus-one or logarithm is not an independent uncertainty in the physical source.

2.1 Recovered G11555 and the exact domain

The original G11555 Python has now been recovered on the T500. Its raw SHA-256 matches the identity in the retained CR075 manifest. The code starts from the source coordinate and parent-ledger admissibility rule, calculates the exterior logarithmic road, and labels its finite infaller quantity as a GR comparison proxy. The recovered review retains `PASS_MAGIC_BELL_THEOREM` and `NO_COMPLETED_PARENT_LEDGER_CROSSING` in that source scope. No original runner was restarted. The complete original seal and output receipt set is not included here; no replacement receipts are manufactured. The force/progress law for the stronger traveler-local-duration hypothesis remains a separate dynamical object.

The historical implementation uses a 10^{-12} numerical band about unity. Preserve that implementation while stating the exact successor predicate in the directly stored gap $\delta = 1-A$:

$$\delta > 0: \text{ exterior}, \qquad \delta = 0: \text{ closure},$$
$$\delta < 0: \text{ rejected parent write}.$$

An uncertain interval is assigned a strict side only when its entire interval lies there. A singleton zero is closure; an interval containing zero and another value is unresolved. Exact rational tests retain nonzero gaps down to 10^{-40} without converting them to a unit endpoint. This is a source-domain predicate, not an observed crossing.

2.2 A divergent road and its power measure

The recovered G11555 review also supplies a useful work distinction. A physical power density must come with its actual clock:

$$dW = P_t\,dt = P_\tau\,d\tau, \qquad P_\tau = P_t\frac{dt}{d\tau}.$$

On the exterior null-road comparison, $|dt/dx| = t_s(1+x)/x$. If $0 \le P_t(x) \le P_0 x^a$ with $a > 0$, then

$$W_{0<x\le\varepsilon} \le t_s P_0\left(\frac{\varepsilon^a}{a} + \frac{\varepsilon^{a+1}}{a+1}\right).$$

For a positive matching asymptotic, finiteness is equivalent to $a > 0$; equality is logarithmic. On the distinct constant-optical-road-speed traveler, $|dt/dx|$ scales as $x^{-3/2}$ and $|d\tau/dx|$ as x^{-1}. The absolute values define positive inward duration measures. Its coordinate-power tail $P_t \sim P_0 x^a$ is finite exactly when $a > 1/2$, while its proper-power tail $P_\tau \sim Q_0 x^b$ is finite exactly when $b > 0$. The clock Jacobian maps $b = a - 1/2$ and makes these criteria agree. Signed net work, nonnegative apparatus expenditure and accumulated proper acceleration retain separate accounts. No propulsion power is selected by a divergent traversal integral alone.

3. Strong fields, arrival times, and the surface A = 1

A divergence in a travel-time formula is meaningful only after specifying whose clock measures that time. Three questions

must be kept distinct: whether a static-coordinate time diverges, whether a falling clock reaches a limiting radius in finite proper time, and whether a regular spacetime extension exists across that radius. None is a substitute for the others. In particular, the notation $A = 1$ does not itself establish either a horizon or its absence.

3.1 Metric hypotheses and the clocks being compared

Let $r_s > 0$ be the fixed source-normalization length, $A = r_s/r$, and consider an exterior static, spherically symmetric metric in areal radius,

$$ds^2 = -F(r)c^2dt^2 + H(r)dr^2 + r^2d\Omega^2, \qquad r > r_s, \quad (3)$$

with positive, sufficiently differentiable F, H on this open region. If an asymptotically flat clock normalization is intended, set $F \to 1$ at infinity. These are kinematic assumptions, not field equations. A static clock satisfies $d\tau_{\text{stat}} = \sqrt{F}\,dt$; radial distance on a constant-t slice is $d\ell = \sqrt{H}\,dr$. Static observers need not exist at the limiting surface. Neither their lapse nor their slice distance is automatically the proper time or distance measured by an infaller.

For a radial null ray, $ds^2 = 0$ gives

$$\frac{dt}{dr} = \pm\frac{1}{c}\sqrt{\frac{H}{F}}, \qquad \ell = \int_{r_1}^{r_2}\sqrt{H}\,dr, \qquad \Delta t_{\text{null}} = \frac{1}{c}\int_{r_1}^{r_2}\sqrt{\frac{H}{F}}\,dr. \quad (4)$$

The positive integral is the coordinate travel duration between ordered radii. Light has zero proper time; the first integral is not a photon-clock reading. For a receiver held at r_o, differences in arrival coordinate times become receiver proper-time differences through $\sqrt{F(r_o)}$.

For a radial timelike geodesic, the time-translation Killing symmetry gives the dimensionless conserved energy per unit rest energy $e = F\,dt/d\tau > 0$. Timelike normalization then yields

$$\frac{dr}{d\tau} = -c\sqrt{\frac{e^2 - F}{FH}}, \qquad \frac{dt}{d\tau} = \frac{e}{F}. \quad (5)$$

The minus sign selects inward motion. On an inward segment without a turning point, $e^2 > F$, the elapsed times down to $r_1 < r_2$ are

$$\Delta\tau = \frac{1}{c}\int_{r_1}^{r_2}\frac{\sqrt{FH}}{\sqrt{e^2 - F}}\,dr, \quad (6)$$

$$\Delta t_{\text{fall}} = \frac{1}{c}\int_{r_1}^{r_2}\frac{e\sqrt{H/F}}{\sqrt{e^2 - F}}\,dr. \quad (7)$$

These expressions distinguish the clock carried by the particle from the static time label. A turning point elsewhere may prevent an orbit from entering this segment; local endpoint tests do not rule that out.

3.2 Endpoint classification

Write $x = (r - r_s)/r_s > 0$ and suppose, as $x \downarrow 0$,

$$F \quad fx^p, \qquad H \quad hx^{-q}, \qquad f, h > 0. \quad (8)$$

Here $\sim$ means the ratio tends to one. Thus positive comparison bounds hold sufficiently close to the endpoint, making the following convergence tests rigorous. Since $dr = r_s dx$, null coordinate time has integrand proportional to $x^{-(p+q)/2}$ and static slice distance has integrand proportional to $x^{-q/2}$. Therefore

$$\Delta t_{\text{null}} = \infty \iff p + q \geq 2, \quad (9)$$

$$\ell = \infty \iff q \geq 2. \quad (10)$$

Equality produces logarithmic divergence; a strict inequality produces power-law divergence. The conclusions concern the improper integrals from a fixed exterior radius to the endpoint, not finite exterior intervals.

For infall additionally assume $p > 0$ and a fixed finite nonzero e. Then $F \to 0$, $\sqrt{e^2 - F} \to e$, and the radial branch is allowed sufficiently near the endpoint. Consequently

$$\left|\frac{d\tau}{dx}\right| \sim \frac{r_s\sqrt{fh}}{ce}\,x^{(p-q)/2}, \qquad \left|\frac{dt}{dx}\right| \sim \frac{r_s}{c}\sqrt{\frac{h}{f}}\,x^{-(p+q)/2}. \quad (11)$$

It follows that

$$\Delta\tau_{\text{fall}} = \infty \iff q - p \geq 2, \qquad \Delta t_{\text{fall}} = \infty \iff p + q \geq 2. \quad (12)$$

For $p = 0$ these tests also hold if $e^2 > f$, but the coefficients differ. The threshold case $e^2 = f$ requires subleading terms. For $p < 0$, F diverges and a fixed finite-energy timelike orbit cannot approach this endpoint through an allowed radial branch. These qualifications matter: a lapse alone does not determine timelike accessibility.

Finite infaller proper time establishes approach to the boundary in finite clock time. Crossing requires a suitable extension of the geometry and worldline. Conversely, infinite static-coordinate time establishes the behavior of that coordinate. Curvature, causal structure, regularity, and the actual strong-field dynamical law must be examined before assigning a physical interpretation to $A = 1$.

3.3 Schwarzschild as a comparison benchmark

For the Schwarzschild benchmark only, identify $r_s = 2GM/c^2$ and take

$$F = 1 - A = 1 - \frac{r_s}{r}, \qquad H = F^{-1}. \quad (13)$$

This is an independently specified general-relativistic example, not a strong-field law derived from SAM.[7, 8] Here $p = q = 1$: null and infall coordinate times diverge logarithmically, whereas radial static distance and infall proper time to r_s are finite.

Define the tortoise radius

$$r_* = r + r_s\log\left(\frac{r}{r_s} - 1\right). \quad (14)$$

Radial null rays satisfy $ct = \pm r_* + \text{constant}$. For inward geodesics with $e = 1$, corresponding to release from rest at infinity, $dr/d\tau = -c\sqrt{r_s/r}$. Writing $y = \sqrt{r/r_s} > 1$ gives

$$\tau_H - \tau(r) = \frac{2}{3c\sqrt{r_s}}\left(r^{3/2} - r_s^{3/2}\right), \tag{15}$$

$$t(r) = C - \frac{r_s}{c}\left[\frac{2}{3}y^3 + 2y + \log\left(\frac{y-1}{y+1}\right)\right]. \tag{16}$$

The first is the finite remaining proper time from any finite exterior radius; it does not assert finite falling time from infinity. The second diverges as $r \downarrow r_s$.

An explicit change of coordinates establishes what the divergent static chart cannot decide. Set $v = t + r_*/c$. Then

$$ds^2 = -Fc^2dv^2 + 2c\,dv\,dr + r^2d\Omega^2. \tag{17}$$

Its radial block has determinant $-c^2$, including at r_s, and its coefficients are regular there. Inward timelike motion satisfies

$$\frac{dv}{d\tau} = \frac{e - \sqrt{e^2 - F}}{F} = \frac{1}{e + \sqrt{e^2 - F}} \longrightarrow \frac{1}{2e}. \tag{18}$$

Thus Schwarzschild admits regular inward crossing despite static-time divergence. This benchmark neither establishes nor excludes the same extension for another proposed strong-field metric.

3.4 Received clock rates and the finite source-age window

Consider outward radial photons emitted at r and received by a static observer at a fixed $r_o > r$, with $F_o = F(r_o) > 0$. A static emitter gives

$$\frac{\nu_o}{\nu_{\text{em}}} = \sqrt{\frac{F}{F_o}}. \tag{19}$$

An inward geodesic emitter has both gravitational and motion-induced redshift. Conservation of photon Killing energy, contracted with each observer's four-velocity, gives instead

$$\frac{\nu_o}{\nu_{\text{em}}} = \frac{F}{\sqrt{F_o}\,[e + \sqrt{e^2 - F}]} = \frac{e - \sqrt{e^2 - F}}{\sqrt{F_o}}. \tag{20}$$

Equivalently, differentiate the arrival map

$$\begin{aligned} t_o(\tau) &= t(\tau) + \frac{1}{c}\int_{r(\tau)}^{r_o} \sqrt{H/F}\,dr, \\ \frac{d\tau_o}{d\tau} &= \sqrt{F_o}\,\frac{e + \sqrt{e^2 - F}}{F}. \end{aligned} \tag{21}$$

The frequency ratio is its reciprocal for neighboring emitted ticks. Thus a static-lapse formula cannot be applied unchanged to a falling clock. These are radial geometric-optics rates; brightness additionally depends on emission, propagation, and detector sensitivity.

In Schwarzschild, $F \sim x$, so the falling-clock rate is $x/(2e\sqrt{F_o})$, whereas the static-clock rate is $\sqrt{x/F_o}$. Moreover,

$$\tau_H - \tau \sim \frac{r_s x}{ce}, \qquad \tau_o = -\frac{2r_s\sqrt{F_o}}{c}\log x + O(1). \tag{22}$$

Arbitrarily late idealized received signals therefore encode a finite interval of the original falling clock's age, approaching τ_H from below. They do not display unbounded aging of that clock. A source that stops emitting earlier has a correspondingly smaller age window; a finite pulse train or finite detector threshold need not yield an indefinitely observable signal. The divergent arrival label must not be mistaken for an assertion about unlimited visible source history.

3.5 Why weak-field agreement does not settle the endpoint

Even matching both metric functions to first order leaves strong-field behavior undetermined. For example, $F = e^{-A}$, $H = e^A$ and the Schwarzschild pair both satisfy $F = 1 - A + O(A^2)$, $H = 1 + A + O(A^2)$, but the exponential pair is regular and positive at $A = 1$. More generally, on $0 < A < 1$,

$$F = e^{(p-1)A}(1-A)^p, \qquad H = e^{(1-q)A}(1-A)^{-q} \tag{23}$$

has those same first-order expansions for arbitrary p, q, while realizing the independent endpoint exponents above. These are kinematic counterexamples to inference from weak-field coefficients, not proposed solutions of any particular field equations. A defensible Chapter V conclusion must supply the independently justified strong-field functions, identify the relevant clocks, and establish accessibility and extension with those functions. Neither fitting a weak-field law nor finding a static-chart divergence supplies that missing proof.

4. Circular orbits: what the strong-field adapter fixes

The source coordinate $A = r_s/r$, with $r_s = 2GM/c^2$, labels a compactness; it does not by itself determine a spacetime. Section 4 of the unified source makes this distinction explicitly: its photon-sphere and innermost-stable-circular-orbit results are inherited through a Schwarzschild exterior adapter. We retain that evidence boundary.[3, 8] First derive what an arbitrary static spherical metric would predict; then insert the benchmark adapter. In particular, this calculation does not claim that a SAM action has already derived exact metric geodesics.

Use the areal radius r, an asymptotically normalized time t, and

$$ds^2 = -F(A)c^2dt^2 + H(A)dr^2 + r^2d\Omega^2, \qquad A = \frac{r_s}{r}. \tag{24}$$

The circular-orbit calculation applies in a regular static region where $F > 0$ and $0 < H < \infty$. Assume a C^2-controlled weak-field expansion of the standard exterior through first order means $F = 1 - A + O(A^2)$ and $H = 1 + A + O(A^2)$. These are assumptions on an adapter, not a derivation of its higher orders. Spherical symmetry permits an equatorial orbit. For a massive test body define dimensionless specific energy $E = F\,dt/d\tau$ and specific angular momentum $L = r^2 d\phi/d\tau$. Thus the physical conserved energy and angular momentum are mc^2E and mL. The normalization of the four-velocity gives

$$FH\frac{\dot r^2}{c^2} + V(r;L) = E^2, \qquad V = F\quad 1 + \frac{L^2}{c^2r^2}\quad . \tag{25}$$

A prime below denotes d/dr, including the radial dependence through A; a subscript A denotes d/dA.

4.1 Circularity and radial stability

At a circular orbit $\dot r = 0$ and $V' = 0$ at fixed L. Writing $D = 2F - rF'$, the second condition gives $F' + (L^2/c^2)(F'/r^2 - 2F/r^3) = 0$, hence

$$E^2 = \frac{2F^2}{2F - rF'}, \qquad \frac{L^2}{c^2} = \frac{r^3F'}{2F - rF'}, \qquad \Omega^2 = \left(\frac{d\phi}{dt}\right)^2 = \frac{c^2F'}{2r}. \tag{26}$$

For the ordinary nonzero-angular-momentum timelike branch one needs $D > 0$ and $F' > 0$ as well as regularity. The case $F' = 0$ can instead support a static geodesic with $L = 0$ and requires separate interpretation. These existence conditions matter: an algebraic root outside the static region is not an exterior circular orbit.

Differentiate the potential again before substituting the circular value of L. The resulting curvature is

$$V''_{\rm circ} = \frac{2}{rD}\underbrace{\left(rFF'' + 3FF' - 2rF'^2\right)}_{S(r)}. \tag{27}$$

Linear radial perturbations satisfy $\delta\ddot r + [c^2V''_{\rm circ}/(2FH)]\delta r = 0$. Consequently $S > 0$ is stable, $S < 0$ unstable, and $S = 0$ marginal on this regular timelike branch. A marginal root is an ISCO only after checking the surrounding existence and stability intervals; disconnected stable bands need not have a single universal inner boundary. The function H drops out of circular locations and the sign criterion, but it sets proper radial distances and perturbation time scales. The A versions useful for direct comparison are

$$D = 2F + AF_A, \qquad S = \frac{A}{r}\left(AFF_{AA} - FF_A - 2AF_A^2\right). \tag{28}$$

For a null geodesic let $\mathscr{E} = Fc^2dt/d\lambda$ and $\mathscr{J} = r^2d\phi/d\lambda$, and define its impact parameter $b = c\mathscr{J}/\mathscr{E}$. The radial equation is $FH\dot r^2 = \mathscr{E}^2/c^2 - F\mathscr{J}^2/r^2$. The circular-null condition and its impact parameter are therefore

$$\frac{d}{dr}\left(\frac{F}{r^2}\right) = 0 \quad\Longleftrightarrow\quad rF' = 2F, \qquad 2F + AF_A = 0, \qquad b^2 = \frac{r^2}{F}. \tag{29}$$

A maximum of F/r^2 is radially unstable and a minimum stable when FH is positive and regular. Calling its impact parameter a capture threshold also requires an absorbing interior or suitable global completion; circularity alone does not establish a black-hole shadow.

The same formulas recover the appropriate weak-field limits without assuming the Schwarzschild completion. For $A \ll 1$, $F' = r_s/r^2 + O(r_s^2/r^3)$, so $\Omega^2 = GM/r^3 + O(c^2r_s^2/r^4)$ and $L^2 = GMr[1 + O(A)]$. Expanding the specific energy gives $E = 1 - A/4 + O(A^2)$; therefore $mc^2(E-1) = -GMm/(2r) + O(mc^2A^2)$, the Newtonian circular binding energy. The cancellation leading to these limits explains why the weak-field tests can agree even when strong-field landmarks differ. Neither the first-order coefficient nor the definition of A fixes F_{AA} at finite compactness, although that derivative enters the stability test directly.

The time normalization is equally important. Here Ω is an angular frequency with respect to the asymptotic time coordinate, not the frequency on a local static clock. A static observer at the orbit has proper time $d\tau_{\rm stat} = \sqrt{F}\,dt$ and measures $v_{\rm stat} = r\Omega/\sqrt{F}$, so $v_{\rm stat}^2/c^2 = rF'/(2F)$. The timelike condition $D > 0$ therefore says precisely that this local orbital speed is less than c. Its approach to unity identifies the null-circle limit. This check also prevents a coordinate speed from being mistaken for a local violation of the light-speed bound.

4.2 The Schwarzschild benchmark in the source coordinate

Now, and only now, choose

$$F = 1 - A, \qquad H = (1-A)^{-1}, \qquad 0 < A < 1. \tag{30}$$

The null condition becomes $2 - 3A = 0$. Its unstable photon sphere is $r_{\rm ph} = 3r_s/2$, or $A_{\rm ph} = 2/3$, with $b_{\rm crit} = 3\sqrt{3}\,r_s/2$. For an asymptotically distant observer and a black-hole completion, the associated diameter in the impact parameter plane is $2b_{\rm crit} = 3\sqrt{3}\,r_s$. The timelike family is

$$E^2 = \frac{2(1-A)^2}{2-3A}, \qquad \frac{L^2}{c^2r_s^2} = \frac{1}{A(2-3A)}, \qquad \frac{\Omega^2r_s^2}{c^2} = \frac{A^3}{2}. \tag{31}$$

It exists for $0 < A < 2/3$. Since $S = (A/r)(1-3A)$, the stable part has $A < 1/3$ and the unstable part $1/3 < A < 2/3$. Thus

$$A_{\rm ISCO} = \frac{1}{3}, \quad r_{\rm ISCO} = 3r_s, \quad E_{\rm ISCO} = \sqrt{\frac{8}{9}}, \qquad \frac{L_{\rm ISCO}}{cr_s} = \sqrt{3}, \qquad \frac{\Omega_{\rm ISCO}r_s}{c} = \frac{1}{\sqrt{54}}. \tag{32}$$

A further, distinct landmark follows from the marginal binding condition $E = 1$, with rest energy normalized at infinity. The equation $2(1-A)^2 = 2 - 3A$ has the asymptotic root $A = 0$ and the finite circular-orbit root $A_{\rm mb} = 1/2$, or $r_{\rm mb} = 2r_s$. It is unstable, with $L/(cr_s) = 2$. Finally $F = 0$ and $H^{-1} = 0$ at $A_H = 1$ locate the Schwarzschild Killing horizon, which is an event horizon in the standard black-hole completion. None of these results identifies marginal binding, marginal stability, the photon sphere, and the horizon with one another. Their ordered compactnesses are $1/3$, $1/2$, $2/3$, and 1.

4.3 Why the weak field does not select these landmarks

Compare two mathematical completions: the Schwarzschild pair above and the exponential pair

$$F_{\rm exp} = e^{-A}, \qquad H_{\rm exp} = e^{A}. \tag{33}$$

Both match $F = 1 - A + O(A^2)$ and $H = 1 + A + O(A^2)$, and both have $FH = 1$. No field equation or matter model is being asserted for the second pair; one counterexample to uniqueness is sufficient. For it, $D = e^{-A}(2 - A)$ and $S = (A/r)e^{-2A}(1 - A)$. Circular timelike orbits consequently exist for $0 < A < 2$ and are stable for $A < 1$. Its marginally stable orbit is at $A = 1$, its unstable null circle at $A = 2$, and its finite marginally bound orbit solves $2e^{-A} = 2 - A$, giving $A \simeq 1.593624260$. The lapse and inverse radial metric never vanish at any finite positive A, including $A = 1$. This excludes a finite-radius Killing horizon in that metric domain; claims about an event horizon or the endpoint $r = 0$ would require a global analysis not supplied here.

The exponential landmarks beyond unity refer to the broader domain $r > 0$ of this comparison metric. If the SAM parent-ledger domain is restricted to $A < 1$, they are outside that domain, and no photon circle occurs inside it. At unity this example remains locally regular. Thus the coordinate statement $A(r_s) = 1$ cannot select a horizon, a divergence of traversal time, or a terminal physical state by itself. A metric law or an additional physical completion rule must do that work. The source's unit-closure interpretation may be retained as an explicit extra SAM commitment; it must not be presented as a consequence of these weak-field expansions or of circular-orbit algebra.

4.4 Measure the radial metric through the same chain

Retain $D = 2F - rF'$ and $S = rFF'' + 3FF' - 2rF'^2$. Linear radial perturbations about a stable circular orbit have

$$\omega_{r,\tau}^2 = \frac{c^2 S}{rDFH}, \qquad \omega_{r,t}^2 = \frac{c^2 S}{2rFH}. \tag{34}$$

The time conversion uses the circular value $dt/d\tau = E/F$ and $E^2 = 2F^2/D$. Away from the marginal root,

$$H = \frac{c^2 S}{2rF\omega_{r,t}^2} = c^2 F[T'_{\text{null}}(r)]^2. \tag{35}$$

Static clock ratios determine relative F; normalization fixes its scale. The equality tests common metric coupling using the same clock, route and orbital perturbation. Known areal radii and the null/test-body actions are inputs. A separately assigned optical index must retain its own hypothesis.

An exact control changes only the radial metric:

$$F = 1 - A, \qquad H_\lambda = \frac{1 + \lambda A^2}{1 - A}, \qquad \lambda > -1.$$

All static rates, circular radii, E, L, Ω, and photon impact parameter are unchanged. Nevertheless,

$$\omega_{r,t}^2 = \frac{\Omega^2(1 - 3A)}{1 + \lambda A^2},$$
$$O := c^2[T'_{\text{null}}]^2 = \frac{1 + \lambda A^2}{(1 - A)^2}. \tag{36}$$

The leading radial-road logarithm acquires $\sqrt{1 + \lambda}$. The photon-circle instability exponent is

$$\Lambda_t^2 = \frac{c^2(2F - r^2 F'')}{2Hr^2} = \frac{4c^2}{27r_s^2(1 + 4\lambda/9)}.$$

Connecting this exponent to a wave, image or echo amplitude still uses the corresponding wave and boundary problem.

4.5 Finite-window inference and an exact allowance

On a compact known areal-radius interval require positive lower bounds for $F, S, \omega_{r,t}^2$. Outward intervals then give, at an exact declared r,

$$H \in \left[\frac{c^2 S_-}{2rF_+ k_+}, \frac{c^2 S_+}{2rF_- k_-}\right], \qquad k_\pm \text{ bound } \omega_{r,t}^2.$$

Use interval multiplication for the optical estimate in Eq. (35); include radius uncertainty in the same positive products when present. No independence assumption is needed. If derivatives are estimated by centered steps h, absolute sample errors δ_F, δ_T and supplied smoothness ceilings give

$$|F' - D_h F| \le \delta_F/h + M_3 h^2/6,$$
$$|F'' - D_h^2 F| \le 4\delta_F/h^2 + M_4 h^2/12,$$
$$|T' - D_h T| \le \delta_T/h + M_{T,3} h^2/6.$$

Interval evaluation of S composes these errors. The smoothness ceilings and phase-to-time calibration are explicit inputs; the original interacting transient remains in that calibration.

For the H_λ control set $z = \omega_{r,t}^2/\Omega^2$ and $q = 1 - 3A$. The two estimates are

$$R_{\text{dyn}} = q/z, \qquad R_{\text{opt}} = (1 - A)^2 O, \qquad \lambda = (R - 1)/A^2.$$

An exact worked allowance uses one $\lambda = 1/2$ at $A = 1/12, 1/10, 1/8, 1/6, 1/5, 1/4$. Give each of F, O, Ω^2 and $\omega_{r,t}^2$ relative error at most $\varepsilon = 10^{-4}$ in the same declared time/radius chart. Then

$$R_{\text{dyn}} \in \left[\frac{1 - \varepsilon}{1 + \varepsilon}, \frac{1 + \varepsilon}{1 - \varepsilon}\right] R_{\text{true}},$$
$$R_{\text{opt}} \in [(1 - \varepsilon)^3, (1 + \varepsilon)^3] R_{\text{true}}.$$

Mapping both intervals to λ and intersecting contains $1/2$ and excludes zero at all six radii. At $A = 1/6$ specifically,

$$F = \tfrac{5}{6}, \quad H = \tfrac{73}{60}, \quad O = \tfrac{73}{50}, \quad \Omega^2 = \tfrac{1}{432}, \quad \omega_{r,t}^2 = \tfrac{1}{876}$$

in $c = r_s = 1$ units, and the outward joint interval is contained in $[0.49270072, 0.50730074]$. These are declared measurement allowances; the apparatus must supply them. Shared estimates must retain their covariance. Using the static observer's local radial frequency without the lapse conversion changes the common metric identity from one to $1/F$, with exact residual $A/(1 - A)$.

The exact coefficient task evaluates the complete degree-five stability polynomial in an eight-coordinate faithful polynomial representation. The six-radius native task evaluates the common optical/dynamic identity on literal comparison points. The analytical endpoint and force arguments below supply their universal statements; finite matrix samples retain their own scope. Portable Fraction checks and outward intervals are supplied with the source packet, separately from the native execution receipts.

5. An additional decimal nine is an operational comparison

Let $t_s = r_s/c$ and use one common radial gap $x = (r - r_s)/r_s$. For a fixed outer radius $r_o = r_s(1+x_o)$, the benchmark radial coordinate light time is exactly

$$\frac{T(x \to x_o)}{t_s} = x_o - x + \log\frac{x_o}{x}. \tag{37}$$

Replacing x by $x/10$ adds $t_s[\log 10 + 0.9x]$, approaching $t_s \log 10$. A decimal nine in A, however, means replacing $\delta = 1 - A$ by $\delta/10$. The exact dimensionless increment is

$$\frac{\Delta T}{t_s} = \frac{\delta}{1-\delta} - \frac{\delta}{10-\delta} + \log\frac{10-\delta}{1-\delta}. \tag{38}$$

It has the same limit, but differs at finite gap. A threshold table must state which gap it uses. Logarithms of the directly stored gap avoid subtracting a rounded A from one when that gap becomes smaller than floating-point resolution.

For scale only, freeze the same $G = 6.6743 \times 10^{-11}$ SI and $M_\odot = 1.98847 \times 10^{30}$ kg used in the recovered packet. These are comparison operands, not new mass measurements. The limiting one-way coordinate-road increment per decimal nine is about 0.00022683 seconds at $10M_\odot$, 97.54 seconds at $4.3 \times 10^6 M_\odot$, and 1.7065 days at $6.5 \times 10^9 M_\odot$. The last mass is an illustrative scale motivated by the frozen 2019 M87 result, whose inference has its own uncertainty and emission-model calibration.[12]

These increments are not a traveler's diary intervals. A stationary observer at r_o multiplies a static one-way interval by $\sqrt{F_o}$. An ideal reflected radar trip between that observer and a fixed mirror has twice the one-way delay. Successive outward signals from an *infalling* emitter instead have the near-boundary arrival scale $2t_s\sqrt{F_o}\log 10$ per decimal gap, because the emitter's later emission time and the longer outward route both contribute. Static redshift, null flight time, radar time and infalling-signal time must not be assigned the same numerical coefficient.

6. The original interacting clock remains an explicit input

The geometry supplies a proper-time history only after a metric and worldline have been selected. Applying the Chapter VII clock then requires the complete coupling and common calibration retained in Chapter IV. Write $\zeta = E_u/I_u$ for the common conversion and

$$\eta_j = \zeta(\tau_j - \tau_{j,0}), \qquad \theta_j = \theta_{l,0}(\eta_j). \tag{39}$$

Under ideal phase encoding with known transfer phase, the rate of a received source phase relative to the receiver's same-lane clock is

$$\mathscr{R}_l = \frac{d\tau_E}{d\tau_O}\frac{\nu_{l,0}(\eta_E)}{\nu_{l,0}(\eta_O)}. \tag{40}$$

The first factor is the geometric frequency ratio derived above; the second retains the interacting transient. An equal-coordinate-time or arbitrarily late reception comparison does not remove it. The complete Hamiltonian coupling is a stated physical hypothesis here, not a consequence of the test-particle metric.

The inherited phase/current bounds apply only on their certified internal-age interval, ending at one bare cycle $T_0 = 573049\pi$. A finite remaining source proper time can lie inside that interval only after the common unit conversion and preparation epoch are checked. An arbitrarily late receiving observer need not remain inside its own certified window. The geometrical arrival asymptotics are therefore not a new all-time microscopic clock theorem. If a finite source-age interval is certified and $0 \le \nu_{l,0} \le u_l$, its remaining phase satisfies $|\Delta\theta_l| \le u_l\zeta\Delta\tau_E$; stretching its reception across unbounded observer time does not turn it into an unbounded source history.

7. An original-state finite-gap clock protocol

The completed Chapter VII clocks can already be carried through a declared strong-field event map without replacing them by bare two-state oscillators. The result is conditional on a precise ideal-probe law; finite-size geometry and the remaining curved-field theorem are recorded separately.

7.1 The unchanged clock and the stated coupling

For every original lane $l = (a,b)$ retain the exact interacting hard-branch superposition with soft vacuum, signed D, C, both tensor polarizations, complete sixteen-state source, even energy and both shell counterterms. Its actual current satisfies

$$z_l = \langle Z_l \rangle, \quad \theta_l = -\arg z_l, \qquad \dot{z}_l = -\mathrm{i}\Delta z_l + 2g_l w_l.$$

The joint correlator w_l remains. The inherited certificate gives, for $0 \le \eta \le T_0 = 573049\pi$,

$$\ell_l \le \nu_l(\eta) \le u_l, \qquad |z_l| \ge m_l = M_{l,-} - E_l(T_0) > 999/1000.$$

The hard-reference disk bounds the actual phase; it does not replace it by $\Omega_l\eta$. With the original full joint Hamiltonian H_0, assume the conditional point-clock law

$$\mathrm{i}\frac{\mathrm{d}\Psi}{\mathrm{d}\tau} = s_0 H_0 \Psi, \qquad \eta = s_0\tau, \quad s_0 = e_u/I_u.$$

Its exact evolution is the unchanged $\Psi_0(\eta)$. Every Hamiltonian term receives the same proper-time conversion. This is a stated moving-probe coupling, not a completed covariant action for the extended source and field.

The source normalization can be checked on the original real Q in its full address-pair basis. With $Q_L = Q \otimes I$, $Q_R = I \otimes Q$, $D = Q_L - Q_R$, $C = \mathrm{i}SD$ and $H_{\mathrm{pair}} = S + Q_L + Q_R + Q_L Q_R$,

$$Z_{\mathrm{raw}} = D - SD, \qquad [H_{\mathrm{pair}}, Z_{\mathrm{raw}}] = -2Z_{\mathrm{raw}},$$
$$D^2 + C^2 = D^2 - (SD)^2 = 2D^2.$$

The two nonzero full-source matrices are the native R1 operands. The actual clock equation still contains its field correlator and the four dark states.

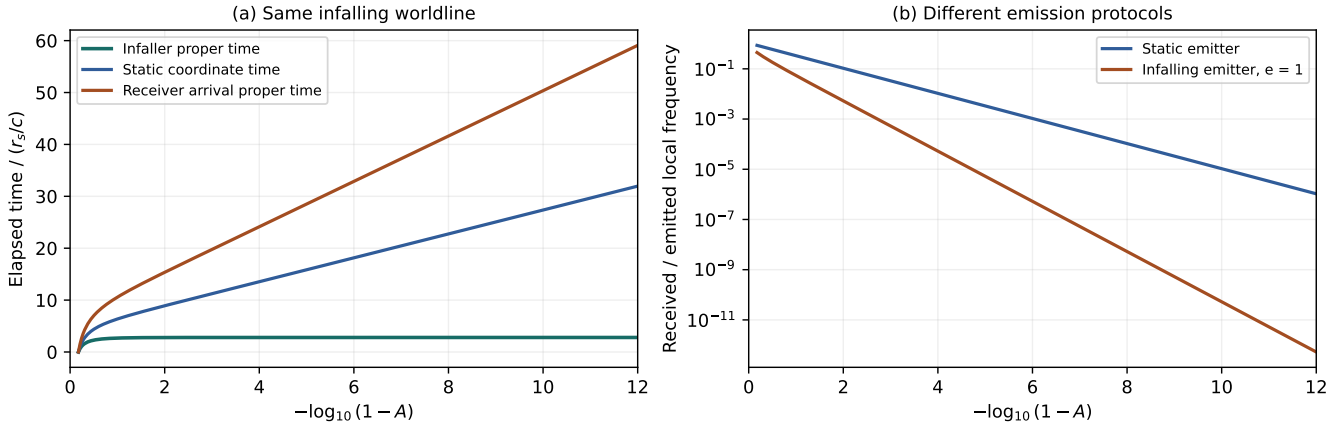


Figure 1: Declared Schwarzschild benchmark, $r_i = 3r_s$, radial geodesic energy $e = 1$, and an observer at $r_o = 10r_s$. The horizontal variable is $-\log_{10}(1-A)$, with the exact conversion to r/r_s retained. Left: elapsed infaller proper time approaches a finite limit while static-coordinate and receiver-arrival times diverge. Each curve has its initial value subtracted; times are in r_s/c. Right: gravitational redshift for a static emitter and the full outgoing redshift for an infalling emitter have different near-boundary powers. The latter includes motion. These are mathematical controls, not trajectories or measurements of a particular black hole.

7.2 One shared conversion and an exact finite arrival map

Write $\bar{t} = ct/r_s$, $\bar{\tau} = c\tau/r_s$ and $y = \sqrt{r/r_s}$. Use one declared fixture for every lane and both experiments:

$$e = 1, \quad y_0 = 5/4, \quad y_1 = 1001/1000,$$
$$r_o/r_s = 25/9, \quad \sqrt{F_o} = 4/5, \qquad \zeta = s_0 r_s/c = T_0/16.$$

This conversion is not fitted by lane. Prepare emitter and receiver clocks at coordinate $t = 0$. Hold one emitter at $r_0 = 25r_s/16$; let the other follow the inward $e = 1$ geodesic. Compare equal emitter proper ages,

$$\bar{\tau}_e(y) = \frac{2}{3}(y_0^3 - y^3), \qquad y_j = y_0 - \frac{j}{8}(y_0 - y_1), \quad j = 0, \ldots, 8.$$

Use ideal tagged radial-null events. This specifies the causal timing benchmark; a continuous physical phase encoder needs its own coupling, nonzero received amplitude and transfer calibration.

The first flight and the two exact arrival maps are

$$D_0 = \frac{175}{144} + \log\frac{256}{81}, \qquad \frac{\eta_{o,S}}{T_0} = \frac{D_0}{20} + \frac{\bar{\tau}_e}{12},$$

$$\bar{t}_{o,F}(y) = \frac{365}{96} + \frac{25}{9} - \left(\frac{2}{3}y^3 + y^2 + 2y\right) + \log\frac{16}{81(y-1)^2}, \qquad \frac{\eta_{o,F}}{T_0} = \frac{\bar{t}_{o,F}}{20}.$$

The emitted proper tick factors are

$$g_S = \frac{3}{4}, \quad g_F(y) = \frac{1 - 1/y}{\sqrt{F_o}},$$
$$g_F(y_0) = \frac{1}{4}, \qquad g_F(y_1) = \frac{5}{4004}.$$

The final finite gap is $F_1 = 2001/1002001$. Directed exact evaluation gives

clock age	η/T_0 at final emission/arrival
emitter, either experiment	0.039588416625
held receiver	0.171084858879245…
falling-experiment receiver	0.755042078465470…

All prefixes remain inside the original certificate, including receiver age. The phase-event relation uses the actual $\theta_l(\eta_e)$ and $\theta_m(\eta_o)$. For a calibrated phase carrier its received phase-rate ratio to the receiver's own clock is

$$g\,\frac{\nu_l(\eta_e)}{\nu_m(\eta_o)}.$$

Both interacting transients survive. The same-source arrival-generator contrast and the exact six-lane inverse lower-derivative matrix are the native R2 operands; they do not evolve a replacement field or source.

7.3 Preparation, shots, timing and unwrapping

Measure $X_l = P_l D P_l/d_l$ and $Y_l = P_l C P_l/d_l$, extended by zero outside the lane. Both have norm one and outcomes in $\{-1,0,1\}$; leakage outcomes remain in the mean. Each quadrature at each event uses fresh original joint preparations. No first readout disturbs a later clock or quadrature. A joint preparation infidelity p gives expectation bias at most$2\sqrt{p}$. Its energy account is a separate apparatus requirement.

There are at most480 scalar means: six lanes, four emitter/receiver histories, ten slots and two quadratures. Independent bounded repetitions within each mean give

$$P(\text{any statistical miss}) \le 960e^{-nd^2_{\text{stat}}/2}.$$

Two sufficient designs have statistical error$1/200$ or$1/2000$; preparation and readout biases are respectively$1/400$ or$1/4000$ each. They require preparation infidelities at most$1/640000$

or$1/64000000$, and10^6 or10^8 shots per scalar mean. Total independent copies are 4.8×10^8 or4.8×10^{10}. Both union bounds are below0.01. These are sufficient instrument conditions, not achieved precision or minimal counts.

If d is the summed scalar error, the complex current disk has radius less than$3d/2$. Its phase error is bounded by

$$\arcsin\frac{3d/2}{m_l}\le\frac{3d/2}{999/1000-3d/2}.$$

Adding timing and transfer-phase allowances of$1/1000$ each in the coarse design, or$1/4000$ each in the fine design, gives total phase allowances $\beta=1/50$ or$1/400$ radians.

The finite padded range has$g_F\ge1/1000$. Fine emission-age uncertainty at most$1/4000000$ in$\bar\tau_e$, plus residual propagation/receiver timing uncertainty at most$1/4000$ in$\bar\tau_o$, yields $\delta\bar\tau_o\le1/2000$. Together with a shared relative conversion error at most$1/200000$, the actual rate bounds fit the fine timing phase allowance. These requirements include geometry and trigger uncertainty; precision arithmetic alone does not meet them.

The executable bounds prove that every sampled phase increment, including preparation to first reception, plus2β is below$3<\pi$. Continuous unwrapping is consequently unique on the simultaneous confidence event. An endpoint qubit phase is not a winding record; preparation, control and reference payments remain in their own accounts.

For measured phase increments$\Delta\hat\theta_e,\Delta\hat\theta_o$, the elapsed proper-time ratio has the interval

$$\frac{\Delta\tau_e}{\Delta\tau_o}\ge\frac{(\Delta\hat\theta_e-2\beta)_+\ell_l}{(\Delta\hat\theta_o+2\beta)u_l},$$
$$\frac{\Delta\tau_e}{\Delta\tau_o}\le\frac{(\Delta\hat\theta_e+2\beta)u_l}{(\Delta\hat\theta_o-2\beta)_+\ell_l},$$

when the denominator lower bound is positive. The common conversion cancels. This is a finite-window quantity. Instantaneous frequency requires a further window/geometry bound. The planned coarse whole-window envelopes separate held and falling histories in every lane: approximately$[0.40991,1.30632]$ versus$[0.04135,0.08385]$. The fine design separates all eight adjacent windows as well. These are confidence-resolution guarantees, not acquired data. The calculation pays4β when allowing every possible noisy data realization and then forming its2β confidence interval.

7.4 Static size and freely falling tidal geometry

With the same$c_*=1$ calibration, the original shell has $R_{\rm phys}/r_s<1.454\times10^{-6}$. A declared proper source/apparatus tube$L/r_s=10^{-4}$ contains it. This does not truncate the infinite field. For a static center$\rho_c=r_c/r_s$, put$\lambda=L/r_s$, $\rho_-=\rho_c-\lambda>1$. The exact local bound is

$$\left|\log\frac{N}{N_c}\right|\le\frac{\lambda}{2\rho_-^2\sqrt{1-1/\rho_-}}.$$

It is below$1/25000$ at launch and below$1/800$ for a held tube at the falling endpoint. Static lapse variation grows as$L/(2r_s\sqrt{F})$ toward closure. The exact inward proper distance is

$$\ell_H/r_s=\sqrt{x(1+x)}+\operatorname{asinh}\sqrt{x}.$$

A wholly exterior static ball must have$L<\ell_H$; this is a size condition, not an infaller crossing theorem. No Chapter IV small-Plummer constant is extrapolated to$F\to0$.

The orthonormal Schwarzschild curvature remains finite in the radial freefall frame: a radial boost cancels through$\gamma^2-\sinh^2\alpha=1$. The spatial scale$L^2r_s/r_1^3<10^{-8}$ measures local tidal metric variation, but is not a completed finite-clock error estimate. For initially comoving radial geodesic separation$J$, the exact first-variation equation and bounds are

$$J''=\frac{c^2r_s}{r(\tau)^3}J,$$

$$\cosh\left(\sqrt{\frac{c^2r_s}{r_0^3}}\Delta\tau\right)\le\frac{J}{J_0}\le\cosh\left(\sqrt{\frac{c^2r_s}{r_1^3}}\Delta\tau\right).$$

The exact solution sharpens this comparison bracket. With $\rho=r/r_s=y^2$,

$$\frac{J}{J_0}=\frac15\left(\frac{y}{y_0}\right)^4+\frac{4y_0}{5y},\qquad J_{ss}=\rho^{-3}J.$$

It has $J(0)=J_0$ and zero initial relative velocity. At the retained endpoints,

$$\frac{J_1}{J_0}-1=\frac{992802978755001}{12219238281250000}=0.08124917084876092\ldots.$$

The strain is exactly about 8.1249% within linear geodesic deviation.[1] A bound original quantum clock is not an unbound geodesic congruence. Its restoring/tidal coupling must be calculated; small metric variation alone does not bound its long-time dynamics.

7.5 A same-mass short-window bridge

The preceding calibration has$r_s=T_0/16$ model lengths. The wave-boundary fixture instead has$r_s=10^{10}$ model lengths; these are distinct fixtures. A separate bridge uses$c=1$, $\zeta=10^{10}$ for all six original clocks,

$$r_0/r_s=1+2\times10^{-6},\quad r_o=r_0+100,$$
$$y_1=\sqrt{1+2\times10^{-6}}-10^{-8}.$$

Exact directed evaluation gives final emitter age$200.000398\ldots$, first receiver age$70710.675539\ldots$, held final receiver age $70911.175313\ldots$ and falling final receiver age $355687.588344\ldots$, all belowT_0. The endpoint tick factor is $0.000698291817502\ldots$. A separately specified$\beta=10^{-4}$ design separates the whole-window ratios in all six lanes; it requires10^{12} shots for each of144 scalar means. These demanding counts are sufficient apparatus conditions, not achieved data. This short-window bridge shares the wave fixture's mass/unit choice without changing the source or silently combining the two calibrations.

7.6 Original whole-field localization

The original field has infinite support. The preceding spatial-tail estimate at the long-fall tube $L_{\mathrm{model}} = \zeta/10000$ clipped to the complete positive-energy ceiling in every lane. The centered calculation below retains annihilation/source-shift cancellation before taking norms, evaluates larger apparatus scales, and carries both propagated current insertions. No finite tube deletes the exterior field.

8. Centered localization of the original clock field

The original source shift cancels the leading hard-field term before any norm is taken. Keeping this cancellation gives a useful spatial bound for the actual Chapter VII preparations. This section retains their signed couplings, all sixteen source states, both tensor polarizations, the hard eigenbranches with soft vacuum, the full field and both shell counterterms. The spatial quantity is the positive residual-energy receiver of Chapter IV; its identification with curved canonical stress is a separate problem.

8.1 A first form moment suffices

Write $K = \|T_g\Psi\|^2$, where

$$(T_g\Psi)_\alpha(k) = \sqrt{k}\,a_\alpha(k)\Psi - \frac{g u_R(k)}{\sqrt{k}}\sigma_\alpha\Psi,$$
$$u_R(k) = \sqrt{\frac{2k}{\kappa\pi}}\,j_2(kR), \qquad \kappa = 4/5.$$

Here $\varepsilon = \Delta/2$, S is the original swap/involution (the active lane's remaining Pauli direction), and

$$\mu = \int_0^\infty u_R(k)^2/k\,\mathrm{d}k = \frac{1}{5\kappa R} = \frac{1}{4R}, \qquad C_{\mathrm{ct}} = 2g^2\mu.$$

The source's full even block energy is retained. It commutes with the lane-preserving field receiver and cancels in same-lane comparisons. For an original hard branch Φ_j, let $B_j = H_h - E_j \geq -\Delta$ and $v = \sigma_\alpha\Phi_j$. At the original split $\lambda = 64\Delta$, pull-through gives the centered expression

$$(T_g\Phi_j)_\alpha(k) = -\frac{g u_R(k)}{\sqrt{k}} B_j(B_j+k)^{-1}v, \qquad k \geq \lambda.$$

All these spectral statements hold in the original invariant unordered-pair source block containing Φ_j and $\sigma_\alpha\Phi_j$. Other source blocks and dark states remain orthogonal retained sectors; their even energies are not covered by this same-lane spectral gap. No vector $B_j v$ is assumed. Spectral calculus instead gives

$$\|B_j(B_j+k)^{-1}v\|^2 \leq \frac{\langle v,(B_j)_+ v\rangle}{4k} + \frac{\Delta^2\|v\|^2}{(k-\Delta)^2}.$$

The positive part follows from $b^2/(b+k)^2 \leq b/(4k)$ for $b \geq 0$; the negative part uses $-\Delta \leq b < 0$. These are form-domain statements.

Peel off the two retained low hard branches, using their original isolated two-branch gap certificate. If Q_h is their orthogonal complement, then $Q_h B_j Q_h \geq 0$ and

$$\|Q_h\sigma_\alpha\Phi_j\|^2 \leq 1 - M_-^2.$$

Set $b_2 = \int u_R^2/k^2 \leq 5/72$, $h = 64/63$, $n_+ = 2g^2b_2h^2/(1+2g^2b_2h^2)$ and

$$F_* = 4|g|\sqrt{\mu}\left(\sqrt{C_{\mathrm{ct}}} + \sqrt{2K_{\mathrm{prep}}}\right).$$

The actual source-conjugation identity, the hard-branch source charge and the certified $0 < \Omega \leq \Delta$ give positive first form moments

$$M_{Q,0} \leq F_* + \Delta - \Omega_- M_-^2,$$
$$M_{Q,1} \leq F_* + 2\Delta n_+.$$

Nonnegative upper ceilings are used if an interval endpoint needs them. The exact full-source check is $DH_{\mathrm{pair}}D - D^2H_{\mathrm{pair}} = -2SD^2$; the field contribution keeps the other polarization. Thus this is a source-bound first-moment estimate, without an ultraviolet operator moment. The new centered whole-energy ceiling is intersected with the stronger inherited K_{prep} whenever appropriate. It improves lanes $01, 02, 12$; no uniform improvement is claimed for the other three lanes.

8.2 Spatial tails and both instantaneous insertions

Use the original regular Bessel/TT isometry J and radial split at L: $K_{\mathrm{out}}[\Psi] = \|1_{r>L}JT_g\Psi\|^2$. For the centered hard amplitude, split the Bessel integral at $k_0 = 1/R$, integrate each oscillatory term once by parts, and keep its endpoint at λ. On the positive complement use either its norm ceiling or its first form moment, taking the better valid bound for each integral. The complete nine high-frequency Bessel products and the entire soft shift are retained. Explicit coefficients and directed arithmetic are in the accompanying derivation and executable certificate.

Source charge makes the radial square form diagonal between the original two hard branches. Low-branch and complementary spectral amplitudes are also orthogonal. Their squares therefore add before the remaining hard/soft triangle inequality. The fully coupled original flat evolution then obeys

$$\eta(T,L) \leq \min\{\mathscr{K}_*, D_K(T,L)^2\},$$
$$D_K(T,L) = \sqrt{p_{\mathrm{out,prep}}(L)} + \sqrt{2(1+C_{\mathrm{ct}})d_A^2(T)},$$
$$\mathscr{K}_* = K_{\mathrm{prep}} + \varepsilon E(T),$$
$$d_A^2(T) \leq 8T^2g^2v_{\mathrm{soft}}\left[1 + \left(\sqrt{C_{\mathrm{ct}}} + \sqrt{\mathscr{K}_*}\right)^2 + \lambda\right].$$

Here $E(T) \leq (T/T_0)^2E(T_0)$ for $T \leq T_0$. The hard/free-soft comparison generator includes the retained soft counterterm, so its global reference phase is matched in the vector-distance estimate. No field is truncated at the displayed radius. Over the complete original window $T_0 = 573049\pi$, the certified tail fractions are

lane	$L = T_0/160000$	$L = 10^4$	$L = 10^6$
01	1	0.0234440	0.000262687
02	1	0.0232842	0.000246210
03	1	0.0340807	0.000342323
12	1	0.0234231	0.000260502
13	1	0.0340646	0.000342224
23	1	0.0340450	0.000342101

The entries are upper ceilings, not measured fractions or tail lower bounds. The first radius is about 11.25 in the original model chart; there the certificate still clips at the whole energy. At 10^6 it no longer does so.

For $Z = \sigma_z + \mathrm{i}\sigma_y$ and its adjoint, the exact identity $T_\alpha Z = ZT_\alpha - g(u_R/\sqrt{k})[\sigma_\alpha, Z]$ gives

$$\sqrt{K_{\mathrm{out}}[Z\Psi]},\ \sqrt{K_{\mathrm{out}}[Z^\dagger\Psi]} \le 2\sqrt{K_{\mathrm{out}}[\Psi]} + \sqrt{8/5}\,\frac{C_F(L)}{L^{5/2}},$$

where $\|F_\alpha(r)\| \le C_F(L)/r^4$ is the complete shell shift. This is an instantaneous bound. It does not identify the tails of backward-propagated insertions with instantaneous tails.

8.3 A finite current disk with propagated insertions

There is a finite comparison which needs only a global form ceiling for the propagated insertions. Specify the receiving Hamiltonian

$$H_W = \varepsilon S + T_g^* W T_g, \qquad m_W I \le W \le M_W I, \quad m_W > 0,$$

with time-independent W on the common closed form domain $\mathscr{Q} = D(H_f^{1/2})$. The field receiver is the identity on source blocks. The common even source energy is retained and cancels from this comparison. Let the difference form obey

$$|r[\Phi,\Psi]| \le \varepsilon_{\mathrm{near}}\sqrt{K_{\mathrm{in}}[\Phi]K_{\mathrm{in}}[\Psi]} + B_{\mathrm{far}}\sqrt{K_{\mathrm{out}}[\Phi]K_{\mathrm{out}}[\Psi]}.$$

Put the perturbed backward observable $O_W(t) = U_W(T-t)^\dagger Z U_W(T-t)$ against the *original* reference state $\Psi_0(t)$. Energy conservation, form coercivity and $K[Z\xi], K[Z^\dagger\xi] \le 4(\sqrt{K[\xi]} + \sqrt{C_{\mathrm{ct}}}\|\xi\|)^2$ give, for both insertions,

$$K[O_W(t)\Psi_0(t)] \le K_B,$$
$$K[O_W(t)^\dagger\Psi_0(t)] \le K_B,$$
$$K_B = \frac{4M_W}{m_W}\left(\sqrt{\frac{M_W\mathscr{K}_* + 2\varepsilon}{m_W}} + \sqrt{C_{\mathrm{ct}}}\right)^2 + \frac{8\varepsilon}{m_W}.$$

The weak Heisenberg identity now yields the finite, nonperturbative disk

$$|z_W(T) - z_0(T)| \le \mathscr{E}_z,$$
$$\mathscr{E}_z = 2T\sqrt{K_B}\left(\varepsilon_{\mathrm{near}}\sqrt{\mathscr{K}_*} + B_{\mathrm{far}}\sqrt{\eta(T,L)}\right).$$

Both O_W and $O_W^\dagger$ occur in its commutator. No unknown perturbed-state spatial tail has been replaced by a reference tail. If $|z_0| \ge a_* > \mathscr{E}_z$, the continuous unwrapped phase error is at most $\arcsin(\mathscr{E}_z/a\ )$.

For one explicit smooth bounded assay, take

$$N_L(r) = 1 + 10^{-3} + (1/2 - 10^{-3})\,\chi\left(\frac{r-L}{L}\right), \qquad W = J^* N_L J,$$

where $\chi(s) = 0$ for $s \le 0$, $\chi(s) = 1$ for $s \ge 1$, and in between $\chi(s) = e^{-1/s}/(e^{-1/s} + e^{-1/(1-s)})$. This specifies the entire profile, including the exterior. Use the conservative class $m_W = 1/2, M_W = 3/2$, $\varepsilon_{\mathrm{near}} = 10^{-3}, B_{\mathrm{far}} = 1/2$, $L = 10^6$. For the retained same-mass bridge, all model ages are below $T = 355689$. The six phase-error upper bounds are

l	$\lvert\delta\theta_l\rvert$ upper bound
01	$3.421\,10^{-7}$
02	$1.533\,10^{-6}$
03	$4.911\,10^{-4}$
12	$3.945\,10^{-7}$
13	$4.434\,10^{-4}$
23	$3.957\,10^{-4}$

in radians, each below 10^{-3}. The largest current disk is below $4.908\,10^{-4}$. At the full T_0, the largest phase bound is below $2.495\,10^{-3}$ instead. These are actual-original-state results for the stated positive receiver, with the original interacting transient.

8.4 The remaining physical obstruction is quantitative

The assay above is not the full Schwarzschild lapse or a curved canonical Hamiltonian. In the same-mass geometry, $r_s = 10^{10}$, $r_0/r_s = 1 + 2\,10^{-6}$, the radius $L = 10^6$ remains smaller than a rigorous endpoint proper distance to the horizon, $2.8142\,10^7$. Nevertheless the normalized static lapse has near variation bounded by 0.035534, far contrast exceeding 706, and global infimum zero. Thus it fails the displayed coercivity class. Even optimistically keeping that class's constants, lane 03 at the short window would need a far-tail fraction below approximately $8.72\,10^{-10}$ if its near error were zero; the present upper certificate is about $3.41\,10^{-4}$. This comparison describes a gap in the proof, not a lower bound on the actual physical error.

A curved canonical completion must additionally control its pair kernel, vacuum convention, source-shell vertices/counterterms and all angular complement returns on the original state and its propagated insertions. The tempting direct equation for $T_g U_0(t)$ contains $[H_0, \sigma_\alpha]$ and the shell field $\phi(u_R)$. Here $u_R \notin L^2$ although $u_R/\sqrt{k} \in L^2$; its Hilbert-space commutator norm cannot be assumed from finite K alone. The first-form method avoids that unavailable norm and the finite receiver theorem avoids propagating an instantaneous spatial estimate as though it were a two-time estimate. Neither step supplies the still missing curved joint propagator.

9. A critical curve is not a photograph of the boundary

For a static observer at r_o in a spherical metric, a local orthonormal measurement of a null ray gives

$$\sin\alpha = \frac{b\sqrt{F_o}}{r_o}. \tag{41}$$

Indeed the local photon energy is the conserved Killing energy divided by $\sqrt{F_o}$, whereas its transverse momentum is its angular momentum divided by r_o. Their ratio yields the displayed angle. For an exterior observer $r_o > r_{\rm ph}$ in the Schwarzschild black-hole benchmark, the critical-cone angle uses $b = b_{\rm crit}$ and the appropriate inward sky branch. At large r_o, its angular diameter approaches $2b_{\rm crit}/r_o$.

The areal photon-sphere radius, the impact-parameter critical curve, and the observed emission ring are different objects. A dark capture region additionally needs an absorbing/global completion and a specified illumination model. Emission, absorption, lensing, source motion, polarization and instrument response determine the bright image. The 2019 EHT mass inference calibrates an observed emission diameter against a model family; it is not a direct measurement of material crossing $A = 1$.[12, 13]

A no-completed-crossing model must therefore supply its boundary response: absorption, reflection, re-emission or another causal rule, with energy and record transfer retained. Agreement with the exterior photon-sphere scale alone cannot select among these possibilities. If a proposed surface lies at $r_b = r_s(1+\varepsilon)$ and an exterior reflecting/scattering region is represented at $r_p > r_b$, the benchmark radial round-trip interval between them is

$$\Delta t_{\rm rt} = \frac{2}{c}\left[r_*(r_p) - r_*(r_b)\right] = -2t_s\log\varepsilon + O(1). \tag{42}$$

This conditional time scale is not an echo prediction. It says nothing about excitation, reflection amplitude, wave potential, attenuation or detector observability until those are specified. A prescribed boundary that absorbs all relevant signals need not produce such a return, and a wave-scattering peak is not automatically a physical mirror.

9.1 A declared tensor-mode wave and boundary law

The retained carrier has both original signed tensor polarizations, the whole-sphere TT-image isometry, the full positive channel metric $\kappa = 2B/5$, and driven angular eigenvalue six. Its tensor block has $\kappa = 4/5$. A curved propagation equation needs a further choice. We make *hypothesis RM*: on the prescribed static spherical metric, retain the original mode space and nonnegative angular operator $\mathscr{L}$, self-adjoint and nonnegative in the κ inner product (or commuting with κ in the ordinary convention), and choose

$$S_{\rm RM} = \frac{1}{4}\int dt\,dr\left[\frac{r^2\sqrt{H}}{Nc^2}\phi_t^*\kappa\phi_t - \frac{r^2N}{\sqrt{H}}\phi_r^*\kappa\phi_r - N\sqrt{H}\,\phi^*\kappa\mathscr{L}\phi\right], \qquad N=\sqrt{F}. \tag{43}$$

At $F = H = 1$ this is the original radial tensor action. RM is a stated extension of those mode amplitudes, with their original tensor incidence; it does not select a unique covariant spin-two theory. A curvature coupling, spin connection or constrained field completion would require its own derivation. No scalar carrier or general-relativistic perturbation potential is silently substituted.

Set $dx/dr = \sqrt{H/F}$, $a = dr/dx = \sqrt{F/H}$ and $u = r\phi$. Variation and a radial integration by parts give

$$\frac{u_{tt}}{c^2} - u_{xx} + Vu = 0, \qquad V = \frac{F\mathscr{L}}{r^2} + \frac{(F/H)'}{2r}I. \tag{44}$$

Nonnegative $\mathscr{L}$ does not alone make V nonnegative on an arbitrary metric: $(F/H)'/(2r)$ can be negative. The complete coercive form and surface terms must be checked there. The driven quadrupole of the Schwarzschild comparison has $V = F(6/r^2 + r_s/r^3)$. The other angular and channel directions remain; RM's spherical coefficients preserve their sectors. An added angular coupling must propagate its full complement and possible return.

The surface term is part of the energy. With real parts and both polarizations understood,

$$\begin{aligned} E_u &= \frac{\kappa}{4}\int\left(\frac{|u_t|^2}{c^2} + |u_x|^2 + u^*Vu\right)dx, \\ E_\phi &= E_u - \frac{\kappa}{4}\left[\frac{a}{r}|u|^2\right]_b^p, \\ j_\phi &= -\frac{\kappa}{2}\,\mathrm{Re}(u_t^*u_x) + \frac{\kappa}{4}\partial_t\left(\frac{a}{r}|u|^2\right). \end{aligned} \tag{45}$$

These identities preserve the original $1/4$ action normalization.

9.2 Passive termination, physical storage and causal response

At $x_b = 0$ choose the explicit termination

$$u_x = \frac{u_t + \sigma u}{c}, \qquad \sigma > 0. \tag{46}$$

Its canonical spring energy is $E_s = \kappa\sigma|u_b|^2/(4c)$ and its absorbed power is $P_d = \kappa|u_{b,t}|^2/(2c)$. In the original field variables the physical spring and outer-port storage are

$$\begin{aligned} E_{s,\phi} &= \frac{\kappa}{4}\left(\frac{\sigma}{c} - \frac{a_b}{r_b}\right)|u_b|^2, \qquad E_{\rm port} = \frac{\kappa a_p}{4r_p}|u_p|^2, \\ E_\phi + E_{s,\phi} + E_{\rm port} &= E_u + E_s. \end{aligned} \tag{47}$$

Thus the physical spring is positive when $\sigma > ca_b/r_b$. For a characteristic matched launch/readout port at $x_p = L$, write $u_p = u_{\rm in} + u_{\rm out}$ and impose $u_t + cu_x = 2u_{{\rm in},t}$. The complete balance is

$$\frac{d}{dt}(E_\phi + E_{s,\phi} + E_{\rm port}) + P_d = \frac{\kappa}{2c}(|u_{{\rm in},t}|^2 - |u_{{\rm out},t}|^2). \tag{48}$$

Incoming fields add their full energies and interference terms. The port retains the exterior recipient; it does not declare the actual spacetime outside r_p free of scattering.

For zero potential the boundary reflection is exactly

$$R(s) = -\frac{\sigma}{2s+\sigma}, \qquad (Rf)(t) = -\frac{\sigma}{2}\int_{-\infty}^{t} e^{-\sigma(t-v)/2} f(v)\,dv. \tag{49}$$

Its pole is stable, the response is causal, its impulse-response L^1 norm is one and $|R(i\omega)| \le 1$. A return therefore has an amplitude, a storage history and an exponential tail, in addition to delay $2L/c$. This is a selected boundary law, not a thermal-state assertion.

9.3 A complete original-pulse finite-gap experiment

Use the original maintained compact pulse, both signed polarizations, all six g_{ab} and radius balls, $\Delta = 2/573049$, and $c = c_* = 1$ in the shared model chart. With $z = \Delta R$ and $\mu(w) = 15(1-w^2)^2/16$ on $[-1,1]$, the complete original carrier is

$$F_p(v) = \sin^4(v/8)e^{-iv}, \quad 0 \le v \le 8\pi, \qquad \psi_{\text{original}}(t) = A_0 C_z F_p''(\Delta t), \qquad A_0 = \frac{2g_{ab}z^2}{15\kappa}, \qquad C_z h(v) = \int_{-1}^{1} \mu(w) h(v - zw)\,dw. \tag{50}$$

F_p is zero outside its stated support. This exact positive averaging retains both compact edges. Its original energy deficit relative to the small-source value is below 6.633×10^{-14}.

The launcher subsystem retains the original source even energy, full local field, drive $-\mathrm{Re}(f^*\phi(R))$ and counterterm $|f|^2/(4R)$. At a uniform local lapse N_p, the whole subsystem and its switching ledger scale by N_p. The calibrated isometric launch $u_{\text{in}}(t) = \psi_{\text{original}}(N_p t)$ then supplies Killing energy $E_{\text{in}} = N_p W_{\text{original}}$. This is a declared composite maintained experiment, not a microscopic central source placed within the macroscopic r_s, nor an autonomous four-cycle quantum-clock certificate. Finite launcher size requires the separate worldtube error bound.

A genuinely nonuniform central-shell coupling would instead require the full curved static inverse: if its free form is $\langle\phi, A\phi\rangle/4$ and its drive is $-\langle j,\phi\rangle$, completing the square requires $H_{\text{ct}} = \langle j, A^{-1} j\rangle$, when that source functional has a finite inverse-form value. The flat quadrupole gives $1/(5\kappa R)$ and recovers the original counterterm. It is not generally correct to retain that numerical coefficient after changing the entire radial geometry.

Choose one finite gap and boundary rate before evaluating the pulse:

$$r_s = 10^{10}, \quad \delta = 10^{-6}, \quad r_b = r_s(1+\delta), \quad r_p = r_s(1+2\delta), \quad N_p^2 = \frac{1}{500001}, \quad \sigma = N_p\Delta, \quad L = r_s(\delta + \log 2), \quad 0 \le V \le V_{\max} = \frac{14\delta}{r_s^2}. \tag{51}$$

The microscopic source radius remains approximately 0.163544 in this chart. Here $\sigma \simeq 4.935746 \times 10^{-9}$, whereas $a_b/r_b \simeq 10^{-16}$; the original-variable spring is strictly positive. No SI calibration is inferred from the selected model lengths.

For the zero-potential response to F_p'', its exact five frequencies are $\omega = (1/2, 3/4, 1, 5/4, 3/2)$, with $c = (1/16, -1/4, 3/8, -1/4, 1/16)$. On $[0,U]$, $U = 8\pi$,

$$H(v) = \sum_j h_j e^{-i\omega_j v} - Se^{-v/2}, \qquad h_j = \frac{\omega_j^2 c_j}{1 - 2i\omega_j}, \quad S = \sum_j h_j, \qquad H(v) = S(1-e^{-4\pi})e^{-(v-U)/2}, \quad v > U. \tag{52}$$

The initial transient and final tail are required by the compact pulse. Direct integration gives

$$I_{\text{in}} = \int |F_p'''|^2 dv = \frac{14245\pi}{4096}, \qquad I_{\text{ret}} = \int |H'|^2 dv = \frac{2128297\pi}{3860480} - \frac{243(1-e^{-4\pi})}{5821603840}. \tag{53}$$

Thus $I_{\text{ret}}/I_{\text{in}} = 0.1585215864253203$ and the absorbed fraction is 0.8414784135746797. Positive source smoothing changes this fraction by less than 10^{-13}. These are exact-pulse boundary-model results, not a monochromatic approximation.

The nonzero potential and every further return. Let G_0 be the free retarded Green kernel with the stated passive termination and matched outer port. The complete potential is retained by

$$u = u_0 - c^2 \int_0^t \int_0^L G_0(t-s;x,y) V(y) u(s,y)\,dy\,ds. \tag{54}$$

The direct kernel plus its causal reflected image obeys $|G_0| \le 1/c$. For $B = 17|A_0|/16$, define the exact positive integrated potential

$$J_V = 6\left(\frac{1}{r_b} - \frac{1}{r_p}\right) + \frac{r_s}{2}\left(\frac{1}{r_b^2} - \frac{1}{r_p^2}\right), \qquad q = cJ_V T.$$

The complete Volterra equation uses this L^1 norm. Gronwall and differentiated delayed kernels yield

$$\|u - u_0\|_\infty \le 2B(e^q - 1), \qquad \|u_t - u_{0,t}\|_\infty \le 2cJ_V B e^q, \qquad \|u_x - u_{0,x}\|_\infty \le 2J_V B e^q. \tag{55}$$

The exact original compact edges satisfy the weak-derivative requirements. Potential backscatter can arrive before the inner-boundary round trip; that earlier response is not a signal launched from the inner boundary.

Take $T = (8\pi + 2z_{\max} + 32)/\sigma + 2L/c$. The interval certificate gives $q < 1.781 \times 10^{-5}$, returned-waveform error below $3.784 \times 10^{-5}|A_0|$, and finite-window returned-energy error below $8.226 \times 10^{-6} E_{\text{in},0}$. It also retains the entire late field, potential, physical spring and port storage: $(E_u(T) +$

$E_s(T))/E_{\mathrm{in},0} < 6.582\times10^{-11}$. Passivity bounds any later returned energy by this residual. The all-time lower energy endpoint also subtracts the free reference pulse's post-window tail; a sufficient allowance is $6.756\times10^{-23}E_{\mathrm{in},0}$.[1] After also normalizing to the true smoothed input, all six preparations satisfy

$$0.15851336 \le \frac{E_{\mathrm{ret}}}{E_{\mathrm{in}}} \le 0.15852982. \tag{56}$$

The amplitudes below retain their original signs; the incident energies use the common model normalization; each preparation has weight $1/6$.

pair	signed A_0	$E_{\mathrm{in},0}$
01	$-5.18100\,10^{-18}$	$5.79019\,10^{-43}$
02	$-1.07380\,10^{-17}$	$2.48723\,10^{-42}$
03	$-1.33669\,10^{-16}$	$3.85414\,10^{-40}$
12	$-5.55704\,10^{-18}$	$6.66119\,10^{-43}$
13	$-1.28488\,10^{-16}$	$3.56116\,10^{-40}$
23	$-1.22931\,10^{-16}$	$3.25978\,10^{-40}$

The exact input balls and outward intervals are retained in the replay. Absorbed plus returned energy equals incident energy after the residual leaves or dissipates. During the experiment the residual stays in the ledger. The launcher pays the original maintained-source work and switching payments; zero net counterterm endpoint change does not establish zero apparatus cost.

The decay limit also has a short certificate for this regular finite interval. A nonzero-frequency undamped mode has zero damping at the matched port, hence $u(L)=u_x(L)=0$ and is identically zero. A static mode obeys $\int(|u_x|^2+V|u|^2)dx+(\sigma/c)|u(0)|^2=0$, so it too vanishes. The smooth compact input leaves a precompact energy orbit under the contractive evolution; its invariant limit is therefore zero. No numerical decay rate is assumed in the finite-time residual bound above.

9.4 Static and infalling escape cones

For isotropic emission by a static source at $r_s<r<3r_s/2$ in the comparison geometry,

$$\sin\alpha_c=\frac{b_{\mathrm{crit}}\sqrt{F}}{r},\qquad f_{\mathrm{esc,stat}}=\frac{1-\cos\alpha_c}{2}\sim\frac{27F}{16}. \tag{57}$$

The fraction counts all 4π directions; an outward-hemisphere normalization doubles it. For steady emission the total escaped power at a complete outer static sphere has the further energy/rate factor F/F_o. Finite pulses need their angular travel-time distribution.

For inward geodesic emission let $\beta=\sqrt{1-F/e^2}$ and $\mu'=\cos\alpha'$ in the emitter's rest frame. Aberration and energy conversion give

$$\begin{aligned}\mu&=\frac{\mu'-\beta}{1-\beta\mu'},\qquad \frac{E_o}{E_{\mathrm{em}}}=\frac{e(1-\beta\mu')}{\sqrt{F_o}},\\ \mu_c'&=\frac{\mu_c+\beta}{1+\beta\mu_c},\qquad f_{\mathrm{esc,fall}}=\frac{1-\mu_c'}{2}\sim\frac{27F^2}{64e^2}.\end{aligned} \tag{58}$$

The radial ray recovers the earlier infaller redshift. A total received luminosity additionally needs emission, angular acceptance and the angle-dependent event maps; the radial factor alone does not supply it.

9.5 The static inverse with the actual apparatus endpoints

The supplied review gives a useful closed inverse for hypothesis RM. Its endpoint choice must agree with the physical question. For a Schwarzschild angular sector,

$$A_l=\kappa[-\partial_r r(r-r_s)\partial_r+l(l+1)]. \tag{59}$$

Here angular positivity is in the κ inner product; alternatively conjugate by $\kappa^{1/2}$. On an arbitrary metric it does not alone imply positivity of the transformed V, whose radial term can have either sign. The original complete form and its surface terms control that issue. The selected Schwarzschild annulus is positive and regular.

For the actual finite experiment, retain the inner spring and the zero-input static limit of the matched outer port:

$$\begin{aligned}u_x(r_b)&=\frac{\sigma}{c}u(r_b),\qquad u_x(r_p)=0,\\ \mathscr{B}_b h&=F_b(h+r_bh')-\frac{\sigma r_b}{c}h,\qquad \mathscr{B}_p h=F_p(h+r_ph').\end{aligned} \tag{60}$$

The positive original-field spring and outer-port storage remain in this domain. On the finite interval the complete form controls H^1 and its trace evaluations, so the source inverse-form value is finite.

Put $y=2r/r_s-1$. For $l=2$,

$$P=\frac{3y^2-1}{2},\qquad Q=\frac{P}{2}\log\frac{y+1}{y-1}-\frac{3y}{2}. \tag{61}$$

Their Wronskian obeys $r(r-r_s)(PQ'-P'Q)=-r_s/2$. Define $v_b=P\mathscr{B}_bQ-Q\mathscr{B}_bP$ and $v_p=P\mathscr{B}_pQ-Q\mathscr{B}_pP$. Then

$$\begin{aligned}C&=-r(r-r_s)(v_bv_p'-v_b'v_p)>0,\\ G_l(r,r')&=\frac{v_b(r_<)v_p(r_>)}{\kappa C},\\ G_\phi(r_p,r_p)&=\frac{v_b(r_p)}{\kappa r_pF_p[v_b(r_p)+r_pv_b'(r_p)]}.\end{aligned} \tag{62}$$

The jump is $r(r-r_s)[G_r^+-G_r^-]=-1/\kappa$. The complete positive form establishes the inverse and excludes a zero denominator; the Wronskian sets its normalization.

For the unchanged r_s,r_b,r_p,σ of the pulse experiment, outward rational logarithm and square-root intervals give

Source and endpoints	G_ϕ
Actual outer port, finite apparatus	$8.91755323\,10^{-11}$
Areal test shell $r_a=(r_b+r_p)/2$, same apparatus	$5.32154769\,10^{-11}$
Same r_a, horizon regular and infinity decaying	$1.30128154\,10^{-9}$

The areal test shell is not the microscopic off-center launcher's radius R. The original local $1/(4R)$ counterterm therefore

remains distinct. For a source drive $-\mathrm{Re}(f^*\phi)$ with free form $\langle\phi,A\phi\rangle/4$, square completion requires $H_{\mathrm{ct}} = |f|^2 G_\phi$. Here G is the inverse-form kernel; the physical static minimizer is $\phi = 2Gf$. The actual incoming characteristic gives the effective port force $j_u = \kappa u_{\mathrm{in},t}/c$, or $f_\phi = r_p j_u$; it does not identify the microscopic g_{ab} with a macroscopic static force. Promoting this controlled port force to a Hamiltonian source requires its own counterterm and paid preparation. A prepared incoming guide state's energy is already in the full guide Hamiltonian and is not charged twice.

Angular mixing requires the complete inverse. In block notation its retained block is $(A_{PP} - A_{PQ}A_{QQ}^{-1}A_{QP})^{-1}$, which can exceed A_{PP}^{-1}. Discarding the complement can remove a positive return.

The separately declared horizon-regular/infinity-decaying kernel is $2P_l(y_<)Q_l(y_>)/(\kappa r_s)$. At fixed central areal radius its flat limit is $1/[\kappa(2l+1)r_a]$, and for $l=2$ its near-boundary diagonal is $[\log(r_s/(r_a-r_s)) - 3 + o(1)]/(\kappa r_s)$. The actual finite apparatus has a different flat limit: hold its radii and σ fixed, send $r_s \to 0$ and use the same Robin conditions on r^l, r^{-l-1}. Its outer-port limit is $2.53265981 \times 10^{-12}$. The domain, rather than a coordinate label, selects the counterterm.

9.6 The exact integrated potential sharpens the unchanged pulse

The supplied review observes that the complete causal proof needs $\int V dx$, rather than $V_{\max}L$. For the actual quadrupole,

$$J_V = 6\left(\frac{1}{r_b} - \frac{1}{r_p}\right) + \frac{r_s}{2}\left(\frac{1}{r_b^2} - \frac{1}{r_p^2}\right). \tag{63}$$

With $B = 17|A_0|/16$, the same exact Volterra equation implies

$$\begin{gathered}\|u-u_0\|_\infty \le 2B(e^{cJ_V T}-1),\\ \|u_t-u_{0,t}\|_\infty \le 2cJ_V B e^{cJ_V T}, \qquad \|u_x-u_{0,x}\|_\infty \le 2J_V B e^{cJ_V T}.\end{gathered} \tag{64}$$

Use J_V in the residual potential energy too. Every scattering order, both compact edges and all six original signed amplitudes/radius balls remain. For the unchanged final time the resulting ceilings are

Quantity	Bound
$cJ_V T$	$1.780673\,10^{-5}$
Waveform error$/\lvert A_0\rvert$	$3.783964\,10^{-5}$
Finite-window energy error$/E_{\mathrm{in},0}$	$8.225496\,10^{-6}$
Late field/spring/port energy$/E_{\mathrm{in},0}$	$6.581033\,10^{-11}$
Free post-window return tail$/E_{\mathrm{in},0}$	$6.755624\,10^{-23}$

The last positive term must be subtracted from a finite-window lower comparison to the reference's all-time returned energy. Applying source smoothing and normalizing every term once to the true positive input gives

$$\boxed{0.15851336 \le E_{\mathrm{ret}}/E_{\mathrm{in}} \le 0.15852982.} \tag{65}$$

This narrows the enclosure about 28% for the same model. Amplitudes retain their original signs; incident energies use the common positive normalization. No measured reflection fraction is asserted.

9.7 A Hamiltonian reservoir behind the passive boundary

The review's two-port completion can be realized by an actual conservative wave system. Add one lossless reservoir guide $v(y)$ per original tensor polarization, $y \ge 0$. At the common trace $q = u(0) = v(0)$ impose

$$u_x(0) + v_y(0) = \frac{\sigma}{c}q. \tag{66}$$

The exterior guide retains the full RM potential on the annulus and joins the calibrated external lead. With $\pi_u = \kappa u_t/(2c^2)$ and likewise for v, the Hamiltonian is

$$\begin{aligned}H = &\int\left[\frac{c^2|\pi_u|^2}{\kappa} + \frac{\kappa}{4}(|u_x|^2 + V|u|^2)\right]dx\\ &+\int\left[\frac{c^2|\pi_v|^2}{\kappa} + \frac{\kappa}{4}|v_y|^2\right]dy + \frac{\kappa\sigma}{4c}|q|^2.\end{aligned} \tag{67}$$

Both polarizations and the original physical-field surface corrections are understood. The positive closed form on H^1 pairs with matching traces defines a self-adjoint nonnegative graph operator K. Its natural junction condition is exactly Eq. (66); the boundary symplectic form cancels. This supplies conservative canonical evolution and storage, rather than deleting a dissipative bath.

For a purely outgoing initially unexcited reservoir, $v_y = -q_t/c$, recovering the original spring/damper law. With independent incoming derivative amplitudes a,b, dimensionless time σt gives $\dot q = a + b - q/2$, $a_{\mathrm{out}} = b - q/2$ and $b_{\mathrm{out}} = a - q/2$. Consequently

$$|a|^2 + |b|^2 - |a_{\mathrm{out}}|^2 - |b_{\mathrm{out}}|^2 = \mathrm{Re}(q^*\dot q). \tag{68}$$

This is the complete flux/storage identity with independent bath input and both outputs. Exterior absorption becomes outgoing reservoir energy.

The local free-lead junction has

$$S(s) = \begin{pmatrix} R & T \\ T & R\end{pmatrix}, \qquad R = -\frac{\sigma}{2s+\sigma}, \quad T = \frac{2s}{2s+\sigma}. \tag{69}$$

It is unitary on the frequency axis. The actual outer-port matrix also contains the retained annular potential; its conservative propagation composes into the full unitary S_{eff}. The pulse bounds above retain that distinction.

A specified quantum state and finite energy account. Rescale to canonical wave fields and quantize K with $H_{\mathrm{q}} = \mathrm{d}\Gamma(\hbar c\sqrt{K})$ relative to its stated vacuum reference. The incoming state is the coherent displacement of the unchanged original compact pulse in both polarizations, with independent reservoir scattering vacuum. It has finite energy and particle number; the pulse's low-frequency moments vanish and its derivative energy is finite. For the local junction,

$$a_{\mathrm{out}}(\omega) = Ra_{\mathrm{in}}(\omega) + Tb_{\mathrm{in}}(\omega), \qquad [a_{\mathrm{out}}, a_{\mathrm{out}}^\dagger] = \delta, \tag{70}$$

using $|R|^2 + |T|^2 = 1$. At the actual outer port use S_{eff}. For independent stationary free-lead inputs, $n_{\mathrm{out}} = |R|^2 n_a + |T|^2 n_b$;

the bath state $n_b = 0$ is vacuum, not a thermal-emission law. A thermal state and its energy supply remain additional physical inputs.

The complete excitation energy obeys

$$E_{\text{ext,out}} + E_{\text{bath,out}} = E_{\text{ext,in}} + E_{\text{bath,in}} \tag{71}$$

after the field and spring empty. A preloaded finite-energy reservoir can supply only its charged initial energy. Coherent energy differences agree with the classical storage account; split pointwise vacuum energies require a common regulator or smeared measurement. They are not silently assigned zero absolute apparatus cost.

A finite closed guide returns energy after $2L_b/c$. It can realize the same classical coherent-increment response before that return, but cannot inherit the semi-infinite model's all-time return fraction. Its actual vacuum correlations and late stored energy must be retained. A Neumann far end keeps the same static spring compliance, while a Dirichlet end adds stiffness $1/L_b$; static compliance alone does not select reservoir dynamics.

9.8 Escaping flash energy and a variable-gap power interface

The supplied review closes the event-energy integral. For an instantaneous isotropic infalling rest-frame flash, put $f = (1-\mu_c')/2$ and $\beta = \sqrt{1-F/e^2}$. Collecting every escaping direction at a complete outer static sphere yields

$$\frac{E_{\text{esc},o}}{E_{\text{em}}} = \frac{e}{2\sqrt{F_o}} \int_{\mu_c'}^{1} (1-\beta\mu')\, d\mu' = \frac{ef}{\sqrt{F_o}}[(1-\beta)+\beta f],$$
$$\frac{E_{\text{esc},o}}{E_{\text{em}}} \sim \frac{27F^3}{128e^3\sqrt{F_o}}. \tag{72}$$

The escape-number fraction is $O(F^2)$, but the total event energy is $O(F^3)$. Static-flash energy instead scales as $27F^{3/2}/(16\sqrt{F_o})$. These fluences are not instantaneous luminosities; angular reception delays and detector acceptance remain.

In the separately declared outgoing null-fluid comparison allow $r_b = 2\mu(1+\delta(U))$, $U = cu$, and set $Q = d\tau_b/du$. For purely outward radial emission with no intervening scattering, storage or incoming flux, $L_\infty = Q^2 P_{\text{local}}$. Writing $\lambda = GP_{\text{local}}/c^5$ gives

$$Q^2 = \frac{\delta/(1+\delta) + 4\mu\,\delta_U}{1+4\lambda(1+\delta)}. \tag{73}$$

The positive numerator is the causal condition. A shrinking gap costs margin, but positive luminosity vanishing sufficiently rapidly is allowed. For constant gap every finite nonnegative local power obeys the strict earlier luminosity bound; bounded local power gives $L_\infty = O(\delta)$. This radial protocol must not replace the different isotropic flash or count deposited reservoir energy again as escaped luminosity. Surface stress, depletion, interior matching and a thermal state remain inputs.

9.9 Finite-step temperature and recovered thermal chronology

Under the retained area-entry premises, write $T_H(M) = K/M$ and $M_j = M\sqrt{1-j/N}$ for a permitted entry count $0 < j \le N$. Then

$$\frac{\Delta E_j}{jk_B} = \frac{2K}{M+M_j} = \frac{2}{1/T_H(M) + 1/T_H(M_j)}. \tag{74}$$

For positive final mass this is exactly the harmonic mean of the endpoint temperature scales. The event energies telescope; they assign no probabilities, widths, times, species or spectrum. The formal final zero mass does not establish a valid thermal endpoint approximation.

The complete local LC11 output package and runner have now been retained with their exact custody: 12 files, historical 100 PASS rows and eight controls. Its eleven output hashes reconcile, and 31 listed local source snapshots are retained; twenty original Windows-source inputs remain unrecovered. Fourteen exact QGA025/026 file/hash/revision routes are retained without synthesizing their missing source or gap-register contents. These are recovered historical receipts, not a new execution of LC11 or a derivation of a thermal state and radiation rate.

10. Unit closure: source coordinate, domain and physical claim

The strong-field coordinate in this chapter is the source quantity

$$A(r) = \frac{r_s}{r}, \qquad r_s = \frac{2GM}{c^2},$$

with r the areal radius and M the declared spherical source mass. Thus $A(r_s) = 1$ is an exact identity within this definition. It is not the condition $A_0 + A = 1$. The introductory total account $A_{\text{tot}} = A_0 + A$, when applicable to a specified weak-field operation, has a different domain from the source coordinate used here. Substituting that total into the closure condition would move the boundary without supplying a strong-field equation that authorizes the move. Conversely, retaining the source condition does not establish that a background has no gravitational effect. A background-aware strong-field theory must determine its own geometry, reference normalization and boundary conditions.

Three statements must now be distinguished. The exterior adapter becomes singular in its static coordinates at $A = 1$. An exterior ledger may exclude completed records labeled $A > 1$. A physical traveler may be unable to reach or cross the boundary in finite elapsed local time. These statements concern, respectively, a representation, an allowed record domain and a dynamical prediction. None becomes the next merely by reusing the word "closure."

Additional SAM postulate: traveler closure. The unified synthesis, Section 4.4, explicitly advances the stronger physical proposition: every stage $A < 1$ is attainable with a finite traversal requirement, but completing the approach to $A = 1$ requires

unbounded traveler-local clock accumulation, and no completed parent-ledger crossing into $A > 1$ occurs. The traveler's clock is not assumed to malfunction; the proposed divergence belongs to the physical traversal requirement. This is retained here as an *additional SAM postulate*, pending a dynamical derivation from a fully specified strong-field law. It is not inferred from divergence of Schwarzschild coordinate time.

This qualification preserves, rather than weakens, the intended distinction from general relativity. A timelike geodesic of the Schwarzschild geometry can cross its future horizon in finite proper time. Therefore that complete geodesic dynamics cannot simultaneously supply a divergent traveler proper time for the same trajectory and initial data. SAM must specify what physical law, metric, coupling, admissible history or progress observable differs. Restricting an exterior reporting domain alone does not provide that difference.

The recovered CR008 package checks the exterior lapse, redshift and logarithmic outside-ledger traversal under declared premises. Its verdict remains `BOUNDARY`, and its explicit rejection of an infalling-observer overclaim is material. It cannot be promoted into a traveler-proper-time theorem. The original G11555 code is now authenticated, and its source-domain and road assumptions are stated in Section 2. Its full original receipt packet is not included. Those assumptions and the newly derived conditional force theorems preserve the distinction between the source rule and a selected traveler evolution law.

10.1 A conditional infinite-road completion and its dynamical cost

The proposition that the traveler's own intervals diverge can be made mathematically precise, but it requires a relation between the chosen road measure and the traveler's proper time. Consider the exterior benchmark $F = 1 - r_s/r$, $H = 1/F$, and fix an initial radius $r_0 > r_s$. Define the inward optical-road length

$$\begin{aligned} L(r) &= \int_r^{r_0} \frac{d\rho}{F(\rho)} = r_*(r_0) - r_*(r), \\ L(r) &\longrightarrow +\infty \quad (r \downarrow r_s). \end{aligned} \tag{75}$$

This equals c times the radial null coordinate travel duration. It is not the static spatial distance $\int dr/\sqrt{F}$, and it is not a proper-time reading. Its normalization inherits that of the static time.

Conditional completion theorem. Suppose a future-directed inward timelike trajectory is defined in this exterior and an additional dynamical rule imposes

$$0 \le \frac{dL}{d\tau} \le v_{\max} < \infty \tag{76}$$

along it, with fixed $v_{\max}$. Then $L(\tau) - L(\tau_0) \le v_{\max}(\tau - \tau_0)$, so the trajectory cannot approach r_s at finite proper time. If it approaches r_s at all, its proper-time duration is infinite. This conclusion follows from the bound plus the divergent road integral; it does not follow from the road integral alone. The bound is an additional SAM-type dynamical postulate if adopted for that purpose, not a consequence of Schwarzschild geodesic motion. In fact an inward geodesic has $dL/d\tau = c\sqrt{e^2 - F}/F \to \infty$.

A concrete trajectory saturating a finite road-speed bound is

$$\begin{aligned} u(r) := \frac{dr}{d\tau} &= -v_0 F(r), \qquad \beta_0 = v_0/c, \quad v_0 > 0, \\ \frac{dt}{d\tau} &= \frac{\sqrt{1+\beta_0^2 F}}{\sqrt{F}}. \end{aligned} \tag{77}$$

The second equation follows from timelike normalization, and $dL/d\tau = v_0$. Thus

$$\begin{aligned} \tau - \tau_0 = \frac{L(r)}{v_0} &= -\frac{r_s}{v_0}\log x + O(1), \\ t(r) &= \frac{2r_s}{v_0} x^{-1/2} + O(1). \end{aligned} \tag{78}$$

For this nongeodesic trajectory both times diverge, with different asymptotic laws. Its speed relative to a local static observer is $|v_{\text{local}}| = v_0\sqrt{F}/\sqrt{1+\beta_0^2 F} \to 0$: constant optical-road speed means progressively slower local inward motion in this particular geometry.

The dynamical price is explicit. For any differentiable radial trajectory specified by $u(r) = dr/d\tau$ in this metric, the radial four-acceleration is $a^r = uu' + c^2F'/2$, and orthogonality to the four-velocity gives the invariant proper acceleration

$$a = \sqrt{a_\mu a^\mu} = \frac{|uu' + c^2F'/2|}{\sqrt{F + u^2/c^2}}. \tag{79}$$

For the constant-road-speed example this becomes

$$a = \frac{F'(c^2/2 + v_0^2 F)}{\sqrt{F + \beta_0^2 F^2}} \sim \frac{c^2}{2r_s\sqrt{F}} \longrightarrow \infty. \tag{80}$$

Hence this construction is a consistent timelike kinematic example at all exterior points, but it is not force-free falling and it does not supply a finite-force realization. It establishes the conditional arrival statement while identifying the next physical obligation: derive the road-speed bound from an independently justified dynamical law and explain its force or energy budget, or replace the benchmark geometry and derive the corresponding law anew. It does not prove that every trajectory obeys the bound, that all bounded-road-speed trajectories have this same acceleration asymptotic, or that the underlying spacetime has no extension to $A > 1$.

10.2 Force, endpoint and radial metric closure

The source-only coordinate remains $A = r_s/r$, with r the areal radius and $r_s = 2GM/c^2$. The original microscopic radius R and interacting clock are separate inputs. In this section the static metric

$$ds^2 = \quad Fc^2dt^2 + H\,dr^2 + r^2 d\Omega^2$$

is explicitly declared, with $F,H>0$ and asymptotically normalized t. The radial null road is $c\,dt=\sqrt{H/F}\,dr$; its Schwarzschild value $dr/(1-A)$ contains the clock and spatial factors once each. The recovered G11555 ledger rule is retained separately from the test-body and apparatus laws examined here. No new SAM field equation, stress law or quantum state is selected by these comparisons.

10.2.1 Bounded force and a sharp integrated cost

Consider a future timelike radial C^2 trajectory with $u=dr/d\tau<0$ on a final monotone exterior segment. Its instantaneous dimensionless Killing energy is

$$e=F\frac{dt}{d\tau}=\sqrt{F(1+Hu^2/c^2)}.$$

It need not be conserved. Direct differentiation and four-velocity orthogonality give

$$\begin{aligned}a^r&=uu'+\frac{c^2F'}{2H}\left(\frac{e}{F}\right)^2+\frac{H'}{2H}u^2,\\ a_s&=\frac{\sqrt{FH}}{e}a^r,\qquad e'=\frac{\sqrt{FH}}{c^2}a_s,\qquad a=|a_s|.\end{aligned}\tag{81}$$

Here a_s is the signed outward proper acceleration. This is the mechanical work relation for a fixed-rest-mass probe, whose Killing energy is mc^2e. Apparatus power still needs its actual clock measure, $dW=p_t\,dt=p_\tau\,d\tau$.

Bounded-acceleration theorem. In the Schwarzschild exterior, suppose this trajectory approaches r_s while $a\le a_{\max}$ on a final segment. Since $FH=1$, Eq. (81) makes e Lipschitz in r; it has a finite limit $e_H\ge0$. If $e_H=0$, integration gives

$$e(r)\le\frac{a_{\max}}{c^2}(r-r_s),\qquad e(r)\ge\sqrt{F}\sim\sqrt{\frac{r-r_s}{r_s}}.$$

These inequalities are incompatible near r_s. Thus $e_H>0$ and $u=-c\sqrt{e^2-F}\to-ce_H$. The remaining proper-time integral is finite. An infinite-proper-time monotone radial approach in this fixed geometry therefore requires unbounded proper acceleration. This conclusion uses neither a geodesic assumption nor a power ansatz for the trajectory; an extension of the actual worldline remains a further question about the chosen completion.

The argument also covers $F\sim fx^p$, $H\sim hx^{-q}$ with $p>0,q<2$, where $x=(r-r_s)/r_s$. The factor $\sqrt{FH}$ is integrable. A bounded acceleration and $e_H=0$ would imply

$$e(r)\le\frac{a_{\max}}{c^2}\int_{r_s}^{r}\sqrt{FH}\,d\rho.$$

The ratio of $\sqrt{F}$ to this integral is proportional to $x^{q/2-1}\to\infty$. Hence again $e_H>0$, and $\int\sqrt{FH}\,dr/(ce_H)$ is finite. This extension uses value asymptotics only.

Accumulated proper impulse. For any such static spherical metric define the inward optical road $L(r)=\int_r^{r_0}\sqrt{H/F}\,d\rho$. If $0<dL/d\tau\le v_{\max}<\infty$, then, with $\beta_{\max}=v_{\max}/c$,

$$\frac{dL}{d\tau}=\frac{c\sqrt{e^2-F}}{F},\qquad\sqrt{e^2-F}\le\beta_{\max}F\le\beta_{\max}e^2.\tag{82}$$

Combining the work identity with $d\tau=\sqrt{FH}\,|dr|/[c\sqrt{e^2-F}]$ yields

$$\begin{aligned}I:=\int a\,d\tau&=c\int\frac{|e'|}{\sqrt{e^2-F}}\,|dr|\\&\ge\frac{c^2}{v_{\max}}\int\frac{|e'|}{e^2}\,|dr|\\&\ge\frac{c^2}{v_{\max}}\left|\frac{1}{e_{\rm end}}-\frac{1}{e_{\rm start}}\right|.\end{aligned}\tag{83}$$

The last step is total variation of $1/e$, without requiring e to be monotone. Also $e^2\le F+\beta_{\max}^2F^2$, so $F\to0$ forces $I\to\infty$, with the explicit lower bound

$$I\ge\frac{c^2}{v_{\max}}\left[\frac{1}{\sqrt{F_{\rm end}+\beta_{\max}^2F_{\rm end}^2}}-\frac{1}{e_{\rm start}}\right].$$

The original constant-road-speed example is sharp at leading order: $I\sim c^2/(v_0\sqrt{F_{\rm end}})$. Its $e\to0$ leaves a finite signed Killing-energy change. Infinite proper impulse, fuel, support energy and paid reference work are different accounts; no propulsion law is supplied by this inequality.

10.2.2 An endpoint theorem without a power ansatz

Assume $F\to0$ and $F'>0$ on a final segment, and define

$$\kappa_{\rm red}=\frac{c^2F'}{2\sqrt{FH}}>0.$$

This is a redshifted-acceleration quantity; its surface-gravity interpretation also needs a normalized regular Killing horizon. Changing variable to F gives

$$\begin{aligned}\mathscr{A}_{\rm aff}:=\int\sqrt{FH}\,dr&=\frac{c^2}{2}\int\frac{dF}{\kappa_{\rm red}(F)},\\ \Delta\tau_e&=\frac{c}{2}\int\frac{dF}{\kappa_{\rm red}(F)\sqrt{e^2-F}},\\ \Delta t_{\rm null}&=\frac{c}{2}\int\frac{dF}{\kappa_{\rm red}(F)F},\\ \Delta\ell_{\rm stat}&=\frac{c^2}{2}\int\frac{dF}{\kappa_{\rm red}(F)\sqrt{F}}.\end{aligned}\tag{84}$$

A radial null affine parameter differs from $\mathscr{A}_{\rm aff}$ by a fixed positive normalization. For a timelike geodesic, $e>0$ is fixed and $0<F\le F_0<e^2$. The bounded positive factor $1/\sqrt{e^2-F}$ therefore establishes the exact equivalence

$$\Delta\tau_e<\infty\iff\mathscr{A}_{\rm aff}<\infty\iff\int_0^{F_0}\frac{dF}{\kappa_{\rm red}(F)}<\infty.$$

If $\kappa_{\rm red} \geq \kappa_{\rm min} > 0$, in particular,

$$\Delta\tau_e \leq \frac{c}{\kappa_{\rm min}}[e - \sqrt{e^2 - F_0}] < \infty.$$

Infinite proper approach requires loss of every positive uniform lower bound on $\kappa_{\rm red}$. A zero limit alone does not settle convergence. The explicit non-power family

$$\kappa_{\rm red}(F) = \kappa_0 F[\log(1/F)]^\beta, \qquad 0 < F < e^{-1},$$

tends to zero for every finite β, while the proper/affine integral is finite exactly for $\beta > 1$. Substitution $y = \log(1/F)$ reduces it to $\int_1^\infty y^{-\beta} dy$. Optical time and static distance diverge for every member of this comparison family.

For the power class additionally require the derivative control $xF_x/F \to p > 0$. Then

$$\kappa_{\rm red} \sim \frac{c^2 p}{2r_s}\sqrt{\frac{f}{h}}x^{(p+q)/2-1}.$$

Infinite geodesic proper/affine approach, $q - p \geq 2$, implies $q > 2$ and $\kappa_{\rm red} \to 0$. Conversely a finite nonzero limit in this class has $p + q = 2$, hence $q - p < 2$ and finite proper/affine approach. These are endpoint convergence statements; the global completion and quantum state remain separate inputs.

10.2.3 A finite-force comparison with calculable source cost

The already declared $p = q = 2$ comparison is

$$F = e^A(1-A)^2, \qquad H = 1/F.$$

Its constant-road trajectory $u = -v_0 F$ has

$$\tau \sim \frac{r_s}{v_0 \exp(1) x}, \qquad a \longrightarrow \frac{c^2\sqrt{\exp(1)}}{r_s}, \qquad \kappa_{\rm red} \longrightarrow 0.$$

Infinite proper approach with finite limiting acceleration is thus possible in a changed geometry. Its accumulated proper impulse still diverges by Eq. (83). Geodesics instead approach in finite proper time since $FH = 1$; the imposed road rule is not made universal by this example. The ingoing radial metric block is regular at r_s, with determinant $-c^2$.

Write $F = e^A P$. The exact derivative factors are

$$P = (1-A)^2, \quad Q = P + P' = A^2 - 1,$$
$$R = Q + Q' = A^2 + 2A - 1.$$

At $A = 1$, $F = F' = 0$, $F'' = 2\exp(1)/r_s^2$, and direct curvature calculation gives

$$\mathscr{R} r_s^2 = 2 - 2\exp(1),$$
$$R_{\mu\nu}R^{\mu\nu} r_s^4 = 2 + 2\exp(2),$$
$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} r_s^4 = 4 + 4\exp(2).$$

These endpoint invariants are finite.

As a conditional source diagnostic only, choose $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$. Multiplying its effective energy density and principal pressures by $8\pi G r^2/c^4$ gives

$$\widehat{\rho} = 1 - e^A(1-A)(1+A^2),$$
$$\widehat{p}_r = -\widehat{\rho}, \qquad \widehat{p}_t = \tfrac{1}{2}e^A A^2(A^2 + 2A - 1). \tag{85}$$

This conditional tensor is distinct from the original microscopic quantum source. On $0 < A \leq 1/4$,

$$\frac{d}{dA}[e^A(1-A)(1+A^2)] = e^A A(1 - 2A - A^2) > 0.$$

The quadratic's Bernstein coefficients on $[0, 1/4]$ are $(1, 3/4, 7/16)$. Its primitive starts at one, so $\widehat{\rho} < 0$ throughout this interval, with $\widehat{\rho} = -A^2/2 + O(A^3)$. The nonzero sign crossing lies at $A \simeq 0.584724386479634$. The finite-force comparator consequently makes its source-law obligation explicit rather than assigning a new SAM stress law.

This diagnostic also supplies a complete integrated account. Its $m(r) = c^2 r(1-F)/(2G)$ satisfies $dm/dr = 4\pi r^2\rho/c^2$, with ρ the effective energy density. Both endpoint masses are M. Nonnegative exterior density would make m nondecreasing between equal endpoints, forcing $m \equiv M$ and the Schwarzschild F. A changed $H = 1/F$ profile with the same horizon radius and ADM mass therefore requires compensating source energy under these hypotheses. For this comparison, the unique density sign crossing A_ρ gives

$$\frac{m_{\max}}{M} = \frac{1 + A_\rho}{1 + A_\rho^2} \simeq 1.180953354484870.$$

The positive inner and negative outer exterior mass contributions each have magnitude approximately $0.180953354484870M$. Outward rational enclosures are retained in the packet. This conditional source account is not an additional SAM energy condition.

Its orbit packet is also calculable. The exact source coefficients give

$$2P + AQ = (A-1)(A^2 + 3A - 2),$$
$$APR - PQ - 2AQ^2 = (1-A)^2[2 - (1+A)^3]. \tag{86}$$

Thus the ordinary timelike branch ends at $A_{\rm ph} = (\sqrt{17} - 3)/2 \simeq 0.561552812808830$, while

$$A_{\rm ISCO} = 2^{1/3} - 1 \simeq 0.259921049894873.$$

The branch is stable below this ISCO and unstable up to its photon circle. The null potential has one interior maximum between its zero endpoint values. These are new exact comparison-metric consequences, not a fit of the SAM landmark packet.

10.3 Finite impulse closes the general approach question

In the declared static metric, write

$$e = F\frac{\mathrm{d}t}{\mathrm{d}\tau} > 0, \qquad q = \overline{e^2 - F}, \qquad J_0 = \int_{r_H}^{r_0}\sqrt{FH}\,\mathrm{d}r. \tag{87}$$

For a strictly inward radial C^2 trajectory, signed outward proper acceleration a_s obeys

$$e' = \frac{\sqrt{FH}}{c^2} a_s, \qquad I := \int a \, \mathrm{d}\tau = c \int \frac{|e'|}{q} \, |\mathrm{d}r|. \tag{88}$$

Suppose r_H is finite, $F \to 0$, and $J_0 < \infty$. Because $q \le e$,

$$I \ge c \, \mathrm{Var}(\log e), \qquad e \ge e_{\min} := e_0 \exp(-I/c) > 0. \tag{89}$$

If $I < \infty$, then near the endpoint $F \le e_{\min}^2/2$, so

$$\Delta\tau \le \frac{\sqrt{2}}{c e_{\min}} \int \sqrt{FH} \, \mathrm{d}r < \infty. \tag{90}$$

Earlier compact strictly inward segments also have finite duration. Thus infinite proper approach at a finite-null-affine endpoint requires infinite accumulated proper impulse, even when acceleration has spikes. This establishes an approach-time statement, not a completed ledger crossing or an extension theorem. The affine hypothesis matters: $F \sim x$, $H \sim x^{-3}$ permits a zero-impulse geodesic with infinite approach time.

There is also a geometric sufficient condition without a power ansatz. Put $N = \sqrt{F}$ and $\mathrm{d}\ell = \sqrt{H}\,\mathrm{d}r$. Then

$$\left| \frac{\mathrm{d}\log e}{\mathrm{d}\ell} \right| = \frac{N}{e} \frac{a}{c^2} \le \frac{a}{c^2}. \tag{91}$$

Finite static distance and $\int a \, \mathrm{d}\ell < \infty$ imply $e_H > 0$ and $J_0 = \int N \, \mathrm{d}\ell < \infty$. Hence they imply finite proper approach. In particular this applies to bounded acceleration at every finite-static-distance zero-lapse endpoint, without differentiating an asymptotic expansion or assuming a monotone lapse.

10.4 The optimal road bound and a causal apparatus budget

Let the inward optical road speed satisfy $0 < |\mathrm{d}L/\mathrm{d}\tau| \le \beta c$. Since $|\mathrm{d}L/\mathrm{d}\tau| = cq/F$, its exact consequence is

$$q \le q_{\max}(e) = \frac{2\beta e^2}{\sqrt{1+4\beta^2 e^2}+1}. \tag{92}$$

Define the primitive

$$\Psi_\beta(e) = c \left\{ \frac{\sqrt{1+4\beta^2 e^2}+1}{2\beta e} - \operatorname{arsinh}(2\beta e) \right\}. \tag{93}$$

Direct differentiation gives $\Psi'_\beta = -c/q_{\max}$; the terms $2\beta/\sqrt{1+4\beta^2 e^2}$ cancel. Therefore

$$I \ge \mathrm{Var}(\Psi_\beta(e)) \ge |\Psi_\beta(e_{\mathrm{end}}) - \Psi_\beta(e_0)|. \tag{94}$$

Equality is attained when the road ceiling is saturated wherever e changes and e is monotone. Constant road speed on a monotone-F segment supplies such a trajectory with compatible endpoint data. The expansion $\Psi_\beta/c = 1/(\beta e) - \beta e + O(e^3)$ recovers the leading impulse divergence while retaining its finite remainder.

Now declare an apparatus: an isolated compact test rocket with rest mass m, positive surviving payload, no incoming power or momentum, and future-causal exhaust. In its instantaneous rest frame,

$$ma \, \mathrm{d}\tau \le -c \, \mathrm{d}m, \qquad \log \frac{m_0}{m_{\mathrm{end}}} \ge \frac{I}{c}. \tag{95}$$

This follows from $|p_{\mathrm{exhaust}}| \le E_{\mathrm{exhaust}}/c$; collimated photon exhaust attains the bound. A finite mass ratio $R = m_0/m_{\mathrm{payload}}$ therefore excludes infinite proper approach at a finite-affine endpoint for this apparatus. On a decreasing-e bounded-road segment,

$$\Psi_\beta(e_{\mathrm{end}}) - \Psi_\beta(e_0) \le c \log R. \tag{96}$$

The simpler preceding primitive also gives the explicit weaker gap

$$\begin{aligned} B &= e_0^{-1} + \beta \log R, \\ F_{\mathrm{end}} &\ge \frac{\sqrt{1+4\beta^2/B^2}-1}{2\beta^2} > 0. \end{aligned} \tag{97}$$

These are isolated-rocket statements, not prices for tethers, externally powered equipment or an unspecified SAM apparatus. Variable-mass energy accounting must retain both terms in $\mathrm{d}(mc^2 e) = mc^2 \mathrm{d}e + c^2 e \, \mathrm{d}m$.

10.5 A worked apparatus candidate and its actual clock chain

The following apparatus candidate is developed in campaign voli_chapter5_localization2_20261016 (GEN6 Duck run chapter5-localization-20261016-01, Duck 1.2.3, core 0.10.2) on a fixed background; its exhaust and residual energy–momentum balance was checked by one native operator-word task (r2_source_law_photon_rocket_energy_recipient). It is not a selected SAM source–geometry law. Use the declared Schwarzschild comparator, $F = 1 - r_s/r$, $H = 1/F$, the same central M, and the source-only $r_s = 2GM/c^2$. Impose outward braking with

$$\begin{aligned} \dot r = -\beta c F, \quad \beta = \tfrac{2}{3}, \quad r_0 = 4r_s, \quad \dot r_0 = -c/2, \\ e_0 = 1, \qquad m_0/m_{\mathrm{payload}} = 2. \end{aligned} \tag{98}$$

The control is a geodesic with these same launch position, velocity, mass and retained clock preparation. Its thrust and exhaust vanish. For the candidate, the closed background equations are

$$\begin{aligned} &e = \sqrt{F + \beta^2 F^2}, \quad q = \beta F, \\ &a = \frac{c^2 F'(1/2 + \beta^2 F)}{e}, \quad \dot m = -ma/c, \quad P = mac. \end{aligned} \tag{99}$$

If $U \cdot U = -c^2$ and n is outward unit spacelike with $n \cdot U = 0$, then $k = U/c - n$ is future inward null and

$$\frac{\mathrm{D}(mU)}{\mathrm{d}\tau} = -\frac{P}{c} k, \qquad \nabla_\mu (T_{\mathrm{rocket}} + T_{\mathrm{exhaust}})^{\mu\nu} = 0. \tag{100}$$

The second equation is distributional, with emitted photons following null geodesics. Reparametrization-invariant massive/null particle actions and momentum-conserving emission

vertices implement this local exchange; the feedback and mass equations above specify the apparatus rule. They are not derived from the original microscopic source action.

The exact mass curve is

$$\begin{aligned}\Phi(F) &= \frac{\sqrt{1+\beta^2 F}}{\beta\sqrt{F}} - 2\,\mathrm{arsinh}(\beta\sqrt{F}),\\ m(F) &= m_0 \exp[-\Phi(F)+\Phi(3/4)],\\ \Phi(3/4) &= 2-\log 3.\end{aligned} \tag{101}$$

Solving $\Phi(F_*)-\Phi(3/4)=\log 2$ with outward rational intervals gives

$$\begin{aligned}F_* &\in [0.446222641815801,\ 0.446222641815802],\\ r_*/r_s &\in [1.805779859398618,\ 1.805779859398619],\\ e_* &\in [0.731244172673343,\ 0.731244172673344].\end{aligned} \tag{102}$$

The imposed road rule exhausts its declared fuel at this finite gap. Switching off thrust leaves a geodesic payload with remaining proper approach less than $3r_s/c$. The control reaches the horizon after $14r_s/(3c)$ from the shared launch. The necessary fuel account already screens unlimited progress; additional controller and reference costs remain unpriced apparatus accounts.

The inward exhaust carries Killing factor $e+q$. Thus

$$\frac{\mathrm{d}E_{\mathrm{rocket}}}{\mathrm{d}\tau} = -P(e+q), \qquad E_{\mathrm{exhaust}} = m_0c^2 - m_{\mathrm{payload}}c^2 e_*. \tag{103}$$

Its fraction of initial rest energy lies in $[0.634377913663328,\, 0.634377913663329]$; the payload carries the complement. This inward energy has a central recipient in a black-hole completion. It is not also an escaping luminosity. Its backreaction is of test order m_0/M and is not evolved here: fixed background M and an exact claimed central-mass gain cannot be counted simultaneously.

For observation put $\rho = r/r_s$ and choose one static receiver at $\rho_o = 5$. The candidate proper time and tagged arrival are

$$\begin{aligned}\frac{c\tau_R}{r_s} &= \frac{4-\rho+\log[3/(\rho-1)]}{\beta},\\ \frac{c\tau_o}{r_s} &= \sqrt{\frac{4}{5}}\left\{\frac{ct}{r_s}+5-\rho+\log\frac{4}{\rho-1}\right\}.\end{aligned} \tag{104}$$

An exact coordinate-time primitive, supplied with its derivative in the replay, is obtained with $w=\sqrt{1/F+\beta^2}$ and $b^2=1+\beta^2$:

$$\begin{aligned}\mathscr{T}(w) &= \frac{2}{\beta}\left\{w - \frac{w}{2(w^2-b^2)} + \frac{2b^2+1}{4b}\log\frac{w-b}{w+b}\right\},\\ ct_R/r_s &= \mathscr{T}(w) - \mathscr{T}(4/3).\end{aligned} \tag{105}$$

It obeys $\mathscr{T}' = 2w^2(w^2-\beta^2)/[\beta(w^2-b^2)^2]$. The control uses the preceding exact $e=1$ geodesic primitive. The table gives conservative rounded enclosures at the same r_*, in units r_s/c; full directed endpoints are in the replay.

Elapsed/event time	Enclosure, units r_s/c
Traveler, rocket	$[5.26316569, 5.26316570]$
Traveler, control	$[3.71560367, 3.71560368]$
Coordinate, rocket	$[7.57362949, 7.57362950]$
Coordinate, control	$[5.84889110, 5.84889111]$
Receiver, rocket	$[11.06414368, 11.06414369]$
Receiver, control	$[9.52149076, 9.52149077]$

The final geometric received tick factors are respectively enclosed by $[0.484961119057537, 0.484961119057538]$ and $[0.286035372326617, 0.286035372326618]$.

Use one common calibration $\zeta = T_0/16$ and the explicitly conditional ideal point-clock transport

$$|\Psi(\tau)\rangle = \exp(-\mathrm{i}H_0\eta)\,|\Psi_0\rangle\,, \qquad \eta = \zeta c\tau/r_s. \tag{106}$$

Every source and receiver age above is below the original T_0 window. H_0 and the preparations retain all sixteen source states, all six original interacting hard-eigenbranch/soft-vacuum lanes, signed interactions, both polarizations, even energy and counterterms. The clock functions are the actual $\theta_l(\eta)$ with inherited $\ell_l \le \nu_l \le u_l$ and $|z_l| \ge m_l$, rather than constant-gap phases. The replay encloses their phases and the received ratios $g[\ell_l/u_l, u_l/\ell_l]$ for every lane.

Tagged stages $F = 3/4, 7/10, 13/20, 3/5, 11/20, 1/2, F_*$, together with preparation, keep every phase increment plus twice the inherited $1/50$-radian allowance below $3 < \pi$. This defines a unique lifted history under the stated preparation, finite-shot, readout and timing conditions. An endpoint qubit phase alone does not supply that history; the allowances remain apparatus requirements, not acquired data. Tagged current readouts already define an event comparison; transmission of an encoded phase needs its own calibrated transfer. Extended curved quantum-clock coupling remains to be supplied.

The retained predicate still gives $A < 1$ exterior writes, $A = 1$ terminal closure and no completed parent-ledger write at $A > 1$. It supplies no fuel or thrust law. This declared candidate fails as a mechanism for unlimited traveler duration before reaching $A = 1$; it does not falsify an unselected SAM constitutive law.

10.6 Conditional stress diagnostics and the missing source operand

Under the separately declared Einstein relation, orthonormal energy density ρ and radial pressure p_r satisfy

$$(\log FH)' = \frac{8\pi G}{c^4} rH(\rho + p_r). \tag{107}$$

Radial NEC makes FH nondecreasing outward. On a finite exterior radial interval it gives $J_0 \le \sqrt{FH(r_0)}(r_0 - r_H) < \infty$. Consequently every approaching finite-energy radial geodesic and every finite-impulse radial trajectory has finite proper approach. Equivalently, radial null focusing gives $r_{\lambda\lambda} = -4\pi G r T_{kk}/c^4 \le 0$: an initially inward radial null ray cannot stall at positive radius. This is neither an extension theorem nor a NEC statement about the original quantum state. Global positivity of K does not imply local NEC.

For general H, the mass function

$$m = \frac{c^2 r}{2G}(1 - H^{-1}), \qquad m' = \frac{4\pi r^2}{c^2}\rho \tag{108}$$

shows that $H^{-1} \to 0$ at $r_H = 2GM/c^2$, $m(\infty) = M$, and $\rho \geq 0$ force $m \equiv M$, hence the Schwarzschild H. This alone does not determine F. Adding $\rho + p_r = 0$ and asymptotic normalization does; DEC also suffices since $\rho = 0$ then forces every pressure to vanish.

The earlier identifiability control $F = 1 - A$, $FH = 1 + \lambda A^2 =: R$, has for $\lambda > 0$

$$\frac{8\pi G r^2}{c^4}(\rho + p_r) = -\frac{2\lambda A^2(1-A)}{R^2} < 0. \tag{109}$$

Its full effective Einstein stresses nevertheless obey conservation:

$$\begin{aligned} \frac{8\pi G}{c^4}\rho &= -\frac{\lambda A^4(1 - 2A - \lambda A^2)}{r_s^2 R^2}, \\ \frac{8\pi G}{c^4}p_r &= -\frac{\lambda A^4}{r_s^2 R}, \\ \frac{8\pi G}{c^4}p_t &= \frac{\lambda A^4(1 - A/2)}{r_s^2 R^2}, \\ p_r' + \frac{F'}{2F}(\rho + p_r) - \frac{2(p_t - p_r)}{r} &= 0. \end{aligned} \tag{110}$$

The native successor certificate inserts the actual first-round null-stress return into this full anisotropic constraint. Its exact coefficient arrays use $r_s = 1$; the displayed physical stresses restore r_s^{-2}. Conservation and an energy-condition branch are distinct tests.

The worked rocket is not a unified source–geometry completion. Such a proposal must specify, for example,

$$\begin{aligned} S_{\text{trial}} &= S_{\text{EH}} + S_{\text{central}} + S_{\text{original}}[g, \Phi, \text{shell}, Q] \\ &\quad + S_{\text{rocket}} + S_{\text{exhaust}} + S_{\text{reference}}, \\ T^{\text{original}}_{\mu\nu} &= -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{original}}}{\delta g^{\mu\nu}}, \end{aligned} \tag{111}$$

using inverse-metric variation, followed by the Einstein, original state/field and controller/null-transport equations. The necessary operand is the complete local renormalized metric variation in the actual hard-branch/soft-vacuum state: shell transport, pair/vacuum/reference terms, both counterterms and angular complement return all matter. The existing integrated positive K is not this operand.

In a separately justified static spherical reduction the displayed mass, null-stress and anisotropic-conservation equations are the required constraints. The original angular content and time-dependent flux cannot be discarded to impose that reduction; otherwise their evolution equations must be retained. A common unit chart must map the full source/reference energy to gravitating M, rather than identify M with a convenient bare gap. The flat source action and static-lapse extensions do not yet select the curved action or its renormalized stress. If a calculated source satisfies radial NEC, the finite-impulse theorem already screens this infinite-approach route; otherwise its actual violation or flux is the quantity to propagate. This is the remaining source-law calculation, not a parameter to fit.

11. An invariant question about the boundary

A divergence in one time coordinate does not diagnose trapping. For a general static spherical exterior, write

$$ds^2 = -F(r)c^2dt^2 + H(r)dr^2 + r^2 d\Omega^2, \qquad F > 0, \quad H > 0.$$

Here $4\pi r^2$ is the area of each symmetry sphere. The scalar

$$\chi := g^{ab}(\partial_a r)(\partial_b r) = \frac{1}{H}, \qquad m_{\text{MS}} := \frac{c^2 r}{2G}(1 - \chi) \tag{112}$$

packages the radial geometry and its Misner–Sharp mass.[9] This definition does not, by itself, establish field equations or an interpretation of m_{MS} as a particular matter integral. Under the Schwarzschild adapter, $H = (1-A)^{-1}$, so $\chi = 1 - A$ and $m_{\text{MS}} = M$. In a different geometry, declaring the same source label $A = r_s/r$ would not force these relations.

Choose future-directed radial null normals l^a, n^a to a symmetry sphere, normalized by $l \cdot n = -2$, and define their expansions by fractional area change:

$$\theta_l = \frac{2}{r} l^a \partial_a r, \qquad \theta_n = \frac{2}{r} n^a \partial_a r.$$

On the two-dimensional normal space, $g_\perp^{ab} = -(l^a n^b + n^a l^b)/2$. Consequently

$$\theta_l \theta_n = -\frac{4\chi}{r^2}. \tag{113}$$

A reciprocal rescaling $l \mapsto \alpha l$, $n \mapsto n/\alpha$, with $\alpha > 0$, preserves both the normalization and this product. A different cross-normalization changes the numerical coefficient and must be declared.

In the ordinary static exterior, the outgoing expansion is positive and the ingoing expansion negative. A future-trapped sphere has both negative; a past-trapped sphere has both positive. Thus a positive product alone does not select future trapping. A vanishing product identifies a candidate marginal sphere, but identifying a future outer trapping horizon additionally requires the individual expansions, time orientation and the appropriate transverse variation. These local distinctions are central to the trapping-horizon framework.[10]

For the adopted exterior, $\chi \to 0$ as $r \to r_s^+$. Extending that limiting statement into a trapping claim requires a regular geometry and null normals at the boundary. Static-coordinate components that vanish or diverge there are insufficient. Nor does a local marginal-sphere calculation establish a global event horizon, whose definition depends on the complete causal spacetime and its future asymptotics. If SAM retains only an exterior domain, it must specify whether its limiting surface is a physical boundary, an asymptotic end or part of an extendible geometry, together with the allowed causal behavior. The source coordinate cannot decide this on its own.

11.1 A spherical mass–flux and moving-boundary interface

A time-dependent mass must enter an actual field/flux law. Merely replacing M by $M(t)$ in static expressions misses derivative terms and energy payments. Even hypothesis RM has explicit metric work. At zero shift set $a = r^2\sqrt{H}/(Nc^2)$, $b = r^2N/\sqrt{H}$ and $d = N\sqrt{H}$; differentiating the Hamiltonian at fixed canonical momentum gives

$$P_{\text{metric}} = \frac{\kappa}{4}\int \left[-a_t|\phi_t|^2 + b_t|\phi_r|^2 + d_t\phi^*\mathscr{L}\phi\right] dr. \tag{114}$$

A shift or moving interface contributes its own canonical work. The field, boundary, source and apparatus accounts must retain those recipients.

One minimal explicit comparison is the *Einstein/null-fluid interface* in retarded length $U = cu$, with $\mu(U) = GM(U)/c^2$:

$$ds^2 = -\left(1 - \frac{2\mu(U)}{r}\right) dU^2 - 2\,dU\,dr + r^2 d\Omega^2. \tag{115}$$

Direct evaluation gives

$$\begin{gathered} G_{UU} = -\frac{2\mu_U}{r^2}, \quad T_{UU} = -\frac{c^4\mu_U}{4\pi G r^2}, \\ L_\infty = -\frac{c^5}{G}\mu_U = -c^2\frac{dM}{du}. \end{gathered} \tag{116}$$

For example, in the two-dimensional normal metric, $\Gamma^r_{UU} = (F_U + FF_r)/2$ and $\nabla^a\nabla_a r = F_r$. The static terms cancel in the Einstein tensor, leaving $G_{UU} = F_U/r$; this is precisely the term absent from a static substitution. Positive outgoing energy requires $\mu_U \le 0$. Einstein dynamics and the null-fluid stress are explicitly chosen comparison assumptions here, rather than a derived SAM source equation.

For a selected finite-gap surface $r_b(U) = 2\mu(U)(1+\delta)$ with fixed $\delta > 0$,

$$\frac{ds_b^2}{dU^2} = -[F_b + 2r_{b,U}], \qquad F_b = \frac{\delta}{1+\delta}. \tag{117}$$

Its tangent is timelike only if

$$-\mu_U < \frac{\delta}{4(1+\delta)^2}, \qquad L_\infty < \frac{c^5}{G}\frac{\delta}{4(1+\delta)^2}. \tag{118}$$

Thus a surface-gap prescription and luminosity already have a concrete causal compatibility test. A constitutive law must determine the luminosity from source depletion, emission, absorption, heat and mechanical work. The passive wave absorber's deposited heat cannot also be removed as escaped luminosity. Boundary stress, proper-time evolution and feedback remain inputs to that completion.

The invariant spherical diagnostics continue to apply: $\chi = g^{ab}\partial_a r\partial_b r$, $m_{\text{MS}} = c^2 r(1-\chi)/(2G)$ and $\theta_l\theta_n = -4\chi/r^2$ for $l\cdot n = -2$. In the retarded comparison choose $l = \partial_r$, $n = 2\partial_U - F\partial_r$; then $\theta_l = 2/r$ and $\theta_n = -2F/r$. In an advanced comparison one may instead choose $l = \partial_V + (F/2)\partial_r$, $n = -2\partial_r$, giving $\theta_l = F/r$ and $\theta_n = -4/r$. The individually oriented expansions distinguish the marginal surfaces. A changing compactness or vanishing product alone does not determine a future-trapped boundary, its global extension or the parent-ledger completion rule.

12. Surface gravity and a conditional thermal interface

The near-boundary gradient is useful, but its geometric meaning requires the metric. Normalize the static Killing time so that $F \to 1$ at the asymptotically flat reference infinity. The proper acceleration magnitude of a static observer is

$$a_{\text{stat}} = \frac{c^2|F'|}{2F\sqrt{H}}.$$

For a regular outer Killing horizon with the usual outward orientation, the acceleration redshifted to that reference yields

$$\kappa = \frac{c^2}{2}\lim_{r\to r_H^+}\frac{F'}{\sqrt{FH}}. \tag{119}$$

The sign convention is replaced by an absolute value when only the magnitude is intended. Existence and regularity of the limit are assumptions to check, not consequences of the symbol r_H.

For $F = 1 - A = 1 - r_s/r$ and $H = 1/F$, the product $FH = 1$ and $F' = r_s/r^2$. Hence

$$\kappa = \frac{c^2}{2r_s} = \frac{c^4}{4GM}.$$

The shortcut $(c^2/2)|dA/dr|_{r_s}$ works here because of these specific metric relations and the Killing-time normalization. Differentiating A alone does not determine surface gravity in a general completion. In particular, dA/dr is an areal-coordinate derivative; the static-slice proper spatial gradient has magnitude $|A'|/\sqrt{H}$, which tends to zero in this exterior limit. The finite surface gravity instead combines acceleration and redshift as above. Calling all three quantities simply "the gradient" would conceal the needed geometry.

A further, explicitly conditional interface is

$$T_H = \frac{\hbar\kappa}{2\pi c k_B} = \frac{\hbar c^3}{8\pi GMk_B}. \tag{120}$$

This is the standard stationary Schwarzschild temperature scale when the relevant quantum-field, state and horizon assumptions apply. Hawking's particle-creation calculation supplies a quantum argument, not merely dimensional conversion of a classical derivative.[11] A predicted detected flux additionally needs field content, propagation and transmission through the exterior potential. A SAM completion with different causal structure or boundary behavior must specify its quantum state and field evolution before asserting the same radiation. The radiation prediction requires the quantum state and field evolution specified above.

The Chapter IV operator boundary. The common-form argument of Chapter IV requires a uniform positive bound $W \ge mI > 0$ in its declared operator domain. On an exterior compact range $A \le 1-\delta$, with fixed $\delta > 0$, the lapse remains bounded away from zero; the argument can be applied when its other hypotheses hold. That restricted result does not carry

uniform constants to $A = 1$ as $N \to 0$. Any lapse-dependent weights must be checked in that limit, rather than assigned the earlier coercivity bound by continuity. This limitation is not a universal claim that every quantum evolution fails at zero lapse. For the special spatially uniform law $H(t) = N(t)H_0$, a self-adjoint H_0 and a real locally integrable N still define spectral evolution through the time change $s(t) = \int^t N(u)\,du$, even when N vanishes. That special construction supplies neither a nonuniform quantum coupling nor the traveler's physical progress law. It therefore cannot close the missing strong-field dynamics on its own.

12.1 The original area-entry rule at a finite decrement

The retained SAMA thermal source preserves the G18/G59/G60 chain: the four-cell ledger entry, its horizon-channel and gravity-bridge commitments, the paired emission/mass change, and the conditional entropy contribution k_B per entry. Under these stated source premises,

$$N = \frac{\mathscr{A}_H}{4\ell_P^2} = \frac{4\pi G M^2}{\hbar c}, \qquad S_H = k_B N. \tag{121}$$

The original differential energy-area step gives $k_B T_H(M)$. Its exact one-entry continuation, for $N \geq 1$, is

$$\begin{gathered} M' = M\sqrt{1-\frac{1}{N}}, \qquad \Delta E = Mc^2\left(1-\sqrt{1-\frac{1}{N}}\right), \\ \frac{\Delta E}{k_B T_H(M)} = \frac{2}{1+\sqrt{1-1/N}} \in [1,2]. \end{gathered} \tag{122}$$

This follows directly from $N(M') = N(M) - 1$ and energy conservation; it replaces a differential decrement by a finite one under the same premises. The associated integral first law gives $\Delta S_H = k_B$.

For $q = 1/N$, the ratio $c(q)$ obeys $c = 1 + qc^2/4$, hence

$$c(q) = 1 + \frac{q}{4} + \frac{q^2}{8} + \frac{5q^3}{64} + \frac{7q^4}{128} + \cdots. \tag{123}$$

Its coefficient of q^j is $\binom{2j}{j}/[4^j(j+1)]$. All coefficients are positive and at most one. Therefore for $N \geq 2$ the remainder after degree m is bounded by $q^{m+1}/(1-q)$. The leading relative correction to the original differential step is $1/(4N)$.

The area-entry, channel and pair premises retain their original source status. This new exact algebra has its own portable and native checks. It does not assign a microscopic degeneracy e^N, quantum state, transition matrix element, detailed-balance law, event rate, species, greybody coefficient or terminal state. Those inputs are needed to turn the thermal scale and exact event energy into a predicted spectrum and absolute luminosity. The microscopic clock gap Δ is not silently identified with this thermal emission energy.

12.2 Recovered thermal source and current custody

The thermal occupation input can now be recovered from an actual retained source. HOME-QW1 (H000848, retained in the current Starbreaker record)[14] implements a finite boundary-Write kernel from HOME-PC1 custody and its stated SAM_QM/A-gradient composition. The matching original SAM_QM PDF, QW1/PC1 code and current formation/exterior-custody sources are preserved in the companion packet. The geometric amplitude weights are QW1's specified composition; Born normalization alone does not select them.

The controlling H000797 lifecycle[15] returns a failed inward option to the same unresolved receipt, permits parent mass loss from realized outward post-completion exterior custody, and retains the formation mass/radius floor. Its receipt state machine does not assign physical waiting times. The older LC11/SUK061 inside-write interpretation and zero-remnant endpoint therefore retain historical status, with their thermal arithmetic preserved.

The direct recovered-kernel replay passes 242 checks, including 60 source hashes, sixteen thermal kernels and three explicitly fresh scalar fixtures. Its three retained calibration ratios $\varepsilon_{\mathrm{QW}}/\omega$ are $8, 32, 32$ in natural units, so the physical energy/frequency join requires attention. The old QGA025/026 originals, QW1 frozen contract/release JSON, and original PC1 trace remain listed by their actual routes and digests; historical campaign runners were not executed. The following calculations use the recovered occupation law and declare their additional quantum preparation and carrier calibration.

12.3 A recovered finite thermal kernel and its state boundary

The later HOME-QW1 source already supplies thermal occupation probabilities; the open register must be narrowed accordingly. In its natural-unit chart,

$$q = e^{-\omega/T_A}, \qquad K = \lfloor C/\varepsilon_Q \rfloor, \qquad p_n = \frac{q^n}{Z_K}, \quad Z_K = \sum_{n=0}^{K} q^n. \tag{124}$$

Here K is recoverable-custody capacity, distinct from the area count $N_A = \mathscr{A}_H/(4\ell_P^2)$. An admitted realization debits $n\varepsilon_Q$ from recoverable custody and credits it once to the parent; formation inventory is fixed. The controlling completed-Home lifecycle keeps failed IN on the same unresolved receipt and allows successful OUT from post-completion exterior custody only. Its formation mass/radius floor is retained. A zero QW1 occupation is not silently identified with an IN option or a successful inward Write.

QW1 fixes occupation probabilities, not all off-diagonal state data. The pure vector $\sum_n \sqrt{p_n} e^{i\phi_n} |n\rangle$ and the diagonal $\rho_K = \sum_n p_n |n\rangle\langle n|$ have the same draws, but different entropy and generally different stationarity. At $K = 1, q = 1/4$, their purities can be 1 and $17/25$. An orthogonal-partner purification or paid dephasing would specify the diagonal state; the source's conjugate Born weights and partner history do not themselves establish that operation.

The ledger quantum ε_Q and the kernel's ω are independent arguments in the actual source code. After a common dimensional lift, write $E_\omega = \hbar\omega_{\mathrm{phys}}$ and $q = e^{-E_\omega/(k_B T_A)}$. The diagonal

law is Gibbs for $H_Q = \varepsilon_Q n$ at

$$T_{\text{eff}} = \frac{\varepsilon_Q}{E_\omega} T_A. \tag{125}$$

Thus the physical carrier/state join at T_A requires $\varepsilon_Q = \hbar\omega_{\text{phys}}$; neither this calibration nor its identification with the original clock's ε is automatic. The exact finite mean and its infinite-law error are

$$\begin{aligned} \langle n\rangle_K &= \frac{q}{1-q} - \frac{(K+1)q^{K+1}}{1-q^{K+1}}, \\ \|p_K - p_\infty\|_{\text{TV}} &= q^{K+1}. \end{aligned} \tag{126}$$

The diagonal finite law is Gibbs for a physically truncated ladder. Embedding it in the full guide Fock space does not give an exact full oscillator KMS state; its missing top transition must be included in that comparison. The original sixteen-state source and six interacting clock preparations remain separate retained operands.

12.4 Finite entropy fixes a conditional ratio, with a controlled correction

For the separately declared nonrotating mass/area relation put

$$E = Mc^2, \qquad a = \frac{4\pi G}{\hbar c^5}, \qquad N_A = aE^2, \qquad S_H/k_B = N_A. \tag{127}$$

An allowed area step $N \to N - j \geq N_{\text{form}} > 0$ has $\Delta S/k_B = -j$ and the previously derived harmonic-mean temperature $T_{\text{step}} = \Delta E_j/(jk_B)$. Integer j is an area-step predicate; an arbitrary admitted mass receipt instead has its actual real area change. The floor excludes a formation-core evaporation endpoint.

If actual microcanonical shells, uniform preparation and precisely reversed matched channels are additionally supplied, let $\mathscr{A}_{if}$ be their common summed transition strength. Then

$$\begin{aligned} \Gamma_{\text{em}} &= \frac{\mathscr{A}_{if}}{\Omega_i}, \qquad \Gamma_{\text{abs}} = \frac{\mathscr{A}_{if}}{\Omega_f}, \\ \frac{\Gamma_{\text{em}}}{\Gamma_{\text{abs}}} &= e^{-j}, \end{aligned} \tag{128}$$

provided $\Omega = e^{S/k_B}$ in the specified common shell measure. Unknown density prefactors contribute their ratio; unequal bins or different inclusive angular/spin sums need their own matching proof. No integer degeneracy e^N or microscopic shell is manufactured from the thermodynamic entropy. This absorption comparator is a reverse external channel, not a successful IN crossing.

The exact exponent at the initial gradient temperature differs:

$$\frac{\Delta E_j}{k_B T_H(E)} = j + \frac{j^2}{N[1+\sqrt{1-j/N}]^2} > j. \tag{129}$$

Hence a unit entry has the conditional ratio e^{-1} at T_{step}, with a finite correction to a fixed-initial-T_H Boltzmann factor. For an energy-calibrated QW1 occupation, the analogous conditional area-entropy weights on $0 \leq n \leq K$ are

$$p_n \propto q^n e^{a(n\varepsilon_Q)^2}, \qquad q = e^{-\varepsilon_Q/(k_B T_H(E))}. \tag{130}$$

The fixed-temperature QW1 law omits the finite-backreaction quadratic term. On this same finite support,

$$\|\widetilde{p} - p\|_{\text{TV}} \leq \tanh\left(\frac{a(K\varepsilon_Q)^2}{4}\right). \tag{131}$$

For completeness, an exponential tilt in $[0,D]$ has weights in $[1,R]$, $R = e^D$, with mean m. Convex interpolation bounds its total variation by $(R-m)(m-1)/[m(R-1)]$; maximizing at $m = \sqrt{R}$ gives $\tanh(D/4)$. This is a finite bound, not a discarded remainder. The checked conditional fixture $a = 1, E = 100, \varepsilon_Q = 1/100, C = 1/50$ has $K = 2$, positive formation floor $E_{\text{form}} = 4999/50$, and bound less than 0.000099999999667. It assigns no new SAM mass calibration.

12.5 The transition clock remains an independent source operand

The new state information does not fix absolute transition strengths. On a resonant pair $|E_i, 0\rangle$, $|E_f, 1\rangle$ with $\varepsilon_Q = E_i - E_f$, total energy is $E_i I$. Every rotation of that pair conserves total energy, while its emission probability $\sin^2\theta$ varies. Two exact checked rotations on the algebraic area gap $N = 16 \to 9$ give $16/25$ and $144/169$, with the same energies and opposite source/carrier energy transfers. These are nonuniqueness witnesses, not a replacement for the original dressed source.

Likewise the finite generators $W_{n\to n+1} = \gamma q(n+1)$ and $W_{n+1\to n} = \gamma(n+1)$ all have the same stationary p_n and detailed balance for every $\gamma > 0$. Their clock is free. This mathematical family is not QW1's irreversible receipt/depletion law. Actual QW1 realization is externally supplied, its initial kernel is retained after depletion, and overdraw is rejected; a repeated-draw or rethermalization rule is not selected. A hypothetical attempt intensity λ would give initial gross power $\lambda\varepsilon_Q\langle n\rangle_K$, but the source supplies no λ. The following subsection specifies a guide preparation and its carrier transition. Selecting that preparation and its realization clock from the original source remains the connected physical task.

12.6 A quantum state and an outgoing finite-custody protocol

The recovered HOME-QW1 law supplies actual finite weights

$$\begin{aligned} T_A &= \frac{1}{4\pi r_s}, \qquad q = e^{-\omega_*/T_A}, \\ p_n &= \frac{q^n}{Z_K(q)}, \qquad Z_K(q) = \sum_{n=0}^{K} q^n, \\ K &= \lfloor C/\varepsilon_{\text{Q}} \rfloor. \end{aligned}$$

in its declared natural units; ε_{Q} denotes its ledger quantum, distinct from the original clock's $\varepsilon = \Delta/2$. The following additional preparation bridge makes those weights a definite guide state:

$$\rho_K = \sum_{n=0}^{K} p_n |n_f\rangle\langle n_f|.$$

Here f is a normalized incoming reservoir mode; an explicitly dephased preparation, or orthogonal history purification, supplies this density operator. QW1's probability list alone does

not specify that dephasing. The guide remains at the declared finite gap $r_b > r_s$; this is a preparation interface to the retained boundary model, rather than a literal launch from its unit-A source boundary.

The unchanged positive graph Hamiltonian has a useful immediate consequence: its scattering vacuum gives zero autonomous outgoing photon flux. The outgoing annihilators contain incoming annihilators only. A nonvacuum state must carry its prepared energy; the original coherent pulse and a populated reservoir are separate charged inputs.

The full annulus fixes the transmission. For $k = \omega/c$, let f_k solve the actual outgoing Jost problem

$$f_k'' + (k^2 - V) f_k = 0, \qquad f_k(L) = 1, \quad f_k'(L) = \mathrm{i}k.$$

At the junction put

$$D(k) = f_k'(0) - (\sigma/c - \mathrm{i}k) f_k(0), \qquad \tau_{\mathrm{eff}}(\omega) = \frac{4k^2}{|D(k)|^2}.$$

Wronskian conservation gives $0 < \tau_{\mathrm{eff}} \le 1$. Thus the actual boundary action already determines the transmission matrix; no additional transition probability is fitted. The complete RM potential is retained. For the original annulus its positive integral and length are

$$J_V = 6\left(\frac{1}{r_b} - \frac{1}{r_p}\right) + \frac{r_s}{2}\left(\frac{1}{r_b^2} - \frac{1}{r_p^2}\right),$$
$$L = r_p - r_b + r_s \log 2.$$

The outgoing Volterra equation yields the uniform, all-frequency comparison

$$\delta_D \le \frac{cJ_V}{\sigma} e^{LJ_V} + \frac{9}{8}(e^{LJ_V} - 1) < 5.600361\,10^{-6},$$
$$\frac{\tau_0}{(1+\delta_D)^2} \le \tau_{\mathrm{eff}} \le \frac{\tau_0}{(1-\delta_D)^2}, \qquad \tau_0 = \frac{4\omega^2}{\sigma^2 + 4\omega^2}.$$

This controls the full potential instead of substituting the free junction.

Finite occupation gives an exact count law. The passive scattering unitary maps a single packet creation operator to

$$b_f^\dagger \longmapsto \sqrt{\eta_f}\, a_g^\dagger + \sqrt{1-\eta_f}\, b_h^\dagger, \qquad \eta_f = \int \tau_{\mathrm{eff}}(\omega) |f(\omega)|^2 \,\mathrm{d}\omega.$$

The two outgoing packet shapes include their actual phases and are normalized separately. With external vacuum, the exact transmitted count is

$$P(m \mid n) = \binom{n}{m} \eta_f^m (1-\eta_f)^{n-m},$$
$$G(z) = \frac{Z_K(q[1 - \eta_f + \eta_f z])}{Z_K(q)}, \qquad \langle m \rangle = \eta_f \langle n \rangle_K.$$

The reflected number is retained. For an ideal energy eigenmode the identification $\varepsilon_{\mathrm{Q}} = \hbar\omega$ makes its transmitted credit $m\varepsilon_{\mathrm{Q}}$ and reflected energy $(n-m)\varepsilon_{\mathrm{Q}}$ exact. A normalizable finite-band packet instead has its actual frequency-resolved energy account. If its support is $[\omega_-, \omega_+]$, its m-photon energy lies in $[m\hbar\omega_-, m\hbar\omega_+]$. A fixed-quantum ledger then needs that bandwidth allowance or a resolved energy register.

For a concrete source-compatible preparation fixture, retain $r_s = 10^{10}$, $\sigma = 2/(573049\sqrt{500001})$, and declare $\omega_* = 10^{-11}$, $K = 4$ and $\varepsilon_{\mathrm{Q}} = \hbar\omega_*$. The ideal eigenmode has $C_{\mathrm{ideal}} = 4\varepsilon_{\mathrm{Q}}$. For a flat normalized packet in the band $\omega_*(1 \pm 10^{-3})$, charge $C_{\mathrm{packet}} = K\hbar\omega_+ = 4.004\varepsilon_{\mathrm{Q}}$. Its integer QW1 capacity remains four, and every outcome fits this worst-case energy budget with frequency-resolved output credits. It has

$$q = e^{-0.4\pi} \simeq 0.2846095433,$$
$$1.638601\,6\,10^{-5} < \eta_f < 1.645206\,0\,10^{-5},$$
$$6.365694\,8\,10^{-6} < \langle m \rangle < 6.391351\,4\,10^{-6}.$$

The packet occupies one declared normalized mode in the original two-polarization space; its orthogonal polarization and angular complement remain in vacuum. The packet budget is charged once, including its bandwidth reserve. Independently populating both original guides requires both energy allocations.

This count law is an all-time packet result. For an ideal positive-frequency time measurement, any gate of coordinate duration T captures at most $(\omega_+ - \omega_-)T/(2\pi)$ of such a band packet. The original one-cycle outer-clock window $T = T_0/N_p$ gives a fraction below $4.052073\,10^{-6}$, hence expected external counts below $2.590\,10^{-11}$ per prepared fixture in that one window. Its physical launch and measurement schedule remain declared apparatus inputs; an autonomous attempt clock is not inferred from p_n.

Finite QW1 weights and a KMS bath are distinct state choices. For the ideal single-frequency oscillator on the guide's full bosonic Fock space the finite state satisfies

$$q\langle aa^\dagger \rangle - \langle a^\dagger a \rangle = (K+1)qp_K > 0.$$

Thus it is a valid finite-energy state, with a quantified departure from exact oscillator KMS detailed balance. A truncated ladder restores finite Gibbs balance but changes its commutator; it cannot replace the guide CCR. As $K \to \infty$ the defect vanishes and the retained Planck occupation $n_A(\omega) = (e^{\omega/T_A} - 1)^{-1}$ follows.

If that limiting occupation is explicitly prepared in both original bath guides, with external vacuum, their stationary spectral flux is

$$\frac{\mathrm{d}\dot{N}}{\mathrm{d}\omega} = \frac{\tau_{\mathrm{eff}} n_A}{\pi}, \qquad \frac{\mathrm{d}P}{\mathrm{d}\omega} = \frac{\hbar\omega\,\tau_{\mathrm{eff}} n_A}{\pi}.$$

For the retained parameters, rigorous full-potential bounds give, in the declared natural model units,

$$6.331087\,8\,10^{-17} < \dot{N} < 6.331911\,2\,10^{-17},$$
$$1.360766\,2\,10^{-27} < P < 1.361062\,8\,10^{-27}.$$

This stationary comparison charges its reservoir supply and is separate from the finite $K = 4$ custody packet. Equal-temperature inputs at both ports give zero net flux; a global equilibrium state does not supply a net luminosity. The two original tensor guides do not count unselected species or angular populations. Physical selection of the QW1 state, its frequency/energy calibration, replenishment and depletion still require a dynamical source law. The finite count distribution and the boundary transmission are now specified without assigning those remaining inputs.

13. Current closures and the remaining source law

The review-derived finite-impulse and finite-static-distance theorems now screen an entire class of endpoint laws. Exact tidal strain, the selected curved inverse, complete integrated-potential pulse payment, flash energy and conservative reservoir account sharpen the earlier interfaces. Their proofs and stated domains accompany the finite native certificates.

Centered localization must retain the cancellation in the original prepared state before bounding it, and a finite-time current comparison must carry both backward-propagated insertions. The new state-specific bounds here are distinguished from a completed moving curved canonical propagator. A small source radius, or an instantaneous insertion lemma, supplies only part of that calculation.

The isolated finite-fuel rocket is an explicit progress candidate under one declared apparatus model. Its common-launch control and complete energy payments make the budget obstruction quantitative. It is not the missing source-selected SAM geometry. That completion requires the same curved tensor/source action to determine the full local stress, geometry, traveler evolution, clock response and boundary supply.

Physical target	Required connected operand
Source and metric	A specified curved tensor/source action and renormalized original-state stress, with constraint and conservation checks. The radial NEC is a conditional diagnostic, not an assigned quantum property.
Extended clock	Complete near/far current-form bound with both propagated insertions, moving source/shell, angular return, pair/vacuum terms, counterterms and boundary transport.
Physical boundary	Reservoir Hamiltonian and state are explicit for the conservative model. Source-driven reservoir supply, absorption, heat and gravitational mass feedback must share one energy account.
Thermal emission	HOME-QW1 occupation weights, a calibrated finite-packet state and graph transmission now give explicit conditional counts and a stationary supplied-bath spectrum. A source-selected quantum preparation, repeated realization clock, source vertex, depletion/geometry feedback and species dynamics remain connected inputs.

The retained original clocks, sectors and source evidence provide the common starting point for these remaining calculations.

14. A joint original-source measurement protocol

This continuation joins Chapters IV and V through the original interacting source. It retains the six preparations, signed couplings, all16 source states, both field polarizations, source even energy, both counterterms, initial hard dressing and the field/angular complements. The phase-bearing signal below is derived from that source. The earlier maintained compact carrier remains its separately labelled comparison protocol.

The chain has two levels of completion. The original local source supplies an exact field identity and a nonzero finite-mode signal with a finite measurement budget. A specified passive interface then carries that mode through the strong-field annulus and a calibrated detector. The interface, phase reference and receiving gate are apparatus inputs. A curved source-selected vertex and self-consistent stress/geometry evolution remain named physical operands in the ledger below.

14.1 The original interacting clock already drives a phase-bearing field

The signal can be derived from the original source rather than supplied by a bare oscillator encoder. Retain the full sixteen-state source, signed $D, C = \mathrm{i}SD$, even energies, both tensor polarizations, both counterterms and the original hard-branch superposition with soft vacuum. Within each unchanged active lane, $g = \varepsilon(q_a - q_b)$, $e = q_a + q_b + q_a q_b$, and

$$H_0 = \varepsilon e I + \varepsilon S + H_f - g(\sigma_z \phi_+ + \sigma_y \phi_\times) + C_{\mathrm{ct}},$$
$$C_{\mathrm{ct}} = 2g^2\mu, \quad \mu = 1/(5\kappa R) = 1/(4R), \quad \kappa = 4/5.$$

Write $z(t) = \langle \sigma_z + \mathrm{i}\sigma_y \rangle$ and $u_R(k) = \sqrt{2k/(\kappa\pi)}\, j_2(kR)$. The exact field equation is

$$a_\alpha(k,t) = e^{-\mathrm{i}kt} a_\alpha(k,0) + \mathrm{i}g u_R(k) \int_0^t e^{-\mathrm{i}k(t-s)} \sigma_\alpha(s)\, \mathrm{d}s.$$

Its circular mean consequently receives the *actual* interacting $z(s)$; no source–field factorization is used. The full-source identities

$$Z_{\mathrm{raw}} = D + \mathrm{i}C = D - SD, \qquad [H_{\mathrm{pair}}, Z_{\mathrm{raw}}] = -2Z_{\mathrm{raw}},$$
$$Z_{\mathrm{raw}}^\dagger Z_{\mathrm{raw}} = 2D^2(I+S), \qquad Z_{\mathrm{raw}} Z_{\mathrm{raw}}^\dagger = 2D^2(I-S)$$

retain both noise orderings. The bare commutator is only one term of the interacting equation; the actual source frequency remains its certified Ω, with its all-order residual.

Initial dressing and the completed physical receiver. The hard field is not initially vacant. Exact branch pull-through gives

$$\langle a_+(k) + \mathrm{i}a_\times(k) \rangle_0 = \frac{gMu_R(k)}{k - \Omega}, \qquad k \geq \lambda = 64\Delta.$$

The opposite helicity supplies its $k + \Omega$ denominator. Together they reproduce the retained nonzero incoming field and momentum. At the original physical receiver $r = R + 64$,

$$F_r(t) = F_{\mathrm{free},0}(t) + g \int_0^t \chi_{Rr}(t-s) z(s)\, \mathrm{d}s,$$
$$\chi_{Rr}(s) = \frac{P_2((R^2 + r^2 - s^2)/(2Rr))}{\kappa Rr}, \quad 64 < s < 64 + 2R,$$

with zero kernel outside that interval. The completed original receiver result retains both terms and gives additional phase ceilings below $5.734\,10^{-10}$ radians in all six lanes. Its source-induced first advance is64; the initially dressed total field can vary earlier. These are reused completed results, distinguished from the new mode certificate below.

14.2 A nonzero soft-mode signal with an actual brightness bound

Select a detector observable in the original soft band without deleting any physical hard field. The same normalized filter is used in every lane:

$$f_0(k) = \frac{2}{\sqrt{3\Delta}} \cos^2 \frac{\pi(k - \Delta)}{2\Delta}, \quad 0 < k < 2\Delta,$$
$$f_T(k) = e^{\mathrm{i}kT/2} f_0(k),$$
$$b_T = \int f_T(k) \frac{a_+(k,T) + \mathrm{i}a_\times(k,T)}{\sqrt{2}}\, \mathrm{d}k.$$

It satisfies $[b_T, b_T^\dagger] = 1$. The original soft vacuum gives zero homogeneous mean exactly. For $T = T_0/4$ or $T_0/2$,

$$\alpha_T = \langle b_T \rangle = \frac{\mathrm{i}g}{\sqrt{2}} \int f_T u_R \int_0^T e^{-\mathrm{i}k(T-s)} z(s)\, \mathrm{d}s\, \mathrm{d}k,$$
$$\alpha_{h,T} = \frac{\mathrm{i}gMT}{\sqrt{2}} e^{-\mathrm{i}\Omega T/2} \int f_0 u_R \operatorname{sinc} \frac{(k - \Omega)T}{2}\, \mathrm{d}k.$$

The second integral is strictly positive. The first is bounded about it by the original $|z(s) - Me^{-\mathrm{i}\Omega s}| \leq E(T)$ certificate. Its phase therefore tracks the actual source current at $T/2$, with known signed orientation $\arg(\mathrm{i}g)$. The two source tags are $T_0/8, T_0/4$; their phase increments stay below π, permitting unwrapping with the stated measurement allowances.

At $T = T_0/2$, rigorous outward bounds give

lane	$\lvert\alpha_T\rvert$ lower	$\langle b_T^\dagger b_T \rangle$ upper
01	$5.309\,10^{-18}$	$1.494\,10^{-34}$
02	$1.100\,10^{-17}$	$6.417\,10^{-34}$
03	$1.369\,10^{-16}$	$9.944\,10^{-32}$
12	$5.694\,10^{-18}$	$1.719\,10^{-34}$
13	$1.316\,10^{-16}$	$9.188\,10^{-32}$
23	$1.259\,10^{-16}$	$8.410\,10^{-32}$

The largest intrinsic phase discrepancy from the actual source current is below $1.442\,10^{-21}$ radians. The photons' two-time source covariance has not been factorized: total occupation is bounded above, whereas $|\alpha_T|^2$ is a coherent-brightness lower bound. In particular,

$$\frac{|\alpha_T|^2}{1 + N_T} \leq \Pr(n_T \geq 1) \leq N_T,$$
$$N_T \leq 2g^2T^2 \left(\int |f_T u_R|\, \mathrm{d}k \right)^2.$$

This is source-created finite-age soft occupation; an escaping luminosity also needs its physical outgoing channel and extraction account.

14.3 What a phase measurement must supply

For either normalized-mode quadrature the conservative variance bound is $1+2N_T$. Independent fresh joint preparations and finite-variance median-of-means estimation give a simultaneous 1 milliradian design for the twelve lane/age settings and two quadratures. The sufficient counts at $T_0/2$ range from about $7.10\,10^{41}$ to $4.73\,10^{44}$ copies per quadrature. These large counts follow the original tiny quadrupole brightness; no detector gain has been substituted for a source amplitude. Preparation, readout bias, timing, local-oscillator phase and its energy must satisfy their separate explicit allowances.

The soft-band observable is a precise canonical mode, not automatically a finite local time gate. Its frequency filter has temporal tails. The channel calculation must specify source-to-port mode transfer, buffering or finite-gate leakage, vacuum/thermal noise and detector backaction. A diagonal QW1 thermal state has zero coherent mean; it does not supply this clock phase. The phase-bearing mean above comes directly from the original interacting source. Physical curved canonical coupling and autonomous preparation/repeated-emission timing remain distinct joins.

14.4 The original clock packet through the complete complex channel

The actual source calculation supplies a normalized circular soft mode, with the common smooth filter

$$f_0(k)=\frac{2}{\sqrt{3\Delta}}\cos^2\frac{\pi(k-\Delta)}{2\Delta},\qquad 0<k<2\Delta,$$
$$f_T=e^{ikT/2}f_0,\qquad T=T_0/2.$$

Its mean is generated by the original signed coupling and interacting current; its encoded current age is $T/2=T_0/4$. No coherent brightness is assigned. The source preparation retains the initial hard field; only the measured soft band is initially vacuum. With $\omega=N_pk$ and $\sigma=N_p\Delta$, the inherited annular Jost problem gives

$$D=f'(0)-(\sigma-i\omega)f(0),\qquad T_{\rm eff}=\frac{2i\omega}{D},$$
$$D_0=(2i\omega-\sigma)e^{-i\omega L}.$$

Here $c=1$ and the complete positive potential is retained. The previous uniform bound $|D/D_0-1|\le\delta_D$ implies

$$\left|\frac{T_{\rm eff}}{T_{\rm free}}-1\right|\le\frac{\delta_D}{1-\delta_D}<5.601\,10^{-6},$$
$$|\arg(T_{\rm eff}/T_{\rm free})|\le\arcsin\delta_D<5.601\,10^{-6}.$$

This controls the entire finite packet in norm, including all edges of the unchanged compact comparison pulse. Differentiating the outgoing Volterra kernel, using $|\partial_k[\sin(kd)/k]|\le d^2/2$, gives

$$\left|\tau_g-\left[L+\frac{2\sigma}{\sigma^2+4\omega^2}\right]\right|<94486.$$

The proof uses $J=\int V\,dx$, $E=e^{LJ}$ and the finite bounds $A=(L+L^2JE/2)E$, $B=L^2JE/2+LJA$, $C=J(LE+A)$:

$$|\partial_k(D/D_0-1)|\le C/\sigma+B+(E-1)/\sigma+(L+2/\sigma)\delta_D.$$

The propagation time exceeds five remote source-cycle windows. A paid receiver reference and the launch/arrival event map must therefore accompany the phase record. Group delay is not the causal-front timestamp.

14.5 A specified finite gate, thermal input and detector requirement

The source-to-annulus interface is a declared apparatus hypothesis: a one-way mode sorter and buffer extract this normalized mode, retain all other original outputs, and preserve a calibrated phase and energy account. It does not identify the microscopic source shell with an areal annular boundary or prove the missing curved joint Hamiltonian. Its effect on the source during the certified interval must be bounded separately. Norm preservation alone does not prove that a frequency-mixing mode sorter preserves free energy; its energy change is paid apparatus work.

For the worked protocol a fixed balanced passive combiner joins the source and the actual independent, number-diagonal finite QW1 packet: $d=(s+t)/\sqrt{2}$, $e=(s-t)/\sqrt{2}$. The unused output e is retained. Thus the signal amplitude is divided by $\sqrt{2}$ and the total output energy equals the two input energies. The old $K=4$ thermal packet near 10^{-11} is not replaced by a new packet at σ; its old capacity cannot buy one photon at that carrier. Since f_0 extends to zero, exact spectral orthogonality is not assumed.

The packet transmission satisfies

$$\eta=\int|T_{\rm eff}|^2|f_{\rm remote}|^2d\omega\ge\frac{2/5+2/(3\pi)}{(1+\delta_D)^2}>0.61219973369.$$

A calibrated number-preserving output mode processor takes its transmitted normalized shape to the stated smooth mode, with phase fixed by the carrier port reference independently of the unknown source phase. Its frequency conversion work is retained. Parseval gives its temporal variance $\pi^2/(3\sigma^2)$, so a centered gate $G=2\pi/\sigma$ collects at least $2/3$ of that mode norm. Normalized positive-frequency gate projection therefore has real overlap at least $\sqrt{2/3}$, and

$$|\alpha_{\rm det}|\ge\sqrt{\eta/3}\,|\alpha_f|.$$

This gate statement applies after the specified extraction and mode processing; it is not automatically a gate bound on the full dressed field. The ideal all-time matched thermal contamination is below $4.779\,10^{-26}$ photons. A safe finite-gate bound is half its actual total external output number, below $3.196\,10^{-6}$ photons. Thermal occupation has no coherent mean under this declared diagonal preparation.

For an explicit readout use the canonical heterodyne POVM $\Pi(d^2\beta) = |\beta\rangle\langle\beta| d^2\beta/\pi$. For every finite-number state,

$$\mathrm{E}\beta = \langle a\rangle, \qquad \mathrm{E}|\beta - \langle a\rangle|^2 = \langle N\rangle + 1 - |\langle a\rangle|^2 \le V.$$

The vacuum term remains. Independent fresh preparations, divided into 19 blocks with at least $\lceil 16V/\varepsilon^2\rceil$ records each, and coordinatewise median block means give complex mean error at most ε with probability above 0.99. Chebyshev bounds each block and Hoeffding bounds the majority; no bounded quadrature is assumed. With $\varepsilon = 0.000899\,\alpha_{\mathrm{det},-}$, apparatus phase budget $9\,10^{-5}$ rad and the source and annulus phase disks, the total error is below 0.995 mrad. The sufficient requirements are:

lane	detected mean lower	fresh preparations
01	$2.39836\,10^{-18}$	$6.53918\,10^{43}$
02	$4.97081\,10^{-18}$	$1.52231\,10^{43}$
03	$6.18775\,10^{-17}$	$9.82404\,10^{40}$
12	$2.57244\,10^{-18}$	$5.68414\,10^{43}$
13	$5.94791\,10^{-17}$	$1.06323\,10^{41}$
23	$5.69067\,10^{-17}$	$1.16153\,10^{41}$

These are sufficient ideal-instrument requirements, not achieved measurements or optimal lower bounds. The same per-lane block lengths with 25 blocks give simultaneous six-lane confidence above 0.99, since $6e^{-225/32} < 0.01$; 27 blocks give $12e^{-243/32} < 0.01$ for twelve lane/age estimates using their own block lengths. Preparation leakage, mode mismatch, reference work, detector gain, timing, disturbance and replenishment of finite inputs require their own budgets. For unbounded quadratures a mode error δ_{mode} needs an affected-number bound: its mean bias is at most $\delta_{\mathrm{mode}}\sqrt{N_{\mathrm{leak}}}$; trace distance alone is inadequate.

14.6 What the amplitude record identifies

For a known source and receiver calibration write $\beta = K(d)\alpha_f$. If the actually calibrated radiation-zone aperture law is $|K(d)| = d_*/d$, a measured disk $|\beta - \widehat{\beta}| \le \varepsilon$ and source/gain calibration $A_\pm$ give

$$\frac{d_* A_-}{|\widehat{\beta}| + \varepsilon} \le d \le \frac{d_* A_+}{|\widehat{\beta}| - \varepsilon}, \qquad |\widehat{\beta}| > \varepsilon.$$

Otherwise the derived receiver law must be inverted directly. A normalized guide-mode amplitude does not itself imply geometric $1/d$ dilution. The present source amplitude enclosures are about 15 percent wide, so more shots alone do not narrow that calibration uncertainty. Unknown source brightness or detector gain remains degenerate with distance. At the original carrier the normalized local amplitude-distance/phase Jacobian and its inverse are

$$J_0 = \frac{1}{5}\begin{pmatrix} -4 & 2 \\ 2 & 4 \end{pmatrix}, \qquad J_0^{-1} = \begin{pmatrix} -1 & 1/2 \\ 1/2 & 1 \end{pmatrix}.$$

The native two-port certificates retain both polarizations, commutators, reflected output and vacuum/noise coefficients. They certify finite coefficient maps. The end-to-end apparatus error, cosmological Q and a positive physical relaxation spectrum remain distinct derivations; a positive Euclidean measure does not provide the latter.

14.7 Intrinsic energy, apparatus custody and the cost of repeated tags

The signal comes from the original autonomous interacting source, with its correlated hard-branch/soft-vacuum preparation. It is not the old maintained bright pulse. On the full sixteen-state source,

$$\begin{aligned} H_0 = \varepsilon H_{\mathrm{pair}} + H_f - \lambda(D\Phi_+ + C\Phi_\times) \\ + \lambda^2\mu(D^2 + C^2), \qquad \lambda = \varepsilon. \end{aligned} \tag{132}$$

H_0 generates the original local model age τ. The exact source identities give

$$\frac{\mathrm{d}\langle \varepsilon H_{\mathrm{pair}}\rangle}{\mathrm{d}\tau} = 2\lambda\varepsilon\langle C\Phi_+ - D\Phi_\times\rangle. \tag{133}$$

This is a joint expectation. The even source energy and both counterterm expectations are conserved at fixed coupling and radius; they remain in the absolute energy account. The interaction and hard dressing generally persist at the readout endpoint. Neither can be removed when equating an emitted packet with source disturbance.

The initially soft vacuum yields a useful state-uniform quantum ceiling. For each original polarization,

$$a_\alpha(k,T)\Psi_0 = \mathrm{i}g u_R(k)\int_0^T e^{-\mathrm{i}k(T-s)}\sigma_\alpha(s)\Psi_0\,\mathrm{d}s. \tag{134}$$

Using $\|\sigma_\alpha(s)\Psi_0\| = 1$ and $|j_2(kR)| \le (kR)^2/15$, both polarizations give

$$\begin{aligned} \langle N_{<b}(T)\rangle &\le \frac{4g^2T^2R^4b^6}{1350\kappa\pi}, \\ \langle E_{<b}(T)\rangle &\le \frac{4g^2T^2R^4b^7}{1575\kappa\pi}. \end{aligned} \tag{135}$$

For the whole soft field use $b = 64\Delta$, $T = T_0$; for the observed band use $b = 2\Delta$, $T = T_0/2$. These bounds retain the full source and are uniform over its initial relative phase. They do not factorize two-time correlations or identify total photon energy with its coherent mean. Initially hard field energy and any physical hard-band leakage remain separate entries.

Lane	E_{soft} upper	N_{band} upper	Copies lower
01	$6.196637\,10^{-28}$	$7.535720\,10^{-34}$	$3.186158\,10^{32}$
02	$2.661819\,10^{-27}$	$3.237033\,10^{-33}$	$7.417286\,10^{31}$
03	$4.124683\,10^{-25}$	$5.016021\,10^{-31}$	$4.786663\,10^{29}$
12	$7.128776\,10^{-28}$	$8.669293\,10^{-34}$	$2.769545\,10^{32}$
13	$3.811135\,10^{-25}$	$4.634715\,10^{-31}$	$5.180469\,10^{29}$
23	$3.488605\,10^{-25}$	$4.242487\,10^{-31}$	$5.659416\,10^{29}$

Energy is in original local model units. With the declared $\tau = N_p t$, the channel-time generator and photon energies carry

the same factor N_p. No gravitating-mass calibration follows from this chart.

The copy column is a necessary information bound, not a sufficient measurement protocol. If every phase-conditioned observed-band state has mean number at most $\bar{n}$, its vacuum probability is at least $1-\bar{n}$. For N independent copies, distance to the common vacuum is at most $\sqrt{N\bar{n}}$, and the distance between two phase hypotheses is at most $2\sqrt{N\bar{n}}$. The Helstrom bound therefore requires

$$N \geq \frac{(1-2\delta)^2}{4\bar{n}} = \frac{2401}{10000\bar{n}} \quad (\delta = 1/100). \tag{136}$$

Identical independent thermal, LO and reference ancillas, subsequent processing and joint readout cannot improve this necessary bound. An independently correlated reference already carrying the phase would violate that premise. Multiplying by the full source preparation energy gives prepared-energy throughput across fresh copies, not net work, irrecoverable heat or simultaneously stored energy. The actual preparation, reset, filtering and readout payments remain required.

Every output and work recipient remains. The balanced combiner keeps $d = (s+t)/\sqrt{2}$ and its unused $e = (s-t)/\sqrt{2}$, so their summed energy is exactly the two inputs' energy, including cross terms. The clock mean is attenuated by $1/\sqrt{2}$. QW1 custody is charged once; formation inventory stays fixed. The two-port channel retains reflected energy and physical field/spring/port storage. With $E_{SF} = \langle H_{\text{source}} + H_f + V + H_{\text{ct}} \rangle$, one complete coordinate-time ledger reads

$$\begin{aligned} &\Delta E_{SF} + \Delta E_{\text{guide}} + \Delta E_{\text{reservoir}} + \Delta E_{\text{detector}} \\ &\quad + \Delta E_{\text{LO/ref}} + \Delta E_{\text{added coupling}} + E_{\text{escaped}} \\ &\quad = W_{\text{adapter}} + W_{\text{readout/ref}} + W_{\text{metric}}. \end{aligned} \tag{137}$$

An explicit battery is counted by its energy change or as external work, once. Generic spectral shape conversion preserves number but may exchange frequencies; its actual energy change is processor work. Heterodyne LO work is not intrinsic clock radiation. Original source switching is absent; if added, both $-\dot{\lambda}\langle D\Phi_+ + C\Phi_\times\rangle$ and $2\lambda\dot{\lambda}\mu\langle D^2 + C^2\rangle$ must be paid, with moving-shell work if the radius changes. Added physical linear sources need their own actual-domain counterterm, not one borrowed from another boundary or radius.

The flight delay exceeds one original receiver cycle. A fresh receiver may be prepared around a stated coarse arrival-prior window; no source certificate is extended through an unproved long wait. Reference waiting is not reference-off: the retained paid protocol has $H_{\text{ref}} = -2E_{\text{ref}}P_{\text{excited}}$ on its source-wait interval. Ramps need their areas, and phase lifting needs tagged prefixes and turn storage. Signed work cancellation supplies no zero apparatus cost. An output-field instrument leaves the instantaneous unconditioned source marginal unchanged, but not generally its conditional state or subsequent joint clock evolution. A physical coupler/backaction certificate is still needed beyond the declared packet interface.

14.8 A flux-driven geometry control, without a prescribed static mass

The original microscopic radius and field receiver are not the annulus's areal endpoints. A full moving SAM metric is not derived here. Under the separate outgoing Einstein/null-fluid comparison,

$$\begin{aligned} ds^2 &= -\left(1 - \frac{2\mu(U)}{r}\right) \mathrm{d}U^2 - 2\mathrm{d}U\,\mathrm{d}r + r^2\mathrm{d}\Omega^2, \\ L_\infty &= -\frac{c^5}{G}\mu_U, \qquad \mu(U) = \mu_0 - \frac{G}{c^4}E_\infty(U). \end{aligned} \tag{138}$$

The mass aspect follows actual outward flux. Detector/control work is subtracted before inferring that source flux; inward energy, anisotropic stress and storage require their corresponding evolution terms. Gross OUT and simultaneous inflow are distinct, and $M \geq M_{\text{form}}$ remains.

For the same fixed areal endpoints and retarded null event U, put $\chi = (M_0 - M)/M_0$ and $B_i = A_{i0}/F_{i0}$. Their exact photon ratio is

$$\begin{aligned} \frac{g^2}{g_0^2} &= \frac{1 + B_b\chi}{1 + B_p\chi}, \\ \frac{g^2}{g_0^2} - 1 - (B_b - B_p)\chi &= -\frac{(B_b - B_p)B_p\chi^2}{1 + B_p\chi}. \end{aligned} \tag{139}$$

The retained finite-gap radii give $B_b = 10^6$, $B_p = 5\,10^5$ and $(\mathrm{d}\log g/\mathrm{d}\chi)_0 = 250000$. The displayed exact remainder controls the small-depletion approximation and exposes near-gap amplification. It does not substitute $M(t)$ into a static travel integral. Given an actual common physical energy unit E_u and source-mass map, the selected soft band would contribute at most $\chi_{\text{band}} \leq E_u N_p E_{\text{band}}/(M_0 c^2)$ in this conditional chart. That mass map and the complete local quantum stress remain independent physical operands.

14.9 The shared dependency ledger after this continuation

Link	Result and remaining physical operand
Original source	Exact interacting-current field identity, dressed initial contribution and six nonzero finite-mode means. Original finite-age certificates are authenticated inputs.
Port and propagation	Full annular complex-transfer error, phase and group-delay control. The calibrated passive source-to-port interface retains its geometry, unused outputs and feedback boundary.
Detector	An explicit quantum measurement, finite-gate and thermal-noise budget, repeated-preparation count, calibration and phase branch. These are sufficient apparatus requirements, without a claim of measured performance.
Energy and geometry	Source/field/near-storage/reservoir/detector/reference accounts, output ceilings and conditional inventory updates. A source-to-mass calibration and self-consistent curved stress law remain required for an autonomous geometry.
Thermal timing	The retained finite thermal packet contributes zero phase mean and specified noise. Its weights do not select a repeated realization clock or prepare the phase signal.
Distance dynamics	The calibrated amplitude and timing Jacobians are locally invertible under their stated source/channel model. Cosmological $Q(z)$, a real-time relaxation spectrum and population/selection dynamics remain separate physical derivations.

14.10 Reproduction of the joint additions

The joint source packet contains both editable successor chapters, the unchanged predecessor source archives, frozen original operands, detailed derivations, independent standard-library checks and the exact native request/return maps. Run `python3 ops/reproduce.py`. Supply the separate native-evidence directory with `-native-evidence` to authenticate the completed execution, every matrix/solve, all local retrieval excerpts and helper custody. Reproduction uses a fresh directory with spaces and compares regenerated result bytes and both assembled sources. Earlier chapter checks retain their original source packets and scope; this joint replay does not silently recalculate the historical spectral or continuum certificates. Native retrieval and direct source recovery have separate provenance. The qualified continuation uses six controllers, three helpers and two180-second admission windows, its own CPU locks and no global startup lease or automatic third round.

The test result suggests strong contact with the concept.

A. Gap arithmetic and finite-input uncertainty

For the exact A gap $\delta = 1 - A$, the road in Eq. (37) satisfies

$$\frac{dT}{d\delta} = -\frac{t_s}{\delta(1 \quad \delta)^2}. \tag{140}$$

If a supplied gap lies in $0 < \delta_- \le \delta \le \delta_+ < 1$, the mean-value theorem bounds a gap perturbation by

$$|\Delta T| \le \frac{t_s|\Delta\delta|}{\delta_-(1-\delta_+)^2}. \tag{141}$$

Endpoint uncertainty therefore becomes important near closure even when a formula can be evaluated to many digits. These bounds concern the declared radial metric law; they do not include metric-model, trajectory, emission or detector uncertainty. At fixed dimensionless geometry $T \propto M$, so a fractional source-mass error contributes the same fractional time error. Fixed physical endpoints require recomputing their dimensionless geometry as the mass changes.

B. Reproduction and evidence boundaries

The independent check scripts differentiate the radial primitives, verify normalization and received-rate identities, test the endpoint exponent logic, derive the conditional-road acceleration, recover the orbital landmarks and check the counterexample completion. The figure uses exact gap conversion and high-precision intermediate arithmetic. The illustrative mass table freezes the source packet's constants. These are mathematical and arithmetic checks, not observed black-hole trajectories or an independent execution of the original clock's spectral enclosure.

The recovered source set contains 55 files authenticated against the pinned Git blob identities. The saved manifest identifies each source path and revision. CR008's clock/road gap conventions and CR009's control naming remain visible in the original snapshots. In CR009 the control named as a photon orbit at $2r_s$ actually shares the inverse-square control implementation and evaluates its root at $\sqrt{2}r_s$; the present chapter does not count that row as an independent test of a $2r_s$ photon orbit. Neither an authentication check nor a dependency appeal changes the physical scope of the underlying computation.

A substantive next campaign should specify the strong-field equations, source and initial data before evaluation. It should return three separate objects: an admissible worldline and its proper time; the causal signal map to a receiver; and the parent-ledger completion rule. Wrong controls should include a static-time substitution for proper time, the square-root static redshift substituted for an infalling emitter, a shifted floor boundary, and weak-field optical extrapolation. Failure of one such control is informative only for the particular observable and assumptions tested.

B.1 The unchanged predecessor replay

The full H002656 source packet is preserved as `inputs/PREDECESSOR_SOURCE.zip`. Its manifest, report, verification and delivered-artifact digests are retained separately. Inside that archive, `inputs/predecessor` preserves the earlier source snapshots and historical clock inputs. The recovered G11555 Python has SHA-256 prefix `10f33d297627e80a`, matching the retained CR075

manifest. Its original missing seal/output set has not been regenerated.

The unchanged H002656 native companion contains its fifteen mathematical returns and four local retrievals; its identity remains in `inputs/PREDECESSOR_FINAL_ARTIFACTS.json`. The present source and native companions are separate successors. Their new requests, independent readback, actual result counts and complete custody are described below and in the accompanying README.

B.2 Review and successor reproduction

The original nineteen-page H002656 manuscript and its frozen native evidence remain unchanged. The supplied twelve-page review, editable checks and manifest are retained. Its four programs replayed successfully with the existing SymPy environment and regenerated their three saved JSON outputs byte-for-byte; no historical or native campaign was restarted.

The successor provides separate standard-library checkers for the new source/progress, localization and boundary calculations. Its compact source bundle retains the unchanged predecessor source ZIP, editable base manuscript and focused inserts, original input snapshots, exact and outward coefficient files, proofs and commands. The full native companion retains every accepted request/return, source binding, actual worker receipt, research excerpt and helper custody file. Independent rational readback checks complete matrices and solve enclosures from those bytes.

The new campaign uses the qualified installed native engine with six controllers and three helpers over two original 180-second admission windows. Only its own CPU cores are reserved; there is no global startup lease or third round. Review, final organization and manuscript assembly are outside the admission clocks. Source retrieval, finite native algebra, analytical proofs and achieved measurements retain their separate scope.

The thermal follow-up also recovers the actual HOME-QW1/PC1 kernel and current formation/exterior-custody law. Its direct function replay and new finite-state, backreaction and full-potential quantum-channel checks are separate standard-library computations. They add no round or result to the frozen twelve-return native campaign. Missing old contract/release files are listed rather than reconstructed from their historical digests.

Data and code availability

The accompanying compact source/check bundle contains the editable chapter, figure generator, independent mathematical checks and results, and a manifest of the pinned strong-field source snapshots. The recovered public inputs are the Courtroom strong-field branch and SAMA chapters cited below. The successor also retains the authenticated G11555 code, all new derivations and portable checks, and a separate immutable native-evidence companion. The new native run uses six controllers and three helpers over two original 180-second admission windows. Source snapshots and historical receipts preserve their original scope. Detailed replay commands, dependency versions and source/worker maps are included in the packet.

References

[1] Supplied Chapter V review. *What is now closed, what we can derive next*. Twelve-page review, editable source, checks and handoff, 16 October 2026. Review-derived analytical results are distinguished from the successor's new calculations.

[2] H. S. Reall. *Black Holes*. Cambridge Part III lecture notes, 3 March 2020, Sections 4.8–4.10, printed pp. 57–59. https://www.damtp.cam.ac.uk/user/hsr1000/black_holes_lectures_2020.pdf.

[3] *SAM Unified Technical Synthesis—Volume I: Substrate and Accumulation*. Supplied manuscript, strong-field Sections 4–4.4, printed pp. 5–6, and Appendix A.7, p. 12.

[4] Same authors. *Clocks, signals, and calibrated observation*. Combined Volume I, Chapter IV, physical source and local coupling development, 16 October 2026. Completed 45-page susceptibility continuation; its retained clock certificates and conditional receiving laws retain their original evidence boundaries.

[5] SAM Research Project. Courtroom strong-field branch, revision b5e914f71377e86ef4c67e199973d9300795cda1: CR007 landmark selector, CR008 clock/traversal closure, CR009 photon-sphere contact, CR010 ISCO contact and CR011 support zipper. https://github.com/iwtbotiwtwot/The_Courtroom/tree/b5e914f71377e86ef4c67e199973d9300795cda1/05_STRONG_FIELD_AND_HORIZON_CLOSURE.

[6] SAM Research Project. SAMA strong-field chapters and G11555 reference index, revision 4c2b5dc1649da5900c90b90d0ce56c223dbb7717. https://github.com/iwtbotiwtwot/SAMA/tree/4c2b5dc1649da5900c90b90d0ce56c223dbb7717.

[7] M. D. Kruskal. Maximal Extension of Schwarzschild Metric. *Physical Review* **119**, 1743–1745 (1960). https://doi.org/10.1103/PhysRev.119.1743.

[8] D. Tong. *General Relativity*. University of Cambridge lecture notes, Sections 1.3 and 6.1; geodesics and regular horizon coordinates. https://www.damtp.cam.ac.uk/user/tong/gr/grhtml/S1.html; https://www.damtp.cam.ac.uk/user/tong/gr/grhtml/S6.html.

[9] S. A. Hayward. Gravitational energy in spherical symmetry. *Physical Review D* **53**, 1938–1949 (1996). https://doi.org/10.1103/PhysRevD.53.1938; https://arxiv.org/abs/gr-qc/9408002.

[10] S. A. Hayward. General laws of black-hole dynamics. *Physical Review D* **49**, 6467–6474 (1994). doi:10.1103/PhysRevD.49.6467.

[11] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics* **43**, 199–220 (1975). https://doi.org/10.1007/BF02345020.

[12] Event Horizon Telescope Collaboration. First M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole. *Astrophysical Journal Letters* **875**, L6 (2019). https://doi.org/10.3847/2041-8213/ab1141; https://arxiv.org/abs/1906.11243.

[13] S. E. Gralla, D. E. Holz and R. M. Wald. Black hole shadows, photon rings, and lensing rings. *Physical Review D* **100**, 024018 (2019). https://doi.org/10.1103/PhysRevD.100.024018; https://arxiv.org/abs/1906.00873.

[14] SAM Research Project. *Starbreaker HOME-QW1 thermal boundary-Write kernel*. History H000848, 2 September 2026; recovered `qw1/kernel.py`, README and technical map. Source snapshots and direct replay accompany this chapter.

[15] SAM Research Project. *Completed-Home collapse lifecycle, exterior balance and radius floor*. History H000797, 31 August 2026; retained in the current RH and Starbreaker live records.

[16] C. W. Gardiner and M. J. Collett. Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation. *Physical Review A* **31**, 3761–3774 (1985). https://doi.org/10.1103/PhysRevA.31.3761.

SAM VOLUME I

Quantum routes, shared fields and retained physical records

Chapter VI

Abstract. Coherent alternatives, physical correlation and a readable record are distinct objects. We integrate the original Chapter VI with its completed continuum, source-decoder and detector/memory supplements. The full original source, signed currents, even energy, two field polarizations, both counterterms, hard dressing, soft input and angular complements are retained. A new driven receiving construction places two prescribed moving source arms in one continuum Fock field. Its moving-frame generator includes field momentum, admits common-form dynamics, and gives an energy-based error relative to the isolated proper-age reference. A closed-arm experiment with a finite, paid coherent source encoding joins that dynamics to the actual local detector and retained record. New exact six-lane arithmetic gives ready error below 0.0016 under the stated budgets. This encoded example is separate from an unforced gravitational age signal. Multi-history coherence, repeated-read error, all outcomes, latency and work are carried explicitly. Born probabilities, unitarity and instruments remain receiving premises; microscopic routing/actuator calibration and a distant intrinsic source port remain named physical assignments.

Reading map. The chapter follows phase and correlation, readable records, restoration, the original source clock, its source-only decoder, the actual detector, and the new shared-field/coherent join. The prior supplement statements are integrated as mathematical sections, not restarted campaigns. Their historical provenance and original receiving assumptions remain explicit. New constructions and finite parameter choices in this chapter are developed in campaigns voli_chvi_foundation1, voli_chvi_source_decoder1, voli_chvi_detector_record1 and voli_chvi_shared_routes1 (17 October 2026; H002700, H002701, H002706), with finite matrices checked by native SLC-GEN5-R1 calls. Provenance trace: Courtroom II case voli_provenance/01.

1. A route can carry phase before it carries a record

A detector need not be watched for it to change an experiment. A physical interaction can correlate a route with another degree of freedom, and that correlation can change the interference seen when the route is recombined. The further question is whether the correlation remains readable, by which apparatus, and for how long. These are physical questions about a carrier and its dynamics, not about an observer's awareness.

Unified Volume I calls the transition to a durable accessible record a substrate write.[1] Its quantum section retains the Born rule and action phase, while deferring the detailed write algebra to the computational volumes. The present chapter expands that interface. It does not infer an amplitude evolution from a route label or turn an accessibility convention into a new collapse law.

Chapters II, III and VII provide an explicit interacting source, its clock current, and a source-to-field interface. Chapter IV separates local clock evolution from signal transport. Chapter V distinguishes a traveler's history from what a receiver can obtain and from what a prescribed boundary allows.[2, 3, 4, 5, 6] The quantum continuation asks what happens when alternatives remain coherent through part of that chain.

Three questions guide the calculation:

1. What phase and amplitude does a declared dynamics assign to each coherent alternative?
2. Which physical degrees of freedom become correlated with those alternatives, and how does that change interference?
3. What record can a specified domain actually decode, retain and transmit?

A free photon is not unwritten because it is massless. It can carry energy, momentum and phase without yet leaving the kind of persistent record being discussed. Nor does an inaccessible correlation stop affecting a reduced quantum state. Accessibility concerns which records a domain can use; the full joint evolution still contains correlations outside that domain.

For an orthonormal set of exclusive route-readout alternatives, normalized amplitudes give

$$P(j) = \frac{|a_j|^2}{\sum_k |a_k|^2}. \tag{1}$$

This is the retained Born premise.[11] Coherent alternatives reaching the same complete final outcome contribute to a common amplitude before its squared magnitude is taken. Arbitrary decompositions of one state do not define additional mutually exclusive outcomes. After a record forms, subsequent and conditional measurements still obey the same probability rule.

2. What the recovered source record establishes

The recovered CR075 package reports a scoped structural ledger-write pass. Its runner inspects source text, manifests, disclosures and named bridge entries. The retained QP reports supply constructed route tables and threshold selectors. Their presence is evidence of a specified computational interpretation, not a derivation of arbitrary complex amplitudes or a physical detector dynamics.[7]

This distinction matters for the probability account. QP014 normalizes constructed capacity, coherence-survival and access weights. It does not establish that these weights equal $|a_j|^2$ for arbitrary quantum preparations. QP001, QP010 and QP016 likewise do not establish universal physical decoherence or outcome thresholds. This chapter therefore uses their route/write

distinction while deriving its interference calculations from explicitly stated quantum laws.

The later exact-write sources supply reversible finite address updates and separately retained history data. A signed update followed by its inverse can restore an address while a retained event list remains different. That is a useful exact algebraic example of restoration without deletion. It is not yet a microscopic measurement interaction.[8]

The lifecycle source also keeps an important boundary open: the maps from its computational `A_OPERATOR` to contact and to `X1` are not supplied as a completed physical law. That typed operator must not be silently identified with the scalar accumulation coordinate $A(r)$ or with a Hamiltonian coupling.[9] The task here is to state what such a bridge would need to reproduce.

All recovered repository bytes are pinned and authenticated in the companion package. The complete CR075 directory was inspected; its original 242-input upstream environment was not recovered or rerun. Historical reported control flags are not recast as newly executed perturbed quantum experiments. The derivations and checks below are a new independent receiving calculation, not an upgrade of those historical receipts.

3. Three phases and one consistent action account

3.1 A signed Write integral is not a complete history

The original lifted source phases define

$$\begin{aligned} \gamma_{\mathrm{W}} &= \phi_a + \phi_b - 2\phi_p, \\ \chi &= \phi_p, \end{aligned} \tag{2}$$

with the legal-path one-form

$$\omega_{\mathrm{Write}} = \hbar(\mathrm{d}\gamma_{\mathrm{W}} + 3\mathrm{d}\chi). \tag{3}$$

Consequently its integrated signed action is

$$S_{\mathrm{Write}} = \hbar(\Delta\gamma_{\mathrm{W}} + 3\Delta\chi). \tag{4}$$

The lifted endpoints, wrapped phases, ordered event word and elapsed duration preserve different information. Chapter VII's retained histories can have equal endpoint and equal signed action while retaining different positive durations and event orders.[2] An exact differential does not reconstruct an ordered word from its integral.

The interacting phase $\theta = -\arg\langle Z\rangle$ is a separate observable coordinate. An interferometer's relative phase is another operand, obtained from its full controlled amplitude. No equality among γ_{W}, θ and the observed fringe phase follows simply because all three are called phases. Their identification requires an explicit coupling and readout.

3.2 An explicit massive-route comparison

For a declared test-body action, the massive point contribution is

$$S_{\mathrm{mass},j} = -m_0c^2 \int_{\gamma_j} \mathrm{d}\tau. \tag{5}$$

Here m_0c^2 must be a part of the energy not already counted in the internal generator. In the weak static comparison with $F = 1 - A$ to first order, $A \ll 1$, and $v^2/c^2 \ll 1$,

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = 1 - \frac{A}{2} - \frac{v^2}{2c^2} + O(A^2, Av^2/c^2, v^4/c^4). \tag{6}$$

For two routes with the same coordinate duration, the common rest phase cancels and

$$\Delta\varphi_{\mathrm{mass}} = \frac{m_0}{2\hbar} \int \left[c^2(A_1 - A_0) + (v_1^2 - v_0^2)\right] \mathrm{d}t + \text{higher orders}. \tag{7}$$

This is the ordinary phase of kinetic energy minus Newtonian potential energy with $\Phi = -c^2A/2$. It is conditional on the stated action and geometry, not a derivation of an A-dependent quantum law from the source grammar.

An interferometer also needs the phases of its preparation, splitting, routing and recombination operations, with recoil and every record retained by those controls. A comparison of two held clocks is not automatically a coherent recombination experiment. For static segments of duration T, the proper-age difference is $T(\sqrt{F_1} - \sqrt{F_0})$ under the declared metric; the apparatus must still make both alternatives and their interference available.

Equation (5) is not a photon phase prescription: a null route has zero proper time. A photon requires its wave equation and boundary-value or eikonal phase, with the source and receiver controls included. The earlier photon-road delay does not by itself define a quantum-state amplitude.

4. Quantum routes and physical records

Evidence status. The calculations below are consequences of standard finite-dimensional quantum mechanics. Born probabilities, Hilbert-space composition, and quantum instruments are premises. The SAM contribution at this stage is a proposed interpretation and a specification of what a substrate write must implement, not a new probability law or an experimentally distinct prediction.

The route language of Unified Volume I, §5, separates an unresolved alternative from a physical record. Its useful content becomes testable only after the record carrier, decoder, accessible domain, and retention interval are declared. A route can carry phase without leaving a readable mark. Conversely, correlations can suppress local interference even when no observer can read their remote carrier. Accessibility therefore qualifies the *record available to a domain*; it does not switch ordinary quantum entanglement or decoherence on and off.

4.1 A controlled marker and its observable fringe

Let S be a two-route system with orthogonal states $|0\rangle, |1\rangle$, and let M be a marker initially in $|m\rangle$. Write $q = 1 - p$, $0 \le p \le 1$, and prepare

$$|\psi\rangle_S = \sqrt{p}\,|0\rangle + e^{i\phi}\sqrt{q}\,|1\rangle. \tag{8}$$

A path-preserving marker coupling has the form

$$\begin{aligned} U &= |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1, \\ |m_j\rangle &= U_j|m\rangle. \end{aligned} \tag{9}$$

The joint state and overlap convention are

$$|\Psi\rangle = \sqrt{p}\,|0\rangle|m_0\rangle + e^{i\phi}\sqrt{q}\,|1\rangle|m_1\rangle, \tag{10}$$
$$\gamma = \langle m_1|m_0\rangle. \tag{11}$$

Tracing out M, rather than assuming a collapse, gives

$$\rho_S = \begin{pmatrix} p & \sqrt{pq}\,e^{-i\phi}\gamma \\ \sqrt{pq}\,e^{i\phi}\gamma^* & q \end{pmatrix}. \tag{12}$$

For a balanced output analyser with projectors onto $|\pm_\theta\rangle = (|0\rangle \pm e^{i\theta}|1\rangle)/\sqrt{2}$, the Born rule yields

$$P_\pm(\theta) = \tfrac{1}{2} \pm \sqrt{pq}\,\mathrm{Re}[e^{i(\theta-\phi)}\gamma]. \tag{13}$$

Thus the scanned single-port visibility is

$$V = \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}} = 2\sqrt{pq}\,|\gamma|. \tag{14}$$

These expressions assume a lossless two-route preparation, a balanced analyser, a phase scan, and no outcome-dependent filtering. Unequal route populations already limit V without a marker. A global phase in the overlap shifts the fringe; its magnitude controls the contrast. A change in contrast is consequently not, by itself, proof of a durable record.

The action-phase interface remains the ordinary one: for two coherent routes, $\Delta\phi = (S_1 - S_0)/\hbar$. An A-dependent action must specify its Lagrangian, integration parameter, and units before it predicts a fringe shift. The schematic symbol L_A in the source text supplies no independent microscopic Hamiltonian and does not derive the Born rule. In particular, probabilities continue to govern subsequent measurements after a record has formed; "before the write" describes the proposed accounting language, not a restriction on the applicability of quantum probability.

4.2 Distinguishability, with the priors retained

Consider the operational task of guessing a route from its marker, given priors p, q. For any binary POVM E_0, E_1 on M,

$$P_{\text{guess}} = p\,\mathrm{Tr}(E_0\rho_0) + q\,\mathrm{Tr}(E_1\rho_1), \qquad \rho_j = |m_j\rangle\langle m_j|. \tag{15}$$

Set $H = p\rho_0 - q\rho_1$. Since $E_1 = I - E_0$, maximizing $q + \mathrm{Tr}(E_0 H)$ selects the positive spectral subspace of H. Hence

$$P_{\text{guess}}^{\text{opt}} = \tfrac{1}{2}(1 + D), \qquad D = \|p\rho_0 - q\rho_1\|_1. \tag{16}$$

This is weighted Helstrom distinguishability,[15] not the unweighted trace distance unless $p = q = 1/2$. On the span of the two pure markers, $\mathrm{Tr}\,H = p - q$ and $\det H = -pq(1 - |\gamma|^2)$. The two eigenvalues have opposite signs or one is zero, so

$$D = \sqrt{1 - 4pq|\gamma|^2}, \qquad D^2 + V^2 = 1. \tag{17}$$

Identical marker states give $D = |p - q|$: success then comes only from the prior imbalance. Orthogonal markers give $D = 1$, $V = 0$. Equation (17) is an equality for this pure joint state with full marker access and the specified analyser. It is not an unrestricted equality for every apparatus. It belongs to the established two-way duality framework; see Englert.[12]

There is a short general proof of the relevant inequality. Let ρ_{SM} be any joint state and purify it with E. For $0 < p < 1$ the purification can be written

$$|\Omega\rangle = \sqrt{p}\,|0\rangle|u\rangle_{ME} + \sqrt{q}\,|1\rangle|v\rangle_{ME}, \tag{18}$$

with normalized conditional vectors. If $c = \langle 0|\rho_S|1\rangle$, then $c = \sqrt{pq}\langle v|u\rangle$ and the same analyser gives $V = 2|c|$. Weighted pure-state discrimination on ME gives $D_{ME} = \sqrt{1 - 4|c|^2}$. Partial trace is a trace-norm contraction on Hermitian operators, yielding

$$D_M \le D_{ME}, \qquad D_M^2 + V^2 \le 1. \tag{19}$$

The endpoints $p = 0, 1$ follow directly. This proof also covers a restricted accessible subsystem of the marker. It does not license comparing visibility in one postselected ensemble with distinguishability in another. These are the standard channel and discrimination conventions.[10]

4.3 Reversal, erasure, and inaccessible markers

An exact reversible two-qubit fixture uses $|m\rangle = |0\rangle$, $U_0 = I$, and

$$U_1 = R(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}. \tag{20}$$

Then $\gamma = \cos\alpha$. At $p = 1/2$, $V = |\cos\alpha|$ and $D = |\sin\alpha|$. At $\alpha = \pi/2$ the marker is orthogonal and the reduced route state has no coherence. Applying the *joint controlled inverse* $U^\dagger$ before analysis restores $|\psi\rangle|0\rangle$ and the original visibility. This is a reversal of the correlation-generating interaction. A marker-only unitary cannot accomplish it, since it preserves $\langle m_1|m_0\rangle$.

A quantum eraser is a different operation. For the orthogonal marker, measure M in $|\pm\rangle_M = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Each outcome has probability $1/2$ and prepares the conditional route state

$$|\psi_\pm\rangle = \sqrt{p}\,|0\rangle \pm e^{i\phi}\sqrt{q}\,|1\rangle. \tag{21}$$

Its fringe visibility is $2\sqrt{pq}$, but the two fringes are shifted by π. Averaging their distributions cancels the interference. Conditional coincidence sorting requires the marker outcome; it does not restore an unconditional remote fringe. Outcome-dependent feed-forward can align the phases, but uses a communicated outcome and an operation on S. This distinction is central to the original quantum-eraser proposal of Scully and Drühl and to conditional coincidence experiments.[13, 14]

Environmental copies constrain reversal. For branch states of the factorized form $|m_j\rangle|e_j\rangle$, the coherence multiplier is

$$\gamma_{ME} = \langle m_1|m_0\rangle\langle e_1|e_0\rangle. \tag{22}$$

Reversing the controlled S–M coupling while leaving E intact does not remove the environmental factor. Orthogonal $|e_j\rangle$ maintain zero local coherence even when E is inaccessible. Generic environmental states need not factorize; the purification

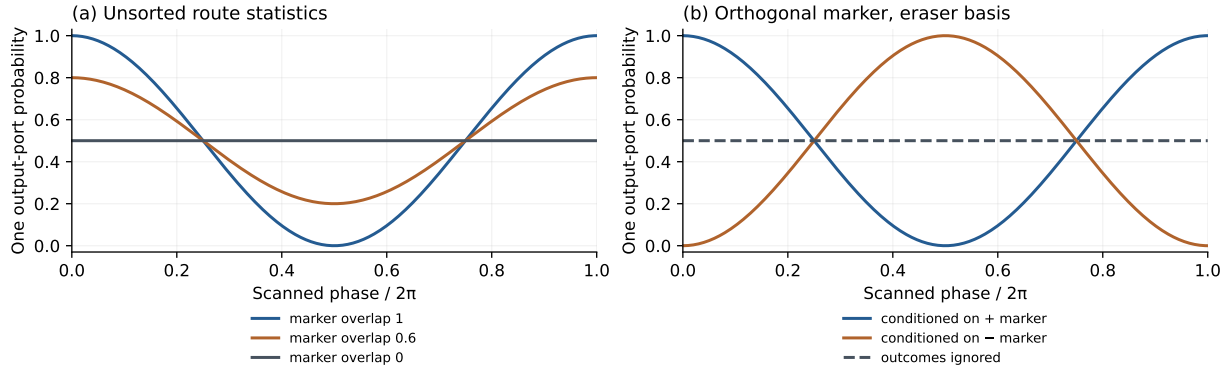


Figure 1: Exact balanced two-route comparator. (a) Real marker overlap changes unsorted interference through $P_+ = (1+\gamma\cos\varphi)/2$. (b) For orthogonal markers, measuring the complementary marker basis gives two conditional fringes, each with probability one half. Ignoring that outcome yields the unchanged flat marginal. These analytic finite-system curves illustrate the stated protocol; they are not measured SAM data or an original-field simulation.

argument above replaces this product formula in that case. No assertion about a gravitational horizon follows merely from tracing out E.

More generally, let Λ_M be any trace-preserving local channel, including an unread measurement or reset. For every bipartite state,

$$\rho_S' = \mathrm{Tr}_M[(I\otimes\Lambda_M)(\rho_{SM})] = \rho_S, \tag{23}$$

$$T(\rho_S',\rho_S) = \tfrac{1}{2}\|\rho_S'-\rho_S\|_1 = 0. \tag{24}$$

The equality follows from $\Lambda_M^*(I) = I$: every local expectation on S is unchanged. Thus no route-only POVM can reveal the remote channel choice. Selective outcomes can change a conditional state, but the selection information must be transmitted. This is the precise no-signalling control on any proposed accessibility or erasure narrative.

4.4 A durable accessible record is an operational claim

A useful record definition specifies more than entanglement. Let $\{\mathscr{I}_j\}$ be a quantum instrument: each map is completely positive and trace non-increasing, and $\sum_j \mathscr{I}_j$ is trace preserving. Its probabilities are $p_j = \mathrm{Tr}\,\mathscr{I}_j(\rho)$, and its conditional output is $\mathscr{I}_j(\rho)/p_j$ for $p_j > 0$. The instrument can describe a declared path-marking task, but must not be read as establishing a pre-existing route value in every coherent preparation. If only selected outcomes are retained, the omitted outcomes must remain in the complete instrument; renormalizing the retained branch describes postselection, not a new unconditional probability law.

For a proposed memory R, declare (i) the operational labels j, (ii) the conditional memory states σ_R^j, (iii) the storage dynamics $\mathscr{N}_t$, (iv) the channel $\mathscr{A}_{\mathscr{D},t}$ delivering data to domain $\mathscr{D}$, and (v) an allowed decoder POVM $\{F_j(t)\}$. Call it an (ε,T) readable record for that domain when a predeclared physically implementable decoder satisfies

$$\inf_{0\le t\le T}\ \min_{j:p_j>0}\ \mathrm{Tr}\Big[F_j(t)\mathscr{A}_{\mathscr{D},t}\mathscr{N}_t(\sigma_R^j)\Big] \ge 1-\varepsilon. \tag{25}$$

Here $0 \le \varepsilon < 1/2$ for a binary record. The minimum over labels prevents a strongly biased source from passing by always guessing the common label. The time origin is the end of writing, and the interval tests one read performed at any chosen time; repeated non-destructive reading requires an additional back-action specification. Fixed decoders are a valid special case. Access may incorporate transport delay, controllable operations, or a restricted measurement algebra; the protocol must state which. Without those declarations, "accessible" is not a quantitative physical property.

Four different statements must remain separate. Entanglement concerns the joint state. Decoherence concerns suppressed off-diagonal terms in a specified reduced state and basis. A single registered outcome is described operationally by an instrument and the Born rule. A persistent readable record additionally passes Eq. (25). Unitary correlation and partial trace alone do not select one unique global outcome or solve the measurement problem.[16] Nor does a record require awareness: every element of the definition is a physical preparation, channel, or decoder.

4.5 Controls and the boundary of the claim

The preceding equations prescribe concrete wrong-controls. Treating a marker-only rotation as a joint inverse incorrectly predicts an unconditional restored fringe. Tracing out an orthogonal environmental marker and then declaring it absent incorrectly reinstates coherence. Sorting eraser outcomes and comparing that conditional visibility with unsorted which-way data mixes ensembles. Treating D as marker-acquired information without subtracting or reporting prior predictability misreads biased preparations. Treating a returned local state as a globally erased event discards the history carrier. Finally, labelling a reduced density matrix as one selected outcome imports an additional postulate without saying so.

These controls are standard quantum benchmarks. They discipline the route/write terminology and identify what an executable substrate account must reproduce. No new SAM prediction follows from renaming the instrument outcome a write. Such a prediction would require an additional, quantitatively specified dynamics or access rule, its parameter domain, and an experiment separating it from the retained quantum

premises.

5. Readable records and retained route coherence

The following statements use the quantum state, channel, and measurement premises already declared in Chapter VI. They make the relation between a readable record and an unsorted interference experiment quantitative. The record labels must refer to the same routes and the same ensemble as the visibility; a reliable record of an unrelated label supplies no such bound.

5.1 A same-state, worst-label theorem

Let S be the two-route system and let M contain the remaining physical marker degrees of freedom. At the chosen observation time write an arbitrary normalized joint state as

$$\begin{aligned} \rho_{SM} &= \sum_{j,k=0}^{1} |j\rangle\langle k| \otimes R_{jk}, \\ p = \mathrm{Tr} R_{00}, &\quad q = \mathrm{Tr} R_{11} = 1-p. \end{aligned} \tag{26}$$

For $p,q>0$, the conditional marker states are $\sigma_M^0 = R_{00}/p$ and $\sigma_M^1 = R_{11}/q$. Their labels specify the two diagonal route alternatives used by this operational test; this notation does not select a unique outcome in a coherent joint state.

Declare a label-independent channel $\Lambda_t : M \longrightarrow D$ that includes the storage and access operations available to the reading domain at time t. It is completely positive and trace preserving, and acts on the marker rather than on S. Let $F_0(t), F_1(t)$ be one physically allowed decoder POVM on D, with $F_0 + F_1 = I_D$. The decoder may depend on the declared time and controls, but receives no copy of the unknown label. Define

$$\begin{aligned} \tau_D^j(t) &= \Lambda_t(\sigma_M^j), \\ r_j(t) &= \mathrm{Tr}[F_j(t)\tau_D^j(t)]. \end{aligned} \tag{27}$$

Storage and access can instead be specified jointly on an enlarged marker containing a correlated environment, a transport carrier, and retained classical records. No product-state or independent-noise assumption is required. An environment correlated with the marker must be included in that joint channel description when the proposed reduced channel would otherwise depend on its correlations. Conditional states for the theorem are always the actual conditional states of Eq. (26) under that description.

Let $c = \langle 0|\rho_S|1\rangle$. For a lossless two-route preparation and the declared balanced, variable-phase analyser, with no outcome-dependent filtering, the unsorted single-port visibility is

$$V = 2|c|. \tag{28}$$

The local, unselected channel Λ_t does not change ρ_S. Consequently a decoder can be analysed on D while the visibility refers to this same route marginal. A joint controlled inverse or feed-forward operation on S would define a different experiment and must be included before applying the theorem to its resulting state.

Theorem. If, at the chosen time, both conditional decoder successes satisfy

$$\begin{aligned} \min\{r_0(t), r_1(t)\} &\geq 1-\varepsilon, \\ 0 \leq \varepsilon &< \tfrac{1}{2}, \end{aligned} \tag{29}$$

then

$$\boxed{V \leq 4\sqrt{pq\,\varepsilon(1-\varepsilon)} \leq 2\sqrt{\varepsilon(1-\varepsilon)}.} \tag{30}$$

For balanced priors the two bounds agree. For unequal priors the first retains their additional restriction on the fringe. At $p=0$ or $q=0$, positivity already gives $c=0$ and $V=0$; a conditional success statement about an absent label must be supplied by a separately declared calibration preparation if it is wanted.

Derivation. Apply the decoder to $(I_S \otimes \Lambda_t)(\rho_{SM})$ and retain its two outcomes. The unnormalized route state for outcome y is positive, with diagonal entries pa_y, qb_y, where

$$\begin{aligned} a_y &= \mathrm{Tr}[F_y \tau_D^0], \\ b_y &= \mathrm{Tr}[F_y \tau_D^1]. \end{aligned} \tag{31}$$

If its off-diagonal entry is c_y, positivity of this 2×2 matrix and preservation of the unselected marginal give

$$\begin{aligned} |c_y| &\leq \sqrt{pq\,a_y b_y}, \\ c &= \sum_y c_y, \\ |c| &\leq \sqrt{pq}\sum_y \sqrt{a_y b_y}. \end{aligned} \tag{32}$$

This argument retains arbitrary off-diagonal marker blocks and arbitrary correlations with inaccessible degrees of freedom.

Put $a = a_1$ and $b = b_0$, the two conditional error probabilities. The readability hypothesis gives $0 \leq a, b \leq \varepsilon$, and the last sum in Eq. (32) is

$$\sqrt{(1-a)b} + \sqrt{a(1-b)}. \tag{33}$$

Write $a = \sin^2\alpha$, $b = \sin^2\beta$, with $0 \leq \alpha, \beta \leq \arcsin\sqrt{\varepsilon} < \pi/4$. This sum is $\sin(\alpha+\beta)$, hence it is at most $2\sqrt{\varepsilon(1-\varepsilon)}$. Multiplying by $2\sqrt{pq}$ gives Eq. (30), including $\varepsilon = 0$ by continuity or direct substitution.

The weaker bound also follows immediately from weighted discrimination: the stated decoder has prior-weighted success at least $1-\varepsilon$, so $D_D = \|p\tau_D^0 - q\tau_D^1\|_1 \geq 1-2\varepsilon$. The same-state inequality $D_D^2 + V^2 \leq 1$ then gives $V \leq 2\sqrt{\varepsilon(1-\varepsilon)}$. The Bhattacharyya derivation uses both separate error constraints and therefore retains the stronger prior-dependent factor. No optimal decoder needs to be physically available: the actual allowed decoder in Eq. (29) is enough.

Tight qubit example. For any $0 \leq \varepsilon < 1/2$, take

$$|m_0\rangle = \sqrt{1-\varepsilon}|0\rangle + \sqrt{\varepsilon}|1\rangle, \tag{34}$$

$$|m_1\rangle = \sqrt{\varepsilon}|0\rangle + \sqrt{1-\varepsilon}|1\rangle, \tag{35}$$

$$|\Psi\rangle = \sqrt{p}\,|0\rangle|m_0\rangle + e^{i\phi}\sqrt{q}\,|1\rangle|m_1\rangle. \tag{36}$$

The computational-basis decoder succeeds with probability $1-\varepsilon$ on each label, and $\langle m_1|m_0\rangle = 2\sqrt{\varepsilon(1-\varepsilon)}$. Its visibility saturates the first bound in Eq. (30) for every p,q. For $\varepsilon = 0$ the marker is perfectly readable and the route marginal has zero visibility, while the pure joint state still retains its coherent superposition. A later joint reversal can recover interference while changing the record. The theorem bounds the specified unsorted fringe; it does not identify loss of its contrast with deletion of global coherence.

5.2 Arrival, retention, and the time of reading

An (ε, W_D) readable record for domain D requires the worst-label condition at every allowed reading time $t \in W_D$, with the channel, allowed decoder, and time origin declared. The theorem then bounds the contemporaneous visibility at every such time for which the same route/marker description applies. The statement concerns one read at a chosen time. A sequence of reads requires the actual reading instruments and their disturbance; the interval condition alone supplies no nondestructive repeated-read guarantee.

There is a precise causal pre-arrival restriction. Suppose a newly encoded label has no label-dependent receiver state in its causal past and the declared channel has not delivered any label-dependent degree of freedom by time t. Then

$$\tau_D^0(t) = \tau_D^1(t) = \tau_D(t). \tag{37}$$

Every binary decoder obeys $r_0 + r_1 = 1$ on this common state, so

$$\min_j r_j(t) \le \tfrac{1}{2}. \tag{38}$$

Thus it cannot satisfy the declared $\varepsilon < 1/2$ readability before arrival. This conclusion needs the explicit identical-state premise, provided physically by the causal/no-signalling access model. A receiver that already holds a correlated copy of the label is a different resource configuration. Unequal priors can improve an average guess before arrival, but cannot pass the worst-label condition.

The storage window and receiver window are consequently different objects. Let the end of writing be $t = 0$, and suppose the stored encoding is guaranteed at the memory for emission times $t_e \in [0, T_M]$. For a fixed latency τ and a separately budgeted readout time τ_{dec}, the corresponding idealized completion times are

$$\begin{aligned} t_D &= t_e + \tau + \tau_{\mathrm{dec}}, \\ W_D &= [\tau + \tau_{\mathrm{dec}},\, T_M + \tau + \tau_{\mathrm{dec}}]. \end{aligned} \tag{39}$$

This translated window assumes that the access channel actually samples the memory at t_e and that its transport, intervening storage, and decoder errors have been paid. A protocol needing the memory throughout a finite read instead requires that entire memory-side interval inside its retention window. Variable travel times use the declared retarded emission map $t_e(t_D)$ and its admissible interval. Starting a receiver guarantee at the end of writing would incorrectly include times before delivery unless delivery was already available. Choosing the domain time origin at arrival is valid when that convention is stated.

5.3 Composable error budgets and retained failed preparations

Worst-label budgets compose without independent errors. One convenient quantum formulation includes all classical outcome flags and relevant environmental references in the joint state. Let the ideal decoder have error at most ε_0 on each label. Suppose a hybrid comparison of the actual preparation, storage, transport, and readout gives trace-distance errors δ_k on the relevant joint conditional states at stage k. Each stage bound must hold for the hybrid inputs actually used, including retained correlations; an entangled reference is included in the comparison when present. Equivalently, a uniform channel bound can be supplied on that declared joint input class.

Contractivity under the subsequent trace-preserving channels and the triangle inequality give a final joint-state distance at most $\sum_k \delta_k$. A fixed POVM outcome probability changes by no more than that distance. Consequently

$$\begin{aligned} r_j^{\mathrm{actual}}(t) &\ge 1 - \varepsilon_0 - \sum_k \delta_k, \\ \varepsilon_{\mathrm{tot}} &= \varepsilon_0 + \sum_k \delta_k. \end{aligned} \tag{40}$$

The readability and visibility theorem applies when the declared total is below $1/2$. Isolated marginal error estimates cannot be substituted for these joint-state bounds when unaccounted correlations change the channel. The hybrid channels and flags also preserve any feedback used by the actual instrument.

A classical alternative is an explicit sequence of bad-event flags whose conditional probability, on every reached prior history and each label, is at most δ_k. If the final decoder error on the all-good branch is at most ε_0, the union bound gives the same additive budget. This argument likewise needs no independence. It does need the stated conditional bounds; adding unrelated average failure rates would not establish a worst-label guarantee.

Preparation failure belongs in the full instrument. Suppose label j has failure probability $f_j \le f$, and the successful preparation has worst-label reading error at most $\varepsilon_{\mathrm{succ}}$. If an explicit failure or abstention is counted as an unsuccessful attempted read, then

$$r_j^{\mathrm{attempt}} \ge (1 - f_j)(1 - \varepsilon_{\mathrm{succ}}) \ge 1 - f - \varepsilon_{\mathrm{succ}}. \tag{41}$$

Success, failure, and abstention flags remain in the normalized all-outcome state and in the physical resource account. A larger decoder alphabet can be retained explicitly, or its failure outcomes can be assigned a stated binary guess without discarding them.

Alternatively a claim may be conditioned on a heralded success. Its normalization, conditional record states, and conditional visibility must then all use that same accepted ensemble. In a route-preserving heralded preparation the accepted priors become

$$\begin{aligned} p_{\mathrm{acc}} &= \frac{p(1-f_0)}{p(1-f_0) + q(1-f_1)}, \\ q_{\mathrm{acc}} &= 1 - p_{\mathrm{acc}}, \end{aligned} \tag{42}$$

when the denominator is positive. Outcome-dependent filtering can also change the coherence, so the accepted joint state itself

must be used. A conditional high-visibility ensemble cannot be compared with a readable record certified only in a different, unselected ensemble. No reset of failed preparations or deletion of their records is supplied by this probability accounting.

5.4 A finite readable-label capacity consequence

For N declared labels, suppose the full accessible record input has finite Hilbert dimension d and one POVM $\{F_j\}_{j=1}^N$ satisfies $\mathrm{Tr}(F_j\sigma_j) \geq 1-\varepsilon$ for every normalized conditional state σ_j. Since $0 \leq \sigma_j \leq I$,

$$N(1-\varepsilon) \leq \sum_{j=1}^{N} \mathrm{Tr}(F_j\sigma_j) \leq \sum_{j=1}^{N} \mathrm{Tr}\,F_j = d. \tag{43}$$

Thus $d \geq \lceil N(1-\varepsilon)\rceil$. Any correlated side carrier available to the decoder belongs in that accessible dimension; alternatively the decoder is pulled back to a POVM on the declared input space. This bound concerns N physically decodable labels. Merely naming additional ordered histories does not add distinguishable record states. It assigns neither an energy per label nor a thermal erasure cost.

6. Restoration of a local state and retention of a physical history

The source interface is

$$W8 + X1 \longrightarrow W9 \longrightarrow W8'. \tag{44}$$

Here $W8$ names the resting, non-contact local substrate state; $X1$ names the supplied interaction; and $W9$ names the temporary write state. The prime marks return after interaction, not a new substrate atom. The source's intended distinction is that a local state can return while an event remains recorded elsewhere. This section gives that distinction an explicit quantum-information comparator. It does not identify the symbols $W8$, $X1$, and $W9$ with Hilbert-space dimensions, derive their exact SAM/SLC grammar, or prove a substrate microdynamics.

Chapter II already distinguishes an address from the history of its use. Returning to a retained address does not by itself say whether a record survives. Nor does a finite alphabet supply unlimited storage: a physical history requires distinguishable states of some specified recipient. The same discipline applies here. We must name the restored subsystem, the other degrees of freedom, and the operations by which a record can be read.

6.1 What a unitary reset can and cannot do

Let L be a local record cell and let E include every other degree of freedom participating in a closed finite-dimensional evolution. Suppose $\{|m_i\rangle_L\}_{i=1}^n$ are orthonormal pointer states and E starts in a fixed pure state $|e_0\rangle$. An exact reset of L has the form

$$U\big(|m_i\rangle_L|e_0\rangle_E\big) = |0\rangle_L|e_i\rangle_E. \tag{45}$$

Reset–recipient proposition. If U is unitary, then $\langle e_i|e_j\rangle = \delta_{ij}$. Consequently E cannot also return to one fixed state when $n > 1$. The recipient needs at least n mutually orthogonal states.

Proof. Preservation of inner products gives

$$\delta_{ij} = \langle m_i, e_0 | m_j, e_0\rangle = \langle 0, e_i | 0, e_j\rangle = \langle e_i | e_j\rangle . \tag{46}$$

An exact pure local output factors from the rest, so this argument also covers a purported reset whose output was initially written as a general joint pure state. A mixed initial environment may be purified by adding its reference to E; the conclusion then concerns that enlarged recipient. It does not guarantee a record readable from an arbitrarily chosen proper part of E. □

There is an important scope condition. If a separate system S already contains a correlated copy, the input is not the fixed-environment input of Eq. (45). In the more general pure-state mapping

$$|s_i\rangle|m_i\rangle|e_0\rangle \longmapsto |s_i'\rangle|0\rangle|e_i\rangle, \tag{47}$$

unitarity requires, for $i \neq j$,

$$\langle s_i'|s_j'\rangle\langle e_i|e_j\rangle = 0. \tag{48}$$

If S returns to one common pure state, the recipient states must be orthogonal. If S instead retains orthogonal alternatives, it can itself carry the distinction; a new record in E is not compulsory. For example, reversing a controlled write resets the cell while retaining the source bit. The assertion that "every reset must write a new history register" would therefore be false. The invariant is distinguishability of the complete closed state, not the permanence of a selected local mark. For mixed or entangled subsystems, surviving information may be in joint correlations rather than readable in any one marginal.

6.2 Three registers: a write, a restoration, and a retained marker

Use three qubits: route S, local cell L, and history recipient H. The history register is initially blank. Define

$$\begin{aligned} C|r,\ell,h\rangle &= |r,\ell\oplus r,h\rangle,\\ T &= \mathrm{SWAP}_{LH}, \end{aligned} \tag{49}$$

where $r,\ell,h \in \{0,1\}$ and $\oplus$ is addition modulo two. Both maps permute the full eight-state computational basis, so both are unitary without a restriction to the intended input states. Their composition is

$$\begin{aligned} U &= TC,\\ U|r,\ell,h\rangle &= |r,h,\ell\oplus r\rangle. \end{aligned} \tag{50}$$

For a normalized coherent route input,

$$\begin{aligned} |\Psi_{\mathrm{in}}\rangle &= (\alpha|0\rangle+\beta|1\rangle)_S|0\rangle_L|0\rangle_H,\\ |\Psi_{\mathrm{write}}\rangle &= \alpha|000\rangle+\beta|110\rangle, \end{aligned} \tag{51}$$

$$\begin{aligned} |\Psi_{\mathrm{restored}}\rangle &= \alpha|000\rangle+\beta|101\rangle\\ &= |0\rangle_L\big(\alpha|00\rangle_{SH}+\beta|11\rangle_{SH}\big). \end{aligned} \tag{52}$$

The local cell is exactly restored. Conditioned on the route alternative, the history states are $|0\rangle_H$ and $|1\rangle_H$, which can be perfectly discriminated. For a coherent input, however, the final state is an entangled superposition, not a derivation of one selected outcome. The comparator exhibits a readable pointer correlation and local restoration. It does not derive collapse or a unique realized history.

The analogy to Eq. (44) is deliberately limited: resting cell, interaction-enabled write, restored cell with a retained recipient correlation. The interaction $X1$ is not treated as an extra qubit

or an arithmetical increment. The temporary $W9$ designation may refer to the interaction configuration; it need not mean that the cell alone has changed in every branch. In particular, the $r = 0$ branch leaves its pointer value at zero even during the controlled write.

The inverse exists on every input: $U^{-1} = CT$. With joint control of S, L, H, it returns Eq. (52) to $|\Psi_{\text{in}}\rangle$. The phase information was never deleted from the complete state. By contrast, tracing out the record gives

$$\rho_S = |\alpha|^2 |0\rangle\langle 0| + |\beta|^2 |1\rangle\langle 1| \tag{53}$$

both after the write and after the restoration. Thus the swap does not restore route-only interference. Global coherence is preserved throughout; its observability depends on access to the correlated degrees of freedom. Joint reversal, or suitable measurements of the record with conditioned route statistics, can reveal that coherence. An operation on the inaccessible record alone cannot change the unconditioned route marginal.

For example, take $\alpha = 1/\sqrt{2}$ and $\beta = e^{i\phi}/\sqrt{2}$. The joint observable $X_S X_H$ has expectation $\cos\phi$ after restoration, whereas every route-only expectation is independent of ϕ. This is the precise distinction between preservation in the closed system and accessibility in a restricted domain.

This device stores one binary marker. It does not store an arbitrary sequence of writes. If H is already occupied, the swap returns that previous content to L, rather than resetting L to zero. Repeated restoration needs additional blank capacity, a deliberate reversible compression when possible, or an explicitly modeled disposal channel. Perfectly distinguishing N possible complete histories requires N orthogonal global record states and hence a record-space dimension at least N. This is a storage statement, not a heat formula.

6.3 Pointer copying is not universal quantum cloning

For a declared orthogonal pointer basis, a controlled operation can copy its label:

$$|r\rangle|0\rangle \longmapsto |r\rangle|r\rangle. \tag{54}$$

Applied to a superposition, it creates the correlated state in Eq. (51); it does not produce two copies of $\alpha|0\rangle + \beta|1\rangle$. Indeed, let the overlap of two normalized inputs be z and allow the cloner to finish in input-dependent auxiliary states $|a_\psi\rangle$ and $|a_\varphi\rangle$. Inner-product preservation would require $z = z^2\langle a_\psi|a_\varphi\rangle$. For $0 < |z| < 1$, this implies $1 \le |z|$, a contradiction. Universal cloning is therefore impossible even when the auxiliary state is allowed to change.

Transfer is different. A swap implements

$$|\psi\rangle_L|0\rangle_H \longmapsto |0\rangle_L|\psi\rangle_H \tag{55}$$

for every $|\psi\rangle$, preserving its entanglement with any reference. The input remains present once, at its new location. Resetting L while leaving both the recipient and all other degrees of freedom fixed would instead delete an unknown input, contrary to inner-product preservation. These elementary statements suffice for this interface; no universal copying or unexplained destruction of quantum information is being assumed. Related no-deleting and no-hiding results have their own hypotheses; no claim that information must reside only in correlations is used here.[19, 18]

6.4 A physical record needs an accessible decoder

The decoder criterion of Eq. (25) applies to the nominated recipient, not merely the restored cell. For equal priors, the three-register example has trace distance one between its conditional history states and zero between its conditional local-cell states. The history permits perfect route decoding; the restored cell permits only chance performance. The Helstrom optimum is attainable only if its measurement belongs to the allowed set.

Durability adds a time and noise specification. A useful requirement is that a stated decoder recover the label with error at most ε after a declared storage interval and channel. Orthogonality at the moment of writing does not by itself establish durability. For example, encoding a classical bit in an odd number n of identical pointer cells and reading by majority vote gives

$$P_{\text{err}} \le \exp\{-2n(\tfrac{1}{2} - p)^2\}, \qquad p < \tfrac{1}{2}, \tag{56}$$

if, conditional on the stored bit, the cell errors are independent and each has probability at most p. For error indicators E_j and $t = 1/2 - p > 0$, independence and the bounded-variable exponential-moment inequality give

$$\Pr\left\{ \sum_j (E_j - \mathbb{E}E_j) \ge nt \right\} \le \inf_{\lambda > 0} e^{-\lambda nt + n\lambda^2/8} = e^{-2nt^2}. \tag{57}$$

Majority failure is contained in the displayed event. Correlated failures need not satisfy it: one common flip of all cells leaves a majority decoder with the same error probability as a single cell. This example is an optional classical storage mechanism, not a derived SAM durability law.

Finally, a history description is not an additional quantum observable. If two proposed histories yield the same accessible density operator, every accessible POVM gives the same outcome probabilities. Calling one state $W8$ and the other $W8'$ cannot change this. If a history can be read, its evidence must reside in distinguishable accessible states or correlations, with a specified decoding operation. This is the physical content of the prime: local return is compatible with a retained record, but the notation alone neither guarantees that record nor makes it accessible.

No thermodynamic cost is inferred from these unitary maps. A Landauer bound belongs to a separately declared cyclic erasure protocol with a thermal reservoir, temperature, initial logical ensemble, and work/heat accounting. Neither the number of local addresses nor the logarithm of a record-space dimension is, by itself, dissipated heat. The present result is narrower: a local restoration can transfer a physical distinction; it cannot establish the disappearance of that distinction from a closed unitary account.

7. The original interacting clock on coherent routes

7.1 The full interacting source as a route marker

A route-dependent clock requires a receiving rule connecting the local interacting dynamics to spacetime transport. The following rule is conditional [17]: the same isolated, co-moving source–field Hamiltonian acts along each prescribed timelike route, with its Chapter VII time chart replaced by accumulated proper time. External beam splitters, recombination, routing forces, and the spacetime geometry are controls; they are not derived from the source Hamiltonian. Tidal changes of the internal Hamiltonian, radiation caused by the controls, and inaccessible environmental records require additional terms or a larger marker system. A field shared nonlocally by both arms does not automatically admit this isolated co-moving reduction.

Retain the complete source space $\mathbb{C}^4 \otimes \mathbb{C}^4$ and the two polarizations of the continuum Fock field. In the Chapter VII dimensionless chart the rest generator is

$$\begin{aligned} H_0 &= \varepsilon(S + Q_L + Q_R + Q_L Q_R) + H_f \\ &\quad - \varepsilon(D\phi_+ + C\phi_\times) \\ &\quad + \varepsilon^2 \mu (D^2 + C^2), \\ D &= Q_L - Q_R, \qquad C = iSD, \qquad \mu = (4R)^{-1}. \end{aligned} \tag{58}$$

This is the original interacting form-sum Hamiltonian, not a bare two-level substitution. Its six two-state lanes and four unforced diagonal states are all retained. In a chosen prepared lane $a < b$, $d = q_a - q_b < 0$, $g = \varepsilon d$, and the restriction is $H_0 = \varepsilon e I + \varepsilon S + H_f - g(\sigma_z \phi_+ + \sigma_y \phi_\times) + cI$, where $e = q_a + q_b + q_a q_b$ and $c = 2g^2\mu$. Both the even energy and the two counterterms remain in the generator.

For route j, define

$$\begin{aligned} \eta_j &= s_0 \int_{\gamma_j} d\tau_{\text{proper}}, \qquad U_j = e^{-iH_0\eta_j}, \\ U_{\text{ctrl}} &= \sum_{j=1}^{2} |j\rangle\langle j| \otimes U_j. \end{aligned} \tag{59}$$

Here $s_0 = 1/t_u$ in the declared Chapter VII chart; no new SI calibration is inferred. With $I_* = \hbar$, the physical rest energy is $(\hbar/t_u)H_0$. Conditional on this transport law, the actual clock coordinate obeys $d\theta/dt = \nu(\eta_j)(d\tau_{\text{proper}}/dt)/t_u$ wherever its certified inverse exists. This does not extend the Chapter VII finite-window rate certificate beyond its stated preparation and interval.

Let the route input be $\sqrt{p_1}|1\rangle + e^{i\alpha}\sqrt{p_2}|2\rangle$, $p_1 + p_2 = 1$, initially uncorrelated with a marker state ρ. Tracing the marker gives route coherence proportional to

$$\begin{aligned} \Gamma(\delta\eta) &= \mathrm{Tr}(\rho U_2^\dagger U_1) = \mathrm{Tr}(\rho e^{-iH_0\delta\eta}), \\ \delta\eta &= \eta_1 - \eta_2. \end{aligned} \tag{60}$$

An ideal variable-phase recombiner measures fringe visibility $\mathscr{V} = 2\sqrt{p_1 p_2}|\Gamma|$. The standard Born measurement rule is assumed in this calculation, not derived from the original Write histories. The phase of Γ shifts the fringes; its modulus controls their contrast. Extra uncontrolled route records multiply or otherwise modify this coherence and must be included explicitly.

For balanced priors, the operational which-route distinguishability in the full marker is

$$\begin{aligned} \mathscr{D} &= \tfrac{1}{2}\|U_1 \rho U_1^\dagger - U_2 \rho U_2^\dagger\|_1, \\ \mathscr{D}^2 + |\Gamma|^2 &\le 1. \end{aligned} \tag{61}$$

Equality holds for a pure initial full marker with unrestricted joint measurements. For a mixed marker the inequality follows by purifying ρ: the two purified states have overlap Γ, and partial trace contracts trace distance. Restricted access to the source alone can only decrease $\mathscr{D}$. For unequal priors the corresponding statement uses $\|p_1\rho_1 - p_2\rho_2\|_1$ and $\mathscr{V}$. In particular, if $[\rho, H_0] = 0$, then $\rho_1 = \rho_2$ and $\mathscr{D} = 0$, although $|\Gamma|$ may be small. Mixed-state phase averaging is not itself accessible which-route information in that marker. Nor does the phase-clock expectation $z = \langle Z \rangle$ determine Γ: one is an expectation of an evolved observable, the other a full-state propagator overlap.

7.2 The needed spectral operand and its domain

Let P_{H_0} be the spectral measure of the full interacting Hamiltonian. The new original-source operand needed for exact route contrast is

$$\begin{aligned} d\mathscr{P}_\rho(E) &= \mathrm{Tr}[\rho P_{H_0}(dE)], \\ \Gamma(s) &= \int e^{-iEs}\, d\mathscr{P}_\rho(E). \end{aligned} \tag{62}$$

Chapter VII's bounds on M, Ω, z, and ν do not provide this entire measure or the exact Loschmidt amplitude. A new spectral enclosure, a controlled evaluation of this matrix element, or an overlap enclosure with a stated error is needed for a numerical full-continuum fringe curve. The matched hard mean energy $\bar{E}$ is also required for its absolute reference phase; a gap enclosure alone does not determine that phase.

If $\int E^2 d\mathscr{P}_\rho(E) < \infty$, set $\sigma_H^2 = \int (E - \bar{E})^2 d\mathscr{P}_\rho(E)$. Then

$$\begin{aligned} |\Gamma(s)|^2 &= 1 - \sigma_H^2 s^2 + o(s^2), \\ |\Gamma(s)| &= 1 - \tfrac{1}{2}\sigma_H^2 s^2 + o(s^2). \end{aligned} \tag{63}$$

These coefficients use the full prepared joint-energy distribution. Neither the bare exchange gap 2ε nor the calibration expectation $E_{\text{ref}} = 573049$ is an energy variance. The latter belongs to a different placement-basis calibration preparation. For a pure state the requisite condition is $\Psi \in D(H_0)$, not merely the energy form domain $D(H_f^{1/2})$. Chapter VII explicitly has ultraviolet-divergent $\int_0^\infty |u_R|^2 d\omega$; consequently a generic bare vacuum argument cannot supply this second moment.

When this finite-variance condition holds, the standard quantum speed bound [20, 21] follows because the projective Hilbert-space speed of a purification evolving under H_0 is σ_H. Thus

$$\begin{aligned} \arccos|\Gamma(s)| &\le \sigma_H |s|, \\ |\Gamma(s)| &\ge \cos(\sigma_H |s|), \quad \sigma_H |s| \le \pi/2. \end{aligned} \tag{64}$$

Any zero of the amplitude requires $|s| \ge \pi/(2\sigma_H)$. For a mixed initial marker this is a bound on the specified amplitude (or its purification), not a claim that the reduced marker becomes distinguishable at that time. The full pure-state marker also obeys $\mathscr{D} \quad \sin(\sigma_H\, s\,)$ on this interval.

A finite full-H second moment for the actual declared preparation. The ultraviolet qualification does not invalidate the particular hard-branch preparation that Chapter VII actually declares. Let H_{ref} be its matched hard-interacting Hamiltonian, with the same even scalar and soft counterterm as H_0, and let

$$\begin{aligned} \psi_k &= \Phi_k \otimes |0_s\rangle, & \Psi_0 &= (\psi_0+\psi_1)/\sqrt{2}, \\ H_{\mathrm{ref}}\psi_k &= E_k\psi_k, & E_1-E_0 &= \Omega. \end{aligned} \tag{65}$$

The normalized exact hard eigenvectors are operator-domain vectors of their hard Hamiltonian. The reference preparation is therefore in $D(H_{\mathrm{ref}})$. Write

$$\begin{aligned} V_s &= H_0 - H_{\mathrm{ref}} \\ &= -g(\sigma_z\phi_{+,s} + \sigma_y\phi_{\times,s}), \\ v_s &= \int_0^\lambda |u_R(\omega)|^2 \mathrm{d}\omega \le v_s^{\mathrm{up}} = \frac{R^4\lambda^6}{675\kappa\pi}. \end{aligned} \tag{66}$$

On this soft vacuum, $V_s\Psi_0$ is a Hilbert-space vector with exactly one soft photon. The two polarization insertions are orthogonal and both Pauli operators preserve norm, giving $\|V_s\Psi_0\|^2 = 2g^2v_s$. The form identity against every test vector in the common form domain is represented by the Hilbert vector $H_{\mathrm{ref}}\Psi_0 + V_s\Psi_0$. The representation theorem for the closed form therefore proves $\Psi_0 \in D(H_0)$ and $H_0\Psi_0 = H_{\mathrm{ref}}\Psi_0 + V_s\Psi_0$. Orthogonality of zero and one soft photon sectors now gives the new exact identity

$$\begin{aligned} \bar{E} &= \tfrac{1}{2}(E_0+E_1), \\ \boxed{\sigma_H^2 = \frac{\Omega^2}{4} + 2g^2v_s} &\le \frac{\Omega^2}{4} + 2g^2v_s^{\mathrm{up}}. \end{aligned} \tag{67}$$

This is a full interacting variance, despite its expression in matched reference data. It relies on the exact hard eigenbranch preparation; it is not a general domain assertion about arbitrary coherent source states. It also does not assert $\Psi_0 \in D(H_0^2)$ or existence of higher moments. The signed g is retained in the Hamiltonian; its square appears here because this is a norm. A rigorous numerical speed enclosure follows by using the original certificate's upper bounds for Ω, $|g|$, R, and λ, with their declared dependencies, rather than rounded lane numbers.

Six-lane consequences of the retained inputs. Table 1 uses the original rational enclosures for Ω, signed g, v_s^{up} and $T_0 = 2\pi/\Delta$. The new rational operations are recorded in the companion check output; each printed bound is verified outward by exact rational comparison. These are independently derived enclosures, not a new native-certificate run. The lower orthogonality ratio is slightly above one half in every lane, while the reference-amplitude error at T_0 is at most 4.54×10^{-11}. This does not imply that an exact zero or a perfect revival occurs.

An independent overlap enclosure from the same preparation. The reference overlap is $\Gamma_{\mathrm{ref}}(s) = e^{-i\bar{E}s}\cos(\Omega s/2)$. The reference orbit is a finite sum of operator-domain eigenvectors, and its soft insertion is Hilbert-space continuous. Strong

Lane	$10^6\sigma_H$ upper	error at T_0 upper	$s_\perp/T_0$ lower
01	1.745051466079	1.759×10^{-12}	0.500000001207
02	1.745051452191	3.644×10^{-12}	0.500000005186
03	1.745048665211	4.536×10^{-11}	0.500000803726
12	1.745051465446	1.886×10^{-12}	0.500000001389
13	1.745048878445	4.360×10^{-11}	0.500000742629
23	1.745049097788	4.172×10^{-11}	0.500000679781

Table 1: Full interacting hard-branch preparation: variance-speed and matched-reference bounds. The error is an amplitude error, and $s_\perp$ is any first possible overlap zero, in the dimensionless proper-time chart.

Hilbert-space Duhamel, with the full interacting propagator left intact, gives

$$\begin{aligned} &\|(e^{-isH_0} - e^{-isH_{\mathrm{ref}}})\Psi_0\| \\ &\quad\le \int_0^{|s|} \|V_s e^{-iuH_{\mathrm{ref}}}\Psi_0\| \mathrm{d}u = |s|\sqrt{2g^2v_s}. \end{aligned} \tag{68}$$

The same orthogonal soft-sector norm is constant for every reference time; negative s is handled by reversing the integral. Consequently

$$\begin{aligned} \varepsilon_s &= |s|\sqrt{2g^2v_s^{\mathrm{up}}}, \\ |\Gamma(s) - e^{-i\bar{E}s}\cos(\Omega s/2)| &\le \varepsilon_s, \\ |\Gamma(s)| &\ge \max\{0, |\cos(\Omega s/2)| - \varepsilon_s\}. \end{aligned} \tag{69}$$

This is a newly derived overlap estimate using Chapter VII's exact preparation, not an inference from its phase-rate certificate. It preserves all photon sectors in the actual propagation. The bound can become uninformative at long times; it does not certify phase-clock invertibility there.

For another preparation lacking a verified second moment, a safe alternative is spectral truncation. For a bounded interval I, let $p = \mathrm{Tr}[\rho P_{H_0}(I)] > 0$ and normalize its restricted spectral measure. Its amplitude Γ_I satisfies $|\Gamma - p\Gamma_I| \le 1-p$ and hence $|\Gamma| \ge \max\{0, p|\Gamma_I| - (1-p)\}$. A finite-variance speed bound for the restricted measure may then be used, but the tail probability $1-p$ must itself be established for the original state. A finite matrix cutoff without that tail is a comparator, not a continuum certificate.

7.3 Action phases, energy offsets, and route records

The coherent amplitude of a controlled route may contain a scalar factor $e^{iS_{\mathrm{ext},j}/\hbar}U_j$. Its scalar interference phase is $(S_{\mathrm{ext},1} - S_{\mathrm{ext},2})/\hbar$, combined with the phase of Γ. This accounting must partition the generator once. If an assigned rest mass m_0 is excluded from H_0, its point-particle contribution is $S_{\mathrm{mass},j} = -m_0c^2\int_{\gamma_j} d\tau_{\mathrm{proper}}$. If that same energy is already included in the internal generator, adding this phase again double counts it. The same warning applies to an even source-energy offset and the scalar counterterm in a fixed lane.

Explicitly, replacing H_0 by $H_0 + bI$ changes $\Gamma(s)$ to $e^{-ibs}\Gamma(s)$ without changing visibility, the same-history expectation z, or which-route distinguishability. Thus an energy offset invisible to the original clock-rate equation can produce a relative route phase $-b(\eta_1 - \eta_2)$. An unchanged total amplitude requires the compensating assignment $S_{\mathrm{ext},j} \mapsto S_{\mathrm{ext},j} + \hbar b\eta_j$. A lane-dependent even operator is not a common scalar on a superposition of lanes and can affect its full spectral distribution. The

original signed Write integral, the quantum rest-generator phase, and a relativistic point-particle action are therefore distinct operands until a receiving identification and a non-overlapping energy account are supplied.

Finally, distinct ordered route or Write records need not imply distinct accessible quantum states. If two histories induce the same observable state and leave no coupled record, an extra history label alone changes no measurement probability. If a physical record is produced, it belongs in the marker or environment and its overlap controls the residual coherence. This allows different histories with equal endpoint and equal signed action without silently creating an additional observable or a new clock law.

Independent finite-source comparator. The companion executable `check_clock_routes.py` retains the exact 16-state source matrices, signed D, C, the full even source energy, both polarizations, and both discrete counterterms. It checks source calibration, controlled-route coherence, pure and mixed duality, the short-time coefficient, the hard/soft variance identity, the Duhamel enclosure, and scalar-phase accounting. Its finite oscillator modes are explicitly assigned comparison data. Their tests neither replace the continuum smearing nor certify its spectral measure; the continuum results above follow from the stated form-domain and hard-branch arguments.

8. A quadratic overlap enclosure for the declared continuum preparation

Provenance and scope. The quadratic refinement and robustness corollaries in this section are developed in campaign voli_chvi_foundation1_20261017 (H002700), with two native SLC-GEN5-R1 operator-word checks. They use the retained Chapter VI Hamiltonian, exact hard-eigenbranch preparation and continuum form construction. They do not replace that preparation by bare source levels, change its hard–soft split, or assert a new native certificate or a laboratory preparation. Route interpretation continues to use the chapter's declared proper-time controlled evolution.

8.1 Hamiltonian, preparation and operator-domain hypotheses

Retain the complete sixteen-state source and both continuum polarizations:

$$\begin{aligned} H_0 &= \varepsilon(S+Q_L+Q_R+Q_LQ_R)+H_f-\varepsilon(D\phi_+ + C\phi_\times) \\ &\quad +\varepsilon^2\mu(D^2+C^2), \\ D &= Q_L-Q_R, \quad C=iSD, \quad \mu=(4R)^{-1}. \end{aligned} \tag{70}$$

The result below concerns one invariant, separately prepared lane $a<b$, with signed $d=q_a-q_b<0$, $g=\varepsilon d$, and $e=q_a+q_b+q_aq_b$. The restriction of this same generator is

$$H_0 = \varepsilon eI+\varepsilon\sigma_x+H_f-g(\sigma_z\phi_+ +\sigma_y\phi_\times)+2g^2\mu I. \tag{71}$$

The lane restriction specifies the preparation sector; it does not discard the remaining source sectors from the model. No claim about an arbitrary superposition of different lanes is made here.

Use the chapter's declared split $\lambda=64\Delta$ and factor the field into hard and soft Fock factors, including both polarizations in each. Let H_{ref} retain the hard interaction, free soft field, even source energy and the *same full counterterm* $2g^2\mu I$ as H_0. Thus their closed quadratic forms have difference

$$\begin{aligned} V_s &= -g(\sigma_z\phi_{+,s}+\sigma_y\phi_{\times,s}), \\ v_s &= \int_0^\lambda |u_R(\omega)|^2\,\mathrm{d}\omega \le v_s^{\mathrm{up}} = \frac{R^4\lambda^6}{675\kappa\pi}. \end{aligned} \tag{72}$$

In particular, no counterterm is subtracted a second time. The signed g remains in the interaction; its square below comes from a norm.

Both Hamiltonians are the self-adjoint, semibounded operators associated with the retained closed forms on the common form domain $\mathscr{Q}=D(H_f^{1/2})$, with the finite source factor understood. The full ultraviolet smearing need not be square integrable. The soft smearing in Eq. (72) is square integrable, and the chapter's weighted field estimates define the common forms. We do not treat V_s as a bounded operator on the full Fock space or subtract unbounded operators on an unspecified domain.

Let the normalized orthogonal exact hard eigenbranches, including the matched scalar terms, give

$$\begin{aligned} \psi_k &= \Phi_k\otimes|0_s\rangle \in D(H_{\mathrm{ref}}), \\ H_{\mathrm{ref}}\psi_k &= E_k\psi_k, \\ \Psi_0 &= \frac{\psi_0+\psi_1}{\sqrt{2}}, \\ \Omega &= E_1-E_0>0. \end{aligned} \tag{73}$$

Here $|0_s\rangle$ is the joint soft vacuum of both polarizations. Set

$$\begin{aligned} U(t) &= e^{-itH_0}, \\ R(t) &= e^{-itH_{\mathrm{ref}}}, \\ b^2 &= 2g^2v_s, \\ \bar{E} &= \frac{E_0+E_1}{2}. \end{aligned} \tag{74}$$

The symbol $R(t)$ denotes the reference propagator, whereas the scalar R in the form factor is the retained source radius.

For every real u, the form action of V_s on $R(u)\Psi_0$ is represented by the Hilbert-space vector $f(u)=V_sR(u)\Psi_0$. It lies in the one-soft-photon sector and is continuous in u. The two polarization insertions are orthogonal, while σ_z and σ_y preserve the source norm. Consequently

$$\begin{aligned} \|f(u)\|^2 &= b^2, \\ \langle\Psi_0, R(v)f(u)\rangle &= 0 \quad (u,v\in\mathbb{R}). \end{aligned} \tag{75}$$

The last identity uses the zero-soft-photon reference orbit, not a truncation of the actual evolution.

For completeness, the representation theorem for the common closed form gives, against every test vector in $\mathscr{Q}$,

$$\begin{aligned} R(u)\Psi_0 &\in D(H_0), \\ H_0R(u)\Psi_0 &= H_{\mathrm{ref}}R(u)\Psi_0+f(u). \end{aligned} \tag{76}$$

The right side is Hilbert-space continuous, since the reference orbit is a finite sum of eigenvectors and f is continuous. These are sufficient strong-Duhamel hypotheses. At $u = 0$ they also give

$$\begin{aligned} \langle H_0\rangle_{\Psi_0} &= \bar{E}, \\ \sigma_H^2 := \|(H_0-\bar{E})\Psi_0\|^2 &= \frac{\Omega^2}{4} + b^2. \end{aligned} \tag{77}$$

This uses $\Psi_0 \in D(H_0)$; neither $D(H_0^2)$ nor a fourth energy moment is assumed. No numerical value of $\bar{E}$ is supplied here.

8.2 The scalar cancellation

Theorem (quadratic scalar Duhamel enclosure). Under Eqs. (70)– (76), define

$$\begin{aligned} \Gamma_0(s) &= \langle \Psi_0, U(s)\Psi_0\rangle, \\ \Gamma_{\mathrm{ref}}(s) &= e^{-i\bar{E}s}\cos(\Omega s/2). \end{aligned} \tag{78}$$

Then, for every real s,

$$\boxed{|\Gamma_0(s) - \Gamma_{\mathrm{ref}}(s)| \le \frac{b^2 s^2}{2} = g^2 v_s s^2 \le q(s) := g^2 v_s^{\mathrm{up}} s^2.} \tag{79}$$

The actual propagator on the left retains the complete continuum and all photon sectors. The statement concerns the scalar survival amplitude; it does not replace the first-order bound on the state-vector difference.

Proof. Write $\mathscr{E}(t) = U(t) - R(t)$. Equation (76) permits differentiating $U(t-u)R(u)\Psi_0$ and integrating in the Hilbert space. With the oriented integral for negative t,

$$\begin{aligned} \mathscr{E}(t)\Psi_0 &= -i\int_0^t U(t-u) f(u)\,\mathrm{d}u, \\ \|\mathscr{E}(t)\Psi_0\| &\le b|t|. \end{aligned} \tag{80}$$

For $t \ge 0$, insert the second identity in Eq. (75) into the scalar matrix element to obtain

$$\langle \Psi_0, \mathscr{E}(t)\Psi_0\rangle = -i\int_0^t \langle \Psi_0, \mathscr{E}(t-u) f(u)\rangle\,\mathrm{d}u. \tag{81}$$

Here $\mathscr{E}(v)^* = \mathscr{E}(-v)$, so Cauchy–Schwarz and Eq. (80) give

$$\begin{aligned} |\langle \Psi_0, \mathscr{E}(t)\Psi_0\rangle| &\le \int_0^t \|\mathscr{E}(-(t-u))\Psi_0\|\,\|f(u)\|\,\mathrm{d}u \\ &\le b^2 \int_0^t (t-u)\,\mathrm{d}u = \frac{b^2 t^2}{2}. \end{aligned} \tag{82}$$

For negative t, self-adjointness gives the complex conjugate scalar difference at $-t$, with the same modulus. Finally, Eq. (73) evaluates the reference matrix element as the displayed cosine. This establishes Eq. (79). □

The chapter's previous radius was $\varepsilon_s = |s|\sqrt{2g^2 v_s^{\mathrm{up}}}$. The new radius is $q(s) = \varepsilon_s^2/2$. Both bounds remain valid; one may use $\min\{2, \varepsilon_s, q(s)\}$ at long times. This improvement does not certify an exact zero, a revival, a phase-clock inverse, or an extended physical two-arm coupling.

8.3 Preparation error and timing uncertainty

Corollary (full-marker preparation error). Let ρ be a normalized density operator on the *same complete* source–field Hilbert space, initially uncorrelated with the route system, and suppose

$$T(\rho, |\Psi_0\rangle\langle\Psi_0|) := \tfrac{1}{2}\|\rho - |\Psi_0\rangle\langle\Psi_0|\|_1 \le \delta_{\mathrm{prep}}. \tag{83}$$

For the unchanged actual generator, set $\Gamma_\rho(s) = \mathrm{Tr}(\rho U(s))$. Then

$$|\Gamma_\rho(s) - e^{-i\bar{E}s}\cos(\Omega s/2)| \le q(s) + 2\delta_{\mathrm{prep}}. \tag{84}$$

Proof. Trace/operator-norm duality and $\|U(s)\| = 1$ bound the change of amplitude by $2\delta_{\mathrm{prep}}$; add Eq. (79). □

The factor two follows from the stated trace-distance convention. Closeness of only the source marginal does not supply this full-marker preparation hypothesis. Trace-distance closeness also does not, by itself, give a finite energy variance for ρ. Neither inference is needed for Eq. (84).

Corollary (centered-energy timing bound). Let s be the nominal proper-time difference in the chapter's dimensionless chart and let the actual difference be $s+\xi$, with $|\xi| \le a$. Define the centered amplitude

$$G_\rho(t) = e^{i\bar{E}t}\Gamma_\rho(t) = \mathrm{Tr}[\rho e^{-i(H_0-\bar{E})t}]. \tag{85}$$

Then

$$\boxed{|G_\rho(s+\xi) - \cos(\Omega s/2)| \le q(s) + 2\delta_{\mathrm{prep}} + \sigma_H a.} \tag{86}$$

A certified upper bound for σ_H may replace σ_H. If the two route ages have errors $|\xi_j| \le a_j$, take $\xi = \xi_1 - \xi_2$ and $a = a_1 + a_2$; a common equal age error cancels.

Proof. For $A = H_0 - \bar{E}$, the spectral theorem and $|e^{-ix} - 1| \le |x|$ give

$$\|(e^{-iA\xi} - I)\Psi_0\| \le |\xi|\,\|A\Psi_0\| = \sigma_H|\xi|. \tag{87}$$

Consequently $|G_0(s+\xi) - G_0(s)| \le \sigma_H|\xi|$. Compare $G_\rho(s+\xi)$ first with $G_0(s+\xi)$, then with $G_0(s)$, and finally with the reference cosine. The three errors are bounded by $2\delta_{\mathrm{prep}}$, $\sigma_H a$ and $q(s)$, respectively. Thus only the exact preparation's variance is used. □

Centering is an explicit energy-phase assignment, not an instruction to discard an observable route phase. In raw amplitudes the conclusion is

$$\left|\Gamma_\rho(s+\xi) - e^{-i\bar{E}(s+\xi)}\cos(\Omega s/2)\right| \le q(s) + 2\delta_{\mathrm{prep}} + \sigma_H a. \tag{88}$$

If one instead compares with the *nominal* raw reference $e^{-i\bar{E}s}\cos(\Omega s/2)$, add the energy-phase error

$$2|\cos(\Omega s/2)|\,|\sin(\bar{E}\xi/2)| \le |\cos(\Omega s/2)|\min\{2, |\bar{E}|a\}. \tag{89}$$

Thus variance controls the centered timing error and the modulus error; a numerical absolute-phase bound additionally needs the mean energy or a specified compensating control phase. No numerical $\bar{E}$ is inferred from the gap Ω.

Under a common scalar reassignment $H_0 \mapsto H_0 + \beta I$ and $H_{\mathrm{ref}} \mapsto H_{\mathrm{ref}} + \beta I$, G_ρ, σ_H and q are unchanged, whereas $\Gamma_\rho(t) \mapsto e^{-i\beta t}\Gamma_\rho(t)$. Preserving a complete controlled-route amplitude requires the corresponding external action assignment $S_{\mathrm{ext},j} \mapsto S_{\mathrm{ext},j} + \hbar\beta\eta_j$ at the actual route ages. Even energy and counterterms remain counted once.

8.4 Modulus and phase disks

Corollary (readable amplitude and phase enclosure). Suppose any of the preceding estimates gives a complex disk $|z-c| \le r$ for a normalized amplitude z. Writing $m = |c|$,

$$\max\{0, m-r\} \le |z| \le \min\{1, m+r\}. \tag{90}$$

If $r < m$, the amplitude is nonzero and its circular phase distance obeys

$$\mathrm{dist}_{\mathbb{S}^1}(\arg z, \arg c) \le \arcsin(r/m). \tag{91}$$

Proof. The modulus bounds are the triangle and reverse-triangle inequalities, together with $|z| \le 1$. Rotate c to $m > 0$. For $r < m$ the disk lies in the open right half-plane; the distance from m to the ray through z is $m|\sin\theta| \le r$ with $|\theta| < \pi/2$. This gives Eq. (91). □

For exact timing use $c = e^{-i\bar{E}s}\cos(\Omega s/2)$ and $r = q(s) + 2\delta_{\mathrm{prep}}$. For the centered jitter estimate use $c = \cos(\Omega s/2)$ and $r = q(s) + 2\delta_{\mathrm{prep}} + \sigma_H a$. The modulus enclosure is then also an enclosure of the raw amplitude's modulus, and an ideal recombiner has visibility $2\sqrt{p_1 p_2}|z|$ under its stated assumptions. For the raw jittered phase, restore $-\bar{E}(s+\xi)$ explicitly. If the cosine is negative its reference phase includes π; at its zeros, or whenever $r \ge m$, this disk yields no phase enclosure. A valid local phase branch is therefore confined to an interval on which the disk excludes zero. Additional beam-splitter, routing and recombination phases must still be included in an experimental fringe.

8.5 Six-lane quadratic scalar-overlap bounds

The retained foundation native return evaluates the scalar cancellation above at the original cycle endpoint, on the same continuum preparation:

Lane	Previous linear remainder upper	New quadratic remainder upper
01	$1.758036931864852 \times 10^{-12}$	1.545347×10^{-24}
02	$3.643671179298703 \times 10^{-12}$	6.638170×10^{-24}
03	$4.535709727166836 \times 10^{-11}$	1.028634×10^{-21}
12	$1.885634247433851 \times 10^{-12}$	1.777809×10^{-24}
13	$4.359906033980351 \times 10^{-11}$	9.504391×10^{-22}
23	$4.171342609236966 \times 10^{-11}$	8.700050×10^{-22}

This amplitude error is distinct from the linear state-vector distance and the later preparation, transport and apparatus errors.

9. Preparation, timing, and causal retention

The purpose of this section is to specify the common operands for the next source-only decoder and detector-to-record calculations. Numerical laboratory values are not assigned to an unspecified preparation, control, or memory. The estimates below apply to explicitly declared channels and states and can be filled with source-derived bounds as those constructions are completed.

9.1 A preparation specification with retained failure

Let $\rho_* = |\Psi_0\rangle\langle\Psi_0|$ denote the exact Chapter VI hard-eigenbranch/soft-vacuum preparation in one declared lane. A physical preparation must return a complete instrument with a ready flag and all failure flags. For each operational input label j, write

$$\widehat{\rho}_j = (1-f_j)|\mathrm{ready}\rangle\langle\mathrm{ready}| \otimes \rho_j^{\mathrm{r}} + f_j|\mathrm{fail}\rangle\langle\mathrm{fail}| \otimes \rho_j^{\mathrm{f}}. \tag{92}$$

Multiple failure outcomes can be retained as separate blocks. Specify the ready-branch trace-distance error

$$\delta_{\mathrm{prep},j} = \tfrac{1}{2}\|\rho_j^{\mathrm{r}} - \rho_{*,j}\|_1. \tag{93}$$

When $\rho_{*,j}$ is pure, a fidelity floor $\langle\Psi_{*,j}|\rho_j^{\mathrm{r}}|\Psi_{*,j}\rangle \ge 1-\zeta_j$ gives $\delta_{\mathrm{prep},j} \le \sqrt{\zeta_j}$ by the standard trace-distance/fidelity relation [24]. This requires fidelity with the full specified source–field preparation; a source marginal fidelity alone does not certify the joint field state.

If failure and abstention are counted as unsuccessful decoding, the complete attempt law obeys the exact identity

$$e_j^{\mathrm{all}} = f_j + (1-f_j)e_j^{\mathrm{r}}. \tag{94}$$

Consequently $f_j \le \bar{f}$ and $e_j^{\mathrm{r}} \le \bar{e}$ imply $e_j^{\mathrm{all}} \le \bar{f} + (1-\bar{f})\bar{e}$. Conditioning on the ready flag defines a different, explicitly heralded protocol. Its success probability and retained failed attempts remain part of the resource account. A single experiment must choose whether it reports the all-attempt or ready-conditional record criterion.

9.2 A channel error budget that permits correlations

For each label j, let Φ_k and $\widetilde{\Phi}_k$ be reference and actual trace-preserving channels for the successive writing, switching, storage, transport, access, and decoding stages. Classical flags are included in their outputs. Registers needed to retain environmental correlations are included in the stage input. The allowed states must be rich enough that each bound below holds for the states on which that stage is actually used; if an arbitrary reference is retained, use the corresponding bound with that reference. Define

$$\delta_k = \sup_{j,t,\omega\in\mathscr{S}_{k,j,t}} \tfrac{1}{2}\|[(\widetilde{\Phi}_k - \Phi_k)\otimes I](\omega)\|_1. \tag{95}$$

The time supremum is over the declared receiving interval. A half-diamond-norm bound is sufficient; a bound uniform on the relevant reachable states also suffices. An environment with memory is retained as part of the joint state; independent noise between stages or cells is not assumed.

Proposition 1 (Composed readout error)**.** *If the reference decoder has worst-label error at most $e_*(t)$ and all stage bounds apply on the declared time interval, then the actual ready-branch decoder satisfies*

$$\sup_{j,t} e_j^{\mathrm{r}}(t) \le \min\{1,\ \sup_t e_*(t) + \delta_{\mathrm{prep}} + \sum_k \delta_k\}. \tag{96}$$

Proof. Insert the reference preparation and each reference channel one at a time into the composition. The triangle inequality bounds the final trace distance by $\delta_{\mathrm{prep}} + \sum_k \delta_k$; subsequent trace-preserving channels contract each difference. The success effect has spectrum in $[0,1]$, so its probability changes by at most that trace distance. Apply the bound to every label and time, then take the supremum. This argument also applies to a channel whose output contains both a classical instrument label and the corresponding quantum state. □

Equation (94) then adds retained preparation failure when the all-attempt criterion is used. Do not count that same failure a second time inside an all-attempt channel-distance term.

9.3 Timing errors and the energy account

All internal ages use the Chapter VII dimensionless chart. Let the actual readout ages differ from the nominal ages by ξ_1, ξ_2. If $|\xi_j| \le a_j$, the overlap time difference changes by at most $a_1 + a_2$. The centered-amplitude estimate in the preceding overlap section uses the full prepared energy variance. It is insensitive to a common scalar energy reassignment only after the compensating external action phase is included. Absolute fringe phase requires the matched hard mean energy and the scalar control-action convention. A gap interval alone does not fix that absolute phase.

For a switched or noncommuting control Hamiltonian, replacing the entire experiment by an age shift is a separate assertion. Instead, a common-domain strong Duhamel estimate can bound a declared perturbation $\Delta K_j(t)$:

$$r_j \le \int_0^{t_f} \|\Delta K_j(u)\Psi_{\mathrm{ref},j}(u)\|\, du. \tag{97}$$

The forcing must exist and be integrable in Hilbert norm. The reference orbit and domain assumptions must be stated for the actual singular-source control law. A nominal pulse duration by itself supplies no value of r_j. For normalized pure outputs, the resulting state trace distance is at most $\min\{1, r_j\}$, so it contributes directly to the decoder error budget. For branch overlap it contributes $r_1 + r_2$ by the triangle inequality.

If the good-timing event has probability at least $1-\beta$ for every label, and the conditional good-event decoding error is at most e_{good}, then the unconditional error is bounded by

$$e \le \beta + (1-\beta)e_{\mathrm{good}}. \tag{98}$$

The conditional input on that event must satisfy the preparation and control bounds used for e_{good}. Independence between timing, preparation, and detector noise is not needed when the bounds are uniform or formulated on their joint conditioned state.

9.4 The specification to carry into the next calculation

The following ledger distinguishes a defined mathematical operand from an unassigned implementation value.

Operand	Meaning	Present input or next receiving construction
$\rho_*, \Omega, M, v_s^{\mathrm{up}}$	Joint reference and certified bounds	Retained Chapter VI/Chapter VII package; exact hard branches are defined.
$\bar{E}, \Phi_0, \Phi_1$	Absolute energy and explicit hard state operands	Recover certified hard-state data for numerical absolute phase and preparation.
$f_j, \delta_{\mathrm{prep},j}$	Failure probability and ready-state accuracy	Bound from a complete finite preparation instrument.
a_j, β, r_j	Age uncertainty, timing tail, control mismatch	Derive from the specified clock, switch, and actuator.
$\mathscr{N}_t$	Storage including correlated noise and read disturbance	Choose a physical memory and retained environment.
$\mathscr{A}_{\mathscr{D},t}$	Transport and permitted receiving access	State receiver region, causal arrival, and available controls.
$F_j(t), e_*(t)$	Fixed experiment's decoder and reference error	Next source-only quadrature benchmark, then actual detector bins.
$\mathscr{T}_{\mathrm{mem}}, \mathscr{T}_{\mathrm{read}}$	Memory and receiving windows	Use a causal onset and uniform retention/readout bounds.

For repeated reads, the channel sequence includes every earlier read and its retained output. Uniform single-read performance alone specifies no bound on a sequence of disturbed reads. Initial source–apparatus correlations, finite control references, and failure recipients enter the declared state and channels rather than being discarded at the composition step.

The interface to the later two-arm transport task is fixed now: one full source–field state, declared branch controls and receiving regions, both polarizations, all source sectors, even energy and counterterms, and the same preparation/clock chart. The next detector calculation can then return a complete conditional law and its error budget under the stated transport law; actual two-arm transport will supply its additional controlled discrepancy.

10. A source-only decoder for the retained interacting clock

Provenance and receiving assumptions. The decoder and bounds below are developed in campaign voli_chvi_source_decoder1_20261017 (H002701) from the retained Chapter VII clock observable and its Chapter VI route interpretation, with four native SLC-GEN5-R1 calls. The source equations are in `CHAPTER_I_MODEL.md`, Sections 2 and 5: the exact source-lane restrictions, $Z = (D+iC)/d$, the reference expectation $h(t) = Me^{-i\Omega t}$, and the actual bound $|z(t) - h(t)| \le E(t)$. The original signed coupling, both field polarizations, even source energy and counterterms remain in the actual evolution that supplies z. The ideal POVM at apparatus entry is source-only. During a finite physical readout

the same full Hamiltonian can continue to correlate source and field; an implemented joint instrument must then approximate this ideal POVM on the specified entry states. No direct field readout is prescribed. Section 12 supplies that finite control and pointer construction with its explicit drift allowance.

We retain the chapter's conditional route law: the complete marker starts in its declared exact hard-eigenbranch/soft-vacuum preparation Ψ_0, uncorrelated with the route label, and route j supplies $e^{-iH_0\eta_j}\Psi_0$. The source output is the partial trace over the *complete* field. No field fragment is measured by this decoder.

10.1 The quadrature embedded in all sixteen source states

Let $|q_a\rangle$ be the normalized ordered eigenvectors of the original four-state source matrix Q. For one chosen lane $a<b$, abbreviate $|ab\rangle = |q_a\rangle\otimes|q_b\rangle$ and define operators on $\mathbb{C}^4\otimes\mathbb{C}^4$ by

$$\begin{aligned} P &= P_{ab}^{\mathrm{lane}} = |ab\rangle\langle ab| + |ba\rangle\langle ba|,\\ Z_{ab} &= |ab\rangle\langle ab| - |ba\rangle\langle ba|,\\ X_{ab} &= PSP, \qquad Y_{ab} = iSZ_{ab}, \end{aligned} \tag{99}$$

where S is the original full source swap. On the ordered lane $(|ab\rangle,|ba\rangle)$ these are $I,\sigma_z,\sigma_x,\sigma_y$; Z_{ab}, X_{ab} and Y_{ab} vanish on the other fourteen source states. In particular Y_{ab} is Hermitian, $Z_{ab}^2 = Y_{ab}^2 = P$, and $Z_{ab}Y_{ab} + Y_{ab}Z_{ab} = 0$. The complex clock observable embedded in this lane is

$$\begin{aligned} Z_{ab}^{\mathrm{clk}} = Z_{ab} + iY_{ab} = \frac{P(D+iC)P}{d_{ab}},\\ d_{ab} = q_a - q_b < 0. \end{aligned} \tag{100}$$

Every displayed source operator acts as its tensor product with the identity on the field. Thus this is the original source observable, not an observable requiring joint access to a dressed hard-field state.

The projector P commutes with Q_L, Q_R, S, D, C and preserves the common energy form domain. It consequently reduces the retained form-sum Hamiltonian. The exact lane preparation stays in this lane under H_0. Writing its reduced source state at age t as $\rho_S(t)$, we therefore have

$$\begin{aligned} z(t) &= \mathrm{Tr}[\rho_S(t)Z_{ab}^{\mathrm{clk}}],\\ O_\chi &= \frac{e^{-i\chi}Z_{ab}^{\mathrm{clk}} + e^{i\chi}(Z_{ab}^{\mathrm{clk}})^*}{2} = \cos\chi\, Z_{ab} + \sin\chi\, Y_{ab},\\ O_\chi^2 &= P, \qquad \|O_\chi\| = 1, \qquad m_j(\chi) = \mathrm{Re}[e^{-i\chi}z(\eta_j)]. \end{aligned} \tag{101}$$

For an explicitly complete binary measurement, assign every state outside the nominated lane to label zero:

$$\begin{aligned} E_1(\chi) &= \frac{P - O_\chi}{2},\\ E_0(\chi) &= I_{16} - E_1(\chi). \end{aligned} \tag{102}$$

These are orthogonal projections and sum to I_{16}. On the lane they are $(P\mp O_\chi)/2$; on its complement $E_0 = I$ and $E_1 = 0$. The full binary observable is accordingly $E_0 - E_1 = (I_{16} - P) + O_\chi$, rather than O_χ on the complement. No population has been dropped, and no success-conditioned renormalization is used. For lane-supported states the conditional correct-label probabilities are

$$\begin{aligned} p(0|0) &= \frac{1+m_0(\chi)}{2},\\ p(1|1) &= \frac{1-m_1(\chi)}{2}. \end{aligned} \tag{103}$$

These probabilities and their worst-label bounds do not require balanced label priors.

10.2 General ages and a source-only discrimination witness

Proposition (quadrature contrast). For two exact lane-supported source outputs, let $z_j = z(\eta_j)$ and $D_S = \frac{1}{2}\|\rho_S(\eta_0) - \rho_S(\eta_1)\|_1$. Then

$$D_S \geq \frac{|z_0 - z_1|}{2}. \tag{104}$$

If $z_0 \neq z_1$, the quadrature maximizing $m_0 - m_1$ is $\chi = \arg(z_0 - z_1)$ modulo 2π. This maximizes contrast within the stated quadrature family; it is not an assertion that this family is the unrestricted Helstrom measurement or that it maximizes worst-label success for arbitrary unequal-radius outputs.

Proof. For any real χ, trace/operator-norm duality gives $|m_0 - m_1| \leq 2D_S\|O_\chi\| = 2D_S$. Maximizing $\mathrm{Re}[e^{-i\chi}(z_0 - z_1)]$ gives $|z_0 - z_1|$. On lane-supported inputs the completed binary measurement also attains the corresponding average success $1/2 + |z_0 - z_1|/4$ for balanced priors. □

For the reference expectations $h_j = Me^{-i\Omega\eta_j}$ with the retained $M > 0$, put $\bar\eta = (\eta_0+\eta_1)/2$ and $v = \sin[\Omega(\eta_1 - \eta_0)/2]$. When $v \neq 0$, choose

$$\chi_* = \arg(h_0 - h_1) = \frac{\pi}{2} - \Omega\bar\eta + \begin{cases} 0, & v > 0,\\ \pi, & v < 0, \end{cases} \qquad (\mathrm{mod}\ 2\pi). \tag{105}$$

The reference quadrature means are then $+M|v|$ and $-M|v|$. For a predeclared calibrated implementation of this axis, the retained actual-clock bound implies

$$\begin{aligned} \min_j p(j|j) &\geq \frac{1 + M|v| - \max_j E(\eta_j)}{2},\\ D_S &\geq \max\left\{0, M|v| - \frac{E(\eta_0)+E(\eta_1)}{2}\right\}. \end{aligned} \tag{106}$$

The second bound follows directly from $|z_0 - z_1| \geq |h_0 - h_1| - E(\eta_0) - E(\eta_1)$. If the reference contrast vanishes, the displayed distance lower bound is zero and no orientation is selected by this contrast criterion. Choosing an axis from reference data before a trial does not presume knowledge of its unknown label. Parameter uncertainty in that chosen axis must be included in its calibration error.

10.3 A fixed-axis, quarter-cycle protocol

The following protocol avoids choosing its nominal axis from an absolute hard energy or from the measured trial output. Use

the retained chart

$$\begin{aligned} T_0 &= \frac{2\pi}{\Delta}, \\ \eta_0 &= \frac{T_0}{4}, \\ \eta_1 &= \frac{3T_0}{4}, \\ \chi &= -\frac{\pi}{2}, \\ d &= 1-\frac{\Omega}{\Delta}. \end{aligned} \tag{107}$$

The exact retained lane data satisfy $0<\Omega\le\Delta$, so $d\ge 0$. The new symbol d in Eq. (107) denotes the relative gap deficit; it is distinct from the signed source difference d_{ab} in Eq. (100). The ages describe the two candidate marker histories at a common apparatus entry, not two label-dependent readout triggers. The same rotation, pointer test and outcome assignment are used for both hypotheses. The measured lane quadrature is $O_{-\pi/2}=-Y_{ab}$. Its reference means, including their signs, are

$$\begin{aligned} m_{0,\mathrm{ref}} &= M\cos(\pi d/2), \\ m_{1,\mathrm{ref}} &= -M\cos(3\pi d/2). \end{aligned} \tag{108}$$

Thus the two *correct-label margins* are positive cosines, whereas the raw means have opposite signs. The retained comparison error gives

$$\begin{aligned} p(0|0) &\ge \frac{1+M\cos(\pi d/2)-E(\eta_0)}{2}, \\ p(1|1) &\ge \frac{1+M\cos(3\pi d/2)-E(\eta_1)}{2}. \end{aligned} \tag{109}$$

Let outward inputs satisfy

$$\begin{aligned} 0\le m_*\le M\le 1, \\ 0\le d\le d_*, \\ \pi\le\pi_*, \\ \Omega\le\Omega_*, \\ E(t)\le E_* \\ (0\le t\le T_0). \end{aligned} \tag{110}$$

Here m_* is a lower bound for the retained M, not a rounded display. For example $d_*\ge 1-\Omega_{\mathrm{lower}}/\Delta_{\mathrm{upper}}$ is safe with positive denominator and the stated $\Omega\le\Delta$ premise. E_* can be supplied by the retained cycle enclosure of $E(t)=\exp(4g^2v_s^{\mathrm{up}}t^2)-1$, which is nondecreasing on this interval. The signed g and the original split remain unchanged.

Set $\Theta_0=3\pi_* d_*/2$. Whenever $\Theta_0\le\sqrt{2}$,

$$\boxed{\varepsilon_{\mathrm{clock}} := \max_j [1-p(j|j)] \le \frac{1-m_*[1-\Theta_0^2/2]+E_*}{2}.} \tag{111}$$

Indeed $\cos x\ge 1-x^2/2$ for real x. Both angular offsets in Eq. (108) have modulus at most Θ_0. The condition makes the common lower cosine bound nonnegative, so replacing M by its lower bound preserves the inequality. Without that condition the same substitution need not be valid; the universal fallback is the trivial error bound one.

10.4 Timing, axis and preparation robustness

Allow actual ages $t_j=\eta_j+\xi_j$ with $|\xi_j|\le a$ and $0\le t_j\le T_0$, and let the actual quadrature axis be $\chi=-\pi/2+\zeta$ with $|\zeta|\le b$. A sufficient age-window condition is $a\le T_0/4$. These are dimensionless ages and angular axis errors; no SI time calibration is introduced. The signed reference means now obey

$$\begin{aligned} m_{0,\mathrm{ref}} &= M\cos(-\pi d/2+\Omega\xi_0+\zeta), \\ -m_{1,\mathrm{ref}} &= M\cos(-3\pi d/2+\Omega\xi_1+\zeta). \end{aligned} \tag{112}$$

Both correct-label angular errors are bounded by

$$\Theta = \frac{3\pi_* d_*}{2}+\Omega_* a+b. \tag{113}$$

Consequently, for $\Theta\le\sqrt{2}$,

$$\max_j [1-p(j|j)] \le \frac{1-m_*[1-\Theta^2/2]+E_*}{2}. \tag{114}$$

The timing and axis losses enter through a squared displacement from the reference antipodal extrema. The square includes the gap-deficit and cross terms; it is not a claim that each uncertainty contributes an independent additive quadratic error. This bound is on a source probability, so it needs no absolute hard mean energy $\bar{E}$ or absolute route amplitude phase. Indeed a common scalar shift of the generator multiplies each branch propagator by a scalar phase, which cancels in $U_j\rho U_j^*$ and hence in its source measurement probabilities. That invariance does not remove the scalar contribution from an interferometric comparison of the route amplitudes themselves.

Suppose an imperfect complete marker preparation ρ satisfies

$$\tfrac{1}{2}\|\rho-|\Psi_0\rangle\langle\Psi_0|\|_1\le\delta_{\mathrm{prep}}, \tag{115}$$

with the same generator and declared ages. Unitary evolution and partial trace do not increase this distance. Each completed-POVM outcome probability therefore changes by at most δ_{prep}: for an effect $0\le F\le I$, the trace-distance variational bound is $|\mathrm{Tr}[F(\rho-\sigma)]|\le T(\rho,\sigma)$. It follows that

$$\boxed{\begin{gathered}\varepsilon_{\mathrm{dec}}\le\min\left\{1,\frac{1-m_*[1-\Theta^2/2]+E_*}{2}+\delta_{\mathrm{prep}}\right\}, \\ \Theta\le\sqrt{2}.\end{gathered}} \tag{116}$$

This includes leakage into the other fourteen source states and source–field preparation errors covered by Eq. (115). The penalty is δ_{prep} for an outcome probability, rather than the $2\delta_{\mathrm{prep}}$ bound for a general unitary amplitude. If an apparatus construction subsequently bounds an additional per-label outcome-probability error by δ_{read}, it may be added to the right side before clipping at one. That implementation error is not established by the ideal measurement algebra alone. In particular, the finite-control Kraus operators may act jointly on source and field even when all external controls and the pointer couple only to the source. Their approximation to the ideal entry POVM, rather than an exact source–field factorization during the pulse, is the required estimate.

If Eq. (115) is established only on a heralded ready branch, its resulting error bound is conditional on that branch. With per-label failure probability at most f, counting a failure as unsuccessful gives the all-attempt bound $f+(1-f)\varepsilon$, where $\varepsilon \le 1$ is the complete ready-branch error including preparation and readout. The failure outcome remains in the instrument; this formula does not assert an additional implemented flag or silently discard unsuccessful attempts.

10.5 Rotation sign and the complete binary readout

For a general nominal axis let $R_\chi = e^{-i\chi S/2}$. The original swap satisfies $S^2 = I_{16}$ and commutes with P, while it anticommutes with Z_{ab} on the lane. Hence

$$\begin{aligned} R_\chi^* Z_{ab} R_\chi &= \cos\chi\, Z_{ab} + \sin\chi\, Y_{ab} = O_\chi, \\ E_1(\chi) &= R_\chi^* |ba\rangle\langle ba| R_\chi. \end{aligned} \tag{117}$$

In particular

$$\begin{aligned} R &= e^{+i\pi S/4}, \\ R^* Z_{ab} R &= -Y_{ab}, \\ E_1 &= R^* |ba\rangle\langle ba| R, \\ E_0 &= I_{16} - E_1. \end{aligned} \tag{118}$$

Thus rotating by R and testing for $|ba\rangle$ realizes the specified *ideal POVM*; every other outcome is assigned label zero. Equivalently, a three-valued Z_{ab} read after the rotation can be coarse-grained by assigning its zero eigenspace to label zero. An exchange-angle error $R = e^{i(\pi/4+\alpha)S}$ gives quadrature-axis error $\zeta = -2\alpha$; a bound on $|\alpha|$ must therefore be doubled when used as the axis budget b. Other errors caused by a concurrently running drift or field interaction need their own control estimate.

For a separately declared leakage-calibration experiment one may instead use the three effects

$$\begin{aligned} &R^* |ab\rangle\langle ab| R, \\ &R^* |ba\rangle\langle ba| R, \\ &I_{16} - P. \end{aligned} \tag{119}$$

The third effect is a failure/leakage outcome and the three sum to I_{16}. This is a different stated output alphabet; its existence as a POVM does not supply a second physical flag apparatus, and conditioning away its third outcome would require a separate probability account.

The present section supplies the complete source-only POVM and its worst-label error budget. Section 12 implements its rotation and eigenbasis test with finite bounded controls and a two-state pointer. The durable memory, causal receiving channel and microscopic control actuator form the next apparatus join.

11. Exact source-bound values for the six original lanes

The original source certificate supplies the exact bare gap $\Delta = 2/573049$ and $\varepsilon = 1/573049$. For each lane use its retained outward endpoints for Ω, M, c and K_{prep}. Define entirely rational enclosing inputs

$$\begin{aligned} d_* &= 1 - \Omega_{\mathrm{lower}} \frac{573049}{2}, \\ \pi_* &= \frac{\Delta_{\mathrm{upper}} T_{0,\mathrm{upper}}}{2}, \\ q_* &= g^2 v_s^{\mathrm{up}} T_0^2 \text{ (retained outward rational)}, \\ E_* &= 8 q_*. \end{aligned} \tag{120}$$

All six lanes have $0 \le 4q_* \le 1/2$, so $e^{4q_*} - 1 \le 8q_*$ by the positive exponential series and $e^x \le (1-x)^{-1}$ for $0 \le x < 1$. The exact q_* values are inherited from the foundation capsule; they use its original continuum soft moment and upper cycle endpoint. The certificate endpoints enclose the exact bare gap, making $\pi_* \ge \pi$. The exact gap is used in d_* rather than a rounded gap display. The independent check verifies all sign, interval and nonnegative-cosine conditions before making a bound.

The largest ideal-entry bound occurs in lane 03. For the budgeted example declare

$$\begin{aligned} \Omega_* a &= \frac{1}{100}, \\ b &= 0, \\ \delta_{\mathrm{prep}} &= \frac{1}{10000}, \\ r_{\mathrm{ctrl}} &\le \frac{1}{1000}. \end{aligned} \tag{121}$$

The phase allowance here is entirely an age/timing allowance; the nominal analyser axis is exact. The resulting worst-label ready-branch bound is

$$\begin{aligned} \varepsilon_{\mathrm{ready}} \le \min\Bigg\{ 1, \frac{1 - m_*[1 - (3\pi_* d_*/2 + 1/100)^2/2] + E_*}{2} \\ + \frac{1}{10000} + \frac{1}{1000} \Bigg\} \le 0.0011254539 \end{aligned} \tag{122}$$

across all six lanes. The age allowance lies inside the original entry window in every lane. These values specify a receiving model; they do not assign an achieved laboratory preparation, timing or pointer accuracy. An ideal blank pointer and its ideal final binary observation are assumed in this benchmark. Additional initialization/readout errors can instead be supplied as joint-state or outcome-probability bounds. Per-label ready failure at most f gives all-attempt error at most $f+(1-f)\varepsilon_{\mathrm{ready}}$, with the failure output retained.

12. Finite source controls and a two-state readout pointer

This construction specifies a source-selective receiving model. Its controls act on the original sixteen-state source and a finite pointer; the complete continuum field remains active. It supplies a finite mathematical implementation of the stated source effects with an explicit drift and switching-work budget. The actuator implementing these source operators, its timing and calibration, and any autonomous replacement for the driven controls remain physical apparatus assignments.

Lane	Ideal-entry error upper	Finite-pulse r upper	Budgeted error upper
01	$6.2305241 \times 10^{-10}$	0.00014722126	0.0011250007
02	2.6763754×10^{-9}	0.00015791109	0.0011250030
03	4.1473953×10^{-7}	0.00082431540	0.0011254539
12	$7.1677609 \times 10^{-10}$	0.00014772605	0.0011250008
13	3.8321098×10^{-7}	0.00079335659	0.0011254194
23	3.5077944×10^{-7}	0.00076019657	0.0011253839

Table 2: Outward bounds for each original lane. The ideal-entry column uses exact preparation and nominal ages. The finite-pulse column uses the explicit mathematical controls below. The budgeted column adds the declared timing, preparation and 10^{-3} control allowances in Eq. (121). These allowances are receiving inputs.

12.1 The retained source energy is enough for short pulses

In the declared prepared lane retain

$$\begin{aligned} H_0 &= \varepsilon e I + \varepsilon S + K, \\ K &= H_f + H_{\mathrm{int}} + cI \geq 0, \\ c &= 2g^2\mu. \end{aligned} \tag{123}$$

Here S is the original Swap restriction, $\|S\| = 1$, and the even energy, both physical polarizations, both counterterms, hard dressing, soft vacuum, and all field complements are retained. On the original common form domain the annihilation form estimate gives

$$|\langle H_{\mathrm{int}} \rangle| \leq 2\sqrt{c\langle H_f \rangle} \leq \langle H_f \rangle + c. \tag{124}$$

Consequently, with the lane scalar removed only from the comparison phase,

$$\begin{aligned} &L = H_0 - \varepsilon e I + \varepsilon I \geq 0, \\ &L \leq 2\varepsilon I + 2H_f + 2cI \quad \text{as quadratic forms.} \end{aligned} \tag{125}$$

The scalar $\varepsilon e - \varepsilon$ is still part of the physical generator and its elapsed phase; removing its common phase from this comparison does not subtract its energy or any relative route phase.

Precisely, the positivity and scalar subtraction in Eq. (125) apply on the selected invariant lane tensored with the full Fock space and pointer. They are not a global all-sixteen-state positivity claim after subtracting one lane's e. The actual H_0 and bounded source controls are defined on all sixteen source states. Swap and P_{ba} preserve each invariant source lane, so the finite-energy proof restricts to the reference input lane without discarding the other states from the physical dynamics.

Suppose the declared reference source–field input to the readout has $\langle H_f \rangle \leq F$ at its declared entry age. Adjoin the blank finite pointer. Let V be an ideal source–pointer unitary and B a self-adjoint contraction, both preserving the selected lane and commuting with H_f. The controls used below have these properties. Then for $\xi = V\psi$ the form bounds imply

$$\begin{aligned} \langle \xi, L\xi \rangle &\leq C, \\ \langle B\xi, LB\xi \rangle &\leq C, \\ C &= 2\varepsilon + 2F + 2c. \end{aligned} \tag{126}$$

The second expression is not normalized: $\|B\xi\| \leq 1$ and $B\xi, H_f B\xi \qquad F$. Commutation with H_f preserves its form domain and is used here; preservation of $D(H_0)$ by B or V is neither assumed nor needed. A mixed input is covered by a purification, with the same expectation bounds and an untouched reference.

The retained source estimate can be inserted before selecting pulse times. For example, if the declared input-age bound is

$$F \leq 2c + 2K_{\mathrm{prep}} + 2\varepsilon E(T_0), \tag{127}$$

then one may use the outward source-bound ceiling

$$C \leq C^{\mathrm{up}} = 2\varepsilon + 6c + 4K_{\mathrm{prep}} + 4\varepsilon E(T_0). \tag{128}$$

This substitution inherits the preparation, input-age window and definitions of its source bound. It does not assign a new measured accuracy or an energy second moment.

12.2 A bounded-control drift lemma

For $L \geq 0$ and any vector in its form domain, the spectral theorem gives

$$\|(e^{-itL} - I)\xi\|^2 = \int 2[1 - \cos(tE)]\, d\mathscr{P}_\xi(E) \leq 2|t|\langle \xi, L\xi \rangle. \tag{129}$$

Thus a finite first energy form suffices, even when $L\xi$ is not a Hilbert-space vector.

Let a pulse of duration $\tau > 0$ have actual generator

$$\begin{aligned} H(t) &= L + \lambda(t)B, \\ A_{\mathrm{abs}} &= \int_0^\tau |\lambda(t)|\, dt < \infty, \end{aligned} \tag{130}$$

with B as above. The finite pointer is degenerate during this modelled control, and $\lambda(t)$ is real. In the interaction picture put

$$\begin{aligned} B_H(t) &= e^{itL} B e^{-itL}, \\ V(t) &= \exp\left[-iB \int_0^t \lambda(u)\, du \right]. \end{aligned} \tag{131}$$

The ideal $V(t)$ commutes with H_f. For $\xi_t = V(t)\psi$, split

$$\begin{aligned} (B_H(t) - B)\xi_t = e^{itL} B (e^{-itL} - I)\xi_t \\ + (e^{itL} - I) B \xi_t. \end{aligned} \tag{132}$$

Equations (126) and (129) therefore give

$$\begin{aligned} &\|(B_H(t) - B)V(t)\psi\| \leq 2\sqrt{2tC}, \\ &0 \quad t \quad \tau. \end{aligned} \tag{133}$$

All free factors in this calculation are unitary on the Hilbert space. No commutator $[L,B]$, derivative $LB\xi_t$, or operator-domain preservation is required.

The actual interaction-picture generator $\lambda(t)B_H(t)$ is bounded and strongly measurable with integrable norm. Its Dyson propagator exists on the full Hilbert space. Bounded-generator Duhamel, applied to the ideal reference vectors only, gives

$$\|(U_I(\tau)-V(\tau))\psi\| \le 2\int_0^\tau |\lambda(t)|\sqrt{2tC}\,dt \le 2A_{\rm abs}\sqrt{2\tau C}. \tag{134}$$

The actual Schrödinger propagator is $U(\tau)=e^{-i\tau L}U_I(\tau)$. Adding its final free factor, and using the form bound on $V(\tau)\psi$, yields the finite-pulse estimate

$$\boxed{\|(U(\tau)-V(\tau))\psi\| \le (1+2A_{\rm abs})\sqrt{2\tau C}.} \tag{135}$$

This is a state-specific finite-energy estimate. For rectangular pulses, $L+\lambda B$ is self-adjoint on $D(L)$ by bounded perturbation and has the same form domain after a finite scalar shift. Its unitary preserves that form domain. This also makes the endpoint mean energies and bounded switching payments well defined for the stated input. The proof of Eq. (135) itself uses bounded Dyson evolution and the reference form estimates, not an assumed invariance of $D(L)$.

12.3 Exact ideal rotation and phase-free pointer gate

Take the ideal source rotation

$$R = e^{i\pi S/4}. \tag{136}$$

It is generated by $B_{\rm rot}=S$ with signed area $\int\lambda_{\rm rot}=-\pi/4$, hence $A_{\rm rot}=\pi/4$. The full source operator is the original six-lane Swap with its four unforced directions; it is not replaced by a bare Hamiltonian during the pulse.

Let P_{ba} be the declared source-only projector onto the ordered state selected by the receiver, and let $X_{\rm ptr}$ be the pointer Pauli matrix. Assign the bounded projector generator

$$B_{\rm gate} = P_{ba}\otimes\frac{I-X_{\rm ptr}}{2}. \tag{137}$$

At area $\int\lambda_{\rm gate}=\pi$,

$$\begin{aligned} G = e^{-i\pi B_{\rm gate}} &= I-2B_{\rm gate} \\ &= (I-P_{ba})\otimes I + P_{ba}\otimes X_{\rm ptr}. \end{aligned} \tag{138}$$

Thus a blank pointer $|0\rangle$ is flipped precisely on the selected source subspace, with no extra conditional phase. Both generators have norm at most one and commute with H_f. Their finite source–pointer matrices act on all 16×2 coordinates while the entire field remains in the dynamics.

For reference, put $P^{\rm ord}_{ab}=|ab\rangle\langle ab|$. In the ordered prepared-lane basis $|ab\rangle,|ba\rangle$ define $Z=P^{\rm ord}_{ab}-P_{ba}$, $X=S$, and $Y=-i|ab\rangle\langle ba|+i|ba\rangle\langle ab|$. Then

$$\begin{aligned} P_{ba} &= \frac{I-Z}{2}, \\ R^\dagger P_{ba} R &= \frac{I+Y}{2}. \end{aligned} \tag{139}$$

The two corresponding ideal signed pointer means are $\langle Z\rangle = 1-2P(1)$ without R and $\langle Y\rangle = 2P(1)-1$ after R. These signs belong to this explicit ordered basis. Any source-current convention used elsewhere must be translated by its actual source operator map.

12.4 Two-pulse error, finite switching work and event time

Let $U_{\rm rot},U_{\rm gate}$ be the actual rectangular pulses and R,G the ideal ones. Unitary invariance and one hybrid intermediate state give

$$\begin{aligned} &\|U_{\rm gate}U_{\rm rot}\psi - GR\psi\| \\ &\quad\le \|(U_{\rm rot}-R)\psi\| + \|(U_{\rm gate}-G)R\psi\| \\ &\quad\le r, \\ r &= \left(1+\frac{\pi}{2}\right)\sqrt{2C\tau_{\rm rot}} + (1+2\pi)\sqrt{2C\tau_{\rm gate}}. \end{aligned} \tag{140}$$

The second pulse is tested on the ideal state $R\psi$, whose H_f expectation is still bounded by F. The estimate does not require a new field-energy estimate on the actual intermediate state. A pointer-only readout omitting the rotation uses the second term alone.

For any final pointer event, the probability discrepancy is at most r, because pure-state trace distance is no larger than the vector difference. Partial trace and purification give the same bound for a mixed input. A binary pointer value in $\{-1,+1\}$ consequently has mean error at most $2r$. Pointer initialization and detector errors, when present, must be added using their declared all-outcome probability or joint-state bounds. These inequalities introduce no additional measurement rule.

If the actual preparation is known only to satisfy $T(\rho_{\rm act},\rho_{\rm ref})\le\delta_{\rm prep}$, apply the same full actual pulse and pointer channel to both states. Trace-distance contractivity adds at most $\delta_{\rm prep}$ to each output probability error. The reference pulse estimate therefore gives

$$|P_{\rm act}(E)-P_{\rm ideal,ref}(E)| \le \delta_{\rm prep}+r. \tag{141}$$

No field-energy or form-domain assumption is thereby made on $\rho_{\rm act}$, which may have a component outside the reference lane. The C bound and the pulse-drift proof concern the ideal reference states. Finite source-energy claims for an actual preparation, when needed for a resource account, require its own supplied energy bound.

Calibrated areas use their actual absolute areas. The nominal analyser angle is $\chi=-\pi/2$, corresponding to the source area $\chi/2=-\pi/4$. If a rectangular rotation has $|\delta\chi|\le b$, its frozen-unitary error relative to R is at most $b/2$, and its absolute area is at most $\pi/4+b/2$. If the pointer area is $\pi+\delta_g$ with $|\delta_g|\le u$, its frozen-unitary error relative to G is at most u and its absolute area is at most $\pi+u$. A sufficient full bound is therefore

$$\begin{aligned} r_{\rm cal} \le{}& \frac{b}{2}+u \\ &+\left(1+\frac{\pi}{2}+b\right)\sqrt{2C\tau_{\rm rot}} \\ &+(1+2\pi+2u)\sqrt{2C\tau_{\rm gate}}. \end{aligned} \tag{142}$$

This follows from $\|e^{-i\delta B}-I\|\le|\delta|$ for a self-adjoint contraction and the same hybrid comparison. A net-area uncertainty

does not bound the absolute area of a pulse with uncontrolled sign reversals; that case must supply A_{abs} itself. Nominal-area feasibility values use $b = u = 0$ in the pulse estimate. A separate timing or accumulated-phase allowance, for example $|\Omega\,\delta\eta|$, is not an analyser-axis error unless its declared control map makes it one. General calibrated work bounds likewise replace the nominal areas in Eq. (145) by these actual absolute-area ceilings.

There are two equivalent ways to allocate the rotation-axis error, and they must not be added together. Equation (142) compares every pulse with the *nominal* ideal rotation; the separate decoder phase/age error then excludes the same b. Alternatively compare the finite rotation with its *actual frozen* area and include b in the decoder's ideal-axis phase allowance. The finite-pulse part in that allocation is

$$\begin{aligned} r_{\text{actual axis}} \le u + \left(1 + \frac{\pi}{2} + b\right)\sqrt{2C\tau_{\mathrm{rot}}} \\ + (1 + 2\pi + 2u)\sqrt{2C\tau_{\mathrm{gate}}}, \end{aligned} \tag{143}$$

with no additional $b/2$. Both allocations use the actual absolute areas in their drift bounds. A nominal-axis benchmark with $b = 0$, whose stated phase allowance comes entirely from age/timing, has no such axis payment.

For a desired positive vector-error allowance $r_{\mathrm{rot}} + r_{\mathrm{gate}} = r_*$, one sufficient finite choice is

$$\begin{aligned} \tau_{\mathrm{rot}} &\le \frac{r_{\mathrm{rot}}^2}{2C^{\mathrm{up}}(1+\pi/2)^2}, \\ \tau_{\mathrm{gate}} &\le \frac{r_{\mathrm{gate}}^2}{2C^{\mathrm{up}}(1+2\pi)^2}. \end{aligned} \tag{144}$$

Choose strictly positive times. Their finite rectangular control heights are $\pi/(4\tau_{\mathrm{rot}})$ and π/τ_{gate}; tightening the error allowance charges these larger amplitudes explicitly.

At each on/off jump the changed part of the Hamiltonian is a bounded operator. The sum of absolute mean switching payments is therefore at most

$$\begin{aligned} W_{\mathrm{abs}}^{\mathrm{switch}} &\le 2\left(\frac{A_{\mathrm{rot}}}{\tau_{\mathrm{rot}}} + \frac{A_{\mathrm{gate}}}{\tau_{\mathrm{gate}}}\right) \\ &= \frac{\pi}{2\tau_{\mathrm{rot}}} + \frac{2\pi}{\tau_{\mathrm{gate}}}. \end{aligned} \tag{145}$$

Directly changing from one rectangular pulse to the next satisfies this same upper bound by the triangle inequality. This is an absolute bound on the declared driven switching work, not a deduction of autonomous controller capital. Initial source/field energy, the supplied blank pointer, actuator resources, retained outcomes, and any eventual reset remain separate recipients. Time and energy use the original model chart; no new SI assignment is supplied.

The selected projector control need not commute with the original source–field charge. Its actuator and phase/action reference are part of this external receiving model. The switching-work estimate does not by itself supply an autonomous charge-conserving actuator or its microscopic SAM realization.

The state to which the ideal readout refers is the state at pulse entry. Its ideal H_f bound is used throughout the comparison because the reference controls preserve H_f. The actual classical pointer result is available only after the latency $\tau_{\mathrm{rot}} + \tau_{\mathrm{gate}}$ and its subsequent registered readout. Calling it the source value at completion would require the corresponding event-time and phase-advance map. If the complete physical readout must lie inside the original certified window, choose its entry and positive pulse durations accordingly. No rate certificate is extended merely by omitting this finite latency.

The same fixed rotation, pointer gate, durations and output convention are used for all declared input-age hypotheses $\rho(\eta_j)$. The unknown label is not supplied to a control schedule. Its physical preparation and the two-arm arrival map are separate receiving-model data; a label-dependent ideal timing choice cannot be substituted for a single implemented decoder.

13. The instrument and an explicit finite pulse choice

Write $P_1 = P_{ba}$ and $P_0 = I_{16} - P_{ba}$. In the ideal frozen-control model with a blank pointer, the joint source–field Kraus operators and the measurement channel are

$$\begin{aligned} K_y &= (P_y R) \otimes I_f, \\ \mathscr{J}_y(\rho) &= K_y \rho K_y^*, \\ \sum_y K_y^* K_y &= I, \\ \mathscr{J}(\rho) &= \sum_{y=0}^{1} |y\rangle\langle y| \otimes \mathscr{J}_y(\rho). \end{aligned} \tag{146}$$

Thus $K_y^* K_y = E_y \otimes I_f$. The post-readout source can be disturbed by the projector; the field is retained in each output rather than replaced by a scalar marker. With the actual full drift and pulses, $\widetilde{K}_y = \langle y|U_{\mathrm{gate}}U_{\mathrm{rot}}|0\rangle$ is in general a joint source–field operator. Unitarity still gives $\sum_y \widetilde{K}_y^* \widetilde{K}_y = I$. The complete ideal and actual postmeasurement channels, with outcomes and quantum output retained, are within trace distance r on the reference input because the corresponding premeasurement pure states are within vector distance r. Applying the same channel to a preparation within trace distance δ_{prep} adds at most δ_{prep} to this joint-output distance. The two instruments therefore carry an all-outcome bound, not only a contrast estimate.

Use rectangular nominal pulses of equal finite duration

$$\begin{aligned} \tau_{\mathrm{rot}} = \tau_{\mathrm{gate}} = \tau = \frac{1}{32768}, \\ C^{\mathrm{up}} = 2\varepsilon^{\mathrm{up}} + 6c^{\mathrm{up}} + 4K_{\mathrm{prep}}^{\mathrm{up}} + 4\varepsilon^{\mathrm{up}}E_*. \end{aligned} \tag{147}$$

The exact native comparison evaluates

$$r^2 \le 2C^{\mathrm{up}}\tau\left(2 + \frac{5\pi_*}{2}\right)^2 \le 6.7949588 \times 10^{-7} < 10^{-6} \tag{148}$$

for all six lanes. In particular $r \le 0.00082431540$ across the six lanes, leaving the declared 10^{-3} control allowance sufficient for the nominal mathematical pulses. Their strengths and absolute

switching-work ceiling are finite and explicitly charged:

$$\begin{aligned} |\lambda_{\mathrm{rot}}| &\le \frac{\pi_*}{4\tau} < 25735.928, \\ |\lambda_{\mathrm{gate}}| &\le \frac{\pi_*}{\tau} < 102943.71, \\ W_{\mathrm{abs}}^{\mathrm{switch}} &\le \frac{5\pi_*}{2\tau} < 257359.28. \end{aligned} \tag{149}$$

These are original model units, without an assigned SI conversion or an observed control calibration. They display the strength/work cost of this short-pulse construction. An actuator/reference implementing the spectral projector and exchange drive must supply its charge, phase, timing and energy resources. The two pulses take the positive latency $1/16384$ before subsequent pointer registration. The ideal observable remains the observable at pulse entry.

14. A written bit on the actual nonuniform detector pointer

The completed Chapter IV detector supplies an actual continuum measurement and a finite uncentered second moment. Its positive intrinsic cap mean remains a separate source-to-field transport operand. The construction below uses the Chapter VI source decoder to write a calibrated displacement onto that actual detector pointer. The unknown continuum signal is retained in the background noise, rather than assigned a positive mean. Thus this is an explicit added bit-to-meter receiving law, not a certificate that the unmodified radiation alone distinguishes the two source ages.

The original sixteen source states, six signed lanes, both field polarizations and counterterms, hard dressing, soft input and all angular complements remain. One nominated lane and one of the actual local pointer coordinates suffice for the binary receiving protocol. If both compatible physical pointers are operated, the other pointer and its outcomes are retained; no independence between their records is inferred.

14.1 Re-key the field energy after the source decoder

Let the complete source-decoder output retain its two-state bit B and the source–field system. It can be correlated or entangled across these recipients. In the selected reference lane use

$$\begin{aligned} L &= H_0 - \varepsilon e I + \varepsilon I \ge 0, \\ H_f &\le 2L + 2cI, \\ c &= 2g^2\mu. \end{aligned} \tag{150}$$

These are the original completed-square form inequalities; the scalar subtraction is lane-specific. Suppose the reference decoder input has $\langle L\rangle \le C_{\mathrm{dec}}$ and its finite rectangular controls have total absolute switching-work bound W_{dec}. Their bounded Hamiltonian jumps and conservation between jumps imply

$$\begin{aligned} \langle L\rangle_{\mathrm{out}} &\le C_{\mathrm{dec}} + W_{\mathrm{dec}}, \\ E_{\mathrm{rec}} := \langle H_f\rangle_{\mathrm{out}} &\le 2(C_{\mathrm{dec}} + W_{\mathrm{dec}}) + 2c. \end{aligned} \tag{151}$$

The nonselective source-bit observation, or merely retaining the coherent bit, does not increase this source–field expectation. Do not condition this energy bound on a correct decoder outcome. All bit outcomes remain in the joint state. The old Chapter IV accepted-state energy and its numerical noise table are not substituted after the new driven decoder.

For an actual preparation supplied only within trace distance δ_{prep} of that reference, apply the same entire receiving channel to the two inputs and add δ_{prep} once at the final probability bound. No energy or form-domain assumption on the actual preparation follows from trace distance alone. Equation (151) and the ensuing second-moment calculation concern the finite-energy reference input and its complete, unselected decoder output.

14.2 The actual continuum gate on this new input

After the source bit is isolated, the actual nonuniform source–field propagator and detector gate act trivially on B. For a prescribed source-excluding local gate let X be its actual integrated field observable in the unmeasured joint Heisenberg picture. The original retained reflection and causal commutator statements give the scalar self-Magnus coefficient c_h. Write

$$\begin{aligned} C_f &\le 1.00301, \\ c_{\max} &= \frac{C_f^2 V_c}{2}, \\ |c_h| &\le c_{\max}. \end{aligned} \tag{152}$$

Here μ_c, V_c, D_c are the full compact-cap energy-dual, vacuum and half-energy squared norms, and $\mu_{\mathrm{sh}} = 1/(4R_s)$ is the original source-shell norm. The source radius R_s is distinct from the bit register B. Let t_* bound the actual new elapsed source-to-gate epoch, including the declared receiving chronology. The backward smooth-test estimate gives

$$\begin{aligned} \langle X^2\rangle &\le M_{\mathrm{rec}}, \\ M_{\mathrm{rec}} &= 6C_f^2\mu_c E_{\mathrm{rec}} + 3C_f^2 V_c \\ &\quad + 48g^2 C_f^2 n_+^2 \mu_{\mathrm{sh}} t_*^2 D_c, \qquad n_+ = \frac{2003}{2000}. \end{aligned} \tag{153}$$

The input may be mixed and correlated with B: the field estimate uses its unselected initial field energy and the unchanged bounded source-current norms. The bit is a spectator in this evolution. No final old-Fock energy, angular-vacuum factorization, Gaussian field state or positive cap mean is inserted. The signed g remains in the source law; its square appears only in this moment bound.

Let Q, P be the selected detector pointer with $[Q,P] = i$, and let kP^2, $k > 0$, be its declared kinetic Hamiltonian. The chosen k retains the Chapter IV field-form completion margin, or that margin is supplied by the additional quadratic stabilizer below. The gate has duration ℓ and normalized positive profile h. A small positive pointer mass coefficient alone does not stabilize the bilinear field gate.

For clarity, make the gate's form certificate explicit. In the static normal frame or the common moving frame used to construct the actual joint propagator, let $F_t \ge 0$ be the reference

field-energy form and suppose

$$\begin{aligned} H_{\mathrm{sf}}(t) &\geq a_0 F_t - b_0 I, \\ a_0 &> 0, \\ \|a_t(f)\psi\|^2 &\leq \mu_* F_t[\psi]. \end{aligned} \tag{154}$$

The latter energy-dual bound is in the same frame, with $\Phi_t(f) = a_t(f) + a_t(f)^*$. The finite coefficients a_0, b_0, μ_* belong to that actual compact-profile common-form certificate. They are not a claim that its conjugating implementer preserves the old $Q(H_f)$. If a fixed old-energy frame with the stated lower bound is available, one may take $F_t = H_f$ and $\mu_* = \mu_c$; that substitution must be certified, rather than inferred from the initial-energy bound. On the product form domain, energy-dual Cauchy–Schwarz gives

$$|h\langle P\Phi_t(f)\rangle| \leq \frac{a_0}{2}\langle F_t\rangle + \frac{2\mu_*}{a_0}h^2\langle P^2\rangle. \tag{155}$$

Thus add a known smooth compact pointer control $s(t)P^2$, with

$$s(t) \geq \frac{2\mu_*}{a_0}h(t)^2. \tag{156}$$

The total gate form retains the lower bound $(a_0/2)F_t + kP^2 - b_0 I$; a slightly larger s gives a strict relative form margin for the common-form construction. This is a declared receiving actuator, not a change of the original source counterterms. The compact actual frame supplies a finite common-form certificate; numerical stabilizer resources require its actual coefficients.

For the retained receiving geometry there is a concrete specialization. The original receiver radius is below 65, the prescribed quarter-age gate ends before T_0, and $65 + n_+ t_* < 2 \cdot 10^6$. The retained compact auxiliary is identically physical on that entire backward cap worldtube. This is the causal-agreement statement used for the incoming homogeneous transport estimate, so its exterior does not alter the local field gate. Use this same fresh postdecoder stage, with its gate still before T_0; the later translation and hold do not need to fit inside that interval. The Chapter IV source-energy certificate has $Q_\tau \geq (7/8)X$, $X \geq H_f/2$, and physical lapse at least $n_- = 1997/2000$. After the retained scalar shift its coefficient is at least $7n_-/16 > 1/4$, so one may use $a_0 = 1/4$ with a strict margin. Its inverse-frame weighted operator bound is

$$\begin{aligned} \|\omega^{-1/2}(p \pm q - I)\omega^{1/2}\| &\leq 401A, \\ A &= 2^{-83}. \end{aligned} \tag{157}$$

For a real coordinate cap test the pulled smearing is $(p \pm q)f$. Consequently $\mu_* \leq (1 + 401A)^2 \mu_c < C_f^2 \mu_c$, and a sufficient choice is

$$s(t) = 8C_f^2 \mu_c h(t)^2. \tag{158}$$

This specialization uses the weighted inverse-frame bound, not an assertion of old-Fock form-domain preservation. Source-region agreement alone would be insufficient: the full detector-worldtube agreement is used here explicitly. A different geometry or longer field gate requires its own version of that causal and common-form certificate. Here $S_g = 8C_f^2\mu_c \quad h(t)^2\,dt$, and the peak and total variation are the same coefficient times $\|h\|_\infty^2$ and $\mathrm{TV}(h^2)$, respectively.

Put $S_g = \int_{\mathrm{gate}} s(t)\,dt$. Since P^2 commutes with the field gate interaction at every time, this addition changes only a known free pointer shear. Its exact pointer equation is

$$Q_{\mathrm{gate}} = Q + 2(k\ell + S_g + c_h)P + X. \tag{159}$$

The initially prepared pointer is independent of the entire B–source–field state. This pointer preparation is the only product preparation used in the following Gaussian convolution; the bit and continuum need not be independent.

14.3 An exact finite controlled translation

Put $Z_B = |0\rangle\langle 0| - |1\rangle\langle 1|$. After the field gate has ended, assign a local pointer-coordinate control of duration $\tau > 0$,

$$H_w(t) = kP^2 + \lambda(t)Z_B P + \frac{\lambda(t)^2}{4k}I = k\left(P + \frac{\lambda(t)Z_B}{2k}\right)^2 \geq 0. \tag{160}$$

The real pulse has area $\int_0^\tau \lambda(t)\,dt = a > 0$. All displayed generators commute, so

$$U_w = \exp\left[-ik\tau P^2 - iaZ_B P - i\int_0^\tau \frac{\lambda(t)^2}{4k}\,dt\right]. \tag{161}$$

This is a direct sum of translated massive-pointer Hamiltonians on the two bit sectors. The unbounded momentum is handled by its spectral calculus and positive quadratic completion, rather than called a bounded bit control. On constant pulse segments the operators are self-adjoint on the momentum-squared domain; their form domain requires only finite $\langle P^2\rangle$. The exponential is defined on the whole Hilbert space. This adds a specified bit-to-meter interaction, not a derived microscopic SAM actuator or an autonomous controller.

Focus the initial Gaussian pointer for the known combined gate-and-write duration: set

$$\begin{aligned} Y &= Q + 2[k(\ell + \tau) + S_g]P, \\ \langle Y\rangle &= \langle P\rangle = 0, \\ \mathrm{Var}\,Y = \sigma^2, &\quad \mathrm{Var}\,P = \frac{1}{4\sigma^2}, \\ \mathrm{Cov}_{\mathrm{sym}}(Y,P) &= 0. \end{aligned} \tag{162}$$

After writing and a further free hold $s \geq 0$, the exact output coordinate in the initial picture is

$$Q_{\mathrm{out}}(s) = Y + X + 2(c_h + ks)P + aZ_B. \tag{163}$$

The write drift and the stabilizer's known shear have been included in the focus. Simply reusing a focus for the field gate alone would leave additional drift in the noise. The stabilizer leaves the actual X, c_h and source backreaction law unchanged. The bit commutes with the actual joint field gate and with X, even when its state is correlated with the field.

14.4 A finite retention window without an intrinsic mean floor

Declare one read at any chosen hold time $s \in [0, T_A]$. The background coordinate $Q_b(s) = Y + X + 2(c_h + ks)P$ has uncentered

second moment

$$\langle Q_b(s)^2\rangle \le M_{\rm rec} + \sigma^2 + \frac{(c_{\max}+kT_A)^2}{\sigma^2}. \tag{164}$$

The cross expectations between the centered initial Gaussian variables and X vanish by their initial product preparation. This statement does not factor an expectation involving Z_B and X. Choose the finite width

$$\sigma^2 = c_{\max} + kT_A > 0. \tag{165}$$

Then the following uniform bound holds:

$$\begin{aligned} B_A &= M_{\rm rec} + 2(c_{\max}+kT_A) \\ &= 6C_f^2\mu_c E_{\rm rec} + 4C_f^2 V_c \\ &\quad + 48g^2C_f^2 n_+^2 \mu_{\rm sh} t_*^2 D_c + 2kT_A, \\ \sup_{0\le s\le T_A} \langle Q_b(s)^2\rangle &\le B_A. \end{aligned} \tag{166}$$

The positive width follows from the nonzero cap test $V_c > 0$. Thus the construction gives a finite noisy position record for a declared finite hold interval. It does not assume an indefinitely stable free pointer or an unmodelled sequence of nondestructive reads.

Decode pointer positions $q > 0$ as bit zero and $q \le 0$ as bit one. Since Z_B and $Q_b(s)$ commute, their spectral probabilities are an ordinary joint probability distribution, with arbitrary correlations. If the pointer sign disagrees with the stored bit, then $|Q_b(s)| \ge a$. Spectral Markov therefore gives

$$P(\widehat{r} \ne r) \le \frac{\langle Q_b(s)^2\rangle}{a^2} \le \frac{B_A}{a^2}. \tag{167}$$

No conditional moment divided by the probability of a correct source bit is used. For each true source-age label j, if the source decoder has error at most $\varepsilon_{\rm src}^{\rm ref}$ on the reference input, the union of the source-error and transfer-error events gives

$$\varepsilon_{\rm final}^{\rm ref} \le \min\left\{1, \varepsilon_{\rm src}^{\rm ref} + \frac{B_A}{a^2}\right\}. \tag{168}$$

This inequality needs neither independence of errors nor a known mean of the actual field. Any desired positive transfer allowance η has the finite sufficient displacement $a \ge \sqrt{B_A/\eta}$.

For a subsequent smooth sign-controlled memory write, let its trusted positive and negative position bands start at w and $-w$, with $0 < w < a$. The probability of being in the wrong band or its unresolved central region obeys the same joint-event argument:

$$P(\text{position outside its bit's trusted band}) \le \frac{B_A}{(a-w)^2}. \tag{169}$$

This is a one-time position probability at the specified memory-entry time. Motion during a later memory gate requires that gate's own bound; it is not an assertion that the pointer is parked.

If the signed displacement magnitude differs from a by at most δ_a and the threshold has position error at most δ_q, replace the denominator in Eq. (167) by $(a \quad \delta_a \quad \delta_q)^2$, with a strictly positive margin. Nearest-bin rounding of width w can be included by increasing δ_q by $w/2$. This is a finite threshold margin, not a demand for an infinitely fine position ruler. A finite aperture $|q| \le L_q$ can retain overflow as an explicit failure outcome. Its additional probability is at most

$$P(\text{overflow}) \le \frac{(\sqrt{B_A}+a+\delta_a)^2}{L_q^2}. \tag{170}$$

The estimate uses the second-moment triangle inequality, with no factorization of bit and background. No overflow is silently discarded.

14.5 The complete receiving POVM and instrument

The actual Gaussian pointer variance at hold time s is $\nu(s) = \sigma^2 + (c_h + ks)^2/\sigma^2$. Let Γ_ν be its centered Gaussian probability measure and let E_X be the actual spectral measure of X. With $I_0 = (0,\infty)$, $I_1 = (-\infty, 0]$ and $\Pi_r = |r\rangle\langle r|_B$, the joint bit/output effects in the unmeasured source–field picture are

$$\begin{aligned} F_{r,y} &= \Pi_r \int \Gamma_{\nu(s)}\big(I_y - x - a(-1)^r\big)\, E_X(dx), \\ F_y &= \sum_r F_{r,y}. \end{aligned} \tag{171}$$

These positive effects sum to the identity. They are the actual Chapter IV Gaussian-convolution law with the added controlled displacement. They retain the state-dependent continuum spectral law; it has not been replaced by a fitted cap mean. Unknown c_h has its stated uniform bound, while its actual value defines the exact instrument.

For completeness let $U_{\rm all}(s)$ denote the actual nonuniform joint evolution, field gate, write unitary and free hold, including the unmeasured source–field drift. For the supplied independent Gaussian pointer state σ_D, define

$$\begin{aligned} \mathscr{I}_y(\rho) &= (I\otimes \mathscr{P}_y) U_{\rm all}(s)(\rho\otimes\sigma_D) U_{\rm all}(s)^*(I\otimes\mathscr{P}_y), \\ \mathscr{P}_y &= \mathbf{1}_{I_y}(Q). \end{aligned} \tag{172}$$

Here ρ includes the bit, source and full field, and the pointer is kept in the quantum output. The maps are completely positive and their sum is trace preserving. A classical outcome register may be added as $\sum_y |y\rangle\langle y| \otimes \mathscr{I}_y(\rho)$. The overflow refinement likewise includes every outcome. Tracing some recipients for a reported marginal does not remove them from the physical resource account.

Equation (172) is the ideal sharp position-event instrument. A discontinuous position projection generally does not preserve finite pointer kinetic energy, so it is not assigned a finite registration-work implementation. A smooth finite-energy target instrument can instead use real smooth multipliers

$$\begin{aligned} m_0(q) &= \cos\vartheta(q), \\ m_1(q) &= \sin\vartheta(q), \\ \vartheta(q) &= \begin{cases} \pi/2, & q \le -w, \\ 0, & q \ge w, \end{cases} \\ 0 &\le \vartheta \le \pi/2, \end{aligned} \tag{173}$$

with bounded derivatives and a smooth connection inside $(\quad w, w)$. Replace each $\mathscr{P}_y$ in the retained-pointer map by

$m_y(Q)$. The resulting maps again form a complete instrument. Their effects on the bit–source–field input are the convolution of m_y^2 with the same displaced actual continuum Gaussian law. Outside the central band they return the correct sign with certainty, so the bit-transfer error is bounded by Eq. (169). For any finite pointer momentum second moment, their nonselective output obeys

$$\begin{aligned}\sum_y \|Pm_y(Q)\psi\|^2 &= \|P\psi\|^2 + \langle\psi, \vartheta'(Q)^2\psi\rangle \\ &\le \|P\psi\|^2 + \|\vartheta'\|_\infty^2\|\psi\|^2.\end{aligned} \tag{174}$$

This identity extends to mixed and entangled inputs. It describes the real-multiplier target instrument. The positive memory pulse in the next section has the same effects but includes its coordinate-dependent common phase: with $\vartheta = \pi A_w/2$, its Kraus operators are $e^{-i\vartheta}\cos\vartheta$ and $ie^{-i\vartheta}\sin\vartheta$. For that instrument the corresponding identity is $\sum_y \|PK_y\psi\|^2 = \|(P-\vartheta')\psi\|^2 + \langle\vartheta'^2\rangle$. Its finite-write estimate below includes this phase through the full multiplier derivative bounds. A two-state register has the bounded frozen-coordinate target dilation generated by $\vartheta(Q)$ times its Pauli rotation. Its finite-duration physical implementation must retain the active kP^2 drift and charge the corresponding memory-gate error; the smooth target is not a hidden pointer-parking assumption.

If the original preparation is within δ_{prep} of the reference and the specified access, storage or registration channel has a further worst-label joint-state error budget δ_{access}, contractivity of this same complete channel gives

$$\varepsilon_{\text{final}} \le \min\left\{1, \varepsilon_{\text{src}}^{\text{ref}} + \frac{B_A}{a^2} + \delta_{\text{prep}} + \delta_{\text{access}}\right\}. \tag{175}$$

Use the calibrated denominator and overflow term when those outputs are part of the declared protocol. The preparation distance is paid once on the entire channel; it is not also included in $\varepsilon_{\text{src}}^{\text{ref}}$. If a previously quoted source ready-error budget already includes that distance, reassemble its stated additive components rather than adding the same preparation allowance twice. Ready failures remain explicit, and counting a per-label failure at most f as unsuccessful gives the all-attempt error $f + (1-f)\varepsilon_{\text{final}}$.

14.6 Finite energy, switching work and causal event map

The focused Gaussian has finite pointer momentum energy $k\langle P^2\rangle = k/(4\sigma^2)$ and finite initial position variance $\sigma^2 + [k(\ell+\tau)+S_g]^2/\sigma^2$. Momentum is conserved through the field gate, translation and hold. For a rectangular write with $\lambda = a/\tau$, the two jumps of Eq. (160) have total absolute mean work bounded by

$$W_{\text{write}}^{\text{abs}} \le 2|\lambda|\sqrt{\langle P^2\rangle} + \frac{\lambda^2}{2k} = \frac{a}{\tau\sigma} + \frac{a^2}{2k\tau^2}. \tag{176}$$

This includes the positive scalar completion on both jumps. The signed write payment telescopes for this commuting control; its absolute borrow, controller capital and pointer preparation are still charged. The actual field-gate switching, deposited energy and backreaction retain the Chapter IV ledger. The added stabilizer has finite peak expected capital $\|s\|_\infty/(4\sigma^2)$ and additional absolute work at most

$$W_{\text{stab}}^{\text{abs}} \le \langle P^2\rangle \quad |s'(t)|\,dt = \frac{\mathrm{TV}(s)}{4\sigma^2}, \tag{177}$$

including endpoint jumps if used. Its known focus preparation also has the finite position variance just displayed. No global renormalized vacuum-ramp work or thermal reset cost is inferred from Eq. (176).

One explicit finite model choice is

$$\begin{aligned} a &= 2^{10}, \\ k &= 2^{-40}, \\ \tau &= 2^{20}, \\ \lambda &= 2^{-10}, \\ T_A &= 2^{20}. \end{aligned} \tag{178}$$

Its pointer mass in the convention $k = 1/(2M_D)$ is 2^{39}, and its write completion height is $\lambda^2/(4k) = 2^{18}$. The focused variance is $\sigma^2 = c_{\max} + 2^{-20}$ and $B_A = B_{\mathrm{N2}}(E_{\mathrm{rec}}, t_*) + 2^{-19}$. With trusted-band width $w = a/2$, Eq. (169) is $4B_A/a^2$. These numbers exhibit finite amplitude, capital and latency in model units; they are not measured SI calibration. The field-gate stabilizer still pays its actual certified S_g, peak and variation separately.

The source decoder, actual continuum gate and added translation have separate finite durations. The retention window starts after writing; the receiving result exists only after the chosen hold and registered read. A spatially local Z_BP actuator requires the bit carrier to be co-located with the selected meter. If the decoded bit is delivered from elsewhere, its causal transport, delay and possible errors belong in the declared access map and the new t_* chronology. It is not supplied instantaneously to a remote detector. The same fixed schedule and classification rule apply to both unknown source-age hypotheses. The source-clock phase certificate is needed before the bit is decoded. The new t_* concerns the finite actual continuum stage through its field gate; the long write and hold occur after that coupling is off and do not extend a source-clock or field-gate certificate to later times.

This supplies a complete continuum-backed receiving instrument and a finite noisy pointer-retention model under its stated actuator and access laws. The intrinsic nonuniform cap-mean transfer, microscopic actuator, achieved calibration and a physical carrier's durable-memory/access performance remain distinct operands; none is replaced by the added known displacement.

15. A finite pointer-to-memory instrument with causal local access

Source and scope. The retained SAM_land donor readout construction supplies neutral record bits with degenerate rest channels and positive bounded couplings proportional to $I - X$; its finite mark/write/unmark law retains the complete complementary cloud. Its final record is held, with reset treated separately (`samland_donor_readout1/DERIVATION.md`, Section 4, and its `ENERGY_ACCOUNT.md`). We reuse that kind of record recipient below, with a new, explicitly specified smooth coordinate response. The physical memory covariance of `volii_physical_memory1` and the colored classical

field bath of `voli_radiative_memory1` retain their own source laws; neither supplies a spin-register flip rate here.

The new instrument, finite-write estimate and correlated-hold bounds are developed in campaign voli_chvi_detector_record1_20261017, with four native SLC-GEN5-R1 calls. They join a decoded bit to the actual detector pointer under the stated receiving law. The coordinate-to-qubit coupling specified below is an explicit apparatus assignment; it is not derived here from a microscopic material actuator. They do not extract a new positive cap mean from the old field, assign a microscopic material or temperature, or replace the later two-arm transport calculation. Local controls in this section require the decoded bit, detector pointer and memory to be physically co-located, or a previously specified causal delivery channel bringing the bit to that location.

15.1 A correlated loaded pointer and its complete binary conversion

Let B be the isolated decoder-output qubit from the source-only protocol, with $P_b = |b\rangle\langle b|$, $b \in \{0,1\}$. For each operational input label j, let ρ_j be the finite-energy reference output of the actual decoder pulses on the ideal prepared marker. Its error obeys $\mathrm{Tr}(P_{1-j}\rho_j) \le e_{\mathrm{dec,ref},j}$, excluding the marker preparation distance δ_{prep}. The field can remain correlated with this bit. Let the actual analog detector noise observable be N, acting on the pointer/source/field factors outside B, so $[N,P_b]=0$. Assume the retained full detector calculation supplies the per-label moment bound

$$\begin{gathered}\mathrm{Tr}(\rho_j N^2) \le \mathscr{B}_{\mathrm{ret}} \\ \text{uniformly over the declared} \\ \text{analog receiving interval.}\end{gathered} \tag{179}$$

The joint state ρ_j may correlate B, N and all other degrees of freedom. No conditioning on an exceptionally rare bit value is used to claim the same moment bound.

After the detector field gate, the specified bit-controlled translations load the same pointer with signs $+a$ for $b=0$ and $-a$ for $b=1$. Its loaded coordinate is therefore

$$\begin{gathered}Q = N + a(P_0 - P_1), \\ a > 0.\end{gathered} \tag{180}$$

Equivalently, in a Schrödinger description the loading unitary translates the ordinary pointer coordinate and the displayed equation specifies its input Heisenberg observable. The source-defined field distribution and its correlations remain in N. The added sign is the decoded bit's paid apparatus displacement, not an assumed intrinsic mean of N.

Choose $0 < w < a$. With $x = (q+w)/(2w)$ define the C^1 threshold response

$$\begin{gathered}A_w(q) = \begin{cases} 1, & q \le -w, \\ 1 - 3x^2 + 2x^3, & -w < q < w, \\ 0, & q \ge w. \end{cases} \\ 0 \le A_w \le 1, \\ G := \|A_w'\|_\infty = \frac{3}{4w}.\end{gathered} \tag{181}$$

This A_w is an apparatus response, not the accumulation scalar $A(r)$ or the computational `A_OPERATOR`.

For a finite registration aperture, take $L_q > w$ and a C^1 multiplier χ_{L_q} equal to one on $|q| \le L_q$ and zero on $|q| \ge 2L_q$, with cubic shoulders and $\|\chi_{L_q}'\|_\infty \le 3/(2L_q)$. Use $A_{w,L_q} = \chi_{L_q} A_w$ in the physical memory pulse. Its response remains in $[0,1]$ and has compact coordinate support. The two derivative supports are disjoint, so $\|A_{w,L_q}'\|_\infty \le \max\{G, 3/(2L_q)\}$. The finite-pulse derivative bounds below apply with that derivative bound. The uncut threshold formulas acquire the following additional wrong-bit probability, charged once through the final aperture/read allowance:

$$\varepsilon_{\mathrm{ap}} \le \frac{(\sqrt{\mathscr{B}_{\mathrm{ret}}} + a)^2}{L_q^2} \le \frac{2(\mathscr{B}_{\mathrm{ret}} + a^2)}{L_q^2}. \tag{182}$$

This follows from $Q = N + aZ_B$ and the second-moment triangle inequality. The trusted-band bound is supplemented by this aperture term; its output is still part of the complete binary experiment. For $L_q = 2^{20}$ and $w = 2^9$, the original G is unchanged. The final budget can absorb this probability into a stated combined aperture/read allowance.

Supply one memory qubit R, initially $|0\rangle_R$, with degenerate rest Hamiltonian, and set $X_- = (I - X_R)/2$ and $D_w = A_w(Q) \otimes X_-$. The ideal positive write pulse has area π:

$$\begin{gathered}V_\theta = e^{-i\theta D_w}, \quad 0 \le \theta \le \pi, \\ U_w = V_\pi.\end{gathered} \tag{183}$$

It is exactly the identity on the memory for $q \ge w$, and exactly X_R for $q \le -w$. At every coordinate it is unitary, including the finite transition band. Reading the memory in its Z_R basis gives the two Kraus operators on the retained incoming system

$$\begin{aligned}K_0(Q) &= e^{-i\pi A_w(Q)/2}\cos[\pi A_w(Q)/2], \\ K_1(Q) &= ie^{-i\pi A_w(Q)/2}\sin[\pi A_w(Q)/2],\end{aligned} \tag{184}$$

with $K_0^* K_0 + K_1^* K_1 = I$. Both outcomes and their conditional residual source/pointer/field states are retained. The transition band is not postselected away; its possibly erroneous binary output is included. The memory need not be read immediately: Eq. (183) first creates a coherent memory correlation, to which the hold channel below can be applied.

Proposition (threshold error without independence). For each label j, the ideal write obeys

$$e_{\mathrm{thresh,ref},j} \le \min\left\{1, e_{\mathrm{dec,ref},j} + \frac{\mathscr{B}_{\mathrm{ret}}}{(a-w)^2}\right\}. \tag{185}$$

Proof. On the simultaneous spectral subspace where the decoded bit equals j and $|N| < a - w$, Eq. (180) lies strictly beyond the correct side of the transition band. The wrong memory effect vanishes there. Since it is a positive contraction, the commuting spectral calculus gives

$$K_{1-j}(Q)^* K_{1-j}(Q) \le P_{1-j} + \mathbf{1}_{\{|N| \ge a-w\}}. \tag{186}$$

The spectral second-moment inequality bounds the second term's expectation by $\mathscr{B}_{\mathrm{ret}}/(a-w)^2$. Taking the expectation in the full joint state establishes Eq. (185), with no factorization of bit and detector noise. □

15.2 A finite positive pulse with the pointer kinetic energy retained

Let the pointer's actual rest Hamiltonian be $H_P = kP^2$, $k > 0$, $P = -i\partial_Q$, and retain it throughout the write. A rectangular pulse of duration $\tau_w > 0$ is

$$H_{\mathrm{write}} = kP^2 + \frac{\pi}{\tau_w} D_w, \qquad U_{\mathrm{write}} = e^{-i\tau_w H_{\mathrm{write}}}. \tag{187}$$

Both terms are positive; there is no kinetic parking. The actual source–field generator $H_{\mathrm{sf}}(t)$ is a spectator commuting with these pointer and memory operators. It factors from the comparison even when its state is entangled with the pointer. Its own evolution and energy are retained. Equation (187) specifies the write-stage memory law: any additional noncommuting bath coupling during that stage needs a separate write-error bound. Activating a different hold coupling afterward retains its switching work rather than parking the pointer.

Suppose the joint write-entry state has $\langle P^2\rangle \le p_2$. In the intended pipeline the detector gate and bit-controlled translations depend on the analog pointer momentum and commute with P^2; its free analog hold does also. Thus the declared Gaussian input moment $p_2 = 1/(4\sigma^2)$ travels unchanged through those stages. A further control that does not preserve this moment needs its own bound. Purification includes arbitrary correlations with the remaining systems. The bounded C^1 multiplier A_w preserves $D(P)$, and for a normalized purification ψ,

$$\|PV_\theta\psi\| \le \sqrt{p_2} + |\theta| G, \qquad \|PD_w V_\theta \psi\| \le \sqrt{p_2} + (|\theta|+1)G. \tag{188}$$

These follow from $[P, D_w] = -iA_w'(Q)X_-$ and differentiation of the commuting multiplication exponential. In particular both ideal orbits needed by the bounded-control comparison have free-energy form expectation at most

$$C_w = k[\sqrt{p_2} + (\pi+1)G]^2 \le C_w^{\mathrm{up}} := 2k[p_2 + (\pi_*+1)^2 G^2], \qquad \pi_* \ge \pi. \tag{189}$$

No $D(P^2)$ assumption on the initial state is required for this estimate. The self-adjoint Hamiltonian in Eq. (187) is the nonnegative free Hamiltonian plus a bounded self-adjoint control.

Proposition (finite-write discrepancy). For the specified reachable input family,

$$\|(U_{\mathrm{write}} - U_w)\psi\| \le r_w := (1+2\pi)\sqrt{2\tau_w C_w} \le (1+2\pi_*)\sqrt{2\tau_w C_w^{\mathrm{up}}}. \tag{190}$$

The corresponding complete joint-output trace distance and each memory-outcome probability error are at most $\min\{1, r_w\}$.

Proof. Set $F(t) = e^{-itH_P}$ and move the actual pulse into its free interaction picture. Its bounded generator is $(\pi/\tau_w)F(t)^* D_w F(t)$; the ideal one is $(\pi/\tau_w)D_w$. For any form-domain vector ξ, the spectral theorem gives

$$\|(F(t)-I)\xi\|^2 \le 2|t|\langle \xi, H_P \xi\rangle. \tag{191}$$

Expanding $F(t)^* D_w F(t) - D_w = F(t)^* D_w (F(t)-I) + (F(t)^* - I)D_w$ and using Eq. (189) bounds its action on the ideal orbit by $2\sqrt{2tC_w}$. Bounded-generator Duhamel therefore bounds the interaction-picture difference by $2\pi\sqrt{2\tau_w C_w}$, using $t \le \tau_w$ under the integral. Returning from that picture costs at most $\sqrt{2\tau_w C_w}$ on the final ideal orbit. Add the two bounds. Purifying a mixed input and then taking partial traces gives the same output trace-distance conclusion. □

15.3 Passive hold and correlated transverse disturbances

After writing, turn off the write pulse. The memory rest law is the same degenerate channel used by the retained SAM_land record construction. A more general admitted passive hold is

$$H_d(t) = |0\rangle\langle 0|_R \otimes L_0(t) + |1\rangle\langle 1|_R \otimes L_1(t), \qquad [H_d(t), Z_R] = 0, \tag{192}$$

with well-defined block propagators on the retained environment. This environment includes the pointer, source, field and every noise reference needed to describe correlations. The memory may dephase or acquire joint environmental records, but each Z_R population is exactly constant. Thus its readable binary record is held for every declared finite interval under this law. No refresh or measurement is needed during the hold.

For an actual transverse disturbance, split any admitted additional interaction into its diagonal part, absorbed in H_d, and its bounded off-diagonal part

$$V_{\mathrm{off}}(t) = \Pi_0 V(t)\Pi_1 + \Pi_1 V(t)\Pi_0, \quad \Pi_b = |b\rangle\langle b|_R \otimes I, \qquad \Lambda_T = \int_0^T \|V_{\mathrm{off}}(u)\|\,\mathrm{d}u. \tag{193}$$

The norm is on the full retained joint system, not an average of independent cell errors. Interaction-picture bounded-perturbation Duhamel gives a joint-channel trace-distance error at most $\min\{1, \Lambda_T\}$ on every input and reference. Consequently the wrong-label population can increase by at most $\min\{1, \Lambda_T\}$ uniformly for $0 \le t \le T$.

A sharper population envelope is also available. If its initial value is e_0, then

$$e(t) \le \sin^2\left(\min\left\{\frac{\pi}{2}, \arcsin\sqrt{e_0} + \Lambda_t\right\}\right). \tag{194}$$

To see this, purify the complete state. In the diagonal interaction picture, the wrong-sector probability obeys, almost everywhere, $|\dot e| \le 2\|V_{\mathrm{off}}\|\sqrt{e(1-e)}$ by Cauchy–Schwarz between the two orthogonal sectors. Integrating the bound on $\arcsin\sqrt{e}$ gives Eq. (194), with endpoint values obtained by continuity. For $e_0 = 0$ this becomes $e(t) \le \sin^2(\min\{\pi/2, \Lambda_t\})$. Use this population envelope or the additive channel error in a budget; do not charge the same disturbance twice.

This is a passive finite-spin receiving model, not an assigned Markov flip rate. For example a finite correlated thermal environment may be declared by a positive Ising Hamiltonian

$$H_E = \sum_\ell h_\ell \frac{I - Z_\ell}{2} + \sum_\ell J_\ell \frac{I - Z_\ell Z_{\ell+1}}{2}, \qquad h_\ell, J_\ell \ge 0, \quad \rho_E = \frac{e^{-\beta H_E}}{\mathrm{Tr}\, e^{-\beta H_E}}, \tag{195}$$

and a static coupling $V_{\mathrm{off}} = X_R \otimes B_x + Y_R \otimes B_y$. Its actual matrices give $\Lambda_T \le T(\|B_x\| + \|B_y\|)$, regardless of thermal correlations. Diagonal Z_R couplings may be retained in Eq. (192). If a nonnegative finite total bath Hamiltonian is desired, include the common offset $\|V_{\mathrm{off}}\| I$ (and the norm of any additional bounded diagonal coupling) in its energy account. This does not change the memory probabilities. The bath preparation, β, couplings and energy offsets would be new receiving operands; no such numerical thermal model is inferred from an isotope covariance or inserted into the example below. No thermal activation advantage is claimed by this state-independent norm bound.

15.4 Readout, complete outcomes and causal access

For local readout supply a blank qubit A. A positive bounded pulse

$$\begin{aligned} H_{\mathrm{copy}}(t) &= u(t)\Pi_1 \otimes X_{-,A}, \\ u(t) \ge 0, &\quad \int u(t)\,\mathrm{d}t = \pi \end{aligned} \tag{196}$$

copies the memory's orthogonal Z_R label to A. It commutes with the diagonal hold law, even with the full correlated environment retained. The final Z_A measurement uses the retained Born/instrument premise. This specifies its ideal back-action: it preserves memory label populations, while permitting correlation and dephasing. An actual copy/read channel may instead carry a declared outcome error δ_{read} uniform over the receiving interval. A transverse exposure during copying is included either in the extended hold exposure or in that read error, once.

More explicitly, first append the blank memory, apply the finite write, then the full hold channel, append the read qubit and apply its copy unitary. Projection onto $|b\rangle_A$ followed by the read-register partial trace defines a completely positive instrument map $\mathscr{J}_b^t$ on the retained systems. The two maps sum to a trace-preserving channel. No residual source, pointer or environmental cloud is replaced by a pure state merely because a classical outcome has been registered. For an input already carrying the chapter's ready/failure flags, define

$$\begin{aligned} \mathscr{I}_b^t(\widehat{\rho}) &= \mathscr{J}_b^t(P_{\mathrm{ready}}\widehat{\rho}P_{\mathrm{ready}}), \\ \mathscr{I}_{\mathrm{fail}}^t(\widehat{\rho}) &= P_{\mathrm{fail}}\widehat{\rho}P_{\mathrm{fail}}, \end{aligned} \tag{197}$$

with multiple failure flags kept as separate blocks when supplied. Their traces sum to one. Failure is a recorded outcome, not a missing attempt.

For the closed local protocol let the receiving region contain the memory worldtube and read apparatus. Its access channel is the local channel just specified, available after the finite write/copy completion. Retention is tested for one read at any chosen time in its declared interval. Repeated reads require their additional copy gates, blank qubits, disturbance and work accounts; single-read performance does not automatically pay for them.

For a remote region, an additional causal delivery channel is an explicit operand. Before its first allowed arrival t_{arr} the receiver output has a common no-access flag, described by the trace-preserving replacement map $\omega \mapsto \mathrm{Tr}(\omega)|\perp\rangle\langle\perp|$. After arrival, a supplied delivery channel and its error can be composed with the memory readout. The actual Chapter IV hyperbolic propagation constrains source-generated changes outside their causal future; neither this memory coupling nor the word "access" supplies an instantaneous remote control. The local construction is complete without selecting a new remote carrier. Its remote extension must supply arrival and delivery accuracy, rather than infer them from a finite register circuit. Memory survival on $[0, T_{\mathrm{store}}]$ and remote readability after t_{arr} are therefore distinct intervals.

15.5 The resulting uniform error and resource ledger

Let $\delta_{\mathrm{acc}} = \delta_{\mathrm{read}} + \varepsilon_{\mathrm{ap}}$ include the physical aperture term when that response is used. Let δ_{init} bound any additional joint ready-state error for the analog-pointer and blank-memory receiving preparation, relative to the family for which the moment and momentum bounds above hold. Propagate this error as a trace distance through the complete instrument. It does not imply a second-moment bound for an arbitrary nearby actual state, and it is not a second charge for marker preparation. First apply the threshold and finite-write estimates to the finite-energy reference pipeline. The actual marker preparation is within δ_{prep} of that reference at the original input; contraction through the entire actual receiving channel adds δ_{prep} once at its final output. For compact notation write $e^{\mathrm{r}}_{\mathrm{dec}} = e^{\mathrm{r}}_{\mathrm{dec,ref}} + \delta_{\mathrm{prep}}$, as in the inherited decoder budget. This notation includes the final whole-channel preparation payment; it does not transfer a moment bound to an arbitrary nearby actual state. Combining the preceding estimates gives the ready-conditional local bound

$$\boxed{e^{\mathrm{r}}_{\mathrm{rec}} \le \min\left\{1, e^{\mathrm{r}}_{\mathrm{dec}} + \frac{\mathscr{B}_{\mathrm{ret}}}{(a-w)^2} + r_w + \delta_{\mathrm{init}} + \Lambda_{T_{\mathrm{store}}} + \delta_{\mathrm{acc}}\right\}.} \tag{198}$$

All terms are per-label uniform bounds on the states where they are used. The threshold argument handles bit/field correlations directly, and the channel/hold arguments retain their environment. No independence between decoding, analog noise, writing, storage and access is required. For a complete attempt with failure probability at most f, counted as unsuccessful, the bound becomes

$$e^{\mathrm{all}}_{\mathrm{rec}} \le f + (1-f)e^{\mathrm{r}}_{\mathrm{rec}}. \tag{199}$$

If failure was already included in an all-attempt channel error, it must not be charged again by Eq. (199).

A rectangular write has peak π/τ_w, operator-norm action π, and switch-work variation at most $2\pi/\tau_w$ in the common model chart. Indeed its turn-on work lies in $[0, \pi/\tau_w]$ and its turn-off work in $[-\pi/\tau_w, 0]$. For finite-rest-energy inputs the signed total equals the change of pointer rest energy when the controller is zero at both endpoints; the spectator source–field energy is conserved by this coupling. This signed work is not identified with heat. The finite copy pulse has the same action and, at duration τ_r, peak π/τ_r and switch-work variation at most $2\pi/\tau_r$. Its ideal diagonal-hold commuting limit has cancelling signed on/off work; peak and temporary borrow remain charged.

The ledger additionally retains the upstream field-gate and bit-loading translation controls, the pointer kinetic preparation, one fresh blank memory bit and one read bit per supplied protocol, all failed preparation attempts and their ready/failure

preparation accounts, any hold bath and coupling energy, physical delivery when used, and later reset/disposal. A degenerate rest energy is not a claim of free pure-state preparation or a Landauer heat formula. A cyclic erasure bound would require its own reservoir, temperature and logical ensemble. Model energies/work convert through the already declared action/time units; no SI calibration is inferred.

For the proposed arithmetic recipient one may use

$$\begin{aligned} a = 2^{10}, \quad w = a/2 = 2^9, \quad k = 2^{-40}, \\ T_{\mathrm{analog}} = 2^{20}, \quad \tau_w = 2^{-20}, \\ \sigma^2 = C_{\mathrm{prop}}^2 V_{\mathrm{up}}/2 + kT_{\mathrm{analog}}, \\ p_2 = \frac{1}{4\sigma^2}, \\ \mathscr{B}_{\mathrm{ret}} = \mathscr{B}_{\mathrm{new}} + 2kT_{\mathrm{analog}}. \end{aligned} \tag{200}$$

Here the propagation and detector ceilings retain their previously defined source meanings; these equations specify the receiving pointer rather than a measured calibration. The analog contribution is $4\mathscr{B}_{\mathrm{ret}}/a^2$, and the finite-write allowance $r_w \le 10^{-4}$ is used only after checking Eq. (190) with the corresponding outward inputs. A logical hold interval $T_{\mathrm{store}} = 2^{20}$ with $\Lambda_{T_{\mathrm{store}}} \le 10^{-4}$, together with $\delta_{\mathrm{init}} \le 10^{-4}$ and $\delta_{\mathrm{acc}} \le 10^{-4}$, is an explicit receiving specification, not an observed thermal lifetime. The exact local copy/read reference can take $\delta_{\mathrm{read}} = 0$; a finite aperture still retains its separately bounded tail. For the example, $\varepsilon_{\mathrm{ap}} \le 2 \cdot 10^{-6}$ and $\delta_{\mathrm{read}} \le 98 \cdot 10^{-6}$ fit the combined $\delta_{\mathrm{acc}} \le 10^{-4}$ allowance. The diagonal passive hold itself has $\Lambda_T = 0$; a nonzero exposure budget must be established from the actual admitted transverse interaction. These distinctions keep a constructive finite record law, its mathematical robustness, and achieved apparatus calibration as separately identified parts of the same end-to-end account.

16. Six-lane receiving values and a complete finite example

The native calculation first consumes the source decoder's actual form-energy and switching-work returns. It uses

$$\begin{aligned} E_{\mathrm{rec}}^{\mathrm{up}} = 2(C_{\mathrm{dec}}^{\mathrm{up}} + W_{\mathrm{dec}}^{\mathrm{up}}) + 2c^{\mathrm{up}}, \\ E_{\mathrm{rec}}^{\mathrm{up}} < 514719 \end{aligned} \tag{201}$$

in every lane, instead of importing the old accepted-state noise values. The cap operands are the retained upward dyadic enclosures

$$\begin{aligned} C_f &= \frac{100301}{100000}, & \mu_c^{\mathrm{up}} &= \frac{1310994537}{281474976710656}, \\ V_c^{\mathrm{up}} &= \frac{2553359}{70368744177664}, & D_c^{\mathrm{up}} &= \frac{79569}{281474976710656}. \end{aligned} \tag{202}$$

Each original signed coupling is bounded through its retained absolute upper endpoint in the second moment. The source-shell norm and actual elapsed horizon are retained from the same Chapter IV input. The finite-energy estimate applies to the unselected reference output of the actual decoder; preparation error is propagated once through the entire final channel.

Use the explicit pointer, translation and memory parameters

$$\begin{aligned} a = 1024, \quad w = 512, \quad L_q = 2^{20}, \quad k = 2^{-40}, \\ \tau_{\mathrm{tr}} = T_{\mathrm{analog}} = T_{\mathrm{store}} = 2^{20}, \qquad \tau_w = 2^{-20}, \\ \sigma^2 = \frac{C_f^2 V_c^{\mathrm{up}}}{2} + kT_{\mathrm{analog}}, \qquad p_2 = \frac{1}{4\sigma^2} \le 512^2. \end{aligned} \tag{203}$$

The focused Gaussian has $\sigma^2 < 9.7192641 \times 10^{-7}$ and pointer kinetic energy $kp_2 < 2.3394125 \times 10^{-7}$. The actual continuum/analog moment has $B_A < 16$ in all six lanes. Consequently the finite registration aperture contributes

$$\varepsilon_{\mathrm{ap}} \le \frac{2(B_A + a^2)}{L_q^2} < 2 \times 10^{-6}. \tag{204}$$

Its compact response has the same derivative bound $G = 3/(4w)$; the finite memory-pulse calculation is unchanged. The exact reciprocal candidate for p_2 is certified by a native product equal to one, before its use in that calculation.

The finite write discrepancy obeys

$$\begin{aligned} r_w^2 \le 4.7337967 \times 10^{-11}, \\ r_w \le 6.8802592 \times 10^{-6} < 10^{-4}. \end{aligned} \tag{205}$$

For a transparent composable budget retain the parent's declared age-phase 0.01, marker preparation distance 10^{-4} and source-control allowance 10^{-3}. Add receiving initialization, write and transverse-hold allowances each at most 10^{-4}. Allocate the final 10^{-4} access allowance as aperture at most 2×10^{-6} plus local copy/read error at most 98×10^{-6}. The resulting conservative bound is

$$\begin{aligned} e_{\mathrm{rec}}^{\mathrm{r}} \le e_{\mathrm{dec}}^{\mathrm{r}} + \frac{4B_A}{a^2} + \frac{3}{10000} + \frac{98}{10^6} + \frac{2(B_A + a^2)}{L_q^2} \\ \le 0.0015821358 \quad \text{in every original lane.} \end{aligned} \tag{206}$$

The bound uses the 10^{-4} write allowance, retaining the smaller calculated r_w as available margin. It does not silently reassign that margin to another control. All-attempt failure remains $f + (1-f)e_{\mathrm{rec}}^{\mathrm{r}}$. The bit and detector may remain correlated; no independence assumption was needed for these additions.

Resources and completion time. The pointer mass is 2^{39} in the convention $k = 1/(2M_D)$. The translation has $\lambda = 2^{-10}$, positive completion height 2^{18} and absolute switch-work bound 524289. The memory write has peak below 3294198.7 and absolute switch-work bound below 6588397.4. These finite quantities are model-unit payments. The detector stabilizer additionally charges its actual gate profile, quadratic capital and variation; the focused pointer's initial coordinate variance includes the known integrated stabilizer shear. All source even/-counterterm energy, original field-gate work and residual field states remain recipients.

A local read copy can have any declared positive finite duration and area π, with its finite peak and switch work as above. The record becomes readable after the source decode, continuum gate, translation, chosen analog wait, memory write, chosen logical hold and copy/read have finished. The source clock

Lane	Actual detector B_{new} upper	Added tail error upper	Full ready error upper
01	14.471425	0.000055204110	0.0015801122
02	14.473466	0.000055211897	0.0015801222
03	14.883113	0.000056774575	0.0015821358
12	14.471518	0.000055204465	0.0015801127
13	14.851770	0.000056655013	0.0015819817
23	14.819530	0.000056532025	0.0015818233

Table 3: Outward six-lane bounds for the declared receiving example. The tail column charges the pointer's central transition band. The full ready column also charges source decoding/preparation, finite memory writing, initialization, hold and the combined aperture/read allowance. It is uniform over the stated retention interval.

certificate refers to the earlier source entry. The long receiving latency is explicitly retained. Repeated reads, cyclic reset, bath preparation and remote delivery use additional supplied resources and their own errors.

17. Accessibility is a receiving operation

The domain in the phrase "accessible record" must be part of the experiment. A record may be perfectly distinguishable jointly and unreadable from a restricted subsystem. A signal may already have been emitted but not yet have reached a receiver. A memory may survive while its decoder is unavailable. These cases have different operational meanings.

For any two candidate memory states, a trace-preserving physical access channel $\mathscr{A}$ cannot increase their trace distance:

$$\tfrac{1}{2}\left\|\mathscr{A}(\rho_0)-\mathscr{A}(\rho_1)\right\|_1 \le \tfrac{1}{2}\left\|\rho_0-\rho_1\right\|_1 . \tag{207}$$

The proof is the variational characterization of trace distance: every allowed output POVM pulls back through the unital dual channel to an allowed input POVM. An inaccessible fragment can reduce readout performance; calling it inaccessible does not remove its contribution to the global state or restore a lost marginal fringe.

Chapter V adds a causal constraint to this account. An exterior record must reach an exterior receiver through an allowed physical process. Late receipt of signals from a finite source-age interval does not give the emitter an infinite history. A boundary convention that excludes a completed parent-ledger state supplies neither a quantum channel across that boundary nor a rule selecting one outcome. The state, field dynamics and receiving map remain necessary.

The finite tensor-product examples in this chapter are operational models of separated systems. They do not establish an exact factorization of a gravitational or constrained gauge-field Hilbert space across a horizon. If a proposed theory replaces that subsystem structure by an observable algebra, its accessible operations and probabilities must be specified there instead.

These restrictions preserve the source's physical distinction: a write is more than an unresolved possibility. They also prevent the word "write" from performing work that belongs to a Hamiltonian, an instrument, a decoder or a causal signal.

18. Two prescribed arms in one original continuum field

The following receiving construction replaces the isolated-arm substitution by a specified moving-source action. The geometry is a flat laboratory chart with field wave speed one in the original model units. Each rigid source centre follows a prescribed piecewise smooth timelike worldtube $x_j(t)$, with $v_j=\dot{x}_j$, $|v_j|\le 1/4$, and $n_j=\sqrt{1-|v_j|^2}$. Smooth pieces have two continuous derivatives in v_j,n_j; finitely many joins have continuous velocities. Splitting, trajectory control, and recombination are prescribed receiving operations. They are not newly derived gravitational or autonomous actuator laws. The intrinsic source is the original linear-current H_0 in Eq. (58); the distinct Chapter VII quadratic completion candidates are not silently substituted for that source.

18.1 A translated source action and the full field complement

Use one full one-particle space containing the two physical polarizations and every angular mode. Embed the original two normalized quadrupole patterns as orthonormal vectors $e_+,e_\times$. Their radial factor is the original u_R, so $u_a(\omega,\widehat{k})=u_R(\omega)e_a(\widehat{k})$ and $\|u_a/\sqrt{\omega}\|^2=\mu=1/(4R)$. An explicit fixed-axis realization uses the two unit Frobenius-norm trace-free quadrupoles $A_+=\mathrm{diag}(1,-1,0)/\sqrt{2}$ and $(A_\times)_{xy}=(A_\times)_{yx}=1/\sqrt{2}$. With $P=I-\widehat{k}\widehat{k}^T$, put $e_a=\sqrt{5/(8\pi)}[PA_aP-\frac{1}{2}P\,\mathrm{tr}(PA_a)]$. The spherical second/fourth moments give $\int e_a:e_b\,d\Omega=\delta_{ab}$. This fixes the actual angular profile and orientation of this receiving construction. Other orientations are simultaneous tensor rotations; no new amplitude is fitted. All orthogonal angular modes remain in the common Fock space, initially in their retained vacuum. Translation mixes these modes; none is dropped. Let

$$H_f=d\Gamma(\omega),$$
$$P_f=d\Gamma(\omega\widehat{k}),$$
$$T_x=e^{-ix\cdot P_f},$$
$$u_{a,x}(k)=e^{-ik\cdot x}u_a(k).$$

The interaction prescription is the original current paired with the field on that moving worldtube, integrated over its proper

time. Explicitly the boundary part of its action is

$$I_* \int dt\, n_j(t) \left[\varepsilon \sum_a J_a \phi(u_{a,x_j}) - h_s - H_{ct} \right], \quad J_+ = D,\ J_\times = C,$$

where $h_s = \varepsilon H_{\text{pair}}$ and $H_{ct} = \varepsilon^2 \mu (D^2 + C^2)$. The common bulk field action uses laboratory time. The normalization is a declared rigid-quadrupole receiving incidence; boost deformations of the source profile and microscopic routing devices are additional physical operands. This action, rather than an independent bath for each arm, yields

$$H_j^{lab}(t) = H_f + n_j(t) T_{x_j(t)} (H_0 - H_f) T_{x_j(t)}^*. \tag{208}$$

Both original counterterms and all even energy are present. Scaling the interaction by n_j does not authorize replacing the retained source counterterm by a newly fitted static completion.

Put $\psi_j = T_{x_j}^* \Psi_j^{lab}$. Direct differentiation on a regularized field core, then closed-form extension, gives

$$\begin{aligned} K_j(t) &= H_f - v_j(t) \cdot P_f + n_j(t)(H_0 - H_f), \\ U_j^{lab}(t,s) &= T_{x_j(t)} U_{K_j}(t,s) T_{x_j(s)}^*. \end{aligned} \tag{209}$$

In particular the field momentum term is retained. Differentiating the singular translated form factor directly would need an ultraviolet norm not supplied by the original preparation. Equation (209) avoids that operation and preserves the original singular coupling. Because $|\langle \xi, P_f \zeta \rangle| \le \|H_f^{1/2} \xi\| \|H_f^{1/2} \zeta\|$, this new term is a continuous form on the original common energy domain.

18.2 Existence, energy, and a quantitative proper-age comparison

For a nominated original lane write $H_0 = \varepsilon e I + \varepsilon S + H_f + V + cI$, $c = 2g^2\mu$, preserving the sign of g in V. The retained annihilation-map estimate gives

$$|\langle V \rangle| \le 2\sqrt{c \langle H_f \rangle} \le \tfrac{1}{4} \langle H_f \rangle + 4c.$$

Define the positive comparison form

$$\begin{aligned} A_j &= K_j - n_j \varepsilon e I + \varepsilon I + 3cI, \\ b &= \varepsilon + 5c, \\ E_j(t) &= \langle A_j + bI \rangle. \end{aligned}$$

Then $A_j \ge H_f/2$, and its form norm is uniformly equivalent, after a positive shift, to $Q_f = D(H_f^{1/2})$. Its coefficients are piecewise C^2 as forms on this fixed domain. The constant-form-domain propagator theorem therefore supplies unique unitary weak dynamics, energy-domain propagation, and the form energy identity on each smooth piece; concatenate at the finite joins. The same construction on the finite direct sum of every source lane and four diagonal states gives the full source16 propagator. This application uses the common-domain hypotheses of the standard theorem explicitly.[29]

For normalized states the derivative obeys

$$\begin{aligned} |\dot{E}_j| &\le (2|\dot{v}_j| + |\dot{n}_j|) E_j, \\ E_j(t) &\le E_j(0) e^{B_j(t)}, \\ B_j(t) &= \int_0^t (2|\dot{v}_j| + |\dot{n}_j|) ds. \end{aligned}$$

Indeed $\langle H_f \rangle \le 2\langle A_j \rangle$ and $|\langle \varepsilon S + V + c \rangle| \le \langle A_j \rangle/2 + b$, which gives this inequality without requiring $\psi \in D(H_f)$. If the source enters at rest with the retained decoder form ceiling C, then $E_j(0) \le E_* = C + \varepsilon + 8c$.

The proper-age reference has generator $n_j H_0$, age $\eta_j(t) = \int_0^t n_j(s) ds$, and field energy at most $2(C+c)$ throughout. Its difference from the actual moving-frame generator is exactly

$$K_j - n_j H_0 = (1 - n_j) H_f - v_j \cdot P_f.$$

Weak Duhamel between the two energy-domain vectors gives

$$\begin{aligned} &\| [U_{K_j}(t,0) - e^{-iH_0 \eta_j(t)}] \psi \|^2 \\ &\le 2 \int_0^t \kappa_j(s) \sqrt{F_j(s) F_{ref,j}(s)}\, ds, \\ &\kappa_j = |1 - n_j| + |v_j|. \end{aligned} \tag{210}$$

Here $F = \langle H_f \rangle$. To see the norm estimate, differentiate the cross inner product as a Q_f, Q_f^* pairing. The difference form is bounded by $\kappa_j \sqrt{F_j F_{ref,j}}$; integrate it from one and use $\|u - v\|^2 = 2 - 2\,\mathrm{Re}\langle u, v \rangle$. This is an energy-form estimate, not an assumed Hilbert-space action of the singular interaction on arbitrary dressed vectors.

At a common endpoint $x_0(t) = x_1(t)$, the translation factors cancel from the relative overlap. If r_j encloses the square root of Eq. (210), then

$$\begin{aligned} \Gamma_{12} &= \mathrm{Tr}(\rho U_1^* U_0), \\ |\Gamma_{12} - \mathrm{Tr}(\rho e^{-iH_0(\eta_0 - \eta_1)})| &\le r_0 + r_1. \end{aligned} \tag{211}$$

Purification extends the statement to a mixed input with the same energy bound. A noncoincident final position supplies the relative translation $T_{x_1}^* T_{x_0}$ and belongs to an unrecombined experiment. The proper-age expression is exact when the difference form annihilates the relevant evolution; in general its error is now quantified. The actual source-lane clock equation also follows directly:

$$\begin{aligned} \dot{z}_j &= n_j [-i\Delta z_j + 2g w_j], \\ -\frac{d}{dt} \arg z_j &= n_j [\Delta - 2g\, \mathrm{Im}(w_j / z_j)]. \end{aligned}$$

Here $z_j \ne 0$, and w_j is the correlator in the actual common field state. The momentum term commutes with the source clock operator, but changes the field evolution and this correlator. An old isolated clock-rate certificate is therefore not assigned to these new trajectories.

18.3 Stability under additional source-profile and control data

The defined worldtube law has fixed lab-frame quadrupoles. A further physical profile, boost, control or microscopic-source assignment can be compared quantitatively rather than

identified with that law by notation. Suppose two such actual/reference propagators exist on the common energy domain. In their common frame let the difference form be $\delta h(t) - \delta v(t)\cdot P_f - \varepsilon\sum_a J_a\phi(\delta u_a(t))$, with bounded source $\delta h, J_a$ and $m_a(t) = \|\delta u_a(t)/\sqrt{\omega}\|^2 < \infty$. Any additional lapse-current difference can be included in δu_a and δh, retaining its even/counterterm terms. The annihilation form estimate between normalized vectors gives

$$|\delta q[\xi,\zeta]| \le \|\delta h\| + |\delta v|\sqrt{F_\xi F_\zeta} + \varepsilon\sum_a \|J_a\|\sqrt{m_a}(\sqrt{F_\xi}+\sqrt{F_\zeta}).$$

Consequently the squared vector distance is at most twice the integral of this expression along the actual/reference pair, by the same weak Duhamel argument. Finite energy ceilings close that estimate. This supplies an explicit extra error radius for independently supplied profile/control data without requiring an unweighted ultraviolet norm or fitting clock rates. Its work correction must likewise retain the corresponding control variation and energy balance. The numerical experiment below is for the exactly stated receiving action; no numerical profile or microscopic-actuator discrepancy is inferred for it.

18.4 Actual field response and common-arm coherence

At a frequency cutoff, the laboratory Heisenberg equation can be integrated exactly:

$$\begin{aligned} a_\sigma(k,t) &= e^{-i\omega t}a_\sigma(k,0) \\ &\quad + i\varepsilon\sum_{j,a}\Pi_j\int_0^t ds\, n_j(s)e^{-i\omega(t-s)} \\ &\quad \times u_{a,x_j(s),\sigma}(k)J_a(s). \end{aligned} \tag{212}$$

Smearing against an energy-admissible receiving test extends this identity as a form limit. The $J_a(s)$ are the actual interacting operators; their source and field backreaction has not been replaced by a classical force. Both arms use the same initial field operators. In the route-block state the cross block is $U_0\rho U_1^*$, not a product of two bath states. Its trace is the interference operand in Eq. (211). The causal response is retarded in time. A spatially distant port additionally needs its actual local gauge-invariant observable and propagation geometry; a global frequency filter alone is not a local arrival channel.

18.5 Finite mechanical work and the phase reference

Let $E_{lab,j} = \langle H_j^{lab}\rangle$. The moving-frame identity is $E_{lab,j} = \langle K_j\rangle + v_j\cdot\langle P_f\rangle$. For a closed velocity cycle its net work is therefore

$$W_{lab,j} = \int dt\,\left[-\dot v_j\cdot\langle P_f\rangle + \dot n_j\langle H_0 - H_f\rangle\right].$$

This definition follows from endpoint energy balance and the common-form energy identity; it does not differentiate the ultraviolet-singular laboratory field at a moving point. A finite absolute payment ceiling is

$$W^{up}_{route,j} \quad \mathrm{Var}(v_j)F^{up} + \mathrm{Var}(n_j)[\tfrac14 F^{up} + \varepsilon\ e\ + \varepsilon + 5c].$$

The even energy and counterterm remain in that expression. For nonzero endpoint velocities include $[v_j\cdot\langle P_f\rangle]_0^t$. An extra prescribed carrier rest energy m_{ext}, excluded from H_0, contributes its own point action and, for the sinusoidal closed speed cycle below, kinetic-energy variation $4m_{ext}[(1-v_*^2)^{-1/2}-1]$. Beam splitters, traps and their calibration remain prescribed devices with separately supplied resources; the calculation is a driven receiving model, not a microscopic autonomous centre-of-mass theory. Their phases are included once in the total interference amplitude, as in Section 7.3.

19. A complete coherent local detector and record join

The two-arm evolution just derived is an actual source–continuum construction for the stated receiving action. It does not assume that every arm pair produces the quarter-cycle age separation used in the decoder calibration. We give a finite, explicitly paid route-label encoding that supplies a complete worked experiment, while retaining the unforced proper-age comparison as a distinct prediction.

19.1 Closed physical paths and a controlled source encoding

Take mirror paths with

$$\begin{aligned} v_{0,1}(t) &= \pm v_*\sin(2\pi t/T)\widehat{e}, \\ x_{0,1}(t) &= \pm\frac{v_*T}{2\pi}[1-\cos(2\pi t/T)]\widehat{e}, \\ v_* &= 1/64, \\ T &= 2^{-20}. \end{aligned}$$

Both start and return to the same centre at rest. Their proper ages are equal. They still couple at different positions to one common field. Their total variation gives $B(T) \le 8v_* + 4v_*^2 = 129/1024 < 1$, hence $e^{B(T)} \le (1-129/1024)^{-1} = 1024/895$, and $\kappa \le v_* + v_*^2 = 65/4096$. The actual continuum route bound is

$$r^2_{route} \le 4T\,\frac{65}{4096}\,\frac{1024}{895}(C+\varepsilon+8c). \tag{213}$$

For an initially coherent route state and one common prepared marker, the controlled actual-to-reference vector error is at most the largest branch radius; it is not the sum of two preparation errors. With the encoder absent and ideal unrecorded routing controls, equal proper ages give the concrete continuum consequence $|\Gamma_{unencoded} - 1| \le 2r_{route}$. The source and shared field thus supply a near-unit-overlap control for this short closed excursion. This is distinct from the encoded record experiment below, whose route correlation is paid by a different physical operation.

Prepare the original marker near age $T_0/4$, with its retained preparation-distance budget. After the closed excursion apply the positive finite route-controlled source pulse

$$\begin{aligned} B_{enc} &= \Pi_1\otimes(I_{16}-S)/2, \\ B_{enc}^2 &= B_{enc}, \\ \int\lambda(t)dt &= \pi, \\ U^{ideal}_{enc} &= \Pi_0 \quad I + \Pi_1 \quad S. \end{aligned} \tag{214}$$

It acts on all 32 route/source states. It is a quantum controlled operation: the apparatus does not receive a classical copy of the unknown route label. Swap anticommutes with the source-lane Y, so the two reference outputs have opposite $-Y$ expectations and are decoded by the same complete source16 POVM. Swap commutes with H_f and preserves every source lane. The retained first-energy pulse lemma therefore gives

$$r_{enc} \le (1+2\pi)\sqrt{2C\tau_{enc}},$$
$$\tau_{enc} = 2^{-30}.$$

The pulse comparison factors out the same lane scalar phase on both route branches, exactly as in the retained finite-energy lemma. That scalar remains in the physical action and work account. The physical generator during the pulse is $H_0 + \lambda B_{enc}$, with the original even phase retained. Its peak is π/τ_{enc} and absolute switch-work payment is at most $2\pi/\tau_{enc}$. Those large finite controls are recorded explicitly. This encodes a route label; it is not an intrinsic gravitational age signal. In particular the mirror arms have equal proper ages, not the two distinct ages of the earlier calibration example. Without the encoder their overlap is given by the shared-field dynamics alone.

The small reference age increment obeys $\Omega T \le \Delta_{up} T < 0.01$ in every original lane. Thus the same declared phase allowance already present in the parent source decoder covers this reference timing. After ideal encoding the source field-energy ceiling is unchanged, since S commutes with H_f. The actual finite source decoder then obeys its original uniform form-energy and work bounds on either encoded input. No absolute hard energy phase needs to be measured by the source-only quadrature.

19.2 Coherent composition and the noise/work distinction

At the common return location couple the source to the finite decoder bit B, the actual nonuniform Chapter IV pointer, retained memory R and local access bit A. These are the coherent unitary/pulse dilations of the preceding integrated constructions. They act on marker and apparatus, with identity on the route register. The complete source, common field, bit/field/bath correlations and every residual pointer cloud remain.

The scalar response is the actual compact cap background plus the bit-controlled signed displacement. The latter is a transducer actuator, not an assignment of an uncomputed intrinsic remote cap mean. A local station at the common endpoint needs no remote bit-delivery assumption. For a remote station the physical bit or source signal must arrive through a specified causal channel before this interaction; no remote record is being inferred from the local construction.

Use a whole-pipeline hybrid: compare the actual routed and encoded joint input with the proper-age reference followed by ideal swap encoding; then apply exactly the same full detector/storage channel to both. Trace-distance contraction pays $r_{route} + r_{enc}$ once at the final readout. It is not used to transfer an unbounded energy moment. The parent detector moment is recalculated on its reference input using the unchanged field-energy ceiling and actual decoder work. The actual encoder energy has its separate, possibly much larger work ceiling. Both payments remain in the physical energy ledger. A small state error never licenses assigning the actual encoded state the reference energy. There is a separate actual-energy envelope. At the closed return, $\langle L \rangle \le E_* e^{B(T)}$, and bounded-pulse energy balance through encoding and decoding gives

$$F_{actual,in} \le 2\left[E_* e^{B(T)} + W_{enc}^{abs} + W_{dec}^{abs} + c\right]. \tag{215}$$

The physical field-gate resource ledger uses this larger ceiling, not the reference noise moment used in the final probability comparison. For its actual energy-admissible cap test f_t, put $\mu_t = \|f_t/\sqrt{\omega}\|^2$ and $\dot{\mu}_t^{test} = \|\dot{f}_t/\sqrt{\omega}\|^2$. The conserved pointer momentum second moment p_2 and annihilation form estimate bound the controlled cap switching work by

$$W_{cap}^{abs} \le 2\sqrt{p_2}\int dt\,[|\dot{h}|\sqrt{\mu_t} + |h|\sqrt{\dot{\mu}_t^{test}}]\sqrt{F_{actual}(t)} + p_2\,\mathrm{TV}(s).$$

This follows from $|\langle P\phi(f)\rangle| \le 2\sqrt{p_2 \mu_f F_{actual}}$ on the joint state, with no bit/field independence or fourth moment. For the stabilized gate's common-form graph, let $a(t)G \le q_t + b(t)$ and $|\dot{q}_t + \dot{b}(t)| \le \ell(t)G$. Its actual energy satisfies $\mathscr{E}(t) \le \mathscr{E}(0)\exp\int \ell/a$, using Eq. (215) at entry. The retained smooth compact test/gate and positive stabilizer give finite graph coefficients and cap-test norms. This is a finite functional ceiling for actual gate work even if the actual energy greatly exceeds its reference value. Global background vacuum-ramp and material actuator work remain separately specified external payments, as in the parent; they are not set to zero or inferred from the noise ceiling.

Consequently the worst-label final ready error obeys

$$e_{new}^r \le e_{record}^r + r_{route} + r_{enc},$$
$$e_{all} \le f + (1-f)e_{new}^r. \tag{216}$$

Here the parent record bound already includes preparation, source decode, pointer loading, finite writing, initialization, hold, aperture and local read error. Those terms are not charged a second time. The native operand used here is the parent's conservative `full_error` return; its smaller refined aperture/read display is not silently used to improve this budget. An encoder area-calibration error a_{cal} adds at most $|a_{cal}|$ by bounded-generator Duhamel. The numerical example specifies nominal encoder area exactly, just as its nominal trajectory; it assigns no achieved control calibration. The native six-lane evaluation certifies $e_{new}^r < 0.0016$ with the displayed chosen controls; f remains the supplied per-label failure ceiling. The particular $f = 10^{-4}$ line is an explicit budget, not an achieved failure measurement.

For the same coherent ensemble, route visibility and this readable route record obey the integrated tight theorem

$$V \le 4\sqrt{pq\,e_{all}(1-e_{all})}.$$

The marker-only decoder, writing and access operations leave the unsorted route marginal exactly unchanged. They make an existing route correlation readable; they do not additionally suppress that marginal merely because a later observer reads it. Conditional erasure or route feed-forward is a separately specified experiment.

19.3 All outcomes, resources and latency

Retain read results 0 and 1, every preparation/aperture/control failure flag, and all unobserved recipients in one trace-preserving instrument. A ready-branch conditional calculation never discards its failed input weight from the all-attempt ledger. The source decoder and record pulses remain active finite-duration operations with their original positive completions and pointer kinetic energy. The new experiment adds the route work ceiling, encoder capital/peak/switch work and durations $T+\tau_{enc}$ to those parent resources. The long analog and logical holds remain; the source clock certificate applies at source entry, not through the entire long receiving latency. Microscopic actuator, absolute length/time and mass calibration remain physical model assignments.

20. Ordered physical histories and repeated access

These extensions follow the route and coherent-record join. An ordered word is assigned a unitary only after its physical event/route/control incidences have been specified. For such unitaries on the one full marker space define

$$G_{wv} = \mathrm{Tr}(\rho U_v^* U_w),$$
$$(\rho_{route})_{wv} = a_w a_v^* G_{wv}.$$

The matrix G is positive semidefinite with diagonal one: purification expresses it as the overlap matrix of normalized conditional marker vectors (with the indicated index convention). Thus the route density operator is normalized and positive for normalized route amplitudes. Every multi-history interference measurement follows from that same operator and the stated analyser. On a declared accessible domain the conditional states are $\mathscr{A}(U_w \rho U_w^*)$; equality of these states means identical probabilities for its marker-only POVMs. It does not imply equality of the coherent cross blocks, nor does a computational history label alone supply an extra observable. Equal signed action and final address therefore leave a concrete question about the ordered physical propagators rather than an unassigned rule creating additional states. In the moving-source model, $[H_0, P_f]$ need not vanish as a formal core commutator, so ordered velocity controls need not be determined by their integrals. The actual form propagators, not that potentially unbounded commutator, define the history comparison.

20.1 A finite repeated-read theorem with correlated noise

Let Q0,Q1 be the orthogonal populations of the retained memory. An ideal copy-read isometry has $W|j\rangle_R|0\rangle_A = |j\rangle_R|j\rangle_A$, extended unitarily on its complete finite space. A finite positive controlled flip realizes this operation with area π, peak π/τ and absolute switch-work at most $2\pi/\tau$; the memory/bath drift must be charged as in the finite pulse construction. Diagonal interactions conserve Q populations even with a correlated bath. For transverse disturbance $V_\perp$, retain each interval's trace-distance exposure $d_k \le \int \|V_\perp(t)\| dt$. Suppose actual read k differs from the ideal nondestructive read by δ_k on the complete reachable joint state, including all earlier read registers and the bath. Successive hybrid replacement and trace-distance contraction give

$$\Pr\left\{\begin{array}{c}\text{any wrong stored/read label}\\ \text{through read } N\end{array}\right\}$$
$$\le \min\{1, e_{write} + \sum_{k=1}^{N}(d_k+\delta_k)\}.$$

No statistical independence is used. For worst-label uniform bounds $d_k+\delta_k \le b$, an allowed total error e_{max} permits $Nb \le e_{max} - e_{write}$; this is a finite guaranteed-read budget, not a derived thermal decay rate. Actual sequential joint-state bounds are essential: separate one-time marginal read errors do not establish this premise. Classical repetition improves reliability exponentially only under its additional conditional-independence assumptions.

Every read needs a recipient. An occupied register cannot be reused as a blank without an explicit reversible transfer or disposal/reset channel. Exact discrimination of N complete histories requires N orthogonal recipient states; the integrated approximate label-capacity statement gives its finite-error version. Resetting the local cell does not erase its globally retained source/bath/history recipients. A thermal heat claim additionally needs a specified reservoir, initial correlations and cyclic protocol. No universal heat-per-Write number is assigned here.

21. New source-bound receiving values and evidence

All displays round upward; exact native rationals remain in the replay. The all-attempt column uses the explicitly declared budget $f = 10^{-4}$. The additional encoder peak is below 3.374×10^9 and its absolute switch-work ceiling below 6.748×10^9 in original model units. Route mechanical work obeys the full source-dependent bound in Section 18; trajectory carrier mass and splitter/trap calibration are not assigned numerical physical values.

The new native work uses the installed public DomainSession route, SLC-GEN5-R1 / SLC-GEN5-CEV1-R1 and Starbreaker SB-GEN3-ACCUMULATION-R1. One exact six-lane bound calculation and one full-source32 encoder certificate complete. A first 96-column certificate exceeded the installed 64-column word limit; its rejection is retained. A corrected 32-column sum of residual Grams certifies the same three identities, and the completed first bound call was adopted rather than repeated. A subsequent binding-index preparation error started no native work. A native checkpoint and both closed sessions remain. No Duck campaign or installation changed.

The continuum propagation, energy and coherent-join statements are analytical theorems, with independent arranged arithmetic and assigned finite mechanism comparators. Finite field fixtures do not replace the continuum, its ultraviolet domain, or the actual preparation. Offline replay authenticates the used inputs, receipts and exact matrices; it performs no native science. Existing source and supplement results are retained unchanged. The original detector export ZIP was unavailable at its recorded TEMP path; its retained source members are authenticated against the completed manifest, not described as the old ZIP.

Lane	Route radius	Encoder radius	Ready error	All-attempt error
01	6.242E-7	6.012E-7	0.0015814301	0.0016812720
02	6.918E-7	6.448E-7	0.0015815514	0.0016813932
03	0.0000042811	0.0000033658	0.0015898752	0.0016897162
12	6.275E-7	6.032E-7	0.0015814359	0.0016812777
13	0.0000041185	0.0000032394	0.0015894322	0.0016892732
23	0.0000039443	0.0000031040	0.0015889641	0.0016888052

The exact decoder archive was recovered from the detector's retained inputs.

Completed and remaining physical objects. For the specified moving rigid-source action we have full source/shared-field dynamics, a proper-age comparison, actual retarded source response and a coherent co-located detector/record construction. The finite paid encoder is a supplied receiving operation. It is not an independently established primitive route-selection law, a positive intrinsic remote cap mean, or an achieved material calibration. A remote intrinsic-field protocol must specify its actual local observable, spatial propagation, extraction and signal/noise law; a local encoded bit is not substituted for that port. An autonomous quantum centre-of-mass and microscopic control mechanism are also beyond the prescribed-worldtube action. These are exact physical joins to carry forward, alongside every retained counterterm and recipient.

Reproduction. Run `python3 checks/verify.py` from this folder for the independent arithmetic/custody and finite mechanism checks. Build with `python3 build.py` using the retained existing Tectonic executable and cached packages. No network, compiler install, Duck cycle, cloud, Git publication or I9 operation is required.

References

[1] *SAM Unified Technical Synthesis—Volume I: Substrate and Accumulation*, retained source manuscript, §5 and Appendix A.13.

[2] *Original Write histories and an interacting source–field action*, Volume I, Chapter VII (formerly Chapter VII), retained model and exact clock certificate. https://github.com/iwtbotiwtwot/sam_volume_I_i.

[3] *Retention, accumulation and the finite ledger*, Volume I, Chapter II, source-to-field update (2026). https://github.com/iwtbotiwtwot/sam_volume_I_ii.

[4] *The spherical accumulation kernel and the source-to-field interface*, Volume I, Chapter III, retained expanded manuscript (2026).

[5] Volume I, Chapter IV, *Physical source and local coupling development*, retained expanded manuscript, 16 October 2026.

[6] *Strong fields and the approach to unit accumulation*, Volume I, Chapter V, expanded development manuscript and source/replay packet, 16 October 2026.

[7] *The Courtroom*, CR075 Unresolved Path and Ledger Write, result, runner, manifest and precommit; pinned commit b5e914f71377e86ef4c67e199973d9300795cda1. https://github.com/iwtbotiwtwot/The_Courtroom/tree/b5e914f71377e86ef4c67e199973d9300795cda1/11_QUANTUM_MECHANICS_AND_GRAVITY/CR075_UNRESOLVED_PATH_AND_LEDGER_WRITE.

[8] *SAMA*, "Exact algebra and complete exact write" and "Directional history and event-reel computation," Volume III; pinned commit 4c2b5dc1649da5900c90b90d0ce56c223dbb7717. https://github.com/iwtbotiwtwot/SAMA/blob/4c2b5dc1649da5900c90b90d0ce56c223dbb7717/vol_iii/03-writes-and-history/exact-algebra-and-complete-exact-write.md.

[9] *SAMA*, "W8–X1–W9 lifecycle," same pinned commit. https://github.com/iwtbotiwtwot/SAMA/blob/4c2b5dc1649da5900c90b90d0ce56c223dbb7717/vol_iii/03-writes-and-history/w8-x1-w9-lifecycle.md.

[10] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press (2018), Chapters 2–3. Author manuscript: https://cs.uwaterloo.ca/~watrous/TQI/.

[11] M. Born, "Quantenmechanik der Stoßvorgänge," *Zeitschrift für Physik* **38**, 803–827 (1926). https://doi.org/10.1007/BF01397184.

[12] B.-G. Englert, "Fringe Visibility and Which-Way Information: An Inequality," *Physical Review Letters* **77**, 2154–2157 (1996). https://doi.org/10.1103/PhysRevLett.77.2154.

[13] M. O. Scully and K. Drühl, "Quantum eraser: A proposed photon correlation experiment concerning observation and 'delayed choice' in quantum mechanics," *Physical Review A* **25**, 2208 (1982). https://doi.org/10.1103/PhysRevA.25.2208.

[14] Y.-H. Kim, R. Yu, S. P. Kulik, Y. Shih, and M. O. Scully, "Delayed 'Choice' Quantum Eraser," *Physical Review Letters* **84**, 1–5 (2000). https://doi.org/10.1103/PhysRevLett.84.1; https://arxiv.org/abs/quant-ph/9903047.

[15] S. M. Barnett and S. Croke, "Quantum state discrimination," *Advances in Optics and Photonics* **1**, 238–278 (2009). https://doi.org/10.1364/AOP.1.000238; https://arxiv.org/abs/0810.1970.

[16] M. Schlosshauer, "Decoherence, the measurement problem, and interpretations of quantum mechanics," *Reviews of Modern Physics* **76**, 1267–1305 (2005). https://doi.org/10.1103/RevModPhys.76.1267; https://arxiv.org/abs/quant-ph/0312059.

[17] M. Zych, F. Costa, I. Pikovski, and Č. Brukner, "Quantum interferometric visibility as a witness of general relativistic proper time," *Nature Communications* **2**, 505 (2011). https://doi.org/10.1038/ncomms1498; https://arxiv.org/abs/1105.4531.

[18] S. L. Braunstein and A. K. Pati, "Quantum Information Cannot Be Completely Hidden in Correlations: Implications for the Black-Hole Information Paradox," *Physical Review Letters* **98**, 080502 (2007). https://doi.org/10.1103/PhysRevLett.98.080502; https://arxiv.org/abs/gr-qc/0603046.

[19] A. K. Pati and S. L. Braunstein, "Impossibility of deleting an unknown quantum state," *Nature* **404**, 164–165 (2000). https://doi.org/10.1038/35004532; https://arxiv.org/abs/quant-ph/9911090.

[20] G. N. Fleming, "A unitarity bound on the evolution of nonstationary states," *Il Nuovo Cimento A* **16**, 232–240 (1973). https://doi.org/10.1007/BF02819419.

[21] S. Deffner and S. Campbell, "Quantum speed limits: from Heisenberg's uncertainty principle to optimal quantum control," *Journal of Physics A: Mathematical and Theoretical* **50**, 453001 (2017). https://doi.org/10.1088/1751-8121/aa86c6; https://arxiv.org/abs/1705.08023.

[22] *Quantum routes, phase, and physical records*, SAM Volume I, Chapter VI development draft, 16 October 2026. Supplied PDF and source/check package retained unchanged in this supplement.

[23] *Original Write histories and an interacting source–field action*, SAM Volume I, Chapter VII (formerly Chapter VII). Model and exact clock certificate retained by the supplied Chapter VI package. https://github.com/iwtbotiwtwot/sam_volume_I_i.

[24] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press (2018), Chapter 3. Author manuscript: https://cs.uwaterloo.ca/~watrous/TQI/TQI.3.pdf.

[25] *Chapter VI foundation supplement: overlap, readable records, and finite preparation*, 17 October 2026. H002700; unchanged parent capsule retained under inputs/.

[26] *Chapter VI: A source-only clock decoder*, 17 October 2026. H002701; unchanged source/check capsule retained under inputs/.

[27] *Finite preparation and joint composition*, SAM Volume I, Chapter IV, 16 October 2026; actual detector/apparatus continuations measurement9 and measurement10, 17 October 2026. Exact selected derivations and operands retained under inputs/source_selected/.

[28] *SAM_land donor readout*, retained derivation Section 4 and energy account. The degenerate record-bit recipient and positive write coupling are retained inputs to the new smooth conversion.

[29] A. Balmaseda, D. Lonigro and J. M. Perez-Pardo, On the Schrodinger equation for time-dependent Hamiltonians with a constant form domain, Mathematics 10 (2022), 218. Theorem 3.10 and its hypotheses. https://arxiv.org/abs/2112.11063.

SAM VOLUME I

Original Write histories and an interacting source-field action

Chapter VII

Abstract. Original signed Write histories supply phase and action data that can be received into a finite pair source coupled to a two-polarization quadrupole field. This chapter keeps the original rules, the receiving laws, and their derived consequences explicit. A four-component source yields six driven spectral lanes and four unforced states. For a declared coherent hard-field preparation in each lane, a weighted continuum comparison bounds the actual source-field correlator. All six instantaneous phase rates remain positive and below the bare gap throughout one bare cycle. The resulting phase coordinate reparameterizes the complete joint action, retaining the even source energy, both counterterms, and the Berry one-form.

1. The physical question and its inputs

An original Write is a signed event in an ordered source history. A receiving field observes specified combinations of that source. This chapter asks how the original phase incidence, a complete finite source, and a declared field law combine into an interacting action with a usable local phase clock.

Three layers carry different authority. **Original rules** supply the signed Writes, lifted phases, retained tensor map, and source histories. **Receiving laws** specify the quadrupole bulk field, its quantization, the boundary incidence, and the preparation. **Derived compositions** include the source spectrum, conserved charge, actual clock equation, and certified phase reparameterization. An exact composition keeps its inputs identifiable. [1-4]

The construction uses the full source space $\mathbb{C}^4 \otimes \mathbb{C}^4$ and both real field polarizations. All field frequencies and photon sectors belong to the model. The six clock certificates concern six separately prepared lanes. Each certificate carries its own signed root difference, coupling, and hard-branch preparation.

The numerical example receives the unit-strength source chart, the retained boundary radius, and a capacity-normalized model time. A physical exchange strength, the reference capacity and its role, the Write-to-seconds assignment, and the incidence between the source sphere and field boundary remain explicit physical inputs. No inverse-seconds rate is selected by the normalized calculation.

How the source-to-clock argument is organized

A phase becomes a useful local clock only when three separate questions have answers. The phase must belong to an identified physical observable. That observable must remain nonzero so that its argument can be followed continuously. Finally, its phase must advance with a rate that does not vanish or change sign on the interval being used. A frequency written into a Hamiltonian does not by itself answer all three questions for the interacting system.

Here the observable is a bounded source amplitude whose equation contains a joint source–field expectation. The source spectrum fixes the signed lane coupling; the receiving field supplies the correlation that changes the clock rate. The analysis then compares the full continuum trajectory with a specifically prepared reference trajectory in the same field realization. Controlling the difference between those trajectories gives a bound on the correlation and, through the exact clock equation, a uniform bound on the actual phase rate.

The steps have different responsibilities. Sections 2–4 establish the source incidence, the complete pair space and the normalization. Section 5 constructs the joint dynamics and the weighted field pairings that make its continuum expressions meaningful. Section 6 develops the reference comparison and the six lane certificates. Section 7 changes the parameter of the complete action using the resulting phase. The conclusion is a source-to-clock result for the declared receiving model and preparation.

Each certificate concerns one separately prepared lane. The six results do not assert that the phase of an arbitrary coherent combination of lane amplitudes is monotone: interference can change that combined observable. Likewise, the model-time rate becomes a physical inverse-seconds rate only after the stated time assignment is supplied. These distinctions fix which clock has actually been constructed.

2. Lifted phases and history

A phase can be recorded in several ways. A wrapped phase specifies a point on the unit circle, represented by $e^{i\phi}$. A lifted phase is a real number obtained after choosing an initial branch and following the accumulated phase without reducing it modulo 2π. The lift therefore retains winding information that the wrapped value discards. An ordered Write history additionally records which signed events occurred and in what order. These are distinct levels of description, even when they describe the same completed process.

The three coordinates below separate three combinations of the lifted endpoint data. The combination γ_{W} compares the two endpoint phases with their partner; δ resolves the difference between the endpoints; and χ retains the partner phase itself. A common increment of all three original phases leaves γ_{W} and δ unchanged and shifts χ by that increment. Keeping χ is

therefore necessary if the coordinate change is to retain the full lifted phase triple.

For lifted endpoint phases ϕ_a, ϕ_b and partner phase ϕ_p, define

$$\begin{aligned} \gamma_{\mathrm{W}} &= \phi_a + \phi_b - 2\phi_p, \\ \delta &= \phi_a - \phi_b, \qquad \chi = \phi_p. \end{aligned} \tag{1}$$

The invertible incidence is

$$\begin{aligned} \phi_a &= \frac{1}{2}(\gamma_{\mathrm{W}} + \delta) + \chi, \\ \phi_b &= \frac{1}{2}(\gamma_{\mathrm{W}} - \delta) + \chi. \end{aligned} \tag{2}$$

The word "invertible" here has a precise domain. On the real lifted coordinates, the forward map has matrix

$$\begin{pmatrix} \gamma_{\mathrm{W}} \\ \delta \\ \chi \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 & -2 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{M_{\mathrm{inc}}} \begin{pmatrix} \phi_a \\ \phi_b \\ \phi_p \end{pmatrix}, \qquad \det M_{\mathrm{inc}} = -2.$$

The nonzero determinant guarantees that no coordinate of a supplied lifted phase triple is lost when all three transformed coordinates are retained. Equation (2) is its inverse. If a receiving observable uses only γ_{W}, the remaining coordinates still exist; they have simply not been measured by that observable.

For a concrete illustration of the lift convention, consider the triples $(0,0,0)$ and $(2\pi,0,0)$. Their three wrapped phases are identical. Their lifted coordinates are respectively $(\gamma_{\mathrm{W}},\delta,\chi) = (0,0,0)$ and $(2\pi,2\pi,0)$, so the lifted incidence distinguishes them. This is an algebraic example of the information retained by a lift, rather than a claim that either triple is reached by an additional legal Write path. Applying the inverse formula requires the lifted coordinates with their branch information; independently wrapping those coordinates discards information that the real inverse needs.

On the legal piecewise active Write paths, the original action ledger is

$$\omega_{\mathrm{Write}} = \hbar(\mathrm{d}\gamma_{\mathrm{W}} + 3\mathrm{d}\chi). \tag{3}$$

The factor three follows by substituting (2) into the sum of the three phase differentials. The retained tensor orientation obeys $\psi = \psi_0 - \gamma_{\mathrm{W}}/2$. The common phase χ remains part of the action account even when a receiving observable is insensitive to it. [1]

The cancellation of δ from this one-form follows directly from the coordinate change:

$$\mathrm{d}\phi_a + \mathrm{d}\phi_b + \mathrm{d}\phi_p = \mathrm{d}\gamma_{\mathrm{W}} + 3\,\mathrm{d}\chi.$$

Thus an endpoint difference can be retained in the state description while making no separate contribution to this particular sum of phase differentials. Conversely, a common phase increment contributes three times to the sum because it increments each original phase once. This accounts for the coefficient in (3) without identifying the common phase with a receiving clock.

On a supplied legal lifted path from an initial point to a final point, integration gives

$$\int \omega_{\mathrm{Write}} = \hbar\big(\gamma_{\mathrm{W}}^{\mathrm{final}} - \gamma_{\mathrm{W}}^{\mathrm{initial}}\big) + 3\hbar\big(\chi^{\mathrm{final}} - \chi^{\mathrm{initial}}\big).$$

This signed integral can vanish through cancellation. Elapsed duration instead adds the positive time spent on the successive active portions of the history. Determining that duration requires the path and its stated timing calibration. The endpoint action integral alone contains no rule assigning how long the intermediate events take.

Lifted cumulative phase, a wrapped endpoint, and an ordered event word retain different information. The completed H02 and H03 histories reach the same source endpoint with zero total signed phase action, yet have positive durations approximately 0.3156476595 and 0.3181630929 in their stated source calibration. Their difference is a difference of positive durations. [1, §6]

These completed histories illustrate why coordinate invertibility and history reconstruction must be kept separate. The map (1)–(2) acts point by point. Applied to a supplied trajectory, it expresses that trajectory in another set of coordinates. Applied only to a terminal point, it recovers only the terminal lifted phase triple. It does not supply the event word, the intermediate source states, or the durations of the active segments.

Accordingly, the shared endpoint and zero signed action of H02 and H03 do not require their positive durations to agree. The timing information resides in the histories and their source calibration. Keeping those records alongside the phase incidence preserves the distinction between accumulated signed phase and elapsed time that the receiving calculation will need.

The interacting clock phase θ introduced below remains distinct from γ_{W} and χ.

3. A complete four-component source

The original placement codes 46501, 46525, 49061, and 81829 define the retained signed boundary construction. Its 4×78 incidence matrix B gives $T = BB^*$, and the retained signed response supplies K. The receiving family is $Q(g_H) = T + g_H K$. [1, 4]

The incidence matrix and the signed response enter in different ways. The product BB^* is a Gram operator: for every source vector v,

$$\langle v, Tv\rangle = \|B^*v\|^2 \geq 0.$$

Its positivity follows from this construction. The additional operator K carries the retained signed response, and the parameter g_H specifies its weight in the receiving family. Displaying the member with $g_H = 1$ fixes a particular finite calculation; it does not remove g_H from the family or determine a physical exchange rate.

For the displayed member, real symmetry supplies an orthonormal basis of four eigenvectors q_a . The next step is

therefore a finite spectral problem with explicitly given entries. Establishing its polynomial and its invariant pair sectors is an algebraic consequence of the received matrix. The physical interpretation of that matrix continues to depend on the source construction and the receiving choices already stated.

At the inherited diagnostic strength $g_H = 1$, Q is the following real symmetric matrix:

	0	**1**	**2**	**3**
0	756	104	40	0
1	104	2100	0	0
2	40	0	756	40
3	0	0	40	756

With $x = q - 756$, its characteristic polynomial is

$$p(x) = x^4 - 1344x^3 - 14016x^2 + 4300800x + 17305600. \tag{4}$$

The quartic can be obtained without a numerical fit. Subtracting $756I$ from Q gives

$$A_Q = Q - 756I = \begin{pmatrix} 0 & 104 & 40 & 0 \\ 104 & 1344 & 0 & 0 \\ 40 & 0 & 0 & 40 \\ 0 & 0 & 40 & 0 \end{pmatrix}.$$

In the reordered basis $(1,0,2,3)$ this is a real symmetric tridiagonal matrix with successive off-diagonal entries $104, 40, 40$. Let $D_j(x)$ be the determinant of the first j rows and columns of $xI - A_Q$ in that ordering. Expanding along the last row at each step gives

$$\begin{aligned} D_0 &= 1, \qquad D_1 = x - 1344, \\ D_2 &= x(x-1344) - 104^2, \\ D_3 &= xD_2 - 40^2 D_1, \\ D_4 &= xD_3 - 40^2 D_2 = p(x). \end{aligned}$$

Equivalently,

$$p(x) = (x^2 - 1600)\big[x(x-1344) - 10816\big] - 1600x(x-1344).$$

Expanding this expression yields exactly (4).

The same tridiagonal form explains why the four eigenvalues are simple. An eigenvector's first component determines its second component through the first row, and then determines the remaining components successively because each off-diagonal entry is nonzero. If the first component vanished, the recurrence would force the entire vector to vanish. Each eigenspace is consequently one-dimensional. Real symmetry makes the operator diagonalizable, so algebraic and geometric multiplicities agree. This establishes simplicity before any rounded eigenvalue is displayed.

The ordered simple eigenvalues have the following approximate displays. The calculation uses isolated exact roots and native enclosures, rather than these rounded values.

a	q_a
0	697.330492591257
1	752.008183657655
2	810.654350286272
3	2108.006973464815

The source response ledger and the physical strength g_H remain distinct from this completed matrix example. The source record retains T, K, the general polynomial, and the spectral calibration weights. [4, 6]

3.1 Six lanes and four diagonal states

The two tensor factors retain their order. An ordered product $|q_a q_b\rangle$ means $|q_a\rangle \otimes |q_b\rangle$, with the first and second eigenvectors assigned to the left and right factors. There are four choices in each factor, hence sixteen ordered products. For $a \neq b$, the products $|q_a q_b\rangle$ and $|q_b q_a\rangle$ are distinct orthogonal states. The swap operator exchanges these two states and leaves each $|q_a q_a\rangle$ fixed.

This organizes the source without discarding any states. Terms symmetric in the left and right operators act equally on the two orders of a given unequal pair. Their difference changes sign when the two orders are exchanged. The resulting two-dimensional invariant sector is what we call a lane. Its internal dynamics can be written with Pauli matrices while the direct sum of all sectors remains the full source.

Let $Q_L = Q \otimes I$, $Q_R = I \otimes Q$, and $S = \text{Swap}$. The complete pair operators are

$$\begin{aligned} H_{\text{pair}} &= S + Q_L + Q_R + Q_L Q_R, \\ D &= Q_L - Q_R, \qquad C = iSD. \end{aligned} \tag{5}$$

For $a < b$, the ordered lane basis is $(|q_a q_b\rangle, |q_b q_a\rangle)$. Set

$$\begin{aligned} d_{ab} &= q_a - q_b < 0, \\ e_{ab} &= q_a + q_b + q_a q_b. \end{aligned} \tag{6}$$

On that two-dimensional lane,

$$\begin{aligned} S = \sigma_x, \quad D = d_{ab}\sigma_z, \quad C = d_{ab}\sigma_y, \\ H_{\text{pair}} = e_{ab} I + \sigma_x. \end{aligned} \tag{7}$$

To see this explicitly, the two ordered states in a lane are exchanged by S, whereas D multiplies them by d_{ab} and $-d_{ab}$. Thus

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} d_{ab} & 0 \\ 0 & -d_{ab} \end{pmatrix},$$

$$C = iSD = \begin{pmatrix} 0 & -id_{ab} \\ id_{ab} & 0 \end{pmatrix}.$$

The factor i in C gives a Hermitian operator. Indeed, $SD = -DS$, so $(iSD)^* = -iDS = iSD$. This also fixes the sign of the σ_y convention in (7). Replacing d_{ab} by its absolute value would change this signed operator relation in the chosen ordered basis.

The symmetric and antisymmetric lane states are

$$|ab;\pm\rangle = \frac{|q_a q_b\rangle \pm |q_b q_a\rangle}{\sqrt{2}}.$$

They diagonalize S with eigenvalues ± 1. The remaining terms in H_{pair} multiply either ordered state by the same number e_{ab}, which gives the two energies stated below. The common offset and the exchange splitting are therefore separately visible in the lane representation.

Thus the source eigenenergies are $e_{ab} \pm 1$, while $D^2 = C^2 = d_{ab}^2 I$. The sign of d_{ab} remains in the interaction and correlated clock equation. It is retained when positive energy estimates use d_{ab}^2.

The four diagonal states $|q_a q_a\rangle$ have $D = C = 0$ and pair energy $(1+q_a)^2$. Together with the six two-state lanes, they exhaust all sixteen source states.

The completeness count is an orthogonal decomposition, not merely a count of displayed eigenvalues:

$$\mathbb{C}^4 \otimes \mathbb{C}^4 = \bigoplus_{a=0}^{3} \operatorname{span}\{|q_a q_a\rangle\} \oplus \bigoplus_{0 \le a < b \le 3} \operatorname{span}\{|q_a q_b\rangle, |q_b q_a\rangle\}.$$

The six unordered pairs are $(0,1)$, $(0,2)$, $(0,3)$, $(1,2)$, $(1,3)$, and $(2,3)$. Each contributes two ordered products, while the four equal pairs each contribute one. Hence $6 \times 2 + 4 = 16$. Every ordered product appears exactly once, and no additional source sector is needed to complete this basis.

On an equal-pair state, $Q_L - Q_R$ vanishes and swap acts as the identity. The stated energy follows from $1 + 2q_a + q_a^2 = (1+q_a)^2$. Calling these states unforced means that the later boundary interaction built from D and C vanishes on them. The field degrees of freedom still belong to the model. Likewise, a clock certificate in one unequal-pair lane concerns a declared preparation in that invariant sector; the existence of six such certificates does not identify six simultaneous preparations of a single source.

Writing $N_s = (I+S)/2$ gives $H_{\text{pair}} = 2N_s + K_{\text{pair}}$, where $K_{\text{pair}} = Q_L + Q_R + Q_L Q_R - I$ commutes with the source currents. This even remainder is scalar within a lane and varies across the full source. It remains distinct from the independent original common phase χ.

The commuting property follows from the operator structure. Since Q_L and Q_R commute, any polynomial in them commutes with D. The polynomial defining K_{pair} is also symmetric under their interchange, so it commutes with S and therefore with $C = iSD$. Within the ab lane it has the value $(e_{ab} - 1)I$, while $2N_s = I + S$ supplies the two values 2 and 0.

A term scalar on one lane cannot change the relative phase of that lane's two exchange eigenstates. It still contributes to the lane's energy and to the joint action. Moreover, its scalar value changes between lanes, so it need not be a common phase on a superposition spanning different sectors. Keeping this operator is necessary for a complete source action even when it drops out of a particular within-lane clock equation. No equality between this operator contribution and the original coordinate χ has been assumed.

3.2 Observable maps and retained complements

The earlier full18 tensor representation and this sixteen-state pair source are distinct objects. The retained endpoint-to-tensor construction has tensor Gram $32I_2$ and pair-observable Gram $11062272I_2$. Its observable composition reconstructs two tensor coordinates, while a retained complementary map reconstructs the other source coordinates. All sixteen joint C_4 character sectors remain present. [1, §2.2]

That composition preserves the difference between an observable map and a complete Hilbert action. The literal endpoint D/C action comparison retains exact residual Gram entry 54272. The full18 complement and that residual travel with the source construction; the lane clock below uses the explicit operators (5), rather than identifying these distinct state spaces.

The Gram statements identify a well-conditioned two-coordinate observable comparison in the retained normalization. They do not, by themselves, identify every degree of freedom in either underlying representation. A map can reconstruct the coordinates it observes while leaving additional source coordinates to a complementary map. Keeping that complement is how the observable comparison remains compatible with a complete source description.

The nonzero residual entry likewise records information that would be lost by declaring the literal endpoint action and the pair action identical. The full18 comparison, its complement, and its residual are retained as source facts. The subsequent lane dynamics has its own fully specified sixteen-state Hilbert space and operators. This makes it possible to use the observable relation without silently replacing a full action by its two measured coordinates.

4. Source energy and the time chart

An energy normalization requires both an operator and a state in which to evaluate it. The placement basis specifies how the source is initially selected, while the spectral basis of Q specifies how that selected state is resolved into eigenvectors. A placement-basis state will generally be a superposition in the spectral basis. Its energy expectation is the weighted mean of its possible energy values; an eigenenergy is the value associated with a state of sharp energy. The distinction can be checked quantitatively by the variance.

Here the same placement vector is used in both tensor factors. That product is symmetric under swap, which simplifies the calibration calculation while leaving its nontrivial spectral distribution intact.

The original calibration vector $|e_0 e_0\rangle$ is a placement-basis

state. Since $Q_{00} = 756$, its complete pair-energy expectation is

$$\begin{aligned} E_{\mathrm{ref}} &= \langle e_0 e_0 | H_{\mathrm{pair}} | e_0 e_0 \rangle \\ E_{\mathrm{ref}} &= (1+756)^2 = 573049. \end{aligned} \tag{8}$$

This expectation fixes the declared capacity normalization. It is not an eigenenergy. The spectral populations reconstruct (8), while the original pure-state density retains its coherences. Its D and C expectations vanish; its nonzero energy variance is 14384109824. It therefore supplies a different preparation from the interacting clock state below. [4, 6]

Both the expectation and the variance can be obtained directly from the displayed finite matrix. Write $v = |e_0\rangle$ and $f = (I+Q)v$. Because $S|vv\rangle = |vv\rangle$,

$$H_{\mathrm{pair}}|vv\rangle = f \otimes f.$$

The expectation is consequently $\langle v, f\rangle^2 = (1+Q_{00})^2$. For the second moment, Hermiticity gives

$$\langle vv | H_{\mathrm{pair}}^2 | vv\rangle = \|f \otimes f\|^2 = \|f\|^4.$$

The first row of Q gives

$$(Q^2)_{00} = 756^2 + 104^2 + 40^2 = 583952,$$

so

$$\begin{aligned} \|f\|^2 &= 1 + 2(756) + 583952 = 585465, \\ \mathrm{Var}_{vv}(H_{\mathrm{pair}}) &= 585465^2 - 573049^2 \\ &= 14384109824. \end{aligned}$$

The strictly positive result verifies directly that this calibration state does not have a single pair energy. Choosing its mean as the normalization leaves that energy spread present.

The spectral calibration weights have an exact resolvent expression. For $x_a = q_a - 756$ and p from (4),

$$p_a = \frac{(x_a - 1344)(x_a^2 - 1600)}{p'(x_a)}.$$

$$\sum_a p_a = 1, \qquad \sum_a p_a q_a = 756,$$

$$\sum_a p_a q_a^2 = 583952.$$

The four diagonal populations p_a^2 and six pair populations $2p_a p_b$ reconstruct the block-diagonal energy (8). This use of populations retains the coherences in the original pure state. [6]

The weights are the probabilities with which the placement vector occupies the four one-source eigenvectors. If

$$|e_0\rangle = \sum_a c_a |q_a\rangle, \qquad p_a = |c_a|^2,$$

then the scalar resolvent records the complete spectral measure of Q in this vector:

$$G(z) = \langle e_0 | (zI - Q)^{-1} | e_0 \rangle = \sum_a \frac{p_a}{z \quad q_a}.$$

It can also be evaluated by the $(0,0)$ cofactor. Setting $x = z - 756$, deletion of row and column zero from $xI - A_Q$ leaves a one-dimensional factor $x - 1344$ and a two-dimensional factor with determinant $x^2 - 1600$. Therefore

$$G(z) = \frac{(x-1344)(x^2-1600)}{p(x)}.$$

At a simple root x_a, the residue of this rational function is the numerator at x_a divided by $p'(x_a)$, which derives the weight formula above. The spectral representation also shows that these residues are nonnegative: each is the squared magnitude of a component of a normalized state.

The normalization and moment identities follow by expanding the resolvent for large z:

$$\begin{aligned} G(z) &= \frac{1}{z} + \frac{756}{z^2} + \frac{583952}{z^3} + O(z^{-4}) \\ &= \frac{\sum_a p_a}{z} + \frac{\sum_a p_a q_a}{z^2} + \frac{\sum_a p_a q_a^2}{z^3} + O(z^{-4}). \end{aligned}$$

Thus the exact root representation, the displayed matrix moments, and the probability interpretation are three descriptions of the same finite calculation.

For the pair calibration, the product state has the more informative decomposition

$$\begin{aligned} |e_0 e_0\rangle = &\sum_a c_a^2 |q_a q_a\rangle \\ &+ \sum_{a<b} \sqrt{2}\, c_a c_b |ab; +\rangle. \end{aligned}$$

Its antisymmetric lane amplitudes vanish because the two factors are identical. The diagonal weights are therefore p_a^2, and the unequal-pair weights $2p_a p_b$ belong to the symmetric energies $e_{ab} + 1$. In particular,

$$\begin{aligned} E_{\mathrm{ref}} &= \sum_a p_a^2 (1+q_a)^2 \\ &\quad + \sum_{a<b} 2p_a p_b (e_{ab} + 1) \\ &= \left[\sum_a p_a (1+q_a) \right]^2. \end{aligned}$$

Using the zeroth and first moment identities gives (8).

These populations suffice for this energy expectation because the displayed states diagonalize H_{pair}. They are not a replacement of the initial pure state by an incoherent mixture. Products of amplitudes between different components remain in its density operator and can matter for other observables. This is why a spectral calibration by probabilities is compatible with retaining the coherences of the original preparation.

The finite matrix calculation has so far supplied numbers in the unit-strength source convention. To use them in an action, one must specify an action unit and a time unit. The coordinate τ is dimensionless, so a dimensionless Hamiltonian contribution H in the model-time action carries physical energy

scale I_*H/t_u. The coefficient ε converts the finite source operator into that chart. Its definition keeps the Write time and the reference-capacity factor visible rather than assigning an unstated conversion to seconds.

For action unit I_*, physical time unit t_u, Write time t_{SW}, and reference capacity λ_{ref}, the receiving coefficient is

$$\varepsilon = \frac{\hbar}{I_*}\frac{t_u}{t_{\mathrm{SW}}}\frac{\lambda_{\mathrm{ref}}}{E_{\mathrm{ref}}}. \tag{9}$$

Multiplying by the reference expectation makes the normalization transparent:

$$\varepsilon E_{\mathrm{ref}} = \frac{\hbar}{I_*}\frac{t_u}{t_{\mathrm{SW}}}\lambda_{\mathrm{ref}}.$$

Consequently the reference source contribution, when restored to energy units, is

$$\frac{I_*}{t_u}\,\varepsilon E_{\mathrm{ref}} = \frac{\hbar\lambda_{\mathrm{ref}}}{t_{\mathrm{SW}}}.$$

This is a conversion relation for the declared capacity convention. A numerical physical energy or rate still requires the physical inputs on its right-hand side. The convenient choice of units below sets the dimensionless mean source contribution to one.

The normalized chart $I_* = \hbar$ and $t_u = t_{\mathrm{SW}}/\lambda_{\mathrm{ref}}$ gives

$$\varepsilon = \frac{1}{573049}, \qquad \Delta = 2\varepsilon. \tag{10}$$

We use $\tau = t_{\mathrm{phys}}/t_u$ throughout. A displayed angular rate ν is per unit τ. Its conditional physical rate is $(\lambda_{\mathrm{ref}}/t_{\mathrm{SW}})\nu$ in this chart. The source-energy normalization and the signed phase-current gap have separate roles.

In this chart, $\varepsilon E_{\mathrm{ref}} = 1$, whereas the dimensionless exchange splitting within any lane is 2ε. These statements answer different questions. The first specifies the average energy of the calibration preparation in the chosen units. The second specifies the separation between the two exchange eigenvalues of a lane after the same coefficient has been applied. Large lane-dependent even energies remain in the source action even though the unforced exchange splitting is the same in all six lanes.

The conversion of rates follows from the chain rule. For any differentiable phase ϑ,

$$\frac{\mathrm{d}\vartheta}{\mathrm{d}t_{\mathrm{phys}}} = \frac{1}{t_u}\frac{\mathrm{d}\vartheta}{\mathrm{d}\tau} = \frac{\lambda_{\mathrm{ref}}}{t_{\mathrm{SW}}}\frac{\mathrm{d}\vartheta}{\mathrm{d}\tau}$$

in the normalized chart. The signs of the rates and their ratios to a bare rate are unchanged by a positive choice of time unit. Their values in inverse seconds are not fixed until that unit is fixed. This permits a sharp dimensionless clock statement while leaving the physical Write-to-seconds assignment explicit.

5. The complete interacting action

The source, the receiving field, and their interaction must enter one variational account. A source expectation by itself is not that account, and a field driven by an independently prescribed source history is not the joint quantum system used here. The state belongs to the tensor product of the complete sixteen-dimensional source space and the two-polarization field Fock space. In particular, it may be entangled. The six lanes are invariant source blocks of this full model; choosing a lane specifies a preparation and a restricted calculation without deleting the other blocks from the definition of the model.

Let Ψ be a normalized joint source-field state. The model-time action is

$$\begin{aligned} \mathscr{S} &= I_* \int \mathrm{d}\tau\,[\mathscr{B}_\tau - \langle H\rangle], \\ \mathscr{B}_\tau &= \frac{i}{2}(\langle\Psi|\dot\Psi\rangle - \langle\dot\Psi|\Psi\rangle). \end{aligned} \tag{11}$$

Here the dot means differentiation with respect to the same normalized time τ used in (9)–(10). Consequently H is the generator in that time coordinate. The overall factor I_* restores the action unit; it does not insert an additional factor into the normalized Schrödinger equation. The first term is the joint-state Berry term. The second is the expectation of the complete joint Hamiltonian, interpreted as its quadratic form when the state has finite energy but is not in the operator domain.

For continuum trajectories, the derivative and energy pairings use the form domain and its dual, defined in (18). The smooth-core variational expression extends through the closed Hamiltonian form. This distinction will matter below: a finite-energy state need not possess a Hilbert-space-valued $H\Psi$, although its energy and its weak evolution are well defined.

The full form is

$$\begin{aligned} H = {}& \varepsilon H_{\mathrm{pair}} + H_f - \varepsilon(D\phi_+ + C\phi_\times) \\ & + \varepsilon^2\mu(D^2 + C^2), \qquad \mu = \frac{1}{4R}. \end{aligned} \tag{12}$$

Here ϕ denotes the boundary field in the retained receiving normalization. Equation (12) contains the entire joint Hamiltonian; the source and field actions are already included. The four contributions are the complete source energy, the free field energy, the two boundary couplings, and the two static-completion counterterms. None is an additional action to be appended to (11).

In particular, $\langle D\phi_+\rangle$ and $\langle C\phi_\times\rangle$ remain joint correlations. In a product state such an expectation can factor, but interacting evolution is not assumed to preserve a product state. Replacing it by $\langle D\rangle\langle\phi_+\rangle$, for example, would discard the connected correlation and define a different dynamical approximation. The actual clock calculation below keeps the correlations.

What remains when a lane is fixed. On one fixed lane, put $g = \varepsilon d_{ab}$ and $e = e_{ab}$. Then

$$\begin{aligned} H &= \varepsilon eI + \varepsilon S + K, \\ K &= H_f - g(\sigma_z\phi_+ + \sigma_y\phi_\times) + cI, \\ c &= 2g^2\mu = \frac{g^2}{2R}. \end{aligned} \tag{13}$$

Each real polarization contributes $g^2/(4R)$. Both contributions and the even energy εe remain in the action. The source sign also remains: $d_{ab} < 0$ for the ordered lanes, so $g < 0$ in the declared chart. Its square enters the energy completion, while g itself enters the interaction and the clock equation.

The word "scalar" is local to the chosen lane. The quantity e_{ab} is constant on its two source basis states, but differs between lanes. Likewise $D^2 + C^2 = 2d_{ab}^2 I$ on a lane, while the full operator $D^2 + C^2$ is not a common scalar on all sixteen source states. Its value vanishes on the four diagonal states. Thus dropping lane scalars before assembling the full source would also discard relative phases between distinct source blocks.

Even within a fixed lane, commuting with a chosen observable does not mean contributing no energy or action. The scalar $\varepsilon e + c$ drops out of a source commutator, and its effect on a state vector can be represented by a common dynamical phase. Nevertheless, it is present in $\langle H \rangle$ in the declared action. If that phase convention is changed, the Berry term must be changed with it: under $\Psi \mapsto e^{i\alpha(\tau)}\Psi$,

$$\mathscr{B}_\tau \mapsto \mathscr{B}_\tau - \dot{\alpha}.$$

Deleting an energy term while leaving the Berry account untouched is not the same operation. The later change of time coordinate retains both. The alternative decomposition $\varepsilon H_{\text{pair}} = 2\varepsilon N_s + \varepsilon K_{\text{pair}}$ is consistent with (13): on this lane $K_{\text{pair}} = (e-1)I$. It does not identify this commuting source remainder with the original common phase χ.

A positive square without commuting displacements. The positive field block is the square form

$$q_K[\Psi] = \sum_{\alpha=+,\times} \int_0^\infty \mathrm{d}\omega \left\| \sqrt{\omega}\, a_\alpha(\omega)\Psi - \frac{g\sigma_\alpha u_R^*(\omega)}{\sqrt{\omega}}\Psi \right\|^2,$$

where $\sigma_+ = \sigma_z$ and $\sigma_\times = \sigma_y$. The energy-annihilation map and its source shifts act in the same direct-integral space. This defines $K \geq 0$ without requiring the two shifted maps to commute.

Expanding this expression explains every term in (13). The sum of the unshifted squared norms is $\langle H_f \rangle$. The cross terms give $-g\langle \sigma_z \phi_+ + \sigma_y \phi_\times \rangle$ with the stated conjugation convention. The two shift norms are

$$g^2 \int_0^\infty \frac{|u_R(\omega)|^2}{\omega} \|\sigma_\alpha \Psi\|^2 \,\mathrm{d}\omega = g^2 \mu \|\Psi\|^2.$$

Since each Pauli matrix squares to the identity, their sum is exactly $c\|\Psi\|^2$, including both polarizations. There is no mixed $\sigma_z\sigma_y$ term: the squared norms are added in two distinct polarization components of a direct sum.

This positivity argument is not a diagonalization of K. The source matrices obey

$$[\sigma_z, \sigma_y] = \quad 2i\sigma_x \neq 0.$$

Accordingly the two source-dependent shifts need not be canonical annihilation operators for one common displaced Fock vacuum. Positivity of a sum of norm squares requires neither simultaneous diagonalization nor a vector annihilated by both shifted maps. No simultaneous shifted vacuum is assumed.

The same construction exists before restricting to a lane. Replace $g\sigma_\alpha$ by εJ_α, with $J_+ = D$ and $J_\times = C$. The summed shift norm becomes

$$\varepsilon^2 \mu \langle \Psi, (D^2 + C^2)\Psi \rangle,$$

so the field-plus-coupling-plus-counterterm part of (12) is positive on the full source space as well. The finite source Hamiltonian can then be added as a bounded self-adjoint term. The form construction and its domain are made explicit in §5.2.

Variation and the normalization phase. On a smooth core, vary (11) with respect to the bra and ket, with fixed endpoint variations, and impose $\langle \Psi | \Psi \rangle = 1$ by a real multiplier. The resulting equation can be written

$$i\dot{\Psi} = H\Psi + \zeta(\tau)\Psi.$$

The real function ζ changes only the common phase. A choice of that phase sets $\zeta = 0$, giving joint Schrödinger evolution. In the continuum form setting the same equation is understood in the dual of the form domain. This variation uses the complete joint state and therefore does not introduce a separate source-only or mean-field equation. [1, §3; 5, 6]

5.1 Boundary and field normalization

The receiving free-field Lagrangian is

$$\begin{aligned} L_f &= \frac{\kappa}{4} \sum_{a=+,\times} \int_0^\infty dr\, \mathscr{L}_a, \\ \mathscr{L}_a &= r^2 \dot{\varphi}_a^2 - r^2 (\partial_r \varphi_a)^2 - 6\varphi_a^2. \end{aligned} \tag{14}$$

The angular coefficient is $l(l+1) = 6$ for the retained quadrupole sector, $l = 2$. The two real polarizations have the same radial law. The value of κ is not selected by the radial differential equation alone. It is fixed by matching this received bulk law to the specified boundary tensor metric.

Why the static pieces contribute two and three. For one polarization and a fixed boundary coordinate $x = R\varphi(R)$, the static bulk equation away from the boundary is

$$\frac{d}{dr}\big(r^2 \varphi'(r)\big) - 6\varphi(r) = 0.$$

Its two independent solutions are r^2 and r^{-3}. Regularity and finite static energy select the first at the origin; decay and finite static energy select the second at infinity. Continuity of the prescribed boundary value therefore gives

$$\varphi_{\text{in}}(r) = \frac{x r^2}{R^3}, \qquad \varphi_{\text{out}}(r) = \frac{x R^2}{r^3}.$$

The gradient and angular parts are both needed. In the interior their integrands add as

$$r^2(\varphi_{\rm in}')^2+6\varphi_{\rm in}^2=\frac{10x^2r^4}{R^6},$$

and hence the interior integral is $2x^2/R$. In the exterior they add as

$$r^2(\varphi_{\rm out}')^2+6\varphi_{\rm out}^2=\frac{15x^2R^4}{r^6},$$

and the exterior integral is $3x^2/R$. The total is $5x^2/R$. Equivalently, for a static solution,

$$\frac{d}{dr}\big(r^2\varphi\varphi'\big)=r^2(\varphi')^2+6\varphi^2,$$

so the same values follow from the endpoint fluxes. The factors two and three are thus fixed by the regular and decaying quadrupole branches.

For the two-component boundary coordinate $\mathbf{x}$, the retained tensor metric $B_{\rm TT}=2I_2$ assigns static energy

$$\frac{1}{2R}\mathbf{x}^{\mathsf T}B_{\rm TT}\mathbf{x}=\frac{\|\mathbf{x}\|^2}{R}.$$

The bulk energy obtained from (14) is $(5\kappa/4R)\|\mathbf{x}\|^2$. Matching these two specified normalizations gives

$$\frac{5\kappa}{4}=1,\qquad \kappa=\frac{4}{5}. \tag{15}$$

This is a conditional normalization result: it receives the literal tensor metric, the boundary coordinate $x=R\varphi(R)$, and the bulk Lagrangian (14). It does not choose a new physical time scale or derive an alternative field law. Nor does using a static profile here replace the time-dependent continuum by a static field in the subsequent Hamiltonian.

Canonical momentum and the boundary sign. With radial measure r^2dr, the canonical momentum is $p_a=\kappa\dot\varphi_a/2$. With measure dr, the same variable is the density $\Pi_a=r^2p_a$. In particular,

$$\int_0^\infty r^2dr\,p_a\,\delta\varphi_a=\int_0^\infty dr\,\Pi_a\,\delta\varphi_a.$$

These are two ways of writing the same canonical one-form, not two different field normalizations. The associated free Hamiltonian is

$$H_{f,\rm classical}=\frac{\kappa}{4}\sum_a\int_0^\infty dr\,\left[r^2\dot\varphi_a^2+r^2(\varphi_a')^2+6\varphi_a^2\right].$$

The boundary source in the Lagrangian is $+\varepsilon J_a\varphi_a(R)$, where $J_+=D$ and $J_\times=C$. It therefore appears with the opposite sign in the Hamiltonian. Integrating the spatial variation separately on $[0,R]$ and $[R,\infty)$ gives the boundary coefficient

$$\frac{\kappa R^2}{2}\;\;\varphi_a'(R^+)-\varphi_a'(R^-)\;\;+\varepsilon J_a.$$

Its vanishing yields the retained jump

$$[\partial_r\varphi_a]_{R^-}^{R^+}=-\frac{2\varepsilon J_a}{\kappa R^2}. \tag{16}$$

Throughout this chapter the bracket means the exterior value minus the interior value. Reversing that orientation would reverse the left-hand side; it would not be permissible to keep the same signed equation without changing its convention.

The distributional bulk equation gives an independent check:

$$r^2\ddot\varphi_a-(r^2\varphi_a')'+6\varphi_a=\frac{2\varepsilon}{\kappa}J_a\delta(r-R).$$

Integrating it through R gives (16). At the static minimum found below, the profiles also give $\varphi'(R^+)-\varphi'(R^-)=-5x/R^2$; inserting $x=\varepsilon J/2$ and $\kappa=4/5$ reproduces the same sign and coefficient.

From the radial modes to the boundary kernel. The regular radial generalized modes can be normalized as

$$e_\omega(r)=\sqrt{\frac{2}{\pi}}\,\omega\,j_2(\omega r),\qquad \omega>0,$$

using the radial inner product with measure r^2dr. Their generalized orthogonality is

$$\int_0^\infty r^2dr\,e_\omega(r)e_{\omega'}(r)=\delta(\omega-\omega').$$

In this basis each free mode has oscillator Lagrangian $\kappa(\dot q^2-\omega^2q^2)/4$ and effective oscillator mass $\kappa/2$. Its coordinate factor is consequently $1/\sqrt{\kappa\omega}$ in the chosen quantum normalization. Thus the coefficient of $a_a(\omega)+a_a^\dagger(\omega)$ in the formal radial field is

$$\frac{e_\omega(r)}{\sqrt{\kappa\omega}}=\sqrt{\frac{2\omega}{\kappa\pi}}\,j_2(\omega r).$$

Here and below this mode expression is first read with finite frequency cutoffs and then as a form limit. It is not an assertion that the unsmeared boundary field is a Fock-space operator on every vector. The convention $H_f=d\Gamma(\omega)$ fixes the free-vacuum energy to zero, as already specified by the receiving quantization.

Choose the real radial representative $u_R(\omega)=\sqrt{2\omega/(\kappa\pi)}\,j_2(\omega R)$. This fixes the conjugation convention in the square form above. The quantized boundary kernel is

$$\begin{aligned}|u_R(\omega)|^2&=\frac{2\omega\,j_2(\omega R)^2}{\kappa\pi},\\ \mu&=\int_0^\infty\frac{|u_R(\omega)|^2}{\omega}\,d\omega=\frac{1}{4R}.\end{aligned} \tag{17}$$

For completeness, the scalar integral underlying the second line is

$$\int_0^\infty j_2(y)^2\,dy=\frac{\pi}{10}.$$

Indeed $j_2(y)=\sqrt{\pi/(2y)}J_{5/2}(y)$ and $\int_0^\infty J_\nu(y)^2\,dy/y=1/(2\nu)$ for $\nu>0$. Changing variables $y=\omega R$ then gives $\mu=$

$1/(5\kappa R) = 1/(4R)$. The Bessel identity is the equal-order case of the standard product integral.[1]

The retained boundary radius is $R \approx 0.163544171326059$. The static completion pays $\varepsilon^2 J^2/(4R)$ per current. To see the payment directly, first take a fixed real current J and insert the static boundary coordinate into the energy. The bulk and coupling terms give $(x^2 - \varepsilon J x)/R$, whose minimum is $-\varepsilon^2 J^2/(4R)$. Adding the declared counterterm gives

$$\frac{x^2 - \varepsilon J x + \varepsilon^2 J^2/4}{R} = \frac{(x - \varepsilon J/2)^2}{R}.$$

This algebra also explains each operator-valued completion, since source operators commute with the field coordinates on the other tensor factor. It does not make D and C commute with one another, nor does it imply that their two separate minima can be realized by one common source-field vacuum.

The radial and spectral calculations therefore agree: each current contributes the same $\varepsilon^2 \mu J^2$ to the completion, and the same receiving convention gives the sign in (16). Keeping only an exterior static solution, dropping the angular term, or retaining only one polarization would change this account. None of those changes is made here.

5.2 The full continuum and signed charge

Take the bosonic Fock space over $L^2((0,\infty), d\omega; \mathbb{C}^2)$, with $H_f = d\Gamma(\omega)$. Its one-particle space contains every positive frequency and both polarizations; its Fock construction contains every photon sector. The common energy form domain is

$$Q_f = D(H_f^{1/2}), \qquad A = I + H_f. \tag{18}$$

When source factors are present, Q_f denotes the corresponding finite direct sum of this field domain, and H_f and A act as the identity on the source. Its norm is

$$\|\eta\|_{Q_f}^2 = \|\eta\|^2 + \|H_f^{1/2}\eta\|^2 = \|A^{1/2}\eta\|^2.$$

This definition requires finite free-field energy. It does not require finite expectation of the number operator: arbitrarily low-frequency photons can contribute little energy. That distinction is important when interpreting charge conservation below.

Why the energy weight is the right condition. The kernel in (17) satisfies $u_R/\sqrt{\omega} \in L^2(0,\infty)$, with squared norm μ. It is not an unweighted L^2 function. To see the two ends of the integral explicitly, the small-argument expansion $j_2(y) = y^2/15 + O(y^4)$ gives

$$u_R(\omega) = O(\omega^{5/2}) \qquad (\omega \downarrow 0).$$

At high frequency the elementary formula

$$j_2(y) = \left(\frac{3}{y^3} - \frac{1}{y}\right)\sin y - \frac{3}{y^2}\cos y$$

gives

$$|u_R(\omega)|^2 = \frac{2\sin^2(\omega R)}{\kappa \pi R^2 \omega} + O(\omega^{-2}).$$

The unweighted integral diverges logarithmically, whereas division by ω makes the high-frequency integral convergent.[2] Thus the use of an energy form is substantive. Introducing a finite ultraviolet cutoff as the final model would conceal the relevant domain issue rather than resolve it.

For a cutoff kernel $u_R^{(n)}$ supported away from zero and infinity, the energy estimate is just Hilbert-space Cauchy–Schwarz applied to

$$a_a(u_R^{(n)})\eta = \int_0^\infty \frac{u_R^{(n)}(\omega)}{\sqrt{\omega}} \big(\sqrt{\omega}\, a_a(\omega)\eta\big)\, d\omega.$$

The squared norm of the second factor, integrated over frequency, is $\|H_{f,a}^{1/2}\eta\|^2$. The same estimate applied to differences of cutoffs shows convergence as an operator from Q_f to the Hilbert space. Its limit obeys

$$\|a(u_R)\eta\| \le \sqrt{\mu}\, \|H_{f,a}^{1/2}\eta\| \tag{19}$$

for the chosen polarization a. In particular, convergence is controlled by $\|(u_R^{(n)} - u_R^{(m)})/\sqrt{\omega}\|_{L^2}$, which tends to zero. No estimate here uses $\|u_R\|_{L^2}$.

Creation is defined by the adjoint form pairing. With the inner product antilinear in its first entry, the boundary field form is

$$q_{\phi_a}(\xi,\eta) = \langle \xi, a_a(u_R)\eta\rangle + \langle a_a(u_R)\xi, \eta\rangle, \qquad \xi,\eta \in Q_f.$$

It is a continuous sesquilinear form on $Q_f \times Q_f$, or equivalently a continuous map $Q_f \to Q_f^*$. In general it is not a map $Q_f \to \mathscr{H}$. For example, because $u_R \notin L^2$, the formal vector $a_a^\dagger(u_R)|0\rangle$ is not an ordinary one-photon Hilbert-space vector, even though all the displayed energy-form pairings are meaningful. The word "continuum" consequently does not assert that a quantum field has pointwise classical sample values.

For a bounded self-adjoint source current J, source and field factors commute and J preserves Q_f. The field form satisfies

$$|q_{J\phi_a}[\eta]| \le 2\sqrt{\mu}\, \|J\eta\|\, \|H_{f,a}^{1/2}\eta\|.$$

For every $\delta > 0$, Young's inequality therefore gives

$$|\varepsilon q_{J\phi_a}[\eta]| \le \delta \|H_{f,a}^{1/2}\eta\|^2 + \frac{\varepsilon^2 \mu \|J\|^2}{\delta}\|\eta\|^2.$$

Summing over the two polarizations shows that the interaction is infinitesimally bounded relative to the free energy form. The finite source and counterterm operators are bounded. Hence the closed, semibounded-form construction gives a unique self-adjoint Hamiltonian associated with (12), with form domain exactly Q_f.

[1] NIST Digital Library of Mathematical Functions, Eq. 10.22.55; take the two integer indices to be zero and the parameter there to be $3/2$. The spherical-Bessel definition is Eq. 10.47.3.

[2] The explicit formula is NIST DLMF, Eq. 10.49.3; its Taylor expansion also gives the stated small-argument behavior.

The same construction seen as a closed square. There is also a direct way to see why the positive square survives removal of frequency cutoffs. Define the energy-annihilation map by

$$(\mathscr{A}\eta)_a(\omega) = \sqrt{\omega}\, a_a(\omega)\eta,$$

as a map from Q_f into the direct sum, over polarizations, of $L^2((0,\infty), d\omega; \mathscr{H})$. It is closed and has

$$\|\mathscr{A}\eta\|^2 = \|H_f^{1/2}\eta\|^2.$$

The source shift

$$(\mathscr{T}\eta)_a(\omega) = \frac{\varepsilon J_a u_R^*(\omega)}{\sqrt{\omega}}\eta$$

is bounded from $\mathscr{H}$ into that same target space, because $\mu < \infty$ and each J_a is a finite matrix. Therefore $\mathscr{A} - \mathscr{T}$ is closed on the unchanged domain Q_f. Its norm square is the positive field block in (12). Adding the bounded source energy preserves closedness and semiboundedness. This argument makes no reference to commuting shifted annihilation operators.

On a fixed lane the resulting lower bound is

$$H \geq (\varepsilon e - \varepsilon)I,$$

because $K \geq 0$ and S has eigenvalues ± 1. Before restricting to a lane one has the corresponding form bound $H \geq \varepsilon\lambda_{\min}(H_{\text{pair}})I$ in the present positive-ε chart. These statements establish a stable energy form; they do not by themselves establish a ground-state eigenvector or an isolated hard-branch gap. Those spectral and preparation inputs retain their separate provenance in [4–6].

Evolution and the precise Ehrenfest regularity. Let $h_H(\xi,\eta)$ denote the closed Hamiltonian form. Choose a constant b large enough that $H + bI \geq I$. The norm induced by $(H+bI)^{1/2}$ is equivalent to $\|\cdot\|_{Q_f}$, by the form estimates above. The spectral theorem then implies that, for every $\Psi_0 \in Q_f$,

$$\Psi(\tau) = e^{-i\tau H}\Psi_0$$

stays in Q_f and is continuous there. Its weak derivative exists continuously in Q_f^* and satisfies $i\dot{\Psi} = H\Psi$ in that dual space. A Hilbert-space-valued derivative would require the stronger initial condition $\Psi_0 \in D(H)$; that stronger condition is not silently imposed on the continuum preparations used here.

Every bounded source observable O and its adjoint preserve Q_f continuously, since they act only on the finite source factor and commute with H_f. The expectation difference quotient, using the dual pairing, gives the form Ehrenfest identity

$$\frac{d}{d\tau}\langle\Psi, O\Psi\rangle = i\,h_H(\Psi, O\Psi) - i\,h_H(O^*\Psi, \Psi).$$

On an operator core this is $i\langle[H,O]\rangle$. The form expression is the definition that continues to hold for finite-energy states. Since h_H is continuous on $Q_f \times Q_f$ and $\Psi(\tau)$ is continuous in Q_f, its right-hand side is continuous in time. Thus bounded source expectations are continuously differentiable, not merely differentiable almost everywhere.

The same form continuity makes a source–field pairing such as $\langle S\phi_a\rangle$ a continuous function of time. It does not assert that every unbounded field observable has a differentiable expectation. These are precisely the regularity facts needed for (21)–(22): z is continuously differentiable, its form forcing w is continuous, and a nonvanishing z has a continuously differentiable lifted phase. This explains the pointwise meaning of the clock equation for the declared preparations in Q_f. [6]

Circular words and the signed conserved charge. Both real polarizations can be expressed in a circular basis without discarding either. To avoid confusing the real plus polarization with the circular plus mode, write the change of basis initially as

$$a_\pm^{\text{circ}} = \frac{a_{\text{lin},+} \pm i a_{\text{lin},\times}}{\sqrt{2}}, \qquad L = \frac{D + iC}{\sqrt{2}}.$$

The same unitary change of polarization is made at every frequency. Because the two real polarizations have the same real radial kernel, the interaction contains the four words $La_+^\dagger$, $L^\dagger a_+$, $L^\dagger a_-^\dagger$, and La_-, with their common frequency smearing and coupling understood. In these words and in the charge below, the subscripts $+$ and $-$ label the circular modes.

The source commutators follow from (5) and $N_s = (I+S)/2$:

$$[N_s, L] = -L, \qquad [N_s, L^\dagger] = L^\dagger.$$

Thus L lowers the source excitation by one and $L^\dagger$ raises it by one, whenever the corresponding matrix element is nonzero. If $N_\pm$ count the circular photons, each of the four interaction words preserves

$$Q_{\text{ch}} = N_s + N_+ - N_-. \tag{20}$$

The bookkeeping can be checked one word at a time:

word	ΔN_s	$\Delta(N_+ - N_-)$	ΔQ_{ch}
$La_+^\dagger$	-1	$+1$	0
$L^\dagger a_+$	$+1$	-1	0
$L^\dagger a_-^\dagger$	$+1$	-1	0
La_-	-1	$+1$	0

By contrast, the third and fourth words change $N_s + N_+ + N_-$ by $+2$ and -2, respectively. They are the counter-rotating words relative to that unsigned count. Dropping them would preserve an additional unsigned excitation number by changing the interaction; it is not part of the receiving law here.

The algebraic check can be promoted to a continuum conservation statement without assuming finite photon number. Define Q_{ch} as the self-adjoint signed-number generator, using the commuting source and number spectral measures. Its unitary group $U_\beta = e^{i\beta Q_{\text{ch}}}$ preserves Q_f. The factors in each interaction word

acquire opposite phases under U_β, while the free energy, the source Hamiltonian, and the counterterm are invariant. The cutoff field forms are therefore invariant, and their form limits remain invariant:

$$h_H(U_\beta\xi, U_\beta\eta) = h_H(\xi,\eta).$$

Uniqueness of the self-adjoint operator associated with the closed form implies that U_β commutes with H and its propagator. Hence every signed-charge spectral sector is invariant under the exact evolution. When a charge expectation is defined, for example when $\langle|Q_{\rm ch}|\rangle < \infty$, that expectation is conserved as well. Finite free-field energy alone is not used to infer this extra integrability.

The tensor angular action is $2I_*Q_{\rm ch}$ in the retained incidence. The counter-rotating words change the unsigned excitation count by two while preserving this signed charge. No rotating-wave or photon-number truncation defines the receiving law. In particular, a fixed signed-charge sector can still contain arbitrarily many photons; charge conservation is a symmetry statement, not a truncation of the field Hilbert space. [1, §4]

6. The actual interacting clock

The clock is the phase of a bounded source amplitude evaluated in the evolving joint state. Its rate receives the field through a joint correlation. We first derive that equation, then compare the actual evolution with an exactly specified hard-field reference. The comparison controls the instantaneous rate throughout a finite interval and supplies the monotone coordinate used in the action.

Fix one of the six lanes and retain its signed difference $d = d_{ab} < 0$, its coupling $g = \varepsilon d$, and its even source energy $e = e_{ab}$. In this lane define the bounded source amplitude and joint field form

$$\begin{aligned} Z &= (D+iC)/d = \sigma_z + i\sigma_y, \\ z &= \langle Z\rangle, \qquad w = \langle S(\phi_\times - i\phi_+)\rangle. \end{aligned} \tag{21}$$

Expectations here and below are in the joint source-field state. The two real coordinates of z are $\langle\sigma_z\rangle$ and $\langle\sigma_y\rangle$. The field quantity w also contains the source operator S. Knowledge of the separate source and field means does not in general determine it.

6.1 Deriving the exact clock equation

The source algebra fixes both the sign and the factor of two in the clock equation. Since $S = \sigma_x$, the Pauli commutators give

$$\begin{aligned} [S,\sigma_z] &= -2i\sigma_y, \\ [S,i\sigma_y] &= -2\sigma_z, \\ [S,Z] &= -2(\sigma_z + i\sigma_y) = -2Z. \end{aligned}$$

The even energy εeI, the field Hamiltonian, and the scalar counterterm commute with Z. The remaining source commutators are

$$[\sigma_z, Z] = 2S, \qquad [\sigma_y, Z] = 2iS.$$

Consequently, the Hamiltonian (13) obeys the form identity

$$\begin{aligned} i[H,Z] &= i\varepsilon[S,Z] - ig\big([\sigma_z,Z]\phi_+ + [\sigma_y,Z]\phi_\times\big) \\ &= -2i\varepsilon Z + 2gS(\phi_\times - i\phi_+). \end{aligned}$$

The form-domain construction in (18)–(19) justifies taking this expectation along the full continuum evolution. In particular, no derivative of an unbounded field operator is being assumed. The bounded source observable has the continuously differentiable expectation established in Section 5.

With $\Delta = 2\varepsilon$, the exact form equation and actual phase rate are

$$\begin{aligned} \dot z &= -i\Delta z + 2gw, \\ \theta &= -\arg z, \qquad \nu = \dot\theta = -\operatorname{Im}(\dot z/z). \end{aligned} \tag{22}$$

The phase is a continuous lift with $\theta(0) = 0$. Whenever z remains nonzero, write $z = re^{-i\theta}$ with $r = |z| > 0$. Then

$$\frac{\dot z}{z} = \frac{\dot r}{r} - i\dot\theta, \qquad \nu = \Delta - 2g\operatorname{Im}\frac{w}{z}.$$

This separates the amplitude change from the phase change without discarding either. In the absence of the correlator term, a positive bare gap gives the orientation $z \propto e^{-i\Delta\tau}$ and $\dot\theta = \Delta$. With the interaction present, the sign of g and the complex joint correlation both enter the rate. Replacing g by $|g|$ at this stage would change the exact equation; absolute values enter only the error estimates below.

6.2 The preparation and an actual reference expectation

Split the frequencies at $\lambda = 64\Delta$. The declared state is

$$\Psi_0 = \frac{\Phi_0 + \Phi_1}{\sqrt{2}} \otimes |0_s\rangle. \tag{23}$$

Here Φ_0 and Φ_1 are the exact normalized interacting hard-field eigenbranches of signed charges zero and one. The soft field is in its vacuum. Their true hard energy difference is $\Omega > 0$, and their relative phase is chosen so that

$$\langle\Phi_0|Z|\Phi_1\rangle = 2M, \qquad M > 0.$$

Thus the preparation specifies a coherent superposition of two dressed branches. It does not replace those branches by undressed source vectors. [4, 5]

Use a comparison Hamiltonian that keeps the same hard interaction, the same even scalar, the free soft field, and the exact scalar soft counterterm:

$$\begin{aligned} H_{\rm ref} &= \varepsilon e + H_h + H_{f,s} + c_s, \\ c_s &= 2g^2\int_0^\lambda \frac{|u_R(\omega)|^2}{\omega}\,d\omega. \end{aligned}$$

The actual and reference Hamiltonians therefore differ only by the linear soft interaction. If $\mathscr{E}_0$ and $\mathscr{E}_1$ denote the two hard branch energies, then $\mathscr{E}_1 - \mathscr{E}_0 = \Omega$ and

$$\Psi_{\rm ref}(\tau) = \frac{e^{-i(\varepsilon e + c_s)\tau}}{\sqrt{2}}\; e^{-i\mathscr{E}_0\tau}\Phi_0 + e^{-i\mathscr{E}_1\tau}\Phi_1 \;\otimes|0_s\rangle.$$

The common scalar phase has been retained explicitly. It cancels in an expectation, while remaining part of the dynamical action.

The charge rule explains why this trajectory has a single clock frequency. Since Z acts only on the source,

$$[Q_{\mathrm{ch}}, Z] = [N_s, Z] = \tfrac{1}{2}[S, Z] = -Z.$$

It lowers the signed charge by one. Within the two-branch superposition, only the matrix element from the charge-one branch to the charge-zero branch survives. Its coefficient is one half, so the reference amplitude is

$$\langle \Psi_{\mathrm{ref}}(\tau)|Z|\Psi_{\mathrm{ref}}(\tau)\rangle = \tfrac{1}{2}(2M)e^{-i(\mathscr{E}_1-\mathscr{E}_0)\tau} = Me^{-i\Omega\tau}.$$

The soft part of the reference field mean vanishes. Applying the hard-field form Ehrenfest equation to this actual reference expectation therefore gives

$$-i\Omega h = -i\Delta h + 2g w_h.$$

Hence, in the same field realization,

$$h = Me^{-i\Omega\tau}, \qquad w_h = \frac{i(\Delta-\Omega)}{2g}h. \tag{24}$$

The quantity w_h is the expectation of the same form $S(\phi_\times - i\phi_+)$ on the specified reference trajectory. Its relation to h is exact. In particular, it introduces no separately fitted field target or mismatch term.

At $\tau = 0$ the actual and reference states coincide. They have $z(0) = h(0) = M$ and $w(0) = w_h(0)$, so (22) gives $\nu(0) = \Omega$. The three rates now have explicit roles: Δ is the bare source gap, Ω is the hard reference gap, and $\nu(\tau)$ is the instantaneous rate of the full actual evolution.

6.3 Centering the comparison at the dressed gap

Set $\delta z = z - h$ and $\delta w = w - w_h$. Subtract the reference equation from the actual equation and then add $i\Omega z$. The terms proportional to h cancel:

$$\begin{aligned}\dot z + i\Omega z &= -i(\Delta-\Omega)(h+\delta z) + 2g(w_h+\delta w)\\ &= \big[-i(\Delta-\Omega)h + 2g w_h\big] - i(\Delta-\Omega)\delta z + 2g\delta w.\end{aligned}$$

Thus the exact centered residual is

$$\dot z + i\Omega z = -i(\Delta-\Omega)\delta z + 2g\delta w. \tag{25}$$

The error multiplying δz is weighted by the dressed shift $\Delta - \Omega$. The reference already accounts for the hard dressing. The remaining comparison concerns the soft coupling and its propagation by the complete Hamiltonian.

The retained all-order soft estimate and energy bound are

$$\begin{aligned}|\delta z| &\le E(\tau) = e^{4g^2 v_s^{\mathrm{up}}\tau^2} - 1,\\ v_s^{\mathrm{up}} &= \frac{R^4\lambda^6}{675\kappa\pi},\\ F_T &= \Big(\sqrt{c} + \sqrt{K_{\mathrm{prep}} + \varepsilon E(T)}\Big)^2.\end{aligned} \tag{26}$$

Why the soft-field weight has a sixth power

The soft spectral weight in this argument is

$$v_s = \int_0^\lambda |u_R(\omega)|^2\,\mathrm{d}\omega.$$

It is different from the full energy-weighted quantity μ, which includes an additional factor $1/\omega$ and integrates over all frequencies. The spherical-Bessel integral representation [7], at order two and real x, gives

$$j_2(x) = \frac{x^2}{16}\int_{-1}^{1}\cos(xt)(1-t^2)^2\,\mathrm{d}t.$$

Since the weight is nonnegative and its integral is $16/15$, this yields $|j_2(x)| \le x^2/15$. Substituting the boundary kernel from (17) then gives the explicit comparison

$$\begin{aligned}v_s &\le \frac{2R^4}{225\kappa\pi}\int_0^\lambda \omega^5\,\mathrm{d}\omega\\ &= \frac{R^4\lambda^6}{675\kappa\pi} = v_s^{\mathrm{up}}.\end{aligned}$$

Thus the coefficient in (26) is tied to the declared quadrupole kernel and the soft cutoff. This calculation explains the spectral majorant; the retained all-order amplitude estimate $E(\tau)$ and the preparation-energy bound retain their separate source attribution [6]. Here v_s^{up} bounds the soft field weight $\int_0^\lambda |u_R(\omega)|^2 d\omega$. The frequency split is part of the comparison; the actual dynamics still contains the entire continuum.

The preparation quantity K_{prep} is the retained upper bound on $\langle K\rangle$ in the exact state (23), including its full soft vacuum counterterm. The inherited all-time estimate is

$$\langle K\rangle_\tau \le K_{\mathrm{prep}} + \varepsilon E(\tau).$$

These all-order comparison inputs are supplied by the retained derivation [6]. The following argument expands how they lead to a propagated field estimate and a clock certificate. A bound on the single amplitude δz alone would not supply a bound on every source or field expectation.

The square form of K explains the conversion to free field energy. Let $X\Psi$ be the two-polarization energy-annihilation vector and $Y\Psi$ its source shift in the common direct-integral space. For a normalized state,

$$\|X\Psi\|^2 = \langle H_f\rangle, \quad \|Y\Psi\|^2 = c, \quad \|(X-Y)\Psi\|^2 = \langle K\rangle.$$

The triangle inequality yields

$$\sqrt{\langle H_f\rangle} \le \sqrt{\langle K\rangle} + \sqrt{c}.$$

Using the retained energy bounds for both trajectories gives $\langle H_f\rangle \le F_T$ throughout $[0,T]$, for the actual and reference states alike. The monotonicity of $E(\tau)$ for $\tau \ge 0$ permits the single endpoint value $E(T)$ in this interval estimate.

6.4 Propagating the field comparison in the energy norm

The field insertion in the clock is a form rather than a bounded operator on the unweighted Hilbert space. Introduce $A = I + H_f$ and

$$B = S(\phi_\times - i\phi_+), \qquad \|\eta\|_A = \|A^{1/2}\eta\|.$$

The relevant control is on the weighted pairing and on propagation in this energy norm:

$$\begin{aligned} \|A^{-1/2}BA^{-1/2}\| &\le \sqrt{2\mu}, \\ \|A^{1/2}e^{-i\tau H}\eta\| &\le 2\|A^{1/2}\eta\|. \end{aligned} \tag{27}$$

The first inequality means that for $\eta, \xi \in Q_f$,

$$|\langle \eta | B | \xi \rangle| \le \sqrt{2\mu}\, \|\eta\|_A \|\xi\|_A.$$

It follows from the weighted annihilation estimate (19), its adjoint pairing, and the two-polarization combination. It controls exactly the joint form used in w.

For the propagator estimate, first apply the two triangle-square inequalities to the same shifted maps:

$$H_f \le 2K + 2cI, \qquad K \le 2H_f + 2cI,$$

where the inequalities are quadratic-form inequalities. Put

$$H_{\rm red} = \varepsilon S + K, \qquad B_0 = H_{\rm red} + (\varepsilon + c + 1/2)I.$$

Because $-I \le S \le I$, the lower comparison is

$$A = I + H_f \le 2K + (2c+1)I \le 2B_0.$$

For the upper comparison,

$$B_0 \le 2H_f + (2\varepsilon + 3c + 1/2)I \le 2A.$$

The final inequality uses the checked condition $2\varepsilon + 3c + 1/2 \le 2$. Together these give $A \le 2B_0 \le 4A$. Since B_0 is a positive scalar shift of $H_{\rm red}$, its form norm is conserved by the full propagator. The even scalar $\varepsilon e I$ only contributes a common phase. Therefore

$$\begin{aligned} \|e^{-i\tau H}\eta\|_A^2 &\le 2\|B_0^{1/2}e^{-i\tau H}\eta\|^2 \\ &= 2\|B_0^{1/2}\eta\|^2 \le 4\|\eta\|_A^2, \end{aligned}$$

which proves the second inequality in (27).

The difference between the Hamiltonians is the soft interaction

$$V_s = H - H_{\rm ref} = -g(\sigma_z \phi_{+,s} + \sigma_y \phi_{\times,s}).$$

On $\Psi_{\rm ref}$, each soft annihilation term vanishes. Each creation term produces one soft photon, and the two polarization sectors are orthogonal. A photon created at frequency $\omega \le \lambda$ adds at most λ to the free energy. Consequently,

$$\begin{aligned} \|A^{1/2}V_s\Psi_{\rm ref}(s)\|^2 &\le 2g^2 \int_0^\lambda |u_R(\omega)|^2 (1 + F_T + \omega)\, d\omega \\ &\quad 2g^2 v_s^{\rm up}(1 + F_T + \lambda). \end{aligned}$$

The factor of two counts the orthogonal polarization insertions. The source Pauli matrices do not change the free field energy norm.

Write $\delta\Psi(\tau) = \Psi(\tau) - \Psi_{\rm ref}(\tau)$. The two trajectories start from the same vector, so Duhamel propagation gives

$$\delta\Psi(\tau) = -i \int_0^\tau e^{-i(\tau-s)H} V_s \Psi_{\rm ref}(s)\, ds.$$

Using (27) in the energy norm yields, uniformly for $0 \le \tau \le T$,

$$\|\delta\Psi(\tau)\|_A \le 2T\sqrt{2g^2 v_s^{\rm up}(1 + F_T + \lambda)}.$$

Finally, expand the difference of the two actual expectations of B:

$$\delta w = \langle \delta\Psi | B | \Psi \rangle + \langle \Psi_{\rm ref} | B | \delta\Psi \rangle.$$

Both states have energy norm at most $\sqrt{1 + F_T}$. The weighted pairing in (27) therefore contributes the factor $2\sqrt{2\mu(1+F_T)}$. Multiplying the two bounds proves

$$\begin{aligned} \sup_{0 \le \tau \le T} |\delta w(\tau)| &\le W_D(T), \\ W_D(T) &= 8|g|T\sqrt{\mu v_s^{\rm up}(1+F_T)(1+F_T+\lambda)}. \end{aligned} \tag{28}$$

The one-photon statement concerns the forcing vector before propagation. The propagator in the integral is the complete interacting $e^{-i(\tau-s)H}$ and retains all photon sectors. The weighted estimates permit the frequency-cutoff construction to pass to the full form-domain limit. Thus (28) does not impose a photon-number or rotating-wave truncation. [6]

6.5 From the centered residual to a uniform rate

Two quantities must be controlled to divide by z: the residual in (25) and the distance of the amplitude from zero. Since $|h(\tau)| = M$, the reverse triangle inequality gives

$$|z(\tau)| \ge |h(\tau)| - |\delta z(\tau)| \ge M - E(T).$$

If $E(T) < M$, this is a positive lower bound on the entire interval. It also justifies a single continuously differentiable phase lift, rather than unrelated choices of a wrapped argument at each time.

Divide (25) by z. The result isolates the exact dressed frequency:

$$\frac{\dot z}{z} = -i\Omega + \frac{-i(\Delta - \Omega)\delta z + 2g\delta w}{z}.$$

Taking the negative imaginary part and using $|\operatorname{Im}\zeta| \le |\zeta|$, for $0 < \Omega \le \Delta$ one obtains

$$|\nu - \Omega| \le \frac{(\Delta - \Omega)E(T) + 2|g|W_D(T)}{M - E(T)}. \tag{29}$$

This is an instantaneous rate estimate. The time derivative comes from the exact form equation (22). No derivative of the error envelope $E(T)$ has been used.

For clarity, denote the right side of (29) by ρ_T. A sufficient positive-rate condition is $\rho_T < \Omega$. The denominator is positive, so its algebra is reversible:

$$\begin{aligned}(\Delta-\Omega)E(T)+2|g|W_D(T) &< \Omega\big(M-E(T)\big)\\ \iff\quad 2|g|W_D(T)+\Delta E(T) &< \Omega M.\end{aligned}$$

The same true hard gap appears on both sides before simplification. The sufficient positivity condition is therefore

$$2|g|W_D(T)+\Delta E(T) < \Omega M. \tag{30}$$

The cancellation of the $\Omega E(T)$ terms is the reason the final error term uses $\Delta E(T)$. Since $W_D(T) \geq 0$ and $0 < \Omega \leq \Delta$, strict inequality in (30) also implies $E(T) < M$. Keeping the amplitude check explicit remains useful in the numerical certificate.

6.6 Six certified intervals and their inverses

Native outer evaluation satisfies (30) and the positive amplitude condition in every lane for

$$0 \leq \tau \leq T_0 = \frac{2\pi}{\Delta} = 573049\pi. \tag{31}$$

The calculation uses the lane's own d, g, M, Ω, and preparation bound. The common bare interval does not identify the six separately prepared trajectories with one another.

The analytic implication is now explicit: control the soft amplitude error and free energy, propagate the joint insertion, keep the amplitude denominator positive, and use the centered residual to enclose the rate. The native computation supplies the source and hard-branch enclosures needed to evaluate these expressions. Exact roots and outward enclosures enter that evaluation; the rounded displays below communicate its result. [6]

Lane	Lower $10^6\nu$	Upper $10^6\nu$
01	3.490102929541	3.490102934505
02	3.490102893142	3.490102914462
03	3.490095588823	3.490098892473
12	3.490102927880	3.490102933590
13	3.490096147687	3.490099200196
23	3.490096722560	3.490099516734

Table 1: Uniform actual rates per normalized model time. Endpoints are rounded outward to twelve decimal places after multiplication by 10^6. Exact rational outer endpoints remain in the native certificate. [6]

Every lower endpoint is positive, and every upper endpoint lies below $\Delta \approx 3.49010294058623 \times 10^{-6}$. These are two distinct numerical conclusions: the positivity inequality alone would not imply the upper comparison with Δ. The reported native rate enclosures establish both. The achieved W_D upper bounds range from 1.229714×10^{-11} to 3.172702×10^{-10}.

Let ℓ and u be the certified lower and upper rates for any one lane, in the unscaled units of ν. For $0 \leq \tau_1 < \tau_2 \leq T_0$, integrate the pointwise bound:

$$\begin{aligned}\ell(\tau_2-\tau_1) &\leq \theta(\tau_2)-\theta(\tau_1)\\ &\leq u(\tau_2-\tau_1).\end{aligned}$$

The lower inequality proves strict monotonicity throughout the interval. It gives a unique inverse $\tau = \tau(\theta)$ on the image $[0, \theta(T_0)]$. The inverse function theorem and the pointwise regularity of (22) give

$$\frac{d\tau}{d\theta} = \frac{1}{\nu(\tau(\theta))}, \qquad \frac{1}{u} \leq \frac{d\tau}{d\theta} \leq \frac{1}{\ell}.$$

The inverse is continuously differentiable in the interior, with the corresponding one-sided derivatives at the endpoints, and is Lipschitz with constant at most $1/\ell$ on its whole image.

Because $u < \Delta$ and $\theta(0) = 0$,

$$0 < \theta(T_0) \leq uT_0 < \Delta T_0 = 2\pi.$$

Thus T_0 is one bare cycle. The interacting clock has accumulated less than a full 2π phase at that time. The certificate concerns this stated interval; extending it requires the same estimates and positivity checks for the longer window.

6.7 Worked example for lane 01

The ordered spectral roots give $d_{01} = q_0 - q_1 < 0$, and $\varepsilon > 0$ then gives $g_{01} < 0$. Representative values are

Quantity	Approximate value
d_{01}	-54.6776910663982
g_{01}	$-9.54153851876510 \times 10^{-5}$
εe_{01}	0.917631085332839
c_{01}	$2.78337517524759 \times 10^{-8}$

The complete source branch energies are approximately 0.917629340281369 and 0.917632830384310. Their midpoint is the retained even source energy, while their separation is $2\varepsilon = \Delta$. The hard dressed gap below is a separate quantity.

For the declared hard preparation, certified outer displays give

$$\begin{aligned}&0.9999999987538952 < M < 0.9999999987925324,\\ &3.4901029318881649 < 10^6\Omega\\ &\qquad\qquad < 3.4901029321578596.\end{aligned} \tag{32}$$

The amplitude starts on the positive real axis, so the continuous lift begins at $\theta(0) = 0$. Equation (24) then fixes the initial correlation as a purely imaginary number,

$$w(0) = \frac{i(\Delta-\Omega)M}{2g}.$$

Both $S\phi_\times$ and $S\phi_+$ have real expectation values, since source and field factors commute. Comparing the real and imaginary parts with $w = \langle S\phi_\times\rangle - i\langle S\phi_+\rangle$ gives

$$\begin{aligned}\langle S\phi_\times\rangle &= 0,\\ \langle S\phi_+\rangle &= -\frac{(\Delta-\Omega)M}{2g} > 0.\end{aligned} \tag{33}$$

Here $\Delta-\Omega>0$, $M>0$, and $g<0$, which explains the final sign. The plus correlation lies between 4.4166740×10^{-11} and 4.5580007×10^{-11}. Substituting it back into the exact equation checks the initial shift directly:

$$\dot{z}(0)=-i\Delta M+2g\frac{i(\Delta-\Omega)M}{2g}=-i\Omega M.$$

Thus the initial clock rate is the hard gap. The joint correlation is precisely the term that carries this shift. Factoring it into marginal means would change this result.

The first row of the rate table then supplies a uniform statement about the subsequent actual evolution. For example, the ratio of the accumulated phase after one bare cycle to 2π satisfies

$$0.9999999968352702545\le\frac{\theta(T_0)}{2\pi}\le0.9999999982575778725.$$

These two numbers are the exact arithmetic images of the outward displayed lane-01 rate endpoints under multiplication by $573049/2$ and by 10^{-6}. They illustrate the integral consequence of the existing uniform certificate; they are not additional independently evaluated native enclosures.

The completed Duck campaign evaluated 300 native operations with 1,122 passing native checks. A separate checker passed 23,804 checks over all 300 new outputs and 134 matrix readouts. These are computational verification counts; the continuum argument is supplied analytically. The weighted-form, reference, and Duhamel construction receives the original source rules and retained hard preparation (GEN6 Duck run chapter1-lane01-20261015-01, Duck 1.2.2, core 0.10.2; H002630). [6]

7. The action in clock phase

The certified monotone phase can parameterize the same joint history. Write $\tau=\tau(\theta)$ and understand every state, field coordinate, and rate in this section as evaluated on that history. In particular, ν means $\nu(\tau(\theta))$. Uniform positivity supplies the change of differential

$$d\tau=\frac{d\theta}{\nu}.$$

This permits the complete action to be expressed in clock phase on $[0,\theta(T_0)]$ while retaining its source, field, and interaction contributions.

7.1 Energy and field coefficients under the chain rule

For any energy term $\mathscr{E}(\tau)$ in the Hamiltonian expectation,

$$-\int d\tau\,\mathscr{E}(\tau)=-\int d\theta\,\frac{\mathscr{E}(\tau(\theta))}{\nu}.$$

Every energy term therefore receives the same inverse rate. The scalar terms εe and c obey this rule just as the source exchange and the boundary correlation do. Their commutation with Z removes them from the clock commutator, but does not remove them from the action.

Contribution	Coefficient in θ
Field kinetic	$\nu/5$
Field radial gradient	$-1/(5\nu)$
Field angular term	$-6/(5\nu)$
Even source energy	$-\varepsilon e/\nu$
Source exchange	$-\varepsilon/\nu$
Boundary coupling	$+g/\nu$
Both counterterms	$-c/\nu$

Table 2: Coefficients inside the action divided by I_*. Field rows multiply their radial integrals; source and boundary rows multiply the corresponding joint expectations in (13). Field rows use (14); (11) already contains the full source-field Berry form. The full counterterm is $c=g^2/(2R)$.

The kinetic term in the configuration-space field expression (14) also contains a derivative. At fixed radial coordinate,

$$\dot{\varphi}_a=\nu\,\partial_\theta\varphi_a.$$

Using $\kappa/4=1/5$, the same field expression becomes

$$\int d\tau\,L_f=\frac{1}{5}\sum_{a=+,\times}\int d\theta\int_0^\infty dr\left[\nu r^2(\partial_\theta\varphi_a)^2-\frac{r^2(\partial_r\varphi_a)^2}{\nu}-\frac{6\varphi_a^2}{\nu}\right].$$

The kinetic coefficient has one remaining factor of ν: two arise from the squared derivative and one is canceled by the measure. The gradient and angular terms have no time derivatives and retain the inverse rate. This calculation describes the canonical field representation of the change of parameter. The full joint action (11) already contains the field through its Berry form and Hamiltonian; the field expression is not a second action to add to it.

The resulting coefficient family is listed in Table 2. For example, the boundary row multiplies the expectation of $\sigma_z\phi_+ + \sigma_y\phi_\times$. Its displayed plus sign is inherited from subtracting the Hamiltonian, while its coefficient retains the signed $g<0$ of the ordered lane. The counterterm row includes both real polarizations, each of which contributes $g^2/(4R)$ before the change of parameter.

7.2 The Berry one-form is unchanged

The state derivative obeys the same chain rule,

$$\partial_\tau\Psi=\nu\,\partial_\theta\Psi.$$

Consequently,

$$\mathscr{B}_\tau=\nu\mathscr{B}_\theta,\qquad\mathscr{B}_\tau\,d\tau=\mathscr{B}_\theta\,d\theta.$$

The Berry contribution is therefore the unchanged one-form

$$\frac{i}{2}(\langle\Psi|d\Psi\rangle-\langle d\Psi|\Psi\rangle).\tag{34}$$

In the continuum, the derivative pairings retain the form-domain and dual interpretation used in (11) and (18). Positivity and continuity of ν make the change of parameter regular on the certified interval.

Combining this one-form identity with the common inverse-rate factor for the complete Hamiltonian gives

$$\frac{\mathscr{S}}{I_*} = \int d\theta \left[\mathscr{B}_\theta - \frac{\langle H \rangle}{\nu} \right]. \tag{35}$$

Here H is exactly the Hamiltonian (12), with its even source energy and both counterterms. Equation (35) is the same action evaluated along the same history in its certified phase coordinate.

7.3 The coefficients share one actual clock

The entries of the coefficient table are dependent functions of one rate. If $\ell \le \nu \le u$, then

$$\frac{\ell}{5} \le \frac{\nu}{5} \le \frac{u}{5}, \qquad \frac{1}{u} \le \frac{1}{\nu} \le \frac{1}{\ell}.$$

These interval images are useful for bounding individual terms, but the values cannot be chosen independently. At every point of the history, for example,

$$\left(\frac{\nu}{5}\right)\left(-\frac{1}{5\nu}\right) = -\frac{1}{25}, \qquad -\frac{6}{5\nu} = 6\left(-\frac{1}{5\nu}\right).$$

The same common inverse rate also fixes the ratios of the source, coupling, and counterterm coefficients. Selecting unrelated endpoints from their separate intervals would generally violate these identities.

The clock is derived from the actual state through (21)–(22). Equation (35) is a trajectory and history reparameterization on the monotone interval. If one uses it in a variational treatment, the dependence of ν and of the clock coordinate on the joint history must be retained. Varying the transformed expression while holding a separately prescribed ν fixed does not, by itself, establish an equivalent variational principle. The established result transports the original joint action into the actual phase coordinate; it does not introduce an independent clock-rate control.

The local inverse also has a different role from a finite-window mean. For $\tau > 0$,

$$\theta(\tau) = \int_0^\tau \nu(s)\, ds, \qquad \frac{\tau}{\theta(\tau)} = \left[\frac{1}{\tau} \int_0^\tau \nu(s)\, ds \right]^{-1}.$$

Thus $\tau/\theta(\tau)$ is the inverse of the mean rate on a window, while $d\tau/d\theta = 1/\nu$ is the pointwise Jacobian required by the action. They coincide for a constant-rate history and need not coincide during interacting evolution.

Finally, for a conserved energy expectation E, the positive dynamical energy account has the equivalent expressions $I_* E\tau = I_* \int_0^{\theta(\tau)} E\, d\theta'/\nu$. The full variational action contains the negative energy term together with the Berry contribution. Keeping that distinction preserves the complete source, field, and interaction account under the same change of parameter.

8. Chapter conclusion

The signed source construction, complete joint action, continuum form realization, and six uniform clock certificates close the source-to-clock argument of this chapter for the stated receiving law and preparation. **The test result suggests strong contact with the concept.** The centered target in the refinement note [2] is now supplied by the achieved bound (28) and its native evaluation. The historical derivations remain preserved in their original execution records.

The next source interface is the occupied-path relation between the original ledger $\hbar(d\gamma_W + 3d\chi)$ and the quantum generator $H_{\mathrm{pair}}/2$. That work must carry the original common phase, the commuting even operator, and the preparation class. The six lane clocks established here provide the completed receiving calculation for that continuation. Radial geometry and entropy-area results belong to their later chapters. [3]

Data availability

The repository for this chapter is `https://github.com/iwtbotiwtwot/sam_volume_I_i`.

Source register

[1] *Original Write histories and a coupled source-field action*, v1.1, 15 October 2026. Supplied manuscript; GEN6 Duck runs write-coupling-clock-20261015-01 and chapter1-lane01-20261015-01 (H002628, H002630).

[2] *A Reference-Centered Clock Positivity Condition*, 15 October 2026. Supplied refinement note.

[3] *SAM Volume I Chapter Map and Research Plan*. Supplied roadmap; its earlier manuscript references are superseded here by [1] and [6].

[4] *Original Write source coupling and six clocks*, H002628. Completed source campaign, 15 October 2026.

[5] *Resolved interacting clock*, retained source and hard-branch derivation, samland_resolved_clock1.

[6] *Chapter I lane01 and six uniform correlated clocks* (this chapter, now Chapter VII), H002630, 15 October 2026; continuum, reference, and weighted-propagation derivations with native certificates. Exact source paths and hashes are retained with this draft.

[7] NIST Digital Library of Mathematical Functions, §10.54, Eq. 10.54.1, spherical-Bessel integral representation. `https://dlmf.nist.gov/10.54.E1`. Accessed 15 October 2026.

SAM VOLUME I

Horizon thermodynamics and accessible information

Chapter VIII

Abstract. The completed boundary connects exterior geometry, finite thermodynamics, and accessible records. This chapter derives the surface gravity and first law under the explicitly inherited metric, temperature, and quarter-area premises; develops exact finite-entry transitions, their Catalan expansion, and conditional detailed balance; and separates thermal occupation from an absolute emission rate. The retained Si29 calculation is reformulated as a Hessian-level obstruction to scalar area–information proportionality, while its entropy–area pair provides local coordinates on the specified two-parameter source family. New analytic checks clarify finite-capacity errors, transmitted count fluctuations, entropy balance, and the distinction between a causal horizon and a material surface. The remaining physical joins are stated as testable tasks: a source-selected boundary response, energy and clock calibration, species-resolved dynamics, a certified information map, and consistency with compact-surface observations. No new native source execution or absolute luminosity is claimed.

Status, 25 October 2026 (Part 2). This chapter adds to Part 1 the thermal premise: a static metric, a normalized Killing time, the Hawking conversion and the quarter-area entry convention (register T-13). The finite-entry staircase and its Catalan series are exact conditional mathematics; the entry convention is declared, not derived. One run touches it: volii_gate_1 (ironclad H002782), whose reading is that a step temperature named from a hazard is a change of name (Arrhenius through the saddle, $T_A \to 1/\beta$), not an identification.

1. Purpose, premises, and the boundary question

The unit-accumulation surface joins three subjects that must remain logically distinct: exterior geometry, thermodynamic bookkeeping, and accessible information. Chapter V supplies a static exterior adapter and an exact finite ledger. Chapter VI supplies the operational meaning of a record and the contraction of distinguishability under restricted access. This chapter assembles those results, extends their elementary consequences, and identifies the additional physical input needed for an absolute radiation law.

The principal outcome is conditional but precise. Given the Schwarzschild exterior adapter, the usual normalization of time at infinity, the Hawking temperature conversion, and the retained quarter-area entropy rule, the horizon temperature and differential first law agree with semiclassical black-hole thermodynamics. Given the integer-entry continuation, the exact energy of a finite emission follows algebraically. Neither statement derives the adapter, the quarter-area rule, or an emission clock from the microscopic source. Keeping these dependencies visible prevents an agreement inherited from an input from being counted a second time as an independent prediction.

The finite Si29 example provides a different kind of result. Its reported relative-entropy Hessian is not proportional to its enclosing-area Hessian, and entropy together with area locally distinguishes two source directions. This is a useful obstruction to a particular finite-source information map. It is not a disproof of a continuum horizon identity formulated for a different area, state family, and perturbation order.

Throughout this chapter, $A(r)$ denotes dimensionless accumulation, $\mathscr{A}_H$ denotes physical horizon area, and N denotes the dimensionless entropy-entry count. These are different objects. In particular, $A(r_H) = 1$ does not mean that the physical area is one square metre or one Planck area. Entropies written as s are dimensionless, with physical entropy $S = k_B s$. This convention keeps logarithmic quantum-information expressions separate from thermodynamic units.

1.1 Evidence levels

Four levels of statement recur. An *input* is an adopted bridge or constitutive premise. A *conditional derivation* follows from such inputs by explicit mathematics. A *retained calculation* is reported in the supplied source record but has not been rerun here. An *open closure* is a missing physical map, experiment, or computation. The expanded calculations below are analytic consequences of printed premises; they are not new native source runs.

The closure surface also requires a causal interpretation. The divergence of a static coordinate travel time does not by itself prove that an infaller cannot cross a regular horizon. In the adopted metric, radial free-fall trajectories can reach the horizon in finite proper time. SAM's rule about completion of the parent ledger therefore needs its own operational statement: which event, record, or custody transfer is forbidden, and with respect to which observer? A restriction on exterior record access is weaker than a reflecting material wall. The distinction will matter both for radiation and for observational limits on compact surfaces.

2. Surface gravity from a specified exterior

2.1 Three gradients, three different meanings

Consider the static, spherically symmetric line element

$$ds^2 = -F(r)c^2dt^2 + H(r)dr^2 + r^2d\Omega^2, \qquad F > 0, \quad H > 0, \tag{1}$$

with Killing time normalized so that $F \to 1$ at infinity. Chapter V adopts

$$A(r) = \frac{r_s}{r}, \qquad r_s = \frac{2GM}{c^2}, \qquad F = 1 - A, \qquad H = F^{-1}. \tag{2}$$

The areal-coordinate derivative $|A'| = r_s/r^2$ tends to $1/r_s$. The spatial proper gradient on a static slice is $|A'|/\sqrt{H}$, which tends

to zero. Neither quantity alone is the surface gravity. Surface gravity is the redshifted acceleration required to hold a static observer near the horizon, with a specified normalization of the time-translation generator.

A static worldline has $d\tau = \sqrt{F}\,dt$. Evaluating its four-acceleration gives the positive magnitude

$$a_{\rm stat} = \frac{c^2|F'|}{2F\sqrt{H}}, \qquad \kappa = \lim_{r\downarrow r_s}\sqrt{F}\,a_{\rm stat} = \frac{c^2}{2}\lim_{r\downarrow r_s}\frac{|F'|}{\sqrt{FH}}. \quad (3)$$

The first denominator contains $F\sqrt{H}$, not merely $\sqrt{FH}$. This distinction separates local proper acceleration, which diverges for the adopted adapter, from its finite redshifted limit. Substituting $FH = 1$ yields

$$\kappa = \frac{c^2}{2r_s} = \frac{c^4}{4GM}. \quad (4)$$

The normalization of r_s is inherited from the earlier gravitational bridge; the result also uses the spatial metric adapter. A scalar inverse-radius kernel without a spatial metric does not establish this surface gravity.

The temperature conversion of semiclassical horizon theory is

$$T_H = \frac{\hbar\kappa}{2\pi c k_B} = \frac{\hbar c^3}{8\pi GMk_B}. \quad (5)$$

Hawking's field-theoretic calculation supplies this conversion [VII.1]. Here it is used as a bridge rather than rederived from the finite substrate source. For one solar mass the retained example gives $r_s \simeq 2.953\,\mathrm{km}$, $\kappa \simeq 1.522\times10^{13}\,\mathrm{m\,s^{-2}}$, and $T_H \simeq 6.17\times10^{-8}\,\mathrm{K}$. The scaling with mass is more robust than the final displayed digits: $r_s \propto M$, while κ and T_H are proportional to M^{-1}.

2.2 Local temperature and equilibrium

If a stationary thermal state is in equilibrium with respect to the same Killing time, the locally measured temperature obeys the Tolman redshift relation

$$T_{\rm loc}(r)\sqrt{F(r)} = T_\infty. \quad (6)$$

One way to see the relation is to require the same occupation factor for a quantum described locally or at infinity. Since $E_\infty = \sqrt{F}E_{\rm loc}$, equality of $E/(k_BT)$ gives equation (6). This derivation establishes a consistency condition on equilibrium descriptions; it does not say that an arbitrarily accelerated detector or a freely falling detector measures the same temperature.

The divergence of $T_{\rm loc}$ as $F\to0$ belongs to an ideal family of static observers whose required acceleration also diverges. It must not be interpreted as a measured infinite temperature on a freely falling material thermometer. A real static apparatus has finite strength, finite size, and a backreaction that invalidates the ideal limit. This observer dependence is especially important if the parent-ledger language is later attached to detector records.

3. Entropy entries and the first law

3.1 Coefficient provenance

Set $\ell_P^2 = \hbar G/c^3$. The retained four-cell entry convention reads

$$\mathscr{A}_H = 4\pi r_s^2, \qquad N = \frac{\mathscr{A}_H}{4\ell_P^2} = \frac{4\pi GM^2}{\hbar c}, \qquad S_H = k_B N. \quad (7)$$

The area law is the Bekenstein–Hawking normalization [VII.1–VII.2], adopted here as the retained area-entry premise. The fact that an entry is called four-cell does not independently prove that its microscopic degeneracy has logarithm one. Those are separate physical claims until the counting measure and state space are specified.

Differentiating equation (7) gives

$$dS_H = \frac{8\pi GMk_B}{\hbar c}\,dM, \qquad T_H dS_H = c^2 dM. \quad (8)$$

Thus the adopted temperature and entropy laws are mutually consistent with the first law for a neutral, nonrotating, quasistatic family. Angular momentum, charge, exterior work, and matter fluxes are absent from this restricted calculation. Their thermodynamic potentials cannot be obtained by relabelling M in this expression.

Coefficient	Place in calculation	Logical source
2	$r_s = 2GM/c^2$	Earlier gravitational normalization
$1/2$	Redshifted acceleration	Metric adapter and static observer
$1/(2\pi)$	$T_H = \hbar\kappa/(2\pi c k_B)$	Semiclassical temperature bridge
$1/4$	$N = \mathscr{A}_H/(4\ell_P^2)$	Retained entropy-entry premise
k_B	Entropy per entry	Retained thermodynamic convention

Each coefficient has one place in the argument. Agreement of the final expression does not erase the need to derive the bridges if a microscopic explanation is claimed.

The heat capacity along this family is negative:

$$C_H = \frac{d(Mc^2)}{dT_H} = -\frac{Mc^2}{T_H} = -2k_BN. \quad (9)$$

This follows because decreasing mass raises the temperature. It warns against treating an isolated finite reservoir as a stable canonical bath of unlimited capacity. The finite-entry construction is particularly useful because it does not require that approximation.

3.2 What is, and is not, an integer count

Equation (7) assigns a real number N to every positive mass. The additional statement that allowed states lie on integer N is a quantization hypothesis. On that hypothesis,

$$M_N = \sqrt{\frac{\hbar c}{4\pi G}}\sqrt{N}, \qquad E_N = \sqrt{\frac{\hbar c^5}{4\pi G}}\sqrt{N}. \quad (10)$$

The mathematical sequence exists for $N = 0, 1, 2, \ldots$. Its applicability to Planck-scale endpoints is not established by a semiclassical temperature law. In particular, a formation floor or remnant condition from Chapter XII may restrict the admissible sequence to $N \geq N_{\text{floor}}$ and remove some terminal transitions entirely.

There is also a microscopic counting subtlety. If $S_H = k_B \log g_N = k_B N$ were an exact entropy of uniformly weighted, finite microcanonical states, it would imply $g_N = e^N$, which is not an integer for positive integer N. Consequently the entry entropy cannot simultaneously be interpreted as an exact logarithm of a finite integer degeneracy with that literal normalization, unless further structure changes the interpretation. It can instead be a thermodynamic entropy, an effective shell weight, or an asymptotic count. The chapter's conditional balance formulas use such weights; they do not supply the missing integer degeneracy. This distinction also matters when comparing alternative discrete-area proposals.

4. Exact finite transitions

4.1 One-entry emission

For an allowed transition from N to $N-1$, take the emitted energy positive and write $\Delta S_{\text{loss}} = k_B$. The horizon's signed entropy change is $-k_B$. Direct subtraction of equation (10) gives

$$\Delta E = Mc^2 \left(1 - \sqrt{1 - \frac{1}{N}}\right), \qquad \frac{\Delta E}{k_B T_H(M)} = \frac{2}{1 + \sqrt{1 - 1/N}}. \tag{11}$$

The rationalized form avoids numerical cancellation at large N. It also proves immediately that the ratio lies in $[1, 2]$, approaches one at large N, and equals two at $N = 1$. These statements are exact properties of the assumed sequence, not a measured endpoint spectrum.

Let $q = 1/N$ and $c_*(q) = \Delta E/(k_B T_H)$. Multiplication by the conjugate, followed by rearrangement, gives

$$c_* = 1 + \frac{q}{4} c_*^2, \qquad c_*(q) = \sum_{j=0}^{\infty} \frac{C_j}{4^j} q^j, \qquad C_j = \frac{1}{j+1} \binom{2j}{j}. \tag{12}$$

The branch satisfying $c_*(0) = 1$ is selected. The first terms are $1 + q/4 + q^2/8 + 5q^3/64 + \cdots$. Since $0 < C_j/4^j \leq 1$, for $0 \leq q < 1$ the remainder after degree m obeys

$$0 \leq c_*(q) - \sum_{j=0}^{m} \frac{C_j}{4^j} q^j \leq \frac{q^{m+1}}{1-q}. \tag{13}$$

This is a deliberately simple bound rather than an optimal one. It is uniform on every interval $0 \leq q \leq q_0 < 1$ and provides a reproducible error estimate for the large-ledger approximation.

The ratios at $N = 1, 2, 10, 100$ are respectively 2, 1.17157, 1.02633, and 1.00251. The solar-mass example has approximately 1.05×10^{77} entries, so its correction is of order 10^{-78}. The finite law therefore has mathematical endpoint content while giving no practical temperature correction for a stellar-mass hole.

4.2 Multiple entries and a finite first law

For $0 < j \leq N$ the same calculation gives

$$\begin{aligned} \Delta E_j &= Mc^2 \left(1 - \sqrt{1 - j/N}\right), \\ \frac{\Delta E_j}{k_B T_H(M)} &= \frac{2j}{1 + \sqrt{1 - j/N}} \\ &= j + \frac{j^2}{N(1 + \sqrt{1 - j/N})^2}. \end{aligned} \tag{14}$$

The initial temperature underestimates the energy per unit entropy lost because the reservoir heats while it loses mass. The exact finite first law is an integral,

$$\Delta E_j = \int_{N-j}^{N} k_B T_H(n)\, dn, \qquad T_{\text{step}} = \frac{\Delta E_j}{j k_B}. \tag{15}$$

For $j < N$, monotonicity gives $T_H(N) \leq T_{\text{step}} \leq T_H(N-j)$. At the endpoint the upper limit diverges in the extrapolated temperature formula, but the integral stays finite. Defining a step temperature in this manner is an exact thermodynamic average, not an independent thermometer prescription.

This distinction clarifies the retained phrase "the final entry carries twice its thermal energy." It means twice $k_B T_H$ evaluated at the initial level $N = 1$. It does not mean twice the temperature of a nonexistent final black hole, and it does not by itself specify the number of particles sharing that energy. One ledger transition could distribute its energy among multiple coupled channels if the source dynamics allows it.

5. Detailed balance and its assumptions

Suppose initial and final microcanonical shells have effective weights proportional to e^{S_H/k_B}, and suppose the emission and reversed absorption channels have matched dynamical matrix elements and compatible measures. Then the ratio of shell-averaged transition weights is

$$\frac{\Gamma_{\text{em}}}{\Gamma_{\text{abs}}} = \exp\left[\frac{S_H(E - \varepsilon) - S_H(E)}{k_B}\right]. \tag{16}$$

For an allowed j-entry transition the exponent is exactly $-j$. This proves the conditional ratio e^{-j} under the stated channel and counting assumptions. It does not prove equality of unmatched scattering channels, equal matrix elements for all transitions, or a total emission rate.

Writing $S_H(E)/k_B = aE^2$ with $a = 4\pi G/(\hbar c^5)$ makes finite backreaction transparent:

$$\frac{\Delta S_H}{k_B} = -2aE\varepsilon + a\varepsilon^2 = -\frac{\varepsilon}{k_B T_H(E)} + a\varepsilon^2. \tag{17}$$

The correction is positive relative to a fixed-initial-temperature Boltzmann exponent. Its sign is another expression of negative heat capacity. Evaluated on the discrete energy gaps, the entire exponent remains $-j$; one must not apply the e^{-j} factor and then add this backreaction a second time.

For a finite ladder $\varepsilon_n = n\varepsilon_Q$, $0 \leq n \leq K$, the correction to the log weight lies in an interval of width $a(K\varepsilon_Q)^2$. A stan-

dard bounded-likelihood-ratio argument gives the retained total-variation estimate

$$\|p_{\mathrm{back}} - p_{\mathrm{fixed}}\|_{\mathrm{TV}} \le \tanh\left[\frac{a(K\varepsilon_Q)^2}{4}\right]. \tag{18}$$

Indeed, reweighting a probability law by e^f with $\max f - \min f = L$ has extremal variation bounded by $\tanh(L/4)$; the extremum is obtained by concentrating weight at the endpoints and optimizing their initial probabilities. The bound requires that the same finite support and channel measure are used on both sides. It is not a claim about arbitrary changes of species, boundary state, or detector acceptance.

6. From finite occupations to transmitted counts

6.1 The truncated geometric law

The retained HOME-QW1 occupation law has

$$p_n = \frac{q^n}{Z_K}, \quad 0 \le n \le K, \qquad Z_K = \frac{1-q^{K+1}}{1-q}, \qquad 0 < q < 1. \tag{19}$$

Its exact mean follows by differentiating $\log Z_K$:

$$\langle n\rangle_K = q\frac{d}{dq}\log Z_K = \frac{q}{1-q} - \frac{(K+1)q^{K+1}}{1-q^{K+1}}. \tag{20}$$

The infinite law is $(1-q)q^n$. The finite law is that law conditioned on $n \le K$, so their total-variation distance equals the omitted probability q^{K+1}. This elementary proof supplies a transparent finite-capacity error criterion: choose K such that q^{K+1} is below the required tolerance. It is not enough merely to say that K is large, because convergence is slow when q is close to one.

An occupation distribution specifies diagonal number statistics but not the complete density operator. States with the same probabilities can differ in phase coherence, inter-guide correlations, and entanglement with the retained reservoir. For a thermal interpretation the basis, Hamiltonian, and state preparation must all be named. In particular, the model relation $q = e^{-\omega/T_A}$ uses source units. Converting it into $e^{-\hbar\omega_{\mathrm{phys}}/k_BT_H}$ requires the energy and clock bridge discussed below.

6.2 Transmission and noise

Chapter V reports the annular Jost transmission

$$\tau_{\mathrm{eff}}(k) = \frac{4k^2}{|D(k)|^2}, \qquad 0 < \tau_{\mathrm{eff}} \le 1, \tag{21}$$

for its specified real, passive scattering problem and positive open-channel wave number. Wronskian conservation establishes flux conservation in that problem. A different dissipative boundary, coupled species, or unequal asymptotic channel velocities would require a corresponding flux normalization and scattering calculation. The retained formula is therefore a result for its stated annulus, not automatically every black-hole greybody factor.

For a number state entering a passive loss channel with the other input in vacuum, transmitted counts obey

$$P(m \mid n) = \binom{n}{m}\eta_f^m(1-\eta_f)^{n-m}. \tag{22}$$

Here η_f is the transmission and detection efficiency for the selected wave packet. For an input mixture, the law of total expectation and variance yields

$$\langle m\rangle = \eta_f\langle n\rangle, \tag{23}$$

$$\mathrm{Var}(m) = \eta_f(1-\eta_f)\langle n\rangle + \eta_f^2\,\mathrm{Var}(n). \tag{24}$$

Thus transmission changes both the mean and counting noise. These equations provide a stronger check than mean flux alone. They cease to be the complete answer when the unused port is populated or correlated with the incident mode; in that case interference and stimulated contributions must be retained.

The stationary two-guide convention retained in Chapter V gives $d\dot{N}/d\omega = \tau_{\mathrm{eff}} n_A/\pi$. The factor $1/\pi$ belongs to that channel normalization and population choice. A physical mode sum is often organized instead as

$$L_\infty = \sum_{s,\ell,m}\int_0^\infty \frac{d\omega}{2\pi}\hbar\omega\,\Gamma_{s\ell m}(\omega)\,n_{s\ell m}(\omega), \tag{25}$$

with the channel measure and degeneracies explicitly stated. Equation (25) is a template, not a newly computed SAM luminosity. It demonstrates exactly where the source state, energy conversion, species content, and transmission must meet.

7. Area and information in Si29

7.1 Relative entropy has no linear term

For a faithful finite reference state σ, define $K_\sigma = -\log\sigma$ and

$$\begin{aligned} D(\rho\|\sigma) &= \mathrm{Tr}\,\rho(\log\rho - \log\sigma) \\ &= \Delta\langle K_\sigma\rangle - \Delta s_{\mathrm{vN}}, \\ s_{\mathrm{vN}} &= -\mathrm{Tr}\,\rho\log\rho. \end{aligned} \tag{26}$$

Take a smooth normalized family $\rho(x)$ with $\rho(0) = \sigma$. Its entropy differential at the reference is $ds_{\mathrm{vN}} = \mathrm{Tr}(d\rho K_\sigma)$, since $\mathrm{Tr}\,d\rho = 0$. Hence $dD = 0$. Relative entropy is stationary at its reference for general structural reasons. If an area observable has a nonzero first derivative, a proposed equality $\Delta\overline{\mathscr{A}} = cD$ fails immediately along that direction.

That observation is generic. It does not test a continuum identity in which both flux and area response are quadratic in an excitation amplitude. Dorau and Much analyze coherent excitations and horizon energy flux under an entropy-area assumption [VII.4]. Their geometrical area is not automatically the finite-source enclosing area. A valid comparison must align the state family, reference, perturbation parameter, geometric map, and order before comparing coefficients. This is the required correction to the earlier first-order interpretation.

7.2 The substantive Hessian comparison

The retained Si29 record uses a four-by-eighteen tensor source, seed $(3,10)$, the original Q/G matrices, reference phases 0.007 and 0.013, and two perturbations $x = (u,v)$. Its common-energy enclosing area is $\overline{\mathscr{A}} = \mathscr{A}/(4\pi a^2)$ at $e = 4$. The record reports 34 signed perturbations and the exact decomposition (26). Numerical matrices quoted here are its rounded reported results, not new interval certificates.

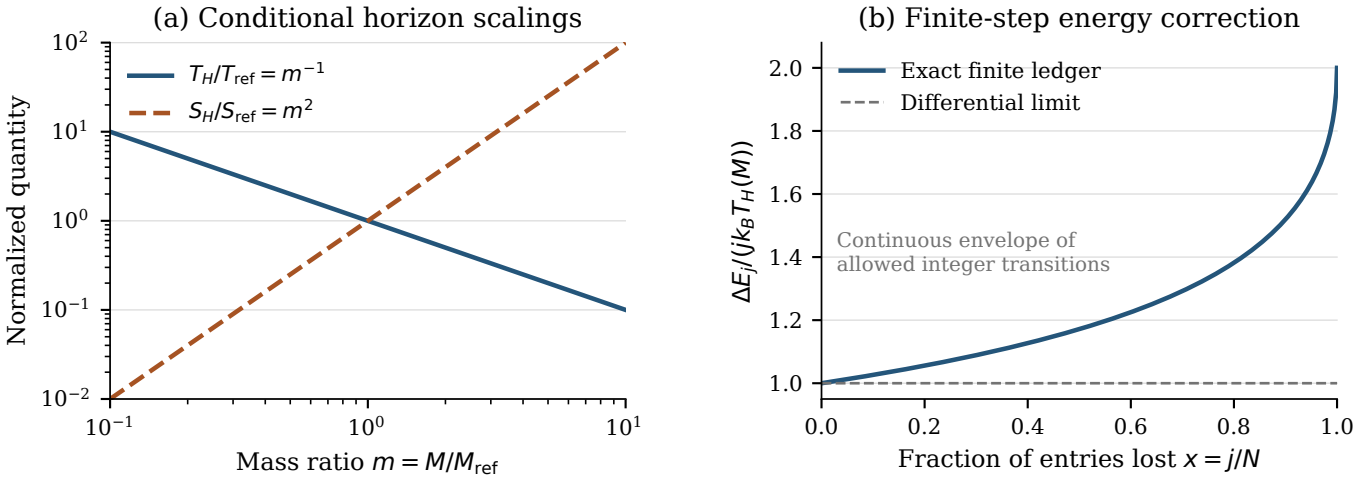


Figure 1: Conditional horizon thermodynamics and the finite ledger. (a) The adopted Schwarzschild geometry, Hawking temperature conversion and quarter-area entropy rule give $T_H/T_{\mathrm{ref}} = (M/M_{\mathrm{ref}})^{-1}$ and $S_H/S_{\mathrm{ref}} = (M/M_{\mathrm{ref}})^2$, with both reference quantities evaluated at M_{ref}. (b) The exact entry ledger gives $\Delta E_j/(jk_BT_H) = 2/[1+\sqrt{1-j/N}]$. The continuous curve is an envelope of integer transitions; its value at zero is the differential limit. Extension to complete entry loss is conditional and need not be physically allowed by a formation floor. These analytic curves are neither an emission spectrum nor a newly simulated source result.

With G the relative-entropy Hessian and $H_{\mathscr{A}}$ the area Hessian, the two references give

$$G_{.007} = \begin{pmatrix} 252.754 & -785.399 \\ -785.399 & 17867.308 \end{pmatrix}, \tag{27}$$

$$H_{.007} = \begin{pmatrix} 7.753 & -6.015 \\ -6.015 & 445.393 \end{pmatrix}, \tag{28}$$

$$G_{.013} = \begin{pmatrix} 110.500 & -516.332 \\ -516.332 & 7083.046 \end{pmatrix}, \tag{29}$$

$$H_{.013} = \begin{pmatrix} 4.095 & -11.510 \\ -11.510 & -24.842 \end{pmatrix}. \tag{30}$$

At the first reference, the ratios of corresponding diagonal entries already disagree. At the second reference, G is positive definite while $H_{\mathscr{A}}$ is indefinite: its diagonal entries have opposite signs. No real scalar multiple of a positive-definite matrix is indefinite. Therefore the quadratic forms are not proportional at either reference.

For a strict second-order test, subtract the linear area variation and compare

$$\overline{\mathscr{A}}(x) - \overline{\mathscr{A}}(0) - \nabla\overline{\mathscr{A}}(0)\cdot x = \tfrac{1}{2}x^T H_{\mathscr{A}} x + O(\|x\|^3) \tag{31}$$

with $D = \frac{1}{2}x^T Gx + O(\|x\|^3)$. This is a coordinate-explicit statement about the chosen finite family. A physical horizon map could contain extra terms or different observables; the present calculation constrains such a map but does not exclude all possible maps.

7.3 Two observables and a local inverse

The retained local phase direction preserves entropy, whereas the interaction phase changes the reduced spectrum. Setting $y = (s_{\mathrm{vN}}, \overline{\mathscr{A}})$ gives

$$J = \frac{\partial y}{\partial x} = \begin{pmatrix} 0 & s_v \\ \overline{\mathscr{A}}_u & \overline{\mathscr{A}}_v \end{pmatrix}, \qquad \det J = -s_v\overline{\mathscr{A}}_u. \tag{32}$$

The source reports determinant enclosures excluding zero, with magnitudes about 3.3636 and 3.5531. If the reported differentiability and enclosure hypotheses hold, the inverse-function theorem establishes an open neighborhood on which these two observables are valid local coordinates. It does not establish global uniqueness, cover periodic phase identifications, or reconstruct all states in the full tensor space.

The reported inverse errors decrease by approximately four when the grid step is halved: from 7.88×10^{-5} to 1.97×10^{-5} at the first reference, and from 1.02×10^{-5} to 2.56×10^{-6} at the second. This supports second-order local behavior on that sample. A certified neighborhood still needs uniform derivative bounds. One sufficient criterion is

$$\sup_{x\in B_r} \|J(0)^{-1}[J(x)-J(0)]\| < 1, \tag{33}$$

on a convex ball B_r. It controls departure from the invertible linear map and gives an injectivity estimate there. Selecting r requires native bounds not printed in the present source.

Because $\nabla D = 0$ at the reference, its Hessian transforms as a metric under this coordinate change:

$$G_y = J^{-T}GJ^{-1}, \qquad g_{tt} = w^T Gw, \qquad w = (1+8v_{\mathrm{Si}}, v_{\mathrm{Si}}). \tag{34}$$

No extra Hessian-of-coordinate term survives when the gradient vanishes. Mixed entries are expected and encode joint sensitivity; discarding them would lose information. The tangent w belongs to the retained tensor dynamics. Its use as a physical time response still needs a time calibration and a boundary coupling.

8. Accessible records and boundary custody

For a completely positive trace-preserving access channel $\mathscr{E}$, trace distance contracts:

$$\tfrac{1}{2}\|\mathscr{E}(\rho_0)-\mathscr{E}(\rho_1)\|_1 \le \tfrac{1}{2}\|\rho_0-\rho_1\|_1. \tag{35}$$

This is the operational content of Chapter VI's accessibility result. A receiver with restricted access cannot distinguish two alternatives better than a receiver supplied with the full state. It is a statement about obtainable records, not a proof that inaccessible global alternatives cease to exist. Conversely, postselected conditioning is not generally a trace-preserving map, so its apparent amplification of distinguishability must include the success flag and its probability.

The boundary pair-write proposes an outward and inward alternative while they are unresolved. The outward member may become an exterior record through allowed transmission and reception. The inward member is unavailable to that exterior observer under the proposed causal cut. Born weighting assigns probabilities once the state and measurement are defined. Neither the accessibility theorem nor the word "pair" fixes a pair-production Hamiltonian, a pair rate, or an entanglement spectrum.

The completed-Home custody laws retained from H000794–H000797 add constraints: failed inward transfers return their receipts unchanged; retained exterior receipts have finite inward/outward schedules ending in a parent-side mass-loss event; gross emission and net growth can coexist; and formation mass and radius provide a floor. These are lifecycle commitments to be reconciled with the admissible transitions. If $N_{\mathrm{floor}} > 0$, the terminal $N = 1 \to 0$ extrapolation cannot be advertised as a realized decay without an additional rule explaining how it avoids the floor.

A useful correlation test would specify joint observables and compare the full outgoing state with a mixture having identical one-mode occupations. Equal Planck-like marginals do not imply equal mutual information or equal conditional records. The target should therefore be a connected correlation, a channel-resolved covariance, or a recovery fidelity with an explicit reference system. At present these are defined measurement opportunities, not computed predictions of the retained occupation law.

9. A horizon is not automatically a material surface

A no-completed-crossing statement admits several mathematical implementations. A purely causal horizon can absorb exterior perturbations into degrees of freedom inaccessible to exterior receivers. A timelike surface can reflect, absorb and later re-emit, or store energy. These possibilities have different Green functions and observational consequences. A boundary condition must specify which implementation is intended.

For a radial mode near a boundary in tortoise coordinate r_*, write schematically

$$\psi \sim e^{-i\omega r_*} + \mathscr{R}(\omega)e^{+i\omega r_*}. \tag{36}$$

An ingoing horizon condition has $\mathscr{R} = 0$ in the appropriate classical exterior problem. A reflecting wall has nonzero $\mathscr{R}$. An absorbing response requires a causal, passive frequency-dependent impedance or additional open channels. This is a parametrization of the missing boundary law, not a new fitted SAM coefficient. The two-polarization source must select the response before either greybody transmission or Chapter XI's ringdown poles can be called native predictions.

Visible surfaces are constrained by accretion energetics. Broderick, Loeb, and Narayan's Sgr A* analysis combines compactness and infrared limits to constrain thermalized surface emission [VII.5]. Its quoted bound places putative visible emission below 0.4% of the accretion luminosity under that analysis's stationary assumptions. This is relevant to a material surface that retains and reradiates accretion energy; it is not a universal exclusion of every causal boundary model. SAM must state its thermalization efficiency, storage time, escape probability, and outflow budget before applying such a bound. Invoking redshift alone does not close the energy account.

10. Energy conservation, entropy balance, and the missing evolution law

A radiation model must distinguish instantaneous power from the entropy carried by that radiation. For a parent whose exterior mass is $M(t)$, an idealized energy balance in the infinity-normalized clock is

$$c^2\frac{dM}{dt} = P_{\mathrm{in}} - L_\infty - P_{\mathrm{other}}. \tag{37}$$

Here all fluxes refer to the same energy normalization and boundary of the account. P_{other} denotes explicitly modeled outward channels not included in the radiation luminosity. It must not serve as an unconstrained sink inserted to repair disagreement. If energy is stored temporarily in exterior fields or a boundary reservoir, its time derivative belongs on the left or in a separately tracked equation. An emission event can therefore coexist with net growth when inward power is larger than total outward power, exactly as the retained custody language allows.

For slowly evolving neutral configurations, the instantaneous first law gives

$$\dot{S}_H = \frac{c^2\dot{M}}{T_H}. \tag{38}$$

This is not the total entropy-production law. Exterior matter, radiation, and correlations also contribute. A generalized-entropy candidate has the form $S_{\mathrm{gen}} = S_H + S_{\mathrm{outside}}$, with a declared subtraction or regulator if continuum field entanglement is involved. The sign of $\dot{S}_{\mathrm{gen}}$ cannot be read off from $\dot{S}_H$ alone. During emission S_H decreases, while the exterior entropy and correlations may increase. Proving a nondecreasing generalized entropy requires dynamics and a state, not merely the area coefficient.

The same point appears in a single finite transition. Knowing that the horizon loses jk_B of thermodynamic entropy does not imply that a measured photon string carries exactly jk_B of Shannon entropy. The record may be conditioned on detection, share correlations with other channels, or have an entropy dependent on coarse graining. An explicit channel state would

allow the von Neumann entropy before and after the interaction to be computed with compatible subsystem definitions. Without it, identifying entropy loss with accessible information is an additional hypothesis.

A minimal dynamical completion could be expressed through populations $P_N(t)$ and rates $W_{N\to N-j}$:

$$\dot{P}_N = \sum_j \left[P_{N+j}W_{N+j\to N} - P_N W_{N\to N-j}\right] \tag{39}$$

for a purely emitting process, with absorption terms added when relevant. The boundary conditions on this equation must enforce normalization, positivity, and any formation floor. The energy expectation then evolves as $\sum_N E_N \dot{P}_N$, which must equal the integrated outgoing flux computed from the same transitions. This provides a direct conservation test joining the ledger and the spectrum.

Equation (39) is a specification of what a completion must supply; no rates have been inferred here. Detailed balance constrains ratios of forward and reverse rates in a suitable equilibrium setting, whereas multiplication of every rate by a common constant leaves those ratios unchanged. More general modifications can preserve occupation statistics while changing temporal correlations. Consequently a source-selected realization clock is necessary even if every equilibrium factor has already been derived.

A successful calculation would report both the occupation relaxation times and the counting correlations in a finite observation window. It would show which reservoir approximation makes a Markov equation adequate, or retain memory effects if that approximation fails. In the small-N regime a single jump can change the temperature substantially, so rates evaluated permanently at the initial mass are not self-consistent. Updating the energy gaps and transmission after each allowed transition is the simplest conditional feedback scheme; deriving that feedback from the dressed source remains the stronger goal.

11. Worked finite-ledger examples

11.1 A three-step energy account

The exact ledger can be inspected without introducing a physical mass calibration. Define the energy unit $E_* = \sqrt{\hbar c^5/(4\pi G)}$, so that $E_N = E_*\sqrt{N}$ and $k_B T_N = E_*/(2\sqrt{N})$. Consider the sequence $4 \to 3 \to 2 \to 1$, assuming that every indicated transition is allowed. Its three emitted energies in units of E_* are

$$\begin{aligned} \Delta e_{4\to3} &= 2-\sqrt{3} \simeq 0.267949, \\ \Delta e_{3\to2} &= \sqrt{3}-\sqrt{2} \simeq 0.317837, \\ \Delta e_{2\to1} &= \sqrt{2}-1 \simeq 0.414214. \end{aligned} \tag{40}$$

Their sum is exactly one. The entropy lost by the horizon is $3k_B$, and the initial and final temperatures are respectively $E_*/(4k_B)$ and $E_*/(2k_B)$. The energy gaps grow as the reservoir shrinks, so a constant-gap approximation would fail visibly in this small example.

The same endpoints can be joined by one allowed three-entry transition. Equation (14) gives the same total energy E_*, as conservation requires. Relative to the initial thermal scale, its ratio is $\Delta E/(k_B T_4) = 4$, whereas a naive differential estimate $j = 3$ would give only three. The finite step temperature is $E_*/(3k_B)$, between the endpoint values. This calculation illustrates why a finite first law is an integral rather than a single initial temperature multiplied by an entropy difference.

Energy additivity does not imply dynamical equivalence of the one-jump and three-jump paths. Their transition amplitudes, waiting times, linewidths, and emitted correlations can differ. Detailed balance supplies the same net entropy factor e^{-3} when three matched e^{-1} ratios are multiplied, but it does not say that the corresponding absolute path probabilities are equal. A path probability includes the product of transition kernels and the probabilities of visiting each intermediate state. These factors depend on the source Hamiltonian and observation schedule.

The example also supplies an exact numerical unit test for depletion feedback. An implementation that subtracts three copies of the first energy gap emits only $3(2-\sqrt{3})E_*$, failing the endpoint energy difference. Updating the mass after each transition restores telescoping energy conservation. If the model has a formation floor at $N = 2$, the last step is forbidden and the allowed emitted total is instead $(2-\sqrt{2})E_*$. A floor is therefore an actual boundary condition on evolution, not a verbal qualification appended after an evaporation curve is computed.

11.2 A reversible reference process and entropy production

For comparison, imagine a quasistatic reversible exchange between the ledger and a sequence of external reservoirs, each matched to the instantaneous horizon temperature. This is a thermodynamic reference construction, not a claim that an isolated hole performs such a process. During $N_i \to N_f$, the reservoirs receive

$$\Delta S_{\rm res} = \int_{N_f}^{N_i} \frac{1}{T_H}\frac{dE}{dN}\,dN = k_B(N_i - N_f), \tag{41}$$

which exactly cancels the horizon entropy change. Reversing the process with the same variable-temperature reservoirs returns both the energy and entropy. There is no net work obtainable from a reversible closed cycle without an additional temperature difference or other resource.

A single fixed-temperature bath is different. If it receives the entire energy ΔE while remaining at T_b, its entropy gain in an ideal quasistatic bath description is $\Delta E/T_b$. The combined entropy change is

$$\Delta S_{\rm tot} = \frac{\Delta E}{T_b} - jk_B. \tag{42}$$

For the $4 \to 1$ example and $T_b = T_4$, this is k_B. For $T_b = T_{\rm step}$ it is zero at the level of endpoint thermodynamics, but matching an average temperature does not ensure a reversible path: local temperature differences can still produce entropy. If a chosen spontaneous-process model predicts a negative total entropy under its complete accounting, then either the process is forbidden or a missing resource, work channel, or correlation must be included. The endpoint calculation is a consistency test, not a substitute for identifying those channels.

Negative heat capacity further limits naive equilibrium intuition. A reservoir that loses energy becomes hotter, so heat exchange can amplify rather than damp a perturbation. A stationary occupation factor can still arise in a suitable open-system setting, but its stability needs the coupled reservoir dynamics. One must not infer stable canonical equilibrium simply from a formal Boltzmann factor attached to one transition.

11.3 An exactly soluble count experiment

Choose the illustrative occupation parameters $q = 1/2$ and $K = 3$. Equation (19) then gives

$$(p_0, p_1, p_2, p_3) = \frac{1}{15}(8,4,2,1), \tag{43}$$

with mean $11/15$ and variance $194/225$. Its total-variation distance from the infinite geometric law is $1/16$. This finite tail is significant enough that replacing the state by the infinite thermal distribution changes more than rounding error.

Send this diagonal mixture through the vacuum-port loss channel with $\eta_f = 1/2$. Summing the conditional binomial distributions gives

$$(P_0, P_1, P_2, P_3) = \frac{1}{120}(85,27,7,1). \tag{44}$$

The output mean is $11/30$ and variance is $359/900$, exactly matching equations (23)–(24). The output Fano factor is $359/330$, slightly greater than one. A Poisson process with the same mean would have variance equal to the mean and would therefore be distinguishable with enough repeated trials.

This small example separates three operations that are often conflated: preparing an occupation law, transmitting a mode, and counting its detected quanta. None specifies how often the preparation occurs. Repeating the experiment once per second or once per century preserves the per-trial probabilities while changing the count rate enormously. The example is consequently a direct demonstration that an absolute luminosity requires a realization clock in addition to a thermal occupation and a transmission coefficient.

12. A worked accessibility channel

12.1 Distinguishability without a geometric identification

A two-level model gives an explicit, non-gravitational illustration of the accessibility theorem. Take

$$\begin{aligned} \rho(x,z) &= \frac{1}{2}(I + x\sigma_x + z\sigma_z), \\ r &= \sqrt{x^2+z^2} < 1, \qquad \sigma = I/2. \end{aligned} \tag{45}$$

The Pauli operators fix the laboratory axes, and the eigenvalues of ρ are $(1 \pm r)/2$. Its relative entropy to the maximally mixed reference is

$$\begin{aligned} D(\rho\|\sigma) &= \tfrac{1}{2}[(1+r)\log(1+r) + (1-r)\log(1-r)] \\ &= \tfrac{1}{2}r^2 + \tfrac{1}{12}r^4 + O(r^6). \end{aligned} \tag{46}$$

This explicitly verifies stationarity at the reference and gives the information Hessian $G = I$ on the (x,z) plane. The entropy is $s_{\mathrm{vN}} = \log 2 - D$. No horizon area has been assigned, so this is a channel example rather than a substitute for Si29 or a new physical boundary model.

Let the accessible observer receive only the state after complete dephasing in the σ_z basis. The channel removes x and preserves z. The alternatives $\rho_+ = \rho(x,0)$ and $\rho_- = \rho(-x,0)$ have trace distance $|x|$ before access and zero after access. Their optimal equal-prior discrimination success falls from $(1+|x|)/2$ to $1/2$. Global distinguishability was present, but this receiver cannot read it through the chosen channel.

Another allowed channel flags an erasure with probability $1-\lambda$, while transmitting the state intact with probability λ. Its output is a block-diagonal state on the transmitted space plus an orthogonal erasure flag. Because the flag probabilities are the same for both alternatives, direct evaluation gives

$$\begin{aligned} D(\mathscr{E}_\lambda(\rho)\|\mathscr{E}_\lambda(\sigma)) &= \lambda D(\rho\|\sigma), \\ \tfrac{1}{2}\|\mathscr{E}_\lambda(\rho_+) - \mathscr{E}_\lambda(\rho_-)\|_1 &= \lambda|x|. \end{aligned} \tag{47}$$

Conditioning on successful transmission recovers the original state, but success occurs only with probability λ. Discarding the failures while claiming an unconditional increase in accessible information would omit the resource cost and the probability accounting. The theorem applies to the full trace-preserving channel including its flag.

12.2 Why two coordinates do not imply complete recovery

The qubit example also clarifies local information maps. Let a hypothetical scalar readout be $a = a_0 + \alpha x + \beta z$, and pair it with the entropy $s(r)$. The Jacobian of (s,a) with respect to (x,z) has determinant proportional to $\beta x - \alpha z$ away from $r = 0$. It is locally invertible where that expression is nonzero and the entropy radial derivative is nonzero. At the maximally mixed reference the entropy gradient vanishes, so the determinant vanishes even though the relative-entropy Hessian remains positive definite.

Thus a positive information metric does not guarantee that any arbitrarily chosen pair of observables is a valid coordinate chart. Conversely, a valid local chart on a restricted two-parameter family does not reconstruct an arbitrary quantum state. Allowing a σ_y component would introduce a third direction not covered by those two scalar readouts. The Si29 inverse should be understood with precisely the same discipline: its claim concerns the supplied two-direction family near specified reference points.

The distinction between local coordinates and global recovery is also topological. Even a nonzero Jacobian on a small region does not prevent distant points from sharing the same observable pair. Phase periodicity and discrete symmetries can create multiple branches. A boundary-information claim therefore needs both a local stability bound and a domain statement. Without the latter, an inverse used to reconstruct a source could silently return the wrong branch while fitting both measured scalars.

These examples turn the chapter's information language into concrete operations: state preparation, restriction, measurement, conditioning, and reconstruction. They show what accessibility can remove, what an inverse can recover, and which claims

require additional dimensions or dynamics. They do not identify inaccessible information with annihilated information, nor do they establish recovery from an evaporating horizon. Those stronger questions require the source-selected global channel that remains to be derived.

12.3 Regularity checks for a proposed physical channel

A proposed horizon channel must pass checks beyond fitting one occupation law. It should map every allowed input density operator to a positive output with unit trace, including inputs entangled with a reference system. That stronger requirement is why complete positivity, rather than positivity only on an isolated subsystem, is the appropriate channel condition. A formula verified on diagonal occupation mixtures may fail on coherent superpositions or reference-entangled states.

Energy accounting must also be stated at the channel level. An apparently passive exterior map can exchange energy with unobserved boundary degrees of freedom, so the channel's dilation or an equivalent work-and-heat account is needed before claiming that it conserves energy. If the map is explicitly time dependent, the control mechanism can perform work. Such work cannot be omitted merely because the exterior output resembles a thermal state.

For a family of stationary scattering channels, analyticity and causality constrain the frequency response. Choosing a separate transmission number at every frequency without a causal response function can reproduce a desired spectrum while failing to define a physical time-domain evolution. Likewise, demanding unit transmission in one calculation and perfect reflection in another is not a single boundary theory. The response has to be compatible across the observables to which it is applied.

These conditions provide practical rejection tests for future completions. A map producing a negative eigenvalue on an allowed input is invalid. A channel that emits more energy than the source loses without identifying an external reservoir has an incomplete account. A response whose chosen boundary condition cannot generate the claimed scattering solution must be revised before interpreting its poles or flux. None of these tests depends on agreement with a particular astronomical dataset; they are internal requirements for a well-defined physical model.

13. Closure program and falsifiable outcomes

The remaining work is finite and testable in structure, even where it is technically demanding. First, derive the causal boundary response for the actual two-polarization field. The same response must be used in the radiation and ringdown calculations. A spectrum obtained with an absorbing horizon cannot be combined with an echo prediction obtained by silently replacing that horizon with a reflector.

Second, identify the boundary density operator, transition strengths, and clock from the original dressed source. The retained finite occupation law leaves all three incomplete. An attempt clock multiplied by an arbitrary constant rescales luminosity without changing equilibrium occupation. Absolute power therefore cannot be inferred from occupation probabilities alone.

Third, derive the energy join $\varepsilon_Q = \hbar\omega_{\mathrm{phys}}$ in one chart. The source's retained calibration ratios are 8, 32, and 32 for the quoted model quantities. They are evidence that the ledger energy and kernel frequency cannot simply be declared identical without conversion. A complete bridge must state physical time per model time, physical energy per model energy, and the action relation linking them.

Fourth, complete the Si29 neighborhood certificate and physical area map. The most direct failure condition is a verified mismatch between the mapped boundary response and the transported information quadratic form on an independently specified family. The finite enclosing-area obstruction already rules out its simplest scalar identification, but it leaves physically richer maps open.

Fifth, count species and angular populations without double-counting the two original guides. A successful sum must conserve energy, recover the single-channel limit, and supply count fluctuations as well as the mean. Sixth, state whether the entry spectrum is an effective thermodynamic discretization or a microscopic level spectrum. Bekenstein and Mukhanov offer an important discrete-area comparison [VII.3]; the distinctive SAM content would be a source-selected transition law and linewidth, rather than the existence of square-root mass spacing alone.

This chapter therefore delivers a consistent conditional thermodynamic account and a sharper finite information test. It does not yet deliver an absolute luminosity, evaporation time, species-resolved radiation spectrum, or unitary information-recovery theorem. Those conclusions require the missing physical joins, and their absence is a precise research boundary rather than an invitation to insert phenomenological rates.

References and provenance

[VII.1] S. W. Hawking, "Particle creation by black holes," *Communications in Mathematical Physics* **43**, 199–220 (1975). https://doi.org/10.1007/BF02345020.

[VII.2] J. D. Bekenstein, "Black holes and entropy," *Physical Review D* **7**, 2333 (1973). https://doi.org/10.1103/PhysRevD.7.2333.

[VII.3] J. D. Bekenstein and V. F. Mukhanov, "Spectroscopy of the quantum black hole," *Physics Letters B* **360**, 7–12 (1995). https://arxiv.org/abs/gr-qc/9505012.

[VII.4] P. Dorau and A. Much, "Quantum Relative Entropy implies the Semiclassical Einstein Equations," arXiv:2510.24491. https://arxiv.org/abs/2510.24491.

[VII.5] A. E. Broderick, A. Loeb, and R. Narayan, "The Event Horizon of Sagittarius A*," *Astrophysical Journal* **701**, 1357 (2009). https://arxiv.org/abs/0903.1105.

[VII.6] *SAM Unified Technical Synthesis, Volume I* (1 September 2026), Sections 4.2, 5, 6, A.7, A.13; *Original Write Histories and a Coupled Source–Field Action*, v1.1 (15 October 2026), Section 8 and Appendix E. Internal provenance retained from the supplied volume.

[VII.7] Chapters V–VI of the supplied volume; LC11; HOME-QW1; H000794–H000797; and the closing review dated 19 October 2026. Native numerical enclosures are quoted from those records and were not rerun for this expansion.

SAM VOLUME I

Cosmic inventory and the Hubble cascade

Chapter IX

Abstract. The selected SAM inventory is developed as a reproducible chain from dimensionless container assignments to physical baryon density and a conditional Hubble constant. The clean and effective matter ledgers are kept separate, the baryonic allocation and abundance exponent are identified as unresolved physical selections, and the horizon projection's dimensional cancellation is restricted to its declared algebraic convention. The thermal conversion is derived explicitly, including its dependence on temperature, mass per baryon, and the critical-density relation. Sensitivity calculations distinguish numerical precision from theoretical uncertainty. The closing review's grammar enumeration is retained with its limits, while observational comparisons are framed as model-dependent tests rather than independent derivations. A frozen-packet protocol connects recombination, distance measurements, halos, and cosmic fate, with radiation-aware closure and an explicit dark-energy law. Independent abundance tests and input manifests provide a route to prospective falsification without inventing numerical runs or concealing previous selection choices.

Status, 25 October 2026 (Part 2). This chapter adds the inventory premise: the floor $A_0 = 1/(12\pi)$, the weight and the count $\hat{d} = 3$ (register T-01, T-02, T-05), from which $\Omega_m = 1/\pi$ and the Hubble cascade follow as identities. Five runs of 24 October (voli_ceiling_1–5, ironclad H002776–H002778, H002780, H002781) gave the road four exact languages, a lawfulness gate for staged alternatives, one chart for the floored reservoir and the line of sight, and the exponent as a count of accumulation directions; none derived the floor, the weight or the count. Their content is summarized in Section 18 below.

1. Purpose and status of the inventory

The cosmic inventory assigns a small set of dimensionless substrate expressions to cosmological density fractions. A thermal conversion then turns the baryon-to-photon ratio into a physical baryon density and, together with the assigned baryon fraction, into a Hubble constant. This chapter expands that chain into an explicit dependency calculation, separates arithmetic closure from physical selection, and fixes the information that a joint observational test must carry.

The retained clean packet is

$$\Omega_m = \frac{1}{\pi} = 0.3183098862, \tag{1}$$

$$\Omega_b = 2A_0(1-\chi) = 0.0492990113, \tag{2}$$

$$\Omega_c = \Omega_m - \Omega_b = 0.2690108749, \tag{3}$$

$$\Omega_{\Lambda,\mathrm{ledger}} = 1 - \Omega_m = 0.6816901138. \tag{4}$$

Here $A_0 = 1/(12\pi)$ and $\chi = 8A_0/3$. The subscript on the final expression is intentional: an exact matter-complement ledger is not automatically a complete present-day Friedmann budget including radiation, curvature, and a declared neutrino convention.

The baryon-to-photon assignment and thermal conversion give

$$\eta_{\mathrm{SAM}} = \frac{7}{4}A_0^6 = 6.0960895245 \times 10^{-10}, \tag{5}$$

followed, under the conversion assumptions below, by $\omega_b \simeq 0.0222960$ and $H_{0,\mathrm{SAM}} \simeq 67.2504\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. These numbers are reproducible arithmetic consequences of the selected rules and the temperature anchor. They are not independent evidence that the allocation and exponent rules are uniquely selected by a microscopic dynamics.

The closing review supplies an important methodological correction. The expressions were selected within a discrete grammar after relevant target values were known. They have no continuously adjustable cosmological parameters in the frozen packet, but a selection among expressions is still a model choice. Accordingly this chapter distinguishes *conditional derivation* of downstream quantities from *retrospective matching* of upstream identities and *prospective testing* on data that did not select them. That distinction makes the inventory more testable rather than less useful.

2. Typed primitives and their roles

Use $\widehat{h} = 2$ for binary arity and $\widehat{d} = 3$ for spatial support. To avoid confusing a retained integer with gravitating mass, denote the retained count by M_{count}. The declared objects are

$$R = \widehat{h}^2\widehat{d} = 12, \qquad S = \widehat{h}^{\widehat{d}} = 8, \tag{6}$$

$$\Theta = \widehat{h}\widehat{d}^2 = 18, \qquad M_{\mathrm{count}} = \frac{7}{8}R^2 = 126, \tag{7}$$

$$L = R^2 + \Theta = 162, \qquad V = \widehat{d}^3 = 27, \tag{8}$$

$$A_0 = \frac{1}{R\pi}, \qquad \chi = \frac{S}{\widehat{d}}A_0 = \frac{2}{9\pi}. \tag{9}$$

All quantities in these equations are dimensionless. Their names describe proposed roles in the substrate partition; the names alone do not identify an experimentally conserved number density. In particular, a ledger count is not a photon count until the mapping to physical modes is specified.

The same floor enters three different operations. The distance-road calculation uses RA_0 as an amplitude. The inventory uses it as the proposed matter fraction. The thermal chain uses powers and ratios of the same containers. Their common ancestry is mathematically clear. Their physical equivalence requires an operator map: a propagation amplitude is not a density merely because its numerical value equals a density fraction. This chapter therefore preserves the shared expression while maintaining the separate operational meanings.

A useful dependency diagram can be read directly as a sequence:

$$\begin{aligned}(\widehat{h},\widehat{d}) &\longrightarrow (R,S,\Theta,M_{\text{count}},L,V,A_0)\\ &\longrightarrow (\Omega_b,\Omega_m,\eta)\longrightarrow(\omega_b,H_0).\end{aligned} \tag{10}$$

The final arrow additionally uses physical constants, a measured temperature, a photon distribution, a mean mass per baryon, and the critical-density convention. Those external inputs must remain visible even though they are not adjustable cosmological fit parameters.

3. The clean matter assignment

3.1 A conditional algebraic identity

The matter rule is $\Omega_m = RA_0$. Since $A_0 = 1/(R\pi)$, cancellation gives $\Omega_m = 1/\pi$ exactly. There is no approximation or numerical fit in this cancellation after the rule is chosen. The unresolved physical question is why a dimensionless substrate product should equal the density ratio $\rho_m/\rho_{\text{crit}}$ for the cosmological epoch and observer convention being used.

The distinction can be made experimentally meaningful. If Ω_m means today's physical matter fraction, the rule predicts one number at today's epoch and does not imply that $\Omega_m(z)$ remains fixed. In a standard expanding model with separately conserved matter, $\rho_m \propto a^{-3}$ and $\rho_{\text{crit}} \propto H^2$, so

$$\Omega_m(z) = \frac{\Omega_{m0}(1+z)^3}{E(z)^2}, \qquad E(z) = \frac{H(z)}{H_0}. \tag{11}$$

If the substrate account instead intends a conserved fraction at all times, it must provide a different evolution or definition. The present frozen packet is interpreted as a present-epoch assignment when exported into conventional cosmological calculations.

3.2 Horizon projection and dimensional continuation

The retained projection writes a dimensionless closure marker $A_{\text{cl}} = 1$ as

$$A_0(D) = \frac{A_{\text{cl}}}{4\pi D}, \qquad R(D) = 2\widehat{h}D. \tag{12}$$

For binary arity this coincides with $R = \widehat{h}^2 D$. The product then satisfies

$$R(D)A_0(D) = \frac{\widehat{h}A_{\text{cl}}}{2\pi} = \frac{1}{\pi}. \tag{13}$$

This is a valid *formal continuation of the declared algebra* with a fixed 4π normalization. It is not a proof that physical horizon areas in arbitrary spatial dimension have the same angular normalization. The area of the unit $(D-1)$-sphere is

$$\mathscr{S}_{D-1} = \frac{2\pi^{D/2}}{\Gamma(D/2)}. \tag{14}$$

If a genuinely dimension-dependent geometric projection replaced 4π by $\mathscr{S}_{D-1}$, the same cancellation of D would leave $2\widehat{h}A_{\text{cl}}/\mathscr{S}_{D-1}$, which generally depends on D. Thus the original dimension-independence statement must be restricted to its stipulated projection convention. At $D = 3$, $\mathscr{S}_2 = 4\pi$, and the two formulations agree.

The projection also uses a dimensionless closure marker rather than the physical horizon area $\mathscr{A}_H$ of Chapter VIII. Substituting square metres into equation (12) would make A_0 dimensional and invalidate the inventory. A future physical derivation must specify the area normalization before relating a completed compact boundary to a universal cosmological floor.

4. The baryonic share and what selects it

4.1 Retained allocation

The first allocation step forms the face-per-direction quotient,

$$\chi = \frac{S}{\widehat{d}}A_0 = \frac{8}{3}\frac{1}{12\pi} = \frac{2}{9\pi} = 0.0707355303. \tag{15}$$

The second step assigns a binary share of the floor reduced by that quotient:

$$\Omega_b = \widehat{h}A_0(1-\chi) = \frac{1}{6\pi}\left(1-\frac{2}{9\pi}\right). \tag{16}$$

This is positive because $0 < \chi < 1$. It is less than the total matter fraction because

$$f_b \equiv \frac{\Omega_b}{\Omega_m} = \frac{\widehat{h}}{R}(1-\chi) = \frac{1-\chi}{6} \simeq 0.1548774. \tag{17}$$

Thus the clean ledger reserves about 15.49% of its matter component for baryons. The complement within matter is the proposed nonbaryonic component. This bookkeeping distinguishes the baryonic fraction of all critical density from the baryonic fraction of matter; conflating them would introduce a factor of approximately six.

The arithmetic establishes closure and positivity, but it does not establish the allocation's uniqueness. To derive the factor $(1-\chi)$ from counting, the theory needs a sample space or set of weighted configurations, a rule that identifies the baryonic subset, and a measure whose excluded share is exactly χ. Since χ contains π, it cannot be the ratio of two unweighted finite integer cardinalities. A continuous measure, geometric weight, limiting procedure, or normalized amplitude could produce such a share, but that additional structure has to be stated.

This observation does not refute the rule. It specifies the kind of derivation that could justify it. A finite enumerator of containers alone cannot turn an irrational weight into an exact unweighted fraction. Conversely, if the containers carry geometric weights, the allocation proof must conserve their total weighted measure and show why the binary baryon branch uses precisely this normalization.

4.2 Alternatives expose the selection problem

Nearby expressions illustrate why the allocation needs a mechanism. The unsuppressed binary floor is $2A_0$. A repeated exclusion would give $2A_0(1-\chi)^2$. A saturating normalization might give $2A_0/(1+\chi)$. These alternatives are examples of algebraic competitors, not proposed replacements. They share the same named ingredients while producing different densities. Declaring that one looks closest to a measured baryon

fraction is a selection rule based on that measurement, not an independent physical derivation.

A legitimate restricted grammar can be much narrower than arbitrary combinations. It could require linear additivity over disjoint branches, one application of a face quotient, and conservation of a specified measure. But each restriction must arise from the model before the target comparison. The publication test is whether another researcher, given the primitives and restrictions but not the observed density, recovers the same expression.

The nonbaryonic remainder

$$\Omega_c = \frac{1}{\pi} - \frac{1}{6\pi}\left(1 - \frac{2}{9\pi}\right) \tag{18}$$

is a density assignment, not a composition theorem. It does not decide whether the component consists of primordial compact objects, field energy, particles, or a mixture. the Supplement (the retired galaxy-halo chapter) must supply clustering and halo behavior; Chapter XII must supply any formation history. Conservation of the sum does not substitute for those dynamics.

5. Clean and effective inventories

The retained directional refinement is

$$\Omega_{m,\mathrm{eff}} = RA_0 - 2A_0(\chi + \widehat{d}\chi^2) = 0.3137609155690572. \tag{19}$$

The parentheses matter. Both the linear and quadratic terms are multiplied by $2A_0$. The baryon row is held fixed by the refinement's definition. Consequently an effective nonbaryonic row, if that operator is used, must be $\Omega_{c,\mathrm{eff}} = \Omega_{m,\mathrm{eff}} - \Omega_b$. One cannot combine an effective total with the clean dark component and still retain additive closure.

There are two scientifically different possibilities. The refinement could be an observational response operator mapping one underlying density to an effective measured parameter. Alternatively it could be a second candidate background model with a different matter density. The first possibility requires an explicit forward map from the clean state to each observable. The second requires separate predictions and separate statistical scores. Neither permits selecting whichever number agrees better with each dataset after inspecting its residual.

For the expanded volume, the clean packet is the default structural ledger. The effective packet remains a separately named conditional lane. A calculation that uses it must state the input, the operator's scope, and the associated complement. The common road amplitude remains RA_0 unless the road operator itself is independently changed. These conventions preserve the original distinction while preventing silent movement between ledgers.

Holding Ω_b and the thermal conversion fixed leaves H_0 unchanged in this particular refinement. It does not leave $H(z)$, the age, growth, matter-radiation equality, or the distance-redshift relation unchanged. Those depend on the matter row and must be recomputed. The importance of the distinction is therefore physical, even though the Hubble cascade itself is unchanged.

6. The baryon-to-photon assignment

The two retained container ratios are

$$\frac{M_{\mathrm{count}}}{\widehat{h}^2\Theta} = \frac{126}{72} = \frac{7}{4}, \qquad \frac{L}{V} = \frac{162}{27} = 6. \tag{20}$$

The selected rule then gives

$$\eta = \frac{M_{\mathrm{count}}}{\widehat{h}^2\Theta} A_0^{L/V} = \frac{7}{4(12\pi)^6}. \tag{21}$$

The simplification is exact. Its physical interpretation is a proposed baryon number per photon after the entropy transfers relevant to the measured late-time photon bath. For standard adiabatic evolution after those transfers, both number densities dilute as a^{-3} and the ratio is constant. A theory with later photon creation, baryon production, or transfer into a hidden sector would need an evolution law for η.

The exponent is a major source of selectivity. Replacing six by five multiplies the result by $1/A_0$, approximately 37.70; replacing six by seven multiplies it by A_0, approximately 0.02653. This demonstrates why the integer is consequential, but does not explain why it is six. A physical derivation would need six independent stages with the same attenuation, a six-dimensional measure, or another explicitly demonstrated multiplicative structure. Merely observing that a ledger-to-volume ratio equals six is not such a demonstration.

Similarly, $7/4$ is a retained-to-carrier ratio in the declared grammar. It is not yet a proof of a baryon asymmetry mechanism, a chemical potential, or a baryogenesis efficiency. The rule fixes a proposed final abundance without supplying the time-dependent production mechanism. This is acceptable as a phenomenological inventory hypothesis, provided the distinction is retained when Chapter X computes a thermal history.

For later tests, one should freeze both η and the conservation assumptions. A light-element calculation can then use the same baryon abundance at nucleosynthesis and compare deuterium and helium yields with observations. If the abundance is adjusted after inspecting those yields, the result becomes another calibration rather than the intended out-of-selection test.

7. The dimensional Hubble cascade

7.1 Photon density from the thermal anchor

A blackbody with two photon polarizations has number density

$$n_\gamma = \frac{2\zeta(3)}{\pi^2}\left(\frac{k_B T_0}{\hbar c}\right)^3. \tag{22}$$

This follows by integrating the occupation $(e^{pc/k_BT_0} - 1)^{-1}$ over momentum phase space, including two polarizations. The dimensionless integral is $\int_0^\infty x^2/(e^x - 1)\,dx = 2\zeta(3)$, obtained by expanding the denominator as a positive exponential series and integrating term by term. The assumed spectrum is therefore part of the conversion, not just a numerical convenience.

The retained ledger uses $T_0 = 2.7255\,\mathrm{K}$, a rounded thermal anchor consistent with Fixsen's combined estimate $2.72548 \pm 0.00057\,\mathrm{K}$ [VIII.1]. The photon density is then approximately

$410.727\,\mathrm{cm}^{-3}$. Using the rounded or unrounded temperature changes only later digits at the precision relevant here; the exact choice must nevertheless be recorded in a reproducible calculation.

7.2 Critical density and mass per baryon

Define

$$H_{100} = 100\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \rho_{\mathrm{crit},100} = \frac{3H_{100}^2}{8\pi G}. \tag{23}$$

If $\overline{m}_b$ is the mean rest mass per counted baryon, then

$$\rho_b = \overline{m}_b \eta n_\gamma, \qquad \omega_b = \Omega_b h^2 = \frac{\rho_b}{\rho_{\mathrm{crit},100}}, \qquad h = \frac{H_0}{H_{100}}. \tag{24}$$

The retained computation approximates $\overline{m}_b$ by the proton mass m_p. This makes the conversion exactly reproducible, but its accuracy should not be confused with the number of printed digits. A physically precise conversion must specify helium abundance, nuclear binding, electron contributions, and the convention used by the cosmological code for baryon mass density. These effects are small compared with order-unity changes of the inventory but matter when claiming precision at the 10^{-4} level.

Introducing the dimensionless conversion factor

$$K = \frac{\rho_{\mathrm{crit},100}}{m_p n_\gamma} \simeq 2.73416 \times 10^{-8} \tag{25}$$

gives $\eta = K\omega_b$, so $\omega_b = \eta/K \simeq 0.0222960$. Finally,

$$h^2 = \frac{\omega_b}{\Omega_b} \simeq 0.452261, \qquad H_0 \simeq 67.2504\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{26}$$

The supplied check script independently evaluates this arithmetic with the constants stated in the script. It is not a rerun of the native cosmological source or a likelihood analysis.

Combining the chain into one expression displays its dependencies:

$$H_0^2 = \frac{8\pi G}{3} \frac{\overline{m}_b \eta n_\gamma}{\Omega_b}. \tag{27}$$

The expansion rate arises because the density fraction is defined relative to a critical density proportional to H_0^2. A theory that rejects that critical-density relation cannot retain this Hubble inference unchanged. Likewise, the calculation uses measured dimensionful constants as well as one measured cosmological thermal anchor. "One anchor" means one cosmological calibration in this chain, not an absence of external metrology or physical assumptions.

8. Sensitivity, uncertainty, and numerical precision

The logarithmic derivative of equation (27) is

$$\begin{aligned} d\log H_0 = {} & \tfrac{1}{2} d\log G + \tfrac{1}{2} d\log \overline{m}_b \\ & + \tfrac{1}{2} d\log \eta - \tfrac{1}{2} d\log \Omega_b \\ & + \tfrac{3}{2} d\log T_0. \end{aligned} \tag{28}$$

with the exact SI constants held fixed. This gives an immediate audit of calibration changes. A fractional temperature shift is amplified by $3/2$ in the inferred expansion rate. A fractional change in the baryon mass convention enters with coefficient $1/2$.

If A_0 is varied diagnostically while the selected allocation functions are retained, $\eta \propto A_0^6$ and $\Omega_b \propto A_0(1 - bA_0)$ with $b = 8/3$. Therefore

$$\frac{d\log H_0}{d\log A_0} = \frac{1}{2}\left(5 + \frac{\chi}{1-\chi}\right) \simeq 2.53806. \tag{29}$$

This is a sensitivity calculation, not a declaration that the floor is a freely fitted parameter. It quantifies why changing one upstream convention requires all downstream lanes to be recomputed.

Propagating only Fixsen's quoted temperature uncertainty would give an expansion-rate uncertainty of approximately $0.0211\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. That is a conditional metrological contribution, not a total theory uncertainty. It omits uncertainty in the allocation, exponent, physical bridges, mass convention, and model selection. The chapter therefore does not attach that small error bar to a claim of comprehensive predictive precision.

A discrete selection uncertainty is not automatically Gaussian. It can be represented by a declared finite set of candidate models with prior weights, by a sensitivity envelope, or by separate hypotheses. Choosing one expression retrospectively and assigning only the thermal measurement error understates the scope of the inference. More printed decimal places improve arithmetic reproducibility; they do not create additional observational evidence.

9. Selection, enumeration, and honest comparison

9.1 What the closing review establishes

The closing review reports a grammar enumeration with multiplicative expressions $2^a 3^b 7^c \pi^d$. Its smallest stated ranges are $|a| \le 8$, $|b| \le 6$, $|c| \le 1$, and $|d| \le 4$, giving $17 \cdot 13 \cdot 3 \cdot 9 = 5967$ exponent tuples. In that enumeration it reports 14 expressions in a one-standard-deviation matter window, 14 in a baryon window, and six in an η_{10} window. These are retained review results, not independently reproduced enumeration outputs in this chapter.

Several qualifications are essential. The target windows and their units must be explicit: $\eta_{10} = 10^{10}\eta$ is not the same numerical target as η. Additive expressions such as $2A_0(1-\chi)$ are not members of the purely multiplicative base grammar, although multiplicative competitors can land near their target value. Counting tuples is not necessarily counting distinct numerical expressions if a richer grammar admits identities. And the fraction of a finite arbitrary list landing inside a window is not automatically a probability or a statistical significance.

The review's wider grammars admit larger exponents, named additive factors, and half-integer powers. Their increasing match counts demonstrate that a match's impressiveness depends on the available search space. They do not determine the

correct prior over physical hypotheses. The required next step is a restricted, role-sensitive grammar fixed before new data are inspected, with canonicalization, range limits, target definitions, and all rejected candidates retained.

9.2 A prospective protocol

A usable protocol has four stages. First, publish the admissible operations and the physical rule selecting each expression. Second, freeze the clean or effective interpretation and every downstream bridge. Third, compute observables on an identified dataset not used to choose the expression. Fourth, report the full residual vector and covariance without revising the grammar in response. A failed prediction can motivate a revised model, but the revision should receive a new version and a new prospective test.

This approach preserves a real advantage of a small grammar: a frozen packet imposes correlations across observables. It also prevents those correlations from being overstated. If H_0 is calculated from η and Ω_b, agreement of all three is not three independent miracles. Their joint significance depends on the data covariance and on which quantities participated in the original selection.

10. Observational placement

Planck's published base-ΛCDM inference gives $H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.315 \pm 0.007$ [VIII.2]. With those rounded comparator values, the clean inventory differs by about -0.30 standard deviations in H_0 and $+0.47$ in Ω_m, treating the SAM point as fixed for this descriptive calculation. These are model-dependent posterior comparisons, not direct measurements of bare substrate quantities, and they are not an independent joint likelihood.

The original draft also compared H_0 with the SH0ES value $73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [VIII.3]. The inventory's lower value is a commitment of its frozen chain. It does not resolve the discrepancy by agreeing with one side. Any proposed resolution must propagate the same cosmology into distance calibration, selection effects, stellar standardization, and the observables used by the relevant analysis.

DESI DR2 supplies an important later test [VIII.4]. Its primary paper finds that the preferred flat-ΛCDM parameters depend on the selected data combination and that allowing evolving dark energy changes the interpretation. The closing review's provisional $\Omega_m \simeq 0.303 \pm 0.004$ was explicitly marked for verification. It is not promoted here into a universal DESI number or a definitive exclusion significance. A valid comparison must name the exact likelihood combination, neutrino assumptions, dark-energy model, and posterior statistic. The clean and effective inventories should then be scored separately against the same setup.

The literature placement is therefore clear without a broad originality claim. Standard analyses infer cosmological parameters from a specified dynamical model and data likelihood. This inventory proposes algebraic restrictions on several of those parameters and then tests the restricted packet. Its distinctive proposal is the particular substrate-to-density assignment. The elementary use of a blackbody photon density and critical-density conversion is shared physical infrastructure. A universal claim that no prior structural proposal resembles these expressions would require a dedicated historical survey and is not established here.

11. A complement is not yet dark-energy dynamics

Setting a ledger remainder equal to $1 - \Omega_m$ supplies a number, not an equation of state. In a conventional homogeneous background,

$$E^2(z) = \Omega_{r0}(1+z)^4 + \Omega_{m0}(1+z)^3 + \Omega_{k0}(1+z)^2 + \Omega_{X0} f_X(z), \tag{30}$$

where a separately conserved dark component obeys

$$f_X(z) = \exp\left[3\int_0^z \frac{1 + w_X(z')}{1+z'}\,dz'\right]. \tag{31}$$

A cosmological constant has $w_X = -1$ and $f_X = 1$. An evolving component requires a specified $w_X(z)$ or field equation. The inventory alone selects neither.

At $z = 0$, exact closure requires $\Omega_{r0} + \Omega_{m0} + \Omega_{k0} + \Omega_{X0} = 1$. Thus in a spatially flat radiation-aware implementation the physical complement is $\Omega_{X0} = 1 - \Omega_{m0} - \Omega_{r0}$, with neutrinos counted consistently between the matter and radiation terms. The clean ledger's $1 - \Omega_m$ is a late-time radiation-neglecting complement unless an alternative accounting convention is explicitly adopted. Radiation is numerically small today but essential for Chapter X and cannot be silently counted twice.

If redshift is attributed in part to a propagation mechanism, the model must predict the transformation of photon energies, arrival intervals, solid angles, and number flux together. Matching a luminosity-distance curve alone does not establish consistency with time dilation, temperature evolution, or distance duality. In the standard metric, photon-conserving case the latter reads $D_L = (1+z)^2 D_A$. A replacement road law must specify which hypotheses it changes and produce a new joint test rather than inherit this relation selectively.

12. One frozen packet across the volume

The inventory value $H_{0,\mathrm{SAM}}$ is the default expansion-rate normalization for a prospective joint test. The retained supernova packet used 73.04 and the retained BAO packet used $68.76\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. They remain historical calibration lanes until rerun with a declared common packet. Merely placing their separate residuals in one table would not make them a joint test.

For fixed dimensionless $E(z)$, distances scale as H_0^{-1} and luminosity distances enter distance moduli logarithmically. The purely kinematic rescaling is

$$\Delta\mu = 5\log_{10}\left(\frac{H_{0,\mathrm{old}}}{H_{0,\mathrm{new}}}\right). \tag{32}$$

This identity is useful for auditing a normalization change, but it is not a substitute for rerunning a lane whose sound horizon, physical densities, or shape also changes. Supernova absolute

magnitude can absorb a global normalization in some analyses; BAO uses a calibrated ruler whose value depends on early-time physics. Those degeneracies must be treated in the appropriate likelihood.

The packet delivered to Chapter X includes η, the density rows, H_0, and explicit radiation and neutrino conventions. The proposed perturbation relations $A_s = \eta\sqrt{R}$, $n_s = 1 - \chi/2$, and $\tau = 2A_0$ remain separately selected assignments, discussed there. The Supplement (the retired galaxy-halo chapter) receives the clean matter row for its stated $X_\infty = (R - \widehat{h})\Omega_m$ relation unless it explicitly declares another operator. Chapter XII receives both the inventory complement and the chosen dynamical interpretation, since a density fraction without an evolution law does not determine a cosmic fate.

A joint run should publish its input manifest, code versions, units, data masks, covariance treatment, nuisance parameters, and residuals. Residuals in incompatible units should not be added by eye. If r is a concatenated data residual with covariance C, a Gaussian diagnostic is $r^T C^{-1} r$ only after cross-correlations and nuisance treatment are specified. The present expansion supplies the protocol; it does not invent the missing joint execution.

13. Independent abundance tests and a reproducible input manifest

The abundance rule becomes most valuable when it commits to a calculation that did not choose its coefficients. Primordial nucleosynthesis is a natural candidate because the baryon-to-photon ratio affects nuclear reaction competition in an expanding plasma. The calculation also depends on the expansion history, neutron lifetime, effective relativistic content, and laboratory reaction rates. Holding only η fixed while adjusting those quantities to force agreement would not be a clean test of the proposed packet.

A prospective abundance analysis should therefore begin with a written input manifest. The manifest specifies whether η refers to the post-electron-annihilation photon bath, which neutrino temperature and effective species count are adopted, and whether the standard thermal relation between cosmic time and temperature is used. It records the treatment of nuclear binding in the baryon mass conversion and distinguishes that conversion from the baryon number that enters reaction equations. It identifies the reaction-rate compilation and its covariance, since several yields share rate uncertainties.

The principal outputs are element abundances with uncertainty budgets separated into nuclear, observational, and cosmological components. Deuterium is useful because its surviving abundance responds strongly to the baryon density, but a numerical scaling law should be obtained from the selected network over its stated range rather than imported without verification. Helium supplies a complementary sensitivity to the expansion and weak-interaction history. Lithium should be reported as well, even where the standard calculation and astrophysical inference have known difficulties; selective reporting of only favorable elements would defeat the prospective protocol.

No BBN network was executed for this expansion. The purpose of the specification is to prevent an analytic inventory from being presented as if it had already passed a reaction-network test. A future result should include the calculated yields, the observed comparison set, the adopted systematic uncertainties, and the change in fit relative to a baseline using the same network. If the abundance rule fails, the analysis should first identify whether the failure is stable under measured rate uncertainties before assigning it to substrate physics.

A parallel manifest is needed for the background and perturbation codes. At minimum it contains T_0, H_0, Ω_b, the chosen total-matter row, the nonbaryonic split, the neutrino masses, spatial curvature, the dark-energy law, and the radiation normalization. Where a code accepts physical densities ω_i rather than fractions Ω_i, all conversions must use the same h. Passing a clean fraction to one module and an effective physical density to another is an input inconsistency even if each number looks individually reasonable.

The manifest should distinguish calculated values from retained source values and from observational anchors. A hash or version identifier makes later changes detectable. It should also record numerical precision, integration tolerances, and the treatment of species crossing between relativistic and nonrelativistic regimes. These details are not extra cosmological freedom when fixed in advance; they are the information required to reproduce the meaning of a result.

Finally, holdout data should remain holdout data. Once an independent test has been examined and used to revise the grammar, it becomes part of the development history. The revised model can still be scientifically valuable, but its subsequent predictive claim belongs to a new comparison. This versioned discipline gives the inventory a route from numerical pattern to constrained physical model without obscuring either successes or failures along the way.

14. Worked dimensional conversion and unit checks

A numerical cascade is easiest to audit when each step has a dimensional meaning. The independent arithmetic check accompanying this chapter uses $c = 299792458\,\mathrm{m\,s^{-1}}$, $k_B = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$, $\hbar = 1.054571817 \times 10^{-34}\,\mathrm{J\,s}$, $G = 6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, and $m_p = 1.67262192595 \times 10^{-27}\,\mathrm{kg}$. These are the chosen numerical inputs to this check, with the displayed rounding of $\hbar$; they are not an independent determination of the constants. The length conversion is $1\,\mathrm{Mpc} = 3.085677581491367 \times 10^{22}\,\mathrm{m}$.

The ratio $k_B T_0/(\hbar c)$ has units of inverse length. Cubing it and multiplying by $2\zeta(3)/\pi^2$ therefore gives a number per cubic metre. The result is $4.10726849 \times 10^8\,\mathrm{m^{-3}}$, which becomes $410.726849\,\mathrm{cm^{-3}}$ because one cubic metre contains one million cubic centimetres. Losing that conversion factor would corrupt every subsequent density by six orders of magnitude while leaving all dimensionless inventory identities untouched.

Next, $H_{100} = 10^5/(3.085677581491367 \times 10^{22})\,\mathrm{s^{-1}}$. Its square divided by G has units of mass density, as required for $\rho_{\mathrm{crit},100}$. The factor K divides that mass density by $m_p n_\gamma$, another mass density, and is therefore dimensionless. Finally,

η/K is a dimensionless physical baryon density parameter, whereas ηn_γ is a physical number density. Distinguishing these two uses of the word density prevents a hidden unit error when the packet is passed to a numerical solver.

The inferred baryon number density is approximately $0.2504\,\mathrm{m}^{-3}$ in this convention. It is an average over cosmological volume, not the density in stars, gas clouds, or galaxies. Multiplying by m_p produces a mean baryonic mass density of order $4.19\times10^{-28}\,\mathrm{kg\,m}^{-3}$. Dividing that result by the critical density at H_0, rather than at H_{100}, recovers Ω_b by construction. The independent script checks this closure in both directions.

There are therefore two equivalent paths through the conversion. One computes K, then ω_b, then h. The other uses equation (27) directly. Agreement between the paths is an implementation check because the same algebra underlies both. It is not a second observational constraint. A useful release should nevertheless retain both paths: one is convenient for cosmological-code inputs and the other exposes dimensional dependencies.

The calculation also shows why the proton-mass approximation deserves explicit treatment. Baryon number counts nucleons, whereas mass density includes the composition-dependent energy of bound nuclei and associated electrons according to the adopted convention. A future high-precision implementation should compute $\overline{m}_b$ from the same primordial composition used in the thermal calculation. Replacing m_p by that mean mass changes H_0 by the square root of their ratio. This correction can be calculated once the composition convention is fixed; it should not be treated as an arbitrary nuisance parameter selected to move the expansion rate toward an observational preference.

15. A bounded alternative-expression experiment

A small transparent calculation illustrates discrete selection without pretending to reconstruct the whole historical grammar. Hold the floor and baryon allocation fixed, and examine only the nine abundance expressions

$$\eta_{w,p} = wA_0^p, \qquad w\in\{1,7/4,2\}, \quad p\in\{5,6,7\}. \tag{33}$$

This is a newly declared diagnostic set, not a claim that these nine candidates were the original admissible universe of theories. It is small enough that every candidate can be listed and its consequence evaluated without fitting.

Relative to the selected expression, each candidate obeys

$$\frac{\eta_{w,p}}{\eta_{7/4,6}} = \frac{4w}{7}A_0^{p-6}, \qquad \frac{H_{0,w,p}}{H_{0,7/4,6}} = \sqrt{\frac{4w}{7}}A_0^{(p-6)/2}. \tag{34}$$

The second identity follows because Ω_b, the thermal anchor, and all dimensional constants were held fixed. If a candidate also changed the baryon allocation, it would require the additional square-root denominator change. This controlled construction isolates the abundance selection alone.

At $p=6$, changing the weight from $7/4$ to one gives $H_0\simeq 50.84\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, while changing it to two gives about 71.89 in the same units. At weight $7/4$, choosing exponent five gives approximately 412.9, while exponent seven gives about 10.95. These examples make the physical leverage of the selected ingredients visible. They do not claim those alternatives are plausible cosmologies; their purpose is to show that "no continuous fit" does not mean "no consequential choice."

The exercise can be made into an auditable selection test by declaring a comparator interval before evaluating the nine entries. The number landing inside the interval then has a definite meaning for this set. It still does not become a universal false-alarm probability, because the set was not drawn from a physical probability distribution and does not include all conceivable expressions. Expanding the grammar after seeing that count changes the experiment. A defensible significance calculation would require an explicit prior over the alternatives or a prospective predictive score.

A role-sensitive derivation could make eight of the nine candidates inadmissible for structural reasons. For example, a demonstrated six-stage independent transfer law could select the exponent before any cosmological target is consulted. A conservation theorem could select the weight. If those arguments are supplied, the numerical comparison becomes much stronger because the expression is no longer selected by its agreement. The present chapter identifies those possible forms of explanation without inventing the missing theorem.

The same logic applies to the clean and effective matter rows. Treating them as two prespecified hypotheses is legitimate. Moving between them separately for Planck, BAO, halos, and fate introduces an undeclared selection at every comparison. A bounded diagnostic should therefore print both complete prediction vectors and apply one scoring rule to each, rather than construct a favorable hybrid after inspecting residuals.

16. Correlated observables and a worked covariance example

The Hubble cascade contains algebraic correlations before any observational covariance is considered. In logarithmic variables, with temperature and constants fixed,

$$\delta\log H_0 = \tfrac{1}{2}\delta\log\eta - \tfrac{1}{2}\delta\log\Omega_b. \tag{35}$$

If a model ensemble or uncertainty prescription assigns a covariance matrix to $\log\eta$ and $\log\Omega_b$, propagation gives

$$\begin{aligned}\mathrm{Var}(\log H_0) = \tfrac{1}{4}\big[&\mathrm{Var}(\log\eta)\\ &+\mathrm{Var}(\log\Omega_b)\\ &-2\,\mathrm{Cov}(\log\eta,\log\Omega_b)\big].\end{aligned} \tag{36}$$

Positive correlation can cancel in the ratio; negative correlation can amplify it. Adding independent error bars in quadrature is therefore wrong unless independence is justified. The formula is meaningful only after a probability model for the upstream uncertainty is specified; it does not assign one by itself.

As an illustration, suppose both logarithmic inputs have standard deviation 0.01. With correlation 0.8, the propagated logarithmic expansion-rate uncertainty is approximately 0.003162, or 0.316%. With correlation -0.8, it is approximately 0.009487, or 0.949%. If one incorrectly assumes independence, the answer is 0.007071, or 0.707%. These are

examples of covariance propagation, not uncertainty estimates for SAM. They demonstrate how the same individual input precision can lead to quite different precision in a derived ratio.

Observational residuals have a related issue. Consider two standardized residuals with correlation matrix

$$C = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad C^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}. \tag{37}$$

For residual vector $(1,1)$ the quadratic score is $2/(1+\rho)$; for $(1,-1)$ it is $2/(1-\rho)$. At $\rho = 0.8$, these are approximately 1.111 and 10, although each individual residual has magnitude one. Ignoring covariance would assign both the value two. A comparison that reports only a list of separate sigma offsets cannot distinguish these cases.

The physical interpretation is straightforward. Positively correlated data commonly move together, so a common-direction displacement is relatively unsurprising. A displacement in opposite directions fights that correlation and is more strongly penalized. Cosmological parameter posteriors and overlapping sky measurements contain precisely this kind of structure, albeit in much larger spaces. It is therefore inappropriate to multiply separate agreement probabilities or add individually quoted sigma values to summarize the inventory's success.

A complete test should operate as close as practical to the actual observable data vector. If compressed parameter posteriors are used instead, their model dependence, covariance, and prior information must be retained. The comparison should also distinguish a fixed prediction from a model whose nuisance quantities were optimized. The number of displayed identities does not determine the effective number of independent tests.

17. A conditional background example from the clean packet

To show what an additional dynamical assumption buys, temporarily adopt a spatially flat, matter-plus-cosmological-constant background and neglect radiation. This is the standard late-time Friedmann example, not the native fate law of Chapter XII. With $\Omega_\Lambda = 1 - \Omega_m$, the age is

$$t_0 = \frac{1}{H_0} \int_0^1 \frac{da}{a\sqrt{\Omega_m a^{-3} + \Omega_\Lambda}} = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \operatorname{arsinh} \sqrt{\frac{\Omega_\Lambda}{\Omega_m}}. \tag{38}$$

The integral follows by setting $u = a^{3/2}\sqrt{\Omega_\Lambda/\Omega_m}$. The resulting Hubble-scaled age depends on the matter row even if the Hubble cascade is unchanged. This is one explicit reason the effective and clean inventories cannot be swapped without recomputing downstream quantities.

Under the same assumptions, the acceleration changes sign when $\rho_m = 2\rho_\Lambda$, giving

$$1 + z_{\mathrm{acc}} = \left(\frac{2\Omega_\Lambda}{\Omega_m}\right)^{1/3}, \qquad H_\infty = H_0\sqrt{\Omega_\Lambda}. \tag{39}$$

These are inherited consequences of pressureless matter and $w = -1$. They do not follow from a density complement alone. If another chapter proposes a different asymptotic power of Ω_Λ, that proposal must be associated with a different evolution law or clock definition; the formulas cannot describe the same $H(t)$ under identical assumptions.

The example is useful as a baseline because it creates a full function $H(z)$ from a small packet and a clear dynamical law. It also makes circularity easy to identify. Agreement with a standard-background age after importing the standard background is a consistency check of the restricted parameter packet. A genuinely new substrate prediction would require deriving or modifying the background law and carrying that modification through distances, growth, and thermal history.

No numerical age is used as a new observational claim here. The companion check evaluates the integral's closed form and the acceleration expression for audit purposes. Radiation, massive-neutrino evolution, and any evolving dark component are omitted by construction. A precision comparison must restore them or bound their effects, rather than present the simple analytic exercise as a complete cosmological computation.

17.1 An explicit audit of shared information

The conditional background example makes it possible to assign each conclusion to its actual inputs. The matter fraction uses the floor and the chosen density identification. The baryon fraction uses the allocation rule. The baryon-to-photon ratio uses the container weight and exponent. The Hubble constant then uses those quantities plus the thermal and critical-density bridges. The age uses the Hubble constant, matter fraction, and an additional expansion law. These dependencies form a directed graph rather than a list of independent successes.

If the same observed CMB posterior helped select the matter rule, baryon allocation, and abundance expression, agreement of the derived age with an age inferred from that posterior is especially correlated. It can expose a transcription error or an inconsistent conversion, but it cannot honestly be counted as a wholly new observation predicted without contact with the data. The route to additional evidence is to test a consequence whose relevant information did not participate in the selections.

There are different strengths of such a test. Reusing the same sky with a different statistic can check an aspect not explicitly fitted, but shared cosmic variance and selection history remain. A different instrument provides different systematics while often measuring the same underlying realization. A light-element abundance or a local distance measurement probes different physical processes, yet still brings model and calibration assumptions. Independence is a graded property to be documented, not a label supplied by a new filename.

This audit also prevents excessive pessimism. A selected expression can become useful when it survives increasingly demanding, genuinely frozen tests. The appropriate record is a sequence of versioned hypotheses and their results. The inventory is not disqualified because its first identification was retrospective; it is simply required to earn prospective status through the next tests. Publishing dependencies and rejected alternatives makes that transition visible to a reader.

18. The $1/\pi$ ceiling as a recurring theme: results of 24 October 2026

The number $1/\pi$ enters this volume three times from one input. Here, $\Omega_m = RA_0 = 1/\pi$ with the floor $A_0 = 1/(12\pi)$ and the container count $R = \hat{h}^2\hat{d} = 12$, so that $\Omega_\Lambda = 1 - 1/\pi$. In Chapter XII the accumulation seen along a line of sight to redshift z is $A_{\rm los}(z) = (1/\pi)\,(1-(1+z)^{-3})$, saturating at $1/\pi$; and Chapter XII joins the two by construction, $A_{\rm los}(\infty) = \Omega_m$, both sides being A_0R. The register of 23 October records $\Omega_m = 1/\pi$ as a selected identification and the road's exponent and plateau as the two selections that fix Chapter IV's family $\Xi(s) = \Xi_\infty + (1-\Xi_\infty)e^{-ps}$, $s = \ln(1+z)$, at $p = \hat{d} = 3$ and $\Xi_\infty = 1 - 1/\pi$. The ceiling is therefore not the spine of the volume; it is a recurring theme, the one selected number that ties the inventory to the distance work and carries a declared test the data did not choose (the ratio of gravitational-wave to electromagnetic luminosity distance following $\Xi(z)$: 0.9906, 0.9208, 0.7760, 0.7215, 0.6935 at $z = 0.01, 0.1, 0.5, 1, 2$; register T-10).

Five idea runs of 24 October 2026 on the installed exact engine (ironclad H002776–H002778, H002780, H002781; four research peers each, the lead outside the window, no external data) asked what the road is as a law. Their results are readings in the engineering register, not claims, and they are recorded here because they change how Sections 3 and the road should be read together.

One sentence in four exact languages. Written in the scale factor $a = (1+z)^{-1}$, the road is $A_{\rm los} = \Omega_m(1-a^3)$ with $dA_{\rm los}/ds = 3(\Omega_m - A_{\rm los})$: the accumulation seen along a line of sight is the matter looked through, counted by comoving volume, in units in which the whole inventory is one. The exponent is $\hat{d} = 3$ because it is a volume; the ceiling is $1/\pi$ because it is all the matter there is. The run wrote this in four exact languages, each certified natively and each naming the same missing law. (a) The mirrored Friedmann reading: $f(z) = \Omega_\Lambda + \Omega_m(1+z)^{-3}$ is the ΛCDM E^2 at a future scale factor $1+z$, the ceiling its de Sitter attractor; a complement with $w \neq -1$ has no plateau, so a plateau at exactly Ω_Λ states that the complement does not dilute. (b) The floored reservoir: $\Xi = \Omega_\Lambda + \Omega_m \operatorname{sech}^2\varphi$ with $\cosh^2\varphi = (1+z)^3$, so the reservoir's log-attenuation along the line of sight is the log comoving volume and its density law $1/q_{\rm res} = (1+z)^3$ is $\rho_m(z)/\rho_m(0)$; the four one-exponent markers of the road (half-saturation at $z = 2^{1/3}-1$, the e-fold at $e^{1/3}-1$, matter equal to complement at $(1+z)^3 = \pi - 1$, and $z = 1$) were read natively on the reservoir to the rational approximants of their times. The ceiling is not in the reservoir, whose $q \to 0$ with no floor; it is the floor Ω_Λ added beside it, and that floor is $1 - 1/\pi$, not A_0. (c) The rate measure: as a floor plus a mixture of memoryless draws, $\Xi(s) = \int e^{-\lambda s}\,d\nu(\lambda)$, the road's coefficients $3/\pi, -6/\pi, 10/\pi, -15/\pi$ are the moments $(1,3,9,27)$ of a measure of Hankel rank one, and within the class the ceiling and the first two coefficients force $\nu = \Omega_\Lambda\delta_0 + \Omega_m\delta_3$: each component dilutes at its own exponent, vacuum at 0 and matter at $\hat{d}$, and the ceiling is the sum of the fractions that dilute. (d) The union law: on $x = A_{\rm los}/\Omega_m$ shells compose by $x \oplus y = x + y - xy$, survival is multiplicative, and each shell is a column-stochastic ledger on (undrawn, drawn) whose drawn column is a floor never debited; the neighbouring velocity-addition law meets the ceiling and the first coefficient but fails the second ($-3/(2\pi)$ against $-6/\pi$) and is not a positive mixture.

Which quantity the volume law counts. In the reservoir's own coordinates, $u = \tanh\varphi$ and $q = 1 - u^2$, the road is $1 - q = u^2 = \Omega_m(1-a^3)$: the quantity drawn by comoving volume is u^2, the squared velocity of the Lorentz dictionary ($u = \beta$, $q = 1/\gamma^2$), not the rapidity φ, not u and not Λ. Each alternative saturates at the same ceiling with the same exponent and fails the road's regression at the first or second coefficient; in the rapidity the road is not analytic at the observer. The reservoir's additive variable is the wrong ledger for the road.

The regression and its blind spot. The ceiling with the coefficients $3/\pi$, $-6/\pi$, $10/\pi$ over-determines the road only inside the memoryless class. Outside it, $G_c(\psi) = 1 - e^{-\Psi}(1 + c\psi^4/24)$ with $\psi = 3s$ passes every declared coefficient and departs at z^4; it is a survival only for $c \leq 8/9$, with tangency at $\psi = 3$, and the admissible member gives 0.7525 at $z = 1$ against the road's 0.7215. A staged ledger matching K conditions of the road is forced to the road unless it has at least K stages, so the smallest lawful alternative has four hidden stages; its memory lives in hidden transitions faster than the road's hazard. The run built the test of lawfulness as three native faces (realisation, monotonicity, slope) and found that lawfulness is global: a lawful five-stage staircase carries the jet of an unlawful law through fourth order, so no finite set of coefficients decides it. The declared siren test of T-10 is therefore a test of memorylessness in the line-of-sight draw, and the shape of the channel ratio between $z \approx 0.5$ and 2 is where a memory law would show.

One object, and what fixes the exponent. With a declared floor and a conserved ruler density along the line of sight installed as constitutive objects of the engine (core 0.11.3, derivation marked open), the floored reservoir at φ and the line of sight at $s = \Lambda(\varphi)/d$ are one object on $\cosh^2\varphi = e^{ds}$; matched-point gaps equal the approximant error exactly, and the light-path delay is the accumulation gap $d\,q_{\rm floor}D = \Lambda_{\rm res} - \Lambda_{\rm obs}$. The tie renames the conservation law for every d ($\Xi\psi = \text{const} \equiv \Xi a^d = \text{const}$) and derives none: the reservoir's two lapses read $d = 1$ or 2, never 3; a volume-measure action with three accumulation directions gives $a = q^{1/3}$ natively and the count $d = \eta_{\rm dir}/\eta_{\rm tot}$ as a declared stiffness isotropy, with shear invisible to the volume law and the floor a declared channel E_χ with weight $E_0/(E_0 + E_\chi)$. Nothing in any action examined fixes d, the floor or the weight.

What this section does not change. The identifications of Section 3 remain selected; the complement remains a complement and not a dark-energy dynamics (Section 11); the ceiling has four exact languages and no law that selects it. What is new is the location of the open question: a continuity current deriving the conserved ruler density from the reservoir action, and a construction that fixes the number of accumulation directions,

would make the ceiling structural. Until then $1/\pi$ is a theme with a test, which is what this chapter has said of it since the register.

19. Results, open closures, and failure conditions

The completed arithmetic establishes exact conditional identities for the clean inventory, its baryonic fraction, the container expression for η, and the thermal conversion to H_0. The formal dimensional cancellation is valid within its fixed angular convention. The finite differences between clean and effective matter rows are explicit. Sensitivity derivatives identify how changes propagate. These are the mathematical closures added here.

Three physical derivations remain central. The baryon allocation needs a measure and a selection theorem for $(1-\chi)$. The abundance needs a mechanism selecting the sixth power and the $7/4$ weight. The cosmological bridge needs a dynamical interpretation of density, critical density, and the complement. None is supplied by repeating an exact decimal match.

Three tests are particularly useful. A frozen BBN calculation tests the abundance outside the original parameter matching, provided reaction rates and their uncertainties are propagated. A single distance and recombination calculation tests whether one expansion-rate ledger can serve all lanes. A specified late-time $w(z)$ or native road law tests the complement's physical interpretation using joint distance and timing observables. Each can fail while the algebra remains correct; that is what makes the physical assignment empirical.

Failure should be recorded at the level of the affected claim. A discrepancy in deuterium could challenge the abundance or thermal assumptions without invalidating $126/72 = 7/4$. A distance mismatch could challenge the expansion history rather than the photon integral. A lack of unique selection would weaken predictive status while leaving a useful phenomenological parametrization. The distinction between arithmetic truth, conditional modeling, and observational adequacy is the organizing discipline of this chapter.

References and provenance

[VIII.1] D. J. Fixsen, "The Temperature of the Cosmic Microwave Background," *Astrophysical Journal* **707**, 916 (2009). https://arxiv.org/abs/0911.1955.

[VIII.2] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* **641**, A6 (2020). https://arxiv.org/abs/1807.06209.

[VIII.3] A. G. Riess et al., "A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s^{-1} Mpc^{-1} Uncertainty from the Hubble Space Telescope and the SH0ES Team," *Astrophysical Journal Letters* **934**, L7 (2022). https://arxiv.org/abs/2112.04510.

[VIII.4] DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," *Physical Review D* **112**, 083515 (2025). https://arxiv.org/abs/2503.14738.

[VIII.5] *SAM Unified Technical Synthesis, Volume I*, Sections 7.5, 8, A.9–A.10; original Chapter VIII (now Chapter IX) and closing review, supplied complete volume dated 19 October 2026. Internal provenance includes CR018, CR019, CR023, CR281; CR003 and CR036 in the thermal-ladder campaign; LC06; and CR004 in the gravitational-wave discrimination map. These records are cited as source provenance, not as newly executed evidence.

SAM VOLUME I

Thermal history, recombination and CMB structure

Chapter X

Abstract. This chapter expands the thermal-history and microwave-background test of the selected SAM inventory. It separates internally implemented hydrogen transport from the standard-physics full-spectrum lane, derives the optical-depth and acoustic bookkeeping, preserves the reported numerical receipts, and corrects their evidential interpretation. The remaining obligations are a converged recombination calculation, a unique drag ruler, a governing perturbation law, and fully specified likelihood comparisons. The native receipt remains a percent-level calculation with documented atomic and time-variable corrections. The full-shape receipts are compatibility checks of a selected packet under inherited transfer physics, rather than independent derivations of that physics. The published second-peak comparator is corrected, and the ACT temperature residual is retained explicitly. New analytical diagnostics distinguish moving recombination endpoints from background-integral changes and show why a visibility peak alone cannot certify the entire ionization history. No historical simulation or likelihood is rerun in this expansion; the numerical execution evidence still required is listed alongside clear validation and failure conditions.

Status, 25 October 2026 (Part 2). This chapter adds the recombination premise and the two rulers, the drag ruler and the sound-horizon ruler, under one background ledger. Its lesson that the ruler is chosen before the comparison is made is the volume's clearest statement that a receiver's weights decide a reading, a point the Reciprocal History work of October 2026 found independently. The native recombination lane and the external full-shape lane answer different questions and stay separate. No run has touched this chapter; the register's T-06 belongs to it.

1. Scope, claims, and the two thermal lanes

The cosmic microwave background tests several parts of a cosmological account at once. Its angular spectra depend on the expansion history, the inertia of the photon–baryon fluid, the ionization history, the gravitational evolution of perturbations, and the projection of those perturbations onto the sky. Agreement with a density fraction alone therefore does not establish agreement with the microwave sky. Conversely, a successful calculation using an established transfer solver is evidence about the supplied parameter packet; it is not automatically a derivation of the transfer physics from the substrate.

This chapter retains the original distinction between two lanes. The *native lane* is the program's internally implemented hydrogen recombination and optical-depth calculation, anchored by the reported record CR001c@19. The word native identifies the implementation and its upstream inventory; it does not mean that the Peebles atomic model or its rate coefficients have been derived from SAM. The *full-shape lane* passes the selected cosmological packet to standard recombination and CAMB. Its reported Planck and ACT comparisons test the propagation of that packet through established thermal and perturbation physics. Both lanes are informative, but they answer different questions.

The numerical receipts reproduced below come from the supplied Volume I draft. The underlying run directories, spectrum arrays, covariance matrices, and executable configurations were not supplied with this chapter's expansion, and no new native execution is claimed. Analytical identities developed here follow from their stated premises. Proposed validation procedures are requirements for future work, not completed tests.

The seven-entry packet is

$$\mathscr{C}_{\rm SAM} = \{\Omega_m, \Omega_b, \Omega_c, H_0, A_s, n_s, \tau_{\rm rei}\}, \qquad \Omega_c = \Omega_m - \Omega_b. \tag{1}$$

Its entries are not seven independent degrees of freedom. The density relation already imposes a constraint, and the expansion calculation adds assumptions about curvature, radiation, neutrinos, and the complementary background component. The reported perturbation selectors are

$$A_s = \eta_{\rm SAM}\sqrt{\mathscr{R}}, \qquad n_s = 1 - \frac{\chi}{2}, \qquad \tau_{\rm rei} = 2A_0. \tag{2}$$

Here $\mathscr{R}$ denotes the upstream radix, avoiding a collision with the baryon-loading variable used below. These are selected identities, not yet solutions of a perturbation-generation equation. Fixing them before a spectrum execution removes continuous retuning during that execution; it does not erase the earlier selection among candidate identities.

2. The background calculation and its premises

2.1 Expansion, closure, and the thermal clock

The background adopted in the original calculation is spatially flat, with nonrelativistic matter, radiation, and a constant-density complement. Under these premises,

$$\begin{gathered} H(a) = H_0 E(a), \qquad \Omega_{\Lambda,\rm bg} = 1 - \Omega_m - \Omega_r, \\ E^2(a) = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_{\Lambda,\rm bg}. \end{gathered} \tag{3}$$

The scale factor is normalized by $a(t_0) = 1$. The clean inventory identity $\Omega_m = 1/\pi$ belongs to the matter/complement partition. Once radiation is present in the Friedmann bookkeeping, inserting $(\pi-1)/\pi$ for the complement without subtracting Ω_r would violate the normalization $E(1) = 1$. This is a small correction numerically, but it is a necessary distinction conceptually.

Equation (3) imports the Friedmann expansion law and a complement with equation of state $w = -1$. A substrate explanation of the complement requires its stress, conservation equation, and perturbative response. Merely naming the complement does not provide those objects. In this chapter they

are assumptions of the comparison model. If a later native law changes $w(a)$, energy exchange, or the relation between redshift and scale factor, the sound horizon and distance must be recomputed rather than retained from this background.

After the relevant entropy-transfer epochs, adiabatically redshifting photons obey $T_\gamma(a) = T_0/a$, with the retained anchor $T_0 = 2.7255\,\mathrm{K}$. The corresponding radiation account is

$$\Omega_r = \Omega_\gamma[1 + 0.2271 N_{\mathrm{eff}}], \quad N_{\mathrm{eff}} = 3.046, \qquad 1 + z_{\mathrm{eq}} = \frac{\omega_m}{\omega_r}, \qquad \omega_i = \Omega_i h^2. \tag{4}$$

The equality formula follows by setting $\rho_m \propto a^{-3}$ equal to $\rho_r \propto a^{-4}$. It assumes that the species counted as radiation are relativistic near equality. A massive-neutrino implementation should use its actual phase-space energy density through the transition rather than simultaneously counting the same species as a fixed radiation term and a fixed matter term. The draft specifies a baseline neutrino mass but does not print its numerical value, hierarchy, or density convention. Those settings belong in the reproducibility manifest.

The simple T_0/a relation should not be extrapolated through every early thermal epoch without accounting for changing entropy degrees of freedom. This matters when Chapter XII connects primordial compact objects to temperatures far above recombination. A reliable recombination clock is not by itself an early-universe formation calculation.

2.2 Baryon loading and two distinct distances

Let $R_b = 3\rho_b/(4\rho_\gamma)$. In tight coupling the baryons supply inertia while photons supply most of the pressure. The resulting adiabatic sound speed and baryon loading are

$$c_s(a) = \frac{c}{\sqrt{3(1+R_b)}}, \qquad R_b(a) = \frac{3\Omega_b}{4\Omega_\gamma} a. \tag{5}$$

The dependence on a is a direct consequence of the different density dilution laws. Increasing baryon loading slows the sound wave while also changing the relative compression and rarefaction amplitudes. Thus an adjustment that repairs peak spacing can disturb peak heights; the same input is used more than once.

The comoving sound horizon, the radial comoving distance in a flat background, and their observable ratio are

$$r_s(a) = \int_0^a \frac{c_s(a')}{a'^2 H(a')}\, da', \tag{6}$$

$$D_M(z) = c\int_0^z \frac{dz'}{H(z')}, \qquad \theta_* = \frac{r_s(a_*)}{D_M(z_*)}, \qquad \ell_A = \frac{\pi}{\theta_*}. \tag{7}$$

Here r_s is a sound horizon, not a Schwarzschild source length. The physical angular-diameter distance is $D_A = D_M/(1+z)$, whereas r_s is comoving. Mixing D_A with a comoving sound horizon would introduce an erroneous factor $1+z$. The expression for r_s follows from a physical propagation distance $c_s dt$ divided by a, with $dt = da/(aH)$.

A useful first-order sensitivity identity is

$$\delta \ln \theta \;=\; \delta \ln r_s \quad \delta \ln D_M. \tag{8}$$

It prevents an erroneous attribution of every acoustic-angle difference to recombination. A change in a_* shifts the endpoints, but a change in $H(a)$ affects both integrals throughout their domains. Background differences therefore survive even after two recombination codes are made identical. Any claim that improved atomic physics will close a particular geometric residual must be demonstrated with matched-background comparisons.

3. Hydrogen recombination as an explicit transport problem

3.1 State variables and equilibrium initialization

Define n_H as the total number density of hydrogen nuclei, ionized or neutral, and $x_e = n_e/n_H$. In a hydrogen-only approximation, charge neutrality gives $n_p = n_e = x_e n_H$ and $n_{1s} \simeq (1-x_e)n_H$. Helium modifies the relation between x_e and the hydrogen ionization fraction; an implementation must say whether helium electrons are included. With a helium mass fraction Y_p, the approximate hydrogen-nucleus density is $n_H = (1-Y_p)\rho_b/m_H$. A change in Y_p affects Thomson opacity even when Ω_b is unchanged.

An equilibrium seed may be obtained from the Saha relation, in a stated degeneracy convention,

$$\frac{x_e^2}{1-x_e} = \frac{1}{n_H}\left(\frac{m_e k_B T}{2\pi\hbar^2}\right)^{3/2} \exp\!\left(-\frac{E_H}{k_B T}\right), \qquad E_H \simeq 13.6\,\mathrm{eV}. \tag{9}$$

Saha equilibrium is an initialization approximation, not a description of the entire recombination transition. Once the reaction time becomes comparable to the expansion time, the ionization history retains memory and must be evolved. Numerically, it is sensible to initialize where equilibrium is accurate and demonstrate that moving the switch to nonequilibrium evolution does not move the final observables beyond the stated numerical tolerance.

3.2 A transparent effective three-level model

One conventional hydrogen-only form of the Peebles effective equation is

$$\frac{dx_e}{dt} = -C\left[\alpha_B(T_m) n_H x_e^2 - \beta_B(T_\gamma)(1-x_e) e^{-E_\alpha/(k_B T_\gamma)}\right], \tag{10}$$

$$C = \frac{1 + K\Lambda_{2s,1s} n_{1s}}{1 + K[\Lambda_{2s,1s} + \beta_B] n_{1s}}, \qquad K = \frac{\lambda_\alpha^3}{8\pi H}. \tag{11}$$

This displayed form states an approximation and its conventions; it is not asserted to be the exact source code of CR001c. The case-B coefficient α_B excludes direct recombination to the ground state. Detailed balance relates β_B to capture into excited states, with a threshold near $E_2 = 3.4\,\mathrm{eV}$. The additional excitation factor contains $E_\alpha \simeq 10.2\,\mathrm{eV}$. At common matter and radiation temperature, their exponential product involves $E_2 + E_\alpha = E_H$, restoring the ground-state equilibrium energy. Rate and degeneracy conventions must be matched consistently when coefficients are imported.

The important lesson is not that every appearance of $3.4\,\mathrm{eV}$ is wrong. That energy belongs in an excited-state photoionization channel. It is wrong when used alone as the net ground-state

ionization threshold. The draft's implementation history identifies precisely this distinction. CR001 established a Saha-seeded optical-depth pipeline. CR001b introduced nonequilibrium evolution but used the excited-state threshold in the net balance and also used a conformal denominator in a proper-time optical-depth integral. CR001c corrected both. These stages should remain in the audit history because they describe changes to the physical calculation, not merely presentation.

The factor C expresses competition between successful arrival in the ground state and reionization from excited states. A minimal model already contains effective Lyman-α escape and two-photon decay. Consequently, the next-stage task is not simply to "add two-photon decay" as though it were wholly absent: it is to document what the effective model contains and then implement the precision corrections and multilevel radiative transfer that it omits. RECFAST and HyRec are reference implementations with different approximation strategies [IX.1–IX.3]. Matching them channel by channel is more informative than changing one phenomenological correction until z_* happens to agree.

3.3 Temperature evolution and a controlled domain

The atomic coefficient depends on the matter temperature T_m, which need not equal T_γ indefinitely. A schematic thermal equation is

$$\frac{dT_m}{dt} = -2HT_m + \Gamma_C(T_\gamma - T_m) + Q_T, \tag{12}$$

where Γ_C is the Compton energy-exchange rate and Q_T represents any separately specified heating per unit temperature-time. Setting $T_m = T_\gamma$ throughout is an approximation whose range must be checked. Adding a substrate heating term without specifying its energy source, spectrum, and coupling would introduce new physics rather than finish an existing numerical calculation. A candidate term must also be propagated to the visibility function and the microwave spectra.

Conservation tests are essential. Hydrogen nuclei must remain conserved; ionization fractions must remain physical; the fully ionized and neutral limits must be stable under refinement; and the stiff integrator must not manufacture negative populations. A solution can appear smooth while violating a balance identity, especially when a large exponential is factored inconsistently. Equilibrium recovery is therefore a unit test of the rate equation before any cosmological comparison is attempted.

4. Optical depth, visibility, and the drag epoch

Define the optical depth remaining between epoch a and today by

$$\tau_\gamma(a) = \int_a^1 \frac{n_e(a')\sigma_T c}{a'H(a')}\,da'. \tag{13}$$

This quantity decreases as a increases. If conformal time is defined by $d\eta = dt/a$, then

$$\frac{d\tau_\gamma}{d\eta} = -an_e\sigma_T c, \qquad g_\eta(\eta) = -\frac{d\tau_\gamma}{d\eta}e^{-\tau_\gamma}. \tag{14}$$

The visibility is a probability density with respect to its specified variable. Under $z = 1/a - 1$, the corresponding density is

$$g_z(z) = e^{-\tau_\gamma(z)}\frac{d\tau_\gamma}{dz} = e^{-\tau_\gamma(z)}\frac{n_e(z)\sigma_T c}{(1+z)H(z)}. \tag{15}$$

Although $g_z\,dz = g_\eta|d\eta|$, the maxima of the two densities need not occur at the same redshift because of the Jacobian. Thus a quoted visibility peak needs an explicit variable convention. The conventional last-scattering epoch associated with a conformal-time visibility peak must be computed and compared in the same convention. This is a reproducibility requirement for the reported native z_*, not evidence that its value has been recomputed here.

The proper-time denominator in Eq. (13) is aH, whereas a comoving propagation distance uses a^2H. Holding $n_e(a)$ fixed, the erroneous optical-depth integrand is larger by $1/a = 1+z$. Around last scattering this is approximately a factor of eleven hundred. This algebraic check explains why the CR001b correction is consequential without claiming a new numerical history.

The normalization supplies another direct check:

$$\int_{\eta_i}^{\eta_0} g_\eta\,d\eta = 1 - e^{-\tau_\gamma(\eta_i)}. \tag{16}$$

It follows by differentiating $e^{-\tau_\gamma}$. An early optically thick boundary gives a result close to unity. Omitting a later reionization contribution changes the late visibility and must be documented separately; $\tau_{\rm rei}$ is not the large optical depth before recombination.

Photon decoupling and baryon drag are distinct. The baryon Euler equation weights momentum exchange by the ratio of photon momentum density to baryon inertia. In the usual approximation a drag optical depth is

$$\begin{aligned} \tau_d(a) &= \int_a^1 \frac{n_e\sigma_T c}{a'H(a')R_b(a')}\,da', \\ \tau_d(a_d) &= 1, \qquad r_d = r_s(a_d). \end{aligned} \tag{17}$$

The integration limits and treatment of late ionization should follow the chosen code convention. Replacing the drag criterion with the photon criterion silently changes the ruler. A precision result must retain both histories and the exact numerical root definition.

5. Reported native results and the single-ruler obligation

The corrected receipt CR001c@19 reports the following values. The comparison column is the original draft's Planck-2018 compressed baseline, not a new fit or an independent measurement of each listed quantity.

Quantity	Native receipt	Comparator	Difference
z_*	1073.063	1089.92	−1.55%
z_d	1055.812	1059.94	−0.39%
$r_s(z_*)$ [Mpc]	143.299	144.43	−0.78%
$D_M(z_*)$ [Mpc]	13532.742	13869.6	−2.43%
$100\theta_*$	1.0589	1.0411	+1.71%
ℓ_A	296.683	301.76	−1.68%
r_d [Mpc]	144.806	147.09	−1.55%

These are useful percent-level diagnostics, but percent-level agreement is not precision-CMB agreement. The native sound-horizon and distance residuals have different signs in their contribution to the angle. Equation (8) explains why a sound horizon only modestly displaced can coexist with an acoustic-angle residual of order one percent. The table does not show that all remaining disagreement resides in atomic recombination; differences in the supplied background and definitions must be separated first.

The original distance packet also quotes $r_d = 147.769\,\mathrm{Mpc}$, whereas the native lane gives 144.806 Mpc. Their difference is 2.963 Mpc, about two percent. They cannot both be used as the same model's uniquely predicted ruler in different data comparisons without a disclosed change of construction. The honest present designation is a native transport ruler and a separate distance-packet ruler. A completed model must choose a physical definition, propagate one value consistently, and explain any approximation connecting the other construction to it.

A controlled comparison should therefore use four calculations: native and reference recombination at the SAM background, followed by the same two calculations at the reference background. The first pair isolates recombination implementation; the second pair tests whether that difference depends on background; comparing backgrounds within one solver isolates the inventory effect. Changing several inputs at once cannot identify the cause of a residual. No convergence toward 147.769 Mpc is promised in advance: the completed atomic calculation may instead show that the distance-packet construction needs revision.

6. From primordial selectors to angular spectra

A common scalar-spectrum convention is

$$\mathscr{P}_{\mathscr{R}}(k) = A_s \left(\frac{k}{k_*}\right)^{n_s-1}, \qquad C_\ell^{XY} = 4\pi \int d\ln k\, \mathscr{P}_{\mathscr{R}}(k)\Delta_\ell^X(k)\Delta_\ell^Y(k). \tag{18}$$

for $X,Y \in \{T,E\}$. This expression separates primordial statistics from transfer physics. It assumes the stated initial-mode content, a pivot k_*, and a transfer convention. CAMB evolves the perturbations and projects them to anisotropy and polarization spectra [IX.4]. The standard linear framework includes coupled species and metric perturbations [IX.5]. Using that framework is legitimate, but it is an inherited dynamical closure of the full-shape lane.

The original packet values are

Parameter	Selected value	Reported centroid offset
A_s	$2.1117473568 \times 10^{-9}$	$+0.392\sigma$
n_s	0.9646322349	-0.064σ
τ_{rei}	0.0530516477	-0.185σ

The centroid comparison is recorded as CR037A. Six alternative amplitude expressions were tested alongside the chosen one: η, $\eta\sqrt{S}$, $\eta\sqrt{\mathscr{R}}/2$, $\eta\sqrt{\Theta}$, $\widehat{d}\eta$, and $\pi\eta$. Thus seven amplitude selectors were examined. A history in which one expression was selected for proximity to a known posterior cannot subsequently be described as an untouched prediction of that posterior.

The useful next theoretical task is to derive a conserved perturbation variable, its evolution equation, its initial state, and its two-point function. A coefficient with the numerical value of A_s is insufficient unless its normalization is tied to $\mathscr{P}_{\mathscr{R}}$ at the stated pivot. Likewise, a tilt formula must explain the scale dependence over the tested interval, and a reionization optical depth must arise from an ionization history or be retained as an empirical selector. Reionization depends on late astrophysical sources; its identification with an upstream constant is a separate physical claim from recombination.

At high multipoles, approximately uniform rescattering suppresses primary power roughly as $A_s e^{-2\tau_{\mathrm{rei}}}$. This exposes a degeneracy: an amplitude comparison alone does not separately validate the amplitude and reionization selectors. Low-multipole polarization and other information are needed to distinguish their roles. A native perturbation account must also specify whether it produces adiabatic or isocurvature modes, Gaussian or non-Gaussian statistics, and a tensor component. These are testable outputs, not optional notation.

7. The retained full-shape receipts

CR037B@19 is reported to have used CAMB 1.6.6, standard recombination, $T_0 = 2.7255\,\mathrm{K}$, $N_{\mathrm{eff}} = 3.046$, a standard pivot, and a fixed baseline neutrino mass, without a cosmological optimizer. Its Planck-PR3 residual statistics are

$$\chi^2_{TT}/N = 1.0339, \quad \chi^2_{TE}/N = 1.0454, \quad \chi^2_{EE}/N = 1.0430. \tag{19}$$

The supplied draft does not provide the complete data-vector sizes, covariance construction, nuisance treatment, or reference-model statistics. Accordingly these are retained comparison statistics, not certified official-likelihood probabilities. The Planck likelihood uses specific spectra, multipole cuts, foreground modeling, and covariance conventions [IX.6]. A plotted residual or a diagonal-error statistic is not interchangeable with that likelihood.

CR037C@19 reports an unchanged packet against ACT DR4 with published windows and full covariance, polarization efficiency fixed at $y_{p2} = 1$, no new parameter, and no Planck files read at execution. The retained values are

ACT component	Reported χ^2/N
TT	1.2996
TE	0.9495
EE	1.0662
Combined	1.1218

ACT provides a valuable cross-instrument comparison with different instrumental systematics and published spectrum products [IX.7]. It is not automatically statistically independent of Planck in every respect: overlapping sky contains common cosmic variance, and a packet selected with Planck knowledge carries that selection history into the ACT test. These qualifications do not erase the test; they define it correctly.

The TT value must remain visible rather than disappear behind the combined number. If N_{TT} is the number of retained TT entries, then its excess relative to the mean of an ideal fixed-model Gaussian χ^2 is $0.2996N_{TT}$. A comparison against a fitted reference instead requires

$$\Delta\chi^2 = \chi^2_{\rm SAM} - \chi^2_{\Lambda{\rm CDM,ref}} \tag{20}$$

on exactly the same data vector and covariance, with an explicit nuisance policy. Neither this difference nor a probability can be recovered from the quoted ratios alone. Unknown N is therefore a substantive missing quantity, not a cosmetic omission.

The original peak-finder history preserves a faulty CR035A specification and a corrected CR035A2 audit root. The reported SAM peaks are $(220,535,812)$. The earlier comparison $(219,520,814)$ must not be retained as the published Planck peak positions. Published Planck values include $(220.0,537.5,810.8)$ [IX.8]. Relative to those central values the second SAM peak differs by about -0.47%. This corrects the transcription comparator; it does not constitute a same-estimator rerun. A fitted peak location in observed bandpowers and an integer maximum of a theoretical spectrum are different estimators. Both spectra must be passed through identical windows, smoothing, and peak definitions before significance is assigned.

CR038@19 is reported to verify exact propagation through CR035A2, CR036, CR036B, CR037A, CR037B, and CR037C. Such propagation checks establish that a packet was not silently altered between stages. They do not independently verify the atomic rates, covariance, or likelihood. Each audit has a domain, and several domains are needed for a complete result.

8. Selection, likelihoods, and a decisive validation program

The original packet lies near the Planck posterior by construction or selection history. Good spectra under the same standard transfer physics are consequently expected to a substantial degree. They remain useful checks of compatibility and implementation, but should not be counted as a wholly independent discovery on top of parameter agreement. A small set of well-motivated discrete choices may eventually be compelling; the relevant evidence requires the allowed grammar, its physical restrictions, the candidates considered, and the information available when each was selected.

For a bandpower vector d and window operator W, a reproducible Gaussian comparison begins with $m = WC(\mathscr{C},\nu)$ and

$$\chi^2 = (d-m)^T\Sigma^{-1}(d-m), \tag{21}$$

where ν contains nuisance quantities. TT, TE, and EE blocks may be correlated, so the combined statistic is not generally the sum of separately inverted block statistics. A complete release should include the ordered data vector, covariance checksum, window convention, multipole range, calibration factors, foreground assumptions, and the treatment of lensing. It must specify whether C_ℓ or $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$ is being convolved. Units and normalization mistakes can otherwise survive a visually plausible plot.

The following completion sequence has definite failure conditions.

1. Freeze the packet, the selector grammar, and all non-packet settings. Record the numerical pivot, neutrino prescription, helium treatment, reionization model, and code versions. A later change is a new model version and must not inherit the earlier test's prospective status.

2. Validate the native ionization solver against equilibrium limits and against a reference at the same background. Refine step size, tolerances, and starting epoch until x_e, z_*, z_d, and both horizons stabilize. Failure of refinement or population conservation is a numerical failure regardless of observational proximity.

3. Publish one physical drag-ruler construction across CMB and distance tests. A discrepancy between constructions is a model inconsistency until its cause is resolved; it is not an uncertainty to be hidden by choosing the favorable ruler in each dataset.

4. Evaluate official Planck and ACT likelihoods with identical frozen cosmology and stated nuisance treatment. Report data counts, log likelihoods, reference differences, and component residuals. Marginalization over instrument nuisance quantities is distinct from cosmological retuning but must still be disclosed.

5. Add independent physical predictions: a full light-element network at the selected baryon abundance, a native perturbation spectrum, and tests of any predicted departures from standard transfer physics. A helium estimate that overshoots is an unresolved result; a larger network is a test, not a guaranteed repair.

A useful error budget separates numerical error, atomic-model truncation, uncertain empirical rate inputs, background assumptions, and selector uncertainty. Tight solver tolerances address only the first. The many digits printed in a selected identity describe arithmetic precision, not demonstrated physical accuracy. Similarly, a percent-level recombination residual cannot be called a precision success merely because it is small on a plot.

The chapter therefore establishes a disciplined intermediate position. The reported packet propagates through standard thermal physics with promising spectrum-level comparisons, and the internally implemented hydrogen lane provides a concrete route for improving the native account. The remaining tasks are sharply defined: derive or explicitly inherit the thermal processes, settle the ruler, expose the likelihood evidence, and replace perturbation selectors with a governing calculation. Until then the full-shape lane is a compatibility result under

established physics and the native lane is a partially completed transport model.

9. Worked diagnostics and reproducible numerical experiments

9.1 Endpoint changes versus changes along the path

A small perturbation of the sound-horizon calculation can be separated into a moving-endpoint term and changes to the integrand. Write $F(a) = c_s(a)/[a^2 H(a)]$. To first order,

$$\delta r_s = F(a_*)\delta a_* + \int_0^{a_*} F(a)\,[\delta \ln c_s(a) - \delta \ln H(a)]\,da. \quad (22)$$

This formula follows directly by varying an integral with a moving upper limit. It is useful because it distinguishes two physically different operations. A recombination-only modification at fixed background moves a_* and changes the visibility width, but it does not change the sound-speed history through a new baryon density. A density modification changes the whole integrand. Their contributions should not be merged into one empirical "recombination correction."

For example, from Eq. (5), a small change in baryon loading produces

$$\delta \ln c_s = -\frac{1}{2}\frac{R_b}{1+R_b}\,\delta \ln R_b. \quad (23)$$

The coefficient tends to zero in the radiation-inertia limit and to $-1/2$ when baryon loading dominates. This is a sign and limiting-case check for numerical sensitivity derivatives. It also demonstrates why a single fractional change in Ω_b does not create a constant fractional change in the sound speed at all times.

A corresponding distance variation contains an endpoint term $c\,\delta z_*/H(z_*)$ and an integral weighted by $-\delta H/H^2$. The two background integrals probe different redshift ranges. Consequently their responses are correlated but not equal. A future error budget should propagate the input covariance through these responses rather than assign unrelated error bars to r_s and D_M. The ratio may be better constrained than either distance separately, or worse constrained under a particular model change; the covariance decides.

9.2 Visibility width and information beyond one epoch

A peak redshift compresses a whole distribution into one number. Two ionization histories can share the same z_* while having different widths or residual ionization tails. Those histories need not have the same diffusion damping or polarization source. A native calculation should therefore publish $x_e(a)$ and the normalized visibility array, not merely its maximum. Useful diagnostics include the mean and variance with respect to the chosen probability measure and the integrated late-time visibility contribution.

This recommendation follows from a simple mathematical fact: fixing the location of a maximum imposes only one local condition on a function. It leaves infinitely many functions with different integrals and widths. Agreement in z_* therefore cannot certify agreement in a line-of-sight integral weighted by that function. The same logic applies to matching a drag epoch without checking the preceding drag history.

A controlled numerical test can deliberately perturb the ionization history in a narrow redshift interval while holding its peak position nearly fixed. Passing that history to a transfer calculation then measures which angular scales respond. Such an experiment diagnoses sensitivity; it does not establish that the perturbation corresponds to valid atomic physics. Physical changes must come from rate equations with a consistent energy and particle account. Keeping these two kinds of experiments separate avoids using a numerical sensitivity kernel as an invented mechanism.

9.3 What a complete execution record must contain

An executable thermal result should be recoverable from a compact manifest rather than from a prose list of successful numbers. At minimum, the manifest needs the input packet at full machine precision, units, code revision, rate-table versions, numerical tolerances, integration interval, transition criteria, and output definitions. It should also retain the solver warnings and failed runs that motivated corrections. A checksum of an output array proves identity with that array; it does not prove the physical correctness of its construction.

Three independent checks are particularly economical. First, integrate optical depth both in a and in z and verify agreement after the analytic change of variables. Second, compute the visibility normalization directly and through the endpoint identity in Eq. (16). Third, calculate the sound horizon in both scale factor and redshift, checking the common integration boundary and units. These are not independent physical models, but they expose common implementation mistakes without requiring observational data.

The matching-reference experiment should retain residual curves rather than only maximum differences. A broad coherent bias suggests a different rate convention or background input; an isolated discontinuity suggests a switch or interpolation artifact; oscillatory differences that shrink with refinement suggest numerical error. These are diagnostic hypotheses to test, not guaranteed classifications. The purpose is to make the next correction attributable and reproducible.

Finally, a failed comparison should have a declared disposition. A numerical failure requires a corrected implementation and a fresh receipt. A mismatch caused by inherited atomic approximations limits the native lane. A discrepancy that remains after matched inputs, converged numerics, and validated atomic physics is evidence against the selected cosmological packet or its assumed background. The chapter becomes stronger when these possibilities are distinguished before the result is known.

10. Worked photon–baryon dynamics

10.1 How an acoustic oscillator emerges

The connection between baryon loading and peaks can be made explicit without pretending to solve the complete microwave sky. Consider a Fourier mode in conformal Newtonian gauge, set $c = 1$ in this subsection, and denote the photon temperature monopole by $\Theta_0 = \delta_\gamma/4$. Let ϑ be the common photon–baryon

velocity divergence in the tight-coupling limit. Choose the convention in which the continuity equation is $\Theta_0' = -\vartheta/3 + \phi'$, with primes denoting conformal-time derivatives and ϕ, ψ the scalar metric potentials. On a short interval where R_b and the potentials can be treated as constant and expansion drag neglected, momentum conservation gives

$$\vartheta' = k^2 \left[\frac{\Theta_0}{1+R_b} + \psi \right]. \tag{24}$$

Combining the two equations produces

$$\Theta_0'' + k^2 c_s^2 \Theta_0 = -\frac{k^2}{3}\psi, \qquad c_s^2 = \frac{1}{3(1+R_b)}. \tag{25}$$

This derivation exhibits the origin of the sound speed used in the background integral. The extra baryonic inertia divides the photon pressure force by $1+R_b$, while the gravitational forcing accelerates the combined fluid.

For initial time η_i, the solution of this constant-coefficient comparator is

$$\begin{aligned} \Theta_0(\eta) = & -(1+R_b)\psi + C\cos[kc_s(\eta-\eta_i)] \\ & + D\sin[kc_s(\eta-\eta_i)]. \end{aligned} \tag{26}$$

The constants C and D encode the initial density and velocity. The locally emitted temperature source contains $\Theta_0 + \psi$, whose equilibrium is $-R_b\psi$. Baryon loading therefore shifts the oscillation centre as well as lowering its frequency. Compression and rarefaction extrema around that shifted centre differ in magnitude. This is the physical reason a baryon-density change affects peak-height patterns, rather than merely moving all peaks horizontally.

Equation (26) is deliberately restricted. In the real calculation R_b changes with time, expansion introduces drag, potentials evolve, neutrinos contribute stress, and tight coupling eventually fails. Those effects modify the phase and amplitude. The analytical oscillator is a unit-level explanation of the response, not a substitute for the hierarchy [IX.9]. The full transfer calculation must retain the missing terms before a residual at a particular multipole can be assigned a cause.

For slowly varying sound speed the accumulated acoustic phase is approximately kr_s. An oversimplified extremum condition would set $kr_s = n\pi$ and project k to $\ell \simeq kD_M$, yielding a spacing near $\pi D_M/r_s = \ell_A$. It does not imply that the first observed peak sits at ℓ_A. Driving, projection, and phase shifts displace the individual extrema. This makes the distinction between an acoustic scale near 300 and a first temperature peak near 220 physically natural. Comparing a peak finder to ℓ_A as though they were the same observable would be a category error.

An instructive perturbation is to increase R_b while holding the potentials and initial displacement fixed. The period grows by the square root of $1+R_b$, but the equilibrium temperature shift grows linearly with R_b. A parameter adjustment that restores the spacing by changing D_M will not generally restore the relative amplitudes. The combined peak-height and peak-spacing information is thus more restrictive than either compressed measurement alone.

10.2 Finite opacity and the damping tail

Tight coupling means that photons scatter often enough for their anisotropic distribution to remain close to the baryon rest frame. It is not exact locking. A finite mean free path permits diffusion and shear, which erase sufficiently short-wavelength perturbations. In the standard tight-coupling expansion, a useful diffusion-scale comparator is

$$k_D^{-2}(\eta) = \int^{\eta} \frac{d\eta'}{6q(1+R_b)} \left[\frac{R_b^2}{1+R_b} + \frac{16}{15} \right], \tag{27}$$

where $q = an_e\sigma_T$ in the $c=1$ convention. The two terms represent the inertia-related slip and radiative-viscosity contributions under the standard polarization treatment [IX.9]. Its use here specifies inherited plasma physics. A native calculation must reproduce or replace it consistently with its own collision operator.

Dimensionally, q is inverse conformal length, so $d\eta/q$ has dimensions of length squared, as required for k_D^{-2}. Lower electron density increases the mean free path and therefore increases the accumulated diffusion length. Approximate Fourier amplitudes acquire a suppression resembling $\exp[-k^2/k_D^2]$ over the regime where this treatment applies. A small change in recombination can consequently matter strongly in the high-multipole tail even when the peak redshift moves little.

The integral also explains why altering helium abundance affects the spectrum without changing the hydrogen binding energy. Helium changes the available free-electron history, hence q, and therefore the diffusion scale. This is an additional reason that z_* alone is not a sufficient benchmark for recombination. The thermal lane should track both acoustic and damping scales, with the same ionization and expansion histories used in each.

11. Two solvable opacity and energy-account checks

11.1 An expanding, constantly ionized toy medium

A solvable opacity model helps distinguish a visibility peak from the condition $\tau = 1$. Consider an interval with $n_e = n_{e0}a^{-3}$, a constant ionized fraction, and matter-dominated expansion $H = H_m a^{-3/2}$. This is a deliberately simplified medium, not a recombination model. Between a and a later endpoint a_f,

$$\tau(a, a_f) = \frac{2n_{e0}\sigma_T c}{3H_m}\left(a^{-3/2} - a_f^{-3/2} \right). \tag{28}$$

The result follows by integrating $a^{-5/2}$ in the proper-time expression. It tests both the power of scale factor and the endpoint sign. Optical depth is nonnegative for $a < a_f$ and vanishes at the upper endpoint, as required.

In this toy model the conformal opacity rate is $q = an_e\sigma_T c \propto a^{-2}$. Since $a' = a^2 H$, differentiation gives

$$\frac{d\ln g_\eta}{d\eta} = -2aH + q. \tag{29}$$

An interior visibility maximum therefore occurs when $n_e\sigma_T c = 2H$. If the later endpoint is far enough away to neglect $a_f^{-3/2}$,

Eq. (28) gives $\tau \simeq 4/3$ at that maximum, rather than exactly one. This is not a correction to the real recombination epoch. It is an explicit counterexample to the assertion that a visibility maximum must always coincide with $\tau = 1$.

Real recombination changes x_e rapidly, adding a large derivative to q and shifting the condition. The toy solution nevertheless supplies a useful integrator test because every term can be evaluated analytically. A code should reproduce it before handling stiff atomic transitions. A discrepancy in this test points to time-variable conversion, normalization, or quadrature, not to missing multilevel atomic physics.

11.2 Photon heating requires both energy and number accounting

The retained thermal clock assumes adiabatic radiation. If a new source injects energy into the photon bath, introduce an energy-density transfer rate Q_γ and a number-density transfer rate S_γ. The homogeneous balance equations are

$$\dot{\rho}_\gamma + 4H\rho_\gamma = Q_\gamma, \qquad \dot{n}_\gamma + 3Hn_\gamma = S_\gamma, \tag{30}$$

where ρ_γ denotes energy density in this subsection. For a blackbody, $\rho_\gamma \propto T^4$ and $n_\gamma \propto T^3$. If the spectrum remains exactly blackbody, these two equations require

$$\frac{\dot{T}}{T} = -H + \frac{Q_\gamma}{4\rho_\gamma} = -H + \frac{S_\gamma}{3n_\gamma}. \tag{31}$$

Thus an arbitrary heat source cannot be added while retaining both $T = T_0/a$ and a fixed blackbody number density. The energy and photon-number rates must satisfy the compatibility relation visible in Eq. (31), or the distribution requires an additional spectral-distortion description. This conclusion follows from the two blackbody scaling laws and conservation; it does not assume a particular injection mechanism.

A transfer between substrate sectors also requires an opposite payment in the donor account. For homogeneous components i, equations of the form $\dot{\rho}_i + 3H(\rho_i + p_i) = Q_i$ use energy units with $c = 1$ and must satisfy $\sum_i Q_i = 0$ when the complete closed system is included. A phenomenological $Q(z)$ inserted only on the receiving side would violate that ledger. Conversely, a microscopic source with an extremely slow radiative rate cannot be declared the origin of an appreciable cosmological Q without calculating the number of sources, their state distribution, available energy, and duration. A rate from one preparation is not automatically a cosmic heating law.

This accounting connects Chapters X and XII. A formation or evaporation proposal may inject photons, particles, or entropy. Its effects must be propagated into the background, ionization history, primordial abundances, and radiation spectrum. Holding all those quantities fixed while adding an unbudgeted injection would double-count the original thermal anchor. The purpose of the balance equations is to make any proposed extension calculable before it is assessed observationally.

12. A small covariance example with a large interpretive lesson

Consider two residual bandpowers of equal magnitude r with covariance $\Sigma = \sigma^2 \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, where $|\rho| < 1$. Direct inversion gives

$$\chi^2_{\text{same}} = \frac{2r^2}{\sigma^2(1+\rho)}, \qquad \chi^2_{\text{opposite}} = \frac{2r^2}{\sigma^2(1-\rho)}. \tag{32}$$

The first expression applies to residuals (r, r), the second to $(r, -r)$. Both have the same pointwise error bars and the same sum of squared standardized residuals if correlations are ignored. For $r = \sigma$ and $\rho = 0.8$, however, their proper statistics are about 1.11 and 10. Correlations make a common-mode deviation inexpensive and an opposing deviation expensive in this example.

This elementary result explains why a near-unity average of pointwise residual squares cannot replace a full covariance likelihood. Calibration and foreground uncertainties can introduce structured directions in data space. Marginalizing a nuisance parameter that shifts both points together may further reduce sensitivity to the common residual while leaving the opposing residual prominent. A model comparison must identify which directions were fitted, integrated over, or held fixed.

The example also separates two uses of a reference cosmology. A frozen reference produces a residual statistic under exactly the same nuisance policy as the SAM packet. An optimized reference has spent additional freedom to lower its statistic. Both comparisons may be useful, but they answer different questions. Reporting the raw difference, the optimization or marginalization policy, and the parameter dimension is more informative than attaching the phrase "no optimizer" to one model alone. No two-point toy calculation supplies the missing Planck or ACT likelihood; it demonstrates why their complete definitions are necessary.

13. Separating numerical accuracy from physical model accuracy

A practical thermal calculation needs a reproducible criterion for deciding when refinement has finished. Let O_h denote an observable computed with a maximum step size h, and suppose a controlled numerical regime admits $O_h = O_* + Ch^p + o(h^p)$. Then two refinements give

$$O_{h/2} - O_h \simeq Ch^p(2^{-p} - 1). \tag{33}$$

With three resolutions one can test whether the observed difference ratio approaches 2^p, rather than merely assuming the integrator's formal order applies through switches and stiffness. If the expected scaling holds, the difference supplies an error estimate. If it fails, the solution may be limited by interpolation, a discontinuous switch, inaccurate rates, or a root-finding tolerance. Those causes require separate checks.

For a visibility maximum, the peak-location error is not controlled solely by the error in g. A broad or flat peak can move appreciably under a small function perturbation. Locally,

if $g'(\eta_*) = 0$ and $g''(\eta_*) \neq 0$, a perturbation gives

$$\delta\eta_* \simeq -\frac{\delta g'(\eta_*)}{g''(\eta_*)}. \tag{34}$$

This follows by linearizing the stationary-point condition. It motivates refining the interpolation and derivative calculation, not simply sampling more points on a plot. The selected conformal-time maximum should then be converted to redshift using the same background map. Maximizing a redshift density instead would change the observable, as discussed earlier.

A drag-epoch root has a similar conditioning test. Linearizing $\tau_d(a_d) = 1$ gives $\delta a_d = -\delta\tau_d(a_d)/\tau_d'(a_d)$. Since the derivative is determined by the positive integrand with a negative lower-limit sign, its magnitude identifies how an opacity error propagates into the epoch. These formulas help allocate numerical tolerances according to the final observable rather than assigning identical relative tolerances to unrelated intermediate variables.

Even perfect numerical convergence leaves the physical approximation intact. A converged hydrogen-only effective atom remains hydrogen-only. A converged inherited perturbation hierarchy remains inherited. A converged likelihood evaluated with an incorrect foreground model remains an evaluation of that incorrect model. The value of a numerical certificate is that it isolates these questions, allowing a remaining discrepancy to be attributed to modeling rather than unresolved arithmetic. It should never be used to suggest that a selected physical interpretation has been proved by additional digits.

14. Sources and provenance

[IX.1] P. J. E. Peebles, "Recombination of the Primeval Plasma," *Astrophysical Journal* **153**, 1 (1968), doi:10.1086/149628.

[IX.2] S. Seager, D. D. Sasselov, and D. Scott, "A New Calculation of the Recombination Epoch," *Astrophysical Journal Letters* **523**, L1 (1999), astro-ph/9909275; see also the authors' RECFAST release page.

[IX.3] Y. Ali-Haïmoud and C. M. Hirata, "HyRec: A fast and highly accurate primordial hydrogen and helium recombination code," *Physical Review D* **83**, 043513 (2011), doi:10.1103/PhysRevD.83.043513.

[IX.4] A. Lewis, A. Challinor, and A. Lasenby, "Efficient Computation of CMB Anisotropies in Closed FRW Models," *Astrophysical Journal* **538**, 473 (2000), astro-ph/9911177; CAMB project documentation.

[IX.5] C.-P. Ma and E. Bertschinger, "Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges," *Astrophysical Journal* **455**, 7 (1995), astro-ph/9506072.

[IX.6] Planck Collaboration, "Planck 2018 results. V. CMB power spectra and likelihoods," *Astronomy & Astrophysics* **641**, A5 (2020), publisher article; "VI. Cosmological parameters," **641**, A6, doi:10.1051/0004-6361/201833910.

[IX.7] S. K. Choi et al., ACT DR4 power-spectrum analysis, *JCAP* **12**, 045 (2020); NASA LAMBDA ACT DR4 spectrum products and documentation.

[IX.8] Planck Collaboration, "Planck 2015 results. I. Overview of products and scientific results," *Astronomy & Astrophysics* **594**, A1 (2016), doi:10.1051/0004-6361/201527101. The peak comparator is used as published; identical-estimator reanalysis remains outstanding.

[IX.9] W. Hu and N. Sugiyama, "Small-Scale Cosmological Perturbations: An Analytic Approach," *Astrophysical Journal* **471**, 542 (1996), astro-ph/9510117. Used for the standard acoustic and tight-coupling comparator.

Internal provenance. Original Chapter IX (now Chapter X), upstream inventory in Chapter IX, and the supplied closing review of 19 October 2026. Historical record identifiers are CR001, CR001b, CR001c, CR035A, CR035A2, CR036, CR036B, CR037A–C, and CR038. Their reported results are preserved as documentary claims, with the missing execution evidence and outstanding tests stated above.

SAM VOLUME I

Carrier radiation and ringdown dynamics

Chapter XI

Abstract. The carrier partition supplies three compact proposals: a radiated-energy ceiling of one eighth of an initial source budget, a fundamental frequency coefficient $3/8$, and a damping coefficient $4/45$. Their arithmetic is exact, but their interpretation as gravitational radiation requires a source-energy map, a physical unit chart and a causal field boundary condition. This chapter proves the ceiling under explicit reservoir premises, derives an illustrative outgoing-flux identity, and formulates the quasinormal-mode problem with a frequency-dependent surface response. It distinguishes the rational matches from the Schwarzschild spectrum, includes the weak-field binary-radiation obligation, and explains why spin, excitation and mass inference matter for observation. A worked finite pulse and independent arithmetic checks clarify which results are newly established analytically and which remain source-record reports. The same boundary law must ultimately account for the thermal behavior of Chapter VIII and the perturbative response studied here.

Status, 25 October 2026 (Part 2). This chapter adds the carrier premise: the released carrier account as a radiation law, the ringdown rationals $\hat{\omega}_R = \Theta\hat{d}/R^2 = 3/8$ and $\hat{\gamma} = R/(L-V) = 4/45$ (register T-08) and the radiated-energy ceiling $\Theta/R^2 = 1/8$ (T-09), the last by the same uniform count-equals-energy weighting that Volume II Chapter XVII uses. A boundary law $R(\omega)$ is the open operand. No run has touched this chapter.

1. Radiation as an interface between the source and the exterior

The earlier chapters give the carrier sector a precise bookkeeping role. With radix $R = 12$, the closed container has $R^2 = 144$ units, of which $\Theta = 18$ are released and $M_{\mathrm{count}} = 126$ are retained. The original notation used M for the retained count and also for a gravitating mass. Here the count receives a subscript; M_f denotes the final compact-object mass. The distinction prevents an exact integer identity from silently becoming a dimensional energy statement.

The working interpretation is that a disturbance of the released sector can propagate away from a changing source. The original chapter describes a merger as exposing carriers before the common boundary closes. An isolated, perfectly spherical formation would have no analogous time-dependent quadrupole in the general-relativistic comparator. These are different statements. The first proposes a microscopic mechanism; the second follows from the symmetry of a particular gravitational theory. A useful carrier theory must specify which source variables excite its outgoing modes and explain the symmetry suppression without relying on a verbal picture alone.

Several pieces of the required interface already exist elsewhere in the volume. Chapter VII provides the source–field action and its two receiving polarizations. Chapter IV retains near-field, induction and radiative contributions in a finite-pulse energy account. Chapter V supplies a causal boundary protocol with surface storage. Chapter VIII asks how the same boundary exchanges energy and information. None of these ingredients should be discarded when constructing a radiation model. Conversely, their existence does not establish that the field has the tensor transformation law, source coupling and asymptotic normalization of gravitational waves.

This chapter therefore has two levels. The first records the original numerical proposals, including the $1/8$ energy fraction and the pair $3/8$, $4/45$. The second asks what action, boundary response and measurement procedure would make those proposals physical predictions. A derivation at the second level must identify the clock, ruler and energy calibration inherited from the earlier source problem. It must also survive the weak-field radiation tests that precede any strong-field ringdown comparison.

2. The count ledger and its exact content

The relevant identities are

$$\begin{aligned} R^2 = 144 = 8\Theta, \qquad & M_{\mathrm{count}} = 126 = 7\Theta, \\ L = 162 = 9\Theta, \qquad & V = 27 = \widehat{d}^3, \\ S = 8, \qquad & \widehat{d} = 3. \end{aligned} \tag{1}$$

The declared closure rule gives

$$R^2 = S\Theta, \qquad S = \widehat{d}^{\,\widehat{d}-1} - 1. \tag{2}$$

For the particular values in (1), the proposed coefficients reduce to

$$\alpha_R = \frac{\Theta\widehat{d}}{R^2} = \frac{\widehat{d}}{S} = \frac{3}{8}, \tag{3}$$

$$\alpha_I = \frac{R}{L-V} = \frac{12}{135} = \frac{4}{45}, \tag{4}$$

$$L - V = 135 = R^2 - \widehat{d}^2. \tag{5}$$

These reductions require no numerical approximation. They establish that several count expressions represent the same rational numbers. They do not identify an eigenfunction, a differential operator, an excitation amplitude or an energy-flux normalization. The distinction is especially important because a rational number may be exact as arithmetic while only approximate as a representation of an independently computed frequency.

The original record CR005 reports 26 exact checks of the ringdown identities. This chapter preserves that report as source provenance; it does not claim to have rerun that record. Similarly, the proposed higher-mode table in CR003b is a retained

research lead rather than an independently verified spectrum. A complete expression grammar would need to specify admissible counts, operations, differences, exponent ranges and a selection rule before the comparison target is inspected.

A role restriction can be scientifically meaningful. For example, requiring a damping coefficient to use a boundary leakage count rather than an unrelated geometric constant may follow from an action. Until that restriction is derived, however, rejecting a close alternative because it uses the wrong "role" is a model-selection choice. The restriction should be published with the candidate, not introduced after seeing which expression is closest.

3. An energy ceiling: theorem, assumptions and scope

3.1 The conditional budget statement

Let E_0 be a declared initial total energy in an asymptotically meaningful accounting convention. Suppose a source class satisfies all of the following premises:

1. Its available carrier reservoir initially obeys $0 \le E_C(0) \le E_0/8$.
2. No retained-sector transfer, external work or incoming radiation replenishes that reservoir during the interval under study.
3. The outgoing gravitational channel is paid entirely from this reservoir.
4. The remaining reservoir, stored boundary energy and other debited outgoing channels are nonnegative in the same reference convention.

Then energy conservation gives

$$E_C(0) = E_C(T) + E_{\mathrm{GW}}(0,T) + E_{\mathrm{other}}(0,T) + \Delta E_{\mathrm{store}}, \qquad (6)$$

where the last three terms apart from E_{GW} are assumed nonnegative in the stated ledger. Consequently,

$$0 \le E_{\mathrm{GW}}(0,T) \le E_C(0) \le \frac{E_0}{8}. \qquad (7)$$

This is a genuine conditional implication. Its proof is subtraction of nonnegative debits from a finite initial reservoir. The physically difficult step is the first premise: a count fraction must be shown to equal an energy fraction for the actual prepared source, including interaction and boundary energy. Counting 18 labels among 144 labels does not prove that their mean energies are equal.

If replenishment is allowed, the correct inequality instead contains its payment:

$$E_{\mathrm{GW}}(0,T) \le E_C(0) + \int_0^T P_{\mathrm{feed}}(t)\,dt, \qquad (8)$$

provided the other debits remain nonnegative. Omitting this term could make an apparent violation merely a change of source class. For that reason a ceiling must declare whether it covers a complete inspiral and merger, only the late merger, or a finite pulse driven by an external controller. Those are not interchangeable experiments.

For a binary initially at large separation, a conventional comparison uses $E_0 = (m_1 + m_2)c^2$ with an explicit treatment of initial kinetic and binding energy. An energetic encounter with substantial incoming kinetic energy cannot automatically be normalized by the rest-mass sum. For a numerical simulation started at finite separation, the initial total energy is the appropriate comparison quantity, with the radiation already emitted before the chosen start accounted for separately. The final mass appearing in ringdown is a different mass again.

3.2 How the original catalog comparison should be read

The source chapter reports a historical selection of sixteen events below 12.5%, with a largest tabulated fraction of 5.98%. Its displayed examples are retained here as the original comparison, not as a new catalog extraction:

Event	$M_{\mathrm{initial}}/M_\odot$	$E_{\mathrm{rad}}/(M_\odot c^2)$	Fraction
GW150914	65.3	3.1	0.0475
GW170729	80.3	4.8	0.0598
GW170817	2.74	0.025	0.0091
GW190521	150	7.6	0.0507
GW200115	6.6	0.025	0.0038

The printed event-class tally in the working draft mixes overlapping categories and does not provide a mutually exclusive sixteen-event list. It should not be used to infer population frequencies. Neutron-star systems also have matter, neutrino and electromagnetic channels; a remnant mass deficit is not automatically all gravitational radiation. A reproducible test must return to event posteriors, document the waveform assumptions and preserve correlations among initial masses, final mass and radiated energy.

Even a perfectly verified set of events below the ceiling would establish consistency, not a measured ceiling. A bound far above all observed values is weakly tested. To reject the proposal, one needs a source demonstrably inside its declared domain with robust support above the bound. A useful statistic is the posterior probability

$$p_> = \Pr(E_{\mathrm{GW}}/E_0 > 1/8 \mid d, \mathscr{M}), \qquad (9)$$

reported together with the likelihood model $\mathscr{M}$, calibration uncertainty, selection effects and alternative waveform families. A strong rejection cannot be based solely on a point estimate obtained under a model whose radiation law is itself under test.

3.3 The general-relativistic comparator

The classical area theorem gives a different kind of bound. Assume two initially nonspinning holes of masses m_1, m_2, initial total ADM energy no greater than $(m_1+m_2)c^2$, no incoming or externally supplied energy, and nonnegative other outgoing energy channels. Then the initial horizon areas are proportional to $m_1^2 + m_2^2$. Under the theorem's classical assumptions, the final irreducible mass obeys $M_{\mathrm{irr},f}^2 \ge m_1^2 + m_2^2$, and $M_f \ge M_{\mathrm{irr},f}$.

Hence

$$\frac{E_{\rm GW}}{(m_1+m_2)c^2} \le 1 - \frac{\sqrt{m_1^2+m_2^2}}{m_1+m_2}. \tag{10}$$

For equal masses this is $1-1/\sqrt{2} \simeq 0.2929$ [XI.1]. This is an upper bound, not the efficiency of a typical merger. It also uses a different set of premises from the carrier-reservoir argument. A value allowed by the area theorem but forbidden by (7) would discriminate only if the carrier assumptions were independently fixed.

The original draft also refers to numerical efficiencies of order ten percent for particular high-spin configurations. No universal numerical-relativity maximum is asserted here: spin, mass ratio, orbital state and incoming energy change the question. Likewise, the fact that isolated spherical collapse is quiet is not unique evidence for the carrier mechanism. A quantitative discriminator would be a predicted amplitude or angular dependence for a nonspherical source, with a well-defined comparison to the established quadrupole limit.

4. What the two-polarization action must supply

The Original Write construction gives two real receiving fields, ϕ_+ and $\phi_\times$, with receiving normalization $\kappa = 4/5$, boundary source terms proportional to $\varepsilon_R J_a \phi_a(R)$, and a counterterm whose cited radial coefficient is $\kappa/4 = 1/5$. The source labels are $J_+ = D$ and $J_\times = C$. Both polarizations survive the circular decomposition. These ingredients identify an interacting model, but the original source and its full energetic complement must remain present when radiation work is computed.

For orientation, consider a separately declared linear comparator action for multipoles $\psi_a(t,x)$,

$$I_{\rm cmp} = \frac{1}{2}\sum_a \int dt\,dx\,\big[(\partial_t\psi_a)^2 - (\partial_x\psi_a)^2 - V_a(x)\psi_a^2\big] + I_{\rm surf} + I_{\rm drive}. \tag{11}$$

This is a mathematical template, not a derivation of the native action. Its bulk equation is

$$\partial_t^2\psi_a - \partial_x^2\psi_a + V_a\psi_a = J_a. \tag{12}$$

For time-independent real V_a, define

$$e_a = \tfrac{1}{2}\left[(\partial_t\psi_a)^2 + (\partial_x\psi_a)^2 + V_a\psi_a^2\right], \qquad j_a = -\partial_t\psi_a\partial_x\psi_a. \tag{13}$$

Multiplying (12) by $\partial_t\psi_a$ and applying the product rule gives

$$\partial_t e_a + \partial_x j_a = J_a\partial_t\psi_a. \tag{14}$$

An outgoing wave $F(t-x)$ has $j = (F')^2 \ge 0$. Integration over a finite interval displays boundary work explicitly; adding the surface-energy equation is what produces a closed balance. Positive outward flux alone does not establish the global $1/8$ bound, because the integrated source work still has to be paid from the stated reservoir.

The normalization of (13) is conventional. To compare with gravitational strain, a physical map from ψ_a to dimensionless strain h_a and a conversion to seconds, metres and joules are required. In the general-relativistic far-zone comparator, the energy flux involves time derivatives of the transverse-traceless strain with its c^3/G normalization. That normalization must be recovered by a carrier theory, rather than assigned solely because the field has two components.

There is also a representation-theory obligation. Two scalar channels can carry two numbers without behaving as the helicity-two polarizations measured by a detector network. The theory must establish the field's transformation under rotations, its coupling to conserved stress, the absence or suppression of extra propagating modes, and its detector response. The two-polarization action is a promising starting point, not a substitute for these tests.

5. The weak-field radiation exam comes first

A successful strong-field proposal must also account for energy loss from slowly moving binaries. In the general-relativistic comparator, the leading circular-orbit luminosity is

$$P_{\rm GR} = \frac{32}{5}\frac{G^4\mu^2 M_T^3}{c^5 a^5}, \qquad M_T = m_1+m_2, \quad \mu = \frac{m_1 m_2}{M_T}, \tag{15}$$

where a is the binary separation. This formula is cited as a benchmark [XI.2], not as a result of the native source action. Combining it with $E_{\rm orb} = -G\mu M_T/(2a)$ gives

$$\dot a = -\frac{64}{5}\frac{G^3\mu M_T^2}{c^5 a^3}. \tag{16}$$

Using Kepler's relation $P_b \propto a^{3/2}$ then yields $\dot P_b/P_b = 3\dot a/(2a)$. Eccentric binaries require the corresponding orbit-averaged enhancement and eccentricity evolution; replacing them by a circular estimate is inadequate for precision pulsar work.

The derivation a native theory owes is not simply the algebra after (15). It is the luminosity itself, including its coefficient, multipolar structure and conservation law. Dipole emission, additional polarizations or a modified propagation speed can enter before the ringdown regime is reached. A carrier-reservoir ceiling does not constrain these effects strongly enough. The closing review correctly identifies binary-pulsar radiation as a separate missing interface.

A sensible calculation begins with a translation-invariant moving-source action, expands its conserved source in velocity, obtains the retarded field, and integrates the outgoing energy current over a distant sphere. It must show why lower radiative multipoles vanish or are unobservable, and how orbital energy loss agrees with the outgoing work. Only then can one state whether the proposed carrier mechanism recovers the weak-field radiation law or predicts a correction to it.

6. The ringdown candidates and their numerical comparison

Use the time dependence $\exp(-i\omega t)$ and write

$$\omega = \omega_R - i\gamma, \qquad \gamma > 0, \qquad \widehat\omega = \frac{GM_f}{c^3}\omega. \tag{17}$$

The proposed fundamental values are $\omega_R = 3/8$ and $\gamma = 4/45$. With the Schwarzschild gravitational quadrupole comparator

$0.37367168 - 0.08896232i$, their fractional residuals are approximately $+0.3555\%$ and -0.0825%. These are residuals relative to a theoretical numerical value; they are not observational standard deviations [XI.3].

The quality factor and physical time scales are

$$Q = \frac{\omega_R}{2\gamma} = \frac{135}{64} = 2.109375, \tag{18}$$

$$f_R = \frac{3c^3}{16\pi G M_f}, \qquad \tau = \gamma^{-1} = \frac{45GM_f}{4c^3}. \tag{19}$$

The comparator quality factor is about 2.1002. The factors of 2π and the distinction between amplitude damping time and energy damping time are essential: an amplitude proportional to $e^{-t/\tau}$ has energy proportional to $e^{-2t/\tau}$. Mixing these conventions can create an apparent factor-of-two agreement or disagreement.

The count identity for one higher example, $(L+\widehat{h})/L = 164/162$ with $\widehat{h} = 2$, was compared in the working draft with the real part of the $\ell = 5$, $n = 0$ Schwarzschild mode. It remains an individually selected candidate. A spectrum requires a rule generating both real and imaginary parts for all specified (ℓ, m, n), with the same action and boundary law. Separate rational fits to each entry are not equivalent to an eigenvalue calculation.

The historical "wrong controls" used R/R^2, R/L, R/M_{count}, $\widehat{d}/(L-V)$ and $\Theta/(L-V)$, with substantially worse damping matches. They demonstrate that the particular expression is not interchangeable with these few neighbors. They do not count all expressions examined during development. The closing review's larger enumeration found several alternatives under wider grammars. Its tolerance windows for the ringdown numbers were chosen comparison bands, not measured 1σ errors. Both the grammar and this distinction belong beside any future significance claim.

7. A well-posed quasinormal-mode problem

7.1 The exterior operator

In the Schwarzschild adapter, set $G = c = 1$, so M_f has dimensions of length and time. The tortoise coordinate is

$$x = r_* = r + 2M_f \log\left(\frac{r}{2M_f} - 1\right). \tag{20}$$

For axial gravitational perturbations, the standard comparison potential is

$$V_\ell^{\text{RW}}(r) = \left(1 - \frac{2M_f}{r}\right)\left[\frac{\ell(\ell+1)}{r^2} - \frac{6M_f}{r^3}\right], \qquad \ell \geq 2. \tag{21}$$

The frequency-domain equation is

$$\frac{d^2\psi_\ell}{dx^2} + \left[\omega^2 - V_\ell(x)\right]\psi_\ell = 0. \tag{22}$$

Writing this equation with V_ℓ^{RW} explicitly adopts the gravitational exterior adapter. The carrier theory must derive its own potential, or explain why its linearized equations reduce to this one. The polar sector supplies an additional consistency check; standard Schwarzschild axial–polar spectral relations depend on the field equations and boundary conditions and need not survive an altered surface response.

7.2 Boundary conditions are part of the theory

At spatial infinity an outgoing mode has $\psi \sim e^{+i\omega x}$. A standard absorbing horizon at $x \to -\infty$ has $\psi \sim e^{-i\omega x}$. These two requirements define a non-self-adjoint resonance problem [XI.3]. They are not ordinary standing-wave boundary conditions in a closed cavity. A numerical implementation must handle the exponential spatial growth of a damped quasinormal mode under analytic continuation, rather than mistake it for a physical instability.

For a surface at finite x_s in a locally free region where the potential is negligible, write locally

$$\psi = A_{\text{in}} e^{-i\omega(x-x_s)} + A_{\text{out}} e^{+i\omega(x-x_s)}, \qquad \mathscr{R}(\omega) = A_{\text{out}}/A_{\text{in}}. \tag{23}$$

A corresponding impedance condition is

$$\left.\frac{\psi'}{\psi}\right|_{x_s} = i\omega\frac{\mathscr{R}(\omega) - 1}{\mathscr{R}(\omega) + 1}, \tag{24}$$

away from zeros of the denominator. This formula follows directly by differentiating the two local waves. It makes the missing datum visible: "no completed crossing" does not determine $\mathscr{R}$. Absorption, reflection and temporary storage are different realizations of a boundary account.

For a passive nonrotating surface in a single propagating channel, real-frequency reflected flux cannot exceed incident flux, so $|\mathscr{R}| \leq 1$. A causal response also constrains the frequency dependence through its time-domain kernel and analyticity. The field may couple to several surface degrees of freedom; eliminating them generally produces a frequency-dependent matrix impedance mixing polarizations. A constant reflectivity is then a phenomenological approximation whose band of validity must be stated.

A minimal storage example makes the point. Let a surface oscillator obey $\ddot{q} + 2\Gamma_s\dot{q} + \Omega_s^2 q = g\psi(x_s, t)$, and couple its response back to the boundary derivative. Its Fourier response contains $(\Omega_s^2 - \omega^2 - 2i\Gamma_s\omega)^{-1}$. Even this simple system introduces resonant memory and absorption. The native theory must fix such frequencies and couplings from its surface action; freely choosing them to reproduce $3/8$ and $4/45$ would fit the desired answer rather than derive it.

8. What a reproducible calculation would establish

A spectral calculation should freeze the action, background, surface location, impedance and unit convention before searching for roots. One then constructs an inner solution satisfying the surface law and an outer solution satisfying the outgoing condition. Their Wronskian at a matching point,

$$W(\omega) = \psi_{\text{in}}\psi'_{\text{out}} - \psi_{\text{out}}\psi'_{\text{in}}, \tag{25}$$

vanishes at a mode. Rescaling either solution changes the normalization of W but not its zeros. Continued-fraction, contour,

spectral or direct-integration methods can be used, provided the chosen method treats the actual boundary law [XI.4].

Verification should include an absorbing Schwarzschild benchmark, changes of matching radius, precision and truncation, and a search for missed or unstable roots. A small value of one residual at one numerical resolution is not a convergence certificate. For a frequency-dependent boundary, poles of the impedance must be distinguished from coupled-system modes. If the eigenproblem is reduced by eliminating surface variables, roots introduced by multiplying denominators must be checked in the unreduced system.

The rational proposal is tested by evaluating whether a converged root equals, approximates or excludes $3/8-4i/45$ at a declared tolerance. If the exterior and both endpoint conditions are exactly the Schwarzschild ones, the resulting spectrum is the Schwarzschild spectrum; the nonzero residuals quoted above remain. An exact absorbing adapter cannot simultaneously make the unequal rational numbers exact eigenfrequencies. They may be a useful compact approximation, but that is a different claim.

If the boundary stores and returns energy, the spectrum and the time-domain waveform can change. A near-horizon surface at $r_s = 2M_f(1+\varepsilon)$ has $x_s \simeq 2M_f \log\varepsilon$ plus finite terms. The approximate round-trip delay between that surface and an exterior potential barrier is

$$\Delta t_{\rm echo} \simeq 2(x_{\rm barrier} - x_s) \sim 4M_f|\log\varepsilon|, \tag{26}$$

when $\varepsilon \ll 1$. This is a geometric comparator estimate, not a native prediction for ε or an assertion that an echo must be observable. Amplitude, dispersion, absorption and excitation determine whether any delayed signal survives [XI.5].

9. Spin, excitation and the detector-facing claim

Astrophysical merger remnants generally rotate. The relevant dimensionless spin is $\chi_f = cJ_f/(GM_f^2)$, and a Kerr comparison uses mode functions $\widehat{\omega}_{\ell mn}(\chi_f)$. A Schwarzschild coefficient describes the zero-spin limit. It cannot be directly fitted to a generic binary remnant without a spin map. The source review's statement that every observed remnant has one particular spin is too strong; the correct requirement is event-specific mass and spin inference with their uncertainties.

Spin is not just a correction to one number. It changes angular eigenfunctions, mode splitting, damping, coupling between labels and the meaning of surface passivity in a rotating frame. A native continuation must specify the angular momentum ledger and its flux along with the energy ledger. If a surface exchanges rotational energy, a reflected amplitude greater than unity in an exterior channel need not represent a violation of total energy conservation; the rotation reservoir must then be included explicitly.

An observed ringdown is also not the bare list of poles. A schematic strain is

$$h(t) = \sum_{\ell mn} A_{\ell mn} e^{-(t-t_0)/\tau_{\ell mn}} e^{-i[\omega_{R,\ell mn}(t-t_0)+\varphi_{\ell mn}]} + h_{\rm tail}(t), \tag{27}$$

with angular projection and detector response understood. The excitation amplitudes and phases depend on the merger, while t_0 controls the interval on which a linear approximation is used. Early starts gain signal but risk nonlinear contamination; late starts reduce contamination but lose signal-to-noise. Overtone studies and catalog tests illustrate why start-time and waveform choices must be reported [XI.6, XI.7]. No new fit to those data is claimed here.

A useful prospective test would publish a single frozen boundary law, its mass–spin spectrum, the expected source classes and the waveform family. It would then compare held-out events or a held-out portion of a numerical waveform, with detector calibration and noise uncertainty propagated. A family with one freely tuned boundary parameter per event would be substantially less predictive than a universal law tied to the source counts.

10. A worked finite-pulse account

An explicit pulse illustrates the difference between finite outgoing energy and a universal reservoir fraction. In the far-zone version of the comparator equation, set $V = 0$ and take an outgoing profile $\psi(t,x) = F(t-x)$, where

$$F(u) = \begin{cases} B\sin^2(\pi u/T), & 0 < u < T, \\ 0, & \text{otherwise.} \end{cases} \tag{28}$$

Here $T > 0$ is the duration in the comparator clock and B is its amplitude in the comparator field normalization. The function and its first derivative vanish continuously at both ends of the pulse. The integrated outgoing energy through any fixed far-zone point is

$$E_{\rm pulse} = \int_{-\infty}^{\infty} (F'(u))^2\,du = \frac{B^2\pi^2}{T^2}\int_0^T \sin^2(2\pi u/T)\,du = \frac{B^2\pi^2}{2T}. \tag{29}$$

Translation of the observing point changes the arrival time but not this integral. The pulse therefore gives a direct example of a finite positive outward energy with an exactly checkable dependence on amplitude and duration. It is a calculation in the declared comparator, not an execution of the Chapter IV native compact-carrier protocol.

If the source reservoir is $E_0/8$, the budget permits this pulse only when

$$B^2 \le \frac{E_0 T}{4\pi^2}. \tag{30}$$

The wave equation does not impose this inequality: it allows any B. The source preparation and coupling must enforce it. This simple example locates the exact logical gap in an argument that jumps from a radiative field to a universal count ceiling. A linear field law supplies propagation and flux; the finite source supplies the allowed amplitudes and total available work.

A shorter pulse of the same amplitude carries more energy in this normalization, because its gradients are larger. A source cannot therefore specify an amplitude independent of bandwidth and still claim a fixed universal energy payment. The physical unit chart and detector response are also necessary before B can be interpreted as an observed strain. The earlier chapters' switching and counterterm budgets become relevant precisely at this point.

11. Stability, scale degeneracy and independent information

11.1 An energy test for a passive exterior

For the homogeneous comparator with $V \geq 0$, its bulk energy integral is nonnegative. Suppose the outer boundary carries only outward flux and the inner boundary, including its storage variables, is passive. Adding the surface energy to the bulk identity yields a total energy that cannot increase after the drive is removed. A finite-energy solution growing exponentially in time would contradict this monotonicity unless it had zero energy or violated one of the premises. This is a useful sufficient stability screen for a proposed boundary model.

The screen must be applied to the complete energy, not only to the exterior wave. Eliminating a driven surface oscillator can make an effective boundary term look negative while its full source account remains well behaved. Conversely, assuming a passive reflection coefficient while omitting an active controller can conceal a real energy supply. The sign and storage of the counterterms inherited from the Original Write action therefore require explicit attention.

Ordinary damped quasinormal eigenfunctions are not square-integrable on the real tortoise line; their boundary behavior under complex continuation is part of the resonance construction. The finite-energy monotonicity argument applies directly to physical wave packets and admissible growing modes, not to a naive norm of every damped resonance. It complements a spectral calculation rather than replacing it. A trustworthy computation reports both the pole search and a time-domain response to a finite, paid pulse.

11.2 What one frequency can and cannot determine

An overall mass change rescales both the real frequency and damping rate by the same factor. If only the real oscillation frequency is measured and the mass is free, a small difference in $\widehat{\omega}_R$ can be absorbed into the fitted mass. For a Schwarzschild comparison, matching that frequency with the rational candidate requires

$$\frac{M_{f,\mathrm{candidate}}}{M_{f,\mathrm{comparator}}} = \frac{3/8}{0.37367168} \simeq 1.003555. \tag{31}$$

The quality factor cannot be changed by this rescaling. Measuring both frequency and damping, or comparing several modes with one common mass, is therefore more informative than comparing one dimensional frequency with a freely chosen mass. A rotating remnant introduces an additional spin dependence and makes the joint inference still more important.

The same logic applies to evidence from exact identities. If the target frequency sets a mass calibration, the resulting dimensional agreement is not an independent test of the count coefficient. If the mass and spin come from an inspiral analysis, the uncertainty and the gravitational assumptions of that analysis must accompany the ringdown comparison. A consistency test within one waveform framework is valuable, but it should not be described as an entirely assumption-free measurement.

A prospective report should separate three uncertainty budgets. Numerical error belongs to the computed eigenproblem. Parameter uncertainty belongs to source mass, spin, orientation, calibration and noise. Structural uncertainty belongs to the choice of exterior action, boundary law and excitation model. Shrinking the first budget to many significant figures does not shrink the other two. The small rational residuals are interesting precisely because they invite a clean calculation, but the claimed precision must refer to the quantity actually constrained.

12. An exactly energy-normalized surface example

12.1 Field, storage and dissipation in one account

A solvable boundary model makes the role of the missing impedance more concrete. Consider a real scalar comparator field $u(t,x)$ on the half-line $x \geq 0$, with unit propagation speed and bulk action $\frac{1}{2}\int dt\,dx\,(u_t^2 - u_x^2)$. Attach the endpoint value $q(t) = u(t,0)$ to a surface mass $m_s > 0$ and spring constant $k_s \geq 0$. Here "mass" means an oscillator inertia in this normalized model, not the astrophysical mass M_f. Add a nonnegative damping coefficient b_s whose dissipated energy is recorded in a separate bath.

Variation of the field-plus-surface action, followed by the stated damping law, gives

$$u_{tt} - u_{xx} = 0, \qquad u_x(t,0) = m_s\ddot{q} + b_s\dot{q} + k_s q. \tag{32}$$

The sign of the derivative is important because the field occupies the right half-line. Define

$$E_{\mathrm{bulk}} = \frac{1}{2}\int_0^X (u_t^2 + u_x^2)\,dx, \qquad E_s = \frac{1}{2}m_s\dot{q}^2 + \frac{1}{2}k_s q^2. \tag{33}$$

An integration by parts gives

$$\frac{d}{dt}(E_{\mathrm{bulk}} + E_s) = -j(t,X) - b_s\dot{q}^2, \qquad j = -u_t u_x. \tag{34}$$

If the bath energy grows at rate $b_s\dot{q}^2$, the enlarged account conserves energy apart from the outer flux. The equation therefore distinguishes reversible surface storage from irreversible transfer to an unobserved bath. Both can reduce the immediately reflected signal, but they have different time-dependent consequences.

This example supplies the normalization missing from a bare reflectivity statement. The incoming wave carries a definite energy current; the endpoint motion stores a definite quadratic energy; and the bath receives a nonnegative payment. No unexplained sink is needed. A native boundary construction should provide an analogous identity, including its actual counterterms and quantum state, before thermal and ringdown interpretations are joined.

12.2 Reflection and absorption calculated explicitly

For a real positive frequency, write $u = \mathrm{Re}[(A_{\mathrm{in}}e^{-i\omega x} + A_{\mathrm{out}}e^{i\omega x})e^{-i\omega t}]$. Equation (32) gives the displacement impedance

$$Z_s(\omega) = k_s - m_s\omega^2 - ib_s\omega, \qquad \mathscr{R}_s(\omega) = \frac{i\omega + Z_s}{i\omega \quad Z_s}. \tag{35}$$

Because both channels have the same speed and normalization, the reflected power fraction is $|\mathscr{R}_s|^2$. Direct evaluation yields

$$|\mathscr{R}_s|^2 = \frac{(k_s - m_s\omega^2)^2 + (1-b_s)^2\omega^2}{(k_s - m_s\omega^2)^2 + (1+b_s)^2\omega^2}. \tag{36}$$

The missing fraction is consequently

$$\mathscr{A}_s = 1 - |\mathscr{R}_s|^2 = \frac{4b_s\omega^2}{(k_s - m_s\omega^2)^2 + (1+b_s)^2\omega^2} \geq 0. \tag{37}$$

It is also at most one, since the denominator minus its numerator is the sum of squares in the numerator of (36). This is an exact passivity check, not an imposed inequality.

The time-averaged incident power is $\omega^2|A_{\mathrm{in}}|^2/2$. The endpoint amplitude is $(1+\mathscr{R}_s)A_{\mathrm{in}}$, so the average bath payment is $b_s\omega^2|1+\mathscr{R}_s|^2|A_{\mathrm{in}}|^2/2$. Their ratio equals (37). Thus the scattering and energy accounts agree independently. At $k_s = m_s\omega^2$ and $b_s = 1$, the reflected wave vanishes at that frequency. Away from resonance the same surface generally reflects, so perfect absorption at one frequency is not a broadband horizon law.

When $b_s = 0$, all incident power eventually returns in steady state, although surface storage can delay it and change its phase. When $b_s > 0$, the bath must be included in a quantum description; damping without the corresponding bath fluctuations is not a complete quantum thermal model. This is why a classical impedance can organize the problem without supplying Chapter VIII's boundary state or emission occupation by itself.

12.3 Poles and the freedom to reverse-engineer a match

With no incoming wave, the outgoing condition gives the pole equation $i\omega = Z_s(\omega)$. Its roots are

$$\omega_\pm = \frac{-i(1+b_s) \pm \sqrt{4m_sk_s - (1+b_s)^2}}{2m_s}. \tag{38}$$

In the underdamped case the real parts are opposite and the decay rate is $(1+b_s)/(2m_s)$. The additional one in this rate is radiation into the exterior half-line; b_s is the separate bath loss. The system can therefore ring down even with no material damping, because it is open to outgoing waves.

This model also demonstrates why matching two desired numbers is insufficient. In its arbitrary normalized units, choose $m_s = 45/4$, $b_s = 1$ and

$$k_s = m_s\left[(3/8)^2 + (4/45)^2\right]. \tag{39}$$

Equation (38) then has a root exactly $3/8 - 4i/45$. The equality is a deliberate inverse design: the target frequencies selected the surface parameters. It is not evidence that this oscillator is the SAM boundary. A native derivation must determine m_s, b_s, k_s or their actual replacements independently of the target, and must include the exterior potential that this half-line example omits.

There is nonetheless a useful lesson. Frequency and damping can arise from different combinations of inertia, restoring response, radiation loss and bath loss. A count expression assigned only to a "damping shell" cannot establish which of those mechanisms it represents. The complete response and energy identity identify the physical interpretation and expose any double-counting of losses.

13. Analytic distinguishability of two ideal ringdowns

13.1 A normalized template calculation

A second solvable example shows how close spectral values become an observational question. Consider complex analytic templates on $t \geq 0$,

$$s_j(t) = \sqrt{2\gamma_j}\,e^{-\gamma_j t}e^{-i\omega_j t}, \qquad \gamma_j > 0. \tag{40}$$

Use the ideal white-noise inner product $\langle s_1, s_2\rangle = \int_0^\infty s_1^* s_2\,dt$. Each template has unit norm. Integrating the exponential exactly gives

$$|\langle s_1, s_2\rangle| = \frac{2\sqrt{\gamma_1\gamma_2}}{\sqrt{(\gamma_1+\gamma_2)^2 + (\omega_2-\omega_1)^2}}. \tag{41}$$

The modulus allows a constant relative phase to be optimized. The calculation assumes a common fixed start, one mode, analytic complex data, infinite observation time and frequency-independent noise. Actual detector data require a colored-noise inner product and detector projections. The formula is an illustrative sensitivity calculation, not a detection forecast.

For small changes around (ω, γ), write $\delta\omega = \omega_2 - \omega_1$ and $\delta\gamma = \gamma_2 - \gamma_1$. Expanding (41) gives

$$1 - |\langle s_1, s_2\rangle| = \frac{(\delta\omega)^2 + (\delta\gamma)^2}{8\gamma^2} + O\big((|\delta\omega| + |\delta\gamma|)^3/\gamma^3\big). \tag{42}$$

The natural comparison is to the linewidth γ, not simply to the oscillation frequency. A small fractional shift in frequency can still matter for a very long-lived mode; a short-lived mode carries fewer useful cycles. The joint measurement of damping and frequency follows from the shape of the entire decaying signal.

For equal-norm signals of signal-to-noise ratio ρ, the phase-optimized squared difference is $2\rho^2(1 - |\langle s_1, s_2\rangle|)$. This expresses how a waveform mismatch competes with noise in the ideal inner product. It is not a universal threshold for evidence, because a real comparison also fits amplitudes, times, masses, spins and nuisance parameters, and may compare models with different prior volume.

Using the Schwarzschild comparator and the rational candidate at the same fixed mass gives an overlap near 0.999972. The corresponding mismatch is about 2.8×10^{-5}. The calculation is included in the independent checks. It illustrates why a percent-level table of frequencies does not establish that the difference is detectable in an individual observed signal. It also does not establish that it is undetectable: multiple modes, longer-lived rotating modes and independent parameter information change the inference.

13.2 Finite windows and parameter fitting

For a finite interval $0 \leq t \leq T$, the unnormalized cross integral acquires the factor

$$\frac{1 - e^{-[\gamma_1+\gamma_2+i(\omega_2-\omega_1)]T}}{\gamma_1+\gamma_2+i(\omega_2-\omega_1)}. \tag{43}$$

The template norms must also be truncated: $\int_0^T e^{-2\gamma_j t}dt = (1-e^{-2\gamma_j T})/(2\gamma_j)$. Applying the infinite-window normalization to a short stretch would misstate the mismatch. Moving the start time changes the retained mixture of modes and can correlate strongly with overtone amplitudes.

Free physical parameters further reduce independent information. A mass rescaling moves both spectral components along one direction in frequency–damping space. Spin moves them along another model-dependent direction. A boundary correction is informative only to the extent that its waveform change cannot be absorbed into those ordinary parameter shifts over the observed band. This is the geometric reason to fit a common mass and spin to multiple modes and to compare the residual pattern, rather than declaring each small offset a separate anomaly.

A complete data-facing report would therefore state the noise weighting, observation window, fitted parameters, waveform derivatives and prior assumptions. The present analytic example supplies a transparent baseline that can be checked without any detector data. It closes a methodological gap between a numerical match and a claim about measurement while leaving the actual likelihood calculation to the declared physical model.

14. From a normalized field to a strain energy

The oscillator and pulse examples use a unit-normalized field, whereas a gravitational-wave detector reports dimensionless strain. To show why that conversion matters, adopt the conventional far-zone gravitational-wave energy account as a separate comparator. In its weak-wave, short-wavelength regime, with transverse-traceless polarizations normalized so that the polarization tensors have squared norm two, the outward power through a large sphere is

$$P_{\mathrm{GW}} = \frac{c^3 r^2}{16\pi G}\int d\Omega\,\left\langle \dot{h}_+^2 + \dot{h}_\times^2 \right\rangle . \tag{44}$$

The brackets denote the averaging appropriate to the effective stress description. The normalization contains Newton's constant and a polarization convention. It is not supplied by the number of receiving channels alone. This benchmark belongs to the general-relativistic gravitational-wave stress construction [XI.8].

As an angular example, let $h_+ = A(u)\sin^2\theta/r$, $h_\times = 0$, with retarded time $u = t - r/c$. The amplitude A has dimensions of length. The angular integral is

$$\int \sin^4\theta\, d\Omega = 2\pi\int_{-1}^{1}(1-v^2)^2\,dv = \frac{32\pi}{15}. \tag{45}$$

Substitution into (44) gives $P_{\mathrm{GW}} = (2c^3/15G)\langle \dot{A}^2\rangle$. The spherical factor and the polarization normalization therefore change the conversion between an amplitude derivative and total power. A detector located at one angle samples only one projection of this pattern; multiplying its local strain by a universal isotropic factor would be incorrect.

For a complete finite waveform, the corresponding energy functional is $(2c^3/15G)\int \dot{A}^2 du$, with the effective-flux assumptions understood. For the compact envelope $A = B\sin^2(\pi u/T)$ on $0 < u < T$, the formal integral is $c^3B^2\pi^2/(15GT)$. The example is an algebraic normalization audit rather than a physical merger waveform; if the effective-stress averaging is used, a waveform with a well-separated carrier and envelope must be supplied. Its role is to show precisely which constants and angular factors are absent from the unit-normalized pulse calculation.

A native map could take the form $A = \mathscr{C}\psi$ after a specified multipole extraction. Then $\mathscr{C}$ carries the required dimensions and the physical flux is proportional to $\mathscr{C}^2$. Rescaling the field in its action also rescales its source coupling, canonical momentum and counterterms. One cannot choose $\mathscr{C}$ from the desired radiation energy while claiming an independently derived amplitude. The source preparation, coupling and receiving calibration must fix it together.

The same discipline applies to two-polarization sums. A circular basis is a linear change of field coordinates; with a unitary normalization it preserves the quadratic polarization sum. A convention lacking the appropriate square-root factors can introduce an apparent factor of two. Any derivation should state the real and circular field definitions and verify the energy identity in both bases. This is a straightforward but decisive audit before comparing an integer carrier fraction to an observed radiated mass.

Finally, total energy and detector observability are different questions. A signal may have finite energy but negligible projection in a particular instrument, or significant projection in a narrow band while much energy lies elsewhere. To turn a carrier budget into a catalog prediction, the model needs the angular pattern, spectrum, source orientation and detector response as well as the total flux. These are consequences to be computed from a physical source, not nuisance details that can be omitted because the final energy ceiling is dimensionless.

A practical benefit of these solved comparators is that each failure can be localized. An incorrect flux normalization changes the relation between amplitude and energy without moving the oscillator poles. A changed damping bath changes absorption and pole widths even when the restoring frequency is unchanged. A different observation window changes template overlap without changing the underlying field equation. Keeping these operations distinct prevents an improved fit at one interface from being presented as a derivation at another. The source theory must eventually select all three: the dynamics that creates a signal, the energy account that pays for it, and the measurement channel that makes it visible. The checks supplied here deliberately test those interfaces separately. Their agreement is a consistency requirement on the illustrative mathematics; it is not additional independent evidence for the carrier ontology. That distinction leaves a clear path for replacing each comparator by the native calculation when the corresponding source and boundary operands become available.

15. Results, limitations and the shortest route forward

The exact result of this chapter is the count reduction and its derived quality factor. The energy bound is proved only under

its explicit reservoir premises. The differential and boundary equations provide a clear research problem, but no native eigenproblem has been solved in this expansion and no new observational analysis is represented as completed.

Three interfaces have priority. First, derive the physical stress and outgoing work from the same moving-source action that supplies the static field. This connects the carrier claim to binary-pulsar decay and fixes the energy normalization. Second, derive one causal boundary response usable in Chapters V, VIII and XI. Its thermal behavior and ringdown poles should be consequences of the same response, not independent prescriptions. Third, extend that response to a conserved angular momentum account and a spin-dependent exterior.

The possible outcomes are scientifically distinct. An exactly absorbing Schwarzschild limit recovers known ringdown and leaves the rationals as approximations. A derived modified boundary could produce shifted modes, altered damping or delayed returns, all subject to stability and existing observational constraints. A source calculation could instead show that the count energy fraction is not a universal reservoir fraction. Any of these outcomes improves the theory by replacing an ambiguous match with an explicit law and an explicit domain of validity.

Chapter references and provenance

[XI.1] S. W. Hawking, "Gravitational Radiation from Colliding Black Holes," *Physical Review Letters* **26**, 1344 (1971). https://doi.org/10.1103/PhysRevLett.26.1344.

[XI.2] P. C. Peters and J. Mathews, "Gravitational Radiation from Point Masses in a Keplerian Orbit," *Physical Review* **131**, 435 (1963). https://doi.org/10.1103/PhysRev.131.435.

[XI.3] E. Berti, V. Cardoso and A. O. Starinets, "Quasinormal modes of black holes and black branes," *Classical and Quantum Gravity* **26**, 163001 (2009). https://arxiv.org/abs/0905.2975.

[XI.4] E. W. Leaver, "An analytic representation for the quasinormal modes of Kerr black holes," *Proceedings of the Royal Society A* **402**, 285–298 (1985). https://doi.org/10.1098/rspa.1985.0119.

[XI.5] V. Cardoso and P. Pani, "Testing the nature of dark compact objects: a status report," *Living Reviews in Relativity* **22**, 4 (2019). https://arxiv.org/abs/1904.05363.

[XI.6] M. Isi, M. Giesler, W. M. Farr, M. A. Scheel and S. A. Teukolsky, "Testing the no-hair theorem with GW150914," *Physical Review Letters* **123**, 111102 (2019). https://arxiv.org/abs/1905.00869.

[XI.7] LIGO Scientific, Virgo and KAGRA Collaborations, "Tests of General Relativity with GWTC-3." https://arxiv.org/abs/2112.06861.

[XI.8] R. A. Isaacson, "Gravitational Radiation in the Limit of High Frequency. II. Nonlinear Terms and the Effective Stress Tensor," *Physical Review* **166**, 1272 (1968). https://doi.org/10.1103/PhysRev.166.1272.

The original SAM content is preserved from Chapter XI of the supplied assembled volume, its cited CR001–CR005b radiation records, the Original Write source–field account, and Chapters IV–V. The closing review supplies the cautions concerning expression selection, the missing binary-pulsar flux and the spin interface. Referenced internal records are provenance supplied by the source manuscript; they were not independently re-executed for this edition.

SAM VOLUME I

Cosmological fate and the completed boundary

Chapter XII

Abstract. The closing chapter brings together two different structures: a line-of-sight accumulation law with saturation $1/\pi$, and a matter-inventory assignment with the same numerical value. Their equality is exact within the declared count rules, but a cosmological future requires a time-evolution law and a calibrated observable. We distinguish a distance-fit coefficient from the fractional expansion rate, derive the conditions under which the original $\Omega_\Lambda^{3/2}$ expression is meaningful, and formulate finite-redshift tests. The completed-boundary or Home continuation is then developed as an explicit conditional ledger, with conservation, radius-floor and information-channel requirements. A worked receipt sequence demonstrates growth while emitting, and a differential distance test exposes the assumptions hidden in a simple rate rescaling. The resulting closure program connects the thermal boundary, radiation response and cosmological inventory without treating their common numerical factors as a substitute for dynamics. No transition mechanism or future cosmological solution is claimed to have been established by count identities alone.

Status, 25 October 2026 (Part 2). Sections 1–8 carry the road law $A_{\rm los}(z) = (1/\pi)\,(1-(1+z)^{-3})$ (register T-10), the distinction of equal numbers under different operators, the background adapter and the finite-redshift tests. These are the sections the ceiling runs of 24 October used, and the road is now one object with the floored reservoir on $\cosh^2\varphi = e^{ds}$ (Chapter IX, the section "The $1/\pi$ ceiling as a recurring theme: results of 24 October 2026"). Section 6, the fate coefficient (T-11), is an open target that hands forward to the completed Home, which the 25 October edition gives its own chapter: Chapter XIII (T-12, T-14).

1. The two arcs that meet at the end of the volume

Volume I begins with a floor and a source-induced lift. The floor supplies a reference level; the lift describes an accumulation attributed to a particular source. Chapters III–VI and VIII follow the source coordinate toward a completed boundary and ask how clocks, records, fields and thermal exchange behave there. Chapters IX and X and the Supplement use the floor in a proposed cosmic inventory and in the organization of many-source systems. Chapter XI asks whether the released carrier account can become an independently testable radiation law.

The original ending joins these arcs in two ways. First, it identifies the limiting line-of-sight road accumulation with the matter fraction. Second, it proposes that a completed Home can supply a new floor through a change of informational indexing. Both are substantial ideas. Neither should be weakened by asking a numerical identity to establish a causal process that has not yet been specified.

This chapter therefore keeps the original proposals while separating four levels of statement. An identity follows from the chosen counts. A conditional theorem follows after its physical premises are declared. An adapter imports a conventional geometry or evolution law. A conjecture identifies a possible continuation whose mechanism remains to be found. The end of the volume is strongest when these levels remain visible, because they show exactly which result a future calculation would add.

The word "fate" also requires care. The future of a cosmological solution is a statement about evolution in proper time. The saturation of an integral along the present past light cone is a statement about progressively more distant emission events. A change of radial distance convention is yet another operation. The three can be related by a theory, but they cannot be interchanged by notation.

2. The road law and what it actually proves

For the nonnegative redshift domain of the original proposal, define

$$A_{\rm los}(z) = A_0 R\left[1-(1+z)^{-\widehat{d}}\right] = \frac{1}{\pi}\left[1-(1+z)^{-3}\right], \qquad z \geq 0. \tag{1}$$

The model uses $A_0 R = 1/\pi$ and $\widehat{d} = 3$. Direct differentiation gives

$$\frac{dA_{\rm los}}{dz} = \frac{3}{\pi}(1+z)^{-4} > 0, \qquad \frac{d^2A_{\rm los}}{dz^2} = -\frac{12}{\pi}(1+z)^{-5} < 0. \tag{2}$$

Thus the road starts at zero, initially rises with slope $3/\pi$, is monotone and concave, and has the exact limit $1/\pi$. Its survival factor is

$$f(z) = 1 - A_{\rm los}(z) = 1 - \frac{1}{\pi} + \frac{1}{\pi(1+z)^3}. \tag{3}$$

For $z \geq 0$, $1 - 1/\pi < f(z) \leq 1$. No dynamics is needed to prove these statements. They are properties of the proposed function, not yet a derivation of that function from microscopic transport.

At small redshift the expansions are

$$A_{\rm los}(z) = \frac{1}{\pi}\left(3z - 6z^2 + 10z^3 + O(z^4)\right), \tag{4}$$

$$f(z) = 1 - \frac{3z}{\pi} + \frac{6z^2}{\pi} + O(z^3). \tag{5}$$

These coefficients are more directly testable than an asymptotic number, once the road is connected to an observable. At $z = 1$, for example, the proposed accumulation is $7/(8\pi)$ and the survival factor is about 0.7215. Any interpretation that changes a measured distance by this factor has a substantial finite-redshift consequence. It cannot be defended solely by noting that its limiting value resembles a cosmological inventory fraction.

There is a useful alternative variable for the mathematical road. Let $u = \log(1+z)$. Then $A_{\rm los} = (1-e^{-3u})/\pi$ and

$$\frac{dA_{\rm los}}{du} = 3\left(\frac{1}{\pi} - A_{\rm los}\right). \tag{6}$$

This is an exact differential representation of the supplied curve. It is not a future-time relaxation law: u labels lookback redshift along a specified observation cone. Treating it as a clock would add a new identification and must be justified independently.

3. Equal numbers, different operators

The inventory assignment of Chapter IX is

$$\Omega_{m,*} = RA_0 = \frac{1}{\pi}, \qquad \Omega_{\Lambda,*} = 1 - \Omega_{m,*} = \frac{\pi - 1}{\pi}. \tag{7}$$

An asterisk emphasizes that these are the clean count-assigned fractions before corrections or additional present-day components are specified. The original typed join is then

$$A_{\rm los}(\infty) = \Omega_{m,*}, \qquad f(\infty) = \Omega_{\Lambda,*}. \tag{8}$$

The equality is exact as a consequence of the common input RA_0. It does not prove that the road operator measures matter density, or that its complement has negative pressure. An inventory is a weight assigned to sectors of a state. A propagation observable is an operation on a source, path, detector and calibration. Equal expectation values do not make those operations identical.

A stronger version of the join would start with a normalized preparation measure μ on actual source states and two declared observables, $\mathscr{I}_m$ and $\mathscr{R}_{\rm los}$. It would show

$$\int \mathscr{I}_m \, d\mu = \frac{1}{\pi}, \qquad \lim_{z\to\infty} \int \mathscr{R}_{\rm los}(z)\, d\mu = \frac{1}{\pi}, \tag{9}$$

with both normalizations derived from the same preparation and no target-dependent adjustment. This is the source-measure interface identified in the closing review. It would replace a numerical coincidence within a grammar by a common physical cause.

The component accounting also has to close. If a spatially flat present-day Friedmann adapter includes a nonzero radiation density, then $1 = \Omega_{r,0} + \Omega_{m,0} + \Omega_{\Lambda,0}$. Assigning both fractions in (7) exactly while adding radiation makes their sum exceed one. One may work in a late-time approximation neglecting radiation, assign $\Omega_{\Lambda,0} = 1 - \Omega_{m,0} - \Omega_{r,0}$, or change the definition of the clean ledger. The choice must be explicit. A small correction is still a correction when the proposal claims exactness.

4. The background adapter and its conditional late-time limit

The original fate calculation adopts

$$H^2(a) = H_0^2\left(\Omega_{r,0}a^{-4} + \Omega_{m,0}a^{-3} + \Omega_{\Lambda,0}\right), \tag{10}$$

with $a = 1$ today and a consistent normalization of its coefficients. This is a flat radiation–matter–cosmological-constant adapter. It assumes the relevant conservation laws, an expanding branch, and a positive constant vacuum term. Under those assumptions, the dilution terms vanish as $a \to \infty$ and

$$H_{\infty,\rm native} = H_0\sqrt{\Omega_{\Lambda,0}}. \tag{11}$$

The conclusion is valid conditional mathematics, but it inherits the cosmological-constant behavior from the adapter. An inventory complement has not independently derived that behavior.

The approach can also be quantified. Factoring out $\Omega_{\Lambda,0}$ and expanding the square root gives

$$\frac{H(a)}{H_0\sqrt{\Omega_{\Lambda,0}}} = 1 + \frac{\Omega_{m,0}}{2\Omega_{\Lambda,0}}a^{-3} + \frac{\Omega_{r,0}}{2\Omega_{\Lambda,0}}a^{-4} + O(a^{-6}). \tag{12}$$

Along the expanding branch, the limiting equation $\dot{a}/a = H_{\infty,\rm native}$ has exponential solutions. This establishes the late-time behavior of the stated adapter; it does not establish stability against arbitrary changes of the source theory or additional components.

For a component with equation of state $p = w(a)\rho c^2$, separate conservation would instead give

$$\rho(a) = \rho(1)\exp\left[-3\int_1^a (1+w(\widetilde{a}))\, d\log\widetilde{a}\right]. \tag{13}$$

A constant density follows only when $w = -1$, or when a separately specified exchange compensates its evolution. If the Home population feeds an effective background, that exchange must appear in the continuity equations. A successful mechanism might derive it, but the number $(\pi-1)/\pi$ alone contains no pressure, transfer rate or stability information.

Measurements such as Planck's parameter inference and DESI's distance constraints are relevant comparators [XII.1, XII.2]. They do not measure a count identity directly: they constrain an evolution and observation model. Consequently, agreement of an adopted present-day density fraction with one parameter fit does not settle the cosmological future, and a model-specific tension should not be quoted without the exact data combination and likelihood assumptions.

5. Past-cone saturation is not future-time saturation

In an ordinary homogeneous cosmological chart, redshift relates an emission scale factor to the observer's scale factor through $1+z = a(t_o)/a(t_e)$. Keeping t_o fixed while taking $z \to \infty$ sends the emission time toward the early universe. It does not send the observer to the future. If one sets $a(t_o) = 1$ and mechanically substitutes $1+z = 1/a$ into (1), one obtains

$$A_{\rm los} = \frac{1}{\pi}(1 - a^3). \tag{14}$$

For $a > 1$ this becomes negative and tends to minus infinity. The substitution is a useful diagnostic: the supplied past-cone function is not a nonnegative future saturation law on that domain.

A cosmological road should instead carry its arguments explicitly. One possible notation is $A_{\rm los}(t_o, t_e, \mathscr{P})$, where $\mathscr{P}$ identifies the ray or route and the preparation of the medium. A

homogeneous effective road might depend on an integral of a rate along the ray,

$$A_{\rm los} = \mathscr{F}\left(\int_{t_e}^{t_o} \Gamma(t, \mathscr{P})\, dt\right), \tag{15}$$

but the rate, saturation function and conservation law must come from the model. Writing an integral does not by itself identify the microscopic carriers or show that their accumulation is positive.

The future limit also needs a family of observations. Does the observer follow a fixed comoving source, a fixed proper-distance source, a fixed emission-time surface, or the edge of an accessible region? Each family changes the relation among t_o, t_e and distance. A single number called the "remaining road" is insufficient until that family and its clocks are chosen.

As a mathematical example only, a future variable $B(N)$ with $N = \log a$ could obey $dB/dN = \lambda(B_* - B)$ for $\lambda > 0$. Then $B = B_* + [B(0) - B_*]e^{-\lambda N}$, so B_* is an attractor. Choosing $B_* = 1/\pi$ reproduces the desired limit but does not derive it. This toy equation identifies the missing tasks: derive the fixed point, the sign and magnitude of its relaxation, the physically allowed interval and the source of any dissipated energy.

6. The fate coefficient and the meaning of an expansion rate

The working draft uses

$$D_{\rm native} = f D_{\rm readout} \tag{16}$$

and proposes multiplication of the background rate by f_∞. There is one clear operational situation where that multiplication is correct. If a separately fixed velocity-like numerator obeys $v = H_{\rm native} D_{\rm native}$ and one fits the same numerator against $D_{\rm readout}$, then

$$H_{\rm fit} \equiv \frac{v}{D_{\rm readout}} = f H_{\rm native}. \tag{17}$$

Assuming additionally that a future observing family has $f_\infty = \Omega_{\Lambda,*}$, and using the radiation-neglecting adapter, this gives the original expression

$$H_{\rm fit,\infty} = H_0 \Omega_{\Lambda,*}^{3/2} = H_0 \left(\frac{\pi - 1}{\pi}\right)^{3/2} \simeq 0.562834 H_0. \tag{18}$$

This is a conditional distance-fit coefficient. The numerator, clock and calibration must remain fixed exactly as stipulated. At cosmological distances, recession velocity is not generally a directly measured local velocity, so the observation rule cannot be left implicit.

A fractional expansion rate behaves differently. For a comoving separation $D_{\rm native} = a\chi$ with fixed χ, differentiating (16) using the same time coordinate yields

$$\frac{\dot{D}_{\rm readout}}{D_{\rm readout}} = H_{\rm native} - \frac{\dot{f}}{f}. \tag{19}$$

If f approaches a nonzero constant and $\dot{f}/f \to 0$, this rate approaches $H_{\rm native}$, not $f_\infty H_{\rm native}$. A constant rescaling of distance cancels from a logarithmic derivative. This simple identity is essential to the final claim: the original multiplication cannot be described as a future Hubble rate without an operational definition that differs from the usual fractional expansion rate.

Changing the clock would add another factor. If $dt_{\rm readout} = g(t)dt$, then the measured fractional rate becomes $[H_{\rm native} - \dot{f}/f]/g$. Neither g nor its asymptote follows from a distance fraction. Chapter IV's local invariant light speed is compatible with many coordinate or readout conventions; it does not select one of them. The calibration of clocks, rulers and propagation must be supplied together.

7. Numerical values with an honest uncertainty ledger

Using the original clean fractions and the frozen Chapter IX value $H_{0,\rm SAM} = 67.2504\,{\rm km\,s^{-1}\,Mpc^{-1}}$, the native adapter limit is approximately 55.525 in the same units and the conditional fit coefficient is approximately 37.851. The original alternatives $H_0 = 68.76$ and 73.04 give conditional fit values about 38.700 and 41.109. These are rescalings of a selected input, not three independent confirmations of the theory.

The exact dimensionless expression is more portable than any one dimensional value. Even so, its apparent exactness does not remove uncertainty in the choice of inventory, the road interpretation or the dynamical adapter. If $F = H_0 \Omega_\Lambda^{3/2}$, first-order propagation of continuous input errors gives

$$\frac{\delta F}{F} = \frac{\delta H_0}{H_0} + \frac{3}{2}\frac{\delta \Omega_\Lambda}{\Omega_\Lambda}. \tag{20}$$

A variance calculation must preserve the covariance of the inputs. Structural alternatives, such as a different road operator or a nonconstant dark-energy equation of state, are not captured by this linear formula. They require separate model branches.

The closing review cautions that the small propagated temperature uncertainty in the Hubble cascade is not a total theory uncertainty. This chapter adopts that caution. A conditional number may be computed to many decimal places, but the number of printed digits should not imply that discrete choices, calibration assumptions and uncomputed source corrections are known to that precision.

8. Finite-redshift observations and distance duality

A useful road theory must say which distances change. In a metric description with photon number conservation and null-geodesic propagation, luminosity and angular-diameter distances obey

$$D_L = (1+z)^2 D_A. \tag{21}$$

The physical assumptions behind this relation matter more than its use as a convenient identity. Distance-duality tests compare different measurement channels and can reveal an inconsistency in an assigned propagation factor [XII.3].

Suppose a proposed readout changes the two distances by factors $g_L(z)$ and $g_A(z)$ relative to a common metric adapter.

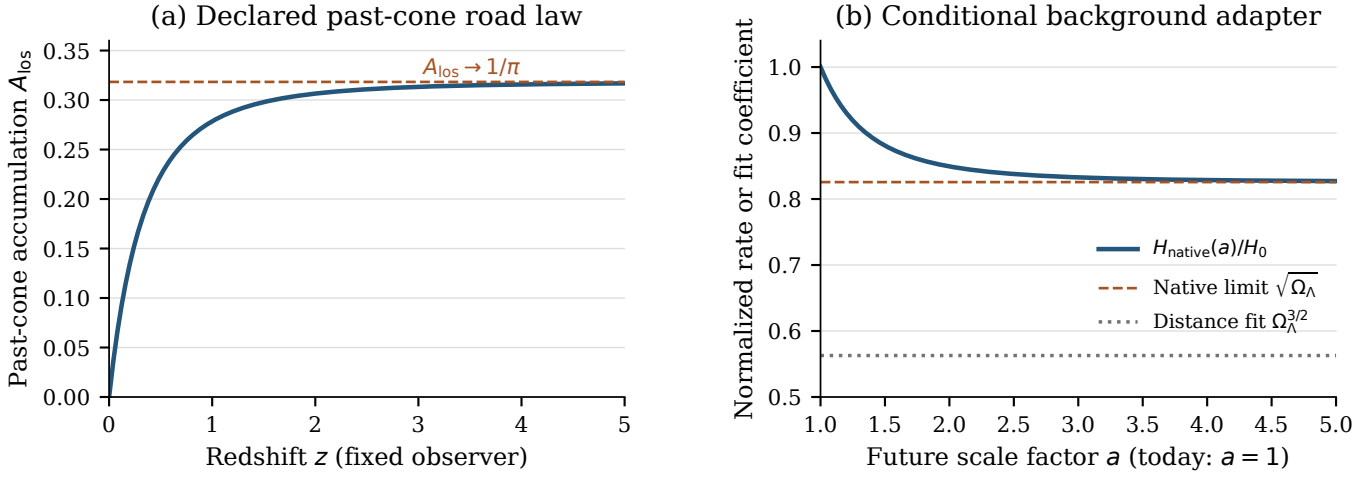


Figure 1: Two mathematically distinct limits. (a) The declared past-cone road law $A_{\mathrm{los}}(z) = \pi^{-1}[1-(1+z)^{-3}]$ approaches $1/\pi$ as redshift increases for a fixed observer. (b) With radiation neglected, $\Omega_m = 1/\pi$ and $\Omega_\Lambda = 1 - 1/\pi$, the adopted flat matter–constant-vacuum adapter gives $H_{\mathrm{native}}(a)/H_0 = \sqrt{\Omega_m a^{-3} + \Omega_\Lambda}$ and tends to $\sqrt{\Omega_\Lambda}$. The lower dotted reference, $\Omega_\Lambda^{3/2}$, is a conditional distance-fit coefficient, not the native fractional-expansion limit; it requires the additional future observing-family and calibration assumptions in the text. Past-cone saturation does not derive a future road limit. These curves are analytic illustrations of declared premises, not observational data or an independently derived SAM future.

Then the diagnostic ratio is

$$\eta_D(z) = \frac{D_{L,\mathrm{readout}}}{(1+z)^2 D_{A,\mathrm{readout}}} = \frac{g_L(z)}{g_A(z)}. \tag{22}$$

If only luminosity distance is multiplied by $1/f$, then $\eta_D = 1/f$. If both distances receive the same factor, the ratio remains one. Thus a road factor is not automatically a distance-duality violation, contrary to an overly broad reading of the closing review. The violation depends on the observation map. A common rescaling still affects other distance and expansion comparisons and is not automatically viable.

For a concrete differential test, assume a flat adapter with $D_C(z) = c\int_0^z dz'/H(z')$, and apply $D_{C,\mathrm{readout}} = D_C/f$. An analyst interpreting this new distance by the same integral relation would infer

$$H_{\mathrm{inferred}}(z) = \frac{c}{d(D_C/f)/dz} = \frac{H(z)f(z)^2}{f(z) - [H(z)D_C(z)/c]f'(z)}. \tag{23}$$

This is not simply fH. The derivative of the road contributes. At $z = 0$, $D_C = 0$ and $f = 1$, so this particular map preserves the local intercept H_0 while changing higher-order distance behavior.

The same point appears in a small-redshift expansion. If q_0 is the adapter deceleration parameter, $D_C = (c/H_0)[z - (1 + q_0)z^2/2 + O(z^3)]$. Dividing by (5) and fitting the same form gives

$$q_{0,\mathrm{inferred}} = q_0 - \frac{6}{\pi}. \tag{24}$$

The shift is large. It is a falsifiable consequence of this particular uncompensated distance map, not a claim that all possible SAM road operators predict it. A viable alternative must explicitly show which other calibration or geometry factors compensate it; silently canceling the factor would erase the claimed new observable.

9. What the completed volume establishes and what it hands forward

The road identities establish a bounded past-cone function and its equality, within the chosen count grammar, to a clean inventory fraction. The standard background adapter gives a conditional native late-time rate. The original $\Omega_\Lambda^{3/2}$ coefficient is retained with the specific distance-fit interpretation that makes its algebra valid; it is not established as the usual fractional expansion rate. The future saturation needed for that coefficient remains a separate dynamical requirement.

The Home recurrence, carried in Chapter XIII, gives a rigorous nonnegative exterior account, a conditional radius floor and a distinction between gross output and net growth. It does not by itself establish an impenetrable surface, a complete evaporation law, a child universe or a source of dark energy. The completed boundary can become a unifying physical object only when one source-selected action supplies those operations with their energy and information costs.

That is the useful ending of Volume I. The early source, clock, field and record constructions have exposed interfaces that the cosmological and boundary proposals must now meet. Volume II's matter and unit calibration can strengthen those interfaces, but should not be invoked as an already completed proof. The program advances when each remaining connection becomes a declared calculation whose success or failure changes the theory.

Chapter references and provenance

[XII.1] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* **641**, A6 (2020). https://arxiv.org/abs/1807.06209.

[XII.2] DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints" (2025). https://arxiv.org/abs/2503.14738.

[XII.3] Euclid Collaboration, "Euclid: Forecast constraints on the cosmic distance duality relation with complementary external probes," *Astronomy & Astrophysics* (2020). https://doi.org/10.1051/0004-6361/202039078. Used for the assumptions and observational formulation of distance duality, not as a new measurement in this chapter.

[XII.4] M. Mars, *Introduction to General Relativity*, Erwin Schrödinger Institute summer-school notes (2014), discussion of Birkhoff's theorem. https://homepage.univie.ac.at/piotr.chrusciel/SummerSchool2014/Marc_Mars_Introduction_to_GR_ESI.pdf.

Original SAM provenance: supplied Chapter XII; the Chapter IX horizon projection; CR002 and CR003 of the thermal-ladder records; historical Home rules H000791 and H000794–H000797; and the supplied closing review. This expansion uses those records as manuscript provenance and does not claim to have independently executed their archived calculations. The distinctions between past-cone and future limits, between distance-fit and fractional rates, and between common and channel-dependent distance factors are explicit analytical clarifications in this edition.

SAM VOLUME I

Home: information on an endless chain of two-dimensional closed spheres

Chapter XIII

Abstract. A Home is a closed two-dimensional ledger with an interior side and a parent-domain exterior side. What an exterior observer calls bulk volume is read as ordered information on that surface, and the surface's own completion can become the floor of a new interior, so that the construction continues as an endless chain of closed spheres, each the exterior of the next. This chapter carries the idea as an idea: the custody recurrence $E_{n+1} = E_n + I_n - O_n$ with its nonnegative exterior account, the conditional radius floor, the distinction between gross output and net growth, surface response, quantum custody and what an exterior observer can learn, the re-indexing of a completed boundary into a new floor, the formation-transfer model and its counterexample to an unsupported inference, the entropy ledger for receipts, and a finite program for deciding the fate proposal. The text is that of Sections 9–19 of the submitted Chapter XII, unchanged, renumbered as a chapter of its own.

The idea, stated once. Every earlier chapter of this volume treats a boundary as a place where accounting changes hands: a source's retained account against its field, a clock's reading against the road it has travelled, a horizon's accessible information against the inventory behind it. The Home proposal takes that habit to its limit. It says that the natural object of the theory is not a volume filled with matter but a closed two-dimensional surface carrying ordered information, with an interior that is itself such a surface's exterior, without end in either direction. Equal intervals and filled geometry are then the exterior observer's conventions, not what the ledger is made of. Nothing in this chapter proves that. What it does is make the proposal precise enough to be wrong: a recurrence whose exterior account is provably nonnegative, a radius floor that follows under a stated closure, a custody rule that forbids free copying, a transfer model that shows what a change of inventory label cannot do by itself, and a finite program of checks with stopping rules. The chapter ends with that program because the proposal's standing is what it has yet to pass.

Status, 25 October 2026 (Part 2). The Home is register target T-12 and the crossing postulate T-14 (*Targets register*, 23 October 2026); neither has been built natively, and the evidence appendix's conditional reading stands. Three results of the record touch the idea without testing it, and are named here as connections, not as support. Volume II's Li-6 receiver algebra, reproduced natively from the archive's exact operands (H002783, 25 October 2026), found the component a receiver discards to be a finite phase register read one plane per Write, carrying a conserved label and a gauge the receiver sees only through part of its response: a closed surface account in the matter language, with nothing of the gate's hidden stage in it. The Volume II gate run (H002782) found retained information monotone and log-convex but not completely monotone, which is the shape a surface ledger with a floor would have. And the Riemann window run (H002784) found the program's prime structure scale-free, a function of ratios and invisible on the equal-interval ruler, which is the same claim about conventions this chapter makes about volume. The chapter's own next step is unchanged from its last section: one surface model shared with Chapters VIII and XI that reproduces the custody recurrence under a source-selected action, with the numerical ledger frozen while it is tested.

1. A completed Home: retained proposal and conditional closure

The second arc of the original chapter describes a Home as a closed two-dimensional ledger with an interior side and a parent-domain exterior side. Apparent bulk volume is interpreted as ordered information on the surface, rather than an additional ontic three-dimensional interior. Tagged child histories retain their custody under recursive maps. These are organizing postulates of the Home proposal, not deductions from the Schwarzschild metric.

The original lifecycle has three stages. A pre-completion collapse state has no filled point interior; localization remains assigned to the forming surface. During formation, the origin-matter lane and the captured outside-origin baryon lane establish a formation inventory. At completion, the source coordinate reaches $A = 1$ and the ledger is sealed under the proposed crossing rule. The historical identifiers H000791 and H000794–H000797 record this construction.

Chapter V already limits how strongly the last step may be stated. Its source coordinate satisfies $A(r_s) = 1$; this is not the condition $A_0 + A = 1$. Its finite-impulse and finite-distance results depend on the source law. In the relevant infinite-approach route, a calculated source satisfying the radial null-energy condition is screened by the theorem; otherwise the actual energy-condition violation or flux must be propagated. Chapter XII cannot turn this conditional result into an unconditional physical law of infinite crossing cost.

Accordingly, the completed Home account below is conditional on a source-selected completion dynamics and boundary law consistent with Chapter V. It may impose an empty post-completion crossing relation as part of that model, but must then derive the stress, work and causal response that realize it. Coordinate-time delay in a Schwarzschild exterior does not by itself establish infinite local crossing cost. The physical statement requires an operational trajectory and energy account.

2. Exterior balance and the radius floor

Once the conditional completion is stipulated, define $M_{\mathrm{form}} > 0$ as the frozen formation inventory and $E_n \geq 0$ as a mutable exterior mass-equivalent inventory. Let $I_n, O_n \geq 0$ be realized incoming and outgoing mass-equivalent receipts over step n. The proposed recurrence is

$$\begin{aligned} E_{n+1} &= E_n + I_n - O_n, \qquad 0 \leq O_n \leq E_n + I_n, \\ M_n &= M_{\mathrm{form}} + E_n. \end{aligned} \tag{1}$$

The explicit upper bound on O_n is what preserves nonnegativity. If $E_0 \geq 0$, induction gives $E_n \geq 0$ for every subsequent step. Summing the recurrence gives the exact cumulative identity

$$E_N = E_0 + \sum_{n=0}^{N-1} I_n - \sum_{n=0}^{N-1} O_n, \qquad \sum O_n \leq E_0 + \sum I_n. \tag{2}$$

These are ledger theorems. They do not specify a physical emission rate or show that every allowed sequence occurs.

If the Schwarzschild mass–radius adapter is adopted at each quasi-static step,

$$R_n = \frac{2G(M_{\mathrm{form}} + E_n)}{c^2}, \qquad R_{\mathrm{form}} = \frac{2GM_{\mathrm{form}}}{c^2}, \tag{3}$$

then $R_n \geq R_{\mathrm{form}}$. An area-based record count with fixed area per unit satisfies

$$\frac{N_n}{N_{\mathrm{form}}} = \left(\frac{R_n}{R_{\mathrm{form}}}\right)^2 = \left(1 + \frac{E_n}{M_{\mathrm{form}}}\right)^2 \geq 1. \tag{4}$$

This consequence requires both the adapter and an unchanged microscopic area unit. If the completion mechanism alters either, the area-memory formula must be rederived.

A worked example clarifies gross output versus net change. In arbitrary mass units choose $M_{\mathrm{form}} = 10$ and $E_0 = 2$. An interval with $I_0 = 3$ and $O_0 = 1$ ends at $E_1 = 4$: the Home emits while its total mass grows from 12 to 14. Next take $I_1 = 0$ and $O_1 = 3$, obtaining $E_2 = 1$ and total mass 11. Finally $I_2 = 0$, $O_2 = 1$ exhausts the exterior account and leaves the floor 10. The cumulative output is 5, paid by initial exterior inventory 2 plus inflow 3. At no step is the formation floor debited.

A positive floor has a sharp thermodynamic consequence. If all incoming supply ceases and exterior inventory drains, the proposed output must eventually cease or change its law. A temperature formula by itself cannot authorize continued emission from a frozen reservoir. The boundary transition strengths and available states must implement the floor. This is where Chapter VIII's thermal account must meet the lifecycle rather than remain a separate analogy.

3. Continuous evolution, vacuum exteriors and surface response

A continuum approximation to the recurrence is

$$\dot{E} = \dot{M}_{\mathrm{in}} - \dot{M}_{\mathrm{out}}, \qquad E \geq 0. \tag{5}$$

At $E = 0$, the boundary rule must prevent $\dot{M}_{\mathrm{out}} > \dot{M}_{\mathrm{in}}$. A formal ordinary differential equation that continues to negative E violates the model's own domain. One can impose a constrained flow, but a microscopic theory should explain its origin through allowed transitions and detailed energy payments.

There is also a geometry requirement. A spherically symmetric vacuum exterior in general relativity has a constant Schwarzschild mass parameter locally; time-varying mass caused by actual inflow or radiation requires nonvacuum stress in the relevant region [XII.4]. Therefore (3) is at most an instantaneous or quasi-static adapter during exchange. It is not a complete spacetime solution for arbitrary receipt sequences. Matching an emitting surface to its exterior requires the energy flux and junction conditions.

The same surface cannot be assigned independently chosen optical, thermal and gravitational-wave properties. A hard reflective surface, a dissipative membrane and an absorbing horizon can share an approximate exterior mass profile while having different responses. Chapter XI's impedance and Chapter VIII's thermal transitions must be compatible with the Home custody rules. A boundary that stores a pulse must say where its energy goes and how a later receipt releases it. A boundary that permanently hides information must specify the channel seen by each side.

These obligations also prevent a purely semantic resolution of the singularity question. Calling the boundary an information surface does not calculate its stress or establish a nonsingular evolution. Conversely, a successful surface model would not need a classical bulk interpretation to be meaningful if it provided a consistent state space, dynamics, observation map and tested exterior limit. Those are the appropriate requirements for the proposal on its own terms.

4. Quantum custody and what an exterior observer can learn

The original Home rules describe unresolved horizon matter as unavailable to either side. A failed incoming attempt retains its exterior receipt; a retained post-completion receipt has a finite sequence of attempted inward operations followed by a realized outward operation. The terminal outward receipt debits scalar parent-visible mass. This is a proposed event grammar, not yet a Hamiltonian or a quantum instrument.

A physical version can represent an allowed step by an instrument $\{\mathscr{E}_r\}$ indexed by receipts r. Each map is completely positive and trace nonincreasing; their sum is trace preserving when all outcomes are included. Probabilities are $p_r = \operatorname{Tr}\mathscr{E}_r(\rho)$, and conditioned states are $\mathscr{E}_r(\rho)/p_r$. Conservation then requires the mean receipt energies and changes in source, surface and control energy to agree. Naming an outcome "OUT" does not supply this equality.

The assertion that each receipt eventually exits also needs a precise quantifier. Every member of a finite set may have a finite exit schedule, yet an indefinitely replenished population need not empty at any finite time. A stochastic model can have eventual exit with probability one while having an infinite mean waiting time. Neither a uniform finite completion time nor a finite luminosity follows from individual eventual realization. The waiting-time distribution is part of the observable dynamics.

The parent-facing scalar quotient severely limits accessible information. If two internal states ρ and σ induce the same output channel state, no measurement on that output distinguishes them. More generally, a completely positive trace-preserving channel Φ contracts trace distance,

$$\frac{1}{2}\|\Phi(\rho)-\Phi(\sigma)\|_1 \le \frac{1}{2}\|\rho-\sigma\|_1. \tag{6}$$

This is the operational frame already used in Chapter VI. A child history that never changes any allowed parent output is not parent-observable. A proposed test must identify which output distribution, response function or conserved scalar actually depends on the child data.

Preserving custody is also different from duplicating an arbitrary quantum state. Side tags and ancestry labels may record classical provenance, but a coherent state cannot be copied freely into independently readable ledgers. The Home map must specify whether it is an isometry, a coarse-graining channel, a conditional instrument or some other well-defined transformation, and carry correlations with the environment accordingly.

5. Re-indexing a completed boundary into a new floor

The original handoff identity is

$$A_0 = \frac{A_H}{4\pi D}, \tag{7}$$

where A_H is the unit completion value in the relevant normalization and D is the spatial-dimension count. To avoid confusion with physical distances elsewhere in this chapter, write $D_{\mathrm{count}} = 3$ here. With $A_H = 1$ and $R = 4D_{\mathrm{count}} = 12$, the identity yields $A_0 = 1/(12\pi)$ and $RA_0 = 1/\pi$. The independence of the product from D_{count} holds only when the relation $R = 4D_{\mathrm{count}}$ is retained.

The arithmetic is clear; the direction and mechanism of the handoff need more definition. The working chapter alternates between a child floor and a floor of the parent accumulation. A physical construction should label the input domain $\mathscr{H}_P$, the boundary system $\mathscr{H}_B$ and the output domain $\mathscr{H}_C$ explicitly. "Re-indexing" might mean a change of description of one state, a new effective subsystem, or actual creation of a causally distinct domain. Those meanings have different conservation and observational requirements.

At minimum, a candidate transition must specify the prepared boundary state, the map acting on it, the conserved quantities, the energy source for its realization and the matching conditions that fix the new inventory. The factor $4\pi D_{\mathrm{count}}$ supplies none of those by itself. A unit closure divided by an angular and dimensional factor is a normalization proposal, not a transition amplitude or a nucleation probability.

If the new domain is causally inaccessible, its internal cosmological history cannot be checked directly from the parent. The strongest available test would be a relation among accessible parent quantities, such as mass, area, response and receipt statistics, fixed by the same map that assigns the child floor. Without such a relation, the continuation remains a possible interpretation of the bookkeeping rather than an empirically distinct cosmological model.

6. Formation history and the primordial-inventory interface

The formation ledger in the original proposal separates origin matter, placed in a primordial-hole or trapped-accumulation lane, from outside-origin baryons captured during formation. Its frozen mass is

$$M_{\mathrm{form}} = M_{\mathrm{origin}} + M_{B,\mathrm{form}}. \tag{8}$$

This is a useful decomposition only when the formation process determines both terms and their distributions. It does not yet produce the mass function used in the Supplement (the retired galaxy-halo chapter), the baryon abundance used in Chapter IX, or the thermal initial conditions used in Chapter X.

A mass window alone is not a formation history. A physical primordial-hole scenario must specify the time or temperature of formation, the relation of object mass to the available horizon or source scale, the initial abundance, subsequent accretion, merging and evaporation, and correlations with the surrounding material. The closing review supplies conditional temperature estimates for particular mass windows. Those depend on a radiation-era horizon-mass adapter and a formation efficiency; they should not be promoted to native SAM outputs without those inputs.

The appropriate cross-chapter test is conservation through the whole chain. Formation receipts establish a distribution; that distribution evolves through the thermal history; the surviving population enters the later inventory and halo model. Energy returned during formation or evaporation must appear in the thermal and radiation budgets. If a radius floor leaves stable remnants, their abundance and mass density must be included. A free choice at each stage would remove the predictive force of the proposed common origin.

7. A solved formation-transfer model

7.1 Changing custody does not automatically create pressure

A homogeneous toy model can distinguish a change of inventory label from a change of cosmic dynamics. Let ρ_u be the mass-equivalent density of unbound nonrelativistic matter and ρ_H that of a population of completed Homes, also treated as nonrelativistic objects on large scales. This approximation deliberately ignores individual strong-field regions and asks only how a declared transfer affects a coarse-grained background. Use $N = \log a$ and a prime for d/dN.

Suppose a fraction $\lambda\, dN$ of the unbound inventory transfers during each logarithmic expansion interval, with $\lambda \ge 0$ constant. If transfer is lossless and both populations have zero effective pressure, the continuity equations are

$$\rho_u' + 3\rho_u = -\lambda\rho_u, \qquad \rho_H' + 3\rho_H = \lambda\rho_u. \tag{9}$$

For initial values at $a = 1$, direct integration gives

$$\rho_u(a) = \rho_{u0} a^{-3-\lambda}, \tag{10}$$

$$\rho_H(a) = a^{-3}\left[\rho_{H0} + \rho_{u0}(1 - a^{-\lambda})\right]. \tag{11}$$

Adding the equations or solutions yields $\rho_u + \rho_H = (\rho_{u0} + \rho_{H0})a^{-3}$. Transfer changes custody but leaves the total dust dilution unchanged. In particular, it does not create a constant-density term or negative pressure simply because the destination is a completed surface.

This is not a derivation of the actual Home formation rate. It is a counterexample to an unsupported inference: a conserved conversion of one pressureless category into another cannot by itself explain cosmic acceleration. A mechanism producing a different effective pressure must supply its stress, work or exchange law. If binding energy is released, the radiation or other receiving sector must appear explicitly instead of being silently removed from the homogeneous account.

7.2 Including a paid radiative fraction

Let a fraction $0 \le \beta \le 1$ of the transferred mass-equivalent energy enter radiation and the remainder enter Homes. Then

$$\rho_u' + 3\rho_u = -\lambda\rho_u, \tag{12}$$

$$\rho_H' + 3\rho_H = (1-\beta)\lambda\rho_u, \tag{13}$$

$$\rho_r' + 4\rho_r = \beta\lambda\rho_u. \tag{14}$$

The first solution remains (10); the second becomes $a^{-3}[\rho_{H0} + (1-\beta)\rho_{u0}(1-a^{-\lambda})]$. For $\lambda \neq 1$, the radiation solution is

$$\rho_r(a) = a^{-4}\left[\rho_{r0} + \frac{\beta\lambda\rho_{u0}}{1-\lambda}(a^{1-\lambda} - 1)\right]. \tag{15}$$

At $\lambda = 1$, the bracket is $\rho_{r0} + \beta\rho_{u0}\log a$, the continuous limit of the same expression. All densities remain nonnegative for $a \ge 1$ and nonnegative initial values. For $\lambda > 1$, both $a^{1-\lambda} - 1$ and $1 - \lambda$ are negative, so their ratio is still nonnegative.

Summing the three equations gives

$$\rho_{\rm tot}' + 3\left(\rho_{\rm tot} + \frac{p_{\rm tot}}{c^2}\right) = 0, \qquad p_{\rm tot} = \frac{1}{3}\rho_r c^2. \tag{16}$$

The exchange terms cancel, while radiation retains its additional expansion work. Consequently the mass-equivalent energy in a comoving volume is not constant when radiation is present; it decreases by the pressure work. That does not represent an unpaid local transfer. It is exactly the homogeneous continuity balance of the adopted geometry.

For an elementary numerical check choose $\rho_{u0} = 1$, $\rho_{H0} = \rho_{r0} = 0$, $\lambda = 1/2$, $\beta = 1/4$ and $a = 4$. The densities are $1/128$, $3/512$ and $1/1024$, respectively. Their sum is $15/1024$, smaller than the lossless dust value $1/64 = 16/1024$ because the produced radiation has performed expansion work. The example uses arbitrary density units and does not assign those parameters to SAM. It supplies an exact test for any implementation claiming to conserve the same transfer ledger.

8. The same present inventory can have different futures

A second solved example shows why a present-day complement cannot determine fate. Adopt a spatially flat homogeneous comparator with pressureless matter and a vacuum-like component of pressure $p_X = -\rho_X c^2$. Permit an explicitly prescribed transfer $Q = 3\xi H\rho_X$ from that component into matter, where $0 < \xi < 1$ is constant. The continuity equations become

$$\rho_X' = -3\xi\rho_X, \qquad \rho_m' + 3\rho_m = 3\xi\rho_X. \tag{17}$$

Their exact solutions are

$$\rho_X(a) = \rho_{X0} a^{-3\xi}, \tag{18}$$

$$\rho_m(a) = \rho_{m0} a^{-3} + \frac{\xi\rho_{X0}}{1-\xi}(a^{-3\xi} - a^{-3}). \tag{19}$$

The right-hand term is nonnegative for $a \ge 1$. At $\xi = 0$ the noninteracting cosmological-constant solution is recovered; at $\xi = 1$ the matter solution acquires the limit $3\rho_{X0}a^{-3}\log a$ in addition to its initial dust term.

The initial fractions can be chosen to equal the same $1/\pi$ and $(\pi-1)/\pi$ used in the original adapter. Nevertheless, for every fixed $0 < \xi < 1$, both densities vanish in the infinite-expansion limit. At late times their ratio approaches $\rho_m/\rho_X = \xi/(1-\xi)$ and the total density is asymptotic to $\rho_{X0}a^{-3\xi}/(1-\xi)$. The effective equation of state approaches $w_{\rm eff} = -(1-\xi)$.

Using the flat Friedmann relation then gives $H \propto a^{-3\xi/2}$. Integrating $\dot{a} = aH$ yields the asymptotic power law

$$a(t) \propto t^{2/(3\xi)}, \qquad H(t) \sim \frac{2}{3\xi t}, \tag{20}$$

up to a shift of the time origin and a positive normalization. Thus H tends to zero, whereas the noninteracting model with the same initial inventory tends to a positive constant. For $0 < \xi < 2/3$ the power exceeds one and expansion still accelerates. Acceleration and a nonzero limiting Hubble rate are therefore distinct properties.

This toy model is not proposed as a fit or a new SAM law. Its exchange was chosen openly to expose a logical dependency. The value of the present inventory does not fix Q, and the future changes when Q changes. A Home-based explanation of the background would have to derive an exchange law, show that it is compatible with local conservation and observations, and state whether its pressure assignment is appropriate. The toy family supplies an exact comparison against which that derivation can be checked.

9. Global causal access requires the whole expansion history

A completed local boundary and a cosmological event horizon are different objects. In a spatially flat homogeneous metric, the comoving radial distance that an outgoing light signal can cover after time t is

$$\chi_{\rm future}(t) = c\int_t^{t_{\rm end}} \frac{dt'}{a(t')}, \tag{21}$$

where t_{end} is the future endpoint of the chosen solution. A finite integral defines a finite future comoving reach for that observer. The corresponding proper event-horizon distance, when applicable, is $a(t)\chi_{\mathrm{future}}(t)$. This is a global statement about an entire future solution, not a local statement at one value of the density fraction.

For exponential expansion $a \propto e^{H_\infty t}$ with constant positive H_∞, the proper distance is c/H_∞. For a future power law $a \propto t^p$ extending to infinite time, the integral converges only when $p > 1$. In that case the proper horizon distance is $ct/(p-1)$, which grows without bound even though the comoving future reach is finite. If $p \le 1$, the integral diverges and this future event horizon is absent.

Applied to (20), the criterion is $\xi < 2/3$. The interacting example can therefore have an event horizon while its Hubble rate tends to zero. Neither the existence of a horizon nor its proper size follows from multiplying a road fraction by a background number. A future recollapse, finite-time endpoint, changing spatial curvature or different propagation law would require a new causal calculation.

This distinction also clarifies the Home continuation. Calling a domain inaccessible to its parent may be a rule of the proposed quantum channel, a feature of the global geometry, or both. A geometric horizon does not automatically impose a particular quantum custody map. A custody map that removes all parent-readable distinctions does not automatically determine a spacetime causal diagram. The model should identify which statement it makes and derive the relation if it claims both.

10. An explicit entropy ledger for receipts

Energy conservation does not determine how much information a receipt contains. A minimal classical example has two possible receipt values with probabilities p and $1-p$. Its dimensionless Shannon entropy is

$$H_2(p) = -p\log p - (1-p)\log(1-p), \tag{22}$$

with the continuous endpoint convention $0\log 0 = 0$. A deterministic outward receipt can carry nonzero energy and zero receipt entropy. A random receipt can carry entropy even when the mean energy is held fixed by suitable channel energies. Thus the scalar mass quotient and the information revealed by its realization are separate observables.

If a classical register R stores the receipt and the conditioned boundary state is ρ_r, the joint state is $\sum_r p_r |r\rangle\langle r| \otimes \rho_r$. Its dimensionless von Neumann entropy is

$$s(BR) = H(p) + \sum_r p_r s(\rho_r). \tag{23}$$

Tracing out the register gives $\overline{\rho}_B = \sum_r p_r \rho_r$, and the mutual information is $I(B:R) = s(\overline{\rho}_B) - \sum_r p_r s(\rho_r)$. Adding the marginal entropies without subtracting this mutual information double-counts correlations. Any claim that the surface preserves or releases information must therefore specify whether it means a joint entropy, a conditioned entropy or an accessible record.

For a quasistatic thermal comparator with Hawking temperature T_H, removal of energy $\delta E > 0$ gives a horizon entropy change approximately $-\delta E/T_H$. An ideal equilibrium massless radiation gas with zero chemical potential satisfies $p = u/3$ and $Ts = u + p$, hence has entropy-to-energy ratio $4/(3T)$, suggesting a positive combined balance when $T = T_H$. That example is not a proof for a filtered, nonthermal or correlated finite pulse. The actual channel state and greybody propagation determine its entropy, and finite reservoir corrections belong to Chapter VIII's exact transition account.

A formation floor makes the bookkeeping more restrictive. Once the mutable exterior energy vanishes, a thermal law cannot continue spending the frozen inventory unless the floor rule is changed. If an external controller resets receipt registers, its work and entropy export also belong in the enlarged account. A complete quantum instrument can preserve global information while making the parent output highly mixed, but that is a statement about the full transformation, not a consequence of the word "custody." The receipt model supplies the variables needed to formulate such a transformation without assuming its answer.

The solved examples identify three different senses of conservation. A finite Home recurrence conserves its receipt account after all inflows and outputs are included. A homogeneous cosmology conserves stress through the continuity equation, including pressure work in an expanding volume. A quantum channel conserves normalization while its accessible entropy can change because correlations move between subsystems. None of these statements can replace the others. A parent mass quotient does not establish a cosmological equation of state, and a conserved global quantum state does not make every detail readable by the parent. Conversely, failure to recover an internal history from one scalar channel does not prove that the full evolution destroys information. A complete continuation mechanism should provide a translation among these three accounts and state the approximation under which each is valid. This requirement is constructive: the receipt sequence, homogeneous transfer solutions and channel-state formulas above give explicit limiting cases against which such a mechanism can be tested.

11. A finite program for deciding the fate proposal

The next calculation should begin with an observation map rather than an asymptote. Define a road observable on finite routes, state its clocks and energy conservation law, and derive its effect on luminosity distance, angular size, radial distance and redshift. Equations the distance-duality ratio of Chapter XII, Section 8 and the inferred-rate relation of Chapter XII, Section 8 then provide immediate consistency checks. A model that only changes the name of a standard parameter will not separate itself from the adapter.

The next dynamical step is to derive an evolution law from the retained source ensemble. It must establish whether $1/\pi$ is an attractor, an initial condition or merely a normalization. It must also identify the pressure and exchange terms that determine $H(a)$. A finite-redshift prediction with an uncertainty and failure criterion is more informative than an unreachable future limit.

For the Home continuation, the shortest route is a single surface model shared with Chapters VIII and XI. It should reproduce the proposed custody recurrence, respect the conditional closure results of Chapter V, and yield both a thermal response and a perturbative response. The absence of one observable does not make the model untestable if another allowed output carries information, but that output must be calculated before a comparison is selected.

Finally, the numerical ledger should remain frozen during each test. If a road factor is changed to preserve distance duality, an inventory is shifted to fit a new cosmological likelihood, or a boundary impedance is adjusted to match ringdown, the change should be recorded as a new model branch. This practice allows the count program to learn from failure without retrospectively labeling every close expression a prediction.

SAM VOLUME I

Evidence, closures and remaining physical assignments

Appendix

Closing synthesis: established results and the physical joins ahead

The strength of this volume lies in keeping a source, its receiving law, and the result of that law in the same account. Its first chapters contain substantial mathematical and computational developments: a complete interacting source–field action, finite-window clock certificates, source-sensitive retention and nonclosure results, spatial receiving constructions, controlled signal and record protocols, and strong-field conditional theorems. Those results survive even where the physical identification of an apparatus, geometry, or cosmic inventory remains unfinished. The later chapters extend the account to thermal boundaries, cosmology, halos, radiation, and the Home continuation. Their standing depends on the additional premises at each extension.

This synthesis replaces the earlier closing review and its interface appendix as the current reading guide. Their chapter-item identifiers are retained below so that no question disappears merely because the edition changes. References identify chapters and sections rather than assembled pages, which changed when the expanded Chapters II–VII were incorporated. Historical numerical outputs are reported from their source records; the present editorial synthesis neither reruns nor newly certifies them. Repository-only claims remain source-reported until their precise hypotheses, operands, and evidence are incorporated.

1. A common standard of claims

A result is *derived within a model* when it follows from its stated premises and domain. A measured value is *recovered under an adapter* when the receiving law supplies the structure responsible for agreement. An expression is *selected or matched* when comparison helped choose it. A *prospective prediction* requires an observable, selection rule, uncertainty treatment, and failure criterion fixed before its test data are examined. A *physical closure* additionally supplies the interface that makes the model quantity the measured one. These descriptions can coexist: an exact conditional calculation can be mathematically complete while its physical coupling remains open.

A certificate concerns its declared preparation, observable, interval, domain, and error allowance. A finite-time estimate is not automatically an asymptotic decay law. A positive global energy form is not a pointwise positive stress tensor. A coordinate singularity is not an observer-independent obstruction to crossing. A retained classical exterior is not a dynamical theory for rotating, moving, or self-gravitating sources. A local encoded record is not an intrinsic remote-field measurement. A finite thermal kernel is not an absolute luminosity without a populated source and a realization clock.

The distinction between numerical error and model uncertainty is equally important. Outward arithmetic enclosures can be very narrow without determining physical calibration, experimental systematics, or the uncertainty associated with a selected formula. Likewise, propagating the measured CMB temperature through a fixed cosmological cascade produces a conditional input-error budget, not a model-independent uncertainty on the Hubble constant. Discrete choices require explicit accounting; they need not be misrepresented as ordinary continuous fit parameters.

2. What the source chapters establish and retain

Chapter VII: the actual interacting clock. The signed Write construction, all six driven lanes, the complete source energy, two real field polarizations, both counterterms, and the correlated hard preparation must remain together. Chapter VII, Sections 6–7, establishes positive actual phase rates on the declared one-bare-cycle interval and transports the complete action into that phase coordinate. The Berry one-form is retained. The rate is history-dependent, and its inverse is a pointwise Jacobian, not an independently chosen control or the inverse accumulated phase. Neither six lane certificates nor their verification counts establish arbitrarily long autonomous operation. The source-to-occupied-path and physical-unit joins remain separate questions.

Chapter II: a receiver-dependent retained account. Present output, predictive state, timed history, and energetic complement are different objects. Equal cardinality between two ledgers does not identify their states or supply a transition map. The finite source and paid reference protocols establish operational distinctions under declared receivers. Nonclosure ranks and finite invariant upper constructions must retain their actual scope; a computed lower bound is not a proven minimal invariant closure. The source-to-space extension requires a complete energy receiver and a physical incidence, including the commuting even energy, interaction, field, reference, and counterterms.

Chapter III: spatial response with its coupling exposed. The Poisson equation and test-body law explicitly import the Newtonian coupling and the passive-to-inertial mass assignment. Their Green functions, source profiles, finite boundaries, energy accounts, and continuum errors are substantive consequences of those assignments. Static potential and an endpoint clock do not determine the relativistic spatial response. The microscopic localization, physical ruler, moving-source completion, and spatial metric coefficient cannot be supplied merely by renaming a retained count as a density.

Chapter IV: complete histories and actual readout. The uniform complete lapse transports the full Hamiltonian through one internal-age map. Local control and donor preparations, unequal accepted branches, paired polarization records, complete failure accounting, and switched-reference constructions are developed further in Sections 79–82. These current results take precedence over the explicitly retained predecessor dependency ledger. Section 82 supplies exact uniform switched-reference and spatial-residual machinery but does not thereby evaluate the physical nonuniform cap mean, angular noise, or external vacuum-work account. Huge finite preparation budgets remain existence constructions under specified apparatus laws. Appendix I distinguishes archived external-contact receipts from new experimental reductions; its binary-pulsar receipt is not an independent timing fit.

Chapter V: strong-field statements with explicit domains. The source lift reaching unity is not a statement that the floor plus lift reaches unity. Static adapter results, approach theorems, curved-port protocols, recovered thermal kernels, and finite area-entry algebra have distinct premises. The source-law discussion, including the radial null-energy-condition alternative, must govern every later reference to infinite approach or sealing. Section 13 and the subsequent developments retain actual source, preparation, backaction, field complement, and resource interfaces. A comparator that supplies an observable orbit does not become a selected SAM stress law.

Chapter VI: records with their physical payments. The record and accessibility results use ordinary quantum states and allowed channels. The shared-field moving-source and coherent detector constructions retain all outcomes and physical complements. The final "Completed and remaining physical objects" statement distinguishes a paid, co-located encoder from a primitive route-selection mechanism and from an intrinsic remote port. Autonomous centre-of-mass dynamics, microscopic routing, calibration, and resource realization remain to be supplied. Later references to information custody must respect this channel dependence rather than imply a new probability law or unobserved signalling.

3. The cross-volume physical interfaces

Five interfaces recur. First, one physical unit chart must connect the model's time, ruler, action, and energy to measured quantities without changing calibration between tests. Second, a self-consistent curved source–field action and renormalized stress must determine geometry and motion, rather than receive them separately. Third, a boundary condition for the actual two-polarization field must connect thermal and ringdown problems. Fourth, microscopic control and apparatus laws must realize the already specified donor, routing, pointer, vacuum-ramp, and reset accounts. Fifth, long-time preparation, replenishment, and cosmological evolution must connect finite-source calculations to macroscopic rates.

These are research interfaces, not permission to erase completed conditional work. A physical unit chart would close a dimensional join but would not derive a spatial metric. A boundary condition would define a spectral problem but would not alone specify its excitation or luminosity. A radius mechanism from a different source lineage can be investigated as a candidate coupling; identifying its radius with Chapter VII's coupling radius requires a map, not numerical proximity.

For gravity, endpoint redshift, light propagation, static second-order motion, frame dragging, orbital radiation, preferred-frame effects, and clock-species universality are separate exams. A Schwarzschild exterior recovers its test-body observables by assumption. It does not establish every parametrized post-Newtonian parameter of a full many-body theory, nor strong-equivalence-principle behavior of self-gravitating bodies. A binary radiation-flux calculation must carry actual source motion, tensor coupling, conservation, and physical normalization into its comparison.

For cosmology, an inventory identity, a background evolution equation, a recombination calculation, and a likelihood comparison are also separate. Numerical proximity of selected parameters to posterior centroids neither derives their physical allocation nor makes the full spectrum test vacuous. It does mean that spectrum agreement cannot be counted as independent evidence for every formula chosen using those same data. Independent instruments help test transportability and systematics, but do not erase selection history.

4. The chapter register

The identifiers below correspond to the earlier review's register. A retained question is described at its present scope; an acknowledged answer is not listed again as though absent. Additional interfaces from the earlier appendix to Chapters I–VII of the 19 October edition (now II–VIII) are carried inside the relevant paragraphs.

Chapter VII: source preparation, isolation, and calibration

I.1–I.2: isolation and radiation. The review attributes a hard-branch isolation proof to the resolved-clock record H002471, cited by Chapter VII. Incorporation requires its exact charge-sector decomposition, thresholds, multiplicities, and separation inequalities. A complement form bound and trial subspace must be used with the relevant min–max statement; a summary alone does not authenticate a new continuum certificate. Existing Chapter VII provenance remains intact. The review also proposes a golden-rule coefficient from Chapter IV, Section 77. Its quoted 3.7×10^{-41} per model time is a reported asymptotic estimate, not a newly established relaxation law. Uniform control of the large-time limit, depletion, repeated preparation, and conversion to physical time are still needed. A finite emitted-number bound cannot close those questions or exclude every cosmological source term.

I.3–I.6: physical source joins. The retained source radius near 0.1635 and the separate radius-mechanism value near 0.151 must be identified by lineage and meaning before comparison. The complete-Hamiltonian coupling principle remains distinct from changing only one source coefficient. One physical calibration must fix time and energy, with a ruler where

spatial observations are used. The occupied-path relation between the original phase ledger and quantum generator must carry common phase, commuting even operator, and preparation. The original connector-to-occupied-source-to-joint-field incidence, physical selection of s and α, modified vacuum, hard-Hamiltonian control port, and actual timing reference remain explicit interfaces. Exact inverse and error-budget constructions do not constitute achieved hardware calibration.

Chapter II: the floor and retained state

II.1–II.2: Berry content and floor normalization. The joint Berry contribution must be evaluated on the clock-active preparation before it can be replaced by a reduced phase formula. The floor drops out of static gradients and shared linear differences, but that does not make it absent from every nonlinear receiver. For the specifically normalized raw-floor square root, a fixed independent gravitational calibration would produce the review's extra linear factor $1/(1-A_0)$. Redshift experiments can test that specified hypothesis. They do not uniquely derive subtraction before every square root, prove universality across all clocks, or prove that the floor is observable only cosmologically. Any claimed exclusion must state the mass, force, and potential calibration held fixed.

II.3–II.4 and retained interfaces. The relation among cadence, interval, distance, and a physical metric remains to be written with its time and length units. Preferred-frame tests require a specified preferred-frame dynamics, not merely the existence of a substrate vocabulary. A physically smeared energy density needs declared incidence for source, field, interaction, boundary, support, and reference. The paid Write/inverse/wait protocol still needs a physical realization. The actual frustrated-spin invariant closure retains its nonclosure bounds and finite upper construction without an unproved minimum. These questions were explicit in the earlier interface appendix and are preserved here.

Chapter III: static gravity and its dynamical extension

III.1–III.4. The spatial-response coefficient κ_s and its connection to γ remain a decisive physical join. The review's criterion $\gamma = m-1$ follows only after its chosen radius–lapse, carrier-speed, and metric identifications. Applying it to another source family requires common definitions, units, route cells, and regime of validity. It is a conditional diagnostic, not a completed derivation or falsification of all SAM carriers. The static Schwarzschild adapter supplies the corresponding test-body second-order lapse and perihelion behavior; a full Nordtvedt prediction needs more than that exterior. A velocity-coupled/vector sector remains necessary for dragging. The nonuniform source energy/stress form, original-state remainder, apparatus errors, spatial metric response, moving-source completion, and common SI chart retain their Chapter III and later-chapter scopes.

Chapter IV: signals, switching, and finite resources

IV.1–IV.3. The sign of an isolated susceptibility must not be inferred from a total response dominated by uniform time change. The raw-floor comparison has the conditional status stated under II.2. Section 77's finite radiation result remains valuable at its established interval; the golden-rule suggestion does not replace the open long-time physical relaxation and cosmological amplitude rows. The positive Euclidean rate measure and its transform also need a detector and a mapping to cosmological variables before becoming a distance-rate law.

IV.4–IV.6. The distance-ratio material should be identified by its receiving assumptions and original provenance; editorial relocation must preserve its actual mathematics and archived benchmark distinctions. Preparation counts of order 10^{58} are finite existence budgets, not feasible apparatus proposals. The present Sections 79–82 must be used to update the quantitative nonuniform cap problem: actual unequal branches, incoming hard field, transported shell return, both signed insertions, full angular noise, and paired records belong in the same estimate. Uniform switched existence and a local record do not replace the missing nonuniform output bound. Microscopic donor/controller realization, common units, throughput and reset, vacuum-ramp work, and long-time repeated preparation remain separate physical obligations.

Chapter V: approach, boundary, and source law

V.1–V.3. The source-coordinate edge, the finite-impulse conditional theorem, and comparator orbit calculations are already present. Their questions should not be reopened as though the chapter omitted them. The retained resolution is conditional: a source satisfying the theorem's radial energy-condition hypotheses is constrained by it; otherwise the actual violation or flux must be calculated and propagated. No universal infinite-cost crossing rule follows independently. Chapter I declares the road-speed postulate (Section 9.2); under that declaration the non-arrival of a massive traveler is unconditional, and without it the resolution is the conditional one stated here. Chapter I declares the road-speed postulate (Section 9.2); under that declaration the non-arrival of a massive traveler is unconditional, and without it the resolution is the conditional one stated here. Comparator photon circles and innermost stable circular orbits are observables of their specified metric, not derived physical stress assignments.

V.4–V.5. A rotating extension is required before importing Kerr comparisons into Chapter XI. The source-to-curved-port coupling must retain support, angular complement, initial dressing, counterterms, reservoir, switching, backaction, and reference energy. A curved source action with renormalized original-state stress must select the traveler and boundary supply. Autonomous recharge, repetition, selection, and stopping remain open. The reported roughly fifteen-percent amplitude–distance interval can be sharpened within its source certificate; repetition alone does not remove that systematic interval.

Chapter VI: accessibility and physical records

VI.1–VI.3. A second-order shared-field velocity/dephasing coefficient must be computed from the actual propagators if it is to replace an existing bound; a bound is not its coefficient. The chapter already states its use of ordinary quantum probability. The remaining physical questions are microscopic routing and control, source-profile calibration using the derived stability radius, autonomous centre-of-mass or boost-deformed action, unforced arm geometry and source-clock preparation, a genuinely remote intrinsic source observable with propagation and extraction, and all vacuum-ramp, actuator, bath, and reset payments. Conditional erasure and feed-forward are different experiments and cannot silently alter the all-attempt account.

Chapter VIII: thermal scale, entropy, and information

VII.1–VII.2. The temperature factor ledger already identifies the static metric, normalized Killing time, Hawking conversion, and retained quarter-area premise. Preserve it prominently. Recovering the Bekenstein–Hawking first law under these premises is distinct from deriving the quarter-area allocation from the microscopic source. Exact finite-entry and Catalan-series consequences remain established conditional mathematics. Their last-entry regime should be compared with discrete-area proposals without claiming an absolute emission rate.

VII.3–VII.5. Relative entropy is stationary at a smooth reference state, so comparing its first derivative with a nonstationary area observable does not test every proposed area–entropy relation. The informative Si29 comparison must align perturbation order and compare Hessians under the same variables and normalization. Nonzero Jacobian enclosures establish local invertibility at their reference points; an explicit quantitative neighborhood and physical boundary-area map remain to be supplied. A material surface and a causal horizon have different emission and accretion implications; surface-flux constraints require a specified model before application. The two-polarization boundary condition, source-selected boundary state, transition strengths, realization clock, depletion, energy join $\varepsilon_Q = \hbar\omega_{\mathrm{phys}}$, species sum, and outward/inward correlations are still the route from finite kernel to radiation law.

Chapter IX: inventory and Hubble cascade

VIII.1–VIII.2. The finite grammar and its selection history must accompany each cosmological identity. The earlier review's broad enumeration is a sensitivity exercise until its script, candidate rules, duplicates, target windows, and normalization are auditable. Candidate counts alone are not calibrated probabilities of accidental agreement. The baryon allocation factor $(1-\chi)$, exponent $\mathscr{L}/V = 6$, and weight $7/4$ require physical allocation or preparation laws; reusing upstream integers does not alone select their cosmological meaning.

VIII.3–VIII.4. The clean $\Omega_m = 1/\pi$ and effective inventory must each have an operator, background role, and comparison protocol fixed before use. One cannot switch between them after seeing which agrees better. A frozen Hubble ledger should list thermal anchor, constants, density convention, radiation and neutrino treatment, and all external inputs. The quoted central value near $67.2504\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is conditional on that cascade. The historical temperature-only error near 0.021 is not total model uncertainty or a claim of superior cosmological precision. Literature comparisons marked "verify" in the earlier review are not retained as established exclusions.

VIII.5–VIII.6. A nucleosynthesis calculation from the selected baryon-to-photon ratio can yield useful independent light-element tests, with nuclear, thermal, and observational errors kept separate. A standard network would test the packet under standard thermal physics, not constitute native substrate nucleosynthesis. The complement $1-\Omega_m$ does not by itself determine a stress tensor or an equation of state. A cosmological-constant adapter and a dynamical $w(z)$ model must be distinguished. Accumulation-based distance or redshift mechanisms must make explicit predictions for spectral redshift, time dilation, temperature evolution, surface brightness, and distance relations before being compared with observations.

Chapter X: recombination and spectrum comparison

IX.1–IX.3. The native recombination lane and the standard-CAMB full-shape lane answer different questions and must remain separate. The packet was frozen before spectrum generation, but its selectors had already been compared with Planck parameters. Report that chronology. The full-shape outputs are source-reported comparisons, not new reductions in this edition. The earlier comparator's second peak, 520, was a transcription error. The published Planck positions are $(220.0\pm0.5,\ 537.5\pm0.7,\ 810.8\pm0.7)$ (Planck 2015 results I, Table 7), against the reported SAM peaks $(220, 535, 812)$, a -0.47% difference at the second peak, as Chapter X now states; an identical-estimator reanalysis remains outstanding. Use one peak finder and windows on both spectra. Report sample sizes, windows, covariance, nuisance handling, and comparison with a baseline on the same data for the ACT TT statistic near 1.30; that number alone does not determine a significance.

IX.4. Complete the native recombination physics, reconcile the drag ruler under one background ledger, and use full likelihoods where warranted. A governing equation for the amplitude/tilt/optical-depth triplet remains distinct from a selected formula for each parameter. Helium, two-photon and Lyman-α processes, additional atomic levels, and a nucleosynthesis network require their own physical inputs. Lack of an optimizer does not remove choices already made before a run.

Supplement (the retired Chapter X): many-source organization and halo tests

X.1–X.3. The observed plateau motivated $10/\pi$, so four-decimal agreement is not a prospective prediction. The review's uncertainty estimate uses a large-sample median approxima-

tion and the reported scatter; it is illustrative until the actual distribution, weighting, correlation, and systematic model are specified. Use a justified resampling or sampling model for the uncertainty. An untouched sample with frozen selection, mass-to-light assumptions, statistic, and rejection rule can supply the prospective CR034 test. The radial organization law, disk geometry, sensitivity to stellar mass-to-light ratio, profile parameters, and like-for-like radial-acceleration comparison remain necessary.

X.4–X.5. A label such as trapped accumulation does not specify a dark component's dynamics. State its compact fraction, mass function, spatial and velocity distributions, baryonic coupling, and formation history. Compact-object, lensing, clustering, and accretion constraints depend on those choices. The heavy-seed ledger and microlensing question remain open; no universal allowed primordial-hole window is assumed here. Many-source nonzero accumulation is not by itself a derived galaxy halo profile.

Chapter XI: radiation and ringdown

XI.1–XI.3. The decisive spectral problem needs the actual field operator, causal boundary condition, excitation, and physical units. A reflecting surface, an absorbing horizon, and a dissipative boundary yield different tests, including possible echoes. Binary-pulsar flux requires a moving-source calculation and energy balance; the retained Chapter IV external-contact receipt does not supply that derivation. Spin must be incorporated before observed rotating remnants are used to assess the nonrotating rational candidates. A nonrotating comparator is useful as a controlled theoretical baseline, not a generic observed remnant.

XI.4–XI.6. The selection histories of $3/8$ and $4/45$ remain explicit. Broader-grammar counts do not turn either into a mode calculation or a probability of correctness. The $1/8$ radiation ceiling needs a precise initial-energy normalization, permitted source class, and accounting of final mass and all outgoing channels before an event can falsify it. Comparisons with area bounds, numerical-relativity efficiencies, or catalog energies require matched assumptions and uncertainties. A flux law, higher-mode structure, spin dependence, and literature placement remain the necessary continuation.

Chapter XII: future road and Home continuation

XII.1–XII.2. The past-light-cone limit $z \to \infty$ does not establish the future-time limit $a \to \infty$. The expression $H_{\infty,\mathrm{readout}} = H_0\Omega_\Lambda^{3/2}$ is conditional on the background adapter, proposed future road limit, and definition of the rate. A constant conversion between two distance coordinates does not by itself change their fractional expansion rates; differentiate the actual time-dependent observable before assigning the extra road factor. The model needs an evolution and conservation law, time coordinate, stability analysis, and finite-redshift distance relations. Distance-duality tests require a stated mapping to luminosity and angular-diameter distances and photon transport. Chapter V's source-law conditional must replace an unconditional physical claim of infinite crossing cost.

XII.3–XII.5. Home re-indexing remains a conjectural continuation. A transition requires a physical process, conserved quantities, matching conditions for child floor and inventory, and energy/information accounting. A parent-readable consequence must pass through an allowed channel under Chapter VI; hidden child labels alone are not an observable. Primordial-hole formation estimates from horizon mass and temperature depend on expansion history, collapse efficiency, particle degrees of freedom, and the mass distribution. The earlier review's approximate epochs are conditional comparators, not derived SAM formation times or universal constraints. Birkhoff-type statements require their gravitational field-equation hypotheses; they cannot be imported during dynamical collapse solely from the static exterior. Any hard-surface claim must specify its own stress, emission, and causal behavior.

5. A coherent route forward

The most economical next work reuses complete source calculations rather than replacing them with numerical identities. A source-measure construction could define inventory and road operators on one declared preparation and ask whether both normalize to A_0R. A boundary instrument could test the baryon survival factor. A correlated completion calculation could test the proposed sixth power rather than assuming independent writes. These are proposed constructions: neither their correction size nor its sign has been established by this volume.

A physical unit chart can then propagate calibration and experimental uncertainty through existing conditional calculations. It must not disguise model-selection uncertainty as arithmetic precision. A single physically selected boundary law could support both the thermal and ringdown problems, with separate preparation and rate calculations. A time-dependent road law could connect finite-redshift observations to a future limit. It cannot be replaced by repeating the shared number $1/\pi$ in two different roles.

For observational contact, prioritize tests whose outcome was not used to select the formula: a predeclared halo sample, a boundary eigenproblem before mode comparison, light-element abundances from a fixed packet, and complete likelihood comparisons under frozen data and nuisance rules. For gravitational contact, prioritize the spatial metric response, moving-source flux, species universality, and the actual nonuniform source-to-detector cap. These choices do not guarantee a discriminator; they make agreement and failure scientifically legible.

The volume therefore closes with a substantial body of exact and conditional source mathematics, not with a completed derivation of cosmology or microscopic matter. Its strongest continuity is methodological: retain the actual state, all physical complements and payments, the receiving law, and the observable at every handoff. Volume II can supply further matter and calibration candidates, but any dependency running back into Volume I must be recorded explicitly rather than presented as an already independent closure.

Provenance

The content of Chapters II–VII was produced in GEN6 Duck runs on the Substrate Ledger Computer (15–17 October 2026; Duck 1.2.2 and 1.2.3, core 0.10.2), recorded in ironclad entries H002628–H002661 and H002700–H002706 and in the campaign folders cited in each chapter's source register; Chapter VI's derivations in the campaigns of 17 October 2026 with native SLC-GEN5-R1 checks; Chapters I and VIII–XIII from the record and from the first Courtroom's sealed CR results of June and July 2026. The co-authorship and in-chapter credits that earlier editions carried were removed on 2 November 2026; the trace is Courtroom II case `voli_provenance/01` (H002821, H002822). Provenance is cited by run, test and folder and will be replaced by SAMA entries as the atlas expands. The editorial synthesis of this appendix is based on the working edition of 19 October 2026 and claims no new native computation.

The chapter source registers remain the authorities for exact operands, archived derivations, receipts, and repository versions. External observational numbers requiring renewed checking are identified as such rather than silently promoted to verified results. Historical review classifications and counts are superseded by the scoped register above: source-reported, conditional, and physically open are deliberately kept distinct.